

第八届（2023 年）
全国高校密码数学挑战赛
参赛论文

对称密码算法的量子电路构造

摘 要

本文研究了基于 NCT 门集的一个优化原位电路实现。问题实质为如何将向量进行映射，使该映射结果与目标结果相同，同时又能保证电路的原位输出和辅助位元素清空，因此，针对目标布尔向量函数的映射问题，将其等价地转换为多个单向量元素构造多项式的问题。

由于目标函数输出与初始输入位的位置不同，对前 128 维进行无消耗的换位操作，保证后续电路的原位实现。根据赛题将目标布尔向量函数分为三个函数，本文考虑将其按照函数划分进行分解和构造。对于 $X_2 \oplus X_3 \oplus X_4$ 函数部分，在后续计算中， X_2 、 X_3 和 X_4 不再调用，故直接在 X_2 的存储位上利用 $CNOT$ 门进行加法运算，保证 32 维向量并行计算，提高电路运行速度。此时电路深度为 2，无辅助位。

对于 $\tau(X_2 \oplus X_3 \oplus X_4)$ 函数部分，其代数标准型属于大型多项式，构造困难，本文考虑利用分块矩阵的思想，将其代数标准型进行分块，把大型多项式分成多个单变量，利用单变量反向构造多项式，使得构造 $\tau(X_2 \oplus X_3 \oplus X_4)$ 函数转化为单变量元素的组合，通过不同的组合搭配得到目标函数。期间，根据组合出现的次数降序排列，以最少的元素进行构造，并且在组合搭配的过程中，利用乘法分配律将加法代替乘法，以此来减少 $Toffoli$ 门，提高整个电路的运行速度。整个构造过程中，为了提高辅助位的利用率，基于共享动态存储的思想，通过清空仅使用一次的 $CNOT$ 门存储位，将辅助位释放，结合部分可组合使用的存储位，进行共享动态存储，有效降低了电路的总宽度。此时电路深度为 22，辅助位有 1216 位，其中 $Toffoli$ 深度为 3。

对于 $L \circ \tau(X_2 \oplus X_3 \oplus X_4)$ 函数部分，该函数属于错位加法运算，构造简单，通过逐次相加与两两相加的并行运算比较，7 层的逐次相加运算，无需可逆运算，未利用辅助位，具有更好的评判指标，因此，选择逐次相加运算构造最终的函数 L 函数。电路深度为 29，辅助位有 1216 位，其中 $Toffoli$ 深度为 3。此时，电路的前 128 位输出，完成了量子电路的映射实现。

本文根据这三部分函数构造电路，在原始存储位上得到了目标输出，最后结合二元域的性质，将其电路可逆计算，以此消除辅助位的元素值，完成目标向量布尔函数的原位输出。最终的电路深度为 51，辅助位有 1216 位，其中 $Toffoli$ 深度为 6。

最后，基于 NCT 门集的优化原位电路由总深度为 51 层（ $Toffoli$ 深度 6）总宽度为 1344 位（辅助位 1216）的电路构造得到，评判指标为 $T - DW - cost = 8064$ 和 $DW - cost = 68544$ 。

关键词：量子电路 NCT 门集 分块矩阵 并行和可逆计算 动态存储

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第一章 引言

为了构造基于 NCT 门集的一个优化原位电路，其问题实质主要为研究：如何将向量进行映射，使该映射得到的多项式与赛题中三个函数的计算结果相同，同时又能保证辅助位上的元素清空，输出存储在初始的输入位上，因此本赛题等价地转换为布尔向量与函数多项式的构造问题。

在赛题二中，主要任务为实现 $L \circ \tau(X_2 \oplus X_3 \oplus X_4)$ ，因此本文考虑将其分成三部分来求解。目标函数与初始输入位的位置不同，故先进行无消耗的换位操作，保证后续电路的原位实现。对于 $X_2 \oplus X_3 \oplus X_4$ 函数，对位相加运算运算量很小，直接利用 $CNOT$ 门进行加法运算得到。电路深度为 2，无辅助位。

对于 $\tau(X_2 \oplus X_3 \oplus X_4)$ 函数，根据其代数标准型，属于大型多项式，构造困难，本文考虑先利用分块矩阵的思想，设计了一种基于分块的大型多项式构造算法，将其代数标准型进行分块，把大型多项式分成多个单变量，使得构造 $\tau(X_2 \oplus X_3 \oplus X_4)$ 函数转化为单变量元素的组合，通过不同的组合搭配得到目标函数，并且在组合搭配的过程中，利用乘法分配律减少乘法的次数，以此来减少 $Toffoli$ 门，同时，结合动态存储，将利用 $CNOT$ 门的辅助位，多次调用存储，提高利用率，以此提高整个电路的运行速度。电路深度为 22，辅助位有 1216 位，其中 $Toffoli$ 深度为 3。算法步骤如图 1-1：

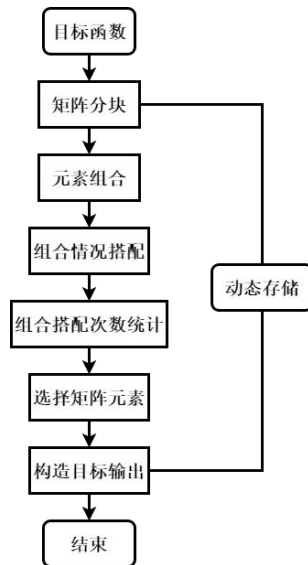


图 1-1 基于分块的大型多项式构造算法

$L \circ \tau(X_2 \oplus X_3 \oplus X_4)$ 函数部分为错位加法，通过逐次相加与两两相加的并行运算比较，逐次相加运算，无需可逆运算，未利用辅助位，具有更好的评判指标。电路深度为 29，辅助位有 1216 位，其中 $Toffoli$ 深度为 3。此时，电路的前 128 位输出，完成了量子电路的映射实现。

本文根据这三部分函数构造电路，最终根据这三部分的电路运行，进行可逆计算，以此消去辅助位的元素值，得到目标向量布尔函数的输出。电路深度为 51，辅助位有 1216 位，其中 *Toffoli* 深度为 6。

最后，通过总深度为 51 层总宽度为 1344 位（辅助位为 1216）的电路完成了构造，实现了基于 *NCT* 门集的优化原位电路，最终， $T - DW - cost = 8064$ ， $DW - cost = 68544$ 。

第二章 电路设计

对于问题的求解，本文使用了分块矩阵的思想简化求解难度，设计了基于分块的大型多项式构造算法，并利用乘法分配律，减少了 *Toffoli* 门个数和计算量。本文将问题实现分为五部分，以下将对每部分进行详细设计表述。

2.1 向量布尔函数分解

由于换位操作看作是无消耗的，故 $F(X_1, X_2, X_3, X_4) = F(X_2, X_3, X_4, X_1)$ 。 (X_2, X_3, X_4, X_1) 为初始的向量，按题目要求需要实现 $(X_2, X_3, X_4, X_1 + L \circ \tau(X_2 \oplus X_3 \oplus X_4))$ ，即实现 $L \circ \tau(X_2 \oplus X_3 \oplus X_4)$ ，即可完成题目要求的电路设计。

根据问题描述， $L \circ \tau(X_2 \oplus X_3 \oplus X_4)$ 由三个函数生成，包括对位相加函数、 τ 函数和 L 函数，因此，整个电路的设计按照这三个函数进行划分，再考虑到辅助位清零的操作，整个电路添加可逆电路实现部分，故该向量布尔函数的电路设计分为四部分。

2.2 线性向量布尔函数对位相加电路设计

$X_2 \oplus X_3 \oplus X_4$ 是 32 个变量 32 维的线性向量布尔函数，由 X_2 、 X_3 和 X_4 每维元素相加得到。由于后续的计算仅利用 $X_2 \oplus X_3 \oplus X_4$ 得到的向量，没有 X_2 、 X_3 和 X_4 单次的调用，因此无需添加辅助位，仅在 X_2 上相加 X_3 和 X_4 ，可以减少一层电路深度和 32 位辅助位的浪费。具体电路描述如下：

#Full depth 1 and *Toffoli* depth 0:

$$\left\{ \begin{array}{l} CNOT_{33,1}, CNOT_{34,2}, CNOT_{35,3}, CNOT_{36,4}, CNOT_{37,5}, CNOT_{38,6}, CNOT_{39,7}, CNOT_{40,8}, \\ CNOT_{41,9}, CNOT_{42,10}, CNOT_{43,11}, CNOT_{44,12}, CNOT_{45,13}, CNOT_{46,14}, CNOT_{47,15}, CNOT_{48,16}, \\ CNOT_{49,17}, CNOT_{50,18}, CNOT_{51,19}, CNOT_{52,20}, CNOT_{53,21}, CNOT_{54,22}, CNOT_{55,23}, CNOT_{56,24}, \\ CNOT_{57,25}, CNOT_{58,26}, CNOT_{59,27}, CNOT_{60,28}, CNOT_{61,29}, CNOT_{62,30}, CNOT_{63,31}, CNOT_{64,32} \end{array} \right\}$$

#Full depth 2 and *Toffoli* depth 0:

$$\left\{ \begin{array}{l} CNOT_{65,1}, CNOT_{66,2}, CNOT_{67,3}, CNOT_{68,4}, CNOT_{69,5}, CNOT_{70,6}, CNOT_{71,7}, CNOT_{72,8}, \\ CNOT_{73,9}, CNOT_{74,10}, CNOT_{75,11}, CNOT_{76,12}, CNOT_{77,13}, CNOT_{78,14}, CNOT_{79,15}, CNOT_{80,16}, \\ CNOT_{81,17}, CNOT_{82,18}, CNOT_{83,19}, CNOT_{84,20}, CNOT_{85,21}, CNOT_{86,22}, CNOT_{87,23}, CNOT_{88,24}, \\ CNOT_{89,25}, CNOT_{90,26}, CNOT_{91,27}, CNOT_{92,28}, CNOT_{93,29}, CNOT_{94,30}, CNOT_{95,31}, CNOT_{96,32} \end{array} \right\}$$

故实现了对位相加部分函数 $F(X_1, X_2, X_3, X_4) = (X_2 \oplus X_3 \oplus X_4, X_3, X_4, X_1)$ 。

电路宽度为 128，辅助位为 0，电路总深度为 2，*Toffoli* 深度为 0。

2.3 非线性向量布尔函数 τ 电路设计

τ 为一个 32 变量 32 维的非线性向量布尔函数,由 4 个 8 变量 8 维的向量布尔函数 S 组成,4 个 S 函数结构完全相同,每个 S 函数由 8 个 y_i 构成,结构类似,因此,以 y_i 为例进行分析。

2.3.1 基于分块矩阵的 τ 函数分解

2.3.1.1 分块矩阵概念

在介绍 τ 函数构造前,先介绍分块矩阵的概念。

把一个大矩阵看成是由一些小矩阵组成的,就如矩阵是由数组成的一样,在运算中,把这些小矩阵当作数一样处理^[1]。例如:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} = \begin{pmatrix} E_2 & O \\ A_1 & E_2 \end{pmatrix}, A_1 = \begin{pmatrix} -1 & 2 \\ 1 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

分块矩阵运算的优点不仅仅在于提高计算效率,还在于其对于内存的利用率更高。由于分块矩阵运算可以将大型矩阵分解成若干个小块,因此可以将这些小块存储在内存中,从而减少内存的占用。这对于处理大型矩阵时尤为重要,因为大型矩阵往往需要占用大量的内存空间。分块矩阵运算是一种高效的矩阵计算方法,它可以大大提高矩阵计算的效率,特别是在处理大型矩阵时更为明显。同时,分块矩阵运算还可以提高内存的利用率,从而减少内存的占用。

2.3.1.2 基于分块的大型多项式构造算法

由分块矩阵的概念,可以联想到,对于大型的多项式可以进行分块构造,将大型的多项式分解成若干个小型的多项式,小型多项式构造简单,那么只需构造多个小型多项式即可,构造难度大大降低。与分块矩阵的优点相同,该方法能提高计算效率,并且存储率更高。

每个 S 函数有 8 个 y_i 函数,8 个 y_i 函数可以并行计算。 y_i 的代数标准型属于大型多项式,由 128 个小型多项式相加构成,要构造 y_i 这样无规律的多项式过于困难,因此,可以考虑将其进行分解,利用分块矩阵的思想构造该多项式。

y_i 函数整体可以看成是一个大的难以构造的矩阵 $y[X]$,那么根据其代数标准型的结构,可

先将其分解成 128×1 的分块矩阵 $y1 \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_{128} \end{bmatrix}$,每块小矩阵 X_i 为一个 8 变量的多项式。故将

构造 y_i 函数转换成构造 128 个 8 变量多项式,然而 8 变量多项式的构造依旧繁琐,为了最简化的构造,利用分块矩阵思想将矩阵 $y[X]$ 分解到不能再分为止,将矩阵化为最小的形式。

$$\begin{aligned}
 y[X] &\rightarrow y1 \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_{128} \end{bmatrix} \rightarrow y2 \begin{bmatrix} X_{1,1}, X_{1,2} \\ X_{2,1}, X_{2,2} \\ \vdots \\ X_{128,1}, X_{128,2} \end{bmatrix} \\
 &\rightarrow y3 \begin{bmatrix} X_{1,1}, X_{1,2}, X_{1,3}, X_{1,4} \\ X_{2,1}, X_{2,2}, X_{2,3}, X_{2,4} \\ \vdots \\ X_{128,1}, X_{128,2}, X_{128,3}, X_{128,4} \end{bmatrix} \rightarrow y4 \begin{bmatrix} X_{1,1}, X_{1,2}, X_{1,3}, X_{1,4}, X_{1,5}, X_{1,6}, X_{1,7}, X_{1,8} \\ X_{2,1}, X_{2,2}, X_{2,3}, X_{2,4}, X_{2,5}, X_{2,6}, X_{2,7}, X_{2,8} \\ \vdots \\ X_{128,1}, X_{128,2}, X_{128,3}, X_{128,4}, X_{128,5}, X_{128,6}, X_{128,7}, X_{128,8} \end{bmatrix}
 \end{aligned}$$

分解成最小分块矩阵 $y4$ 后，构造 y_i 函数，只需选择矩阵 $y4$ 每个元素的数值，而根据代数标准型，可以看出每一列的元素只有 2 种选择(X 和 $X + 1$)。

将上述的分解步骤可逆后，即可看成大型多项式的构造过程。

2.3.2 基于分块矩阵的 τ 函数构造

在将 y_i 函数利用分块矩阵的思想分解成多个最小矩阵后，能很轻易地构造 $y4$ 矩阵，而 $y4$ 矩阵每一列元素只有 2 种选择。

因此，成功地将大矩阵难以构造的问题解决，转化为构造简单的小矩阵。最终，基于分块矩阵的 y_i 函数分解再进行可逆计算，即可实现 y_i 函数构造。

以 y_i 为例，此时电路中含有的基元素有 8 种 $\{y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8\}$ ，因此，需要构造另外的 8 种基元素 $\{y_1 + 1, y_2 + 1, y_3 + 1, y_4 + 1, y_5 + 1, y_6 + 1, y_7 + 1, y_8 + 1\}$ ，并且有 4 个 S 函数，一共需要构造 32 个基元素。具体电路描述如下：

#Full depth 3 and Toffoli depth 0:

$$\left\{ \begin{array}{l} CNOT_{1,129}, CNOT_{2,130}, CNOT_{3,131}, CNOT_{4,132}, CNOT_{5,133}, CNOT_{6,134}, CNOT_{7,135}, CNOT_{8,136}, \\ CNOT_{9,137}, CNOT_{10,138}, CNOT_{11,139}, CNOT_{12,140}, CNOT_{13,141}, CNOT_{14,142}, CNOT_{15,143}, CNOT_{16,144}, \\ CNOT_{17,145}, CNOT_{18,146}, CNOT_{19,147}, CNOT_{20,148}, CNOT_{21,149}, CNOT_{22,150}, CNOT_{23,151}, CNOT_{24,152}, \\ CNOT_{25,153}, CNOT_{26,154}, CNOT_{27,155}, CNOT_{28,156}, CNOT_{29,157}, CNOT_{30,158}, CNOT_{31,159}, CNOT_{32,160} \end{array} \right\}$$

#Full depth 4 and Toffoli depth 0:

$$\left\{ \begin{array}{l} NOT_{129}, NOT_{130}, NOT_{131}, NOT_{132}, NOT_{133}, NOT_{134}, NOT_{135}, NOT_{136}, \\ NOT_{137}, NOT_{138}, NOT_{139}, NOT_{140}, NOT_{141}, NOT_{142}, NOT_{143}, NOT_{144}, \\ NOT_{145}, NOT_{146}, NOT_{147}, NOT_{148}, NOT_{149}, NOT_{150}, NOT_{151}, NOT_{152}, \\ NOT_{153}, NOT_{154}, NOT_{155}, NOT_{156}, NOT_{157}, NOT_{158}, NOT_{159}, NOT_{160} \end{array} \right\}$$

电路宽度为 160，辅助位为 32，电路总深度为 4，*Toffoli*深度为 0。

在基元素构造完成后，即可按 $y4 \rightarrow y3 \rightarrow y2 \rightarrow y1 \rightarrow y$ 的顺序，完成分解后的可逆变换，从而达到实现 y_i 函数的目的。

2.3.3 基于组合的可逆变换

因为每个元素有多种选择，因此，在可逆变换的过程中，需要利用组合的方式实现可逆。 y_4 矩阵不用实现，因为已经处于最简的形式，仅是利用其排列形式构造 y_3 ， y_4 的每一列元素有 2 种选择，基于此建立 y_3 。

2.3.3.1 y_3 实现

以 y_4 中相邻的两个元素为一组进行合并，例如： y_3 第一行的 $X_{1,1}$ 和 $X_{1,2}$ ， $X_{1,3}$ 和 $X_{1,4}$ ， $X_{1,5}$ 和 $X_{1,6}$ ， $X_{1,7}$ 和 $X_{1,8}$ 为一组合并。进而从 $128 * 8$ 的矩阵 τ_4 合并成 $128 * 4$ 的矩阵 τ_3 ，因为每个元素有 2 种情况，因此合并时，每组合并操作要进行 2^2 次，以考虑到所有元素组合的可能。

电路描述如下：

#Full depth 5 and Toffoli depth 1:

$$\left\{ \begin{array}{l} Toffoli_{1,2,161}, Toffoli_{1,130,162}, Toffoli_{129,2,163}, Toffoli_{129,130,164}, \\ Toffoli_{3,4,165}, Toffoli_{3,132,166}, Toffoli_{131,4,167}, Toffoli_{131,132,168}, \\ \vdots \\ Toffoli_{31,32,221}, Toffoli_{31,160,222}, Toffoli_{159,32,223}, Toffoli_{159,160,224} \end{array} \right\}$$

电路宽度为 224，辅助位为 96，电路总深度为 5，*Toffoli* 深度为 1。

具体电路描述见附录。

2.3.3.2 y_2 实现

与 y_3 实现相同，构造 y_2 。从 $128 * 4$ 的矩阵 y_3 合并成 $128 * 2$ 的矩阵 y_2 ， y_3 每个元素有 4 种可能，因此合并时，每组合并操作要进行 2^4 次，以考虑到所有组合的可能。电路描述如下：

#Full depth 6 and Toffoli depth 2:

$$\left\{ \begin{array}{l} Toffoli_{161,165,225}, Toffoli_{161,166,226}, Toffoli_{161,167,227}, Toffoli_{161,168,228}, \\ Toffoli_{162,165,229}, Toffoli_{162,166,230}, Toffoli_{162,167,231}, Toffoli_{162,168,232}, \\ \vdots \\ Toffoli_{220,221,349}, Toffoli_{220,222,350}, Toffoli_{220,223,351}, Toffoli_{220,224,352} \end{array} \right\}$$

电路宽度为 352，辅助位为 224，电路总深度为 5，*Toffoli* 深度为 2。

具体电路描述见附录。

2.3.3.3 基于乘法分配律的 y_1 优化实现

构造所有组合情况的方法构造 y_1 过度依赖于 $Toffoli$ 门的运用，而 $Toffoli$ 门的运行时间过长，在电路设计中应尽可能少的使用。考虑 y_1 元素不再进行任何乘法计算，只利用加法得到最终的 y_i 函数，基于此考虑乘法的性质，进行优化。

乘法是由因数相乘得到的，而当不同的多项式有相同的因数时，可以提取公因式，而剩余的因数相加，即乘法分配律： $ab + ac = (a + b)c$ 。等式左边的乘法次数为 2，等式右边乘法次数为 1，利用乘法分配律可以有效地降低乘法次数。

此时， y_2 为 $128 * 2$ 的矩阵，完全适用于乘法分配律，提取公因式，用多个加法替换乘法，最终利用 $Toffoli$ 门将公因式和合并项相乘，得到优化后的 y_1 。

提取公因式后，剩余的因数进行合并，整理后每个 y_i 函数由 16 个小多项式构成，每个小多项式为一个公因式和一个剩余因数的加法多项式相乘得到。公因式为 y_2 中得到的 4 变量元素，而加法多项式需要根据代数标准型进行构造。

为了快速构造加法多项式，利用并行计算，即将该加法多项式中的元素，两两进行相加，假设合并到一起的个数为 n ，每次都需要以 $\begin{cases} \frac{n}{2}, \text{偶数} \\ \frac{n-1}{2}, \text{奇数} \end{cases}$ 的个数合并。由于每个多项式中含有的元素个数不同，4 变量元素不同，则每个多项式之间可能含有相同的元素，则两两相加构造时极易重复构造，形成资源浪费。因此，对加法多项式中所有两两元素的组合情况进行统计，降序排列。每次选出组合次数最多的搭配情况，在多项式中去除该组合情况后，对于剩余的元素搭配重新计算出现次数。降序排列表见附录，有 4 个 S 函数，每个 S 函数部分有 69 组搭配。

基于降序排列表，选择了使用次数最多的搭配情况，避免了重复构造的可能。两元素组合的电路描述如下：

#Full depth 7 and Toffoli depth 2:

$$\left\{ \begin{array}{l} CNOT_{249,353}, CNOT_{253,354}, CNOT_{244,355}, CNOT_{254,356}, \\ \vdots \\ CNOT_{344,625}, CNOT_{344,626}, CNOT_{341,627}, CNOT_{351,628} \end{array} \right\}$$

#Full depth 8 and Toffoli depth 2:

$$\left\{ \begin{array}{l} CNOT_{247,353}, CNOT_{251,354}, CNOT_{242,355}, CNOT_{250,356}, \\ \vdots \\ CNOT_{341,625}, CNOT_{340,626}, CNOT_{337,627}, CNOT_{342,628} \end{array} \right\}$$

具体电路描述见附录。

根据两元素组合的情况构建加法多项式，由于降序排列表中部分组合情况仅出现一次，共计 100 个，因此，在构建加法多项式时，可以利用组合次数为 1 的搭配，将整个加法多项式存储在该搭配的位置上，从而可以减少 100 个辅助位的使用创建。

加法多项式的构造选择逐次相加，即利用 7 层电路构造，如果选择两两相加构造，则需要增加更多的辅助位，考虑到整个电路的深度大概在 50 层，虽然两两构造加法多项式只需要 4 层电路，但从整体上看，两两相加构造利用了更多的辅助位资源，故导致逐次相加的 $T - DW - cost$ 和 $DW - cost$ 值更低，更加符合题目的要求。电路描述如下：

#Full depth 9 and Toffoli depth 2:

$$\left\{ \begin{array}{l} CNOT_{353,405}, CNOT_{365,406}, CNOT_{354,398}, CNOT_{353,629}, CNOT_{359,407}, CNOT_{242,630}, \\ \vdots \\ CNOT_{339,1035}, CNOT_{350,1036}, CNOT_{560,1037}, CNOT_{563,1038}, CNOT_{342,1039}, CNOT_{339,1040}, \\ \vdots \end{array} \right\}$$

#Full depth 15 and Toffoli depth 2:

$$\left\{ \begin{array}{l} CNOT_{362,674}, CNOT_{375,692}, CNOT_{431,778}, CNOT_{444,796}, \\ CNOT_{500,881}, CNOT_{513,899}, CNOT_{569,984}, CNOT_{582,1002} \end{array} \right\}$$

电路宽度为 1002，辅助位为 874，电路总深度为 15，*Toffoli* 深度为 2。

具体电路描述见附录。

2.3.4 辅助位的动态存储

曹晓丽等人^[2]提出了共享堆栈的动态存储结构，会出现同时使用多个堆栈的情况，将多个堆栈同时共享一个存储空间，进行共享堆栈的动态存储。基于此，本文使用了动态存储的思想，动态调用部分辅助位，存储部分电路信息。

对于辅助位，部分辅助位仅利用一次，在后续的计算中毫无作用，并且占用资源，因此可以将该类型的辅助位清空，用于存储其他数据。由于 *Toffoli* 门计算时间长，因此，对于利用 *Toffoli* 门存储的辅助位不考虑清空。

该类型的辅助位有创建基元素时产生的 32 个辅助位和两元素搭配（出现次数大于 1）产生的 176 个辅助位，清空出的辅助位：

#Full depth 6 and Toffoli depth 2:

$$\left\{ \begin{array}{l} NOT_{129}, NOT_{130}, NOT_{131}, NOT_{132}, NOT_{133}, NOT_{134}, NOT_{135}, NOT_{136}, \\ NOT_{137}, NOT_{138}, NOT_{139}, NOT_{140}, NOT_{141}, NOT_{142}, NOT_{143}, NOT_{144}, \\ NOT_{145}, NOT_{146}, NOT_{147}, NOT_{148}, NOT_{149}, NOT_{150}, NOT_{151}, NOT_{152}, \\ NOT_{153}, NOT_{154}, NOT_{155}, NOT_{156}, NOT_{157}, NOT_{158}, NOT_{159}, NOT_{160} \end{array} \right\}$$

#Full depth 7 and Toffoli depth 2:

$$\left\{ \begin{array}{l} CNOT_{1,129}, CNOT_{2,130}, CNOT_{3,131}, CNOT_{4,132}, CNOT_{5,133}, CNOT_{6,134}, CNOT_{7,135}, CNOT_{8,136}, \\ CNOT_{9,137}, CNOT_{10,138}, CNOT_{11,139}, CNOT_{12,140}, CNOT_{13,141}, CNOT_{14,142}, CNOT_{15,143}, CNOT_{16,144}, \\ CNOT_{17,145}, CNOT_{18,146}, CNOT_{19,147}, CNOT_{20,148}, CNOT_{21,149}, CNOT_{22,150}, CNOT_{23,151}, CNOT_{24,152}, \\ CNOT_{25,153}, CNOT_{26,154}, CNOT_{27,155}, CNOT_{28,156}, CNOT_{29,157}, CNOT_{30,158}, CNOT_{31,159}, CNOT_{32,160} \end{array} \right\}$$

两元素搭配清空出的辅助位，大大减少了公因式和加法多项式需要的辅助位。

#Full depth 16 and Toffoli depth 2:

$$\left\{ \begin{array}{l} CNOT_{249,353}, CNOT_{253,354}, CNOT_{244,355}, CNOT_{254,356}, \\ \vdots \\ CNOT_{347,600}, CNOT_{346,601}, CNOT_{344,602}, CNOT_{343,603} \end{array} \right\}$$

#Full depth 17 and Toffoli depth 2:

$$\left\{ \begin{array}{l} CNOT_{247,353}, CNOT_{251,354}, CNOT_{242,355}, CNOT_{250,356}, \\ \vdots \\ CNOT_{340,600}, CNOT_{341,601}, CNOT_{342,602}, CNOT_{337,603} \end{array} \right\}$$

2.3.5 非线性向量布尔函数 τ 实现

利用清空出的辅助位存储部分公因式和加法多项式的结果，还需要另外补充 304 个辅助位，提取公因式后整理的所有多项式存储如下：

#Full depth 18 and Toffoli depth 3:

$$\left\{ \begin{array}{l} Toffoli_{240,405,129}, Toffoli_{239,406,130}, Toffoli_{238,398,131}, Toffoli_{237,629,132}, \\ \vdots \\ Toffoli_{240,650,353}, Toffoli_{239,651,354}, Toffoli_{238,652,355}, Toffoli_{237,404,356}, \\ \vdots \\ Toffoli_{324,1037,1341}, Toffoli_{323,1038,1342}, Toffoli_{322,1039,1343}, Toffoli_{321,1040,1344} \end{array} \right\}$$

电路宽度为 1344，辅助位为 1216，电路总深度为 18，Toffoli 深度为 3。

具体电路描述见附录。

每个 y_i 函数有 16 个乘法多项式构成，则通过并行计算快速构造 τ 函数，两两相加得到函数 τ 的输出，每层电路的计算将多项式次数减少一半，一共需要 4 成电路。

#Full depth 19 and Toffoli depth 3:

$$\left\{ \begin{array}{l} CNOT_{130,129}, CNOT_{132,131}, CNOT_{134,133}, CNOT_{136,135}, \\ \vdots \\ CNOT_{1338,1337}, CNOT_{1340,1339}, CNOT_{1342,1341}, CNOT_{1344,1343} \end{array} \right\}$$

$\vdots$

#Full depth 22 and Toffoli depth 3:

$$\left\{ \begin{array}{c} CNOT_{137,129}, CNOT_{153,145}, CNOT_{361,353}, CNOT_{377,369}, \\ \vdots \\ CNOT_{1289,1281}, CNOT_{1305,1297}, CNOT_{1321,1313}, CNOT_{1337,1329}, \end{array} \right\}$$

电路宽度为 1344，辅助位为 1216，电路总深度为 22，*Toffoli* 深度为 3。

具体电路描述见附录。

2.4 线性向量布尔函数 L 电路设计

根据 L 函数的代数标准型，进行简单的逐次加法即可得到。与加法多项式的构造相同，逐次相加的 $T - DW - cost$ 和 $DW - cost$ 值更低，两两相加的并行计算花费更多的辅助位，并且在整个电路中，两两相加的方法还需要在后续进行可逆计算，以此来消去辅助位的数值，而逐次相加的方法不需要进行可逆计算，在构造中并没有利用辅助位，直接加到目标位置。

#Full depth 23 and Toffoli depth 3:

$$\left\{ \begin{array}{c} CNOT_{129,97}, CNOT_{145,98}, CNOT_{353,99}, CNOT_{129,100}, \\ \vdots \\ CNOT_{442,125}, CNOT_{458,126}, CNOT_{129,127}, CNOT_{145,128} \end{array} \right\}$$

$\vdots$

#Full depth 29 and Toffoli depth 3:

$$\left\{ \begin{array}{c} CNOT_{1297,97}, CNOT_{1313,98}, CNOT_{1329,99}, CNOT_{1265,100}, \\ \vdots \\ CNOT_{1313,125}, CNOT_{1329,126}, CNOT_{1313,127}, CNOT_{1329,128} \end{array} \right\}$$

具体电路描述见附录。

至此，实现了目标向量布尔函数的要求：

$$F(X_1, X_2, X_3, X_4) = (X_2, X_3, X_4, X_1 + L \circ \tau(X_2 \oplus X_3 \oplus X_4))$$

电路宽度为 1344，辅助位为 1216，电路总深度为 29，*Toffoli* 深度为 3。

2.5 可逆量子电路设计

上述设计的电路实现了赛题要求的函数实现，但是并未对辅助位进行清空，即未实现可逆（量子）电路，可逆电路的实现利用之前的电路对辅助位清空，由于第 23 层到第 29 层构造函数 L 的电路未利用辅助位，直接加到了目标位数上，故从第 22 层电路开始调用，向前逆序逐次调用到第 1 层，即可对辅助位进行清空。

#Full depth 30 and Toffoli depth 3:

$$\left\{ \begin{array}{c} CNOT_{137,129}, CNOT_{153,145}, CNOT_{361,353}, CNOT_{377,369}, \\ \vdots \\ CNOT_{1289,1281}, CNOT_{1305,1297}, CNOT_{1321,1313}, CNOT_{1337,1329} \end{array} \right\}$$

$$\vdots$$

#Full depth 51 and Toffoli depth 6:

$$\left\{ \begin{array}{c} CNOT_{33,1}, CNOT_{34,2}, CNOT_{35,3}, CNOT_{36,4}, CNOT_{37,5}, CNOT_{38,6}, CNOT_{39,7}, CNOT_{40,8}, \\ CNOT_{41,9}, CNOT_{42,10}, CNOT_{43,11}, CNOT_{44,12}, CNOT_{45,13}, CNOT_{46,14}, CNOT_{47,15}, CNOT_{48,16}, \\ CNOT_{49,17}, CNOT_{50,18}, CNOT_{51,19}, CNOT_{52,20}, CNOT_{53,21}, CNOT_{54,22}, CNOT_{55,23}, CNOT_{56,24}, \\ CNOT_{57,25}, CNOT_{58,26}, CNOT_{59,27}, CNOT_{60,28}, CNOT_{61,29}, CNOT_{62,30}, CNOT_{63,31}, CNOT_{64,32} \end{array} \right\}$$

电路宽度为 1344，辅助位为 1216，电路总深度为 51，*Toffoli* 深度为 6。

具体电路描述见附录。

第三章 结论

通过 51 层（*Toffoli* 深度为 6）宽度为 1344（1216 个辅助位）的电路实现了原位电路，具体的实现方案见附录。

评价指标如下：

$$T - DW - cost = 1344 * 6 = 8064$$

$$DW - cost = 1344 * 51 = 68544$$

第四章 参考文献

- [1] 王萼芳, 石生明, 高等代数（第五版）[M]。北京：高等教育出版社, 2019, 121.
- [2] 曹晓丽, 朱正平, 刘君, 岳秋菊, 任志国. 共享堆栈的动态存储结构[J]. 自动化与仪器仪表, 2014(10):8-10.

第五章 附录

两元素组合降序排列表											
介绍： 四变量元素两两组合后，选择多项式中出现次数最多的组合形式作为输出，并在多项式中进行剔除，更新出现次数，重复操作，直至出现次数为 0，将输出的辅助位序号整合得到该表。											
组合 1		次数	组合 2		次数	组合 3		次数	组合 4		次数
249	247	46	281	279	46	313	311	46	345	343	46
253	251	43	285	283	43	317	315	43	349	347	43
244	242	37	276	274	37	308	306	37	340	338	37
254	250	34	286	282	34	318	314	34	350	346	34
255	243	33	287	275	33	319	307	33	351	339	33
256	245	32	288	277	32	320	309	32	352	341	32
252	248	31	284	280	31	316	312	31	348	344	31
246	241	28	278	273	28	310	305	28	342	337	28
255	254	12	287	286	12	319	318	12	351	350	12
250	246	12	282	278	12	314	310	12	346	342	12
247	244	11	279	276	11	311	308	11	343	340	11
245	242	11	277	274	11	309	306	11	341	338	11
251	249	9	283	281	9	315	313	9	347	345	9
243	241	9	275	273	9	307	305	9	339	337	9
256	241	8	288	273	8	320	305	8	352	337	8
254	244	7	286	276	7	318	308	7	350	340	7
252	247	7	284	279	7	316	311	7	348	343	7
252	243	6	284	275	6	316	307	6	348	339	6
246	245	6	278	277	6	310	309	6	342	341	6
256	253	5	288	285	5	320	317	5	352	349	5
256	244	5	288	276	5	320	308	5	352	340	5
249	248	5	281	280	5	313	312	5	345	344	5
243	242	5	275	274	5	307	306	5	339	338	5
253	252	4	285	284	4	317	316	4	349	348	4
253	250	4	285	282	4	317	314	4	349	346	4
252	251	4	284	283	4	316	315	4	348	347	4
256	255	3	288	287	3	320	319	3	352	351	3

256	243	3	288	275	3	320	307	3	352	339	3
255	248	3	287	280	3	319	312	3	351	344	3
252	249	3	284	281	3	316	313	3	348	345	3
252	245	3	284	277	3	316	309	3	348	341	3
251	247	3	283	279	3	315	311	3	347	343	3
242	241	3	274	273	3	306	305	3	338	337	3
255	253	2	287	285	2	319	317	2	351	349	2
254	246	2	286	278	2	318	310	2	350	342	2
254	241	2	286	273	2	318	305	2	350	337	2
253	248	2	285	280	2	317	312	2	349	344	2
253	247	2	285	279	2	317	311	2	349	343	2
251	250	2	283	282	2	315	314	2	347	346	2
251	246	2	283	278	2	315	310	2	347	342	2
251	244	2	283	276	2	315	308	2	347	340	2
250	245	2	282	277	2	314	309	2	346	341	2
248	246	2	280	278	2	312	310	2	344	342	2
247	241	2	279	273	2	311	305	2	343	337	2
256	254	1	288	286	1	320	318	1	352	350	1
256	250	1	288	282	1	320	314	1	352	346	1
255	251	1	287	283	1	319	315	1	351	347	1
255	250	1	287	282	1	319	314	1	351	346	1
255	249	1	287	281	1	319	313	1	351	345	1
255	245	1	287	277	1	319	309	1	351	341	1
255	244	1	287	276	1	319	308	1	351	340	1
255	241	1	287	273	1	319	305	1	351	337	1
254	252	1	286	284	1	318	316	1	350	348	1
254	248	1	286	280	1	318	312	1	350	344	1
253	249	1	285	281	1	317	313	1	349	345	1
253	245	1	285	277	1	317	309	1	349	341	1
252	250	1	284	282	1	316	314	1	348	346	1
252	242	1	284	274	1	316	306	1	348	338	1
251	248	1	283	280	1	315	312	1	347	344	1
251	243	1	283	275	1	315	307	1	347	339	1
250	249	1	282	281	1	314	313	1	346	345	1

250	248	1	282	280	1	314	312	1	346	344	1
250	247	1	282	279	1	314	311	1	346	343	1
250	242	1	282	274	1	314	306	1	346	338	1
249	242	1	281	274	1	313	306	1	345	338	1
248	245	1	280	277	1	312	309	1	344	341	1
248	244	1	280	276	1	312	308	1	344	340	1
245	241	1	277	273	1	309	305	1	341	337	1
255	246	1	287	278	1	319	310	1	351	342	1

量子电路构造过程

介绍:

基于 NCT 门集生成向量布尔函数的过程对应了一个实现目标函数的可逆量子电路

Step1:

$$\left\{ \begin{array}{l} CNOT_{33,1}, CNOT_{34,2}, CNOT_{35,3}, CNOT_{36,4}, CNOT_{37,5}, CNOT_{38,6}, CNOT_{39,7}, CNOT_{40,8}, \\ CNOT_{41,9}, CNOT_{42,10}, CNOT_{43,11}, CNOT_{44,12}, CNOT_{45,13}, CNOT_{46,14}, CNOT_{47,15}, CNOT_{48,16}, \\ CNOT_{49,17}, CNOT_{50,18}, CNOT_{51,19}, CNOT_{52,20}, CNOT_{53,21}, CNOT_{54,22}, CNOT_{55,23}, CNOT_{56,24}, \\ CNOT_{57,25}, CNOT_{58,26}, CNOT_{59,27}, CNOT_{60,28}, CNOT_{61,29}, CNOT_{62,30}, CNOT_{63,31}, CNOT_{64,32} \end{array} \right\}$$

Step2:

$$\left\{ \begin{array}{l} CNOT_{65,1}, CNOT_{66,2}, CNOT_{67,3}, CNOT_{68,4}, CNOT_{69,5}, CNOT_{70,6}, CNOT_{71,7}, CNOT_{72,8}, \\ CNOT_{73,9}, CNOT_{74,10}, CNOT_{75,11}, CNOT_{76,12}, CNOT_{77,13}, CNOT_{78,14}, CNOT_{79,15}, CNOT_{80,16}, \\ CNOT_{81,17}, CNOT_{82,18}, CNOT_{83,19}, CNOT_{84,20}, CNOT_{85,21}, CNOT_{86,22}, CNOT_{87,23}, CNOT_{88,24}, \\ CNOT_{89,25}, CNOT_{90,26}, CNOT_{91,27}, CNOT_{92,28}, CNOT_{93,29}, CNOT_{94,30}, CNOT_{95,31}, CNOT_{96,32} \end{array} \right\}$$

Step3:

$$\left\{ \begin{array}{l} CNOT_{1,129}, CNOT_{2,130}, CNOT_{3,131}, CNOT_{4,132}, CNOT_{5,133}, CNOT_{6,134}, CNOT_{7,135}, CNOT_{8,136}, \\ CNOT_{9,137}, CNOT_{10,138}, CNOT_{11,139}, CNOT_{12,140}, CNOT_{13,141}, CNOT_{14,142}, CNOT_{15,143}, CNOT_{16,144}, \\ CNOT_{17,145}, CNOT_{18,146}, CNOT_{19,147}, CNOT_{20,148}, CNOT_{21,149}, CNOT_{22,150}, CNOT_{23,151}, CNOT_{24,152}, \\ CNOT_{25,153}, CNOT_{26,154}, CNOT_{27,155}, CNOT_{28,156}, CNOT_{29,157}, CNOT_{30,158}, CNOT_{31,159}, CNOT_{32,160} \end{array} \right\}$$

Step4:

$$\left\{ \begin{array}{l} NOT_{129}, NOT_{130}, NOT_{131}, NOT_{132}, NOT_{133}, NOT_{134}, NOT_{135}, NOT_{136}, \\ NOT_{137}, NOT_{138}, NOT_{139}, NOT_{140}, NOT_{141}, NOT_{142}, NOT_{143}, NOT_{144}, \\ NOT_{145}, NOT_{146}, NOT_{147}, NOT_{148}, NOT_{149}, NOT_{150}, NOT_{151}, NOT_{152}, \\ NOT_{153}, NOT_{154}, NOT_{155}, NOT_{156}, NOT_{157}, NOT_{158}, NOT_{159}, NOT_{160} \end{array} \right\}$$

Step5:

Step6:

$\left\{ \begin{array}{l} Toffoli_{1,2,161}, Toffoli_{1,130,162}, Toffoli_{129,2,163}, Toffoli_{129,130,164}, \\ Toffoli_{3,4,165}, Toffoli_{3,132,166}, Toffoli_{131,4,167}, Toffoli_{131,132,168}, \\ Toffoli_{5,6,169}, Toffoli_{5,134,170}, Toffoli_{133,6,171}, Toffoli_{133,134,172}, \\ Toffoli_{7,8,173}, Toffoli_{7,136,174}, Toffoli_{135,8,175}, Toffoli_{135,136,176}, \\ Toffoli_{9,10,177}, Toffoli_{9,138,178}, Toffoli_{137,10,179}, Toffoli_{137,138,180}, \\ Toffoli_{11,12,181}, Toffoli_{11,140,182}, Toffoli_{139,12,183}, Toffoli_{139,140,184}, \\ Toffoli_{13,14,185}, Toffoli_{13,142,186}, Toffoli_{141,14,187}, Toffoli_{141,142,188}, \\ Toffoli_{15,16,189}, Toffoli_{15,144,190}, Toffoli_{143,16,191}, Toffoli_{143,144,192}, \\ Toffoli_{17,18,193}, Toffoli_{17,146,194}, Toffoli_{145,18,195}, Toffoli_{145,146,196}, \\ Toffoli_{19,20,197}, Toffoli_{19,148,198}, Toffoli_{147,20,199}, Toffoli_{147,148,200}, \\ Toffoli_{21,22,201}, Toffoli_{21,150,202}, Toffoli_{149,22,203}, Toffoli_{149,150,204}, \\ Toffoli_{23,24,205}, Toffoli_{23,152,206}, Toffoli_{151,24,207}, Toffoli_{151,152,208}, \\ Toffoli_{25,26,209}, Toffoli_{25,154,210}, Toffoli_{153,26,211}, Toffoli_{153,154,212}, \\ Toffoli_{27,28,213}, Toffoli_{27,156,214}, Toffoli_{155,28,215}, Toffoli_{155,156,216}, \\ Toffoli_{29,30,217}, Toffoli_{29,158,218}, Toffoli_{157,30,219}, Toffoli_{157,158,220}, \\ Toffoli_{31,32,221}, Toffoli_{31,160,222}, Toffoli_{159,32,223}, Toffoli_{159,160,224} \end{array} \right\}$

$\left\{ \begin{array}{l} Toffoli_{161,165,225}, Toffoli_{161,166,226}, Toffoli_{161,167,227}, Toffoli_{161,168,228}, \\ Toffoli_{162,165,229}, Toffoli_{162,166,230}, Toffoli_{162,167,231}, Toffoli_{162,168,232}, \\ Toffoli_{163,165,233}, Toffoli_{163,166,234}, Toffoli_{163,167,235}, Toffoli_{163,168,236}, \\ Toffoli_{164,165,237}, Toffoli_{164,166,238}, Toffoli_{164,167,239}, Toffoli_{164,168,240}, \\ Toffoli_{169,173,241}, Toffoli_{169,174,242}, Toffoli_{169,175,243}, Toffoli_{169,176,244}, \\ Toffoli_{170,173,245}, Toffoli_{170,174,246}, Toffoli_{170,175,247}, Toffoli_{170,176,248}, \\ Toffoli_{171,173,249}, Toffoli_{171,174,250}, Toffoli_{171,175,251}, Toffoli_{171,176,252}, \\ Toffoli_{172,173,253}, Toffoli_{172,174,254}, Toffoli_{172,175,255}, Toffoli_{172,176,256}, \\ Toffoli_{177,181,257}, Toffoli_{177,182,258}, Toffoli_{177,183,259}, Toffoli_{177,184,260}, \\ Toffoli_{178,181,261}, Toffoli_{178,182,262}, Toffoli_{178,183,263}, Toffoli_{178,184,264}, \\ Toffoli_{179,181,265}, Toffoli_{179,182,266}, Toffoli_{179,183,267}, Toffoli_{179,184,268}, \\ Toffoli_{180,181,269}, Toffoli_{180,182,270}, Toffoli_{180,183,271}, Toffoli_{180,184,272}, \\ Toffoli_{185,189,273}, Toffoli_{185,190,274}, Toffoli_{185,191,275}, Toffoli_{185,192,276}, \\ Toffoli_{186,189,277}, Toffoli_{186,190,278}, Toffoli_{186,191,279}, Toffoli_{186,192,280}, \\ Toffoli_{187,189,281}, Toffoli_{187,190,282}, Toffoli_{187,191,283}, Toffoli_{187,192,284}, \\ Toffoli_{188,189,285}, Toffoli_{188,190,286}, Toffoli_{188,191,287}, Toffoli_{188,192,288}, \\ Toffoli_{193,197,289}, Toffoli_{193,198,290}, Toffoli_{193,199,291}, Toffoli_{193,200,292}, \\ Toffoli_{194,197,293}, Toffoli_{194,198,294}, Toffoli_{194,199,295}, Toffoli_{194,200,296}, \\ Toffoli_{195,197,297}, Toffoli_{195,198,298}, Toffoli_{195,199,299}, Toffoli_{195,200,300}, \\ Toffoli_{196,197,301}, Toffoli_{196,198,302}, Toffoli_{196,199,303}, Toffoli_{196,200,304}, \\ Toffoli_{201,205,305}, Toffoli_{201,206,306}, Toffoli_{201,207,307}, Toffoli_{201,208,308}, \\ Toffoli_{202,205,309}, Toffoli_{202,206,310}, Toffoli_{202,207,311}, Toffoli_{202,208,312}, \\ Toffoli_{203,205,313}, Toffoli_{203,206,314}, Toffoli_{203,207,315}, Toffoli_{203,208,316}, \\ Toffoli_{204,205,317}, Toffoli_{204,206,318}, Toffoli_{204,207,319}, Toffoli_{204,208,320}, \\ Toffoli_{209,213,321}, Toffoli_{209,214,322}, Toffoli_{209,215,323}, Toffoli_{209,216,324}, \\ Toffoli_{210,213,325}, Toffoli_{210,214,326}, Toffoli_{210,215,327}, Toffoli_{210,216,328}, \\ Toffoli_{211,213,329}, Toffoli_{211,214,330}, Toffoli_{211,215,331}, Toffoli_{211,216,332}, \\ Toffoli_{212,213,333}, Toffoli_{212,214,334}, Toffoli_{212,215,335}, Toffoli_{212,216,336}, \\ Toffoli_{217,221,337}, Toffoli_{217,222,338}, Toffoli_{217,223,339}, Toffoli_{217,224,340}, \\ Toffoli_{218,221,341}, Toffoli_{218,222,342}, Toffoli_{218,223,343}, Toffoli_{218,224,344}, \\ Toffoli_{219,221,345}, Toffoli_{219,222,346}, Toffoli_{219,223,347}, Toffoli_{219,224,348}, \\ Toffoli_{220,221,349}, Toffoli_{220,222,350}, Toffoli_{220,223,351}, Toffoli_{220,224,352} \\ NOT_{129}, NOT_{130}, NOT_{131}, NOT_{132}, NOT_{133}, NOT_{134}, NOT_{135}, NOT_{136}, \\ NOT_{137}, NOT_{138}, NOT_{139}, NOT_{140}, NOT_{141}, NOT_{142}, NOT_{143}, NOT_{144}, \\ NOT_{145}, NOT_{146}, NOT_{147}, NOT_{148}, NOT_{149}, NOT_{150}, NOT_{151}, NOT_{152}, \\ NOT_{153}, NOT_{154}, NOT_{155}, NOT_{156}, NOT_{157}, NOT_{158}, NOT_{159}, NOT_{160} \end{array} \right\}$

Step7:

$(CNOT_{249,353}, CNOT_{253,354}, CNOT_{244,355}, CNOT_{254,356}, CNOT_{255,357}, CNOT_{256,358}, CNOT_{252,359},$
 $CNOT_{246,360}, CNOT_{255,361}, CNOT_{250,362}, CNOT_{247,363}, CNOT_{245,364}, CNOT_{251,365}, CNOT_{243,366},$
 $CNOT_{256,367}, CNOT_{254,368}, CNOT_{252,369}, CNOT_{252,370}, CNOT_{246,371}, CNOT_{256,372}, CNOT_{256,373},$
 $CNOT_{249,374}, CNOT_{243,375}, CNOT_{253,376}, CNOT_{253,377}, CNOT_{252,378}, CNOT_{256,379}, CNOT_{256,380},$
 $CNOT_{255,381}, CNOT_{252,382}, CNOT_{252,383}, CNOT_{251,384}, CNOT_{242,385}, CNOT_{255,386}, CNOT_{254,387},$
 $CNOT_{254,388}, CNOT_{253,389}, CNOT_{253,390}, CNOT_{251,391}, CNOT_{251,392}, CNOT_{251,393}, CNOT_{250,394},$
 $CNOT_{248,395}, CNOT_{247,396}, CNOT_{256,397}, CNOT_{256,398}, CNOT_{255,399}, CNOT_{255,400}, CNOT_{255,401},$
 $CNOT_{255,402}, CNOT_{255,403}, CNOT_{255,404}, CNOT_{254,405}, CNOT_{254,406}, CNOT_{253,407}, CNOT_{253,408},$
 $CNOT_{252,409}, CNOT_{252,410}, CNOT_{251,411}, CNOT_{251,412}, CNOT_{250,413}, CNOT_{250,414}, CNOT_{250,415},$
 $CNOT_{250,416}, CNOT_{249,417}, CNOT_{248,418}, CNOT_{248,419}, CNOT_{245,420}, CNOT_{255,421}, CNOT_{281,422},$
 $CNOT_{285,423}, CNOT_{276,424}, CNOT_{286,425}, CNOT_{287,426}, CNOT_{288,427}, CNOT_{284,428}, CNOT_{278,429},$
 $CNOT_{287,430}, CNOT_{282,431}, CNOT_{279,432}, CNOT_{277,433}, CNOT_{283,434}, CNOT_{275,435}, CNOT_{288,436},$
 $CNOT_{286,437}, CNOT_{284,438}, CNOT_{284,439}, CNOT_{278,440}, CNOT_{288,441}, CNOT_{288,442}, CNOT_{281,443},$
 $CNOT_{275,444}, CNOT_{285,445}, CNOT_{285,446}, CNOT_{284,447}, CNOT_{288,448}, CNOT_{288,449}, CNOT_{287,450},$
 $CNOT_{284,451}, CNOT_{284,452}, CNOT_{283,453}, CNOT_{274,454}, CNOT_{287,455}, CNOT_{286,456}, CNOT_{286,457},$
 $CNOT_{285,458}, CNOT_{285,459}, CNOT_{283,460}, CNOT_{283,461}, CNOT_{283,462}, CNOT_{282,463}, CNOT_{280,464},$
 $CNOT_{279,465}, CNOT_{288,466}, CNOT_{288,467}, CNOT_{287,468}, CNOT_{287,469}, CNOT_{287,470}, CNOT_{287,471},$
 $CNOT_{287,472}, CNOT_{287,473}, CNOT_{286,474}, CNOT_{286,475}, CNOT_{285,476}, CNOT_{285,477}, CNOT_{284,478},$
 $CNOT_{284,479}, CNOT_{283,480}, CNOT_{283,481}, CNOT_{282,482}, CNOT_{282,483}, CNOT_{282,484}, CNOT_{282,485},$
 $CNOT_{281,486}, CNOT_{280,487}, CNOT_{280,488}, CNOT_{277,489}, CNOT_{287,490}, CNOT_{313,491}, CNOT_{317,492},$
 $CNOT_{308,493}, CNOT_{318,494}, CNOT_{319,495}, CNOT_{320,496}, CNOT_{316,497}, CNOT_{310,498}, CNOT_{319,499},$
 $CNOT_{314,500}, CNOT_{311,501}, CNOT_{309,502}, CNOT_{315,503}, CNOT_{307,504}, CNOT_{320,505}, CNOT_{318,506},$
 $CNOT_{316,507}, CNOT_{316,508}, CNOT_{310,509}, CNOT_{320,510}, CNOT_{320,511}, CNOT_{313,512}, CNOT_{307,513},$
 $CNOT_{317,514}, CNOT_{317,515}, CNOT_{316,516}, CNOT_{320,517}, CNOT_{320,518}, CNOT_{319,519}, CNOT_{316,520},$
 $CNOT_{316,521}, CNOT_{315,522}, CNOT_{306,523}, CNOT_{319,524}, CNOT_{318,525}, CNOT_{318,526}, CNOT_{317,527},$
 $CNOT_{317,528}, CNOT_{315,529}, CNOT_{315,530}, CNOT_{315,531}, CNOT_{314,532}, CNOT_{312,533}, CNOT_{311,534},$
 $CNOT_{320,535}, CNOT_{320,536}, CNOT_{319,537}, CNOT_{319,538}, CNOT_{319,539}, CNOT_{319,540}, CNOT_{319,541},$
 $CNOT_{319,542}, CNOT_{318,543}, CNOT_{318,544}, CNOT_{317,545}, CNOT_{317,546}, CNOT_{316,547}, CNOT_{316,548},$
 $CNOT_{315,549}, CNOT_{315,550}, CNOT_{314,551}, CNOT_{314,552}, CNOT_{314,553}, CNOT_{314,554}, CNOT_{313,555},$
 $CNOT_{312,556}, CNOT_{312,557}, CNOT_{309,558}, CNOT_{319,559}, CNOT_{345,560}, CNOT_{349,561}, CNOT_{340,562},$
 $CNOT_{350,563}, CNOT_{351,564}, CNOT_{352,565}, CNOT_{348,566}, CNOT_{342,567}, CNOT_{351,568}, CNOT_{346,569},$
 $CNOT_{343,570}, CNOT_{341,571}, CNOT_{347,572}, CNOT_{339,573}, CNOT_{352,574}, CNOT_{350,575}, CNOT_{348,576},$
 $CNOT_{348,577}, CNOT_{342,578}, CNOT_{352,579}, CNOT_{352,580}, CNOT_{345,581}, CNOT_{339,582}, CNOT_{349,583},$
 $CNOT_{349,584}, CNOT_{348,585}, CNOT_{352,586}, CNOT_{352,587}, CNOT_{351,588}, CNOT_{348,589}, CNOT_{348,590},$
 $CNOT_{347,591}, CNOT_{338,592}, CNOT_{351,593}, CNOT_{350,594}, CNOT_{350,595}, CNOT_{349,596}, CNOT_{349,597},$
 $CNOT_{347,598}, CNOT_{347,599}, CNOT_{347,600}, CNOT_{346,601}, CNOT_{344,602}, CNOT_{343,603}, CNOT_{352,604},$
 $CNOT_{352,605}, CNOT_{351,606}, CNOT_{351,607}, CNOT_{351,608}, CNOT_{351,609}, CNOT_{351,610}, CNOT_{351,611},$
 $CNOT_{350,612}, CNOT_{350,613}, CNOT_{349,614}, CNOT_{349,615}, CNOT_{348,616}, CNOT_{348,617}, CNOT_{347,618},$
 $CNOT_{347,619}, CNOT_{346,620}, CNOT_{346,621}, CNOT_{346,622}, CNOT_{346,623}, CNOT_{345,624}, CNOT_{344,625},$
 $CNOT_{344,626}, CNOT_{341,627}, CNOT_{351,628}, CNOT_{1,129}, CNOT_{2,130}, CNOT_{3,131}, CNOT_{4,132},$
 $CNOT_{5,133}, CNOT_{6,134}, CNOT_{7,135}, CNOT_{8,136}, CNOT_{9,137}, CNOT_{10,138}, CNOT_{11,139},$
 $CNOT_{12,140}, CNOT_{13,141}, CNOT_{14,142}, CNOT_{15,143}, CNOT_{16,144}, CNOT_{17,145}, CNOT_{18,146},$
 $CNOT_{19,147}, CNOT_{20,148}, CNOT_{21,149}, CNOT_{22,150}, CNOT_{23,151}, CNOT_{24,152}, CNOT_{25,153},$
 $CNOT_{26,154}, CNOT_{27,155}, CNOT_{28,156}, CNOT_{29,157}, CNOT_{30,158}, CNOT_{31,159}, CNOT_{32,160}$

Step8:

$\{$ $CNOT_{247,353}, CNOT_{251,354}, CNOT_{242,355}, CNOT_{250,356}, CNOT_{243,357}, CNOT_{245,358}, CNOT_{248,359},$
 $CNOT_{241,360}, CNOT_{254,361}, CNOT_{246,362}, CNOT_{244,363}, CNOT_{242,364}, CNOT_{249,365}, CNOT_{241,366},$
 $CNOT_{241,367}, CNOT_{244,368}, CNOT_{247,369}, CNOT_{243,370}, CNOT_{245,371}, CNOT_{253,372}, CNOT_{244,373},$
 $CNOT_{248,374}, CNOT_{242,375}, CNOT_{252,376}, CNOT_{250,377}, CNOT_{251,378}, CNOT_{255,379}, CNOT_{243,380},$
 $CNOT_{248,381}, CNOT_{249,382}, CNOT_{245,383}, CNOT_{247,384}, CNOT_{241,385}, CNOT_{253,386}, CNOT_{246,387},$
 $CNOT_{241,388}, CNOT_{248,389}, CNOT_{247,390}, CNOT_{250,391}, CNOT_{246,392}, CNOT_{244,393}, CNOT_{245,394},$
 $CNOT_{246,395}, CNOT_{241,396}, CNOT_{254,397}, CNOT_{250,398}, CNOT_{251,399}, CNOT_{250,400}, CNOT_{249,401},$
 $CNOT_{245,402}, CNOT_{244,403}, CNOT_{241,404}, CNOT_{252,405}, CNOT_{248,406}, CNOT_{249,407}, CNOT_{245,408},$
 $CNOT_{250,409}, CNOT_{242,410}, CNOT_{248,411}, CNOT_{243,412}, CNOT_{249,413}, CNOT_{248,414}, CNOT_{247,415},$
 $CNOT_{242,416}, CNOT_{242,417}, CNOT_{245,418}, CNOT_{244,419}, CNOT_{241,420}, CNOT_{246,421}, CNOT_{279,422},$
 $CNOT_{283,423}, CNOT_{274,424}, CNOT_{282,425}, CNOT_{275,426}, CNOT_{277,427}, CNOT_{280,428}, CNOT_{273,429},$
 $CNOT_{286,430}, CNOT_{278,431}, CNOT_{276,432}, CNOT_{274,433}, CNOT_{281,434}, CNOT_{273,435}, CNOT_{273,436},$
 $CNOT_{276,437}, CNOT_{279,438}, CNOT_{275,439}, CNOT_{277,440}, CNOT_{285,441}, CNOT_{276,442}, CNOT_{280,443},$
 $CNOT_{274,444}, CNOT_{284,445}, CNOT_{282,446}, CNOT_{283,447}, CNOT_{287,448}, CNOT_{275,449}, CNOT_{280,450},$
 $CNOT_{281,451}, CNOT_{277,452}, CNOT_{279,453}, CNOT_{273,454}, CNOT_{285,455}, CNOT_{278,456}, CNOT_{273,457},$
 $CNOT_{280,458}, CNOT_{279,459}, CNOT_{282,460}, CNOT_{278,461}, CNOT_{276,462}, CNOT_{277,463}, CNOT_{278,464},$
 $CNOT_{273,465}, CNOT_{286,466}, CNOT_{282,467}, CNOT_{283,468}, CNOT_{282,469}, CNOT_{281,470}, CNOT_{277,471},$
 $CNOT_{276,472}, CNOT_{273,473}, CNOT_{284,474}, CNOT_{280,475}, CNOT_{281,476}, CNOT_{277,477}, CNOT_{282,478},$
 $CNOT_{274,479}, CNOT_{280,480}, CNOT_{275,481}, CNOT_{281,482}, CNOT_{280,483}, CNOT_{279,484}, CNOT_{274,485},$
 $CNOT_{274,486}, CNOT_{277,487}, CNOT_{276,488}, CNOT_{273,489}, CNOT_{278,490}, CNOT_{311,491}, CNOT_{315,492},$
 $CNOT_{306,493}, CNOT_{314,494}, CNOT_{307,495}, CNOT_{309,496}, CNOT_{312,497}, CNOT_{305,498}, CNOT_{318,499},$
 $\}$

$\{$ $CNOT_{310,500}, CNOT_{308,501}, CNOT_{306,502}, CNOT_{313,503}, CNOT_{305,504}, CNOT_{305,505}, CNOT_{308,506},$
 $CNOT_{311,507}, CNOT_{307,508}, CNOT_{309,509}, CNOT_{317,510}, CNOT_{308,511}, CNOT_{312,512}, CNOT_{306,513},$
 $CNOT_{316,514}, CNOT_{314,515}, CNOT_{315,516}, CNOT_{319,517}, CNOT_{307,518}, CNOT_{312,519}, CNOT_{313,520},$
 $CNOT_{309,521}, CNOT_{311,522}, CNOT_{305,523}, CNOT_{317,524}, CNOT_{310,525}, CNOT_{305,526}, CNOT_{312,527},$
 $CNOT_{311,528}, CNOT_{314,529}, CNOT_{310,530}, CNOT_{308,531}, CNOT_{309,532}, CNOT_{310,533}, CNOT_{305,534},$
 $CNOT_{318,535}, CNOT_{314,536}, CNOT_{315,537}, CNOT_{314,538}, CNOT_{313,539}, CNOT_{309,540}, CNOT_{308,541},$
 $CNOT_{305,542}, CNOT_{316,543}, CNOT_{312,544}, CNOT_{313,545}, CNOT_{309,546}, CNOT_{314,547}, CNOT_{306,548},$
 $CNOT_{312,549}, CNOT_{307,550}, CNOT_{313,551}, CNOT_{312,552}, CNOT_{311,553}, CNOT_{306,554}, CNOT_{306,555},$
 $CNOT_{309,556}, CNOT_{308,557}, CNOT_{305,558}, CNOT_{310,559}, CNOT_{343,560}, CNOT_{347,561}, CNOT_{338,562},$
 $CNOT_{346,563}, CNOT_{339,564}, CNOT_{341,565}, CNOT_{344,566}, CNOT_{337,567}, CNOT_{350,568}, CNOT_{342,569},$
 $CNOT_{340,570}, CNOT_{338,571}, CNOT_{345,572}, CNOT_{337,573}, CNOT_{337,574}, CNOT_{340,575}, CNOT_{343,576},$
 $CNOT_{339,577}, CNOT_{341,578}, CNOT_{349,579}, CNOT_{340,580}, CNOT_{344,581}, CNOT_{338,582}, CNOT_{348,583},$
 $CNOT_{346,584}, CNOT_{347,585}, CNOT_{351,586}, CNOT_{339,587}, CNOT_{344,588}, CNOT_{345,589}, CNOT_{341,590},$
 $CNOT_{343,591}, CNOT_{337,592}, CNOT_{349,593}, CNOT_{342,594}, CNOT_{337,595}, CNOT_{344,596}, CNOT_{343,597},$
 $CNOT_{346,598}, CNOT_{342,599}, CNOT_{340,600}, CNOT_{341,601}, CNOT_{342,602}, CNOT_{337,603}, CNOT_{350,604},$
 $CNOT_{346,605}, CNOT_{347,606}, CNOT_{346,607}, CNOT_{345,608}, CNOT_{341,609}, CNOT_{340,610}, CNOT_{337,611},$
 $CNOT_{348,612}, CNOT_{344,613}, CNOT_{345,614}, CNOT_{341,615}, CNOT_{346,616}, CNOT_{338,617}, CNOT_{344,618},$
 $CNOT_{339,619}, CNOT_{345,620}, CNOT_{344,621}, CNOT_{343,622}, CNOT_{338,623}, CNOT_{338,624}, CNOT_{341,625},$
 $CNOT_{340,626}, CNOT_{337,627}, CNOT_{342,628}$
 $\}$

Step9:

$CNOT_{353,405}$	$CNOT_{365,406}$	$CNOT_{354,398}$	$CNOT_{353,629}$	$CNOT_{359,407}$	$CNOT_{242,630}$	$CNOT_{241,413}$
$CNOT_{355,631}$	$CNOT_{353,419}$	$CNOT_{355,632}$	$CNOT_{356,402}$	$CNOT_{369,633}$	$CNOT_{354,634}$	$CNOT_{354,635}$
$CNOT_{355,397}$	$CNOT_{354,636}$	$CNOT_{254,637}$	$CNOT_{249,638}$	$CNOT_{353,420}$	$CNOT_{358,409}$	$CNOT_{256,639}$
$CNOT_{357,640}$	$CNOT_{354,410}$	$CNOT_{254,641}$	$CNOT_{246,642}$	$CNOT_{249,643}$	$CNOT_{248,644}$	$CNOT_{354,645}$
$CNOT_{251,646}$	$CNOT_{357,647}$	$CNOT_{253,648}$	$CNOT_{248,649}$	$CNOT_{243,650}$	$CNOT_{255,651}$	$CNOT_{241,652}$
$CNOT_{355,404}$	$CNOT_{354,653}$	$CNOT_{255,654}$	$CNOT_{246,655}$	$CNOT_{363,656}$	$CNOT_{353,414}$	$CNOT_{354,657}$
$CNOT_{245,658}$	$CNOT_{353,412}$	$CNOT_{356,659}$	$CNOT_{353,660}$	$CNOT_{248,661}$	$CNOT_{253,662}$	$CNOT_{248,663}$
$CNOT_{254,664}$	$CNOT_{356,665}$	$CNOT_{353,666}$	$CNOT_{353,667}$	$CNOT_{355,668}$	$CNOT_{242,669}$	$CNOT_{252,670}$
$CNOT_{353,671}$	$CNOT_{353,672}$	$CNOT_{256,673}$	$CNOT_{251,674}$	$CNOT_{241,675}$	$CNOT_{245,676}$	$CNOT_{353,677}$
$CNOT_{355,678}$	$CNOT_{253,679}$	$CNOT_{251,680}$	$CNOT_{355,681}$	$CNOT_{243,682}$	$CNOT_{253,683}$	$CNOT_{242,684}$
$CNOT_{242,685}$	$CNOT_{359,686}$	$CNOT_{356,403}$	$CNOT_{356,687}$	$CNOT_{355,688}$	$CNOT_{353,689}$	$CNOT_{353,690}$
$CNOT_{250,691}$	$CNOT_{353,692}$	$CNOT_{247,693}$	$CNOT_{353,416}$	$CNOT_{354,415}$	$CNOT_{247,694}$	$CNOT_{254,695}$
$CNOT_{246,696}$	$CNOT_{251,697}$	$CNOT_{248,698}$	$CNOT_{356,699}$	$CNOT_{353,399}$	$CNOT_{354,700}$	$CNOT_{241,701}$
$CNOT_{353,702}$	$CNOT_{246,703}$	$CNOT_{251,704}$	$CNOT_{355,705}$	$CNOT_{354,706}$	$CNOT_{243,707}$	$CNOT_{354,417}$
$CNOT_{357,418}$	$CNOT_{357,708}$	$CNOT_{355,709}$	$CNOT_{353,408}$	$CNOT_{353,710}$	$CNOT_{353,711}$	$CNOT_{247,712}$
$CNOT_{360,400}$	$CNOT_{252,713}$	$CNOT_{353,421}$	$CNOT_{249,714}$	$CNOT_{250,715}$	$CNOT_{249,716}$	$CNOT_{363,717}$
$CNOT_{356,718}$	$CNOT_{360,401}$	$CNOT_{243,719}$	$CNOT_{353,720}$	$CNOT_{242,721}$	$CNOT_{252,722}$	$CNOT_{244,723}$
$CNOT_{250,724}$	$CNOT_{241,411}$	$CNOT_{353,725}$	$CNOT_{243,726}$	$CNOT_{254,727}$	$CNOT_{353,728}$	$CNOT_{356,729}$
$CNOT_{246,730}$	$CNOT_{243,731}$	$CNOT_{422,474}$	$CNOT_{434,475}$	$CNOT_{423,467}$	$CNOT_{422,732}$	$CNOT_{428,476}$
$CNOT_{274,733}$	$CNOT_{273,482}$	$CNOT_{424,734}$	$CNOT_{422,488}$	$CNOT_{424,735}$	$CNOT_{425,471}$	$CNOT_{438,736}$
$CNOT_{423,737}$	$CNOT_{423,738}$	$CNOT_{424,466}$	$CNOT_{423,739}$	$CNOT_{286,740}$	$CNOT_{281,741}$	$CNOT_{422,489}$
$CNOT_{427,478}$	$CNOT_{288,742}$	$CNOT_{426,743}$	$CNOT_{423,479}$	$CNOT_{286,744}$	$CNOT_{278,745}$	$CNOT_{281,746}$
$CNOT_{280,747}$	$CNOT_{423,748}$	$CNOT_{283,749}$	$CNOT_{426,750}$	$CNOT_{285,751}$	$CNOT_{280,752}$	$CNOT_{275,753}$
$CNOT_{287,754}$	$CNOT_{273,755}$	$CNOT_{424,473}$	$CNOT_{423,756}$	$CNOT_{287,757}$	$CNOT_{278,758}$	$CNOT_{432,759}$
$CNOT_{422,483}$	$CNOT_{423,760}$	$CNOT_{277,761}$	$CNOT_{422,481}$	$CNOT_{425,762}$	$CNOT_{422,763}$	$CNOT_{280,764}$
$CNOT_{285,765}$	$CNOT_{280,766}$	$CNOT_{286,767}$	$CNOT_{425,768}$	$CNOT_{422,769}$	$CNOT_{422,770}$	$CNOT_{424,771}$
$CNOT_{274,772}$	$CNOT_{284,773}$	$CNOT_{422,774}$	$CNOT_{422,775}$	$CNOT_{288,776}$	$CNOT_{283,777}$	$CNOT_{273,778}$
$CNOT_{277,779}$	$CNOT_{422,780}$	$CNOT_{424,781}$	$CNOT_{285,782}$	$CNOT_{283,783}$	$CNOT_{424,784}$	$CNOT_{275,785}$
$CNOT_{285,786}$	$CNOT_{274,787}$	$CNOT_{274,788}$	$CNOT_{428,789}$	$CNOT_{425,472}$	$CNOT_{425,790}$	$CNOT_{424,791}$
$CNOT_{422,792}$	$CNOT_{422,793}$	$CNOT_{282,794}$	$CNOT_{422,795}$	$CNOT_{279,796}$	$CNOT_{422,485}$	$CNOT_{423,484}$
$CNOT_{279,797}$	$CNOT_{286,798}$	$CNOT_{278,799}$	$CNOT_{283,800}$	$CNOT_{280,801}$	$CNOT_{425,802}$	$CNOT_{422,468}$
$CNOT_{423,803}$	$CNOT_{273,804}$	$CNOT_{422,805}$	$CNOT_{278,806}$	$CNOT_{283,807}$	$CNOT_{424,808}$	$CNOT_{423,809}$
$CNOT_{275,810}$	$CNOT_{423,486}$	$CNOT_{426,487}$	$CNOT_{426,811}$	$CNOT_{424,812}$	$CNOT_{422,477}$	$CNOT_{422,813}$
$CNOT_{422,814}$	$CNOT_{279,815}$	$CNOT_{429,469}$	$CNOT_{284,816}$	$CNOT_{422,490}$	$CNOT_{281,817}$	$CNOT_{282,818}$

$CNOT_{281,819}, CNOT_{432,820}, CNOT_{425,821}, CNOT_{429,470}, CNOT_{275,822}, CNOT_{422,823}, CNOT_{274,824},$
 $CNOT_{284,825}, CNOT_{276,826}, CNOT_{282,827}, CNOT_{273,480}, CNOT_{422,828}, CNOT_{275,829}, CNOT_{286,830},$
 $CNOT_{422,831}, CNOT_{425,832}, CNOT_{278,833}, CNOT_{275,834}, CNOT_{491,543}, CNOT_{503,544}, CNOT_{492,536},$
 $CNOT_{491,835}, CNOT_{497,545}, CNOT_{306,836}, CNOT_{305,551}, CNOT_{493,837}, CNOT_{491,557}, CNOT_{493,838},$
 $CNOT_{494,540}, CNOT_{507,839}, CNOT_{492,840}, CNOT_{492,841}, CNOT_{493,535}, CNOT_{492,842}, CNOT_{318,843},$
 $CNOT_{313,844}, CNOT_{491,558}, CNOT_{496,547}, CNOT_{320,845}, CNOT_{495,846}, CNOT_{492,548}, CNOT_{318,847},$
 $CNOT_{310,848}, CNOT_{313,849}, CNOT_{312,850}, CNOT_{492,851}, CNOT_{315,852}, CNOT_{495,853}, CNOT_{317,854},$
 $CNOT_{312,855}, CNOT_{307,856}, CNOT_{319,857}, CNOT_{305,858}, CNOT_{493,542}, CNOT_{492,859}, CNOT_{319,860},$
 $CNOT_{310,861}, CNOT_{501,862}, CNOT_{491,552}, CNOT_{492,863}, CNOT_{309,864}, CNOT_{491,550}, CNOT_{494,865},$
 $CNOT_{491,866}, CNOT_{312,867}, CNOT_{317,868}, CNOT_{312,869}, CNOT_{318,870}, CNOT_{494,871}, CNOT_{491,872},$
 $CNOT_{491,873}, CNOT_{493,874}, CNOT_{306,875}, CNOT_{316,876}, CNOT_{491,877}, CNOT_{491,878}, CNOT_{320,879},$
 $CNOT_{315,880}, CNOT_{305,881}, CNOT_{309,882}, CNOT_{491,883}, CNOT_{493,884}, CNOT_{317,885}, CNOT_{315,886},$
 $CNOT_{493,887}, CNOT_{307,888}, CNOT_{317,889}, CNOT_{306,890}, CNOT_{306,891}, CNOT_{497,892}, CNOT_{494,541},$
 $CNOT_{494,893}, CNOT_{493,894}, CNOT_{491,895}, CNOT_{491,896}, CNOT_{314,897}, CNOT_{491,898}, CNOT_{311,899},$
 $CNOT_{491,554}, CNOT_{492,553}, CNOT_{311,900}, CNOT_{318,901}, CNOT_{310,902}, CNOT_{315,903}, CNOT_{312,904},$
 $CNOT_{494,905}, CNOT_{491,537}, CNOT_{492,906}, CNOT_{305,907}, CNOT_{491,908}, CNOT_{310,909}, CNOT_{315,910},$
 $CNOT_{493,911}, CNOT_{492,912}, CNOT_{307,913}, CNOT_{492,555}, CNOT_{495,556}, CNOT_{495,914}, CNOT_{493,915},$
 $CNOT_{491,546}, CNOT_{491,916}, CNOT_{491,917}, CNOT_{311,918}, CNOT_{498,538}, CNOT_{316,919}, CNOT_{491,559},$
 $CNOT_{313,920}, CNOT_{314,921}, CNOT_{313,922}, CNOT_{501,923}, CNOT_{494,924}, CNOT_{498,539}, CNOT_{307,925},$
 $CNOT_{491,926}, CNOT_{306,927}, CNOT_{316,928}, CNOT_{308,929}, CNOT_{314,930}, CNOT_{305,549}, CNOT_{491,931},$
 $CNOT_{307,932}, CNOT_{318,933}, CNOT_{491,934}, CNOT_{494,935}, CNOT_{310,936}, CNOT_{307,937}, CNOT_{560,612},$
 $CNOT_{572,613}, CNOT_{561,605}, CNOT_{560,938}, CNOT_{566,614}, CNOT_{338,939}, CNOT_{337,620}, CNOT_{562,940},$
 $CNOT_{560,626}, CNOT_{562,941}, CNOT_{563,609}, CNOT_{576,942}, CNOT_{561,943}, CNOT_{561,944}, CNOT_{562,604},$
 $CNOT_{561,945}, CNOT_{350,946}, CNOT_{345,947}, CNOT_{560,627}, CNOT_{565,616}, CNOT_{352,948}, CNOT_{564,949},$
 $CNOT_{561,617}, CNOT_{350,950}, CNOT_{342,951}, CNOT_{345,952}, CNOT_{344,953}, CNOT_{561,954}, CNOT_{347,955},$
 $CNOT_{564,956}, CNOT_{349,957}, CNOT_{344,958}, CNOT_{339,959}, CNOT_{351,960}, CNOT_{337,961}, CNOT_{562,611},$
 $CNOT_{561,962}, CNOT_{351,963}, CNOT_{342,964}, CNOT_{570,965}, CNOT_{560,621}, CNOT_{561,966}, CNOT_{341,967},$
 $CNOT_{560,619}, CNOT_{563,968}, CNOT_{560,969}, CNOT_{344,970}, CNOT_{349,971}, CNOT_{344,972}, CNOT_{350,973},$
 $CNOT_{563,974}, CNOT_{560,975}, CNOT_{560,976}, CNOT_{562,977}, CNOT_{338,978}, CNOT_{348,979}, CNOT_{560,980},$
 $CNOT_{560,981}, CNOT_{352,982}, CNOT_{347,983}, CNOT_{337,984}, CNOT_{341,985}, CNOT_{560,986}, CNOT_{562,987},$
 $CNOT_{349,988}, CNOT_{347,989}, CNOT_{562,990}, CNOT_{339,991}, CNOT_{349,992}, CNOT_{338,993}, CNOT_{338,994},$
 $CNOT_{566,995}, CNOT_{563,610}, CNOT_{563,996}, CNOT_{562,997}, CNOT_{560,998}, CNOT_{560,999}, CNOT_{346,1000},$
 $CNOT_{560,1001}, CNOT_{343,1002}, CNOT_{560,623}, CNOT_{561,622}, CNOT_{343,1003}, CNOT_{350,1004}, CNOT_{342,1005},$
 $CNOT_{347,1006}, CNOT_{344,1007}, CNOT_{563,1008}, CNOT_{560,606}, CNOT_{561,1009}, CNOT_{337,1010}, CNOT_{560,1011},$
 $CNOT_{342,1012}, CNOT_{347,1013}, CNOT_{562,1014}, CNOT_{561,1015}, CNOT_{339,1016}, CNOT_{561,624}, CNOT_{564,625},$
 $CNOT_{564,1017}, CNOT_{562,1018}, CNOT_{560,615}, CNOT_{560,1019}, CNOT_{560,1020}, CNOT_{343,1021}, CNOT_{567,607},$
 $CNOT_{348,1022}, CNOT_{560,628}, CNOT_{345,1023}, CNOT_{346,1024}, CNOT_{345,1025}, CNOT_{570,1026}, CNOT_{563,1027},$
 $CNOT_{567,608}, CNOT_{339,1028}, CNOT_{560,1029}, CNOT_{338,1030}, CNOT_{348,1031}, CNOT_{340,1032}, CNOT_{346,1033},$
 $CNOT_{337,618}, CNOT_{560,1034}, CNOT_{339,1035}, CNOT_{350,1036}, CNOT_{560,1037}, CNOT_{563,1038}, CNOT_{342,1039},$
 $CNOT_{339,1040}$

Step10:

$CNOT_{354,405}, CNOT_{366,406}, CNOT_{355,398}, CNOT_{357,629}, CNOT_{366,407}, CNOT_{353,630}, CNOT_{358,631},$
 $CNOT_{354,419}, CNOT_{357,632}, CNOT_{359,402}, CNOT_{379,633}, CNOT_{356,634}, CNOT_{355,635}, CNOT_{360,397},$
 $CNOT_{357,636}, CNOT_{354,637}, CNOT_{363,638}, CNOT_{354,420}, CNOT_{363,409}, CNOT_{354,639}, CNOT_{358,640},$
 $CNOT_{362,410}, CNOT_{354,641}, CNOT_{354,642}, CNOT_{367,643}, CNOT_{356,644}, CNOT_{355,645}, CNOT_{353,646},$
 $CNOT_{362,647}, CNOT_{355,648}, CNOT_{354,649}, CNOT_{353,650}, CNOT_{354,651}, CNOT_{354,652}, CNOT_{358,404},$
 $CNOT_{360,653}, CNOT_{354,654}, CNOT_{353,655}, CNOT_{394,656}, CNOT_{357,414}, CNOT_{355,657}, CNOT_{353,658},$
 $CNOT_{368,412}, CNOT_{382,659}, CNOT_{355,660}, CNOT_{353,661}, CNOT_{353,662}, CNOT_{353,663}, CNOT_{354,664},$
 $CNOT_{369,665}, CNOT_{355,666}, CNOT_{354,667}, CNOT_{359,668}, CNOT_{354,669}, CNOT_{353,670}, CNOT_{355,671},$
 $CNOT_{355,672}, CNOT_{357,673}, CNOT_{358,674}, CNOT_{353,675}, CNOT_{353,676}, CNOT_{356,677}, CNOT_{358,678},$
 $CNOT_{355,679}, CNOT_{359,680}, CNOT_{362,681}, CNOT_{354,682}, CNOT_{353,683}, CNOT_{353,684}, CNOT_{354,685},$
 $CNOT_{385,686}, CNOT_{358,403}, CNOT_{357,687}, CNOT_{358,688}, CNOT_{355,689}, CNOT_{355,690}, CNOT_{354,691},$
 $CNOT_{357,692}, CNOT_{354,693}, CNOT_{359,416}, CNOT_{357,415}, CNOT_{354,694}, CNOT_{353,695}, CNOT_{355,696},$
 $CNOT_{360,697}, CNOT_{360,698}, CNOT_{358,699}, CNOT_{364,399}, CNOT_{357,700}, CNOT_{354,701}, CNOT_{354,702},$

$CNOT_{356,703}, CNOT_{353,704}, CNOT_{356,705}, CNOT_{356,706}, CNOT_{353,707}, CNOT_{357,417}, CNOT_{360,418},$
 $CNOT_{364,708}, CNOT_{356,709}, CNOT_{357,408}, CNOT_{358,710}, CNOT_{354,711}, CNOT_{354,712}, CNOT_{369,400},$
 $CNOT_{354,713}, CNOT_{356,421}, CNOT_{355,714}, CNOT_{358,715}, CNOT_{354,716}, CNOT_{370,717}, CNOT_{365,718},$
 $CNOT_{373,401}, CNOT_{359,719}, CNOT_{355,720}, CNOT_{356,721}, CNOT_{353,722}, CNOT_{359,723}, CNOT_{354,724},$
 $CNOT_{353,411}, CNOT_{355,725}, CNOT_{358,726}, CNOT_{353,727}, CNOT_{358,728}, CNOT_{357,729}, CNOT_{356,730},$
 $CNOT_{355,731}, CNOT_{423,474}, CNOT_{435,475}, CNOT_{424,467}, CNOT_{426,732}, CNOT_{435,476}, CNOT_{422,733},$
 $CNOT_{427,734}, CNOT_{423,488}, CNOT_{426,735}, CNOT_{428,471}, CNOT_{448,736}, CNOT_{425,737}, CNOT_{424,738},$
 $CNOT_{429,466}, CNOT_{426,739}, CNOT_{423,740}, CNOT_{432,741}, CNOT_{423,489}, CNOT_{432,478}, CNOT_{423,742},$
 $CNOT_{427,743}, CNOT_{431,479}, CNOT_{423,744}, CNOT_{423,745}, CNOT_{436,746}, CNOT_{425,747}, CNOT_{424,748},$
 $CNOT_{422,749}, CNOT_{431,750}, CNOT_{424,751}, CNOT_{423,752}, CNOT_{422,753}, CNOT_{423,754}, CNOT_{423,755},$
 $CNOT_{427,473}, CNOT_{429,756}, CNOT_{423,757}, CNOT_{422,758}, CNOT_{463,759}, CNOT_{426,483}, CNOT_{424,760},$
 $CNOT_{422,761}, CNOT_{437,481}, CNOT_{451,762}, CNOT_{424,763}, CNOT_{422,764}, CNOT_{422,765}, CNOT_{422,766},$
 $CNOT_{423,767}, CNOT_{438,768}, CNOT_{424,769}, CNOT_{423,770}, CNOT_{428,771}, CNOT_{423,772}, CNOT_{422,773},$
 $CNOT_{424,774}, CNOT_{424,775}, CNOT_{426,776}, CNOT_{427,777}, CNOT_{422,778}, CNOT_{422,779}, CNOT_{425,780},$
 $CNOT_{427,781}, CNOT_{424,782}, CNOT_{428,783}, CNOT_{431,784}, CNOT_{423,785}, CNOT_{422,786}, CNOT_{422,787},$
 $CNOT_{423,788}, CNOT_{454,789}, CNOT_{427,472}, CNOT_{426,790}, CNOT_{427,791}, CNOT_{424,792}, CNOT_{424,793},$
 $CNOT_{423,794}, CNOT_{426,795}, CNOT_{423,796}, CNOT_{428,485}, CNOT_{426,484}, CNOT_{423,797}, CNOT_{422,798},$
 $CNOT_{424,799}, CNOT_{429,800}, CNOT_{429,801}, CNOT_{427,802}, CNOT_{433,468}, CNOT_{426,803}, CNOT_{423,804},$
 $CNOT_{423,805}, CNOT_{425,806}, CNOT_{422,807}, CNOT_{425,808}, CNOT_{425,809}, CNOT_{422,810}, CNOT_{426,486},$
 $CNOT_{429,487}, CNOT_{433,811}, CNOT_{425,812}, CNOT_{426,477}, CNOT_{427,813}, CNOT_{423,814}, CNOT_{423,815},$
 $CNOT_{438,469}, CNOT_{423,816}, CNOT_{425,490}, CNOT_{424,817}, CNOT_{427,818}, CNOT_{423,819}, CNOT_{439,820},$
 $CNOT_{434,821}, CNOT_{442,470}, CNOT_{428,822}, CNOT_{424,823}, CNOT_{425,824}, CNOT_{422,825}, CNOT_{428,826},$
 $CNOT_{423,827}, CNOT_{422,480}, CNOT_{424,828}, CNOT_{427,829}, CNOT_{422,830}, CNOT_{427,831}, CNOT_{426,832},$
 $CNOT_{425,833}, CNOT_{424,834}, CNOT_{492,543}, CNOT_{504,544}, CNOT_{493,536}, CNOT_{495,835}, CNOT_{504,545},$
 $CNOT_{491,836}, CNOT_{496,837}, CNOT_{492,557}, CNOT_{495,838}, CNOT_{497,540}, CNOT_{517,839}, CNOT_{494,840},$
 $CNOT_{493,841}, CNOT_{498,535}, CNOT_{495,842}, CNOT_{492,843}, CNOT_{501,844}, CNOT_{492,558}, CNOT_{501,547},$
 $CNOT_{492,845}, CNOT_{496,846}, CNOT_{500,548}, CNOT_{492,847}, CNOT_{492,848}, CNOT_{505,849}, CNOT_{494,850},$
 $CNOT_{493,851}, CNOT_{491,852}, CNOT_{500,853}, CNOT_{493,854}, CNOT_{492,855}, CNOT_{491,856}, CNOT_{492,857},$
 $CNOT_{492,858}, CNOT_{496,542}, CNOT_{498,859}, CNOT_{492,860}, CNOT_{491,861}, CNOT_{532,862}, CNOT_{495,552},$
 $CNOT_{493,863}, CNOT_{491,864}, CNOT_{506,550}, CNOT_{520,865}, CNOT_{493,866}, CNOT_{491,867}, CNOT_{491,868},$
 $CNOT_{491,869}, CNOT_{492,870}, CNOT_{507,871}, CNOT_{493,872}, CNOT_{492,873}, CNOT_{497,874}, CNOT_{492,875},$
 $CNOT_{491,876}, CNOT_{493,877}, CNOT_{493,878}, CNOT_{495,879}, CNOT_{496,880}, CNOT_{491,881}, CNOT_{491,882},$
 $CNOT_{494,883}, CNOT_{496,884}, CNOT_{493,885}, CNOT_{497,886}, CNOT_{500,887}, CNOT_{492,888}, CNOT_{491,889},$
 $CNOT_{491,890}, CNOT_{492,891}, CNOT_{523,892}, CNOT_{496,541}, CNOT_{495,893}, CNOT_{496,894}, CNOT_{493,895},$
 $CNOT_{493,896}, CNOT_{492,897}, CNOT_{495,898}, CNOT_{492,899}, CNOT_{497,554}, CNOT_{495,553}, CNOT_{492,900},$
 $CNOT_{491,901}, CNOT_{493,902}, CNOT_{498,903}, CNOT_{498,904}, CNOT_{496,905}, CNOT_{502,537}, CNOT_{495,906},$
 $CNOT_{492,907}, CNOT_{492,908}, CNOT_{494,909}, CNOT_{491,910}, CNOT_{494,911}, CNOT_{494,912}, CNOT_{491,913},$
 $CNOT_{495,555}, CNOT_{498,556}, CNOT_{502,914}, CNOT_{494,915}, CNOT_{495,546}, CNOT_{496,916}, CNOT_{492,917},$
 $CNOT_{492,918}, CNOT_{507,538}, CNOT_{492,919}, CNOT_{494,559}, CNOT_{493,920}, CNOT_{496,921}, CNOT_{492,922},$
 $CNOT_{508,923}, CNOT_{503,924}, CNOT_{511,539}, CNOT_{497,925}, CNOT_{493,926}, CNOT_{494,927}, CNOT_{491,928},$
 $CNOT_{497,929}, CNOT_{492,930}, CNOT_{491,549}, CNOT_{493,931}, CNOT_{496,932}, CNOT_{491,933}, CNOT_{496,934},$
 $CNOT_{495,935}, CNOT_{494,936}, CNOT_{493,937}, CNOT_{561,612}, CNOT_{573,613}, CNOT_{562,605}, CNOT_{564,938},$
 $CNOT_{573,614}, CNOT_{560,939}, CNOT_{565,940}, CNOT_{561,626}, CNOT_{564,941}, CNOT_{566,609}, CNOT_{586,942},$
 $CNOT_{563,943}, CNOT_{562,944}, CNOT_{567,604}, CNOT_{564,945}, CNOT_{561,946}, CNOT_{570,947}, CNOT_{561,627},$
 $CNOT_{570,616}, CNOT_{561,948}, CNOT_{565,949}, CNOT_{569,617}, CNOT_{561,950}, CNOT_{561,951}, CNOT_{574,952},$
 $CNOT_{563,953}, CNOT_{562,954}, CNOT_{560,955}, CNOT_{569,956}, CNOT_{562,957}, CNOT_{561,958}, CNOT_{560,959},$
 $CNOT_{561,960}, CNOT_{561,961}, CNOT_{565,611}, CNOT_{567,962}, CNOT_{561,963}, CNOT_{560,964}, CNOT_{601,965},$
 $CNOT_{564,621}, CNOT_{562,966}, CNOT_{560,967}, CNOT_{575,619}, CNOT_{589,968}, CNOT_{562,969}, CNOT_{560,970},$
 $CNOT_{560,971}, CNOT_{560,972}, CNOT_{561,973}, CNOT_{576,974}, CNOT_{562,975}, CNOT_{561,976}, CNOT_{566,977},$
 $CNOT_{561,978}, CNOT_{560,979}, CNOT_{562,980}, CNOT_{562,981}, CNOT_{564,982}, CNOT_{565,983}, CNOT_{560,984},$
 $CNOT_{560,985}, CNOT_{563,986}, CNOT_{565,987}, CNOT_{562,988}, CNOT_{566,989}, CNOT_{569,990}, CNOT_{561,991},$
 $CNOT_{560,992}, CNOT_{560,993}, CNOT_{561,994}, CNOT_{592,995}, CNOT_{565,610}, CNOT_{564,996}, CNOT_{565,997},$
 $CNOT_{562,998}, CNOT_{562,999}, CNOT_{561,1000}, CNOT_{564,1001}, CNOT_{561,1002}, CNOT_{566,623}, CNOT_{564,622},$
 $CNOT_{561,1003}, CNOT_{560,1004}, CNOT_{562,1005}, CNOT_{567,1006}, CNOT_{567,1007}, CNOT_{565,1008}, CNOT_{571,606},$

$\{$ $CNOT_{564,1009}, CNOT_{561,1010}, CNOT_{561,1011}, CNOT_{563,1012}, CNOT_{560,1013}, CNOT_{563,1014}, CNOT_{563,1015},$
 $CNOT_{560,1016}, CNOT_{564,624}, CNOT_{567,625}, CNOT_{571,1017}, CNOT_{563,1018}, CNOT_{564,615}, CNOT_{565,1019},$
 $CNOT_{561,1020}, CNOT_{561,1021}, CNOT_{576,607}, CNOT_{561,1022}, CNOT_{563,628}, CNOT_{562,1023}, CNOT_{565,1024},$
 $CNOT_{561,1025}, CNOT_{577,1026}, CNOT_{572,1027}, CNOT_{580,608}, CNOT_{566,1028}, CNOT_{562,1029}, CNOT_{563,1030},$
 $CNOT_{560,1031}, CNOT_{566,1032}, CNOT_{561,1033}, CNOT_{560,618}, CNOT_{562,1034}, CNOT_{565,1035}, CNOT_{560,1036},$
 $CNOT_{565,1037}, CNOT_{564,1038}, CNOT_{563,1039}, CNOT_{562,1040}$ $\}$

Step11:

$\{$ $CNOT_{357,405}, CNOT_{370,398}, CNOT_{358,629}, CNOT_{368,407}, CNOT_{356,630}, CNOT_{366,631}, CNOT_{358,419},$
 $CNOT_{359,632}, CNOT_{391,633}, CNOT_{367,634}, CNOT_{357,635}, CNOT_{384,397}, CNOT_{363,636}, CNOT_{358,637},$
 $CNOT_{386,638}, CNOT_{357,420}, CNOT_{355,639}, CNOT_{359,640}, CNOT_{358,641}, CNOT_{358,642}, CNOT_{368,643},$
 $CNOT_{357,644}, CNOT_{357,645}, CNOT_{360,646}, CNOT_{384,647}, CNOT_{357,648}, CNOT_{356,649}, CNOT_{354,650},$
 $CNOT_{358,651}, CNOT_{355,652}, CNOT_{377,404}, CNOT_{368,653}, CNOT_{359,654}, CNOT_{357,655}, CNOT_{360,414},$
 $CNOT_{358,657}, CNOT_{354,658}, CNOT_{371,412}, CNOT_{356,660}, CNOT_{357,661}, CNOT_{359,662}, CNOT_{357,663},$
 $CNOT_{360,664}, CNOT_{372,665}, CNOT_{361,666}, CNOT_{362,667}, CNOT_{366,668}, CNOT_{358,669}, CNOT_{354,670},$
 $CNOT_{362,671}, CNOT_{356,672}, CNOT_{361,674}, CNOT_{354,675}, CNOT_{355,676}, CNOT_{392,677}, CNOT_{359,678},$
 $CNOT_{356,679}, CNOT_{367,680}, CNOT_{365,681}, CNOT_{356,682}, CNOT_{360,683}, CNOT_{358,684}, CNOT_{358,685},$
 $CNOT_{359,687}, CNOT_{377,688}, CNOT_{357,689}, CNOT_{356,690}, CNOT_{373,691}, CNOT_{358,692}, CNOT_{356,693},$
 $CNOT_{360,416}, CNOT_{364,415}, CNOT_{355,694}, CNOT_{355,695}, CNOT_{356,696}, CNOT_{375,697}, CNOT_{361,698},$
 $CNOT_{359,699}, CNOT_{368,700}, CNOT_{355,701}, CNOT_{356,702}, CNOT_{365,703}, CNOT_{359,704}, CNOT_{359,705},$
 $CNOT_{363,706}, CNOT_{358,707}, CNOT_{358,417}, CNOT_{365,418}, CNOT_{396,708}, CNOT_{359,709}, CNOT_{362,408},$
 $CNOT_{376,710}, CNOT_{356,711}, CNOT_{358,712}, CNOT_{356,713}, CNOT_{359,421}, CNOT_{361,714}, CNOT_{356,716},$
 $CNOT_{371,717}, CNOT_{366,718}, CNOT_{371,719}, CNOT_{357,720}, CNOT_{357,721}, CNOT_{360,722}, CNOT_{366,723},$
 $CNOT_{383,724}, CNOT_{355,411}, CNOT_{359,725}, CNOT_{360,726}, CNOT_{354,727}, CNOT_{359,728}, CNOT_{359,729},$
 $CNOT_{358,730}, CNOT_{356,731}, CNOT_{426,474}, CNOT_{439,467}, CNOT_{427,732}, CNOT_{437,476}, CNOT_{425,733},$
 $CNOT_{435,734}, CNOT_{427,488}, CNOT_{428,735}, CNOT_{460,736}, CNOT_{436,737}, CNOT_{426,738}, CNOT_{453,466},$
 $CNOT_{432,739}, CNOT_{427,740}, CNOT_{455,741}, CNOT_{426,489}, CNOT_{424,742}, CNOT_{428,743}, CNOT_{427,744},$
 $CNOT_{427,745}, CNOT_{437,746}, CNOT_{426,747}, CNOT_{426,748}, CNOT_{429,749}, CNOT_{453,750}, CNOT_{426,751},$
 $CNOT_{425,752}, CNOT_{423,753}, CNOT_{427,754}, CNOT_{424,755}, CNOT_{446,473}, CNOT_{437,756}, CNOT_{428,757},$
 $CNOT_{426,758}, CNOT_{429,483}, CNOT_{427,760}, CNOT_{423,761}, CNOT_{440,481}, CNOT_{425,763}, CNOT_{426,764},$
 $CNOT_{428,765}, CNOT_{426,766}, CNOT_{429,767}, CNOT_{441,768}, CNOT_{430,769}, CNOT_{431,770}, CNOT_{435,771},$
 $CNOT_{427,772}, CNOT_{423,773}, CNOT_{431,774}, CNOT_{425,775}, CNOT_{430,777}, CNOT_{423,778}, CNOT_{424,779},$
 $CNOT_{461,780}, CNOT_{428,781}, CNOT_{425,782}, CNOT_{436,783}, CNOT_{434,784}, CNOT_{425,785}, CNOT_{429,786},$
 $CNOT_{427,787}, CNOT_{427,788}, CNOT_{428,790}, CNOT_{446,791}, CNOT_{426,792}, CNOT_{425,793}, CNOT_{442,794},$
 $CNOT_{427,795}, CNOT_{425,796}, CNOT_{429,485}, CNOT_{433,484}, CNOT_{424,797}, CNOT_{424,798}, CNOT_{425,799},$
 $CNOT_{444,800}, CNOT_{430,801}, CNOT_{428,802}, CNOT_{437,803}, CNOT_{424,804}, CNOT_{425,805}, CNOT_{434,806},$
 $CNOT_{428,807}, CNOT_{428,808}, CNOT_{432,809}, CNOT_{427,810}, CNOT_{427,486}, CNOT_{434,487}, CNOT_{465,811},$
 $CNOT_{428,812}, CNOT_{431,477}, CNOT_{445,813}, CNOT_{425,814}, CNOT_{427,815}, CNOT_{425,816}, CNOT_{428,490},$
 $CNOT_{430,817}, CNOT_{425,819}, CNOT_{440,820}, CNOT_{435,821}, CNOT_{440,822}, CNOT_{426,823}, CNOT_{426,824},$
 $CNOT_{429,825}, CNOT_{435,826}, CNOT_{452,827}, CNOT_{424,480}, CNOT_{428,828}, CNOT_{429,829}, CNOT_{423,830},$
 $CNOT_{428,831}, CNOT_{428,832}, CNOT_{427,833}, CNOT_{425,834}, CNOT_{495,543}, CNOT_{508,536}, CNOT_{496,835},$
 $CNOT_{506,545}, CNOT_{494,836}, CNOT_{504,837}, CNOT_{496,557}, CNOT_{497,838}, CNOT_{529,839}, CNOT_{505,840},$
 $CNOT_{495,841}, CNOT_{522,535}, CNOT_{501,842}, CNOT_{496,843}, CNOT_{524,844}, CNOT_{495,558}, CNOT_{493,845},$
 $CNOT_{497,846}, CNOT_{496,847}, CNOT_{496,848}, CNOT_{506,849}, CNOT_{495,850}, CNOT_{495,851}, CNOT_{498,852},$
 $CNOT_{522,853}, CNOT_{495,854}, CNOT_{494,855}, CNOT_{492,856}, CNOT_{496,857}, CNOT_{493,858}, CNOT_{515,542},$
 $CNOT_{506,859}, CNOT_{497,860}, CNOT_{495,861}, CNOT_{498,552}, CNOT_{496,863}, CNOT_{492,864}, CNOT_{509,550},$
 $CNOT_{494,866}, CNOT_{495,867}, CNOT_{497,868}, CNOT_{495,869}, CNOT_{498,870}, CNOT_{510,871}, CNOT_{499,872},$
 $CNOT_{500,873}, CNOT_{504,874}, CNOT_{496,875}, CNOT_{492,876}, CNOT_{500,877}, CNOT_{494,878}, CNOT_{499,880},$
 $CNOT_{492,881}, CNOT_{493,882}, CNOT_{530,883}, CNOT_{497,884}, CNOT_{494,885}, CNOT_{505,886}, CNOT_{503,887},$
 $CNOT_{494,888}, CNOT_{498,889}, CNOT_{496,890}, CNOT_{496,891}, CNOT_{497,893}, CNOT_{515,894}, CNOT_{495,895},$
 $CNOT_{494,896}, CNOT_{511,897}, CNOT_{496,898}, CNOT_{494,899}, CNOT_{498,554}, CNOT_{502,553}, CNOT_{493,900},$
 $CNOT_{493,901}, CNOT_{494,902}, CNOT_{513,903}, CNOT_{499,904}, CNOT_{497,905}, CNOT_{506,906}, CNOT_{493,907},$
 $CNOT_{494,908}, CNOT_{503,909}, CNOT_{497,910}, CNOT_{497,911}, CNOT_{501,912}, CNOT_{496,913}, CNOT_{496,555},$
 $CNOT_{503,556}, CNOT_{534,914}, CNOT_{497,915}, CNOT_{500,546}, CNOT_{514,916}, CNOT_{494,917}, CNOT_{496,918},$
 $CNOT_{494,919}, CNOT_{497,559}, CNOT_{499,920}, CNOT_{494,922}, CNOT_{509,923}, CNOT_{504,924}, CNOT_{509,925},$ $\}$

$CNOT_{495,926}, CNOT_{495,927}, CNOT_{498,928}, CNOT_{504,929}, CNOT_{521,930}, CNOT_{493,549}, CNOT_{497,931},$
 $CNOT_{498,932}, CNOT_{492,933}, CNOT_{497,934}, CNOT_{497,935}, CNOT_{496,936}, CNOT_{494,937}, CNOT_{564,612},$
 $CNOT_{577,605}, CNOT_{565,938}, CNOT_{575,614}, CNOT_{563,939}, CNOT_{573,940}, CNOT_{565,626}, CNOT_{566,941},$
 $CNOT_{598,942}, CNOT_{574,943}, CNOT_{564,944}, CNOT_{591,604}, CNOT_{570,945}, CNOT_{565,946}, CNOT_{593,947},$
 $CNOT_{564,627}, CNOT_{562,948}, CNOT_{566,949}, CNOT_{565,950}, CNOT_{565,951}, CNOT_{575,952}, CNOT_{564,953},$
 $CNOT_{564,954}, CNOT_{567,955}, CNOT_{591,956}, CNOT_{564,957}, CNOT_{563,958}, CNOT_{561,959}, CNOT_{565,960},$
 $CNOT_{562,961}, CNOT_{584,611}, CNOT_{575,962}, CNOT_{566,963}, CNOT_{564,964}, CNOT_{567,621}, CNOT_{565,966},$
 $CNOT_{561,967}, CNOT_{578,619}, CNOT_{563,969}, CNOT_{564,970}, CNOT_{566,971}, CNOT_{564,972}, CNOT_{567,973},$
 $CNOT_{579,974}, CNOT_{568,975}, CNOT_{569,976}, CNOT_{573,977}, CNOT_{565,978}, CNOT_{561,979}, CNOT_{569,980},$
 $CNOT_{563,981}, CNOT_{568,983}, CNOT_{561,984}, CNOT_{562,985}, CNOT_{599,986}, CNOT_{566,987}, CNOT_{563,988},$
 $CNOT_{574,989}, CNOT_{572,990}, CNOT_{563,991}, CNOT_{567,992}, CNOT_{565,993}, CNOT_{565,994}, CNOT_{566,996},$
 $CNOT_{584,997}, CNOT_{564,998}, CNOT_{563,999}, CNOT_{580,1000}, CNOT_{565,1001}, CNOT_{563,1002}, CNOT_{567,623},$
 $CNOT_{571,622}, CNOT_{562,1003}, CNOT_{562,1004}, CNOT_{563,1005}, CNOT_{582,1006}, CNOT_{568,1007}, CNOT_{566,1008},$
 $CNOT_{575,1009}, CNOT_{562,1010}, CNOT_{563,1011}, CNOT_{572,1012}, CNOT_{566,1013}, CNOT_{566,1014}, CNOT_{570,1015},$
 $CNOT_{565,1016}, CNOT_{565,624}, CNOT_{572,625}, CNOT_{603,1017}, CNOT_{566,1018}, CNOT_{569,615}, CNOT_{583,1019},$
 $CNOT_{563,1020}, CNOT_{565,1021}, CNOT_{563,1022}, CNOT_{566,628}, CNOT_{568,1023}, CNOT_{563,1025}, CNOT_{578,1026},$
 $CNOT_{573,1027}, CNOT_{578,1028}, CNOT_{564,1029}, CNOT_{564,1030}, CNOT_{567,1031}, CNOT_{573,1032}, CNOT_{590,1033},$
 $CNOT_{562,618}, CNOT_{566,1034}, CNOT_{567,1035}, CNOT_{561,1036}, CNOT_{566,1037}, CNOT_{566,1038}, CNOT_{565,1039},$
 $CNOT_{563,1040}$

Step12:

$CNOT_{358,405}, CNOT_{359,629}, CNOT_{389,630}, CNOT_{378,631}, CNOT_{360,419}, CNOT_{372,632}, CNOT_{369,634},$
 $CNOT_{360,635}, CNOT_{370,637}, CNOT_{368,420}, CNOT_{369,639}, CNOT_{363,640}, CNOT_{374,641}, CNOT_{363,642},$
 $CNOT_{365,644}, CNOT_{358,645}, CNOT_{364,646}, CNOT_{362,648}, CNOT_{357,649}, CNOT_{355,650}, CNOT_{359,651},$
 $CNOT_{374,652}, CNOT_{382,653}, CNOT_{363,654}, CNOT_{359,655}, CNOT_{373,414}, CNOT_{360,657}, CNOT_{360,658},$
 $CNOT_{359,660}, CNOT_{362,661}, CNOT_{361,662}, CNOT_{362,663}, CNOT_{374,665}, CNOT_{371,666}, CNOT_{383,667},$
 $CNOT_{377,668}, CNOT_{362,669}, CNOT_{367,670}, CNOT_{386,671}, CNOT_{371,672}, CNOT_{369,674}, CNOT_{355,675},$
 $CNOT_{356,676}, CNOT_{361,678}, CNOT_{370,679}, CNOT_{368,680}, CNOT_{380,681}, CNOT_{364,682}, CNOT_{364,683},$
 $CNOT_{362,684}, CNOT_{387,685}, CNOT_{396,688}, CNOT_{362,689}, CNOT_{360,690}, CNOT_{374,691}, CNOT_{376,692},$
 $CNOT_{358,693}, CNOT_{372,416}, CNOT_{380,694}, CNOT_{360,695}, CNOT_{357,696}, CNOT_{372,698}, CNOT_{366,699},$
 $CNOT_{374,700}, CNOT_{379,701}, CNOT_{370,702}, CNOT_{373,703}, CNOT_{375,704}, CNOT_{365,705}, CNOT_{371,706},$
 $CNOT_{389,707}, CNOT_{359,417}, CNOT_{365,709}, CNOT_{388,710}, CNOT_{360,711}, CNOT_{361,712}, CNOT_{357,713},$
 $CNOT_{364,421}, CNOT_{376,714}, CNOT_{375,716}, CNOT_{393,719}, CNOT_{376,720}, CNOT_{358,721}, CNOT_{361,722},$
 $CNOT_{394,723}, CNOT_{357,411}, CNOT_{388,725}, CNOT_{363,726}, CNOT_{355,727}, CNOT_{361,728}, CNOT_{365,729},$
 $CNOT_{363,730}, CNOT_{395,731}, CNOT_{427,474}, CNOT_{428,732}, CNOT_{458,733}, CNOT_{447,734}, CNOT_{429,488},$
 $CNOT_{441,735}, CNOT_{438,737}, CNOT_{429,738}, CNOT_{439,740}, CNOT_{437,489}, CNOT_{438,742}, CNOT_{432,743},$
 $CNOT_{443,744}, CNOT_{432,745}, CNOT_{434,747}, CNOT_{427,748}, CNOT_{433,749}, CNOT_{431,751}, CNOT_{426,752},$
 $CNOT_{424,753}, CNOT_{428,754}, CNOT_{443,755}, CNOT_{451,756}, CNOT_{432,757}, CNOT_{428,758}, CNOT_{442,483},$
 $CNOT_{429,760}, CNOT_{429,761}, CNOT_{428,763}, CNOT_{431,764}, CNOT_{430,765}, CNOT_{431,766}, CNOT_{443,768},$
 $CNOT_{440,769}, CNOT_{452,770}, CNOT_{446,771}, CNOT_{431,772}, CNOT_{436,773}, CNOT_{455,774}, CNOT_{440,775},$
 $CNOT_{438,777}, CNOT_{424,778}, CNOT_{425,779}, CNOT_{430,781}, CNOT_{439,782}, CNOT_{437,783}, CNOT_{449,784},$
 $CNOT_{433,785}, CNOT_{433,786}, CNOT_{431,787}, CNOT_{456,788}, CNOT_{465,791}, CNOT_{431,792}, CNOT_{429,793},$
 $CNOT_{443,794}, CNOT_{445,795}, CNOT_{427,796}, CNOT_{441,485}, CNOT_{449,797}, CNOT_{429,798}, CNOT_{426,799},$
 $CNOT_{441,801}, CNOT_{435,802}, CNOT_{443,803}, CNOT_{448,804}, CNOT_{439,805}, CNOT_{442,806}, CNOT_{444,807},$
 $CNOT_{434,808}, CNOT_{440,809}, CNOT_{458,810}, CNOT_{428,486}, CNOT_{434,812}, CNOT_{457,813}, CNOT_{429,814},$
 $CNOT_{430,815}, CNOT_{426,816}, CNOT_{433,490}, CNOT_{445,817}, CNOT_{444,819}, CNOT_{462,822}, CNOT_{445,823},$
 $CNOT_{427,824}, CNOT_{430,825}, CNOT_{463,826}, CNOT_{426,480}, CNOT_{457,828}, CNOT_{432,829}, CNOT_{424,830},$
 $CNOT_{430,831}, CNOT_{434,832}, CNOT_{432,833}, CNOT_{464,834}, CNOT_{496,543}, CNOT_{497,835}, CNOT_{527,836},$
 $CNOT_{516,837}, CNOT_{498,557}, CNOT_{510,838}, CNOT_{507,840}, CNOT_{498,841}, CNOT_{508,843}, CNOT_{506,558},$
 $CNOT_{507,845}, CNOT_{501,846}, CNOT_{512,847}, CNOT_{501,848}, CNOT_{503,850}, CNOT_{496,851}, CNOT_{502,852},$
 $CNOT_{500,854}, CNOT_{495,855}, CNOT_{493,856}, CNOT_{497,857}, CNOT_{512,858}, CNOT_{520,859}, CNOT_{501,860},$
 $CNOT_{497,861}, CNOT_{511,552}, CNOT_{498,863}, CNOT_{498,864}, CNOT_{497,866}, CNOT_{500,867}, CNOT_{499,868},$
 $CNOT_{500,869}, CNOT_{512,871}, CNOT_{509,872}, CNOT_{521,873}, CNOT_{515,874}, CNOT_{500,875}, CNOT_{505,876},$
 $CNOT_{524,877}, CNOT_{509,878}, CNOT_{507,880}, CNOT_{493,881}, CNOT_{494,882}, CNOT_{499,884}, CNOT_{508,885},$
 $CNOT_{506,886}, CNOT_{518,887}, CNOT_{502,888}, CNOT_{502,889}, CNOT_{500,890}, CNOT_{525,891}, CNOT_{534,894},$

$CNOT_{500,895}, CNOT_{498,896}, CNOT_{512,897}, CNOT_{514,898}, CNOT_{496,899}, CNOT_{510,554}, CNOT_{518,900},$
 $CNOT_{498,901}, CNOT_{495,902}, CNOT_{510,904}, CNOT_{504,905}, CNOT_{512,906}, CNOT_{517,907}, CNOT_{508,908},$
 $CNOT_{511,909}, CNOT_{513,910}, CNOT_{503,911}, CNOT_{509,912}, CNOT_{527,913}, CNOT_{497,555}, CNOT_{503,915},$
 $CNOT_{526,916}, CNOT_{498,917}, CNOT_{499,918}, CNOT_{495,919}, CNOT_{502,559}, CNOT_{514,920}, CNOT_{513,922},$
 $CNOT_{531,925}, CNOT_{514,926}, CNOT_{496,927}, CNOT_{499,928}, CNOT_{532,929}, CNOT_{495,549}, CNOT_{526,931},$
 $CNOT_{501,932}, CNOT_{493,933}, CNOT_{499,934}, CNOT_{503,935}, CNOT_{501,936}, CNOT_{533,937}, CNOT_{565,612},$
 $CNOT_{566,938}, CNOT_{596,939}, CNOT_{585,940}, CNOT_{567,626}, CNOT_{579,941}, CNOT_{576,943}, CNOT_{567,944},$
 $CNOT_{577,946}, CNOT_{575,627}, CNOT_{576,948}, CNOT_{570,949}, CNOT_{581,950}, CNOT_{570,951}, CNOT_{572,953},$
 $CNOT_{565,954}, CNOT_{571,955}, CNOT_{569,957}, CNOT_{564,958}, CNOT_{562,959}, CNOT_{566,960}, CNOT_{581,961},$
 $CNOT_{589,962}, CNOT_{570,963}, CNOT_{566,964}, CNOT_{580,621}, CNOT_{567,966}, CNOT_{567,967}, CNOT_{566,969},$
 $CNOT_{569,970}, CNOT_{568,971}, CNOT_{569,972}, CNOT_{581,974}, CNOT_{578,975}, CNOT_{590,976}, CNOT_{584,977},$
 $CNOT_{569,978}, CNOT_{574,979}, CNOT_{593,980}, CNOT_{578,981}, CNOT_{576,983}, CNOT_{562,984}, CNOT_{563,985},$
 $CNOT_{568,987}, CNOT_{577,988}, CNOT_{575,989}, CNOT_{587,990}, CNOT_{571,991}, CNOT_{571,992}, CNOT_{569,993},$
 $CNOT_{594,994}, CNOT_{603,997}, CNOT_{569,998}, CNOT_{567,999}, CNOT_{581,1000}, CNOT_{583,1001}, CNOT_{565,1002},$
 $CNOT_{579,623}, CNOT_{587,1003}, CNOT_{567,1004}, CNOT_{564,1005}, CNOT_{579,1007}, CNOT_{573,1008}, CNOT_{581,1009},$
 $CNOT_{586,1010}, CNOT_{577,1011}, CNOT_{580,1012}, CNOT_{582,1013}, CNOT_{572,1014}, CNOT_{578,1015}, CNOT_{596,1016},$
 $CNOT_{566,624}, CNOT_{572,1018}, CNOT_{595,1019}, CNOT_{567,1020}, CNOT_{568,1021}, CNOT_{564,1022}, CNOT_{571,628},$
 $CNOT_{583,1023}, CNOT_{582,1025}, CNOT_{600,1028}, CNOT_{583,1029}, CNOT_{565,1030}, CNOT_{568,1031}, CNOT_{601,1032},$
 $CNOT_{564,618}, CNOT_{595,1034}, CNOT_{570,1035}, CNOT_{562,1036}, CNOT_{568,1037}, CNOT_{572,1038}, CNOT_{570,1039},$
 $CNOT_{602,1040}$

Step13:

$CNOT_{360,629}, CNOT_{361,419}, CNOT_{396,632}, CNOT_{382,635}, CNOT_{391,640}, CNOT_{375,641}, CNOT_{370,642},$
 $CNOT_{385,644}, CNOT_{360,645}, CNOT_{374,648}, CNOT_{360,649}, CNOT_{356,650}, CNOT_{367,654}, CNOT_{364,655},$
 $CNOT_{381,657}, CNOT_{361,658}, CNOT_{366,660}, CNOT_{364,662}, CNOT_{372,663}, CNOT_{378,672}, CNOT_{385,674},$
 $CNOT_{357,675}, CNOT_{360,676}, CNOT_{392,678}, CNOT_{369,682}, CNOT_{381,684}, CNOT_{378,689}, CNOT_{378,690},$
 $CNOT_{359,693}, CNOT_{380,695}, CNOT_{372,696}, CNOT_{390,699}, CNOT_{373,702}, CNOT_{366,705}, CNOT_{387,417},$
 $CNOT_{379,709}, CNOT_{383,711}, CNOT_{395,712}, CNOT_{360,713}, CNOT_{359,721}, CNOT_{364,722}, CNOT_{367,727},$
 $CNOT_{393,728}, CNOT_{381,730}, CNOT_{429,732}, CNOT_{430,488}, CNOT_{465,735}, CNOT_{451,738}, CNOT_{460,743},$
 $CNOT_{444,744}, CNOT_{439,745}, CNOT_{454,747}, CNOT_{429,748}, CNOT_{443,751}, CNOT_{429,752}, CNOT_{425,753},$
 $CNOT_{436,757}, CNOT_{433,758}, CNOT_{450,760}, CNOT_{430,761}, CNOT_{435,763}, CNOT_{433,765}, CNOT_{441,766},$
 $CNOT_{447,775}, CNOT_{454,777}, CNOT_{426,778}, CNOT_{429,779}, CNOT_{461,781}, CNOT_{438,785}, CNOT_{450,787},$
 $CNOT_{447,792}, CNOT_{447,793}, CNOT_{428,796}, CNOT_{449,798}, CNOT_{441,799}, CNOT_{459,802}, CNOT_{442,805},$
 $CNOT_{435,808}, CNOT_{456,486}, CNOT_{448,812}, CNOT_{452,814}, CNOT_{464,815}, CNOT_{429,816}, CNOT_{428,824},$
 $CNOT_{433,825}, CNOT_{436,830}, CNOT_{462,831}, CNOT_{450,833}, CNOT_{498,835}, CNOT_{499,557}, CNOT_{534,838},$
 $CNOT_{520,841}, CNOT_{529,846}, CNOT_{513,847}, CNOT_{508,848}, CNOT_{523,850}, CNOT_{498,851}, CNOT_{512,854},$
 $CNOT_{498,855}, CNOT_{494,856}, CNOT_{505,860}, CNOT_{502,861}, CNOT_{519,863}, CNOT_{499,864}, CNOT_{504,866},$
 $CNOT_{502,868}, CNOT_{510,869}, CNOT_{516,878}, CNOT_{523,880}, CNOT_{495,881}, CNOT_{498,882}, CNOT_{530,884},$
 $CNOT_{507,888}, CNOT_{519,890}, CNOT_{516,895}, CNOT_{516,896}, CNOT_{497,899}, CNOT_{518,901}, CNOT_{510,902},$
 $CNOT_{528,905}, CNOT_{511,908}, CNOT_{504,911}, CNOT_{525,555}, CNOT_{517,915}, CNOT_{521,917}, CNOT_{533,918},$
 $CNOT_{498,919}, CNOT_{497,927}, CNOT_{502,928}, CNOT_{505,933}, CNOT_{531,934}, CNOT_{519,936}, CNOT_{567,938},$
 $CNOT_{568,626}, CNOT_{603,941}, CNOT_{589,944}, CNOT_{598,949}, CNOT_{582,950}, CNOT_{577,951}, CNOT_{592,953},$
 $CNOT_{567,954}, CNOT_{581,957}, CNOT_{567,958}, CNOT_{563,959}, CNOT_{574,963}, CNOT_{571,964}, CNOT_{588,966},$
 $CNOT_{568,967}, CNOT_{573,969}, CNOT_{571,971}, CNOT_{579,972}, CNOT_{585,981}, CNOT_{592,983}, CNOT_{564,984},$
 $CNOT_{567,985}, CNOT_{599,987}, CNOT_{576,991}, CNOT_{588,993}, CNOT_{585,998}, CNOT_{585,999}, CNOT_{566,1002},$
 $CNOT_{587,1004}, CNOT_{579,1005}, CNOT_{597,1008}, CNOT_{580,1011}, CNOT_{573,1014}, CNOT_{594,624}, CNOT_{586,1018},$
 $CNOT_{590,1020}, CNOT_{602,1021}, CNOT_{567,1022}, CNOT_{566,1030}, CNOT_{571,1031}, CNOT_{574,1036}, CNOT_{600,1037},$
 $CNOT_{588,1039}$

Step14:

$CNOT_{377,629}, CNOT_{369,645}, CNOT_{363,649}, CNOT_{359,675}, CNOT_{379,690}, CNOT_{360,693},$
 $CNOT_{364,713}, CNOT_{384,721}, CNOT_{381,727}, CNOT_{446,732}, CNOT_{438,748}, CNOT_{432,752},$
 $CNOT_{428,778}, CNOT_{448,793}, CNOT_{429,796}, CNOT_{433,816}, CNOT_{453,824}, CNOT_{450,830},$
 $CNOT_{515,835}, CNOT_{507,851}, CNOT_{501,855}, CNOT_{497,881}, CNOT_{517,896}, CNOT_{498,899},$
 $CNOT_{502,919}, CNOT_{522,927}, CNOT_{519,933}, CNOT_{584,938}, CNOT_{576,954}, CNOT_{570,958},$
 $CNOT_{566,984}, CNOT_{586,999}, CNOT_{567,1002}, CNOT_{571,1022}, CNOT_{591,1030}, CNOT_{588,1036},$

Step15:

$$\{CNOT_{362,675}, CNOT_{375,693}, CNOT_{431,778}, CNOT_{444,796}, CNOT_{500,881}, CNOT_{513,899}, \\ CNOT_{569,984}, CNOT_{582,1002},\}$$

Step16:

$$\{CNOT_{249,353}, CNOT_{253,354}, CNOT_{244,355}, CNOT_{254,356}, CNOT_{255,357}, CNOT_{256,358}, CNOT_{252,359}, \\ CNOT_{246,360}, CNOT_{255,361}, CNOT_{250,362}, CNOT_{247,363}, CNOT_{245,364}, CNOT_{251,365}, CNOT_{243,366}, \\ CNOT_{256,367}, CNOT_{254,368}, CNOT_{252,369}, CNOT_{252,370}, CNOT_{246,371}, CNOT_{256,372}, CNOT_{256,373}, \\ CNOT_{249,374}, CNOT_{243,375}, CNOT_{253,376}, CNOT_{253,377}, CNOT_{252,378}, CNOT_{256,379}, CNOT_{256,380}, \\ CNOT_{255,381}, CNOT_{252,382}, CNOT_{252,383}, CNOT_{251,384}, CNOT_{242,385}, CNOT_{255,386}, CNOT_{254,387}, \\ CNOT_{254,388}, CNOT_{253,389}, CNOT_{253,390}, CNOT_{251,391}, CNOT_{251,392}, CNOT_{251,393}, CNOT_{250,394}, \\ CNOT_{248,395}, CNOT_{247,396}, CNOT_{281,422}, CNOT_{285,423}, CNOT_{276,424}, CNOT_{286,425}, CNOT_{287,426}, \\ CNOT_{288,427}, CNOT_{284,428}, CNOT_{278,429}, CNOT_{287,430}, CNOT_{282,431}, CNOT_{279,432}, CNOT_{277,433}, \\ CNOT_{283,434}, CNOT_{275,435}, CNOT_{288,436}, CNOT_{286,437}, CNOT_{284,438}, CNOT_{284,439}, CNOT_{278,440}, \\ CNOT_{288,441}, CNOT_{288,442}, CNOT_{281,443}, CNOT_{275,444}, CNOT_{285,445}, CNOT_{285,446}, CNOT_{284,447}, \\ CNOT_{288,448}, CNOT_{288,449}, CNOT_{287,450}, CNOT_{284,451}, CNOT_{284,452}, CNOT_{283,453}, CNOT_{274,454}, \\ CNOT_{287,455}, CNOT_{286,456}, CNOT_{286,457}, CNOT_{285,458}, CNOT_{285,459}, CNOT_{283,460}, CNOT_{283,461}, \\ CNOT_{283,462}, CNOT_{282,463}, CNOT_{280,464}, CNOT_{279,465}, CNOT_{313,491}, CNOT_{317,492}, CNOT_{308,493}, \\ CNOT_{318,494}, CNOT_{319,495}, CNOT_{320,496}, CNOT_{316,497}, CNOT_{310,498}, CNOT_{319,499}, CNOT_{314,500}, \\ CNOT_{311,501}, CNOT_{309,502}, CNOT_{315,503}, CNOT_{307,504}, CNOT_{320,505}, CNOT_{318,506}, CNOT_{316,507}, \\ CNOT_{316,508}, CNOT_{310,509}, CNOT_{320,510}, CNOT_{320,511}, CNOT_{313,512}, CNOT_{307,513}, CNOT_{317,514}, \\ CNOT_{317,515}, CNOT_{316,516}, CNOT_{320,517}, CNOT_{320,518}, CNOT_{319,519}, CNOT_{316,520}, CNOT_{316,521}, \\ CNOT_{315,522}, CNOT_{306,523}, CNOT_{319,524}, CNOT_{318,525}, CNOT_{318,526}, CNOT_{317,527}, CNOT_{317,528}, \\ CNOT_{315,529}, CNOT_{315,530}, CNOT_{315,531}, CNOT_{314,532}, CNOT_{312,533}, CNOT_{311,534}, CNOT_{345,560}, \\ CNOT_{349,561}, CNOT_{340,562}, CNOT_{350,563}, CNOT_{351,564}, CNOT_{352,565}, CNOT_{348,566}, CNOT_{342,567}, \\ CNOT_{351,568}, CNOT_{346,569}, CNOT_{343,570}, CNOT_{341,571}, CNOT_{347,572}, CNOT_{339,573}, CNOT_{352,574}, \\ CNOT_{350,575}, CNOT_{348,576}, CNOT_{348,577}, CNOT_{342,578}, CNOT_{352,579}, CNOT_{352,580}, CNOT_{345,581}, \\ CNOT_{339,582}, CNOT_{349,583}, CNOT_{349,584}, CNOT_{348,585}, CNOT_{352,586}, CNOT_{352,587}, CNOT_{351,588}, \\ CNOT_{348,589}, CNOT_{348,590}, CNOT_{347,591}, CNOT_{338,592}, CNOT_{351,593}, CNOT_{350,594}, CNOT_{350,595}, \\ CNOT_{349,596}, CNOT_{349,597}, CNOT_{347,598}, CNOT_{347,599}, CNOT_{347,600}, CNOT_{346,601}, CNOT_{344,602}, \\ CNOT_{343,603}\}$$

Step17:

$$\{CNOT_{247,353}, CNOT_{251,354}, CNOT_{242,355}, CNOT_{250,356}, CNOT_{243,357}, CNOT_{245,358}, CNOT_{248,359}, \\ CNOT_{241,360}, CNOT_{254,361}, CNOT_{246,362}, CNOT_{244,363}, CNOT_{242,364}, CNOT_{249,365}, CNOT_{241,366}, \\ CNOT_{241,367}, CNOT_{244,368}, CNOT_{247,369}, CNOT_{243,370}, CNOT_{245,371}, CNOT_{253,372}, CNOT_{244,373}, \\ CNOT_{248,374}, CNOT_{242,375}, CNOT_{252,376}, CNOT_{250,377}, CNOT_{251,378}, CNOT_{255,379}, CNOT_{243,380}, \\ CNOT_{248,381}, CNOT_{249,382}, CNOT_{245,383}, CNOT_{247,384}, CNOT_{241,385}, CNOT_{253,386}, CNOT_{246,387}, \\ CNOT_{241,388}, CNOT_{248,389}, CNOT_{247,390}, CNOT_{250,391}, CNOT_{246,392}, CNOT_{244,393}, CNOT_{245,394}, \\ CNOT_{246,395}, CNOT_{241,396}, CNOT_{279,422}, CNOT_{283,423}, CNOT_{274,424}, CNOT_{282,425}, CNOT_{275,426}, \\ CNOT_{277,427}, CNOT_{280,428}, CNOT_{273,429}, CNOT_{286,430}, CNOT_{278,431}, CNOT_{276,432}, CNOT_{274,433}, \\ CNOT_{281,434}, CNOT_{273,435}, CNOT_{273,436}, CNOT_{276,437}, CNOT_{279,438}, CNOT_{275,439}, CNOT_{277,440}, \\ CNOT_{285,441}, CNOT_{276,442}, CNOT_{280,443}, CNOT_{274,444}, CNOT_{284,445}, CNOT_{282,446}, CNOT_{283,447}, \\ CNOT_{287,448}, CNOT_{275,449}, CNOT_{280,450}, CNOT_{281,451}, CNOT_{277,452}, CNOT_{279,453}, CNOT_{273,454}, \\ CNOT_{285,455}, CNOT_{278,456}, CNOT_{273,457}, CNOT_{280,458}, CNOT_{279,459}, CNOT_{282,460}, CNOT_{278,461}, \\ CNOT_{276,462}, CNOT_{277,463}, CNOT_{278,464}, CNOT_{273,465}, CNOT_{311,491}, CNOT_{315,492}, CNOT_{306,493}, \\ CNOT_{314,494}, CNOT_{307,495}, CNOT_{309,496}, CNOT_{312,497}, CNOT_{305,498}, CNOT_{318,499}, CNOT_{310,500}, \\ CNOT_{308,501}, CNOT_{306,502}, CNOT_{313,503}, CNOT_{305,504}, CNOT_{305,505}, CNOT_{308,506}, CNOT_{311,507}, \\ CNOT_{307,508}, CNOT_{309,509}, CNOT_{317,510}, CNOT_{308,511}, CNOT_{312,512}, CNOT_{306,513}, CNOT_{316,514}, \\ CNOT_{314,515}, CNOT_{315,516}, CNOT_{319,517}, CNOT_{307,518}, CNOT_{312,519}, CNOT_{313,520}, CNOT_{309,521}, \\ CNOT_{311,522}, CNOT_{305,523}, CNOT_{317,524}, CNOT_{310,525}, CNOT_{305,526}, CNOT_{312,527}, CNOT_{311,528}, \\ CNOT_{314,529}, CNOT_{310,530}, CNOT_{308,531}, CNOT_{309,532}, CNOT_{310,533}, CNOT_{305,534}, CNOT_{343,560}, \\ CNOT_{347,561}, CNOT_{338,562}, CNOT_{346,563}, CNOT_{339,564}, CNOT_{341,565}, CNOT_{344,566}, CNOT_{337,567}, \\ CNOT_{350,568}, CNOT_{342,569}, CNOT_{340,570}, CNOT_{338,571}, CNOT_{345,572}, CNOT_{337,573}, CNOT_{337,574}, \\ CNOT_{340,575}, CNOT_{343,576}, CNOT_{339,577}, CNOT_{341,578}, CNOT_{349,579}, CNOT_{340,580}, CNOT_{344,581}, \\ CNOT_{338,582}, CNOT_{348,583}, CNOT_{346,584}, CNOT_{347,585}, CNOT_{351,586}, CNOT_{339,587}, CNOT_{344,588}\}$$

$$\left\{ \begin{array}{l} (CNOT_{345,589}, CNOT_{341,590}, CNOT_{343,591}, CNOT_{337,592}, CNOT_{349,593}, CNOT_{342,594}, CNOT_{337,595}, \\ CNOT_{344,596}, CNOT_{343,597}, CNOT_{346,598}, CNOT_{342,599}, CNOT_{340,600}, CNOT_{341,601}, CNOT_{342,602}, \\ CNOT_{337,603} \end{array} \right\}$$

Step18:

$$\left\{ \begin{array}{l} Toffoli_{240,405,129}, Toffoli_{239,406,130}, Toffoli_{238,398,131}, Toffoli_{237,629,132}, Toffoli_{236,407,133}, \\ Toffoli_{235,630,134}, Toffoli_{234,413,135}, Toffoli_{233,631,136}, Toffoli_{232,419,137}, Toffoli_{231,632,138}, \\ Toffoli_{230,402,139}, Toffoli_{229,633,140}, Toffoli_{228,634,141}, Toffoli_{227,635,142}, Toffoli_{226,397,143}, \\ Toffoli_{225,636,144}, Toffoli_{240,637,145}, Toffoli_{239,638,146}, Toffoli_{238,420,147}, Toffoli_{237,409,148}, \\ Toffoli_{236,639,149}, Toffoli_{235,640,150}, Toffoli_{234,410,151}, Toffoli_{233,641,152}, Toffoli_{232,642,153}, \\ Toffoli_{231,643,154}, Toffoli_{230,644,155}, Toffoli_{229,645,156}, Toffoli_{228,646,157}, Toffoli_{227,647,158}, \\ Toffoli_{226,648,159}, Toffoli_{225,649,160}, Toffoli_{240,650,353}, Toffoli_{239,651,354}, Toffoli_{238,652,355}, \\ Toffoli_{237,404,356}, Toffoli_{236,653,357}, Toffoli_{235,654,358}, Toffoli_{234,655,359}, Toffoli_{233,656,360}, \\ Toffoli_{232,414,361}, Toffoli_{231,657,362}, Toffoli_{230,658,363}, Toffoli_{229,412,364}, Toffoli_{228,659,365}, \\ Toffoli_{227,660,366}, Toffoli_{226,661,367}, Toffoli_{225,662,368}, Toffoli_{240,663,369}, Toffoli_{239,664,370}, \\ Toffoli_{238,665,371}, Toffoli_{237,666,372}, Toffoli_{236,667,373}, Toffoli_{235,668,374}, Toffoli_{234,669,375}, \\ Toffoli_{233,670,376}, Toffoli_{232,671,377}, Toffoli_{231,672,378}, Toffoli_{230,673,379}, Toffoli_{229,674,380}, \\ Toffoli_{228,675,381}, Toffoli_{227,676,382}, Toffoli_{226,677,383}, Toffoli_{225,678,384}, Toffoli_{240,679,385}, \\ Toffoli_{239,680,386}, Toffoli_{238,681,387}, Toffoli_{237,682,388}, Toffoli_{236,683,389}, Toffoli_{235,684,390}, \\ Toffoli_{234,685,391}, Toffoli_{233,686,392}, Toffoli_{232,403,393}, Toffoli_{231,687,394}, Toffoli_{230,688,395}, \\ Toffoli_{229,689,396}, Toffoli_{228,690,422}, Toffoli_{227,691,423}, Toffoli_{226,692,424}, Toffoli_{225,693,425}, \\ Toffoli_{240,416,426}, Toffoli_{239,415,427}, Toffoli_{238,694,428}, Toffoli_{237,695,429}, Toffoli_{236,696,430}, \\ Toffoli_{235,697,431}, Toffoli_{234,698,432}, Toffoli_{233,699,433}, Toffoli_{232,399,434}, Toffoli_{231,700,435}, \\ Toffoli_{230,701,436}, Toffoli_{229,702,437}, Toffoli_{228,703,438}, Toffoli_{227,704,439}, Toffoli_{226,705,440}, \\ Toffoli_{225,706,441}, Toffoli_{240,707,442}, Toffoli_{239,417,443}, Toffoli_{238,418,444}, Toffoli_{237,708,445}, \\ Toffoli_{236,709,446}, Toffoli_{235,408,447}, Toffoli_{234,710,448}, Toffoli_{233,711,449}, Toffoli_{232,712,450}, \\ Toffoli_{231,400,451}, Toffoli_{230,713,452}, Toffoli_{229,421,453}, Toffoli_{228,714,454}, Toffoli_{227,715,455}, \\ Toffoli_{226,716,456}, Toffoli_{225,717,457}, Toffoli_{240,718,458}, Toffoli_{239,401,459}, Toffoli_{238,719,460}, \\ Toffoli_{237,720,461}, Toffoli_{236,721,462}, Toffoli_{235,722,463}, Toffoli_{234,723,464}, Toffoli_{233,724,465}, \\ Toffoli_{232,411,491}, Toffoli_{231,725,492}, Toffoli_{230,726,493}, Toffoli_{229,727,494}, Toffoli_{228,728,495}, \\ Toffoli_{227,729,496}, Toffoli_{226,730,497}, Toffoli_{225,731,498}, Toffoli_{272,474,499}, Toffoli_{271,475,500}, \\ Toffoli_{270,467,501}, Toffoli_{269,732,502}, Toffoli_{268,476,503}, Toffoli_{267,733,504}, Toffoli_{266,482,505}, \\ Toffoli_{265,734,506}, Toffoli_{264,488,507}, Toffoli_{263,735,508}, Toffoli_{262,471,509}, Toffoli_{261,736,510}, \\ Toffoli_{260,737,511}, Toffoli_{259,738,512}, Toffoli_{258,466,513}, Toffoli_{257,739,514}, Toffoli_{272,740,515}, \\ Toffoli_{271,741,516}, Toffoli_{270,489,517}, Toffoli_{269,478,518}, Toffoli_{268,742,519}, Toffoli_{267,743,520}, \\ Toffoli_{266,479,521}, Toffoli_{265,744,522}, Toffoli_{264,745,523}, Toffoli_{263,746,524}, Toffoli_{262,747,525}, \\ Toffoli_{261,748,526}, Toffoli_{260,749,527}, Toffoli_{259,750,528}, Toffoli_{258,751,529}, Toffoli_{257,752,530}, \\ Toffoli_{272,753,531}, Toffoli_{271,754,532}, Toffoli_{270,755,533}, Toffoli_{269,473,534}, Toffoli_{268,756,560}, \\ Toffoli_{267,757,561}, Toffoli_{266,758,562}, Toffoli_{265,759,563}, Toffoli_{264,483,564}, Toffoli_{263,760,565}, \\ Toffoli_{262,761,566}, Toffoli_{261,481,567}, Toffoli_{260,762,568}, Toffoli_{259,763,569}, Toffoli_{258,764,570}, \\ Toffoli_{257,765,571}, Toffoli_{272,766,572}, Toffoli_{271,767,573}, Toffoli_{270,768,574}, Toffoli_{269,769,575}, \\ Toffoli_{268,770,576}, Toffoli_{267,771,577}, Toffoli_{266,772,578}, Toffoli_{265,773,579}, Toffoli_{264,774,580}, \\ Toffoli_{263,775,581}, Toffoli_{262,776,582}, Toffoli_{261,777,583}, Toffoli_{260,778,584}, Toffoli_{259,779,585}, \\ Toffoli_{258,780,586}, Toffoli_{257,781,587}, Toffoli_{272,782,588}, Toffoli_{271,783,589}, Toffoli_{270,784,590}, \\ Toffoli_{269,785,591}, Toffoli_{268,786,592}, Toffoli_{267,787,593}, Toffoli_{266,788,594}, Toffoli_{265,789,595}, \\ Toffoli_{264,472,596}, Toffoli_{263,790,597}, Toffoli_{262,791,598}, Toffoli_{261,792,599}, Toffoli_{260,793,600}, \\ Toffoli_{259,794,601}, Toffoli_{258,795,602}, Toffoli_{257,796,603}, Toffoli_{272,485,1041}, Toffoli_{271,484,1042}, \\ Toffoli_{270,797,1043}, Toffoli_{269,798,1044}, Toffoli_{268,799,1045}, Toffoli_{267,800,1046}, Toffoli_{266,801,1047}, \\ Toffoli_{265,802,1048}, Toffoli_{264,468,1049}, Toffoli_{263,803,1050}, Toffoli_{262,804,1051}, Toffoli_{261,805,1052}, \\ Toffoli_{260,806,1053}, Toffoli_{259,807,1054}, Toffoli_{258,808,1055}, Toffoli_{257,809,1056}, Toffoli_{272,810,1057}, \\ Toffoli_{271,486,1058}, Toffoli_{270,487,1059}, Toffoli_{269,811,1060}, Toffoli_{268,812,1061}, Toffoli_{267,477,1062}, \\ Toffoli_{266,813,1063}, Toffoli_{265,814,1064}, Toffoli_{264,815,1065}, Toffoli_{263,469,1066}, Toffoli_{262,816,1067}, \\ Toffoli_{261,490,1068}, Toffoli_{260,817,1069}, Toffoli_{259,818,1070}, Toffoli_{258,819,1071}, Toffoli_{257,820,1072}, \\ Toffoli_{272,821,1073}, Toffoli_{271,470,1074}, Toffoli_{270,822,1075}, Toffoli_{269,823,1076}, Toffoli_{268,824,1077}, \end{array} \right\}$$

*Toffoli*_{267,825,1078}, *Toffoli*_{266,826,1079}, *Toffoli*_{265,827,1080}, *Toffoli*_{264,480,1081}, *Toffoli*_{263,828,1082},
*Toffoli*_{262,829,1083}, *Toffoli*_{261,830,1084}, *Toffoli*_{260,831,1085}, *Toffoli*_{259,832,1086}, *Toffoli*_{258,833,1087},
*Toffoli*_{257,834,1088}, *Toffoli*_{304,543,1089}, *Toffoli*_{303,544,1090}, *Toffoli*_{302,536,1091}, *Toffoli*_{301,835,1092},
*Toffoli*_{300,545,1093}, *Toffoli*_{299,836,1094}, *Toffoli*_{298,551,1095}, *Toffoli*_{297,837,1096}, *Toffoli*_{296,557,1097},
*Toffoli*_{295,838,1098}, *Toffoli*_{294,540,1099}, *Toffoli*_{293,839,1100}, *Toffoli*_{292,840,1101}, *Toffoli*_{291,841,1102},
*Toffoli*_{290,535,1103}, *Toffoli*_{289,842,1104}, *Toffoli*_{304,843,1105}, *Toffoli*_{303,844,1106}, *Toffoli*_{302,558,1107},
*Toffoli*_{301,547,1108}, *Toffoli*_{300,845,1109}, *Toffoli*_{299,846,1110}, *Toffoli*_{298,548,1111}, *Toffoli*_{297,847,1112},
*Toffoli*_{296,848,1113}, *Toffoli*_{295,849,1114}, *Toffoli*_{294,850,1115}, *Toffoli*_{293,851,1116}, *Toffoli*_{292,852,1117},
*Toffoli*_{291,853,1118}, *Toffoli*_{290,854,1119}, *Toffoli*_{289,855,1120}, *Toffoli*_{304,856,1121}, *Toffoli*_{303,857,1122},
*Toffoli*_{302,858,1123}, *Toffoli*_{301,542,1124}, *Toffoli*_{300,859,1125}, *Toffoli*_{299,860,1126}, *Toffoli*_{298,861,1127},
*Toffoli*_{297,862,1128}, *Toffoli*_{296,552,1129}, *Toffoli*_{295,863,1130}, *Toffoli*_{294,864,1131}, *Toffoli*_{293,550,1132},
*Toffoli*_{292,865,1133}, *Toffoli*_{291,866,1134}, *Toffoli*_{290,867,1135}, *Toffoli*_{289,868,1136}, *Toffoli*_{304,869,1137},
*Toffoli*_{303,870,1138}, *Toffoli*_{302,871,1139}, *Toffoli*_{301,872,1140}, *Toffoli*_{300,873,1141}, *Toffoli*_{299,874,1142},
*Toffoli*_{298,875,1143}, *Toffoli*_{297,876,1144}, *Toffoli*_{296,877,1145}, *Toffoli*_{295,878,1146}, *Toffoli*_{294,879,1147},
*Toffoli*_{293,880,1148}, *Toffoli*_{292,881,1149}, *Toffoli*_{291,882,1150}, *Toffoli*_{290,883,1151}, *Toffoli*_{289,884,1152},
*Toffoli*_{304,885,1153}, *Toffoli*_{303,886,1154}, *Toffoli*_{302,887,1155}, *Toffoli*_{301,888,1156}, *Toffoli*_{300,889,1157},
*Toffoli*_{299,890,1158}, *Toffoli*_{298,891,1159}, *Toffoli*_{297,892,1160}, *Toffoli*_{296,541,1161}, *Toffoli*_{295,893,1162},
*Toffoli*_{294,894,1163}, *Toffoli*_{293,895,1164}, *Toffoli*_{292,896,1165}, *Toffoli*_{291,897,1166}, *Toffoli*_{290,898,1167},
*Toffoli*_{289,899,1168}, *Toffoli*_{304,554,1169}, *Toffoli*_{303,553,1170}, *Toffoli*_{302,900,1171}, *Toffoli*_{301,901,1172},
*Toffoli*_{300,902,1173}, *Toffoli*_{299,903,1174}, *Toffoli*_{298,904,1175}, *Toffoli*_{297,905,1176}, *Toffoli*_{296,537,1177},
*Toffoli*_{295,906,1178}, *Toffoli*_{294,907,1179}, *Toffoli*_{293,908,1180}, *Toffoli*_{292,909,1181}, *Toffoli*_{291,910,1182},
*Toffoli*_{290,911,1183}, *Toffoli*_{289,912,1184}, *Toffoli*_{304,913,1185}, *Toffoli*_{303,555,1186}, *Toffoli*_{302,556,1187},
*Toffoli*_{301,914,1188}, *Toffoli*_{300,915,1189}, *Toffoli*_{299,546,1190}, *Toffoli*_{298,916,1191}, *Toffoli*_{297,917,1192},
*Toffoli*_{296,918,1193}, *Toffoli*_{295,538,1194}, *Toffoli*_{294,919,1195}, *Toffoli*_{293,559,1196}, *Toffoli*_{292,920,1197},
*Toffoli*_{291,921,1198}, *Toffoli*_{290,922,1199}, *Toffoli*_{289,923,1200}, *Toffoli*_{304,924,1201}, *Toffoli*_{303,539,1202},
*Toffoli*_{302,925,1203}, *Toffoli*_{301,926,1204}, *Toffoli*_{300,927,1205}, *Toffoli*_{299,928,1206}, *Toffoli*_{298,929,1207},
*Toffoli*_{297,930,1208}, *Toffoli*_{296,549,1209}, *Toffoli*_{295,931,1210}, *Toffoli*_{294,932,1211}, *Toffoli*_{293,933,1212},
*Toffoli*_{292,934,1213}, *Toffoli*_{291,935,1214}, *Toffoli*_{290,936,1215}, *Toffoli*_{289,937,1216}, *Toffoli*_{336,612,1217},
*Toffoli*_{335,613,1218}, *Toffoli*_{334,605,1219}, *Toffoli*_{333,938,1220}, *Toffoli*_{332,614,1221}, *Toffoli*_{331,939,1222},
*Toffoli*_{330,620,1223}, *Toffoli*_{329,940,1224}, *Toffoli*_{328,626,1225}, *Toffoli*_{327,941,1226}, *Toffoli*_{326,609,1227},
*Toffoli*_{325,942,1228}, *Toffoli*_{324,943,1229}, *Toffoli*_{323,944,1230}, *Toffoli*_{322,604,1231}, *Toffoli*_{321,945,1232},
*Toffoli*_{336,946,1233}, *Toffoli*_{335,947,1234}, *Toffoli*_{334,627,1235}, *Toffoli*_{333,616,1236}, *Toffoli*_{332,948,1237},
*Toffoli*_{331,949,1238}, *Toffoli*_{330,617,1239}, *Toffoli*_{329,950,1240}, *Toffoli*_{328,951,1241}, *Toffoli*_{327,952,1242},
*Toffoli*_{326,953,1243}, *Toffoli*_{325,954,1244}, *Toffoli*_{324,955,1245}, *Toffoli*_{323,956,1246}, *Toffoli*_{322,957,1247},
*Toffoli*_{321,958,1248}, *Toffoli*_{336,959,1249}, *Toffoli*_{335,960,1250}, *Toffoli*_{334,961,1251}, *Toffoli*_{333,611,1252},
*Toffoli*_{332,962,1253}, *Toffoli*_{331,963,1254}, *Toffoli*_{330,964,1255}, *Toffoli*_{329,965,1256}, *Toffoli*_{328,621,1257},
*Toffoli*_{327,966,1258}, *Toffoli*_{326,967,1259}, *Toffoli*_{325,619,1260}, *Toffoli*_{324,968,1261}, *Toffoli*_{323,969,1262},
*Toffoli*_{322,970,1263}, *Toffoli*_{321,971,1264}, *Toffoli*_{336,972,1265}, *Toffoli*_{335,973,1266}, *Toffoli*_{334,974,1267},
*Toffoli*_{333,975,1268}, *Toffoli*_{332,976,1269}, *Toffoli*_{331,977,1270}, *Toffoli*_{330,978,1271}, *Toffoli*_{329,979,1272},
*Toffoli*_{328,980,1273}, *Toffoli*_{327,981,1274}, *Toffoli*_{326,982,1275}, *Toffoli*_{325,983,1276}, *Toffoli*_{324,984,1277},
*Toffoli*_{323,985,1278}, *Toffoli*_{322,986,1279}, *Toffoli*_{321,987,1280}, *Toffoli*_{336,988,1281}, *Toffoli*_{335,989,1282},
*Toffoli*_{334,990,1283}, *Toffoli*_{333,991,1284}, *Toffoli*_{332,992,1285}, *Toffoli*_{331,993,1286}, *Toffoli*_{330,994,1287},
*Toffoli*_{329,995,1288}, *Toffoli*_{328,610,1289}, *Toffoli*_{327,996,1290}, *Toffoli*_{326,997,1291}, *Toffoli*_{325,998,1292},
*Toffoli*_{324,999,1293}, *Toffoli*_{323,1000,1294}, *Toffoli*_{322,1001,1295}, *Toffoli*_{321,1002,1296},
*Toffoli*_{336,623,1297}, *Toffoli*_{335,622,1298}, *Toffoli*_{334,1003,1299}, *Toffoli*_{333,1004,1300},
*Toffoli*_{332,1005,1301}, *Toffoli*_{331,1006,1302}, *Toffoli*_{330,1007,1303}, *Toffoli*_{329,1008,1304},
*Toffoli*_{328,606,1305}, *Toffoli*_{327,1009,1306}, *Toffoli*_{326,1010,1307}, *Toffoli*_{325,1011,1308},
*Toffoli*_{324,1012,1309}, *Toffoli*_{323,1013,1310}, *Toffoli*_{322,1014,1311}, *Toffoli*_{321,1015,1312},
*Toffoli*_{336,1016,1313}, *Toffoli*_{335,624,1314}, *Toffoli*_{334,625,1315}, *Toffoli*_{333,1017,1316},
*Toffoli*_{332,1018,1317}, *Toffoli*_{331,615,1318}, *Toffoli*_{330,1019,1319}, *Toffoli*_{329,1020,1320},
*Toffoli*_{328,1021,1321}, *Toffoli*_{327,607,1322}, *Toffoli*_{326,1022,1323}, *Toffoli*_{325,628,1324},
*Toffoli*_{324,1023,1325}, *Toffoli*_{323,1024,1326}, *Toffoli*_{322,1025,1327}, *Toffoli*_{321,1026,1328},
*Toffoli*_{336,1027,1329}, *Toffoli*_{335,608,1330}, *Toffoli*_{334,1028,1331}, *Toffoli*_{333,1029,1332},

$$\left\{ \begin{array}{l} Toffoli_{332,1030,1333}, Toffoli_{331,1031,1334}, Toffoli_{330,1032,1335}, Toffoli_{329,1033,1336}, \\ Toffoli_{328,618,1337}, Toffoli_{327,1034,1338}, Toffoli_{326,1035,1339}, Toffoli_{325,1036,1340}, \\ Toffoli_{324,1037,1341}, Toffoli_{323,1038,1342}, Toffoli_{322,1039,1343}, Toffoli_{321,1040,1344} \end{array} \right\}$$

Step19:

$$\left\{ \begin{array}{l} CNOT_{130,129}, CNOT_{132,131}, CNOT_{134,133}, CNOT_{136,135}, CNOT_{138,137}, CNOT_{140,139}, CNOT_{142,141}, \\ CNOT_{144,143}, CNOT_{146,145}, CNOT_{148,147}, CNOT_{150,149}, CNOT_{152,151}, CNOT_{154,153}, CNOT_{156,155}, \\ CNOT_{158,157}, CNOT_{160,159}, CNOT_{354,353}, CNOT_{356,355}, CNOT_{358,357}, CNOT_{360,359}, CNOT_{362,361}, \\ CNOT_{364,363}, CNOT_{366,365}, CNOT_{368,367}, CNOT_{370,369}, CNOT_{372,371}, CNOT_{374,373}, CNOT_{376,375}, \\ CNOT_{378,377}, CNOT_{380,379}, CNOT_{382,381}, CNOT_{384,383}, CNOT_{386,385}, CNOT_{388,387}, CNOT_{390,389}, \\ CNOT_{392,391}, CNOT_{394,393}, CNOT_{396,395}, CNOT_{423,422}, CNOT_{425,424}, CNOT_{427,426}, CNOT_{429,428}, \\ CNOT_{431,430}, CNOT_{433,432}, CNOT_{435,434}, CNOT_{437,436}, CNOT_{439,438}, CNOT_{441,440}, CNOT_{443,442}, \\ CNOT_{445,444}, CNOT_{447,446}, CNOT_{449,448}, CNOT_{451,450}, CNOT_{453,452}, CNOT_{455,454}, CNOT_{457,456}, \\ CNOT_{459,458}, CNOT_{461,460}, CNOT_{463,462}, CNOT_{465,464}, CNOT_{492,491}, CNOT_{494,493}, CNOT_{496,495}, \\ CNOT_{498,497}, CNOT_{500,499}, CNOT_{502,501}, CNOT_{504,503}, CNOT_{506,505}, CNOT_{508,507}, CNOT_{510,509}, \\ CNOT_{512,511}, CNOT_{514,513}, CNOT_{516,515}, CNOT_{518,517}, CNOT_{520,519}, CNOT_{522,521}, CNOT_{524,523}, \\ CNOT_{526,525}, CNOT_{528,527}, CNOT_{530,529}, CNOT_{532,531}, CNOT_{534,533}, CNOT_{561,560}, CNOT_{563,562}, \\ CNOT_{565,564}, CNOT_{567,566}, CNOT_{569,568}, CNOT_{571,570}, CNOT_{573,572}, CNOT_{575,574}, CNOT_{577,576}, \\ CNOT_{579,578}, CNOT_{581,580}, CNOT_{583,582}, CNOT_{585,584}, CNOT_{587,586}, CNOT_{589,588}, CNOT_{591,590}, \\ CNOT_{593,592}, CNOT_{595,594}, CNOT_{597,596}, CNOT_{599,598}, CNOT_{601,600}, CNOT_{603,602}, CNOT_{1042,1041}, \\ CNOT_{1044,1043}, CNOT_{1046,1045}, CNOT_{1048,1047}, CNOT_{1050,1049}, CNOT_{1052,1051}, CNOT_{1054,1053}, \\ CNOT_{1056,1055}, CNOT_{1058,1057}, CNOT_{1060,1059}, CNOT_{1062,1061}, CNOT_{1064,1063}, CNOT_{1066,1065}, \\ CNOT_{1068,1067}, CNOT_{1070,1069}, CNOT_{1072,1071}, CNOT_{1074,1073}, CNOT_{1076,1075}, CNOT_{1078,1077}, \\ CNOT_{1080,1079}, CNOT_{1082,1081}, CNOT_{1084,1083}, CNOT_{1086,1085}, CNOT_{1088,1087}, CNOT_{1090,1089}, \\ CNOT_{1092,1091}, CNOT_{1094,1093}, CNOT_{1096,1095}, CNOT_{1098,1097}, CNOT_{1100,1099}, CNOT_{1102,1101}, \\ CNOT_{1104,1103}, CNOT_{1106,1105}, CNOT_{1108,1107}, CNOT_{1110,1109}, CNOT_{1112,1111}, CNOT_{1114,1113}, \\ CNOT_{1116,1115}, CNOT_{1118,1117}, CNOT_{1120,1119}, CNOT_{1122,1121}, CNOT_{1124,1123}, CNOT_{1126,1125}, \\ CNOT_{1128,1127}, CNOT_{1130,1129}, CNOT_{1132,1131}, CNOT_{1134,1133}, CNOT_{1136,1135}, CNOT_{1138,1137}, \\ CNOT_{1140,1139}, CNOT_{1142,1141}, CNOT_{1144,1143}, CNOT_{1146,1145}, CNOT_{1148,1147}, CNOT_{1150,1149}, \\ CNOT_{1152,1151}, CNOT_{1154,1153}, CNOT_{1156,1155}, CNOT_{1158,1157}, CNOT_{1160,1159}, CNOT_{1162,1161}, \\ CNOT_{1164,1163}, CNOT_{1166,1165}, CNOT_{1168,1167}, CNOT_{1170,1169}, CNOT_{1172,1171}, CNOT_{1174,1173}, \\ CNOT_{1176,1175}, CNOT_{1178,1177}, CNOT_{1180,1179}, CNOT_{1182,1181}, CNOT_{1184,1183}, CNOT_{1186,1185}, \\ CNOT_{1188,1187}, CNOT_{1190,1189}, CNOT_{1192,1191}, CNOT_{1194,1193}, CNOT_{1196,1195}, CNOT_{1198,1197}, \\ CNOT_{1200,1199}, CNOT_{1202,1201}, CNOT_{1204,1203}, CNOT_{1206,1205}, CNOT_{1208,1207}, CNOT_{1210,1209}, \\ CNOT_{1212,1211}, CNOT_{1214,1213}, CNOT_{1216,1215}, CNOT_{1218,1217}, CNOT_{1220,1219}, CNOT_{1222,1221}, \\ CNOT_{1224,1223}, CNOT_{1226,1225}, CNOT_{1228,1227}, CNOT_{1230,1229}, CNOT_{1232,1231}, CNOT_{1234,1233}, \\ CNOT_{1236,1235}, CNOT_{1238,1237}, CNOT_{1240,1239}, CNOT_{1242,1241}, CNOT_{1244,1243}, CNOT_{1246,1245}, \\ CNOT_{1248,1247}, CNOT_{1250,1249}, CNOT_{1252,1251}, CNOT_{1254,1253}, CNOT_{1256,1255}, CNOT_{1258,1257}, \\ CNOT_{1260,1259}, CNOT_{1262,1261}, CNOT_{1264,1263}, CNOT_{1266,1265}, CNOT_{1268,1267}, CNOT_{1270,1269}, \\ CNOT_{1272,1271}, CNOT_{1274,1273}, CNOT_{1276,1275}, CNOT_{1278,1277}, CNOT_{1280,1279}, CNOT_{1282,1281}, \\ CNOT_{1284,1283}, CNOT_{1286,1285}, CNOT_{1288,1287}, CNOT_{1290,1289}, CNOT_{1292,1291}, CNOT_{1294,1293}, \\ CNOT_{1296,1295}, CNOT_{1298,1297}, CNOT_{1300,1299}, CNOT_{1302,1301}, CNOT_{1304,1303}, CNOT_{1306,1305}, \\ CNOT_{1308,1307}, CNOT_{1310,1309}, CNOT_{1312,1311}, CNOT_{1314,1313}, CNOT_{1316,1315}, CNOT_{1318,1317}, \\ CNOT_{1320,1319}, CNOT_{1322,1321}, CNOT_{1324,1323}, CNOT_{1326,1325}, CNOT_{1328,1327}, CNOT_{1330,1329}, \\ CNOT_{1332,1331}, CNOT_{1334,1333}, CNOT_{1336,1335}, CNOT_{1338,1337}, CNOT_{1340,1339}, CNOT_{1342,1341}, \\ CNOT_{1344,1343} \end{array} \right\}$$

Step20:

$$\left\{ \begin{array}{l} CNOT_{131,129}, CNOT_{135,133}, CNOT_{139,137}, CNOT_{143,141}, CNOT_{147,145}, CNOT_{151,149}, CNOT_{155,153}, \\ CNOT_{159,157}, CNOT_{355,353}, CNOT_{359,357}, CNOT_{363,361}, CNOT_{367,365}, CNOT_{371,369}, CNOT_{375,373}, \\ CNOT_{379,377}, CNOT_{383,381}, CNOT_{387,385}, CNOT_{391,389}, CNOT_{395,393}, CNOT_{424,422}, CNOT_{428,426}, \\ CNOT_{432,430}, CNOT_{436,434}, CNOT_{440,438}, CNOT_{444,442}, CNOT_{448,446}, CNOT_{452,450}, CNOT_{456,454}, \\ CNOT_{460,458}, CNOT_{464,462}, CNOT_{493,491}, CNOT_{497,495}, CNOT_{501,499}, CNOT_{505,503}, CNOT_{509,507}, \\ CNOT_{513,511}, CNOT_{517,515}, CNOT_{521,519}, CNOT_{525,523}, CNOT_{529,527}, CNOT_{533,531}, CNOT_{562,560}, \\ CNOT_{566,564}, CNOT_{570,568}, CNOT_{574,572}, CNOT_{578,576}, CNOT_{582,580}, CNOT_{586,584}, CNOT_{590,588}, \\ CNOT_{594,592}, CNOT_{598,596}, CNOT_{602,600}, CNOT_{1043,1041}, CNOT_{1047,1045}, CNOT_{1051,1049}, \end{array} \right\}$$

$$\left\{ \begin{array}{l}
 CNOT_{1055,1053}, CNOT_{1059,1057}, CNOT_{1063,1061}, CNOT_{1067,1065}, CNOT_{1071,1069}, CNOT_{1075,1073}, \\
 CNOT_{1079,1077}, CNOT_{1083,1081}, CNOT_{1087,1085}, CNOT_{1091,1089}, CNOT_{1095,1093}, CNOT_{1099,1097}, \\
 CNOT_{1103,1101}, CNOT_{1107,1105}, CNOT_{1111,1109}, CNOT_{1115,1113}, CNOT_{1119,1117}, CNOT_{1123,1121}, \\
 CNOT_{1127,1125}, CNOT_{1131,1129}, CNOT_{1135,1133}, CNOT_{1139,1137}, CNOT_{1143,1141}, CNOT_{1147,1145}, \\
 CNOT_{1151,1149}, CNOT_{1155,1153}, CNOT_{1159,1157}, CNOT_{1163,1161}, CNOT_{1167,1165}, CNOT_{1171,1169}, \\
 CNOT_{1175,1173}, CNOT_{1179,1177}, CNOT_{1183,1181}, CNOT_{1187,1185}, CNOT_{1191,1189}, CNOT_{1195,1193}, \\
 CNOT_{1199,1197}, CNOT_{1203,1201}, CNOT_{1207,1205}, CNOT_{1211,1209}, CNOT_{1215,1213}, CNOT_{1219,1217}, \\
 CNOT_{1223,1221}, CNOT_{1227,1225}, CNOT_{1231,1229}, CNOT_{1235,1233}, CNOT_{1239,1237}, CNOT_{1243,1241}, \\
 CNOT_{1247,1245}, CNOT_{1251,1249}, CNOT_{1255,1253}, CNOT_{1259,1257}, CNOT_{1263,1261}, CNOT_{1267,1265}, \\
 CNOT_{1271,1269}, CNOT_{1275,1273}, CNOT_{1279,1277}, CNOT_{1283,1281}, CNOT_{1287,1285}, CNOT_{1291,1289}, \\
 CNOT_{1295,1293}, CNOT_{1299,1297}, CNOT_{1303,1301}, CNOT_{1307,1305}, CNOT_{1311,1309}, CNOT_{1315,1313}, \\
 CNOT_{1319,1317}, CNOT_{1323,1321}, CNOT_{1327,1325}, CNOT_{1331,1329}, CNOT_{1335,1333}, CNOT_{1339,1337}, \\
 CNOT_{1343,1341}
 \end{array} \right\}$$

Step21:

$$\left\{ \begin{array}{l}
 CNOT_{133,129}, CNOT_{141,137}, CNOT_{149,145}, CNOT_{157,153}, CNOT_{357,353}, CNOT_{365,361}, CNOT_{373,369}, \\
 CNOT_{381,377}, CNOT_{389,385}, CNOT_{422,393}, CNOT_{430,426}, CNOT_{438,434}, CNOT_{446,442}, CNOT_{454,450}, \\
 CNOT_{462,458}, CNOT_{495,491}, CNOT_{503,499}, CNOT_{511,507}, CNOT_{519,515}, CNOT_{527,523}, CNOT_{560,531}, \\
 CNOT_{568,564}, CNOT_{576,572}, CNOT_{584,580}, CNOT_{592,588}, CNOT_{600,596}, CNOT_{1045,1041}, \\
 CNOT_{1053,1049}, CNOT_{1061,1057}, CNOT_{1069,1065}, CNOT_{1077,1073}, CNOT_{1085,1081}, \\
 CNOT_{1093,1089}, CNOT_{1101,1097}, CNOT_{1109,1105}, CNOT_{1117,1113}, CNOT_{1125,1121}, \\
 CNOT_{1133,1129}, CNOT_{1141,1137}, CNOT_{1149,1145}, CNOT_{1157,1153}, CNOT_{1165,1161}, \\
 CNOT_{1173,1169}, CNOT_{1181,1177}, CNOT_{1189,1185}, CNOT_{1197,1193}, CNOT_{1205,1201}, \\
 CNOT_{1213,1209}, CNOT_{1221,1217}, CNOT_{1229,1225}, CNOT_{1237,1233}, CNOT_{1245,1241}, \\
 CNOT_{1253,1249}, CNOT_{1261,1257}, CNOT_{1269,1265}, CNOT_{1277,1273}, CNOT_{1285,1281}, \\
 CNOT_{1293,1289}, CNOT_{1301,1297}, CNOT_{1309,1305}, CNOT_{1317,1313}, CNOT_{1325,1321}, \\
 CNOT_{1333,1329}, CNOT_{1341,1337}
 \end{array} \right\}$$

Step22:

$$\left\{ \begin{array}{l}
 CNOT_{137,129}, CNOT_{153,145}, CNOT_{361,353}, CNOT_{377,369}, CNOT_{393,385}, CNOT_{434,426}, CNOT_{450,442}, \\
 CNOT_{491,458}, CNOT_{507,499}, CNOT_{523,515}, CNOT_{564,531}, CNOT_{580,572}, CNOT_{596,588}, CNOT_{1049,1041}, \\
 CNOT_{1065,1057}, CNOT_{1081,1073}, CNOT_{1097,1089}, CNOT_{1113,1105}, CNOT_{1129,1121}, CNOT_{1145,1137}, \\
 CNOT_{1161,1153}, CNOT_{1177,1169}, CNOT_{1193,1185}, CNOT_{1209,1201}, CNOT_{1225,1217}, CNOT_{1241,1233}, \\
 CNOT_{1257,1249}, CNOT_{1273,1265}, CNOT_{1289,1281}, CNOT_{1305,1297}, CNOT_{1321,1313}, CNOT_{1337,1329}
 \end{array} \right\}$$

Step23:

$$\left\{ \begin{array}{l}
 CNOT_{129,97}, CNOT_{145,98}, CNOT_{353,99}, CNOT_{129,100}, CNOT_{145,101}, CNOT_{353,102}, CNOT_{369,103}, \\
 CNOT_{385,104}, CNOT_{129,105}, CNOT_{145,106}, CNOT_{353,107}, CNOT_{369,108}, CNOT_{385,109}, CNOT_{426,110}, \\
 CNOT_{129,111}, CNOT_{145,112}, CNOT_{353,113}, CNOT_{369,114}, CNOT_{129,115}, CNOT_{145,116}, CNOT_{353,117}, \\
 CNOT_{369,118}, CNOT_{129,119}, CNOT_{145,120}, CNOT_{353,121}, CNOT_{369,122}, CNOT_{385,123}, CNOT_{426,124}, \\
 CNOT_{442,125}, CNOT_{458,126}, CNOT_{129,127}, CNOT_{145,128}
 \end{array} \right\}$$

Step24:

$$\left\{ \begin{array}{l}
 CNOT_{353,97}, CNOT_{369,98}, CNOT_{385,99}, CNOT_{369,100}, CNOT_{385,101}, CNOT_{426,102}, CNOT_{442,103}, \\
 CNOT_{458,104}, CNOT_{426,105}, CNOT_{442,106}, CNOT_{458,107}, CNOT_{499,108}, CNOT_{515,109}, CNOT_{531,110}, \\
 CNOT_{442,111}, CNOT_{458,112}, CNOT_{499,113}, CNOT_{515,114}, CNOT_{385,115}, CNOT_{426,116}, CNOT_{442,117}, \\
 CNOT_{458,118}, CNOT_{385,119}, CNOT_{426,120}, CNOT_{442,121}, CNOT_{458,122}, CNOT_{499,123}, CNOT_{515,124}, \\
 CNOT_{531,125}, CNOT_{572,126}, CNOT_{499,127}, CNOT_{515,128}
 \end{array} \right\}$$

Step25:

$$\left\{ \begin{array}{l}
 CNOT_{531,97}, CNOT_{572,98}, CNOT_{588,99}, CNOT_{426,100}, CNOT_{442,101}, CNOT_{458,102}, CNOT_{499,103}, \\
 CNOT_{515,104}, CNOT_{499,105}, CNOT_{515,106}, CNOT_{531,107}, CNOT_{572,108}, CNOT_{588,109}, CNOT_{1041,110}, \\
 CNOT_{572,111}, CNOT_{588,112}, CNOT_{1041,113}, CNOT_{1057,114}, CNOT_{531,115}, CNOT_{572,116}, CNOT_{588,117}, \\
 CNOT_{1041,118}, CNOT_{499,119}, CNOT_{515,120}, CNOT_{531,121}, CNOT_{572,122}, CNOT_{588,123}, CNOT_{1041,124}, \\
 CNOT_{1057,125}, CNOT_{1073,126}, CNOT_{588,127}, CNOT_{1041,128}
 \end{array} \right\}$$

Step26:

$$\left\{ \begin{array}{l}
 CNOT_{1057,97}, CNOT_{1073,98}, CNOT_{1089,99}, CNOT_{1041,100}, CNOT_{1057,101}, CNOT_{1073,102}, CNOT_{1089,103}, \\
 CNOT_{1105,104}, CNOT_{531,105}, CNOT_{572,106}, CNOT_{588,107}, CNOT_{1041,108}, CNOT_{1057,109}, CNOT_{1073,110}, \\
 CNOT_{1057,111}, CNOT_{1073,112}, CNOT_{1089,113}, CNOT_{1105,114}, CNOT_{1073,115}, CNOT_{1089,116}, CNOT_{1105,117},
 \end{array} \right\}$$

$$\begin{aligned}
 & \left\{ \begin{array}{l} CNOT_{1121,118}, CNOT_{1057,119}, CNOT_{1073,120}, CNOT_{1089,121}, CNOT_{1105,122}, CNOT_{1121,123}, \\ CNOT_{1137,124}, CNOT_{1153,125}, CNOT_{1169,126}, CNOT_{1089,127}, CNOT_{1105,128} \end{array} \right\} \\
 \text{Step27:} & \left\{ \begin{array}{l} CNOT_{1121,97}, CNOT_{1137,98}, CNOT_{1153,99}, CNOT_{1105,100}, CNOT_{1121,101}, CNOT_{1137,102}, \\ CNOT_{1153,103}, CNOT_{1169,104}, CNOT_{1121,105}, CNOT_{1137,106}, CNOT_{1153,107}, CNOT_{1169,108}, \\ CNOT_{1185,109}, CNOT_{1201,110}, CNOT_{1089,111}, CNOT_{1105,112}, CNOT_{1121,113}, CNOT_{1137,114}, \\ CNOT_{1121,115}, CNOT_{1137,116}, CNOT_{1153,117}, CNOT_{1169,118}, CNOT_{1137,119}, CNOT_{1153,120}, \\ CNOT_{1169,121}, CNOT_{1185,122}, CNOT_{1201,123}, CNOT_{1217,124}, CNOT_{1233,125}, CNOT_{1249,126}, \\ CNOT_{1185,127}, CNOT_{1201,128} \end{array} \right\} \\
 \text{Step28:} & \left\{ \begin{array}{l} CNOT_{1217,97}, CNOT_{1233,98}, CNOT_{1249,99}, CNOT_{1169,100}, CNOT_{1185,101}, CNOT_{1201,102}, \\ CNOT_{1217,103}, CNOT_{1233,104}, CNOT_{1185,105}, CNOT_{1201,106}, CNOT_{1217,107}, CNOT_{1233,108}, \\ CNOT_{1249,109}, CNOT_{1265,110}, CNOT_{1217,111}, CNOT_{1233,112}, CNOT_{1249,113}, CNOT_{1265,114}, \\ CNOT_{1153,115}, CNOT_{1169,116}, CNOT_{1185,117}, CNOT_{1201,118}, CNOT_{1185,119}, CNOT_{1201,120}, \\ CNOT_{1217,121}, CNOT_{1233,122}, CNOT_{1249,123}, CNOT_{1265,124}, CNOT_{1281,125}, CNOT_{1297,126}, \\ CNOT_{1265,127}, CNOT_{1281,128} \end{array} \right\} \\
 \text{Step29:} & \left\{ \begin{array}{l} CNOT_{1297,97}, CNOT_{1313,98}, CNOT_{1329,99}, CNOT_{1265,100}, CNOT_{1281,101}, CNOT_{1297,102}, \\ CNOT_{1313,103}, CNOT_{1329,104}, CNOT_{1249,105}, CNOT_{1265,106}, CNOT_{1281,107}, CNOT_{1297,108}, \\ CNOT_{1313,109}, CNOT_{1329,110}, CNOT_{1281,111}, CNOT_{1297,112}, CNOT_{1313,113}, CNOT_{1329,114}, \\ CNOT_{1281,115}, CNOT_{1297,116}, CNOT_{1313,117}, CNOT_{1329,118}, CNOT_{1217,119}, CNOT_{1233,120}, \\ CNOT_{1249,121}, CNOT_{1265,122}, CNOT_{1281,123}, CNOT_{1297,124}, CNOT_{1313,125}, CNOT_{1329,126}, \\ CNOT_{1313,127}, CNOT_{1329,128} \end{array} \right\} \\
 \text{Step30:} & \left\{ \begin{array}{l} CNOT_{137,129}, CNOT_{153,145}, CNOT_{361,353}, CNOT_{377,369}, CNOT_{393,385}, CNOT_{434,426}, CNOT_{450,442}, \\ CNOT_{491,458}, CNOT_{507,499}, CNOT_{523,515}, CNOT_{564,531}, CNOT_{580,572}, CNOT_{596,588}, CNOT_{1049,1041}, \\ CNOT_{1065,1057}, CNOT_{1081,1073}, CNOT_{1097,1089}, CNOT_{1113,1105}, CNOT_{1129,1121}, CNOT_{1145,1137}, \\ CNOT_{1161,1153}, CNOT_{1177,1169}, CNOT_{1193,1185}, CNOT_{1209,1201}, CNOT_{1225,1217}, CNOT_{1241,1233}, \\ CNOT_{1257,1249}, CNOT_{1273,1265}, CNOT_{1289,1281}, CNOT_{1305,1297}, CNOT_{1321,1313}, CNOT_{1337,1329} \end{array} \right\} \\
 \text{Step31:} & \left\{ \begin{array}{l} CNOT_{133,129}, CNOT_{141,137}, CNOT_{149,145}, CNOT_{157,153}, CNOT_{357,353}, CNOT_{365,361}, CNOT_{373,369}, \\ CNOT_{381,377}, CNOT_{389,385}, CNOT_{422,393}, CNOT_{430,426}, CNOT_{438,434}, CNOT_{446,442}, CNOT_{454,450}, \\ CNOT_{462,458}, CNOT_{495,491}, CNOT_{503,499}, CNOT_{511,507}, CNOT_{519,515}, CNOT_{527,523}, CNOT_{560,531}, \\ CNOT_{568,564}, CNOT_{576,572}, CNOT_{584,580}, CNOT_{592,588}, CNOT_{600,596}, CNOT_{1045,1041}, \\ CNOT_{1053,1049}, CNOT_{1061,1057}, CNOT_{1069,1065}, CNOT_{1077,1073}, CNOT_{1085,1081}, \\ CNOT_{1093,1089}, CNOT_{1101,1097}, CNOT_{1109,1105}, CNOT_{1117,1113}, CNOT_{1125,1121}, \\ CNOT_{1133,1129}, CNOT_{1141,1137}, CNOT_{1149,1145}, CNOT_{1157,1153}, CNOT_{1165,1161}, \\ CNOT_{1173,1169}, CNOT_{1181,1177}, CNOT_{1189,1185}, CNOT_{1197,1193}, CNOT_{1205,1201}, \\ CNOT_{1213,1209}, CNOT_{1221,1217}, CNOT_{1229,1225}, CNOT_{1237,1233}, CNOT_{1245,1241}, \\ CNOT_{1253,1249}, CNOT_{1261,1257}, CNOT_{1269,1265}, CNOT_{1277,1273}, CNOT_{1285,1281}, \\ CNOT_{1293,1289}, CNOT_{1301,1297}, CNOT_{1309,1305}, CNOT_{1317,1313}, CNOT_{1325,1321}, \\ CNOT_{1333,1329}, CNOT_{1341,1337} \end{array} \right\} \\
 \text{Step32:} & \left\{ \begin{array}{l} CNOT_{131,129}, CNOT_{135,133}, CNOT_{139,137}, CNOT_{143,141}, CNOT_{147,145}, CNOT_{151,149}, CNOT_{155,153}, \\ CNOT_{159,157}, CNOT_{355,353}, CNOT_{359,357}, CNOT_{363,361}, CNOT_{367,365}, CNOT_{371,369}, CNOT_{375,373}, \\ CNOT_{379,377}, CNOT_{383,381}, CNOT_{387,385}, CNOT_{391,389}, CNOT_{395,393}, CNOT_{424,422}, CNOT_{428,426}, \\ CNOT_{432,430}, CNOT_{436,434}, CNOT_{440,438}, CNOT_{444,442}, CNOT_{448,446}, CNOT_{452,450}, CNOT_{456,454}, \\ CNOT_{460,458}, CNOT_{464,462}, CNOT_{493,491}, CNOT_{497,495}, CNOT_{501,499}, CNOT_{505,503}, CNOT_{509,507}, \\ CNOT_{513,511}, CNOT_{517,515}, CNOT_{521,519}, CNOT_{525,523}, CNOT_{529,527}, CNOT_{533,531}, CNOT_{562,560}, \\ CNOT_{566,564}, CNOT_{570,568}, CNOT_{574,572}, CNOT_{578,576}, CNOT_{582,580}, CNOT_{586,584}, CNOT_{590,588}, \\ CNOT_{594,592}, CNOT_{598,596}, CNOT_{602,600}, CNOT_{1043,1041}, CNOT_{1047,1045}, CNOT_{1051,1049}, \\ CNOT_{1055,1053}, CNOT_{1059,1057}, CNOT_{1063,1061}, CNOT_{1067,1065}, CNOT_{1071,1069}, CNOT_{1075,1073}, \\ CNOT_{1079,1077}, CNOT_{1083,1081}, CNOT_{1087,1085}, CNOT_{1091,1089}, CNOT_{1095,1093}, CNOT_{1099,1097}, \\ CNOT_{1103,1101}, CNOT_{1107,1105}, CNOT_{1111,1109}, CNOT_{1115,1113}, CNOT_{1119,1117}, CNOT_{1123,1121} \end{array} \right\}
 \end{aligned}$$

$$\left\{ \begin{array}{l} CNOT_{1127,1125}, CNOT_{1131,1129}, CNOT_{1135,1133}, CNOT_{1139,1137}, CNOT_{1143,1141}, CNOT_{1147,1145}, \\ CNOT_{1151,1149}, CNOT_{1155,1153}, CNOT_{1159,1157}, CNOT_{1163,1161}, CNOT_{1167,1165}, CNOT_{1171,1169}, \\ CNOT_{1175,1173}, CNOT_{1179,1177}, CNOT_{1183,1181}, CNOT_{1187,1185}, CNOT_{1191,1189}, CNOT_{1195,1193}, \\ CNOT_{1199,1197}, CNOT_{1203,1201}, CNOT_{1207,1205}, CNOT_{1211,1209}, CNOT_{1215,1213}, CNOT_{1219,1217}, \\ CNOT_{1223,1221}, CNOT_{1227,1225}, CNOT_{1231,1229}, CNOT_{1235,1233}, CNOT_{1239,1237}, CNOT_{1243,1241}, \\ CNOT_{1247,1245}, CNOT_{1251,1249}, CNOT_{1255,1253}, CNOT_{1259,1257}, CNOT_{1263,1261}, CNOT_{1267,1265}, \\ CNOT_{1271,1269}, CNOT_{1275,1273}, CNOT_{1279,1277}, CNOT_{1283,1281}, CNOT_{1287,1285}, CNOT_{1291,1289}, \\ CNOT_{1295,1293}, CNOT_{1299,1297}, CNOT_{1303,1301}, CNOT_{1307,1305}, CNOT_{1311,1309}, CNOT_{1315,1313}, \\ CNOT_{1319,1317}, CNOT_{1323,1321}, CNOT_{1327,1325}, CNOT_{1331,1329}, CNOT_{1335,1333}, CNOT_{1339,1337}, \\ CNOT_{1343,1341} \end{array} \right\}$$

Step33:

$$\left\{ \begin{array}{l} CNOT_{130,129}, CNOT_{132,131}, CNOT_{134,133}, CNOT_{136,135}, CNOT_{138,137}, CNOT_{140,139}, CNOT_{142,141}, \\ CNOT_{144,143}, CNOT_{146,145}, CNOT_{148,147}, CNOT_{150,149}, CNOT_{152,151}, CNOT_{154,153}, CNOT_{156,155}, \\ CNOT_{158,157}, CNOT_{160,159}, CNOT_{354,353}, CNOT_{356,355}, CNOT_{358,357}, CNOT_{360,359}, CNOT_{362,361}, \\ CNOT_{364,363}, CNOT_{366,365}, CNOT_{368,367}, CNOT_{370,369}, CNOT_{372,371}, CNOT_{374,373}, CNOT_{376,375}, \\ CNOT_{378,377}, CNOT_{380,379}, CNOT_{382,381}, CNOT_{384,383}, CNOT_{386,385}, CNOT_{388,387}, CNOT_{390,389}, \\ CNOT_{392,391}, CNOT_{394,393}, CNOT_{396,395}, CNOT_{423,422}, CNOT_{425,424}, CNOT_{427,426}, CNOT_{429,428}, \\ CNOT_{431,430}, CNOT_{433,432}, CNOT_{435,434}, CNOT_{437,436}, CNOT_{439,438}, CNOT_{441,440}, CNOT_{443,442}, \\ CNOT_{445,444}, CNOT_{447,446}, CNOT_{449,448}, CNOT_{451,450}, CNOT_{453,452}, CNOT_{455,454}, CNOT_{457,456}, \\ CNOT_{459,458}, CNOT_{461,460}, CNOT_{463,462}, CNOT_{465,464}, CNOT_{492,491}, CNOT_{494,493}, CNOT_{496,495}, \\ CNOT_{498,497}, CNOT_{500,499}, CNOT_{502,501}, CNOT_{504,503}, CNOT_{506,505}, CNOT_{508,507}, CNOT_{510,509}, \\ CNOT_{512,511}, CNOT_{514,513}, CNOT_{516,515}, CNOT_{518,517}, CNOT_{520,519}, CNOT_{522,521}, CNOT_{524,523}, \\ CNOT_{526,525}, CNOT_{528,527}, CNOT_{530,529}, CNOT_{532,531}, CNOT_{534,533}, CNOT_{561,560}, CNOT_{563,562}, \\ CNOT_{565,564}, CNOT_{567,566}, CNOT_{569,568}, CNOT_{571,570}, CNOT_{573,572}, CNOT_{575,574}, CNOT_{577,576}, \\ CNOT_{579,578}, CNOT_{581,580}, CNOT_{583,582}, CNOT_{585,584}, CNOT_{587,586}, CNOT_{589,588}, CNOT_{591,590}, \\ CNOT_{593,592}, CNOT_{595,594}, CNOT_{597,596}, CNOT_{599,598}, CNOT_{601,600}, CNOT_{603,602}, CNOT_{1042,1041}, \\ CNOT_{1044,1043}, CNOT_{1046,1045}, CNOT_{1048,1047}, CNOT_{1050,1049}, CNOT_{1052,1051}, CNOT_{1054,1053}, \\ CNOT_{1056,1055}, CNOT_{1058,1057}, CNOT_{1060,1059}, CNOT_{1062,1061}, CNOT_{1064,1063}, CNOT_{1066,1065}, \\ CNOT_{1068,1067}, CNOT_{1070,1069}, CNOT_{1072,1071}, CNOT_{1074,1073}, CNOT_{1076,1075}, CNOT_{1078,1077}, \\ CNOT_{1080,1079}, CNOT_{1082,1081}, CNOT_{1084,1083}, CNOT_{1086,1085}, CNOT_{1088,1087}, CNOT_{1090,1089}, \\ CNOT_{1092,1091}, CNOT_{1094,1093}, CNOT_{1096,1095}, CNOT_{1098,1097}, CNOT_{1100,1099}, CNOT_{1102,1101}, \\ CNOT_{1104,1103}, CNOT_{1106,1105}, CNOT_{1108,1107}, CNOT_{1110,1109}, CNOT_{1112,1111}, CNOT_{1114,1113}, \\ CNOT_{1116,1115}, CNOT_{1118,1117}, CNOT_{1120,1119}, CNOT_{1122,1121}, CNOT_{1124,1123}, CNOT_{1126,1125}, \\ CNOT_{1128,1127}, CNOT_{1130,1129}, CNOT_{1132,1131}, CNOT_{1134,1133}, CNOT_{1136,1135}, CNOT_{1138,1137}, \\ CNOT_{1140,1139}, CNOT_{1142,1141}, CNOT_{1144,1143}, CNOT_{1146,1145}, CNOT_{1148,1147}, CNOT_{1150,1149}, \\ CNOT_{1152,1151}, CNOT_{1154,1153}, CNOT_{1156,1155}, CNOT_{1158,1157}, CNOT_{1160,1159}, CNOT_{1162,1161}, \\ CNOT_{1164,1163}, CNOT_{1166,1165}, CNOT_{1168,1167}, CNOT_{1170,1169}, CNOT_{1172,1171}, CNOT_{1174,1173}, \\ CNOT_{1176,1175}, CNOT_{1178,1177}, CNOT_{1180,1179}, CNOT_{1182,1181}, CNOT_{1184,1183}, CNOT_{1186,1185}, \\ CNOT_{1188,1187}, CNOT_{1190,1189}, CNOT_{1192,1191}, CNOT_{1194,1193}, CNOT_{1196,1195}, CNOT_{1198,1197}, \\ CNOT_{1200,1199}, CNOT_{1202,1201}, CNOT_{1204,1203}, CNOT_{1206,1205}, CNOT_{1208,1207}, CNOT_{1210,1209}, \\ CNOT_{1212,1211}, CNOT_{1214,1213}, CNOT_{1216,1215}, CNOT_{1218,1217}, CNOT_{1220,1219}, CNOT_{1222,1221}, \\ CNOT_{1224,1223}, CNOT_{1226,1225}, CNOT_{1228,1227}, CNOT_{1230,1229}, CNOT_{1232,1231}, CNOT_{1234,1233}, \\ CNOT_{1236,1235}, CNOT_{1238,1237}, CNOT_{1240,1239}, CNOT_{1242,1241}, CNOT_{1244,1243}, CNOT_{1246,1245}, \\ CNOT_{1248,1247}, CNOT_{1250,1249}, CNOT_{1252,1251}, CNOT_{1254,1253}, CNOT_{1256,1255}, CNOT_{1258,1257}, \\ CNOT_{1260,1259}, CNOT_{1262,1261}, CNOT_{1264,1263}, CNOT_{1266,1265}, CNOT_{1268,1267}, CNOT_{1270,1269}, \\ CNOT_{1272,1271}, CNOT_{1274,1273}, CNOT_{1276,1275}, CNOT_{1278,1277}, CNOT_{1280,1279}, CNOT_{1282,1281}, \\ CNOT_{1284,1283}, CNOT_{1286,1285}, CNOT_{1288,1287}, CNOT_{1290,1289}, CNOT_{1292,1291}, CNOT_{1294,1293}, \\ CNOT_{1296,1295}, CNOT_{1298,1297}, CNOT_{1300,1299}, CNOT_{1302,1301}, CNOT_{1304,1303}, CNOT_{1306,1305}, \\ CNOT_{1308,1307}, CNOT_{1310,1309}, CNOT_{1312,1311}, CNOT_{1314,1313}, CNOT_{1316,1315}, CNOT_{1318,1317}, \\ CNOT_{1320,1319}, CNOT_{1322,1321}, CNOT_{1324,1323}, CNOT_{1326,1325}, CNOT_{1328,1327}, CNOT_{1330,1329}, \\ CNOT_{1332,1331}, CNOT_{1334,1333}, CNOT_{1336,1335}, CNOT_{1338,1337}, CNOT_{1340,1339}, CNOT_{1342,1341}, \\ CNOT_{1344,1343} \end{array} \right\}$$

Step34:

*Toffoli*_{240,405,129}, *Toffoli*_{239,406,130}, *Toffoli*_{238,398,131}, *Toffoli*_{237,629,132}, *Toffoli*_{236,407,133},
*Toffoli*_{235,630,134}, *Toffoli*_{234,413,135}, *Toffoli*_{233,631,136}, *Toffoli*_{232,419,137}, *Toffoli*_{231,632,138},
*Toffoli*_{230,402,139}, *Toffoli*_{229,633,140}, *Toffoli*_{228,634,141}, *Toffoli*_{227,635,142}, *Toffoli*_{226,397,143},
*Toffoli*_{225,636,144}, *Toffoli*_{240,637,145}, *Toffoli*_{239,638,146}, *Toffoli*_{238,420,147}, *Toffoli*_{237,409,148},
*Toffoli*_{236,639,149}, *Toffoli*_{235,640,150}, *Toffoli*_{234,410,151}, *Toffoli*_{233,641,152}, *Toffoli*_{232,642,153},
*Toffoli*_{231,643,154}, *Toffoli*_{230,644,155}, *Toffoli*_{229,645,156}, *Toffoli*_{228,646,157}, *Toffoli*_{227,647,158},
*Toffoli*_{226,648,159}, *Toffoli*_{225,649,160}, *Toffoli*_{240,650,353}, *Toffoli*_{239,651,354}, *Toffoli*_{238,652,355},
*Toffoli*_{237,404,356}, *Toffoli*_{236,653,357}, *Toffoli*_{235,654,358}, *Toffoli*_{234,655,359}, *Toffoli*_{233,656,360},
*Toffoli*_{232,414,361}, *Toffoli*_{231,657,362}, *Toffoli*_{230,658,363}, *Toffoli*_{229,412,364}, *Toffoli*_{228,659,365},
*Toffoli*_{227,660,366}, *Toffoli*_{226,661,367}, *Toffoli*_{225,662,368}, *Toffoli*_{240,663,369}, *Toffoli*_{239,664,370},
*Toffoli*_{238,665,371}, *Toffoli*_{237,666,372}, *Toffoli*_{236,667,373}, *Toffoli*_{235,668,374}, *Toffoli*_{234,669,375},
*Toffoli*_{233,670,376}, *Toffoli*_{232,671,377}, *Toffoli*_{231,672,378}, *Toffoli*_{230,673,379}, *Toffoli*_{229,674,380},
*Toffoli*_{228,675,381}, *Toffoli*_{227,676,382}, *Toffoli*_{226,677,383}, *Toffoli*_{225,678,384}, *Toffoli*_{240,679,385},
*Toffoli*_{239,680,386}, *Toffoli*_{238,681,387}, *Toffoli*_{237,682,388}, *Toffoli*_{236,683,389}, *Toffoli*_{235,684,390},
*Toffoli*_{234,685,391}, *Toffoli*_{233,686,392}, *Toffoli*_{232,403,393}, *Toffoli*_{231,687,394}, *Toffoli*_{230,688,395},
*Toffoli*_{229,689,396}, *Toffoli*_{228,690,422}, *Toffoli*_{227,691,423}, *Toffoli*_{226,692,424}, *Toffoli*_{225,693,425},
*Toffoli*_{240,416,426}, *Toffoli*_{239,415,427}, *Toffoli*_{238,694,428}, *Toffoli*_{237,695,429}, *Toffoli*_{236,696,430},
*Toffoli*_{235,697,431}, *Toffoli*_{234,698,432}, *Toffoli*_{233,699,433}, *Toffoli*_{232,399,434}, *Toffoli*_{231,700,435},
*Toffoli*_{230,701,436}, *Toffoli*_{229,702,437}, *Toffoli*_{228,703,438}, *Toffoli*_{227,704,439}, *Toffoli*_{226,705,440},
*Toffoli*_{225,706,441}, *Toffoli*_{240,707,442}, *Toffoli*_{239,417,443}, *Toffoli*_{238,418,444}, *Toffoli*_{237,708,445},
*Toffoli*_{236,709,446}, *Toffoli*_{235,408,447}, *Toffoli*_{234,710,448}, *Toffoli*_{233,711,449}, *Toffoli*_{232,712,450},
*Toffoli*_{231,400,451}, *Toffoli*_{230,713,452}, *Toffoli*_{229,421,453}, *Toffoli*_{228,714,454}, *Toffoli*_{227,715,455},
*Toffoli*_{226,716,456}, *Toffoli*_{225,717,457}, *Toffoli*_{240,718,458}, *Toffoli*_{239,401,459}, *Toffoli*_{238,719,460},
*Toffoli*_{237,720,461}, *Toffoli*_{236,721,462}, *Toffoli*_{235,722,463}, *Toffoli*_{234,723,464}, *Toffoli*_{233,724,465},
*Toffoli*_{232,411,491}, *Toffoli*_{231,725,492}, *Toffoli*_{230,726,493}, *Toffoli*_{229,727,494}, *Toffoli*_{228,728,495},
*Toffoli*_{227,729,496}, *Toffoli*_{226,730,497}, *Toffoli*_{225,731,498}, *Toffoli*_{272,474,499}, *Toffoli*_{271,475,500},
*Toffoli*_{270,467,501}, *Toffoli*_{269,732,502}, *Toffoli*_{268,476,503}, *Toffoli*_{267,733,504}, *Toffoli*_{266,482,505},
*Toffoli*_{265,734,506}, *Toffoli*_{264,488,507}, *Toffoli*_{263,735,508}, *Toffoli*_{262,471,509}, *Toffoli*_{261,736,510},
*Toffoli*_{260,737,511}, *Toffoli*_{259,738,512}, *Toffoli*_{258,466,513}, *Toffoli*_{257,739,514}, *Toffoli*_{272,740,515},
*Toffoli*_{271,741,516}, *Toffoli*_{270,489,517}, *Toffoli*_{269,478,518}, *Toffoli*_{268,742,519}, *Toffoli*_{267,743,520},
*Toffoli*_{266,479,521}, *Toffoli*_{265,744,522}, *Toffoli*_{264,745,523}, *Toffoli*_{263,746,524}, *Toffoli*_{262,747,525},
*Toffoli*_{261,748,526}, *Toffoli*_{260,749,527}, *Toffoli*_{259,750,528}, *Toffoli*_{258,751,529}, *Toffoli*_{257,752,530},
*Toffoli*_{272,753,531}, *Toffoli*_{271,754,532}, *Toffoli*_{270,755,533}, *Toffoli*_{269,473,534}, *Toffoli*_{268,756,560},
*Toffoli*_{267,757,561}, *Toffoli*_{266,758,562}, *Toffoli*_{265,759,563}, *Toffoli*_{264,483,564}, *Toffoli*_{263,760,565},
*Toffoli*_{262,761,566}, *Toffoli*_{261,481,567}, *Toffoli*_{260,762,568}, *Toffoli*_{259,763,569}, *Toffoli*_{258,764,570},
*Toffoli*_{257,765,571}, *Toffoli*_{272,766,572}, *Toffoli*_{271,767,573}, *Toffoli*_{270,768,574}, *Toffoli*_{269,769,575},
*Toffoli*_{268,770,576}, *Toffoli*_{267,771,577}, *Toffoli*_{266,772,578}, *Toffoli*_{265,773,579}, *Toffoli*_{264,774,580},
*Toffoli*_{263,775,581}, *Toffoli*_{262,776,582}, *Toffoli*_{261,777,583}, *Toffoli*_{260,778,584}, *Toffoli*_{259,779,585},
*Toffoli*_{258,780,586}, *Toffoli*_{257,781,587}, *Toffoli*_{272,782,588}, *Toffoli*_{271,783,589}, *Toffoli*_{270,784,590},
*Toffoli*_{269,785,591}, *Toffoli*_{268,786,592}, *Toffoli*_{267,787,593}, *Toffoli*_{266,788,594}, *Toffoli*_{265,789,595},
*Toffoli*_{264,472,596}, *Toffoli*_{263,790,597}, *Toffoli*_{262,791,598}, *Toffoli*_{261,792,599}, *Toffoli*_{260,793,600},
*Toffoli*_{259,794,601}, *Toffoli*_{258,795,602}, *Toffoli*_{257,796,603}, *Toffoli*_{272,485,1041}, *Toffoli*_{271,484,1042},
*Toffoli*_{270,797,1043}, *Toffoli*_{269,798,1044}, *Toffoli*_{268,799,1045}, *Toffoli*_{267,800,1046}, *Toffoli*_{266,801,1047},
*Toffoli*_{265,802,1048}, *Toffoli*_{264,468,1049}, *Toffoli*_{263,803,1050}, *Toffoli*_{262,804,1051}, *Toffoli*_{261,805,1052},
*Toffoli*_{260,806,1053}, *Toffoli*_{259,807,1054}, *Toffoli*_{258,808,1055}, *Toffoli*_{257,809,1056}, *Toffoli*_{272,810,1057},
*Toffoli*_{271,486,1058}, *Toffoli*_{270,487,1059}, *Toffoli*_{269,811,1060}, *Toffoli*_{268,812,1061}, *Toffoli*_{267,477,1062},
*Toffoli*_{266,813,1063}, *Toffoli*_{265,814,1064}, *Toffoli*_{264,815,1065}, *Toffoli*_{263,469,1066}, *Toffoli*_{262,816,1067},
*Toffoli*_{261,490,1068}, *Toffoli*_{260,817,1069}, *Toffoli*_{259,818,1070}, *Toffoli*_{258,819,1071}, *Toffoli*_{257,820,1072},
*Toffoli*_{272,821,1073}, *Toffoli*_{271,470,1074}, *Toffoli*_{270,822,1075}, *Toffoli*_{269,823,1076}, *Toffoli*_{268,824,1077},
*Toffoli*_{267,825,1078}, *Toffoli*_{266,826,1079}, *Toffoli*_{265,827,1080}, *Toffoli*_{264,480,1081}, *Toffoli*_{263,828,1082},
*Toffoli*_{262,829,1083}, *Toffoli*_{261,830,1084}, *Toffoli*_{260,831,1085}, *Toffoli*_{259,832,1086}, *Toffoli*_{258,833,1087},
*Toffoli*_{257,834,1088}, *Toffoli*_{304,543,1089}, *Toffoli*_{303,544,1090}, *Toffoli*_{302,536,1091}, *Toffoli*_{301,835,1092},

*Toffoli*_{300,545,1093}, *Toffoli*_{299,836,1094}, *Toffoli*_{298,551,1095}, *Toffoli*_{297,837,1096}, *Toffoli*_{296,557,1097},
*Toffoli*_{295,838,1098}, *Toffoli*_{294,540,1099}, *Toffoli*_{293,839,1100}, *Toffoli*_{292,840,1101}, *Toffoli*_{291,841,1102},
*Toffoli*_{290,535,1103}, *Toffoli*_{289,842,1104}, *Toffoli*_{304,843,1105}, *Toffoli*_{303,844,1106}, *Toffoli*_{302,558,1107},
*Toffoli*_{301,547,1108}, *Toffoli*_{300,845,1109}, *Toffoli*_{299,846,1110}, *Toffoli*_{298,548,1111}, *Toffoli*_{297,847,1112},
*Toffoli*_{296,848,1113}, *Toffoli*_{295,849,1114}, *Toffoli*_{294,850,1115}, *Toffoli*_{293,851,1116}, *Toffoli*_{292,852,1117},
*Toffoli*_{291,853,1118}, *Toffoli*_{290,854,1119}, *Toffoli*_{289,855,1120}, *Toffoli*_{304,856,1121}, *Toffoli*_{303,857,1122},
*Toffoli*_{302,858,1123}, *Toffoli*_{301,542,1124}, *Toffoli*_{300,859,1125}, *Toffoli*_{299,860,1126}, *Toffoli*_{298,861,1127},
*Toffoli*_{297,862,1128}, *Toffoli*_{296,552,1129}, *Toffoli*_{295,863,1130}, *Toffoli*_{294,864,1131}, *Toffoli*_{293,550,1132},
*Toffoli*_{292,865,1133}, *Toffoli*_{291,866,1134}, *Toffoli*_{290,867,1135}, *Toffoli*_{289,868,1136}, *Toffoli*_{304,869,1137},
*Toffoli*_{303,870,1138}, *Toffoli*_{302,871,1139}, *Toffoli*_{301,872,1140}, *Toffoli*_{300,873,1141}, *Toffoli*_{299,874,1142},
*Toffoli*_{298,875,1143}, *Toffoli*_{297,876,1144}, *Toffoli*_{296,877,1145}, *Toffoli*_{295,878,1146}, *Toffoli*_{294,879,1147},
*Toffoli*_{293,880,1148}, *Toffoli*_{292,881,1149}, *Toffoli*_{291,882,1150}, *Toffoli*_{290,883,1151}, *Toffoli*_{289,884,1152},
*Toffoli*_{304,885,1153}, *Toffoli*_{303,886,1154}, *Toffoli*_{302,887,1155}, *Toffoli*_{301,888,1156}, *Toffoli*_{300,889,1157},
*Toffoli*_{299,890,1158}, *Toffoli*_{298,891,1159}, *Toffoli*_{297,892,1160}, *Toffoli*_{296,541,1161}, *Toffoli*_{295,893,1162},
*Toffoli*_{294,894,1163}, *Toffoli*_{293,895,1164}, *Toffoli*_{292,896,1165}, *Toffoli*_{291,897,1166}, *Toffoli*_{290,898,1167},
*Toffoli*_{289,899,1168}, *Toffoli*_{304,554,1169}, *Toffoli*_{303,553,1170}, *Toffoli*_{302,900,1171}, *Toffoli*_{301,901,1172},
*Toffoli*_{300,902,1173}, *Toffoli*_{299,903,1174}, *Toffoli*_{298,904,1175}, *Toffoli*_{297,905,1176}, *Toffoli*_{296,537,1177},
*Toffoli*_{295,906,1178}, *Toffoli*_{294,907,1179}, *Toffoli*_{293,908,1180}, *Toffoli*_{292,909,1181}, *Toffoli*_{291,910,1182},
*Toffoli*_{290,911,1183}, *Toffoli*_{289,912,1184}, *Toffoli*_{304,913,1185}, *Toffoli*_{303,555,1186}, *Toffoli*_{302,556,1187},
*Toffoli*_{301,914,1188}, *Toffoli*_{300,915,1189}, *Toffoli*_{299,546,1190}, *Toffoli*_{298,916,1191}, *Toffoli*_{297,917,1192},
*Toffoli*_{296,918,1193}, *Toffoli*_{295,538,1194}, *Toffoli*_{294,919,1195}, *Toffoli*_{293,559,1196}, *Toffoli*_{292,920,1197},
*Toffoli*_{291,921,1198}, *Toffoli*_{290,922,1199}, *Toffoli*_{289,923,1200}, *Toffoli*_{304,924,1201}, *Toffoli*_{303,539,1202},
*Toffoli*_{302,925,1203}, *Toffoli*_{301,926,1204}, *Toffoli*_{300,927,1205}, *Toffoli*_{299,928,1206}, *Toffoli*_{298,929,1207},
*Toffoli*_{297,930,1208}, *Toffoli*_{296,549,1209}, *Toffoli*_{295,931,1210}, *Toffoli*_{294,932,1211}, *Toffoli*_{293,933,1212},
*Toffoli*_{292,934,1213}, *Toffoli*_{291,935,1214}, *Toffoli*_{290,936,1215}, *Toffoli*_{289,937,1216}, *Toffoli*_{336,612,1217},
*Toffoli*_{335,613,1218}, *Toffoli*_{334,605,1219}, *Toffoli*_{333,938,1220}, *Toffoli*_{332,614,1221}, *Toffoli*_{331,939,1222},
*Toffoli*_{330,620,1223}, *Toffoli*_{329,940,1224}, *Toffoli*_{328,626,1225}, *Toffoli*_{327,941,1226}, *Toffoli*_{326,609,1227},
*Toffoli*_{325,942,1228}, *Toffoli*_{324,943,1229}, *Toffoli*_{323,944,1230}, *Toffoli*_{322,604,1231}, *Toffoli*_{321,945,1232},
*Toffoli*_{336,946,1233}, *Toffoli*_{335,947,1234}, *Toffoli*_{334,627,1235}, *Toffoli*_{333,616,1236}, *Toffoli*_{332,948,1237},
*Toffoli*_{331,949,1238}, *Toffoli*_{330,617,1239}, *Toffoli*_{329,950,1240}, *Toffoli*_{328,951,1241}, *Toffoli*_{327,952,1242},
*Toffoli*_{326,953,1243}, *Toffoli*_{325,954,1244}, *Toffoli*_{324,955,1245}, *Toffoli*_{323,956,1246}, *Toffoli*_{322,957,1247},
*Toffoli*_{321,958,1248}, *Toffoli*_{336,959,1249}, *Toffoli*_{335,960,1250}, *Toffoli*_{334,961,1251}, *Toffoli*_{333,611,1252},
*Toffoli*_{332,962,1253}, *Toffoli*_{331,963,1254}, *Toffoli*_{330,964,1255}, *Toffoli*_{329,965,1256}, *Toffoli*_{328,621,1257},
*Toffoli*_{327,966,1258}, *Toffoli*_{326,967,1259}, *Toffoli*_{325,619,1260}, *Toffoli*_{324,968,1261}, *Toffoli*_{323,969,1262},
*Toffoli*_{322,970,1263}, *Toffoli*_{321,971,1264}, *Toffoli*_{336,972,1265}, *Toffoli*_{335,973,1266}, *Toffoli*_{334,974,1267},
*Toffoli*_{333,975,1268}, *Toffoli*_{332,976,1269}, *Toffoli*_{331,977,1270}, *Toffoli*_{330,978,1271}, *Toffoli*_{329,979,1272},
*Toffoli*_{328,980,1273}, *Toffoli*_{327,981,1274}, *Toffoli*_{326,982,1275}, *Toffoli*_{325,983,1276}, *Toffoli*_{324,984,1277},
*Toffoli*_{323,985,1278}, *Toffoli*_{322,986,1279}, *Toffoli*_{321,987,1280}, *Toffoli*_{336,988,1281}, *Toffoli*_{335,989,1282},
*Toffoli*_{334,990,1283}, *Toffoli*_{333,991,1284}, *Toffoli*_{332,992,1285}, *Toffoli*_{331,993,1286}, *Toffoli*_{330,994,1287},
*Toffoli*_{329,995,1288}, *Toffoli*_{328,610,1289}, *Toffoli*_{327,996,1290}, *Toffoli*_{326,997,1291}, *Toffoli*_{325,998,1292},
*Toffoli*_{324,999,1293}, *Toffoli*_{323,1000,1294}, *Toffoli*_{322,1001,1295}, *Toffoli*_{321,1002,1296},
*Toffoli*_{336,623,1297}, *Toffoli*_{335,622,1298}, *Toffoli*_{334,1003,1299}, *Toffoli*_{333,1004,1300},
*Toffoli*_{332,1005,1301}, *Toffoli*_{331,1006,1302}, *Toffoli*_{330,1007,1303}, *Toffoli*_{329,1008,1304},
*Toffoli*_{328,606,1305}, *Toffoli*_{327,1009,1306}, *Toffoli*_{326,1010,1307}, *Toffoli*_{325,1011,1308},
*Toffoli*_{324,1012,1309}, *Toffoli*_{323,1013,1310}, *Toffoli*_{322,1014,1311}, *Toffoli*_{321,1015,1312},
*Toffoli*_{336,1016,1313}, *Toffoli*_{335,624,1314}, *Toffoli*_{334,625,1315}, *Toffoli*_{333,1017,1316},
*Toffoli*_{332,1018,1317}, *Toffoli*_{331,615,1318}, *Toffoli*_{330,1019,1319}, *Toffoli*_{329,1020,1320},
*Toffoli*_{328,1021,1321}, *Toffoli*_{327,607,1322}, *Toffoli*_{326,1022,1323}, *Toffoli*_{325,628,1324},
*Toffoli*_{324,1023,1325}, *Toffoli*_{323,1024,1326}, *Toffoli*_{322,1025,1327}, *Toffoli*_{321,1026,1328},
*Toffoli*_{336,1027,1329}, *Toffoli*_{335,608,1330}, *Toffoli*_{334,1028,1331}, *Toffoli*_{333,1029,1332},
*Toffoli*_{332,1030,1333}, *Toffoli*_{331,1031,1334}, *Toffoli*_{330,1032,1335}, *Toffoli*_{329,1033,1336},
*Toffoli*_{328,618,1337}, *Toffoli*_{327,1034,1338}, *Toffoli*_{326,1035,1339}, *Toffoli*_{325,1036,1340},
*Toffoli*_{324,1037,1341}, *Toffoli*_{323,1038,1342}, *Toffoli*_{322,1039,1343}, *Toffoli*_{321,1040,1344}

Step35:

$(CNOT_{247,353}, CNOT_{251,354}, CNOT_{242,355}, CNOT_{250,356}, CNOT_{243,357}, CNOT_{245,358}, CNOT_{248,359},$
 $CNOT_{241,360}, CNOT_{254,361}, CNOT_{246,362}, CNOT_{244,363}, CNOT_{242,364}, CNOT_{249,365}, CNOT_{241,366},$
 $CNOT_{241,367}, CNOT_{244,368}, CNOT_{247,369}, CNOT_{243,370}, CNOT_{245,371}, CNOT_{253,372}, CNOT_{244,373},$
 $CNOT_{248,374}, CNOT_{242,375}, CNOT_{252,376}, CNOT_{250,377}, CNOT_{251,378}, CNOT_{255,379}, CNOT_{243,380},$
 $CNOT_{248,381}, CNOT_{249,382}, CNOT_{245,383}, CNOT_{247,384}, CNOT_{241,385}, CNOT_{253,386}, CNOT_{246,387},$
 $CNOT_{241,388}, CNOT_{248,389}, CNOT_{247,390}, CNOT_{250,391}, CNOT_{246,392}, CNOT_{244,393}, CNOT_{245,394},$
 $CNOT_{246,395}, CNOT_{241,396}, CNOT_{279,422}, CNOT_{283,423}, CNOT_{274,424}, CNOT_{282,425}, CNOT_{275,426},$
 $CNOT_{277,427}, CNOT_{280,428}, CNOT_{273,429}, CNOT_{286,430}, CNOT_{278,431}, CNOT_{276,432}, CNOT_{274,433},$
 $CNOT_{281,434}, CNOT_{273,435}, CNOT_{273,436}, CNOT_{276,437}, CNOT_{279,438}, CNOT_{275,439}, CNOT_{277,440},$
 $CNOT_{285,441}, CNOT_{276,442}, CNOT_{280,443}, CNOT_{274,444}, CNOT_{284,445}, CNOT_{282,446}, CNOT_{283,447},$
 $CNOT_{287,448}, CNOT_{275,449}, CNOT_{280,450}, CNOT_{281,451}, CNOT_{277,452}, CNOT_{279,453}, CNOT_{273,454},$
 $CNOT_{285,455}, CNOT_{278,456}, CNOT_{273,457}, CNOT_{280,458}, CNOT_{279,459}, CNOT_{282,460}, CNOT_{278,461},$
 $CNOT_{276,462}, CNOT_{277,463}, CNOT_{278,464}, CNOT_{273,465}, CNOT_{311,491}, CNOT_{315,492}, CNOT_{306,493},$
 $CNOT_{314,494}, CNOT_{307,495}, CNOT_{309,496}, CNOT_{312,497}, CNOT_{305,498}, CNOT_{318,499}, CNOT_{310,500},$
 $CNOT_{308,501}, CNOT_{306,502}, CNOT_{313,503}, CNOT_{305,504}, CNOT_{305,505}, CNOT_{308,506}, CNOT_{311,507},$
 $CNOT_{307,508}, CNOT_{309,509}, CNOT_{317,510}, CNOT_{308,511}, CNOT_{312,512}, CNOT_{306,513}, CNOT_{316,514},$
 $CNOT_{314,515}, CNOT_{315,516}, CNOT_{319,517}, CNOT_{307,518}, CNOT_{312,519}, CNOT_{313,520}, CNOT_{309,521},$
 $CNOT_{311,522}, CNOT_{305,523}, CNOT_{317,524}, CNOT_{310,525}, CNOT_{305,526}, CNOT_{312,527}, CNOT_{311,528},$
 $CNOT_{314,529}, CNOT_{310,530}, CNOT_{308,531}, CNOT_{309,532}, CNOT_{310,533}, CNOT_{305,534}, CNOT_{343,560},$
 $CNOT_{347,561}, CNOT_{338,562}, CNOT_{346,563}, CNOT_{339,564}, CNOT_{341,565}, CNOT_{344,566}, CNOT_{337,567},$
 $CNOT_{350,568}, CNOT_{342,569}, CNOT_{340,570}, CNOT_{338,571}, CNOT_{345,572}, CNOT_{337,573}, CNOT_{337,574},$
 $CNOT_{340,575}, CNOT_{343,576}, CNOT_{339,577}, CNOT_{341,578}, CNOT_{349,579}, CNOT_{340,580}, CNOT_{344,581},$
 $CNOT_{338,582}, CNOT_{348,583}, CNOT_{346,584}, CNOT_{347,585}, CNOT_{351,586}, CNOT_{339,587}, CNOT_{344,588},$
 $CNOT_{345,589}, CNOT_{341,590}, CNOT_{343,591}, CNOT_{337,592}, CNOT_{349,593}, CNOT_{342,594}, CNOT_{337,595},$
 $CNOT_{344,596}, CNOT_{343,597}, CNOT_{346,598}, CNOT_{342,599}, CNOT_{340,600}, CNOT_{341,601}, CNOT_{342,602},$
 $CNOT_{337,603}$

Step36:

$(CNOT_{249,353}, CNOT_{253,354}, CNOT_{244,355}, CNOT_{254,356}, CNOT_{255,357}, CNOT_{256,358}, CNOT_{252,359},$
 $CNOT_{246,360}, CNOT_{255,361}, CNOT_{250,362}, CNOT_{247,363}, CNOT_{245,364}, CNOT_{251,365}, CNOT_{243,366},$
 $CNOT_{256,367}, CNOT_{254,368}, CNOT_{252,369}, CNOT_{252,370}, CNOT_{246,371}, CNOT_{256,372}, CNOT_{256,373},$
 $CNOT_{249,374}, CNOT_{243,375}, CNOT_{253,376}, CNOT_{253,377}, CNOT_{252,378}, CNOT_{256,379}, CNOT_{256,380},$
 $CNOT_{255,381}, CNOT_{252,382}, CNOT_{252,383}, CNOT_{251,384}, CNOT_{242,385}, CNOT_{255,386}, CNOT_{254,387},$
 $CNOT_{254,388}, CNOT_{253,389}, CNOT_{253,390}, CNOT_{251,391}, CNOT_{251,392}, CNOT_{251,393}, CNOT_{250,394},$
 $CNOT_{248,395}, CNOT_{247,396}, CNOT_{281,422}, CNOT_{285,423}, CNOT_{276,424}, CNOT_{286,425}, CNOT_{287,426},$
 $CNOT_{288,427}, CNOT_{284,428}, CNOT_{278,429}, CNOT_{287,430}, CNOT_{282,431}, CNOT_{279,432}, CNOT_{277,433},$
 $CNOT_{283,434}, CNOT_{275,435}, CNOT_{288,436}, CNOT_{286,437}, CNOT_{284,438}, CNOT_{284,439}, CNOT_{278,440},$
 $CNOT_{288,441}, CNOT_{288,442}, CNOT_{281,443}, CNOT_{275,444}, CNOT_{285,445}, CNOT_{285,446}, CNOT_{284,447},$
 $CNOT_{288,448}, CNOT_{288,449}, CNOT_{287,450}, CNOT_{284,451}, CNOT_{284,452}, CNOT_{283,453}, CNOT_{274,454},$
 $CNOT_{287,455}, CNOT_{286,456}, CNOT_{286,457}, CNOT_{285,458}, CNOT_{285,459}, CNOT_{283,460}, CNOT_{283,461},$
 $CNOT_{283,462}, CNOT_{282,463}, CNOT_{280,464}, CNOT_{279,465}, CNOT_{313,491}, CNOT_{317,492}, CNOT_{308,493},$
 $CNOT_{318,494}, CNOT_{319,495}, CNOT_{320,496}, CNOT_{316,497}, CNOT_{310,498}, CNOT_{319,499}, CNOT_{314,500},$
 $CNOT_{311,501}, CNOT_{309,502}, CNOT_{315,503}, CNOT_{307,504}, CNOT_{320,505}, CNOT_{318,506}, CNOT_{316,507},$
 $CNOT_{316,508}, CNOT_{310,509}, CNOT_{320,510}, CNOT_{320,511}, CNOT_{313,512}, CNOT_{307,513}, CNOT_{317,514},$
 $CNOT_{317,515}, CNOT_{316,516}, CNOT_{320,517}, CNOT_{320,518}, CNOT_{319,519}, CNOT_{316,520}, CNOT_{316,521},$
 $CNOT_{315,522}, CNOT_{306,523}, CNOT_{319,524}, CNOT_{318,525}, CNOT_{318,526}, CNOT_{317,527}, CNOT_{317,528},$
 $CNOT_{315,529}, CNOT_{315,530}, CNOT_{315,531}, CNOT_{314,532}, CNOT_{312,533}, CNOT_{311,534}, CNOT_{345,560},$
 $CNOT_{349,561}, CNOT_{340,562}, CNOT_{350,563}, CNOT_{351,564}, CNOT_{352,565}, CNOT_{348,566}, CNOT_{342,567},$
 $CNOT_{351,568}, CNOT_{346,569}, CNOT_{343,570}, CNOT_{341,571}, CNOT_{347,572}, CNOT_{339,573}, CNOT_{352,574},$
 $CNOT_{350,575}, CNOT_{348,576}, CNOT_{348,577}, CNOT_{342,578}, CNOT_{352,579}, CNOT_{352,580}, CNOT_{345,581},$
 $CNOT_{339,582}, CNOT_{349,583}, CNOT_{349,584}, CNOT_{348,585}, CNOT_{352,586}, CNOT_{352,587}, CNOT_{351,588},$
 $CNOT_{348,589}, CNOT_{348,590}, CNOT_{347,591}, CNOT_{338,592}, CNOT_{351,593}, CNOT_{350,594}, CNOT_{350,595},$
 $CNOT_{343,603}$

Step37:

$$\{CNOT_{362,675}, CNOT_{375,693}, CNOT_{431,778}, CNOT_{444,796}, CNOT_{500,881}, CNOT_{513,899}, \\ CNOT_{569,984}, CNOT_{582,1002},\}$$

Step38:

$$\left\{ \begin{array}{l} CNOT_{377,629}, CNOT_{369,645}, CNOT_{363,649}, CNOT_{359,675}, CNOT_{379,690}, CNOT_{360,693}, \\ CNOT_{364,713}, CNOT_{384,721}, CNOT_{381,727}, CNOT_{446,732}, CNOT_{438,748}, CNOT_{432,752}, \\ CNOT_{428,778}, CNOT_{448,793}, CNOT_{429,796}, CNOT_{433,816}, CNOT_{453,824}, CNOT_{450,830}, \\ CNOT_{515,835}, CNOT_{507,851}, CNOT_{501,855}, CNOT_{497,881}, CNOT_{517,896}, CNOT_{498,899}, \\ CNOT_{502,919}, CNOT_{522,927}, CNOT_{519,933}, CNOT_{584,938}, CNOT_{576,954}, CNOT_{570,958}, \\ CNOT_{566,984}, CNOT_{586,999}, CNOT_{567,1002}, CNOT_{571,1022}, CNOT_{591,1030}, CNOT_{588,1036}, \end{array} \right\}$$

Step39:

$$\left\{ \begin{array}{l} CNOT_{360,629}, CNOT_{361,419}, CNOT_{396,632}, CNOT_{382,635}, CNOT_{391,640}, CNOT_{375,641}, CNOT_{370,642}, \\ CNOT_{385,644}, CNOT_{360,645}, CNOT_{374,648}, CNOT_{360,649}, CNOT_{356,650}, CNOT_{367,654}, CNOT_{364,655}, \\ CNOT_{381,657}, CNOT_{361,658}, CNOT_{366,660}, CNOT_{364,662}, CNOT_{372,663}, CNOT_{378,672}, CNOT_{385,674}, \\ CNOT_{357,675}, CNOT_{360,676}, CNOT_{392,678}, CNOT_{369,682}, CNOT_{381,684}, CNOT_{378,689}, CNOT_{378,690}, \\ CNOT_{359,693}, CNOT_{380,695}, CNOT_{372,696}, CNOT_{390,699}, CNOT_{373,702}, CNOT_{366,705}, CNOT_{387,417}, \\ CNOT_{379,709}, CNOT_{383,711}, CNOT_{395,712}, CNOT_{360,713}, CNOT_{359,721}, CNOT_{364,722}, CNOT_{367,727}, \\ CNOT_{393,728}, CNOT_{381,730}, CNOT_{429,732}, CNOT_{430,488}, CNOT_{465,735}, CNOT_{451,738}, CNOT_{460,743}, \\ CNOT_{444,744}, CNOT_{439,745}, CNOT_{454,747}, CNOT_{429,748}, CNOT_{443,751}, CNOT_{429,752}, CNOT_{425,753}, \\ CNOT_{436,757}, CNOT_{433,758}, CNOT_{450,760}, CNOT_{430,761}, CNOT_{435,763}, CNOT_{433,765}, CNOT_{441,766}, \\ CNOT_{447,775}, CNOT_{454,777}, CNOT_{426,778}, CNOT_{429,779}, CNOT_{461,781}, CNOT_{438,785}, CNOT_{450,787}, \\ CNOT_{447,792}, CNOT_{447,793}, CNOT_{428,796}, CNOT_{449,798}, CNOT_{441,799}, CNOT_{459,802}, CNOT_{442,805}, \\ CNOT_{435,808}, CNOT_{456,486}, CNOT_{448,812}, CNOT_{452,814}, CNOT_{464,815}, CNOT_{429,816}, CNOT_{428,824}, \\ CNOT_{433,825}, CNOT_{436,830}, CNOT_{462,831}, CNOT_{450,833}, CNOT_{498,835}, CNOT_{499,557}, CNOT_{534,838}, \\ CNOT_{520,841}, CNOT_{529,846}, CNOT_{513,847}, CNOT_{508,848}, CNOT_{523,850}, CNOT_{498,851}, CNOT_{512,854}, \\ CNOT_{498,855}, CNOT_{494,856}, CNOT_{505,860}, CNOT_{502,861}, CNOT_{519,863}, CNOT_{499,864}, CNOT_{504,866}, \\ CNOT_{502,868}, CNOT_{510,869}, CNOT_{516,878}, CNOT_{523,880}, CNOT_{495,881}, CNOT_{498,882}, CNOT_{530,884}, \\ CNOT_{507,888}, CNOT_{519,890}, CNOT_{516,895}, CNOT_{516,896}, CNOT_{497,899}, CNOT_{518,901}, CNOT_{510,902}, \\ CNOT_{528,905}, CNOT_{511,908}, CNOT_{504,911}, CNOT_{525,555}, CNOT_{517,915}, CNOT_{521,917}, CNOT_{533,918}, \\ CNOT_{498,919}, CNOT_{497,927}, CNOT_{502,928}, CNOT_{505,933}, CNOT_{531,934}, CNOT_{519,936}, CNOT_{567,938}, \\ CNOT_{568,626}, CNOT_{603,941}, CNOT_{589,944}, CNOT_{598,949}, CNOT_{582,950}, CNOT_{577,951}, CNOT_{592,953}, \\ CNOT_{567,954}, CNOT_{581,957}, CNOT_{567,958}, CNOT_{563,959}, CNOT_{574,963}, CNOT_{571,964}, CNOT_{588,966}, \\ CNOT_{568,967}, CNOT_{573,969}, CNOT_{571,971}, CNOT_{579,972}, CNOT_{585,981}, CNOT_{592,983}, CNOT_{564,984}, \\ CNOT_{567,985}, CNOT_{599,987}, CNOT_{576,991}, CNOT_{588,993}, CNOT_{585,998}, CNOT_{585,999}, CNOT_{566,1002}, \\ CNOT_{587,1004}, CNOT_{579,1005}, CNOT_{597,1008}, CNOT_{580,1011}, CNOT_{573,1014}, CNOT_{594,624}, CNOT_{586,1018}, \\ CNOT_{590,1020}, CNOT_{602,1021}, CNOT_{567,1022}, CNOT_{566,1030}, CNOT_{571,1031}, CNOT_{574,1036}, CNOT_{600,1037}, \\ CNOT_{588,1039} \end{array} \right\}$$

Step40:

$$\left\{ \begin{array}{l} CNOT_{358,405}, CNOT_{359,629}, CNOT_{389,630}, CNOT_{378,631}, CNOT_{360,419}, CNOT_{372,632}, CNOT_{369,634}, \\ CNOT_{360,635}, CNOT_{370,637}, CNOT_{368,420}, CNOT_{369,639}, CNOT_{363,640}, CNOT_{374,641}, CNOT_{363,642}, \\ CNOT_{365,644}, CNOT_{358,645}, CNOT_{364,646}, CNOT_{362,648}, CNOT_{357,649}, CNOT_{355,650}, CNOT_{359,651}, \\ CNOT_{374,652}, CNOT_{382,653}, CNOT_{363,654}, CNOT_{359,655}, CNOT_{373,414}, CNOT_{360,657}, CNOT_{360,658}, \\ CNOT_{359,660}, CNOT_{362,661}, CNOT_{361,662}, CNOT_{362,663}, CNOT_{374,665}, CNOT_{371,666}, CNOT_{383,667}, \\ CNOT_{377,668}, CNOT_{362,669}, CNOT_{367,670}, CNOT_{386,671}, CNOT_{371,672}, CNOT_{369,674}, CNOT_{355,675}, \\ CNOT_{356,676}, CNOT_{361,678}, CNOT_{370,679}, CNOT_{368,680}, CNOT_{380,681}, CNOT_{364,682}, CNOT_{364,683}, \\ CNOT_{362,684}, CNOT_{387,685}, CNOT_{396,688}, CNOT_{362,689}, CNOT_{360,690}, CNOT_{374,691}, CNOT_{376,692}, \\ CNOT_{358,693}, CNOT_{372,416}, CNOT_{380,694}, CNOT_{360,695}, CNOT_{357,696}, CNOT_{372,698}, CNOT_{366,699}, \\ CNOT_{374,700}, CNOT_{379,701}, CNOT_{370,702}, CNOT_{373,703}, CNOT_{375,704}, CNOT_{365,705}, CNOT_{371,706}, \\ CNOT_{389,707}, CNOT_{359,417}, CNOT_{365,709}, CNOT_{388,710}, CNOT_{360,711}, CNOT_{361,712}, CNOT_{357,713}, \\ CNOT_{364,421}, CNOT_{376,714}, CNOT_{375,716}, CNOT_{393,719}, CNOT_{376,720}, CNOT_{358,721}, CNOT_{361,722}, \\ CNOT_{394,723}, CNOT_{357,411}, CNOT_{388,725}, CNOT_{363,726}, CNOT_{355,727}, CNOT_{361,728}, CNOT_{365,729}, \\ CNOT_{363,730}, CNOT_{395,731}, CNOT_{427,474}, CNOT_{428,732}, CNOT_{458,733}, CNOT_{447,734}, CNOT_{429,488}, \\ CNOT_{441,735}, CNOT_{438,737}, CNOT_{429,738}, CNOT_{439,740}, CNOT_{437,489}, CNOT_{438,742}, CNOT_{432,743}, \\ CNOT_{443,744}, CNOT_{432,745}, CNOT_{434,747}, CNOT_{427,748}, CNOT_{433,749}, CNOT_{431,751}, CNOT_{426,752}, \end{array} \right\}$$

$CNOT_{424,753}, CNOT_{428,754}, CNOT_{443,755}, CNOT_{451,756}, CNOT_{432,757}, CNOT_{428,758}, CNOT_{442,483},$
 $CNOT_{429,760}, CNOT_{429,761}, CNOT_{428,763}, CNOT_{431,764}, CNOT_{430,765}, CNOT_{431,766}, CNOT_{443,768},$
 $CNOT_{440,769}, CNOT_{452,770}, CNOT_{446,771}, CNOT_{431,772}, CNOT_{436,773}, CNOT_{455,774}, CNOT_{440,775},$
 $CNOT_{438,777}, CNOT_{424,778}, CNOT_{425,779}, CNOT_{430,781}, CNOT_{439,782}, CNOT_{437,783}, CNOT_{449,784},$
 $CNOT_{433,785}, CNOT_{433,786}, CNOT_{431,787}, CNOT_{456,788}, CNOT_{465,791}, CNOT_{431,792}, CNOT_{429,793},$
 $CNOT_{443,794}, CNOT_{445,795}, CNOT_{427,796}, CNOT_{441,485}, CNOT_{449,797}, CNOT_{429,798}, CNOT_{426,799},$
 $CNOT_{441,801}, CNOT_{435,802}, CNOT_{443,803}, CNOT_{448,804}, CNOT_{439,805}, CNOT_{442,806}, CNOT_{444,807},$
 $CNOT_{434,808}, CNOT_{440,809}, CNOT_{458,810}, CNOT_{428,486}, CNOT_{434,812}, CNOT_{457,813}, CNOT_{429,814},$
 $CNOT_{430,815}, CNOT_{426,816}, CNOT_{433,490}, CNOT_{445,817}, CNOT_{444,819}, CNOT_{462,822}, CNOT_{445,823},$
 $CNOT_{427,824}, CNOT_{430,825}, CNOT_{463,826}, CNOT_{426,480}, CNOT_{457,828}, CNOT_{432,829}, CNOT_{424,830},$
 $CNOT_{430,831}, CNOT_{434,832}, CNOT_{432,833}, CNOT_{464,834}, CNOT_{496,543}, CNOT_{497,835}, CNOT_{527,836},$
 $CNOT_{516,837}, CNOT_{498,557}, CNOT_{510,838}, CNOT_{507,840}, CNOT_{498,841}, CNOT_{508,843}, CNOT_{506,558},$
 $CNOT_{507,845}, CNOT_{501,846}, CNOT_{512,847}, CNOT_{501,848}, CNOT_{503,850}, CNOT_{496,851}, CNOT_{502,852},$
 $CNOT_{500,854}, CNOT_{495,855}, CNOT_{493,856}, CNOT_{497,857}, CNOT_{512,858}, CNOT_{520,859}, CNOT_{501,860},$
 $CNOT_{497,861}, CNOT_{511,552}, CNOT_{498,863}, CNOT_{498,864}, CNOT_{497,866}, CNOT_{500,867}, CNOT_{499,868},$
 $CNOT_{500,869}, CNOT_{512,871}, CNOT_{509,872}, CNOT_{521,873}, CNOT_{515,874}, CNOT_{500,875}, CNOT_{505,876},$
 $CNOT_{524,877}, CNOT_{509,878}, CNOT_{507,880}, CNOT_{493,881}, CNOT_{494,882}, CNOT_{499,884}, CNOT_{508,885},$
 $CNOT_{506,886}, CNOT_{518,887}, CNOT_{502,888}, CNOT_{502,889}, CNOT_{500,890}, CNOT_{525,891}, CNOT_{534,894},$
 $CNOT_{500,895}, CNOT_{498,896}, CNOT_{512,897}, CNOT_{514,898}, CNOT_{496,899}, CNOT_{510,554}, CNOT_{518,900},$
 $CNOT_{498,901}, CNOT_{495,902}, CNOT_{510,904}, CNOT_{504,905}, CNOT_{512,906}, CNOT_{517,907}, CNOT_{508,908},$
 $CNOT_{511,909}, CNOT_{513,910}, CNOT_{503,911}, CNOT_{509,912}, CNOT_{527,913}, CNOT_{497,555}, CNOT_{503,915},$
 $CNOT_{526,916}, CNOT_{498,917}, CNOT_{499,918}, CNOT_{495,919}, CNOT_{502,559}, CNOT_{514,920}, CNOT_{513,922},$
 $CNOT_{531,925}, CNOT_{514,926}, CNOT_{496,927}, CNOT_{499,928}, CNOT_{532,929}, CNOT_{495,549}, CNOT_{526,931},$
 $CNOT_{501,932}, CNOT_{493,933}, CNOT_{499,934}, CNOT_{503,935}, CNOT_{501,936}, CNOT_{533,937}, CNOT_{565,612},$
 $CNOT_{566,938}, CNOT_{596,939}, CNOT_{585,940}, CNOT_{567,626}, CNOT_{579,941}, CNOT_{576,943}, CNOT_{567,944},$
 $CNOT_{577,946}, CNOT_{575,627}, CNOT_{576,948}, CNOT_{570,949}, CNOT_{581,950}, CNOT_{570,951}, CNOT_{572,953},$
 $CNOT_{565,954}, CNOT_{571,955}, CNOT_{569,957}, CNOT_{564,958}, CNOT_{562,959}, CNOT_{566,960}, CNOT_{581,961},$
 $CNOT_{589,962}, CNOT_{570,963}, CNOT_{566,964}, CNOT_{580,621}, CNOT_{567,966}, CNOT_{567,967}, CNOT_{566,969},$
 $CNOT_{569,970}, CNOT_{568,971}, CNOT_{569,972}, CNOT_{581,974}, CNOT_{578,975}, CNOT_{590,976}, CNOT_{584,977},$
 $CNOT_{569,978}, CNOT_{574,979}, CNOT_{593,980}, CNOT_{578,981}, CNOT_{576,983}, CNOT_{562,984}, CNOT_{563,985},$
 $CNOT_{568,987}, CNOT_{577,988}, CNOT_{575,989}, CNOT_{587,990}, CNOT_{571,991}, CNOT_{571,992}, CNOT_{569,993},$
 $CNOT_{594,994}, CNOT_{603,997}, CNOT_{569,998}, CNOT_{567,999}, CNOT_{581,1000}, CNOT_{583,1001}, CNOT_{565,1002},$
 $CNOT_{579,623}, CNOT_{587,1003}, CNOT_{567,1004}, CNOT_{564,1005}, CNOT_{579,1007}, CNOT_{573,1008}, CNOT_{581,1009},$
 $CNOT_{586,1010}, CNOT_{577,1011}, CNOT_{580,1012}, CNOT_{582,1013}, CNOT_{572,1014}, CNOT_{578,1015}, CNOT_{596,1016},$
 $CNOT_{566,624}, CNOT_{572,1018}, CNOT_{595,1019}, CNOT_{567,1020}, CNOT_{568,1021}, CNOT_{564,1022}, CNOT_{571,628},$
 $CNOT_{583,1023}, CNOT_{582,1025}, CNOT_{600,1028}, CNOT_{583,1029}, CNOT_{565,1030}, CNOT_{568,1031}, CNOT_{601,1032},$
 $CNOT_{564,618}, CNOT_{595,1034}, CNOT_{570,1035}, CNOT_{562,1036}, CNOT_{568,1037}, CNOT_{572,1038}, CNOT_{570,1039},$
 $CNOT_{602,1040}$

Step41:

$CNOT_{357,405}, CNOT_{370,398}, CNOT_{358,629}, CNOT_{368,407}, CNOT_{356,630}, CNOT_{366,631}, CNOT_{358,419},$
 $CNOT_{359,632}, CNOT_{391,633}, CNOT_{367,634}, CNOT_{357,635}, CNOT_{384,397}, CNOT_{363,636}, CNOT_{358,637},$
 $CNOT_{386,638}, CNOT_{357,420}, CNOT_{355,639}, CNOT_{359,640}, CNOT_{358,641}, CNOT_{358,642}, CNOT_{368,643},$
 $CNOT_{357,644}, CNOT_{357,645}, CNOT_{360,646}, CNOT_{384,647}, CNOT_{357,648}, CNOT_{356,649}, CNOT_{354,650},$
 $CNOT_{358,651}, CNOT_{355,652}, CNOT_{377,404}, CNOT_{368,653}, CNOT_{359,654}, CNOT_{357,655}, CNOT_{360,414},$
 $CNOT_{358,657}, CNOT_{354,658}, CNOT_{371,412}, CNOT_{356,660}, CNOT_{357,661}, CNOT_{359,662}, CNOT_{357,663},$
 $CNOT_{360,664}, CNOT_{372,665}, CNOT_{361,666}, CNOT_{362,667}, CNOT_{366,668}, CNOT_{358,669}, CNOT_{354,670},$
 $CNOT_{362,671}, CNOT_{356,672}, CNOT_{361,674}, CNOT_{354,675}, CNOT_{355,676}, CNOT_{392,677}, CNOT_{359,678},$
 $CNOT_{356,679}, CNOT_{367,680}, CNOT_{365,681}, CNOT_{356,682}, CNOT_{360,683}, CNOT_{358,684}, CNOT_{358,685},$
 $CNOT_{359,687}, CNOT_{377,688}, CNOT_{357,689}, CNOT_{356,690}, CNOT_{373,691}, CNOT_{358,692}, CNOT_{356,693},$
 $CNOT_{360,416}, CNOT_{364,415}, CNOT_{355,694}, CNOT_{355,695}, CNOT_{356,696}, CNOT_{375,697}, CNOT_{361,698},$
 $CNOT_{359,699}, CNOT_{368,700}, CNOT_{355,701}, CNOT_{356,702}, CNOT_{365,703}, CNOT_{359,704}, CNOT_{359,705},$
 $CNOT_{363,706}, CNOT_{358,707}, CNOT_{358,417}, CNOT_{365,418}, CNOT_{396,708}, CNOT_{359,709}, CNOT_{362,408},$
 $CNOT_{376,710}, CNOT_{356,711}, CNOT_{358,712}, CNOT_{356,713}, CNOT_{359,421}, CNOT_{361,714}, CNOT_{356,716},$
 $CNOT_{371,717}, CNOT_{366,718}, CNOT_{371,719}, CNOT_{357,720}, CNOT_{357,721}, CNOT_{360,722}, CNOT_{366,723},$

$CNOT_{383,724}, CNOT_{355,411}, CNOT_{359,725}, CNOT_{360,726}, CNOT_{354,727}, CNOT_{359,728}, CNOT_{359,729},$
 $CNOT_{358,730}, CNOT_{356,731}, CNOT_{426,474}, CNOT_{439,467}, CNOT_{427,732}, CNOT_{437,476}, CNOT_{425,733},$
 $CNOT_{435,734}, CNOT_{427,488}, CNOT_{428,735}, CNOT_{460,736}, CNOT_{436,737}, CNOT_{426,738}, CNOT_{453,466},$
 $CNOT_{432,739}, CNOT_{427,740}, CNOT_{455,741}, CNOT_{426,489}, CNOT_{424,742}, CNOT_{428,743}, CNOT_{427,744},$
 $CNOT_{427,745}, CNOT_{437,746}, CNOT_{426,747}, CNOT_{426,748}, CNOT_{429,749}, CNOT_{453,750}, CNOT_{426,751},$
 $CNOT_{425,752}, CNOT_{423,753}, CNOT_{427,754}, CNOT_{424,755}, CNOT_{446,473}, CNOT_{437,756}, CNOT_{428,757},$
 $CNOT_{426,758}, CNOT_{429,483}, CNOT_{427,760}, CNOT_{423,761}, CNOT_{440,481}, CNOT_{425,763}, CNOT_{426,764},$
 $CNOT_{428,765}, CNOT_{426,766}, CNOT_{429,767}, CNOT_{441,768}, CNOT_{430,769}, CNOT_{431,770}, CNOT_{435,771},$
 $CNOT_{427,772}, CNOT_{423,773}, CNOT_{431,774}, CNOT_{425,775}, CNOT_{430,777}, CNOT_{423,778}, CNOT_{424,779},$
 $CNOT_{461,780}, CNOT_{428,781}, CNOT_{425,782}, CNOT_{436,783}, CNOT_{434,784}, CNOT_{425,785}, CNOT_{429,786},$
 $CNOT_{427,787}, CNOT_{427,788}, CNOT_{428,790}, CNOT_{446,791}, CNOT_{426,792}, CNOT_{425,793}, CNOT_{442,794},$
 $CNOT_{427,795}, CNOT_{425,796}, CNOT_{429,485}, CNOT_{433,484}, CNOT_{424,797}, CNOT_{424,798}, CNOT_{425,799},$
 $CNOT_{444,800}, CNOT_{430,801}, CNOT_{428,802}, CNOT_{437,803}, CNOT_{424,804}, CNOT_{425,805}, CNOT_{434,806},$
 $CNOT_{428,807}, CNOT_{428,808}, CNOT_{432,809}, CNOT_{427,810}, CNOT_{427,486}, CNOT_{434,487}, CNOT_{465,811},$
 $CNOT_{428,812}, CNOT_{431,477}, CNOT_{445,813}, CNOT_{425,814}, CNOT_{427,815}, CNOT_{425,816}, CNOT_{428,490},$
 $CNOT_{430,817}, CNOT_{425,819}, CNOT_{440,820}, CNOT_{435,821}, CNOT_{440,822}, CNOT_{426,823}, CNOT_{426,824},$
 $CNOT_{429,825}, CNOT_{435,826}, CNOT_{452,827}, CNOT_{424,480}, CNOT_{428,828}, CNOT_{429,829}, CNOT_{423,830},$
 $CNOT_{428,831}, CNOT_{428,832}, CNOT_{427,833}, CNOT_{425,834}, CNOT_{495,543}, CNOT_{508,536}, CNOT_{496,835},$
 $CNOT_{506,545}, CNOT_{494,836}, CNOT_{504,837}, CNOT_{496,557}, CNOT_{497,838}, CNOT_{529,839}, CNOT_{505,840},$
 $CNOT_{495,841}, CNOT_{522,535}, CNOT_{501,842}, CNOT_{496,843}, CNOT_{524,844}, CNOT_{495,558}, CNOT_{493,845},$
 $CNOT_{497,846}, CNOT_{496,847}, CNOT_{496,848}, CNOT_{506,849}, CNOT_{495,850}, CNOT_{495,851}, CNOT_{498,852},$
 $CNOT_{522,853}, CNOT_{495,854}, CNOT_{494,855}, CNOT_{492,856}, CNOT_{496,857}, CNOT_{493,858}, CNOT_{515,542},$
 $CNOT_{506,859}, CNOT_{497,860}, CNOT_{495,861}, CNOT_{498,552}, CNOT_{496,863}, CNOT_{492,864}, CNOT_{509,550},$
 $CNOT_{494,866}, CNOT_{495,867}, CNOT_{497,868}, CNOT_{495,869}, CNOT_{498,870}, CNOT_{510,871}, CNOT_{499,872},$
 $CNOT_{500,873}, CNOT_{504,874}, CNOT_{496,875}, CNOT_{492,876}, CNOT_{500,877}, CNOT_{494,878}, CNOT_{499,880},$
 $CNOT_{492,881}, CNOT_{493,882}, CNOT_{530,883}, CNOT_{497,884}, CNOT_{494,885}, CNOT_{505,886}, CNOT_{503,887},$
 $CNOT_{494,888}, CNOT_{498,889}, CNOT_{496,890}, CNOT_{496,891}, CNOT_{497,893}, CNOT_{515,894}, CNOT_{495,895},$
 $CNOT_{494,896}, CNOT_{511,897}, CNOT_{496,898}, CNOT_{494,899}, CNOT_{498,554}, CNOT_{502,553}, CNOT_{493,900},$
 $CNOT_{493,901}, CNOT_{494,902}, CNOT_{513,903}, CNOT_{499,904}, CNOT_{497,905}, CNOT_{506,906}, CNOT_{493,907},$
 $CNOT_{494,908}, CNOT_{503,909}, CNOT_{497,910}, CNOT_{497,911}, CNOT_{501,912}, CNOT_{496,913}, CNOT_{496,555},$
 $CNOT_{503,556}, CNOT_{534,914}, CNOT_{497,915}, CNOT_{500,546}, CNOT_{514,916}, CNOT_{494,917}, CNOT_{496,918},$
 $CNOT_{494,919}, CNOT_{497,559}, CNOT_{499,920}, CNOT_{494,922}, CNOT_{509,923}, CNOT_{504,924}, CNOT_{509,925},$
 $CNOT_{495,926}, CNOT_{495,927}, CNOT_{498,928}, CNOT_{504,929}, CNOT_{521,930}, CNOT_{493,549}, CNOT_{497,931},$
 $CNOT_{498,932}, CNOT_{492,933}, CNOT_{497,934}, CNOT_{497,935}, CNOT_{496,936}, CNOT_{494,937}, CNOT_{564,612},$
 $CNOT_{577,605}, CNOT_{565,938}, CNOT_{575,614}, CNOT_{563,939}, CNOT_{573,940}, CNOT_{565,626}, CNOT_{566,941},$
 $CNOT_{598,942}, CNOT_{574,943}, CNOT_{564,944}, CNOT_{591,604}, CNOT_{570,945}, CNOT_{565,946}, CNOT_{593,947},$
 $CNOT_{564,627}, CNOT_{562,948}, CNOT_{566,949}, CNOT_{565,950}, CNOT_{565,951}, CNOT_{575,952}, CNOT_{564,953},$
 $CNOT_{564,954}, CNOT_{567,955}, CNOT_{591,956}, CNOT_{564,957}, CNOT_{563,958}, CNOT_{561,959}, CNOT_{565,960},$
 $CNOT_{562,961}, CNOT_{584,611}, CNOT_{575,962}, CNOT_{566,963}, CNOT_{564,964}, CNOT_{567,621}, CNOT_{565,966},$
 $CNOT_{561,967}, CNOT_{578,619}, CNOT_{563,969}, CNOT_{564,970}, CNOT_{566,971}, CNOT_{564,972}, CNOT_{567,973},$
 $CNOT_{579,974}, CNOT_{568,975}, CNOT_{569,976}, CNOT_{573,977}, CNOT_{565,978}, CNOT_{561,979}, CNOT_{569,980},$
 $CNOT_{563,981}, CNOT_{568,983}, CNOT_{561,984}, CNOT_{562,985}, CNOT_{599,986}, CNOT_{566,987}, CNOT_{563,988},$
 $CNOT_{574,989}, CNOT_{572,990}, CNOT_{563,991}, CNOT_{567,992}, CNOT_{565,993}, CNOT_{565,994}, CNOT_{566,996},$
 $CNOT_{584,997}, CNOT_{564,998}, CNOT_{563,999}, CNOT_{580,1000}, CNOT_{565,1001}, CNOT_{563,1002}, CNOT_{567,623},$
 $CNOT_{571,622}, CNOT_{562,1003}, CNOT_{562,1004}, CNOT_{563,1005}, CNOT_{582,1006}, CNOT_{568,1007}, CNOT_{566,1008},$
 $CNOT_{575,1009}, CNOT_{562,1010}, CNOT_{563,1011}, CNOT_{572,1012}, CNOT_{566,1013}, CNOT_{566,1014}, CNOT_{570,1015},$
 $CNOT_{565,1016}, CNOT_{565,624}, CNOT_{572,625}, CNOT_{603,1017}, CNOT_{566,1018}, CNOT_{569,615}, CNOT_{583,1019},$
 $CNOT_{563,1020}, CNOT_{565,1021}, CNOT_{563,1022}, CNOT_{566,628}, CNOT_{568,1023}, CNOT_{563,1025}, CNOT_{578,1026},$
 $CNOT_{573,1027}, CNOT_{578,1028}, CNOT_{564,1029}, CNOT_{564,1030}, CNOT_{567,1031}, CNOT_{573,1032}, CNOT_{590,1033},$
 $CNOT_{562,618}, CNOT_{566,1034}, CNOT_{567,1035}, CNOT_{561,1036}, CNOT_{566,1037}, CNOT_{566,1038}, CNOT_{565,1039},$
 $CNOT_{563,1040}$

Step42:

$\{CNOT_{354,405}, CNOT_{366,406}, CNOT_{355,398}, CNOT_{357,629}, CNOT_{366,407}, CNOT_{353,630}, CNOT_{358,631},\}$
 $\{CNOT_{354,419}, CNOT_{357,632}, CNOT_{359,402}, CNOT_{379,633}, CNOT_{356,634}, CNOT_{355,635}, CNOT_{360,397},\}$

$CNOT_{357,636}, CNOT_{354,637}, CNOT_{363,638}, CNOT_{354,420}, CNOT_{363,409}, CNOT_{354,639}, CNOT_{358,640},$
 $CNOT_{362,410}, CNOT_{354,641}, CNOT_{354,642}, CNOT_{367,643}, CNOT_{356,644}, CNOT_{355,645}, CNOT_{353,646},$
 $CNOT_{362,647}, CNOT_{355,648}, CNOT_{354,649}, CNOT_{353,650}, CNOT_{354,651}, CNOT_{354,652}, CNOT_{358,404},$
 $CNOT_{360,653}, CNOT_{354,654}, CNOT_{353,655}, CNOT_{394,656}, CNOT_{357,414}, CNOT_{355,657}, CNOT_{353,658},$
 $CNOT_{368,412}, CNOT_{382,659}, CNOT_{355,660}, CNOT_{353,661}, CNOT_{353,662}, CNOT_{353,663}, CNOT_{354,664},$
 $CNOT_{369,665}, CNOT_{355,666}, CNOT_{354,667}, CNOT_{359,668}, CNOT_{354,669}, CNOT_{353,670}, CNOT_{355,671},$
 $CNOT_{355,672}, CNOT_{357,673}, CNOT_{358,674}, CNOT_{353,675}, CNOT_{353,676}, CNOT_{356,677}, CNOT_{358,678},$
 $CNOT_{355,679}, CNOT_{359,680}, CNOT_{362,681}, CNOT_{354,682}, CNOT_{353,683}, CNOT_{353,684}, CNOT_{354,685},$
 $CNOT_{385,686}, CNOT_{358,403}, CNOT_{357,687}, CNOT_{358,688}, CNOT_{355,689}, CNOT_{355,690}, CNOT_{354,691},$
 $CNOT_{357,692}, CNOT_{354,693}, CNOT_{359,416}, CNOT_{357,415}, CNOT_{354,694}, CNOT_{353,695}, CNOT_{355,696},$
 $CNOT_{360,697}, CNOT_{360,698}, CNOT_{358,699}, CNOT_{364,399}, CNOT_{357,700}, CNOT_{354,701}, CNOT_{354,702},$
 $CNOT_{356,703}, CNOT_{353,704}, CNOT_{356,705}, CNOT_{356,706}, CNOT_{353,707}, CNOT_{357,417}, CNOT_{360,418},$
 $CNOT_{364,708}, CNOT_{356,709}, CNOT_{357,408}, CNOT_{358,710}, CNOT_{354,711}, CNOT_{354,712}, CNOT_{369,400},$
 $CNOT_{354,713}, CNOT_{356,421}, CNOT_{355,714}, CNOT_{358,715}, CNOT_{354,716}, CNOT_{370,717}, CNOT_{365,718},$
 $CNOT_{373,401}, CNOT_{359,719}, CNOT_{355,720}, CNOT_{356,721}, CNOT_{353,722}, CNOT_{359,723}, CNOT_{354,724},$
 $CNOT_{353,411}, CNOT_{355,725}, CNOT_{358,726}, CNOT_{353,727}, CNOT_{358,728}, CNOT_{357,729}, CNOT_{356,730},$
 $CNOT_{355,731}, CNOT_{423,474}, CNOT_{435,475}, CNOT_{424,467}, CNOT_{426,732}, CNOT_{435,476}, CNOT_{422,733},$
 $CNOT_{427,734}, CNOT_{423,488}, CNOT_{426,735}, CNOT_{428,471}, CNOT_{448,736}, CNOT_{425,737}, CNOT_{424,738},$
 $CNOT_{429,466}, CNOT_{426,739}, CNOT_{423,740}, CNOT_{432,741}, CNOT_{423,489}, CNOT_{432,478}, CNOT_{423,742},$
 $CNOT_{427,743}, CNOT_{431,479}, CNOT_{423,744}, CNOT_{423,745}, CNOT_{436,746}, CNOT_{425,747}, CNOT_{424,748},$
 $CNOT_{422,749}, CNOT_{431,750}, CNOT_{424,751}, CNOT_{423,752}, CNOT_{422,753}, CNOT_{423,754}, CNOT_{423,755},$
 $CNOT_{427,473}, CNOT_{429,756}, CNOT_{423,757}, CNOT_{422,758}, CNOT_{463,759}, CNOT_{426,483}, CNOT_{424,760},$
 $CNOT_{422,761}, CNOT_{437,481}, CNOT_{451,762}, CNOT_{424,763}, CNOT_{422,764}, CNOT_{422,765}, CNOT_{422,766},$
 $CNOT_{423,767}, CNOT_{438,768}, CNOT_{424,769}, CNOT_{423,770}, CNOT_{428,771}, CNOT_{423,772}, CNOT_{422,773},$
 $CNOT_{424,774}, CNOT_{424,775}, CNOT_{426,776}, CNOT_{427,777}, CNOT_{422,778}, CNOT_{422,779}, CNOT_{425,780},$
 $CNOT_{427,781}, CNOT_{424,782}, CNOT_{428,783}, CNOT_{431,784}, CNOT_{423,785}, CNOT_{422,786}, CNOT_{422,787},$
 $CNOT_{423,788}, CNOT_{454,789}, CNOT_{427,472}, CNOT_{426,790}, CNOT_{427,791}, CNOT_{424,792}, CNOT_{424,793},$
 $CNOT_{423,794}, CNOT_{426,795}, CNOT_{423,796}, CNOT_{428,485}, CNOT_{426,484}, CNOT_{423,797}, CNOT_{422,798},$
 $CNOT_{424,799}, CNOT_{429,800}, CNOT_{429,801}, CNOT_{427,802}, CNOT_{433,468}, CNOT_{426,803}, CNOT_{423,804},$
 $CNOT_{423,805}, CNOT_{425,806}, CNOT_{422,807}, CNOT_{425,808}, CNOT_{425,809}, CNOT_{422,810}, CNOT_{426,486},$
 $CNOT_{429,487}, CNOT_{433,811}, CNOT_{425,812}, CNOT_{426,477}, CNOT_{427,813}, CNOT_{423,814}, CNOT_{423,815},$
 $CNOT_{438,469}, CNOT_{423,816}, CNOT_{425,490}, CNOT_{424,817}, CNOT_{427,818}, CNOT_{423,819}, CNOT_{439,820},$
 $CNOT_{434,821}, CNOT_{442,470}, CNOT_{428,822}, CNOT_{424,823}, CNOT_{425,824}, CNOT_{422,825}, CNOT_{428,826},$
 $CNOT_{423,827}, CNOT_{422,480}, CNOT_{424,828}, CNOT_{427,829}, CNOT_{422,830}, CNOT_{427,831}, CNOT_{426,832},$
 $CNOT_{425,833}, CNOT_{424,834}, CNOT_{492,543}, CNOT_{504,544}, CNOT_{493,536}, CNOT_{495,835}, CNOT_{504,545},$
 $CNOT_{491,836}, CNOT_{496,837}, CNOT_{492,557}, CNOT_{495,838}, CNOT_{497,540}, CNOT_{517,839}, CNOT_{494,840},$
 $CNOT_{493,841}, CNOT_{498,535}, CNOT_{495,842}, CNOT_{492,843}, CNOT_{501,844}, CNOT_{492,558}, CNOT_{501,547},$
 $CNOT_{492,845}, CNOT_{496,846}, CNOT_{500,548}, CNOT_{492,847}, CNOT_{492,848}, CNOT_{505,849}, CNOT_{494,850},$
 $CNOT_{493,851}, CNOT_{491,852}, CNOT_{500,853}, CNOT_{493,854}, CNOT_{492,855}, CNOT_{491,856}, CNOT_{492,857},$
 $CNOT_{492,858}, CNOT_{496,542}, CNOT_{498,859}, CNOT_{492,860}, CNOT_{491,861}, CNOT_{532,862}, CNOT_{495,552},$
 $CNOT_{493,863}, CNOT_{491,864}, CNOT_{506,550}, CNOT_{520,865}, CNOT_{493,866}, CNOT_{491,867}, CNOT_{491,868},$
 $CNOT_{491,869}, CNOT_{492,870}, CNOT_{507,871}, CNOT_{493,872}, CNOT_{492,873}, CNOT_{497,874}, CNOT_{492,875},$
 $CNOT_{491,876}, CNOT_{493,877}, CNOT_{493,878}, CNOT_{495,879}, CNOT_{496,880}, CNOT_{491,881}, CNOT_{491,882},$
 $CNOT_{494,883}, CNOT_{496,884}, CNOT_{493,885}, CNOT_{497,886}, CNOT_{500,887}, CNOT_{492,888}, CNOT_{491,889},$
 $CNOT_{491,890}, CNOT_{492,891}, CNOT_{523,892}, CNOT_{496,541}, CNOT_{495,893}, CNOT_{496,894}, CNOT_{493,895},$
 $CNOT_{493,896}, CNOT_{492,897}, CNOT_{495,898}, CNOT_{492,899}, CNOT_{497,554}, CNOT_{495,553}, CNOT_{492,900},$
 $CNOT_{491,901}, CNOT_{493,902}, CNOT_{498,903}, CNOT_{498,904}, CNOT_{496,905}, CNOT_{502,537}, CNOT_{495,906},$
 $CNOT_{492,907}, CNOT_{492,908}, CNOT_{494,909}, CNOT_{491,910}, CNOT_{494,911}, CNOT_{494,912}, CNOT_{491,913},$
 $CNOT_{495,555}, CNOT_{498,556}, CNOT_{502,914}, CNOT_{494,915}, CNOT_{495,546}, CNOT_{496,916}, CNOT_{492,917},$
 $CNOT_{492,918}, CNOT_{507,538}, CNOT_{492,919}, CNOT_{494,559}, CNOT_{493,920}, CNOT_{496,921}, CNOT_{492,922},$
 $CNOT_{508,923}, CNOT_{503,924}, CNOT_{511,539}, CNOT_{497,925}, CNOT_{493,926}, CNOT_{494,927}, CNOT_{491,928},$
 $CNOT_{497,929}, CNOT_{492,930}, CNOT_{491,549}, CNOT_{493,931}, CNOT_{496,932}, CNOT_{491,933}, CNOT_{496,934},$
 $CNOT_{495,935}, CNOT_{494,936}, CNOT_{493,937}, CNOT_{561,612}, CNOT_{573,613}, CNOT_{562,605}, CNOT_{564,938},$

$CNOT_{573,614}, CNOT_{560,939}, CNOT_{565,940}, CNOT_{561,626}, CNOT_{564,941}, CNOT_{566,609}, CNOT_{586,942},$
 $CNOT_{563,943}, CNOT_{562,944}, CNOT_{567,604}, CNOT_{564,945}, CNOT_{561,946}, CNOT_{570,947}, CNOT_{561,627},$
 $CNOT_{570,616}, CNOT_{561,948}, CNOT_{565,949}, CNOT_{569,617}, CNOT_{561,950}, CNOT_{561,951}, CNOT_{574,952},$
 $CNOT_{563,953}, CNOT_{562,954}, CNOT_{560,955}, CNOT_{569,956}, CNOT_{562,957}, CNOT_{561,958}, CNOT_{560,959},$
 $CNOT_{561,960}, CNOT_{561,961}, CNOT_{565,611}, CNOT_{567,962}, CNOT_{561,963}, CNOT_{560,964}, CNOT_{601,965},$
 $CNOT_{564,621}, CNOT_{562,966}, CNOT_{560,967}, CNOT_{575,619}, CNOT_{589,968}, CNOT_{562,969}, CNOT_{560,970},$
 $CNOT_{560,971}, CNOT_{560,972}, CNOT_{561,973}, CNOT_{576,974}, CNOT_{562,975}, CNOT_{561,976}, CNOT_{566,977},$
 $CNOT_{561,978}, CNOT_{560,979}, CNOT_{562,980}, CNOT_{562,981}, CNOT_{564,982}, CNOT_{565,983}, CNOT_{560,984},$
 $CNOT_{560,985}, CNOT_{563,986}, CNOT_{565,987}, CNOT_{562,988}, CNOT_{566,989}, CNOT_{569,990}, CNOT_{561,991},$
 $CNOT_{560,992}, CNOT_{560,993}, CNOT_{561,994}, CNOT_{592,995}, CNOT_{565,610}, CNOT_{564,996}, CNOT_{565,997},$
 $CNOT_{562,998}, CNOT_{562,999}, CNOT_{561,1000}, CNOT_{564,1001}, CNOT_{561,1002}, CNOT_{566,623}, CNOT_{564,622},$
 $CNOT_{561,1003}, CNOT_{560,1004}, CNOT_{562,1005}, CNOT_{567,1006}, CNOT_{567,1007}, CNOT_{565,1008}, CNOT_{571,606},$
 $CNOT_{564,1009}, CNOT_{561,1010}, CNOT_{561,1011}, CNOT_{563,1012}, CNOT_{560,1013}, CNOT_{563,1014}, CNOT_{563,1015},$
 $CNOT_{560,1016}, CNOT_{564,624}, CNOT_{567,625}, CNOT_{571,1017}, CNOT_{563,1018}, CNOT_{564,615}, CNOT_{565,1019},$
 $CNOT_{561,1020}, CNOT_{561,1021}, CNOT_{576,607}, CNOT_{561,1022}, CNOT_{563,628}, CNOT_{562,1023}, CNOT_{565,1024},$
 $CNOT_{561,1025}, CNOT_{577,1026}, CNOT_{572,1027}, CNOT_{580,608}, CNOT_{566,1028}, CNOT_{562,1029}, CNOT_{563,1030},$
 $CNOT_{560,1031}, CNOT_{566,1032}, CNOT_{561,1033}, CNOT_{560,618}, CNOT_{562,1034}, CNOT_{565,1035}, CNOT_{560,1036},$
 $CNOT_{565,1037}, CNOT_{564,1038}, CNOT_{563,1039}, CNOT_{562,1040}$

Step43:

$CNOT_{353,405}, CNOT_{365,406}, CNOT_{354,398}, CNOT_{353,629}, CNOT_{359,407}, CNOT_{242,630}, CNOT_{241,413},$
 $CNOT_{355,631}, CNOT_{353,419}, CNOT_{355,632}, CNOT_{356,402}, CNOT_{369,633}, CNOT_{354,634}, CNOT_{354,635},$
 $CNOT_{355,397}, CNOT_{354,636}, CNOT_{254,637}, CNOT_{249,638}, CNOT_{353,420}, CNOT_{358,409}, CNOT_{256,639},$
 $CNOT_{357,640}, CNOT_{354,410}, CNOT_{254,641}, CNOT_{246,642}, CNOT_{249,643}, CNOT_{248,644}, CNOT_{354,645},$
 $CNOT_{251,646}, CNOT_{357,647}, CNOT_{253,648}, CNOT_{248,649}, CNOT_{243,650}, CNOT_{255,651}, CNOT_{241,652},$
 $CNOT_{355,404}, CNOT_{354,653}, CNOT_{255,654}, CNOT_{246,655}, CNOT_{363,656}, CNOT_{353,414}, CNOT_{354,657},$
 $CNOT_{245,658}, CNOT_{353,412}, CNOT_{356,659}, CNOT_{353,660}, CNOT_{248,661}, CNOT_{253,662}, CNOT_{248,663},$
 $CNOT_{254,664}, CNOT_{356,665}, CNOT_{353,666}, CNOT_{353,667}, CNOT_{355,668}, CNOT_{242,669}, CNOT_{252,670},$
 $CNOT_{353,671}, CNOT_{353,672}, CNOT_{256,673}, CNOT_{251,674}, CNOT_{241,675}, CNOT_{245,676}, CNOT_{353,677},$
 $CNOT_{355,678}, CNOT_{253,679}, CNOT_{251,680}, CNOT_{355,681}, CNOT_{243,682}, CNOT_{253,683}, CNOT_{242,684},$
 $CNOT_{242,685}, CNOT_{359,686}, CNOT_{356,403}, CNOT_{356,687}, CNOT_{355,688}, CNOT_{353,689}, CNOT_{353,690},$
 $CNOT_{250,691}, CNOT_{353,692}, CNOT_{247,693}, CNOT_{353,416}, CNOT_{354,415}, CNOT_{247,694}, CNOT_{254,695},$
 $CNOT_{246,696}, CNOT_{251,697}, CNOT_{248,698}, CNOT_{356,699}, CNOT_{353,399}, CNOT_{354,700}, CNOT_{241,701},$
 $CNOT_{353,702}, CNOT_{246,703}, CNOT_{251,704}, CNOT_{355,705}, CNOT_{354,706}, CNOT_{243,707}, CNOT_{354,417},$
 $CNOT_{357,418}, CNOT_{357,708}, CNOT_{355,709}, CNOT_{353,408}, CNOT_{353,710}, CNOT_{353,711}, CNOT_{247,712},$
 $CNOT_{360,400}, CNOT_{252,713}, CNOT_{353,421}, CNOT_{249,714}, CNOT_{250,715}, CNOT_{249,716}, CNOT_{363,717},$
 $CNOT_{356,718}, CNOT_{360,401}, CNOT_{243,719}, CNOT_{353,720}, CNOT_{242,721}, CNOT_{252,722}, CNOT_{244,723},$
 $CNOT_{250,724}, CNOT_{241,411}, CNOT_{353,725}, CNOT_{243,726}, CNOT_{254,727}, CNOT_{353,728}, CNOT_{356,729},$
 $CNOT_{246,730}, CNOT_{243,731}, CNOT_{422,474}, CNOT_{434,475}, CNOT_{423,467}, CNOT_{422,732}, CNOT_{428,476},$
 $CNOT_{274,733}, CNOT_{273,482}, CNOT_{424,734}, CNOT_{422,488}, CNOT_{424,735}, CNOT_{425,471}, CNOT_{438,736},$
 $CNOT_{423,737}, CNOT_{423,738}, CNOT_{424,466}, CNOT_{423,739}, CNOT_{286,740}, CNOT_{281,741}, CNOT_{422,489},$
 $CNOT_{427,478}, CNOT_{288,742}, CNOT_{426,743}, CNOT_{423,479}, CNOT_{286,744}, CNOT_{278,745}, CNOT_{281,746},$
 $CNOT_{280,747}, CNOT_{423,748}, CNOT_{283,749}, CNOT_{426,750}, CNOT_{285,751}, CNOT_{280,752}, CNOT_{275,753},$
 $CNOT_{287,754}, CNOT_{273,755}, CNOT_{424,473}, CNOT_{423,756}, CNOT_{287,757}, CNOT_{278,758}, CNOT_{432,759},$
 $CNOT_{422,483}, CNOT_{423,760}, CNOT_{277,761}, CNOT_{422,481}, CNOT_{425,762}, CNOT_{422,763}, CNOT_{280,764},$
 $CNOT_{285,765}, CNOT_{280,766}, CNOT_{286,767}, CNOT_{425,768}, CNOT_{422,769}, CNOT_{422,770}, CNOT_{424,771},$
 $CNOT_{274,772}, CNOT_{284,773}, CNOT_{422,774}, CNOT_{422,775}, CNOT_{288,776}, CNOT_{283,777}, CNOT_{273,778},$
 $CNOT_{277,779}, CNOT_{422,780}, CNOT_{424,781}, CNOT_{285,782}, CNOT_{283,783}, CNOT_{424,784}, CNOT_{275,785},$
 $CNOT_{285,786}, CNOT_{274,787}, CNOT_{274,788}, CNOT_{428,789}, CNOT_{425,472}, CNOT_{425,790}, CNOT_{424,791},$
 $CNOT_{422,792}, CNOT_{422,793}, CNOT_{282,794}, CNOT_{422,795}, CNOT_{279,796}, CNOT_{422,485}, CNOT_{423,484},$
 $CNOT_{279,797}, CNOT_{286,798}, CNOT_{278,799}, CNOT_{283,800}, CNOT_{280,801}, CNOT_{425,802}, CNOT_{422,468},$
 $CNOT_{423,803}, CNOT_{273,804}, CNOT_{422,805}, CNOT_{278,806}, CNOT_{283,807}, CNOT_{424,808}, CNOT_{423,809},$
 $CNOT_{275,810}, CNOT_{423,486}, CNOT_{426,487}, CNOT_{426,811}, CNOT_{424,812}, CNOT_{422,477}, CNOT_{422,813},$
 $CNOT_{422,814}, CNOT_{279,815}, CNOT_{429,469}, CNOT_{284,816}, CNOT_{422,490}, CNOT_{281,817}, CNOT_{282,818},$

$CNOT_{281,819}, CNOT_{432,820}, CNOT_{425,821}, CNOT_{429,470}, CNOT_{275,822}, CNOT_{422,823}, CNOT_{274,824},$
 $CNOT_{284,825}, CNOT_{276,826}, CNOT_{282,827}, CNOT_{273,480}, CNOT_{422,828}, CNOT_{275,829}, CNOT_{286,830},$
 $CNOT_{422,831}, CNOT_{425,832}, CNOT_{278,833}, CNOT_{275,834}, CNOT_{491,543}, CNOT_{503,544}, CNOT_{492,536},$
 $CNOT_{491,835}, CNOT_{497,545}, CNOT_{306,836}, CNOT_{305,551}, CNOT_{493,837}, CNOT_{491,557}, CNOT_{493,838},$
 $CNOT_{494,540}, CNOT_{507,839}, CNOT_{492,840}, CNOT_{492,841}, CNOT_{493,535}, CNOT_{492,842}, CNOT_{318,843},$
 $CNOT_{313,844}, CNOT_{491,558}, CNOT_{496,547}, CNOT_{320,845}, CNOT_{495,846}, CNOT_{492,548}, CNOT_{318,847},$
 $CNOT_{310,848}, CNOT_{313,849}, CNOT_{312,850}, CNOT_{492,851}, CNOT_{315,852}, CNOT_{495,853}, CNOT_{317,854},$
 $CNOT_{312,855}, CNOT_{307,856}, CNOT_{319,857}, CNOT_{305,858}, CNOT_{493,542}, CNOT_{492,859}, CNOT_{319,860},$
 $CNOT_{310,861}, CNOT_{501,862}, CNOT_{491,552}, CNOT_{492,863}, CNOT_{309,864}, CNOT_{491,550}, CNOT_{494,865},$
 $CNOT_{491,866}, CNOT_{312,867}, CNOT_{317,868}, CNOT_{312,869}, CNOT_{318,870}, CNOT_{494,871}, CNOT_{491,872},$
 $CNOT_{491,873}, CNOT_{493,874}, CNOT_{306,875}, CNOT_{316,876}, CNOT_{491,877}, CNOT_{491,878}, CNOT_{320,879},$
 $CNOT_{315,880}, CNOT_{305,881}, CNOT_{309,882}, CNOT_{491,883}, CNOT_{493,884}, CNOT_{317,885}, CNOT_{315,886},$
 $CNOT_{493,887}, CNOT_{307,888}, CNOT_{317,889}, CNOT_{306,890}, CNOT_{306,891}, CNOT_{497,892}, CNOT_{494,541},$
 $CNOT_{494,893}, CNOT_{493,894}, CNOT_{491,895}, CNOT_{491,896}, CNOT_{314,897}, CNOT_{491,898}, CNOT_{311,899},$
 $CNOT_{491,554}, CNOT_{492,553}, CNOT_{311,900}, CNOT_{318,901}, CNOT_{310,902}, CNOT_{315,903}, CNOT_{312,904},$
 $CNOT_{494,905}, CNOT_{491,537}, CNOT_{492,906}, CNOT_{305,907}, CNOT_{491,908}, CNOT_{310,909}, CNOT_{315,910},$
 $CNOT_{493,911}, CNOT_{492,912}, CNOT_{307,913}, CNOT_{492,555}, CNOT_{495,556}, CNOT_{495,914}, CNOT_{493,915},$
 $CNOT_{491,546}, CNOT_{491,916}, CNOT_{491,917}, CNOT_{311,918}, CNOT_{498,538}, CNOT_{316,919}, CNOT_{491,559},$
 $CNOT_{313,920}, CNOT_{314,921}, CNOT_{313,922}, CNOT_{501,923}, CNOT_{494,924}, CNOT_{498,539}, CNOT_{307,925},$
 $CNOT_{491,926}, CNOT_{306,927}, CNOT_{316,928}, CNOT_{308,929}, CNOT_{314,930}, CNOT_{305,549}, CNOT_{491,931},$
 $CNOT_{307,932}, CNOT_{318,933}, CNOT_{491,934}, CNOT_{494,935}, CNOT_{310,936}, CNOT_{307,937}, CNOT_{560,612},$
 $CNOT_{572,613}, CNOT_{561,605}, CNOT_{560,938}, CNOT_{566,614}, CNOT_{338,939}, CNOT_{337,620}, CNOT_{562,940},$
 $CNOT_{560,626}, CNOT_{562,941}, CNOT_{563,609}, CNOT_{576,942}, CNOT_{561,943}, CNOT_{561,944}, CNOT_{562,604},$
 $CNOT_{561,945}, CNOT_{350,946}, CNOT_{345,947}, CNOT_{560,627}, CNOT_{565,616}, CNOT_{352,948}, CNOT_{564,949},$
 $CNOT_{561,617}, CNOT_{350,950}, CNOT_{342,951}, CNOT_{345,952}, CNOT_{344,953}, CNOT_{561,954}, CNOT_{347,955},$
 $CNOT_{564,956}, CNOT_{349,957}, CNOT_{344,958}, CNOT_{339,959}, CNOT_{351,960}, CNOT_{337,961}, CNOT_{562,611},$
 $CNOT_{561,962}, CNOT_{351,963}, CNOT_{342,964}, CNOT_{570,965}, CNOT_{560,621}, CNOT_{561,966}, CNOT_{341,967},$
 $CNOT_{560,619}, CNOT_{563,968}, CNOT_{560,969}, CNOT_{344,970}, CNOT_{349,971}, CNOT_{344,972}, CNOT_{350,973},$
 $CNOT_{563,974}, CNOT_{560,975}, CNOT_{560,976}, CNOT_{562,977}, CNOT_{338,978}, CNOT_{348,979}, CNOT_{560,980},$
 $CNOT_{560,981}, CNOT_{352,982}, CNOT_{347,983}, CNOT_{337,984}, CNOT_{341,985}, CNOT_{560,986}, CNOT_{562,987},$
 $CNOT_{349,988}, CNOT_{347,989}, CNOT_{562,990}, CNOT_{339,991}, CNOT_{349,992}, CNOT_{338,993}, CNOT_{338,994},$
 $CNOT_{566,995}, CNOT_{563,610}, CNOT_{563,996}, CNOT_{562,997}, CNOT_{560,998}, CNOT_{560,999}, CNOT_{346,1000},$
 $CNOT_{560,1001}, CNOT_{343,1002}, CNOT_{560,623}, CNOT_{561,622}, CNOT_{343,1003}, CNOT_{350,1004}, CNOT_{342,1005},$
 $CNOT_{347,1006}, CNOT_{344,1007}, CNOT_{563,1008}, CNOT_{560,606}, CNOT_{561,1009}, CNOT_{337,1010}, CNOT_{560,1011},$
 $CNOT_{342,1012}, CNOT_{347,1013}, CNOT_{562,1014}, CNOT_{561,1015}, CNOT_{339,1016}, CNOT_{561,624}, CNOT_{564,625},$
 $CNOT_{564,1017}, CNOT_{562,1018}, CNOT_{560,615}, CNOT_{560,1019}, CNOT_{560,1020}, CNOT_{343,1021}, CNOT_{567,607},$
 $CNOT_{348,1022}, CNOT_{560,628}, CNOT_{345,1023}, CNOT_{346,1024}, CNOT_{345,1025}, CNOT_{570,1026}, CNOT_{563,1027},$
 $CNOT_{567,608}, CNOT_{339,1028}, CNOT_{560,1029}, CNOT_{338,1030}, CNOT_{348,1031}, CNOT_{340,1032}, CNOT_{346,1033},$
 $CNOT_{337,618}, CNOT_{560,1034}, CNOT_{339,1035}, CNOT_{350,1036}, CNOT_{560,1037}, CNOT_{563,1038}, CNOT_{342,1039},$
 $CNOT_{339,1040}$

Step44:

$CNOT_{247,353}, CNOT_{251,354}, CNOT_{242,355}, CNOT_{250,356}, CNOT_{243,357}, CNOT_{245,358}, CNOT_{248,359},$
 $CNOT_{241,360}, CNOT_{254,361}, CNOT_{246,362}, CNOT_{244,363}, CNOT_{242,364}, CNOT_{249,365}, CNOT_{241,366},$
 $CNOT_{241,367}, CNOT_{244,368}, CNOT_{247,369}, CNOT_{243,370}, CNOT_{245,371}, CNOT_{253,372}, CNOT_{244,373},$
 $CNOT_{248,374}, CNOT_{242,375}, CNOT_{252,376}, CNOT_{250,377}, CNOT_{251,378}, CNOT_{255,379}, CNOT_{243,380},$
 $CNOT_{248,381}, CNOT_{249,382}, CNOT_{245,383}, CNOT_{247,384}, CNOT_{241,385}, CNOT_{253,386}, CNOT_{246,387},$
 $CNOT_{241,388}, CNOT_{248,389}, CNOT_{247,390}, CNOT_{250,391}, CNOT_{246,392}, CNOT_{244,393}, CNOT_{245,394},$
 $CNOT_{246,395}, CNOT_{241,396}, CNOT_{254,397}, CNOT_{250,398}, CNOT_{251,399}, CNOT_{250,400}, CNOT_{249,401},$
 $CNOT_{245,402}, CNOT_{244,403}, CNOT_{241,404}, CNOT_{252,405}, CNOT_{248,406}, CNOT_{249,407}, CNOT_{245,408},$
 $CNOT_{250,409}, CNOT_{242,410}, CNOT_{248,411}, CNOT_{243,412}, CNOT_{249,413}, CNOT_{248,414}, CNOT_{247,415},$
 $CNOT_{242,416}, CNOT_{242,417}, CNOT_{245,418}, CNOT_{244,419}, CNOT_{241,420}, CNOT_{246,421}, CNOT_{279,422},$
 $CNOT_{283,423}, CNOT_{274,424}, CNOT_{282,425}, CNOT_{275,426}, CNOT_{277,427}, CNOT_{280,428}, CNOT_{273,429},$
 $CNOT_{286,430}, CNOT_{278,431}, CNOT_{276,432}, CNOT_{274,433}, CNOT_{281,434}, CNOT_{273,435}, CNOT_{273,436},$
 $CNOT_{276,437}, CNOT_{279,438}, CNOT_{275,439}, CNOT_{277,440}, CNOT_{285,441}, CNOT_{276,442}, CNOT_{280,443},$

$CNOT_{274,444}, CNOT_{284,445}, CNOT_{282,446}, CNOT_{283,447}, CNOT_{287,448}, CNOT_{275,449}, CNOT_{280,450},$
 $CNOT_{281,451}, CNOT_{277,452}, CNOT_{279,453}, CNOT_{273,454}, CNOT_{285,455}, CNOT_{278,456}, CNOT_{273,457},$
 $CNOT_{280,458}, CNOT_{279,459}, CNOT_{282,460}, CNOT_{278,461}, CNOT_{276,462}, CNOT_{277,463}, CNOT_{278,464},$
 $CNOT_{273,465}, CNOT_{286,466}, CNOT_{282,467}, CNOT_{283,468}, CNOT_{282,469}, CNOT_{281,470}, CNOT_{277,471},$
 $CNOT_{276,472}, CNOT_{273,473}, CNOT_{284,474}, CNOT_{280,475}, CNOT_{281,476}, CNOT_{277,477}, CNOT_{282,478},$
 $CNOT_{274,479}, CNOT_{280,480}, CNOT_{275,481}, CNOT_{281,482}, CNOT_{280,483}, CNOT_{279,484}, CNOT_{274,485},$
 $CNOT_{274,486}, CNOT_{277,487}, CNOT_{276,488}, CNOT_{273,489}, CNOT_{278,490}, CNOT_{311,491}, CNOT_{315,492},$
 $CNOT_{306,493}, CNOT_{314,494}, CNOT_{307,495}, CNOT_{309,496}, CNOT_{312,497}, CNOT_{305,498}, CNOT_{318,499},$
 $CNOT_{310,500}, CNOT_{308,501}, CNOT_{306,502}, CNOT_{313,503}, CNOT_{305,504}, CNOT_{305,505}, CNOT_{308,506},$
 $CNOT_{311,507}, CNOT_{307,508}, CNOT_{309,509}, CNOT_{317,510}, CNOT_{308,511}, CNOT_{312,512}, CNOT_{306,513},$
 $CNOT_{316,514}, CNOT_{314,515}, CNOT_{315,516}, CNOT_{319,517}, CNOT_{307,518}, CNOT_{312,519}, CNOT_{313,520},$
 $CNOT_{309,521}, CNOT_{311,522}, CNOT_{305,523}, CNOT_{317,524}, CNOT_{310,525}, CNOT_{305,526}, CNOT_{312,527},$
 $CNOT_{311,528}, CNOT_{314,529}, CNOT_{310,530}, CNOT_{308,531}, CNOT_{309,532}, CNOT_{310,533}, CNOT_{305,534},$
 $CNOT_{318,535}, CNOT_{314,536}, CNOT_{315,537}, CNOT_{314,538}, CNOT_{313,539}, CNOT_{309,540}, CNOT_{308,541},$
 $CNOT_{305,542}, CNOT_{316,543}, CNOT_{312,544}, CNOT_{313,545}, CNOT_{309,546}, CNOT_{314,547}, CNOT_{306,548},$
 $CNOT_{312,549}, CNOT_{307,550}, CNOT_{313,551}, CNOT_{312,552}, CNOT_{311,553}, CNOT_{306,554}, CNOT_{306,555},$
 $CNOT_{309,556}, CNOT_{308,557}, CNOT_{305,558}, CNOT_{310,559}, CNOT_{343,560}, CNOT_{347,561}, CNOT_{338,562},$
 $CNOT_{346,563}, CNOT_{339,564}, CNOT_{341,565}, CNOT_{344,566}, CNOT_{337,567}, CNOT_{350,568}, CNOT_{342,569},$
 $CNOT_{340,570}, CNOT_{338,571}, CNOT_{345,572}, CNOT_{337,573}, CNOT_{337,574}, CNOT_{340,575}, CNOT_{343,576},$
 $CNOT_{339,577}, CNOT_{341,578}, CNOT_{349,579}, CNOT_{340,580}, CNOT_{344,581}, CNOT_{338,582}, CNOT_{348,583},$
 $CNOT_{346,584}, CNOT_{347,585}, CNOT_{351,586}, CNOT_{339,587}, CNOT_{344,588}, CNOT_{345,589}, CNOT_{341,590},$
 $CNOT_{343,591}, CNOT_{337,592}, CNOT_{349,593}, CNOT_{342,594}, CNOT_{337,595}, CNOT_{344,596}, CNOT_{343,597},$
 $CNOT_{346,598}, CNOT_{342,599}, CNOT_{340,600}, CNOT_{341,601}, CNOT_{342,602}, CNOT_{337,603}, CNOT_{350,604},$
 $CNOT_{346,605}, CNOT_{347,606}, CNOT_{346,607}, CNOT_{345,608}, CNOT_{341,609}, CNOT_{340,610}, CNOT_{337,611},$
 $CNOT_{348,612}, CNOT_{344,613}, CNOT_{345,614}, CNOT_{341,615}, CNOT_{346,616}, CNOT_{338,617}, CNOT_{344,618},$
 $CNOT_{339,619}, CNOT_{345,620}, CNOT_{344,621}, CNOT_{343,622}, CNOT_{338,623}, CNOT_{338,624}, CNOT_{341,625},$
 $CNOT_{340,626}, CNOT_{337,627}, CNOT_{342,628}$

Step45:

$CNOT_{249,353}, CNOT_{253,354}, CNOT_{244,355}, CNOT_{254,356}, CNOT_{255,357}, CNOT_{256,358}, CNOT_{252,359},$
 $CNOT_{246,360}, CNOT_{255,361}, CNOT_{250,362}, CNOT_{247,363}, CNOT_{245,364}, CNOT_{251,365}, CNOT_{243,366},$
 $CNOT_{256,367}, CNOT_{254,368}, CNOT_{252,369}, CNOT_{252,370}, CNOT_{246,371}, CNOT_{256,372}, CNOT_{256,373},$
 $CNOT_{249,374}, CNOT_{243,375}, CNOT_{253,376}, CNOT_{253,377}, CNOT_{252,378}, CNOT_{256,379}, CNOT_{256,380},$
 $CNOT_{255,381}, CNOT_{252,382}, CNOT_{252,383}, CNOT_{251,384}, CNOT_{242,385}, CNOT_{255,386}, CNOT_{254,387},$
 $CNOT_{254,388}, CNOT_{253,389}, CNOT_{253,390}, CNOT_{251,391}, CNOT_{251,392}, CNOT_{251,393}, CNOT_{250,394},$
 $CNOT_{248,395}, CNOT_{247,396}, CNOT_{256,397}, CNOT_{256,398}, CNOT_{255,399}, CNOT_{255,400}, CNOT_{255,401},$
 $CNOT_{255,402}, CNOT_{255,403}, CNOT_{255,404}, CNOT_{254,405}, CNOT_{254,406}, CNOT_{253,407}, CNOT_{253,408},$
 $CNOT_{252,409}, CNOT_{252,410}, CNOT_{251,411}, CNOT_{251,412}, CNOT_{250,413}, CNOT_{250,414}, CNOT_{250,415},$
 $CNOT_{250,416}, CNOT_{249,417}, CNOT_{248,418}, CNOT_{248,419}, CNOT_{245,420}, CNOT_{255,421}, CNOT_{281,422},$
 $CNOT_{285,423}, CNOT_{276,424}, CNOT_{286,425}, CNOT_{287,426}, CNOT_{288,427}, CNOT_{284,428}, CNOT_{278,429},$
 $CNOT_{287,430}, CNOT_{282,431}, CNOT_{279,432}, CNOT_{277,433}, CNOT_{283,434}, CNOT_{275,435}, CNOT_{288,436},$
 $CNOT_{286,437}, CNOT_{284,438}, CNOT_{284,439}, CNOT_{278,440}, CNOT_{288,441}, CNOT_{288,442}, CNOT_{281,443},$
 $CNOT_{275,444}, CNOT_{285,445}, CNOT_{285,446}, CNOT_{284,447}, CNOT_{288,448}, CNOT_{288,449}, CNOT_{287,450},$
 $CNOT_{284,451}, CNOT_{284,452}, CNOT_{283,453}, CNOT_{274,454}, CNOT_{287,455}, CNOT_{286,456}, CNOT_{286,457},$
 $CNOT_{285,458}, CNOT_{285,459}, CNOT_{283,460}, CNOT_{283,461}, CNOT_{283,462}, CNOT_{282,463}, CNOT_{280,464},$
 $CNOT_{279,465}, CNOT_{288,466}, CNOT_{288,467}, CNOT_{287,468}, CNOT_{287,469}, CNOT_{287,470}, CNOT_{287,471},$
 $CNOT_{287,472}, CNOT_{287,473}, CNOT_{286,474}, CNOT_{286,475}, CNOT_{285,476}, CNOT_{285,477}, CNOT_{284,478},$
 $CNOT_{284,479}, CNOT_{283,480}, CNOT_{283,481}, CNOT_{282,482}, CNOT_{282,483}, CNOT_{282,484}, CNOT_{282,485},$
 $CNOT_{281,486}, CNOT_{280,487}, CNOT_{280,488}, CNOT_{277,489}, CNOT_{287,490}, CNOT_{313,491}, CNOT_{317,492},$
 $CNOT_{308,493}, CNOT_{318,494}, CNOT_{319,495}, CNOT_{320,496}, CNOT_{316,497}, CNOT_{310,498}, CNOT_{319,499},$
 $CNOT_{314,500}, CNOT_{311,501}, CNOT_{309,502}, CNOT_{315,503}, CNOT_{307,504}, CNOT_{320,505}, CNOT_{318,506},$
 $CNOT_{316,507}, CNOT_{316,508}, CNOT_{310,509}, CNOT_{320,510}, CNOT_{320,511}, CNOT_{313,512}, CNOT_{307,513},$
 $CNOT_{317,514}, CNOT_{317,515}, CNOT_{316,516}, CNOT_{320,517}, CNOT_{320,518}, CNOT_{319,519}, CNOT_{316,520},$
 $CNOT_{316,521}, CNOT_{315,522}, CNOT_{306,523}, CNOT_{319,524}, CNOT_{318,525}, CNOT_{318,526}, CNOT_{317,527},$

$\{$ $CNOT_{317,528}, CNOT_{315,529}, CNOT_{315,530}, CNOT_{315,531}, CNOT_{314,532}, CNOT_{312,533}, CNOT_{311,534},$
 $CNOT_{320,535}, CNOT_{320,536}, CNOT_{319,537}, CNOT_{319,538}, CNOT_{319,539}, CNOT_{319,540}, CNOT_{319,541},$
 $CNOT_{319,542}, CNOT_{318,543}, CNOT_{318,544}, CNOT_{317,545}, CNOT_{317,546}, CNOT_{316,547}, CNOT_{316,548},$
 $CNOT_{315,549}, CNOT_{315,550}, CNOT_{314,551}, CNOT_{314,552}, CNOT_{314,553}, CNOT_{314,554}, CNOT_{313,555},$
 $CNOT_{312,556}, CNOT_{312,557}, CNOT_{309,558}, CNOT_{319,559}, CNOT_{345,560}, CNOT_{349,561}, CNOT_{340,562},$
 $CNOT_{350,563}, CNOT_{351,564}, CNOT_{352,565}, CNOT_{348,566}, CNOT_{342,567}, CNOT_{351,568}, CNOT_{346,569},$
 $CNOT_{343,570}, CNOT_{341,571}, CNOT_{347,572}, CNOT_{339,573}, CNOT_{352,574}, CNOT_{350,575}, CNOT_{348,576},$
 $CNOT_{348,577}, CNOT_{342,578}, CNOT_{352,579}, CNOT_{352,580}, CNOT_{345,581}, CNOT_{339,582}, CNOT_{349,583},$
 $CNOT_{349,584}, CNOT_{348,585}, CNOT_{352,586}, CNOT_{352,587}, CNOT_{351,588}, CNOT_{348,589}, CNOT_{348,590},$
 $CNOT_{347,591}, CNOT_{338,592}, CNOT_{351,593}, CNOT_{350,594}, CNOT_{350,595}, CNOT_{349,596}, CNOT_{349,597},$
 $CNOT_{347,598}, CNOT_{347,599}, CNOT_{347,600}, CNOT_{346,601}, CNOT_{344,602}, CNOT_{343,603}, CNOT_{352,604},$
 $CNOT_{352,605}, CNOT_{351,606}, CNOT_{351,607}, CNOT_{351,608}, CNOT_{351,609}, CNOT_{351,610}, CNOT_{351,611},$
 $CNOT_{350,612}, CNOT_{350,613}, CNOT_{349,614}, CNOT_{349,615}, CNOT_{348,616}, CNOT_{348,617}, CNOT_{347,618},$
 $CNOT_{347,619}, CNOT_{346,620}, CNOT_{346,621}, CNOT_{346,622}, CNOT_{346,623}, CNOT_{345,624}, CNOT_{344,625},$
 $CNOT_{344,626}, CNOT_{341,627}, CNOT_{351,628}, CNOT_{1,129}, CNOT_{2,130}, CNOT_{3,131}, CNOT_{4,132},$
 $CNOT_{5,133}, CNOT_{6,134}, CNOT_{7,135}, CNOT_{8,136}, CNOT_{9,137}, CNOT_{10,138}, CNOT_{11,139},$
 $CNOT_{12,140}, CNOT_{13,141}, CNOT_{14,142}, CNOT_{15,143}, CNOT_{16,144}, CNOT_{17,145}, CNOT_{18,146},$
 $CNOT_{19,147}, CNOT_{20,148}, CNOT_{21,149}, CNOT_{22,150}, CNOT_{23,151}, CNOT_{24,152}, CNOT_{25,153},$
 $CNOT_{26,154}, CNOT_{27,155}, CNOT_{28,156}, CNOT_{29,157}, CNOT_{30,158}, CNOT_{31,159}, CNOT_{32,160}$ $\}$

Step46:

$\{$ $Toffoli_{161,165,225}, Toffoli_{161,166,226}, Toffoli_{161,167,227}, Toffoli_{161,168,228},$
 $Toffoli_{162,165,229}, Toffoli_{162,166,230}, Toffoli_{162,167,231}, Toffoli_{162,168,232},$
 $Toffoli_{163,165,233}, Toffoli_{163,166,234}, Toffoli_{163,167,235}, Toffoli_{163,168,236},$
 $Toffoli_{164,165,237}, Toffoli_{164,166,238}, Toffoli_{164,167,239}, Toffoli_{164,168,240},$
 $Toffoli_{169,173,241}, Toffoli_{169,174,242}, Toffoli_{169,175,243}, Toffoli_{169,176,244},$
 $Toffoli_{170,173,245}, Toffoli_{170,174,246}, Toffoli_{170,175,247}, Toffoli_{170,176,248},$
 $Toffoli_{171,173,249}, Toffoli_{171,174,250}, Toffoli_{171,175,251}, Toffoli_{171,176,252},$
 $Toffoli_{172,173,253}, Toffoli_{172,174,254}, Toffoli_{172,175,255}, Toffoli_{172,176,256},$
 $Toffoli_{177,181,257}, Toffoli_{177,182,258}, Toffoli_{177,183,259}, Toffoli_{177,184,260},$
 $Toffoli_{178,181,261}, Toffoli_{178,182,262}, Toffoli_{178,183,263}, Toffoli_{178,184,264},$
 $Toffoli_{179,181,265}, Toffoli_{179,182,266}, Toffoli_{179,183,267}, Toffoli_{179,184,268},$
 $Toffoli_{180,181,269}, Toffoli_{180,182,270}, Toffoli_{180,183,271}, Toffoli_{180,184,272},$
 $Toffoli_{185,189,273}, Toffoli_{185,190,274}, Toffoli_{185,191,275}, Toffoli_{185,192,276},$
 $Toffoli_{186,189,277}, Toffoli_{186,190,278}, Toffoli_{186,191,279}, Toffoli_{186,192,280},$
 $Toffoli_{187,189,281}, Toffoli_{187,190,282}, Toffoli_{187,191,283}, Toffoli_{187,192,284},$
 $Toffoli_{188,189,285}, Toffoli_{188,190,286}, Toffoli_{188,191,287}, Toffoli_{188,192,288},$
 $Toffoli_{193,197,289}, Toffoli_{193,198,290}, Toffoli_{193,199,291}, Toffoli_{193,200,292},$
 $Toffoli_{194,197,293}, Toffoli_{194,198,294}, Toffoli_{194,199,295}, Toffoli_{194,200,296},$
 $Toffoli_{195,197,297}, Toffoli_{195,198,298}, Toffoli_{195,199,299}, Toffoli_{195,200,300},$
 $Toffoli_{196,197,301}, Toffoli_{196,198,302}, Toffoli_{196,199,303}, Toffoli_{196,200,304},$
 $Toffoli_{201,205,305}, Toffoli_{201,206,306}, Toffoli_{201,207,307}, Toffoli_{201,208,308},$
 $Toffoli_{202,205,309}, Toffoli_{202,206,310}, Toffoli_{202,207,311}, Toffoli_{202,208,312},$
 $Toffoli_{203,205,313}, Toffoli_{203,206,314}, Toffoli_{203,207,315}, Toffoli_{203,208,316},$
 $Toffoli_{204,205,317}, Toffoli_{204,206,318}, Toffoli_{204,207,319}, Toffoli_{204,208,320},$
 $Toffoli_{209,213,321}, Toffoli_{209,214,322}, Toffoli_{209,215,323}, Toffoli_{209,216,324},$
 $Toffoli_{210,213,325}, Toffoli_{210,214,326}, Toffoli_{210,215,327}, Toffoli_{210,216,328},$
 $Toffoli_{211,213,329}, Toffoli_{211,214,330}, Toffoli_{211,215,331}, Toffoli_{211,216,332},$
 $Toffoli_{212,213,333}, Toffoli_{212,214,334}, Toffoli_{212,215,335}, Toffoli_{212,216,336},$
 $Toffoli_{217,221,337}, Toffoli_{217,222,338}, Toffoli_{217,223,339}, Toffoli_{217,224,340},$
 $Toffoli_{218,221,341}, Toffoli_{218,222,342}, Toffoli_{218,223,343}, Toffoli_{218,224,344},$
 $Toffoli_{219,221,345}, Toffoli_{219,222,346}, Toffoli_{219,223,347}, Toffoli_{219,224,348},$
 $Toffoli_{220,221,349}, Toffoli_{220,222,350}, Toffoli_{220,223,351}, Toffoli_{220,224,352}$
 $NOT_{129}, NOT_{130}, NOT_{131}, NOT_{132}, NOT_{133}, NOT_{134}, NOT_{135}, NOT_{136},$ $\}$

$\left\{ \begin{array}{l} NOT_{137}, NOT_{138}, NOT_{139}, NOT_{140}, NOT_{141}, NOT_{142}, NOT_{143}, NOT_{144}, \\ NOT_{145}, NOT_{146}, NOT_{147}, NOT_{148}, NOT_{149}, NOT_{150}, NOT_{151}, NOT_{152}, \\ NOT_{153}, NOT_{154}, NOT_{155}, NOT_{156}, NOT_{157}, NOT_{158}, NOT_{159}, NOT_{160} \end{array} \right\}$	
Step47: $\left\{ \begin{array}{l} Toffoli_{1,2,161}, Toffoli_{1,130,162}, Toffoli_{129,2,163}, Toffoli_{129,130,164}, \\ Toffoli_{3,4,165}, Toffoli_{3,132,166}, Toffoli_{131,4,167}, Toffoli_{131,132,168}, \\ Toffoli_{5,6,169}, Toffoli_{5,134,170}, Toffoli_{133,6,171}, Toffoli_{133,134,172}, \\ Toffoli_{7,8,173}, Toffoli_{7,136,174}, Toffoli_{135,8,175}, Toffoli_{135,136,176}, \\ Toffoli_{9,10,177}, Toffoli_{9,138,178}, Toffoli_{137,10,179}, Toffoli_{137,138,180}, \\ Toffoli_{11,12,181}, Toffoli_{11,140,182}, Toffoli_{139,12,183}, Toffoli_{139,140,184}, \\ Toffoli_{13,14,185}, Toffoli_{13,142,186}, Toffoli_{141,14,187}, Toffoli_{141,142,188}, \\ Toffoli_{15,16,189}, Toffoli_{15,144,190}, Toffoli_{143,16,191}, Toffoli_{143,144,192}, \\ Toffoli_{17,18,193}, Toffoli_{17,146,194}, Toffoli_{145,18,195}, Toffoli_{145,146,196}, \\ Toffoli_{19,20,197}, Toffoli_{19,148,198}, Toffoli_{147,20,199}, Toffoli_{147,148,200}, \\ Toffoli_{21,22,201}, Toffoli_{21,150,202}, Toffoli_{149,22,203}, Toffoli_{149,150,204}, \\ Toffoli_{23,24,205}, Toffoli_{23,152,206}, Toffoli_{151,24,207}, Toffoli_{151,152,208}, \\ Toffoli_{25,26,209}, Toffoli_{25,154,210}, Toffoli_{153,26,211}, Toffoli_{153,154,212}, \\ Toffoli_{27,28,213}, Toffoli_{27,156,214}, Toffoli_{155,28,215}, Toffoli_{155,156,216}, \\ Toffoli_{29,30,217}, Toffoli_{29,158,218}, Toffoli_{157,30,219}, Toffoli_{157,158,220}, \\ Toffoli_{31,32,221}, Toffoli_{31,160,222}, Toffoli_{159,32,223}, Toffoli_{159,160,224} \end{array} \right\}$	
Step48: $\left\{ \begin{array}{l} NOT_{129}, NOT_{130}, NOT_{131}, NOT_{132}, NOT_{133}, NOT_{134}, NOT_{135}, NOT_{136}, \\ NOT_{137}, NOT_{138}, NOT_{139}, NOT_{140}, NOT_{141}, NOT_{142}, NOT_{143}, NOT_{144}, \\ NOT_{145}, NOT_{146}, NOT_{147}, NOT_{148}, NOT_{149}, NOT_{150}, NOT_{151}, NOT_{152}, \\ NOT_{153}, NOT_{154}, NOT_{155}, NOT_{156}, NOT_{157}, NOT_{158}, NOT_{159}, NOT_{160} \end{array} \right\}$	
Step49: $\left\{ \begin{array}{l} CNOT_{1,129}, CNOT_{2,130}, CNOT_{3,131}, CNOT_{4,132}, CNOT_{5,133}, CNOT_{6,134}, CNOT_{7,135}, CNOT_{8,136}, \\ CNOT_{9,137}, CNOT_{10,138}, CNOT_{11,139}, CNOT_{12,140}, CNOT_{13,141}, CNOT_{14,142}, CNOT_{15,143}, CNOT_{16,144}, \\ CNOT_{17,145}, CNOT_{18,146}, CNOT_{19,147}, CNOT_{20,148}, CNOT_{21,149}, CNOT_{22,150}, CNOT_{23,151}, CNOT_{24,152}, \\ CNOT_{25,153}, CNOT_{26,154}, CNOT_{27,155}, CNOT_{28,156}, CNOT_{29,157}, CNOT_{30,158}, CNOT_{31,159}, CNOT_{32,160} \end{array} \right\}$	
Step50: $\left\{ \begin{array}{l} CNOT_{65,1}, CNOT_{66,2}, CNOT_{67,3}, CNOT_{68,4}, CNOT_{69,5}, CNOT_{70,6}, CNOT_{71,7}, CNOT_{72,8}, \\ CNOT_{73,9}, CNOT_{74,10}, CNOT_{75,11}, CNOT_{76,12}, CNOT_{77,13}, CNOT_{78,14}, CNOT_{79,15}, CNOT_{80,16}, \\ CNOT_{81,17}, CNOT_{82,18}, CNOT_{83,19}, CNOT_{84,20}, CNOT_{85,21}, CNOT_{86,22}, CNOT_{87,23}, CNOT_{88,24}, \\ CNOT_{89,25}, CNOT_{90,26}, CNOT_{91,27}, CNOT_{92,28}, CNOT_{93,29}, CNOT_{94,30}, CNOT_{95,31}, CNOT_{96,32} \end{array} \right\}$	
Step51: $\left\{ \begin{array}{l} CNOT_{33,1}, CNOT_{34,2}, CNOT_{35,3}, CNOT_{36,4}, CNOT_{37,5}, CNOT_{38,6}, CNOT_{39,7}, CNOT_{40,8}, \\ CNOT_{41,9}, CNOT_{42,10}, CNOT_{43,11}, CNOT_{44,12}, CNOT_{45,13}, CNOT_{46,14}, CNOT_{47,15}, CNOT_{48,16}, \\ CNOT_{49,17}, CNOT_{50,18}, CNOT_{51,19}, CNOT_{52,20}, CNOT_{53,21}, CNOT_{54,22}, CNOT_{55,23}, CNOT_{56,24}, \\ CNOT_{57,25}, CNOT_{58,26}, CNOT_{59,27}, CNOT_{60,28}, CNOT_{61,29}, CNOT_{62,30}, CNOT_{63,31}, CNOT_{64,32} \end{array} \right\}$	
电路具体实现方案	
介绍:	
电路需要将每一步可以同时执行的门详细表示出来, 用 #Full depth i and Toffoli depth j 标记当前步数的深度。	
每个门需要用下述这类代数表达式来描述:	
NOT 门: $y1 = y1+1$; CNOT 门: $y2 = y2 + y1$; Toffoli 门: $y3 = y3 + y1*y2$ 。	
需要实现的函数为:	
f1=x33	
f2=x34	
f3=x35	
f4=x36	
f5=x37	
f6=x38	
f7=x39	
f8=x40	

$f_9=x_{41}$
 $f_{10}=x_{42}$
 $f_{11}=x_{43}$
 $f_{12}=x_{44}$
 $f_{13}=x_{45}$
 $f_{14}=x_{46}$
 $f_{15}=x_{47}$
 $f_{16}=x_{48}$
 $f_{17}=x_{49}$
 $f_{18}=x_{50}$
 $f_{19}=x_{51}$
 $f_{20}=x_{52}$
 $f_{21}=x_{53}$
 $f_{22}=x_{54}$
 $f_{23}=x_{55}$
 $f_{24}=x_{56}$
 $f_{25}=x_{57}$
 $f_{26}=x_{58}$
 $f_{27}=x_{59}$
 $f_{28}=x_{60}$
 $f_{29}=x_{61}$
 $f_{30}=x_{62}$
 $f_{31}=x_{63}$
 $f_{32}=x_{64}$
 $f_{33}=x_{65}$
 $f_{34}=x_{66}$
 $f_{35}=x_{67}$
 $f_{36}=x_{68}$
 $f_{37}=x_{69}$
 $f_{38}=x_{70}$
 $f_{39}=x_{71}$
 $f_{40}=x_{72}$
 $f_{41}=x_{73}$
 $f_{42}=x_{74}$
 $f_{43}=x_{75}$
 $f_{44}=x_{76}$
 $f_{45}=x_{77}$
 $f_{46}=x_{78}$
 $f_{47}=x_{79}$
 $f_{48}=x_{80}$
 $f_{49}=x_{81}$
 $f_{50}=x_{82}$
 $f_{51}=x_{83}$
 $f_{52}=x_{84}$
 $f_{53}=x_{85}$
 $f_{54}=x_{86}$
 $f_{55}=x_{87}$
 $f_{56}=x_{88}$
 $f_{57}=x_{89}$
 $f_{58}=x_{90}$
 $f_{59}=x_{91}$
 $f_{60}=x_{92}$
 $f_{61}=x_{93}$
 $f_{62}=x_{94}$
 $f_{63}=x_{95}$
 $f_{64}=x_{96}$
 $f_{65}=x_{97}$
 $f_{66}=x_{98}$
 $f_{67}=x_{99}$
 $f_{68}=x_{100}$

$f_{69}=x_{101}$
 $f_{70}=x_{102}$
 $f_{71}=x_{103}$
 $f_{72}=x_{104}$
 $f_{73}=x_{105}$
 $f_{74}=x_{106}$
 $f_{75}=x_{107}$
 $f_{76}=x_{108}$
 $f_{77}=x_{109}$
 $f_{78}=x_{110}$
 $f_{79}=x_{111}$
 $f_{80}=x_{112}$
 $f_{81}=x_{113}$
 $f_{82}=x_{114}$
 $f_{83}=x_{115}$
 $f_{84}=x_{116}$
 $f_{85}=x_{117}$
 $f_{86}=x_{118}$
 $f_{87}=x_{119}$
 $f_{88}=x_{120}$
 $f_{89}=x_{121}$
 $f_{90}=x_{122}$
 $f_{91}=x_{123}$
 $f_{92}=x_{124}$
 $f_{93}=x_{125}$
 $f_{94}=x_{126}$
 $f_{95}=x_{127}$
 $f_{96}=x_{128}$
 $f_{97}=x_1+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out1
 $f_{98}=x_2+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out2
 $f_{99}=x_3+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out3
 $f_{100}=x_4+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out4
 $f_{101}=x_5+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out5
 $f_{102}=x_6+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out6
 $f_{103}=x_7+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out7
 $f_{104}=x_8+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out8
 $f_{105}=x_9+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out9
 $f_{106}=x_{10}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out10
 $f_{107}=x_{11}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out11
 $f_{108}=x_{12}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out12
 $f_{109}=x_{13}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out13
 $f_{110}=x_{14}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out14
 $f_{111}=x_{15}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out15
 $f_{112}=x_{16}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out16
 $f_{113}=x_{17}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out17
 $f_{114}=x_{18}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out18
 $f_{115}=x_{19}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out19
 $f_{116}=x_{20}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out20
 $f_{117}=x_{21}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out21
 $f_{118}=x_{22}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out22
 $f_{119}=x_{23}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out23
 $f_{120}=x_{24}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out24
 $f_{121}=x_{25}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out25
 $f_{122}=x_{26}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out26
 $f_{123}=x_{27}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out27
 $f_{124}=x_{28}+\dots$ 详见赛题二附录 SM4_ANF 中 SM4out28

fl25=x29+.....详见赛题二附录 SM4_ANF 中 SM4out29

fl26=x30+.....详见赛题二附录 SM4_ANF 中 SM4out30

fl27=x31+.....详见赛题二附录 SM4_ANF 中 SM4out31

fl28=x32+.....详见赛题二附录 SM4_ANF 中 SM4out32

这里我们利用了 1216 个辅助比特，并且最后列出了如何进行换位操作：

Initialization

y1=x33;y2=x34;y3=x35;y4=x36;y5=x37;y6=x38;y7=x39;y8=x40;y9=x41;y10=x42;y11=x43;y12=x44
;y13=x45;y14=x46;y15=x47;y16=x48;y17=x49;y18=x50;y19=x51;y20=x52;y21=x53;y22=x54;y23=x
55;y24=x56;y25=x57;y26=x58;y27=x59;y28=x60;y29=x61;y30=x62;y31=x63;y32=x64;y33=x65;y34
=x66;y35=x67;y36=x68;y37=x69;y38=x70;y39=x71;y40=x72;y41=x73;y42=x74;y43=x75;y44=x76;y
45=x77;y46=x78;y47=x79;y48=x80;y49=x81;y50=x82;y51=x83;y52=x84;y53=x85;y54=x86;y55=x87
;y56=x88;y57=x89;y58=x90;y59=x91;y60=x92;y61=x93;y62=x94;y63=x95;y64=x96;y65=x97;y66=x
98;y67=x99;y68=x100;y69=x101;y70=x102;y71=x103;y72=x104;y73=x105;y74=x106;y75=x107;y76
=x108;y77=x109;y78=x110;y79=x111;y80=x112;y81=x113;y82=x114;y83=x115;y84=x116;y85=x11
7;y86=x118;y87=x119;y88=x120;y89=x121;y90=x122;y91=x123;y92=x124;y93=x125;y94=x126;y95
=x127;y96=x128;y97=x1;y98=x2;y99=x3;y100=x4;y101=x5;y102=x6;y103=x7;y104=x8;y105=x9;y1
06=x10;y107=x11;y108=x12;y109=x13;y110=x14;y111=x15;y112=x16;y113=x17;y114=x18;y115=x
19;y116=x20;y117=x21;y118=x22;y119=x23;y120=x24;y121=x25;y122=x26;y123=x27;y124=x28;y1
25=x29;y126=x30;y127=x31;y128=x32;y129=0;y130=0;y131=0;y132=0;y133=0;y134=0;y135=0;y13
6=0;y137=0;y138=0;y139=0;y140=0;y141=0;y142=0;y143=0;y144=0;y145=0;y146=0;y147=0;y148=
0;y149=0;y150=0;y151=0;y152=0;y153=0;y154=0;y155=0;y156=0;y157=0;y158=0;y159=0;y160=0;y
161=0;y162=0;y163=0;y164=0;y165=0;y166=0;y167=0;y168=0;y169=0;y170=0;y171=0;y172=0;y17
3=0;y174=0;y175=0;y176=0;y177=0;y178=0;y179=0;y180=0;y181=0;y182=0;y183=0;y184=0;y185=
0;y186=0;y187=0;y188=0;y189=0;y190=0;y191=0;y192=0;y193=0;y194=0;y195=0;y196=0;y197=0;y
198=0;y199=0;y200=0;y201=0;y202=0;y203=0;y204=0;y205=0;y206=0;y207=0;y208=0;y209=0;y21
0=0;y211=0;y212=0;y213=0;y214=0;y215=0;y216=0;y217=0;y218=0;y219=0;y220=0;y221=0;y222=
0;y223=0;y224=0;y225=0;y226=0;y227=0;y228=0;y229=0;y230=0;y231=0;y232=0;y233=0;y234=0;y
235=0;y236=0;y237=0;y238=0;y239=0;y240=0;y241=0;y242=0;y243=0;y244=0;y245=0;y246=0;y24
7=0;y248=0;y249=0;y250=0;y251=0;y252=0;y253=0;y254=0;y255=0;y256=0;y257=0;y258=0;y259=
0;y260=0;y261=0;y262=0;y263=0;y264=0;y265=0;y266=0;y267=0;y268=0;y269=0;y270=0;y271=0;y
272=0;y273=0;y274=0;y275=0;y276=0;y277=0;y278=0;y279=0;y280=0;y281=0;y282=0;y283=0;y28
4=0;y285=0;y286=0;y287=0;y288=0;y289=0;y290=0;y291=0;y292=0;y293=0;y294=0;y295=0;y296=
0;y297=0;y298=0;y299=0;y300=0;y301=0;y302=0;y303=0;y304=0;y305=0;y306=0;y307=0;y308=0;y
309=0;y310=0;y311=0;y312=0;y313=0;y314=0;y315=0;y316=0;y317=0;y318=0;y319=0;y320=0;y32
1=0;y322=0;y323=0;y324=0;y325=0;y326=0;y327=0;y328=0;y329=0;y330=0;y331=0;y332=0;y333=
0;y334=0;y335=0;y336=0;y337=0;y338=0;y339=0;y340=0;y341=0;y342=0;y343=0;y344=0;y345=0;y
346=0;y347=0;y348=0;y349=0;y350=0;y351=0;y352=0;y353=0;y354=0;y355=0;y356=0;y357=0;y35
8=0;y359=0;y360=0;y361=0;y362=0;y363=0;y364=0;y365=0;y366=0;y367=0;y368=0;y369=0;y370=
0;y371=0;y372=0;y373=0;y374=0;y375=0;y376=0;y377=0;y378=0;y379=0;y380=0;y381=0;y382=0;y
383=0;y384=0;y385=0;y386=0;y387=0;y388=0;y389=0;y390=0;y391=0;y392=0;y393=0;y394=0;y39
5=0;y396=0;y397=0;y398=0;y399=0;y400=0;y401=0;y402=0;y403=0;y404=0;y405=0;y406=0;y407=
0;y408=0;y409=0;y410=0;y411=0;y412=0;y413=0;y414=0;y415=0;y416=0;y417=0;y418=0;y419=0;y
420=0;y421=0;y422=0;y423=0;y424=0;y425=0;y426=0;y427=0;y428=0;y429=0;y430=0;y431=0;y43
2=0;y433=0;y434=0;y435=0;y436=0;y437=0;y438=0;y439=0;y440=0;y441=0;y442=0;y443=0;y444=
0;y445=0;y446=0;y447=0;y448=0;y449=0;y450=0;y451=0;y452=0;y453=0;y454=0;y455=0;y456=0;y
457=0;y458=0;y459=0;y460=0;y461=0;y462=0;y463=0;y464=0;y465=0;y466=0;y467=0;y468=0;y46
9=0;y470=0;y471=0;y472=0;y473=0;y474=0;y475=0;y476=0;y477=0;y478=0;y479=0;y480=0;y481=
0;y482=0;y483=0;y484=0;y485=0;y486=0;y487=0;y488=0;y489=0;y490=0;y491=0;y492=0;y493=0;y
494=0;y495=0;y496=0;y497=0;y498=0;y499=0;y500=0;y501=0;y502=0;y503=0;y504=0;y505=0;y50
6=0;y507=0;y508=0;y509=0;y510=0;y511=0;y512=0;y513=0;y514=0;y515=0;y516=0;y517=0;y518=
0;y519=0;y520=0;y521=0;y522=0;y523=0;y524=0;y525=0;y526=0;y527=0;y528=0;y529=0;y530=0;y
531=0;y532=0;y533=0;y534=0;y535=0;y536=0;y537=0;y538=0;y539=0;y540=0;y541=0;y542=0;y54
3=0;y544=0;y545=0;y546=0;y547=0;y548=0;y549=0;y550=0;y551=0;y552=0;y553=0;y554=0;y555=
0;y556=0;y557=0;y558=0;y559=0;y560=0;y561=0;y562=0;y563=0;y564=0;y565=0;y566=0;y567=0;y
568=0;y569=0;y570=0;y571=0;y572=0;y573=0;y574=0;y575=0;y576=0;y577=0;y578=0;y579=0;y58
0=0;y581=0;y582=0;y583=0;y584=0;y585=0;y586=0;y587=0;y588=0;y589=0;y590=0;y591=0;y592=
0;y593=0;y594=0;y595=0;y596=0;y597=0;y598=0;y599=0;y600=0;y601=0;y602=0;y603=0;y604=0;y
605=0;y606=0;y607=0;y608=0;y609=0;y610=0;y611=0;y612=0;y613=0;y614=0;y615=0;y616=0;y61
7=0;y618=0;y619=0;y620=0;y621=0;y622=0;y623=0;y624=0;y625=0;y626=0;y627=0;y628=0;y629=

322=0;y1323=0;y1324=0;y1325=0;y1326=0;y1327=0;y1328=0;y1329=0;y1330=0;y1331=0;y1332=0;
y1333=0;y1334=0;y1335=0;y1336=0;y1337=0;y1338=0;y1339=0;y1340=0;y1341=0;y1342=0;y1343=
0;y1344=0;

#Full depth 1 and Toffoli depth 0

y1=y1+y33;y2=y2+y34;y3=y3+y35;y4=y4+y36;y5=y5+y37;y6=y6+y38;y7=y7+y39;y8=y8+y40;y9=y
9+y41;y10=y10+y42;y11=y11+y43;y12=y12+y44;y13=y13+y45;y14=y14+y46;y15=y15+y47;y16=y1
6+y48;y17=y17+y49;y18=y18+y50;y19=y19+y51;y20=y20+y52;y21=y21+y53;y22=y22+y54;y23=y2
3+y55;y24=y24+y56;y25=y25+y57;y26=y26+y58;y27=y27+y59;y28=y28+y60;y29=y29+y61;y30=y3
0+y62;y31=y31+y63;y32=y32+y64;

#Full depth 2 and Toffoli depth 0

y1=y1+y65;y2=y2+y66;y3=y3+y67;y4=y4+y68;y5=y5+y69;y6=y6+y70;y7=y7+y71;y8=y8+y72;y9=y
9+y73;y10=y10+y74;y11=y11+y75;y12=y12+y76;y13=y13+y77;y14=y14+y78;y15=y15+y79;y16=y1
6+y80;y17=y17+y81;y18=y18+y82;y19=y19+y83;y20=y20+y84;y21=y21+y85;y22=y22+y86;y23=y2
3+y87;y24=y24+y88;y25=y25+y89;y26=y26+y90;y27=y27+y91;y28=y28+y92;y29=y29+y93;y30=y3
0+y94;y31=y31+y95;y32=y32+y96;

#Full depth 3 and Toffoli depth 0

y129=y129+y1;y130=y130+y2;y131=y131+y3;y132=y132+y4;y133=y133+y5;y134=y134+y6;y135=y
135+y7;y136=y136+y8;y137=y137+y9;y138=y138+y10;y139=y139+y11;y140=y140+y12;y141=y141
+y13;y142=y142+y14;y143=y143+y15;y144=y144+y16;y145=y145+y17;y146=y146+y18;y147=y147
+y19;y148=y148+y20;y149=y149+y21;y150=y150+y22;y151=y151+y23;y152=y152+y24;y153=y153
+y25;y154=y154+y26;y155=y155+y27;y156=y156+y28;y157=y157+y29;y158=y158+y30;y159=y15
9+y31;y160=y160+y32;

#Full depth 4 and Toffoli depth 0

y129=y129+1;y130=y130+1;y131=y131+1;y132=y132+1;y133=y133+1;y134=y134+1;y135=y135+1;
y136=y136+1;y137=y137+1;y138=y138+1;y139=y139+1;y140=y140+1;y141=y141+1;y142=y142+1;
y143=y143+1;y144=y144+1;y145=y145+1;y146=y146+1;y147=y147+1;y148=y148+1;y149=y149+1;
y150=y150+1;y151=y151+1;y152=y152+1;y153=y153+1;y154=y154+1;y155=y155+1;y156=y156+1;
y157=y157+1;y158=y158+1;y159=y159+1;y160=y160+1;

#Full depth 5 and Toffoli depth 1

y161=y161+y1*y2;y162=y162+y1*y130;y163=y163+y129*y2;y164=y164+y129*y130;y165=y165+y
3*y4;y166=y166+y3*y132;y167=y167+y131*y4;y168=y168+y131*y132;y169=y169+y5*y6;y170=y1
70+y5*y134;y171=y171+y133*y6;y172=y172+y133*y134;y173=y173+y7*y8;y174=y174+y7*y136;y
175=y175+y135*y8;y176=y176+y135*y136;y177=y177+y9*y10;y178=y178+y9*y138;y179=y179+y
137*y10;y180=y180+y137*y138;y181=y181+y11*y12;y182=y182+y11*y140;y183=y183+y139*y12;
y184=y184+y139*y140;y185=y185+y13*y14;y186=y186+y13*y142;y187=y187+y141*y14;y188=y1
88+y141*y142;y189=y189+y15*y16;y190=y190+y15*y144;y191=y191+y143*y16;y192=y192+y143
*y144;y193=y193+y17*y18;y194=y194+y17*y146;y195=y195+y145*y18;y196=y196+y145*y146;y1
97=y197+y19*y20;y198=y198+y19*y148;y199=y199+y147*y20;y200=y200+y147*y148;y201=y201
+y21*y22;y202=y202+y21*y150;y203=y203+y149*y22;y204=y204+y149*y150;y205=y205+y23*y2
4;y206=y206+y23*y152;y207=y207+y151*y24;y208=y208+y151*y152;y209=y209+y25*y26;y210=y
210+y25*y154;y211=y211+y153*y26;y212=y212+y153*y154;y213=y213+y27*y28;y214=y214+y27
*y156;y215=y215+y155*y28;y216=y216+y155*y156;y217=y217+y29*y30;y218=y218+y29*y158;y2
19=y219+y157*y30;y220=y220+y157*y158;y221=y221+y31*y32;y222=y222+y31*y160;y223=y223
+y159*y32;y224=y224+y159*y160;

#Full depth 6 and Toffoli depth 2

y225=y225+y161*y165;y226=y226+y161*y166;y227=y227+y161*y167;y228=y228+y161*y168;y22
9=y229+y162*y165;y230=y230+y162*y166;y231=y231+y162*y167;y232=y232+y162*y168;y233=y
233+y163*y165;y234=y234+y163*y166;y235=y235+y163*y167;y236=y236+y163*y168;y237=y237
+y164*y165;y238=y238+y164*y166;y239=y239+y164*y167;y240=y240+y164*y168;y241=y241+y1
69*y173;y242=y242+y169*y174;y243=y243+y169*y175;y244=y244+y169*y176;y245=y245+y170*
y173;y246=y246+y170*y174;y247=y247+y170*y175;y248=y248+y170*y176;y249=y249+y171*y17
3;y250=y250+y171*y174;y251=y251+y171*y175;y252=y252+y171*y176;y253=y253+y172*y173;y2
54=y254+y172*y174;y255=y255+y172*y175;y256=y256+y172*y176;y257=y257+y177*y181;y258=
y258+y177*y182;y259=y259+y177*y183;y260=y260+y177*y184;y261=y261+y178*y181;y262=y26

$2+y178*y182; y263=y263+y178*y183; y264=y264+y178*y184; y265=y265+y179*y181; y266=y266+y179*y182; y267=y267+y179*y183; y268=y268+y179*y184; y269=y269+y180*y181; y270=y270+y180*y182; y271=y271+y180*y183; y272=y272+y180*y184; y273=y273+y185*y189; y274=y274+y185*y190; y275=y275+y185*y191; y276=y276+y185*y192; y277=y277+y186*y189; y278=y278+y186*y190; y279=y279+y186*y191; y280=y280+y186*y192; y281=y281+y187*y189; y282=y282+y187*y190; y283=y283+y187*y191; y284=y284+y187*y192; y285=y285+y188*y189; y286=y286+y188*y190; y287=y287+y188*y191; y288=y288+y188*y192; y289=y289+y193*y197; y290=y290+y193*y198; y291=y291+y193*y199; y292=y292+y193*y200; y293=y293+y194*y197; y294=y294+y194*y198; y295=y295+y194*y199; y296=y296+y194*y200; y297=y297+y195*y197; y298=y298+y195*y198; y299=y299+y195*y199; y300=y300+y195*y200; y301=y301+y196*y197; y302=y302+y196*y198; y303=y303+y196*y199; y304=y304+y196*y200; y305=y305+y201*y205; y306=y306+y201*y206; y307=y307+y201*y207; y308=y308+y201*y208; y309=y309+y202*y205; y310=y310+y202*y206; y311=y311+y202*y207; y312=y312+y202*y208; y313=y313+y203*y205; y314=y314+y203*y206; y315=y315+y203*y207; y316=y316+y203*y208; y317=y317+y204*y205; y318=y318+y204*y206; y319=y319+y204*y207; y320=y320+y204*y208; y321=y321+y209*y213; y322=y322+y209*y214; y323=y323+y209*y215; y324=y324+y209*y216; y325=y325+y210*y213; y326=y326+y210*y214; y327=y327+y210*y215; y328=y328+y210*y216; y329=y329+y211*y213; y330=y330+y211*y214; y331=y331+y211*y215; y332=y332+y211*y216; y333=y333+y212*y213; y334=y334+y212*y214; y335=y335+y212*y215; y336=y336+y212*y216; y337=y337+y217*y221; y338=y338+y217*y222; y339=y339+y217*y223; y340=y340+y217*y224; y341=y341+y218*y221; y342=y342+y218*y222; y343=y343+y218*y223; y344=y344+y218*y224; y345=y345+y219*y221; y346=y346+y219*y222; y347=y347+y219*y223; y348=y348+y219*y224; y349=y349+y220*y221; y350=y350+y220*y222; y351=y351+y220*y223; y352=y352+y220*y224; y129=y129+1; y130=y130+1; y131=y131+1; y132=y132+1; y133=y133+1; y134=y134+1; y135=y135+1; y136=y136+1; y137=y137+1; y138=y138+1; y139=y139+1; y140=y140+1; y141=y141+1; y142=y142+1; y143=y143+1; y144=y144+1; y145=y145+1; y146=y146+1; y147=y147+1; y148=y148+1; y149=y149+1; y150=y150+1; y151=y151+1; y152=y152+1; y153=y153+1; y154=y154+1; y155=y155+1; y156=y156+1; y157=y157+1; y158=y158+1; y159=y159+1; y160=y160+1;$

#Full depth 7 and Toffoli depth 2

$y353=y353+y249; y354=y354+y253; y355=y355+y244; y356=y356+y254; y357=y357+y255; y358=y358+y256; y359=y359+y252; y360=y360+y246; y361=y361+y255; y362=y362+y250; y363=y363+y247; y364=y364+y245; y365=y365+y251; y366=y366+y243; y367=y367+y256; y368=y368+y254; y369=y369+y252; y370=y370+y252; y371=y371+y246; y372=y372+y256; y373=y373+y256; y374=y374+y249; y375=y375+y243; y376=y376+y253; y377=y377+y253; y378=y378+y252; y379=y379+y256; y380=y380+y256; y381=y381+y255; y382=y382+y252; y383=y383+y252; y384=y384+y251; y385=y385+y242; y386=y386+y255; y387=y387+y254; y388=y388+y254; y389=y389+y253; y390=y390+y253; y391=y391+y251; y392=y392+y251; y393=y393+y251; y394=y394+y250; y395=y395+y248; y396=y396+y247; y397=y397+y256; y398=y398+y256; y399=y399+y255; y400=y400+y255; y401=y401+y255; y402=y402+y255; y403=y403+y255; y404=y404+y255; y405=y405+y254; y406=y406+y254; y407=y407+y253; y408=y408+y253; y409=y409+y252; y410=y410+y252; y411=y411+y251; y412=y412+y251; y413=y413+y250; y414=y414+y250; y415=y415+y250; y416=y416+y250; y417=y417+y249; y418=y418+y248; y419=y419+y248; y420=y420+y245; y421=y421+y255; y422=y422+y281; y423=y423+y285; y424=y424+y276; y425=y425+y286; y426=y426+y287; y427=y427+y288; y428=y428+y284; y429=y429+y278; y430=y430+y287; y431=y431+y282; y432=y432+y279; y433=y433+y277; y434=y434+y283; y435=y435+y275; y436=y436+y288; y437=y437+y286; y438=y438+y284; y439=y439+y284; y440=y440+y278; y441=y441+y288; y442=y442+y288; y443=y443+y281; y444=y444+y275; y445=y445+y285; y446=y446+y285; y447=y447+y284; y448=y448+y288; y449=y449+y288; y450=y450+y287; y451=y451+y284; y452=y452+y284; y453=y453+y283; y454=y454+y274; y455=y455+y287; y456=y456+y286; y457=y457+y286; y458=y458+y285; y459=y459+y285; y460=y460+y283; y461=y461+y283; y462=y462+y283; y463=y463+y282; y464=y464+y280; y465=y465+y279; y466=y466+y288; y467=y467+y288; y468=y468+y287; y469=y469+y287; y470=y470+y287; y471=y471+y287; y472=y472+y287; y473=y473+y287; y474=y474+y286; y475=y475+y286; y476=y476+y285; y477=y477+y285; y478=y478+y284; y479=y479+y284; y480=y480+y283; y481=y481+y283; y482=y482+y282; y483=y483+y282; y484=y484+y282; y485=y485+y282; y486=y486+y281; y487=y487+y280; y488=y488+y280; y489=y489+y277; y490=y490+y287; y491=y491+y313; y492=y492+y317; y493=y493+y308; y494=y494+y318; y495=y495+y319; y496=y496+y320; y497=y497+y316; y498=y498+y310; y499=y499+y319; y500=y500+y314; y501=y501+y311; y502=y502+y309; y503=y503+y315; y504=y504+y307; y505=y505+y320; y506=y506+y318; y507=y507+y316; y508=y508+y316; y509=y509+y310; y510=y510+y320; y511=y511+y320; y512=y512+y313; y513=y513+y307; y514=y514+y317; y515=y515+y317; y516=y516+y316; y517=y517+y320; y518=y518+y320; y519=y519+y319; y520=y520+y316; y521=y521+y316; y522=y522+y315; y523=y523+y306; y524=y524+y319; y525=y525+$

$y_{318}=y_{526}+y_{318}; y_{527}=y_{527}+y_{317}; y_{528}=y_{528}+y_{317}; y_{529}=y_{529}+y_{315}; y_{530}=y_{530}+y_{315}; y_{531}=y_{531}+y_{315}; y_{532}=y_{532}+y_{314}; y_{533}=y_{533}+y_{312}; y_{534}=y_{534}+y_{311}; y_{535}=y_{535}+y_{320}; y_{536}=y_{536}+y_{320}; y_{537}=y_{537}+y_{319}; y_{538}=y_{538}+y_{319}; y_{539}=y_{539}+y_{319}; y_{540}=y_{540}+y_{319}; y_{541}=y_{541}+y_{319}; y_{542}=y_{542}+y_{319}; y_{543}=y_{543}+y_{318}; y_{544}=y_{544}+y_{318}; y_{545}=y_{545}+y_{317}; y_{546}=y_{546}+y_{317}; y_{547}=y_{547}+y_{316}; y_{548}=y_{548}+y_{316}; y_{549}=y_{549}+y_{315}; y_{550}=y_{550}+y_{315}; y_{551}=y_{551}+y_{314}; y_{552}=y_{552}+y_{314}; y_{553}=y_{553}+y_{314}; y_{554}=y_{554}+y_{314}; y_{555}=y_{555}+y_{313}; y_{556}=y_{556}+y_{312}; y_{557}=y_{557}+y_{312}; y_{558}=y_{558}+y_{309}; y_{559}=y_{559}+y_{319}; y_{560}=y_{560}+y_{345}; y_{561}=y_{561}+y_{349}; y_{562}=y_{562}+y_{340}; y_{563}=y_{563}+y_{350}; y_{564}=y_{564}+y_{351}; y_{565}=y_{565}+y_{352}; y_{566}=y_{566}+y_{348}; y_{567}=y_{567}+y_{342}; y_{568}=y_{568}+y_{351}; y_{569}=y_{569}+y_{346}; y_{570}=y_{570}+y_{343}; y_{571}=y_{571}+y_{341}; y_{572}=y_{572}+y_{347}; y_{573}=y_{573}+y_{339}; y_{574}=y_{574}+y_{352}; y_{575}=y_{575}+y_{350}; y_{576}=y_{576}+y_{348}; y_{577}=y_{577}+y_{348}; y_{578}=y_{578}+y_{342}; y_{579}=y_{579}+y_{352}; y_{580}=y_{580}+y_{352}; y_{581}=y_{581}+y_{345}; y_{582}=y_{582}+y_{339}; y_{583}=y_{583}+y_{349}; y_{584}=y_{584}+y_{349}; y_{585}=y_{585}+y_{348}; y_{586}=y_{586}+y_{352}; y_{587}=y_{587}+y_{352}; y_{588}=y_{588}+y_{351}; y_{589}=y_{589}+y_{348}; y_{590}=y_{590}+y_{348}; y_{591}=y_{591}+y_{347}; y_{592}=y_{592}+y_{338}; y_{593}=y_{593}+y_{351}; y_{594}=y_{594}+y_{350}; y_{595}=y_{595}+y_{350}; y_{596}=y_{596}+y_{349}; y_{597}=y_{597}+y_{349}; y_{598}=y_{598}+y_{347}; y_{599}=y_{599}+y_{347}; y_{600}=y_{600}+y_{347}; y_{601}=y_{601}+y_{346}; y_{602}=y_{602}+y_{344}; y_{603}=y_{603}+y_{343}; y_{604}=y_{604}+y_{352}; y_{605}=y_{605}+y_{352}; y_{606}=y_{606}+y_{351}; y_{607}=y_{607}+y_{351}; y_{608}=y_{608}+y_{351}; y_{609}=y_{609}+y_{351}; y_{610}=y_{610}+y_{351}; y_{611}=y_{611}+y_{351}; y_{612}=y_{612}+y_{350}; y_{613}=y_{613}+y_{350}; y_{614}=y_{614}+y_{349}; y_{615}=y_{615}+y_{349}; y_{616}=y_{616}+y_{348}; y_{617}=y_{617}+y_{348}; y_{618}=y_{618}+y_{347}; y_{619}=y_{619}+y_{347}; y_{620}=y_{620}+y_{346}; y_{621}=y_{621}+y_{346}; y_{622}=y_{622}+y_{346}; y_{623}=y_{623}+y_{346}; y_{624}=y_{624}+y_{345}; y_{625}=y_{625}+y_{344}; y_{626}=y_{626}+y_{344}; y_{627}=y_{627}+y_{341}; y_{628}=y_{628}+y_{351}; y_{129}=y_{129}+y_{130}; y_{130}=y_{130}+y_{131}; y_{131}=y_{131}+y_{132}; y_{132}=y_{132}+y_{133}; y_{133}=y_{133}+y_{134}; y_{134}=y_{134}+y_{135}; y_{135}=y_{135}+y_{136}; y_{136}=y_{136}+y_{137}; y_{137}=y_{137}+y_{138}; y_{138}=y_{138}+y_{139}; y_{139}=y_{139}+y_{140}; y_{140}=y_{140}+y_{141}; y_{141}=y_{141}+y_{142}; y_{142}=y_{142}+y_{143}; y_{143}=y_{143}+y_{144}; y_{144}=y_{144}+y_{145}; y_{145}=y_{145}+y_{146}; y_{146}=y_{146}+y_{147}; y_{147}=y_{147}+y_{148}; y_{148}=y_{148}+y_{149}; y_{149}=y_{149}+y_{150}; y_{150}=y_{150}+y_{151}; y_{151}=y_{151}+y_{152}; y_{152}=y_{152}+y_{153}; y_{153}=y_{153}+y_{154}; y_{154}=y_{154}+y_{155}; y_{155}=y_{155}+y_{156}; y_{156}=y_{156}+y_{157}; y_{157}=y_{157}+y_{158}; y_{158}=y_{158}+y_{159}; y_{159}=y_{159}+y_{160}; y_{160}=y_{160}+y_{32};$

#Full depth 8 and Toffoli depth 2

$y_{353}=y_{353}+y_{247}; y_{354}=y_{354}+y_{251}; y_{355}=y_{355}+y_{242}; y_{356}=y_{356}+y_{250}; y_{357}=y_{357}+y_{243}; y_{358}=y_{358}+y_{245}; y_{359}=y_{359}+y_{248}; y_{360}=y_{360}+y_{241}; y_{361}=y_{361}+y_{254}; y_{362}=y_{362}+y_{246}; y_{363}=y_{363}+y_{244}; y_{364}=y_{364}+y_{242}; y_{365}=y_{365}+y_{249}; y_{366}=y_{366}+y_{241}; y_{367}=y_{367}+y_{241}; y_{368}=y_{368}+y_{244}; y_{369}=y_{369}+y_{247}; y_{370}=y_{370}+y_{243}; y_{371}=y_{371}+y_{245}; y_{372}=y_{372}+y_{253}; y_{373}=y_{373}+y_{244}; y_{374}=y_{374}+y_{248}; y_{375}=y_{375}+y_{242}; y_{376}=y_{376}+y_{252}; y_{377}=y_{377}+y_{250}; y_{378}=y_{378}+y_{251}; y_{379}=y_{379}+y_{255}; y_{380}=y_{380}+y_{243}; y_{381}=y_{381}+y_{248}; y_{382}=y_{382}+y_{249}; y_{383}=y_{383}+y_{245}; y_{384}=y_{384}+y_{247}; y_{385}=y_{385}+y_{241}; y_{386}=y_{386}+y_{253}; y_{387}=y_{387}+y_{246}; y_{388}=y_{388}+y_{241}; y_{389}=y_{389}+y_{248}; y_{390}=y_{390}+y_{247}; y_{391}=y_{391}+y_{250}; y_{392}=y_{392}+y_{246}; y_{393}=y_{393}+y_{244}; y_{394}=y_{394}+y_{245}; y_{395}=y_{395}+y_{246}; y_{396}=y_{396}+y_{241}; y_{397}=y_{397}+y_{254}; y_{398}=y_{398}+y_{250}; y_{399}=y_{399}+y_{251}; y_{400}=y_{400}+y_{250}; y_{401}=y_{401}+y_{249}; y_{402}=y_{402}+y_{245}; y_{403}=y_{403}+y_{244}; y_{404}=y_{404}+y_{241}; y_{405}=y_{405}+y_{252}; y_{406}=y_{406}+y_{248}; y_{407}=y_{407}+y_{249}; y_{408}=y_{408}+y_{245}; y_{409}=y_{409}+y_{250}; y_{410}=y_{410}+y_{242}; y_{411}=y_{411}+y_{248}; y_{412}=y_{412}+y_{243}; y_{413}=y_{413}+y_{249}; y_{414}=y_{414}+y_{248}; y_{415}=y_{415}+y_{247}; y_{416}=y_{416}+y_{242}; y_{417}=y_{417}+y_{242}; y_{418}=y_{418}+y_{245}; y_{419}=y_{419}+y_{244}; y_{420}=y_{420}+y_{241}; y_{421}=y_{421}+y_{246}; y_{422}=y_{422}+y_{279}; y_{423}=y_{423}+y_{283}; y_{424}=y_{424}+y_{274}; y_{425}=y_{425}+y_{282}; y_{426}=y_{426}+y_{275}; y_{427}=y_{427}+y_{277}; y_{428}=y_{428}+y_{280}; y_{429}=y_{429}+y_{273}; y_{430}=y_{430}+y_{286}; y_{431}=y_{431}+y_{278}; y_{432}=y_{432}+y_{276}; y_{433}=y_{433}+y_{274}; y_{434}=y_{434}+y_{281}; y_{435}=y_{435}+y_{273}; y_{436}=y_{436}+y_{273}; y_{437}=y_{437}+y_{276}; y_{438}=y_{438}+y_{279}; y_{439}=y_{439}+y_{275}; y_{440}=y_{440}+y_{277}; y_{441}=y_{441}+y_{285}; y_{442}=y_{442}+y_{276}; y_{443}=y_{443}+y_{280}; y_{444}=y_{444}+y_{274}; y_{445}=y_{445}+y_{284}; y_{446}=y_{446}+y_{282}; y_{447}=y_{447}+y_{283}; y_{448}=y_{448}+y_{287}; y_{449}=y_{449}+y_{275}; y_{450}=y_{450}+y_{280}; y_{451}=y_{451}+y_{281}; y_{452}=y_{452}+y_{277}; y_{453}=y_{453}+y_{279}; y_{454}=y_{454}+y_{273}; y_{455}=y_{455}+y_{285}; y_{456}=y_{456}+y_{278}; y_{457}=y_{457}+y_{273}; y_{458}=y_{458}+y_{280}; y_{459}=y_{459}+y_{279}; y_{460}=y_{460}+y_{282}; y_{461}=y_{461}+y_{278}; y_{462}=y_{462}+y_{276}; y_{463}=y_{463}+y_{277}; y_{464}=y_{464}+y_{278}; y_{465}=y_{465}+y_{273}; y_{466}=y_{466}+y_{286}; y_{467}=y_{467}+y_{282}; y_{468}=y_{468}+y_{283}; y_{469}=y_{469}+y_{282}; y_{470}=y_{470}+y_{281}; y_{471}=y_{471}+y_{277}; y_{472}=y_{472}+y_{276}; y_{473}=y_{473}+y_{273}; y_{474}=y_{474}+y_{284}; y_{475}=y_{475}+y_{280}; y_{476}=y_{476}+y_{281}; y_{477}=y_{477}+y_{277}; y_{478}=y_{478}+y_{282}; y_{479}=y_{479}+y_{274}; y_{480}=y_{480}+y_{280}; y_{481}=y_{481}+y_{275}; y_{482}=y_{482}+y_{281}; y_{483}=y_{483}+y_{280}; y_{484}=y_{484}+y_{279}; y_{485}=y_{485}+y_{274}; y_{486}=y_{486}+y_{274}; y_{487}=y_{487}+y_{277}; y_{488}=y_{488}+y_{276}; y_{489}=y_{489}+y_{273}; y_{490}=y_{490}+y_{278}; y_{491}=y_{491}+y_{311}; y_{492}=y_{492}+y_{315}; y_{493}=y_{493}+y_{306}; y_{494}=y_{494}+y_{314}; y_{495}=y_{495}+y_{307}; y_{496}=y_{496}+y_{309}; y_{497}=y_{497}+y_{312}; y_{498}=y_{498}+y_{305}; y_{499}=y_{499}+y_{318}; y_{500}=y_{500}+y_{310}; y_{501}=y_{501}+y_{308}; y_{502}=y_{502}+y_{306}; y_{503}=y_{503}+y_{313}; y_{504}=y_{504}+y_{305}; y_{505}=y_{505}+y_{305}; y_{506}=y_{506}+y_{308}; y_{507}=y_{507}+y_{311}; y_{508}=y_{508}+y_{307}; y_{509}=y_{509}+y_{309}; y_{510}=y_{510}+y_{317}; y_{511}=y_{511}+y_{308}; y_{512}=y_{512}+y_{312}; y_{513}=y_{513}+y_{306}; y_{514}=y_{514}+y_{316}; y_{515}=y_{515}+y_{314}; y_{516}=y_{516}+y_{315}; y_{517}=y_{517}+y_{319}; y_{518}=y_{518}+y_{307}; y_{519}=y_{519}+y_{312}; y_{520}=y_{520}+y_{313}; y_{521}=y_{521}+y_{309}; y_{522}=y_{522}+y_{311}; y_{523}=y_{523}+y_{305}; y_{524}=y_{524}+y_{317}; y_{525}=y_{525}+y_{310}; y_{526}=y_{526}+y_{305}; y_{527}=y_{527}+y_{312}; y_{528}=y_{528}+y_{311}; y_{529}=y_{529}+y_{314}; y_{530}=y_{530}+y_{310}; y_{531}=y_{531}+y_{308}; y_{532}=y_{532}+y_{309}; y_{533}=y_{533}+y_{310}; y_{534}=y_{534}+y_{305}; y_{535}=y_{535}+y_{318}; y_{536}=y_{536}+y_{314}; y_{537}=y_{537}+y_{315}; y_{538}=y_{538}+y_{314}; y_{539}=y_{539}+y_{313}; y_{540}=y_{540}+y_{309}; y_{541}=y_{541}+y_{308}; y_{542}=y_{542}+y_{315};$

542+y305;y543=y543+y316;y544=y544+y312;y545=y545+y313;y546=y546+y309;y547=y547+y314;
y548=y548+y306;y549=y549+y312;y550=y550+y307;y551=y551+y313;y552=y552+y312;y553=y55
3+y311;y554=y554+y306;y555=y555+y306;y556=y556+y309;y557=y557+y308;y558=y558+y305;y5
59=y559+y310;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+
y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570
=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y3
40;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y
581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;
y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y59
2+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y5
98=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+
y337;y604=y604+y350;y605=y605+y346;y606=y606+y347;y607=y607+y346;y608=y608+y345;y609
=y609+y341;y610=y610+y340;y611=y611+y337;y612=y612+y348;y613=y613+y344;y614=y614+y3
45;y615=y615+y341;y616=y616+y346;y617=y617+y338;y618=y618+y344;y619=y619+y339;y620=y
620+y345;y621=y621+y344;y622=y622+y343;y623=y623+y338;y624=y624+y338;y625=y625+y341;
y626=y626+y340;y627=y627+y337;y628=y628+y342;

#Full depth 9 and Toffoli depth 2

y405=y405+y353;y406=y406+y365;y398=y398+y354;y629=y629+y353;y407=y407+y359;y630=y63
0+y242;y413=y413+y241;y631=y631+y355;y419=y419+y353;y632=y632+y355;y402=y402+y356;y6
33=y633+y369;y634=y634+y354;y635=y635+y354;y397=y397+y355;y636=y636+y354;y637=y637+
y254;y638=y638+y249;y420=y420+y353;y409=y409+y358;y639=y639+y256;y640=y640+y357;y410
=y410+y354;y641=y641+y254;y642=y642+y246;y643=y643+y249;y644=y644+y248;y645=y645+y3
54;y646=y646+y251;y647=y647+y357;y648=y648+y253;y649=y649+y248;y650=y650+y243;y651=y
651+y255;y652=y652+y241;y404=y404+y355;y653=y653+y354;y654=y654+y255;y655=y655+y246;
y656=y656+y363;y414=y414+y353;y657=y657+y354;y658=y658+y245;y412=y412+y353;y659=y65
9+y356;y660=y660+y353;y661=y661+y248;y662=y662+y253;y663=y663+y248;y664=y664+y254;y6
65=y665+y356;y666=y666+y353;y667=y667+y353;y668=y668+y355;y669=y669+y242;y670=y670+
y252;y671=y671+y353;y672=y672+y353;y673=y673+y256;y674=y674+y251;y675=y675+y241;y676
=y676+y245;y677=y677+y353;y678=y678+y355;y679=y679+y253;y680=y680+y251;y681=y681+y3
55;y682=y682+y243;y683=y683+y253;y684=y684+y242;y685=y685+y242;y686=y686+y359;y403=y
403+y356;y687=y687+y356;y688=y688+y355;y689=y689+y353;y690=y690+y353;y691=y691+y250;
y692=y692+y353;y693=y693+y247;y416=y416+y353;y415=y415+y354;y694=y694+y247;y695=y69
5+y254;y696=y696+y246;y697=y697+y251;y698=y698+y248;y699=y699+y356;y399=y399+y353;y7
00=y700+y354;y701=y701+y241;y702=y702+y353;y703=y703+y246;y704=y704+y251;y705=y705+
y355;y706=y706+y354;y707=y707+y243;y417=y417+y354;y418=y418+y357;y708=y708+y357;y709
=y709+y355;y408=y408+y353;y710=y710+y353;y711=y711+y353;y712=y712+y247;y400=y400+y3
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717+y363;y718=y718+y356;y401=y401+y360;y719=y719+y243;y720=y720+y353;y721=y721+y242;
y722=y722+y252;y723=y723+y244;y724=y724+y250;y411=y411+y241;y725=y725+y353;y726=y72
6+y243;y727=y727+y254;y728=y728+y353;y729=y729+y356;y730=y730+y246;y731=y731+y243;y4
74=y474+y242;y475=y475+y434;y467=y467+y423;y732=y732+y422;y476=y476+y428;y733=y733+
y274;y482=y482+y273;y734=y734+y424;y488=y488+y422;y735=y735+y424;y471=y471+y425;y736
=y736+y438;y737=y737+y423;y738=y738+y423;y466=y466+y424;y739=y739+y423;y740=y740+y2
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479+y423;y744=y744+y286;y745=y745+y278;y746=y746+y281;y747=y747+y280;y748=y748+y423;
y749=y749+y283;y750=y750+y426;y751=y751+y285;y752=y752+y280;y753=y753+y275;y754=y75
4+y287;y755=y755+y273;y473=y473+y424;y756=y756+y423;y757=y757+y287;y758=y758+y278;y7
59=y759+y432;y483=y483+y422;y760=y760+y423;y761=y761+y277;y481=y481+y422;y762=y762+
y425;y763=y763+y422;y764=y764+y280;y765=y765+y285;y766=y766+y280;y767=y767+y286;y768
=y768+y425;y769=y769+y422;y770=y770+y422;y771=y771+y424;y772=y772+y274;y773=y773+y2
84;y774=y774+y422;y775=y775+y422;y776=y776+y288;y777=y777+y283;y778=y778+y273;y779=y
779+y277;y780=y780+y422;y781=y781+y424;y782=y782+y285;y783=y783+y283;y784=y784+y424;
y785=y785+y275;y786=y786+y285;y787=y787+y274;y788=y788+y274;y789=y789+y428;y472=y47
2+y425;y790=y790+y425;y791=y791+y424;y792=y792+y422;y793=y793+y422;y794=y794+y282;y7
95=y795+y422;y796=y796+y279;y485=y485+y422;y484=y484+y423;y797=y797+y279;y798=y798+
y286;y799=y799+y278;y800=y800+y283;y801=y801+y280;y802=y802+y425;y468=y468+y422;y803
=y803+y423;y804=y804+y273;y805=y805+y422;y806=y806+y278;y807=y807+y283;y808=y808+y4
24;y809=y809+y423;y810=y810+y275;y486=y486+y423;y487=y487+y426;y811=y811+y426;y812=y
812+y424;y477=y477+y422;y813=y813+y422;y814=y814+y422;y815=y815+y279;y469=y469+y429;

$y816=y816+y284; y490=y490+y422; y817=y817+y281; y818=y818+y282; y819=y819+y281; y820=y820+y432; y821=y821+y425; y470=y470+y429; y822=y822+y275; y823=y823+y422; y824=y824+y274; y825=y825+y284; y826=y826+y276; y827=y827+y282; y480=y480+y273; y828=y828+y422; y829=y829+y275; y830=y830+y286; y831=y831+y422; y832=y832+y425; y833=y833+y278; y834=y834+y275; y543=y543+y491; y544=y544+y503; y536=y536+y492; y835=y835+y491; y545=y545+y497; y836=y836+y306; y551=y551+y305; y837=y837+y493; y557=y557+y491; y838=y838+y493; y540=y540+y494; y839=y839+y507; y840=y840+y492; y841=y841+y492; y535=y535+y493; y842=y842+y492; y843=y843+y318; y844=y844+y313; y558=y558+y491; y547=y547+y496; y845=y845+y320; y846=y846+y495; y548=y548+y492; y847=y847+y318; y848=y848+y310; y849=y849+y313; y850=y850+y312; y851=y851+y492; y852=y852+y315; y853=y853+y495; y854=y854+y317; y855=y855+y312; y856=y856+y307; y857=y857+y319; y858=y858+y305; y542=y542+y493; y859=y859+y492; y860=y860+y319; y861=y861+y310; y862=y862+y501; y552=y552+y491; y863=y863+y492; y864=y864+y309; y550=y550+y491; y865=y865+y494; y866=y866+y491; y867=y867+y312; y868=y868+y317; y869=y869+y312; y870=y870+y318; y871=y871+y494; y872=y872+y491; y873=y873+y491; y874=y874+y493; y875=y875+y306; y876=y876+y316; y877=y877+y491; y878=y878+y491; y879=y879+y320; y880=y880+y315; y881=y881+y305; y882=y882+y309; y883=y883+y491; y884=y884+y493; y885=y885+y317; y886=y886+y315; y887=y887+y493; y888=y888+y307; y889=y889+y317; y890=y890+y306; y891=y891+y306; y892=y892+y497; y541=y541+y494; y893=y893+y494; y894=y894+y493; y895=y895+y491; y896=y896+y491; y897=y897+y314; y898=y898+y491; y899=y899+y311; y554=y554+y491; y553=y553+y492; y900=y900+y311; y901=y901+y318; y902=y902+y310; y903=y903+y315; y904=y904+y312; y905=y905+y494; y537=y537+y491; y906=y906+y492; y907=y907+y305; y908=y908+y491; y909=y909+y310; y910=y910+y315; y911=y911+y493; y912=y912+y492; y913=y913+y307; y555=y555+y492; y556=y556+y495; y914=y914+y495; y915=y915+y493; y546=y546+y491; y916=y916+y491; y917=y917+y491; y918=y918+y311; y538=y538+y498; y919=y919+y316; y559=y559+y491; y920=y920+y313; y921=y921+y314; y922=y922+y313; y923=y923+y501; y924=y924+y494; y539=y539+y498; y925=y925+y307; y926=y926+y491; y927=y927+y306; y928=y928+y316; y929=y929+y308; y930=y930+y314; y549=y549+y305; y931=y931+y491; y932=y932+y307; y933=y933+y318; y934=y934+y491; y935=y935+y494; y936=y936+y310; y937=y937+y307; y612=y612+y560; y613=y613+y572; y605=y605+y561; y938=y938+y560; y614=y614+y566; y939=y939+y338; y620=y620+y337; y940=y940+y562; y626=y626+y560; y941=y941+y562; y609=y609+y563; y942=y942+y576; y943=y943+y561; y944=y944+y561; y604=y604+y562; y945=y945+y561; y946=y946+y350; y947=y947+y345; y627=y627+y560; y616=y616+y565; y948=y948+y352; y949=y949+y564; y617=y617+y561; y950=y950+y350; y951=y951+y342; y952=y952+y345; y953=y953+y344; y954=y954+y561; y955=y955+y347; y956=y956+y564; y957=y957+y349; y958=y958+y344; y959=y959+y339; y960=y960+y351; y961=y961+y337; y611=y611+y562; y962=y962+y561; y963=y963+y351; y964=y964+y342; y965=y965+y570; y621=y621+y560; y966=y966+y561; y967=y967+y341; y619=y619+y560; y968=y968+y563; y969=y969+y560; y970=y970+y344; y971=y971+y349; y972=y972+y344; y973=y973+y350; y974=y974+y563; y975=y975+y560; y976=y976+y560; y977=y977+y562; y978=y978+y338; y979=y979+y348; y980=y980+y560; y981=y981+y560; y982=y982+y352; y983=y983+y347; y984=y984+y337; y985=y985+y341; y986=y986+y560; y987=y987+y562; y988=y988+y349; y989=y989+y347; y990=y990+y562; y991=y991+y339; y992=y992+y349; y993=y993+y338; y994=y994+y338; y995=y995+y566; y610=y610+y563; y996=y996+y563; y997=y997+y562; y998=y998+y560; y999=y999+y560; y1000=y1000+y346; y1001=y1001+y560; y1002=y1002+y343; y623=y623+y560; y622=y622+y561; y1003=y1003+y343; y1004=y1004+y350; y1005=y1005+y342; y1006=y1006+y347; y1007=y1007+y344; y1008=y1008+y563; y606=y606+y560; y1009=y1009+y561; y1010=y1010+y337; y1011=y1011+y560; y1012=y1012+y342; y1013=y1013+y347; y1014=y1014+y562; y1015=y1015+y561; y1016=y1016+y339; y624=y624+y561; y625=y625+y564; y1017=y1017+y564; y1018=y1018+y562; y615=y615+y560; y1019=y1019+y560; y1020=y1020+y560; y1021=y1021+y343; y607=y607+y567; y1022=y1022+y348; y628=y628+y560; y1023=y1023+y345; y1024=y1024+y346; y1025=y1025+y345; y1026=y1026+y570; y1027=y1027+y563; y608=y608+y567; y1028=y1028+y339; y1029=y1029+y560; y1030=y1030+y338; y1031=y1031+y348; y1032=y1032+y340; y1033=y1033+y346; y618=y618+y337; y1034=y1034+y560; y1035=y1035+y339; y1036=y1036+y350; y1037=y1037+y560; y1038=y1038+y563; y1039=y1039+y342; y1040=y1040+y339;$

#Full depth 10 and Toffoli depth 2

$y405=y405+y354; y406=y406+y366; y398=y398+y355; y629=y629+y357; y407=y407+y366; y630=y630+y353; y631=y631+y358; y419=y419+y354; y632=y632+y357; y402=y402+y359; y633=y633+y379; y634=y634+y356; y635=y635+y355; y397=y397+y360; y636=y636+y357; y637=y637+y354; y638=y638+y363; y420=y420+y354; y409=y409+y363; y639=y639+y354; y640=y640+y358; y410=y410+y362; y641=y641+y354; y642=y642+y354; y643=y643+y367; y644=y644+y356; y645=y645+y355; y646=y646+y353; y647=y647+y362; y648=y648+y355; y649=y649+y354; y650=y650+y353; y651=y651+y354; y652=y652+y354; y404=y404+y358; y653=y653+y360; y654=y654+y354; y655=y655+y353; y656=y656+y394;$

$y414=y414+y357; y657=y657+y355; y658=y658+y353; y412=y412+y368; y659=y659+y382; y660=y660+y355; y661=y661+y353; y662=y662+y353; y663=y663+y353; y664=y664+y354; y665=y665+y369; y666=y666+y355; y667=y667+y354; y668=y668+y359; y669=y669+y354; y670=y670+y353; y671=y671+y355; y672=y672+y355; y673=y673+y357; y674=y674+y358; y675=y675+y353; y676=y676+y353; y677=y677+y356; y678=y678+y358; y679=y679+y355; y680=y680+y359; y681=y681+y362; y682=y682+y354; y683=y683+y353; y684=y684+y353; y685=y685+y354; y686=y686+y385; y403=y403+y358; y687=y687+y357; y688=y688+y358; y689=y689+y355; y690=y690+y355; y691=y691+y354; y692=y692+y357; y693=y693+y354; y416=y416+y359; y415=y415+y357; y694=y694+y354; y695=y695+y353; y696=y696+y355; y697=y697+y360; y698=y698+y360; y699=y699+y358; y399=y399+y364; y700=y700+y357; y701=y701+y354; y702=y702+y354; y703=y703+y356; y704=y704+y353; y705=y705+y356; y706=y706+y356; y707=y707+y353; y417=y417+y357; y418=y418+y360; y708=y708+y364; y709=y709+y356; y408=y408+y357; y710=y710+y358; y711=y711+y354; y712=y712+y354; y400=y400+y369; y713=y713+y354; y421=y421+y356; y714=y714+y355; y715=y715+y358; y716=y716+y354; y717=y717+y370; y718=y718+y365; y401=y401+y373; y719=y719+y359; y720=y720+y355; y721=y721+y356; y722=y722+y353; y723=y723+y359; y724=y724+y354; y411=y411+y353; y725=y725+y355; y726=y726+y358; y727=y727+y353; y728=y728+y358; y729=y729+y357; y730=y730+y356; y731=y731+y355; y474=y474+y423; y475=y475+y435; y467=y467+y424; y732=y732+y426; y476=y476+y435; y733=y733+y422; y734=y734+y427; y488=y488+y423; y735=y735+y426; y471=y471+y428; y736=y736+y448; y737=y737+y425; y738=y738+y424; y466=y466+y429; y739=y739+y426; y740=y740+y423; y741=y741+y432; y489=y489+y423; y478=y478+y432; y742=y742+y423; y743=y743+y427; y479=y479+y431; y744=y744+y423; y745=y745+y423; y746=y746+y436; y747=y747+y425; y748=y748+y424; y749=y749+y422; y750=y750+y431; y751=y751+y424; y752=y752+y423; y753=y753+y422; y754=y754+y423; y755=y755+y423; y473=y473+y427; y756=y756+y429; y757=y757+y423; y758=y758+y422; y759=y759+y463; y483=y483+y426; y760=y760+y424; y761=y761+y422; y481=y481+y437; y762=y762+y451; y763=y763+y424; y764=y764+y422; y765=y765+y422; y766=y766+y422; y767=y767+y423; y768=y768+y438; y769=y769+y424; y770=y770+y423; y771=y771+y428; y772=y772+y423; y773=y773+y422; y774=y774+y424; y775=y775+y424; y776=y776+y426; y777=y777+y427; y778=y778+y422; y779=y779+y422; y780=y780+y425; y781=y781+y427; y782=y782+y424; y783=y783+y428; y784=y784+y431; y785=y785+y423; y786=y786+y422; y787=y787+y422; y788=y788+y423; y789=y789+y454; y472=y472+y427; y790=y790+y426; y791=y791+y427; y792=y792+y424; y793=y793+y424; y794=y794+y423; y795=y795+y426; y796=y796+y423; y485=y485+y428; y484=y484+y426; y797=y797+y423; y798=y798+y422; y799=y799+y424; y800=y800+y429; y801=y801+y429; y802=y802+y427; y468=y468+y433; y803=y803+y426; y804=y804+y423; y805=y805+y423; y806=y806+y425; y807=y807+y422; y808=y808+y425; y809=y809+y425; y810=y810+y422; y486=y486+y426; y487=y487+y429; y811=y811+y433; y812=y812+y425; y477=y477+y426; y813=y813+y427; y814=y814+y423; y815=y815+y423; y469=y469+y438; y816=y816+y423; y490=y490+y425; y817=y817+y424; y818=y818+y427; y819=y819+y423; y820=y820+y439; y821=y821+y434; y470=y470+y442; y822=y822+y428; y823=y823+y424; y824=y824+y425; y825=y825+y422; y826=y826+y428; y827=y827+y423; y480=y480+y422; y828=y828+y424; y829=y829+y427; y830=y830+y422; y831=y831+y427; y832=y832+y426; y833=y833+y425; y834=y834+y424; y543=y543+y492; y544=y544+y504; y536=y536+y493; y835=y835+y495; y545=y545+y504; y836=y836+y491; y837=y837+y496; y557=y557+y492; y838=y838+y495; y540=y540+y497; y839=y839+y517; y840=y840+y494; y841=y841+y493; y535=y535+y498; y842=y842+y495; y843=y843+y492; y844=y844+y501; y558=y558+y492; y547=y547+y501; y845=y845+y492; y846=y846+y496; y548=y548+y500; y847=y847+y492; y848=y848+y492; y849=y849+y505; y850=y850+y494; y851=y851+y493; y852=y852+y491; y853=y853+y500; y854=y854+y493; y855=y855+y492; y856=y856+y491; y857=y857+y492; y858=y858+y492; y542=y542+y496; y859=y859+y498; y860=y860+y492; y861=y861+y491; y862=y862+y532; y552=y552+y495; y863=y863+y493; y864=y864+y491; y550=y550+y506; y865=y865+y520; y866=y866+y493; y867=y867+y491; y868=y868+y491; y869=y869+y491; y870=y870+y492; y871=y871+y507; y872=y872+y493; y873=y873+y492; y874=y874+y497; y875=y875+y492; y876=y876+y491; y877=y877+y493; y878=y878+y493; y879=y879+y495; y880=y880+y496; y881=y881+y491; y882=y882+y491; y883=y883+y494; y884=y884+y496; y885=y885+y493; y886=y886+y497; y887=y887+y500; y888=y888+y492; y889=y889+y491; y890=y890+y491; y891=y891+y492; y892=y892+y523; y541=y541+y496; y893=y893+y495; y894=y894+y496; y895=y895+y493; y896=y896+y493; y897=y897+y492; y898=y898+y495; y899=y899+y492; y554=y554+y497; y553=y553+y495; y900=y900+y492; y901=y901+y491; y902=y902+y493; y903=y903+y498; y904=y904+y498; y905=y905+y496; y537=y537+y502; y906=y906+y495; y907=y907+y492; y908=y908+y492; y909=y909+y494; y910=y910+y491; y911=y911+y494; y912=y912+y494; y913=y913+y491; y555=y555+y495; y556=y556+y498; y914=y914+y502; y915=y915+y494; y546=y546+y495; y916=y916+y496; y917=y917+y492; y918=y918+y492; y538=y538+y507; y919=y919+y492; y559=y559+y494; y920=y920+y493; y921=y921+y496; y922=y922+y492; y923=y923+y508; y924=y924+y503; y539=y539+y511; y925=y925+y497; y926=y926+y493; y927=y927+y494; y928=y928+y491; y929=y929+y497; y930=y930+y492; y549$

=y549+y491;y931=y931+y493;y932=y932+y496;y933=y933+y491;y934=y934+y496;y935=y935+y495;y936=y936+y494;y937=y937+y493;y612=y612+y561;y613=y613+y573;y605=y605+y562;y938=y938+y564;y614=y614+y573;y939=y939+y560;y940=y940+y565;y626=y626+y561;y941=y941+y564;y609=y609+y566;y942=y942+y586;y943=y943+y563;y944=y944+y562;y604=y604+y567;y945=y945+y564;y946=y946+y561;y947=y947+y570;y627=y627+y561;y616=y616+y570;y948=y948+y561;y949=y949+y565;y617=y617+y569;y950=y950+y561;y951=y951+y561;y952=y952+y574;y953=y953+y563;y954=y954+y562;y955=y955+y560;y956=y956+y569;y957=y957+y562;y958=y958+y561;y959=y959+y560;y960=y960+y561;y961=y961+y561;y611=y611+y565;y962=y962+y567;y963=y963+y561;y964=y964+y560;y965=y965+y601;y621=y621+y564;y966=y966+y562;y967=y967+y560;y619=y619+y575;y968=y968+y589;y969=y969+y562;y970=y970+y560;y971=y971+y560;y972=y972+y560;y973=y973+y561;y974=y974+y576;y975=y975+y562;y976=y976+y561;y977=y977+y566;y978=y978+y561;y979=y979+y560;y980=y980+y562;y981=y981+y562;y982=y982+y564;y983=y983+y565;y984=y984+y560;y985=y985+y560;y986=y986+y563;y987=y987+y565;y988=y988+y562;y989=y989+y566;y990=y990+y569;y991=y991+y561;y992=y992+y560;y993=y993+y560;y994=y994+y561;y995=y995+y592;y610=y610+y565;y996=y996+y564;y997=y997+y565;y998=y998+y562;y999=y999+y562;y1000=y1000+y561;y1001=y1001+y564;y1002=y1002+y561;y623=y623+y566;y622=y622+y564;y1003=y1003+y561;y1004=y1004+y560;y1005=y1005+y562;y1006=y1006+y567;y1007=y1007+y567;y1008=y1008+y565;y606=y606+y571;y1009=y1009+y564;y1010=y1010+y561;y1011=y1011+y561;y1012=y1012+y563;y1013=y1013+y560;y1014=y1014+y563;y1015=y1015+y563;y1016=y1016+y560;y624=y624+y564;y625=y625+y567;y1017=y1017+y571;y1018=y1018+y563;y615=y615+y564;y1019=y1019+y565;y1020=y1020+y561;y1021=y1021+y561;y607=y607+y576;y1022=y1022+y561;y628=y628+y563;y1023=y1023+y562;y1024=y1024+y565;y1025=y1025+y561;y1026=y1026+y577;y1027=y1027+y572;y608=y608+y580;y1028=y1028+y566;y1029=y1029+y562;y1030=y1030+y563;y1031=y1031+y560;y1032=y1032+y566;y1033=y1033+y561;y618=y618+y560;y1034=y1034+y562;y1035=y1035+y565;y1036=y1036+y560;y1037=y1037+y565;y1038=y1038+y564;y1039=y1039+y563;y1040=y1040+y562;

#Full depth 11 and Toffoli depth 2

y405=y405+y357;y398=y398+y370;y629=y629+y358;y407=y407+y368;y630=y630+y356;y631=y631+y366;y419=y419+y358;y632=y632+y359;y633=y633+y391;y634=y634+y367;y635=y635+y357;y397=y397+y384;y636=y636+y363;y637=y637+y358;y638=y638+y386;y420=y420+y357;y639=y639+y355;y640=y640+y359;y641=y641+y358;y642=y642+y358;y643=y643+y368;y644=y644+y357;y645=y645+y357;y646=y646+y360;y647=y647+y384;y648=y648+y357;y649=y649+y356;y650=y650+y354;y651=y651+y358;y652=y652+y355;y404=y404+y377;y653=y653+y368;y654=y654+y359;y655=y655+y357;y414=y414+y360;y657=y657+y358;y658=y658+y354;y412=y412+y371;y660=y660+y356;y661=y661+y357;y662=y662+y359;y663=y663+y357;y664=y664+y360;y665=y665+y372;y666=y666+y361;y667=y667+y362;y668=y668+y366;y669=y669+y358;y670=y670+y354;y671=y671+y362;y672=y672+y356;y674=y674+y361;y675=y675+y354;y676=y676+y355;y677=y677+y392;y678=y678+y359;y679=y679+y356;y680=y680+y367;y681=y681+y365;y682=y682+y356;y683=y683+y360;y684=y684+y358;y685=y685+y358;y687=y687+y359;y688=y688+y377;y689=y689+y357;y690=y690+y356;y691=y691+y373;y692=y692+y358;y693=y693+y356;y416=y416+y360;y415=y415+y364;y694=y694+y355;y695=y695+y355;y696=y696+y356;y697=y697+y375;y698=y698+y361;y699=y699+y359;y700=y700+y368;y701=y701+y355;y702=y702+y356;y703=y703+y365;y704=y704+y359;y705=y705+y359;y706=y706+y363;y707=y707+y358;y417=y417+y358;y418=y418+y365;y708=y708+y396;y709=y709+y359;y408=y408+y362;y710=y710+y376;y711=y711+y356;y712=y712+y358;y713=y713+y356;y421=y421+y359;y714=y714+y361;y716=y716+y356;y717=y717+y371;y718=y718+y366;y719=y719+y371;y720=y720+y357;y721=y721+y357;y722=y722+y360;y723=y723+y366;y724=y724+y383;y411=y411+y355;y725=y725+y359;y726=y726+y360;y727=y727+y354;y728=y728+y359;y729=y729+y359;y730=y730+y358;y731=y731+y356;y474=y474+y426;y467=y467+y439;y732=y732+y427;y476=y476+y437;y733=y733+y425;y734=y734+y435;y488=y488+y427;y735=y735+y428;y736=y736+y460;y737=y737+y436;y738=y738+y426;y466=y466+y453;y739=y739+y432;y740=y740+y427;y741=y741+y455;y489=y489+y426;y742=y742+y424;y743=y743+y428;y744=y744+y427;y745=y745+y427;y746=y746+y437;y747=y747+y426;y748=y748+y426;y749=y749+y429;y750=y750+y453;y751=y751+y426;y752=y752+y425;y753=y753+y423;y754=y754+y427;y755=y755+y424;y473=y473+y446;y756=y756+y437;y757=y757+y428;y758=y758+y426;y483=y483+y429;y760=y760+y427;y761=y761+y423;y481=y481+y440;y763=y763+y425;y764=y764+y426;y765=y765+y428;y766=y766+y426;y767=y767+y429;y768=y768+y441;y769=y769+y430;y770=y770+y431;y771=y771+y435;y772=y772+y427;y773=y773+y423;y774=y774+y431;y775=y775+y425;y777=y777+y430;y778=y778+y423;y779=y779+y424;y780=y780+y461;y781=y781+y428;y782=y782+y425;y783=y783+y436;y784=y784+y434;y785=y785+y425;y786=y786+y429;y787=y787+y427;y788=y788+y427;y790=y790+y428;y791

=y791+y446;y792=y792+y426;y793=y793+y425;y794=y794+y442;y795=y795+y427;y796=y796+y425;y485=y485+y429;y484=y484+y433;y797=y797+y424;y798=y798+y424;y799=y799+y425;y800=y800+y444;y801=y801+y430;y802=y802+y428;y803=y803+y437;y804=y804+y424;y805=y805+y425;y806=y806+y434;y807=y807+y428;y808=y808+y428;y809=y809+y432;y810=y810+y427;y486=y486+y427;y487=y487+y434;y811=y811+y465;y812=y812+y428;y477=y477+y431;y813=y813+y445;y814=y814+y425;y815=y815+y427;y816=y816+y425;y490=y490+y428;y817=y817+y430;y819=y819+y425;y820=y820+y440;y821=y821+y435;y822=y822+y440;y823=y823+y426;y824=y824+y426;y825=y825+y429;y826=y826+y435;y827=y827+y452;y480=y480+y424;y828=y828+y428;y829=y829+y429;y830=y830+y423;y831=y831+y428;y832=y832+y428;y833=y833+y427;y834=y834+y425;y543=y543+y495;y536=y536+y508;y835=y835+y496;y545=y545+y506;y836=y836+y494;y837=y837+y504;y557=y557+y496;y838=y838+y497;y839=y839+y529;y840=y840+y505;y841=y841+y495;y535=y535+y522;y842=y842+y501;y843=y843+y496;y844=y844+y524;y558=y558+y495;y845=y845+y493;y846=y846+y497;y847=y847+y496;y848=y848+y496;y849=y849+y506;y850=y850+y495;y851=y851+y495;y852=y852+y498;y853=y853+y522;y854=y854+y495;y855=y855+y494;y856=y856+y492;y857=y857+y496;y858=y858+y493;y542=y542+y515;y859=y859+y506;y860=y860+y497;y861=y861+y495;y552=y552+y498;y863=y863+y496;y864=y864+y492;y550=y550+y509;y866=y866+y494;y867=y867+y495;y868=y868+y497;y869=y869+y495;y870=y870+y498;y871=y871+y510;y872=y872+y499;y873=y873+y500;y874=y874+y504;y875=y875+y496;y876=y876+y492;y877=y877+y500;y878=y878+y88+y494;y880=y880+y499;y881=y881+y492;y882=y882+y493;y883=y883+y530;y884=y884+y497;y885=y885+y494;y886=y886+y505;y887=y887+y503;y888=y888+y494;y889=y889+y498;y890=y890+y496;y891=y891+y496;y893=y893+y497;y894=y894+y515;y895=y895+y495;y896=y896+y494;y897=y897+y511;y898=y898+y496;y899=y899+y494;y554=y554+y498;y553=y553+y502;y900=y900+y493;y901=y901+y493;y902=y902+y494;y903=y903+y513;y904=y904+y499;y905=y905+y497;y906=y906+y506;y907=y907+y493;y908=y908+y494;y909=y909+y503;y910=y910+y497;y911=y911+y497;y912=y912+y501;y913=y913+y496;y555=y555+y496;y556=y556+y503;y914=y914+y534;y915=y915+y497;y546=y546+y500;y916=y916+y514;y917=y917+y494;y918=y918+y496;y919=y919+y494;y559=y559+y497;y920=y920+y499;y922=y922+y494;y923=y923+y509;y924=y924+y504;y925=y925+y509;y926=y926+y495;y927=y927+y495;y928=y928+y498;y929=y929+y504;y930=y930+y521;y549=y549+y493;y931=y931+y497;y932=y932+y498;y933=y933+y492;y934=y934+y497;y935=y935+y497;y936=y936+y496;y937=y937+y494;y612=y612+y564;y605=y605+y577;y938=y938+y565;y614=y614+y575;y939=y939+y563;y940=y940+y573;y626=y626+y565;y941=y941+y566;y942=y942+y598;y943=y943+y574;y944=y944+y564;y604=y604+y591;y945=y945+y570;y946=y946+y565;y947=y947+y593;y627=y627+y564;y948=y948+y562;y949=y949+y566;y950=y950+y565;y951=y951+y565;y952=y952+y575;y953=y953+y564;y954=y954+y564;y955=y955+y567;y956=y956+y591;y957=y957+y564;y958=y958+y563;y959=y959+y561;y960=y960+y565;y961=y961+y562;y611=y611+y584;y962=y962+y575;y963=y963+y566;y964=y964+y564;y621=y621+y567;y966=y966+y565;y967=y967+y561;y619=y619+y578;y969=y969+y563;y970=y970+y564;y971=y971+y566;y972=y972+y564;y973=y973+y567;y974=y974+y579;y975=y975+y568;y976=y976+y569;y977=y977+y573;y978=y978+y565;y979=y979+y561;y980=y980+y569;y981=y981+y563;y983=y983+y568;y984=y984+y561;y985=y985+y562;y986=y986+y599;y987=y987+y566;y988=y988+y563;y989=y989+y574;y990=y990+y572;y991=y991+y563;y992=y992+y567;y993=y993+y565;y994=y994+y565;y996=y996+y566;y997=y997+y584;y998=y998+y564;y999=y999+y563;y1000=y1000+y580;y1001=y1001+y565;y1002=y1002+y563;y623=y623+y567;y622=y622+y571;y1003=y1003+y562;y1004=y1004+y562;y1005=y1005+y563;y1006=y1006+y582;y1007=y1007+y568;y1008=y1008+y566;y1009=y1009+y575;y1010=y1010+y562;y1011=y1011+y563;y1012=y1012+y572;y1013=y1013+y566;y1014=y1014+y566;y1015=y1015+y570;y1016=y1016+y565;y624=y624+y565;y625=y625+y572;y1017=y1017+y603;y1018=y1018+y566;y615=y615+y569;y1019=y1019+y583;y1020=y1020+y563;y1021=y1021+y565;y1022=y1022+y563;y628=y628+y566;y1023=y1023+y568;y1025=y1025+y563;y1026=y1026+y578;y1027=y1027+y573;y1028=y1028+y578;y1029=y1029+y564;y1030=y1030+y564;y1031=y1031+y567;y1032=y1032+y573;y1033=y1033+y590;y618=y618+y562;y1034=y1034+y566;y1035=y1035+y567;y1036=y1036+y561;y1037=y1037+y566;y1038=y1038+y566;y1039=y1039+y565;y1040=y1040+y563;

#Full depth 12 and Toffoli depth 2

y405=y405+y358;y629=y629+y359;y630=y630+y389;y631=y631+y378;y419=y419+y360;y632=y632+y372;y634=y634+y369;y635=y635+y360;y637=y637+y370;y420=y420+y368;y639=y639+y369;y640=y640+y363;y641=y641+y374;y642=y642+y363;y644=y644+y365;y645=y645+y358;y646=y646+y364;y648=y648+y362;y649=y649+y357;y650=y650+y355;y651=y651+y359;y652=y652+y374;y653=y653+y382;y654=y654+y363;y655=y655+y359;y414=y414+y373;y657=y657+y360;y658=y658+y360;y660=y660+y359;y661=y661+y362;y662=y662+y361;y663=y663+y362;y665=y665+y374;y666=y666+y371;y667=y667+y383;y668=y668+y377;y669=y669+y362;y670=y670+y367;y671=y671+y386;

$y672=y672+y371; y674=y674+y369; y675=y675+y355; y676=y676+y356; y678=y678+y361; y679=y679+y370; y680=y680+y368; y681=y681+y380; y682=y682+y364; y683=y683+y364; y684=y684+y362; y685=y685+y387; y688=y688+y396; y689=y689+y362; y690=y690+y360; y691=y691+y374; y692=y692+y376; y693=y693+y358; y416=y416+y372; y694=y694+y380; y695=y695+y360; y696=y696+y357; y698=y698+y372; y699=y699+y366; y700=y700+y374; y701=y701+y379; y702=y702+y370; y703=y703+y373; y704=y704+y375; y705=y705+y365; y706=y706+y371; y707=y707+y389; y417=y417+y359; y709=y709+y365; y710=y710+y388; y711=y711+y360; y712=y712+y361; y713=y713+y357; y421=y421+y364; y714=y714+y376; y716=y716+y375; y719=y719+y393; y720=y720+y376; y721=y721+y358; y722=y722+y361; y723=y723+y394; y411=y411+y357; y725=y725+y388; y726=y726+y363; y727=y727+y355; y728=y728+y361; y729=y729+y365; y730=y730+y363; y731=y731+y395; y474=y474+y427; y732=y732+y428; y733=y733+y458; y734=y734+y447; y488=y488+y429; y735=y735+y441; y737=y737+y438; y738=y738+y429; y740=y740+y439; y489=y489+y437; y742=y742+y438; y743=y743+y432; y744=y744+y443; y745=y745+y432; y747=y747+y434; y748=y748+y427; y749=y749+y433; y751=y751+y431; y752=y752+y426; y753=y753+y424; y754=y754+y428; y755=y755+y443; y756=y756+y451; y757=y757+y432; y758=y758+y428; y483=y483+y442; y760=y760+y429; y761=y761+y429; y763=y763+y428; y764=y764+y431; y765=y765+y430; y766=y766+y431; y768=y768+y443; y769=y769+y440; y770=y770+y452; y771=y771+y446; y772=y772+y431; y773=y773+y436; y774=y774+y455; y775=y775+y440; y777=y777+y438; y778=y778+y424; y779=y779+y425; y781=y781+y430; y782=y782+y439; y783=y783+y437; y784=y784+y449; y785=y785+y433; y786=y786+y433; y787=y787+y431; y788=y788+y456; y791=y791+y465; y792=y792+y431; y793=y793+y429; y794=y794+y443; y795=y795+y445; y796=y796+y427; y485=y485+y441; y797=y797+y449; y798=y798+y429; y799=y799+y426; y801=y801+y441; y802=y802+y435; y803=y803+y443; y804=y804+y448; y805=y805+y439; y806=y806+y442; y807=y807+y444; y808=y808+y434; y809=y809+y440; y810=y810+y458; y486=y486+y428; y812=y812+y434; y813=y813+y457; y814=y814+y429; y815=y815+y430; y816=y816+y426; y490=y490+y433; y817=y817+y445; y819=y819+y444; y822=y822+y462; y823=y823+y445; y824=y824+y427; y825=y825+y430; y826=y826+y463; y480=y480+y426; y828=y828+y457; y829=y829+y432; y830=y830+y424; y831=y831+y430; y832=y832+y434; y833=y833+y432; y834=y834+y464; y543=y543+y496; y835=y835+y497; y836=y836+y527; y837=y837+y516; y557=y557+y498; y838=y838+y510; y840=y840+y507; y841=y841+y498; y843=y843+y508; y558=y558+y506; y845=y845+y507; y846=y846+y501; y847=y847+y512; y848=y848+y501; y850=y850+y503; y851=y851+y496; y852=y852+y502; y854=y854+y500; y855=y855+y495; y856=y856+y493; y857=y857+y497; y858=y858+y512; y859=y859+y520; y860=y860+y501; y861=y861+y497; y552=y552+y511; y863=y863+y498; y864=y864+y498; y866=y866+y497; y867=y867+y500; y868=y868+y499; y869=y869+y500; y871=y871+y512; y872=y872+y509; y873=y873+y521; y874=y874+y515; y875=y875+y500; y876=y876+y505; y877=y877+y524; y878=y878+y509; y880=y880+y507; y881=y881+y493; y882=y882+y494; y884=y884+y499; y885=y885+y508; y886=y886+y506; y887=y887+y518; y888=y888+y502; y889=y889+y502; y890=y890+y500; y891=y891+y525; y894=y894+y534; y895=y895+y500; y896=y896+y498; y897=y897+y512; y898=y898+y514; y899=y899+y496; y554=y554+y510; y900=y900+y518; y901=y901+y498; y902=y902+y495; y904=y904+y510; y905=y905+y504; y906=y906+y512; y907=y907+y517; y908=y908+y508; y909=y909+y511; y910=y910+y513; y911=y911+y503; y912=y912+y509; y913=y913+y527; y555=y555+y497; y915=y915+y503; y916=y916+y526; y917=y917+y498; y918=y918+y499; y919=y919+y495; y559=y559+y502; y920=y920+y514; y922=y922+y513; y925=y925+y531; y926=y926+y514; y927=y927+y496; y928=y928+y499; y929=y929+y532; y549=y549+y495; y931=y931+y526; y932=y932+y501; y933=y933+y493; y934=y934+y499; y935=y935+y503; y936=y936+y501; y937=y937+y533; y612=y612+y565; y938=y938+y566; y939=y939+y596; y940=y940+y585; y626=y626+y567; y941=y941+y579; y943=y943+y576; y944=y944+y567; y946=y946+y577; y627=y627+y575; y948=y948+y576; y949=y949+y570; y950=y950+y581; y951=y951+y570; y953=y953+y572; y954=y954+y565; y955=y955+y571; y957=y957+y569; y958=y958+y564; y959=y959+y562; y960=y960+y566; y961=y961+y581; y962=y962+y589; y963=y963+y570; y964=y964+y566; y621=y621+y580; y966=y966+y567; y967=y967+y567; y969=y969+y566; y970=y970+y569; y971=y971+y568; y972=y972+y569; y974=y974+y581; y975=y975+y578; y976=y976+y590; y977=y977+y584; y978=y978+y569; y979=y979+y574; y980=y980+y593; y981=y981+y578; y983=y983+y576; y984=y984+y562; y985=y985+y563; y987=y987+y568; y988=y988+y577; y989=y989+y575; y990=y990+y587; y991=y991+y571; y992=y992+y571; y993=y993+y569; y994=y994+y594; y997=y997+y603; y998=y998+y569; y999=y999+y567; y1000=y1000+y581; y1001=y1001+y583; y1002=y1002+y565; y623=y623+y579; y1003=y1003+y587; y1004=y1004+y567; y1005=y1005+y564; y1007=y1007+y579; y1008=y1008+y573; y1009=y1009+y581; y1010=y1010+y586; y1011=y1011+y577; y1012=y1012+y580; y1013=y1013+y582; y1014=y1014+y572; y1015=y1015+y578; y1016=y1016+y596; y624=y624+y566; y1018=y1018+y572; y1019=y1019+y595; y1020=y1020+y567; y1021=y1021+y568; y1022=y1022+y564; y628=y628+y571; y1023=y1023+y583; y1025=y1025+y582; y1028=y1028+y600; y1029=y1029+y583; y1030=y1030+y565; y1031=y1031+y568; y1032=y1032+y601; y618=y618+y564; y1034=y1034+y595; y1035=y1035+y570; y1036=y1036+y562; y1037=y1037+y568; y10$

38=y1038+y572;y1039=y1039+y570;y1040=y1040+y602;

#Full depth 13 and Toffoli depth 2

y629=y629+y360;y419=y419+y361;y632=y632+y396;y635=y635+y382;y640=y640+y391;y641=y641+y375;y642=y642+y370;y644=y644+y385;y645=y645+y360;y648=y648+y374;y649=y649+y360;y650=y650+y356;y654=y654+y367;y655=y655+y364;y657=y657+y381;y658=y658+y361;y660=y660+y366;y662=y662+y364;y663=y663+y372;y672=y672+y378;y674=y674+y385;y675=y675+y357;y676=y676+y360;y678=y678+y392;y682=y682+y369;y684=y684+y381;y689=y689+y378;y690=y690+y378;y693=y693+y359;y695=y695+y380;y696=y696+y372;y699=y699+y390;y702=y702+y373;y705=y705+y366;y417=y417+y387;y709=y709+y379;y711=y711+y383;y712=y712+y395;y713=y713+y360;y721=y721+y359;y722=y722+y364;y727=y727+y367;y728=y728+y393;y730=y730+y381;y732=y732+y429;y488=y488+y430;y735=y735+y465;y738=y738+y451;y743=y743+y460;y744=y744+y444;y745=y745+y439;y747=y747+y454;y748=y748+y429;y751=y751+y443;y752=y752+y429;y753=y753+y425;y757=y757+y436;y758=y758+y433;y760=y760+y450;y761=y761+y430;y763=y763+y435;y765=y765+y433;y766=y766+y441;y775=y775+y447;y777=y777+y454;y778=y778+y426;y779=y779+y429;y781=y781+y461;y785=y785+y438;y787=y787+y450;y792=y792+y447;y793=y793+y447;y796=y796+y428;y798=y798+y449;y799=y799+y441;y802=y802+y459;y805=y805+y442;y808=y808+y435;y812=y812+y448;y814=y814+y452;y815=y815+y464;y816=y816+y429;y824=y824+y428;y825=y825+y433;y830=y830+y436;y831=y831+y462;y833=y833+y450;y835=y835+y498;y857=y857+y499;y838=y838+y534;y841=y841+y520;y846=y846+y529;y847=y847+y513;y848=y848+y508;y850=y850+y523;y851=y851+y498;y854=y854+y512;y855=y855+y498;y856=y856+y494;y860=y860+y505;y861=y861+y502;y863=y863+y519;y864=y864+y499;y866=y866+y504;y868=y868+y502;y869=y869+y510;y878=y878+y516;y880=y880+y523;y881=y881+y495;y882=y882+y498;y884=y884+y530;y888=y888+y507;y890=y890+y519;y895=y895+y516;y896=y896+y516;y899=y899+y497;y901=y901+y518;y902=y902+y510;y905=y905+y528;y908=y908+y511;y911=y911+y504;y555=y555+y525;y915=y915+y517;y917=y917+y521;y918=y918+y533;y919=y919+y498;y927=y927+y497;y928=y928+y502;y933=y933+y505;y934=y934+y531;y936=y936+y519;y938=y938+y567;y626=y626+y568;y941=y941+y603;y944=y944+y589;y949=y949+y598;y950=y950+y582;y951=y951+y577;y953=y953+y592;y954=y954+y567;y957=y957+y581;y958=y958+y567;y959=y959+y563;y963=y963+y574;y964=y964+y571;y966=y966+y588;y967=y967+y568;y969=y969+y573;y971=y971+y571;y972=y972+y579;y981=y981+y585;y983=y983+y592;y984=y984+y564;y985=y985+y567;y987=y987+y599;y991=y991+y576;y993=y993+y588;y998=y998+y585;y999=y999+y585;y1002=y1002+y566;y1004=y1004+y587;y1005=y1005+y579;y1008=y1008+y597;y1011=y1011+y580;y1014=y1014+y573;y624=y624+y594;y1018=y1018+y586;y1020=y1020+y590;y1021=y1021+y602;y1022=y1022+y567;y1030=y1030+y566;y1031=y1031+y571;y1036=y1036+y574;y1037=y1037+y600;y1039=y1039+y588;

#Full depth 14 and Toffoli depth 2

y629=y629+y377;y645=y645+y369;y649=y649+y363;y675=y675+y359;y690=y690+y379;y693=y693+y360;y713=y713+y364;y721=y721+y384;y727=y727+y381;y732=y732+y446;y748=y748+y438;y752=y752+y432;y778=y778+y428;y793=y793+y448;y796=y796+y429;y816=y816+y433;y824=y824+y453;y830=y830+y450;y835=y835+y515;y851=y851+y507;y855=y855+y501;y881=y881+y497;y896=y896+y517;y899=y899+y498;y919=y919+y502;y927=y927+y522;y933=y933+y519;y938=y938+y584;y954=y954+y576;y958=y958+y570;y984=y984+y566;y999=y999+y586;y1002=y1002+y567;y1022=y1022+y571;y1030=y1030+y591;y1036=y1036+y588;

#Full depth 15 and Toffoli depth 2

y675=y675+y362;y693=y693+y375;y778=y778+y431;y796=y796+y444;y881=y881+y500;y899=y899+y513;y984=y984+y569;y1002=y1002+y582;

#Full depth 16 and Toffoli depth 2

y353=y353+y249;y354=y354+y253;y355=y355+y244;y356=y356+y254;y357=y357+y255;y358=y358+y256;y359=y359+y252;y360=y360+y246;y361=y361+y255;y362=y362+y250;y363=y363+y247;y364=y364+y245;y365=y365+y251;y366=y366+y243;y367=y367+y256;y368=y368+y254;y369=y369+y252;y370=y370+y252;y371=y371+y246;y372=y372+y256;y373=y373+y256;y374=y374+y249;y375=y375+y243;y376=y376+y253;y377=y377+y253;y378=y378+y252;y379=y379+y256;y380=y380+y256;y381=y381+y255;y382=y382+y252;y383=y383+y252;y384=y384+y251;y385=y385+y242;y386=y386+y255;y387=y387+y254;y388=y388+y254;y389=y389+y253;y390=y390+y253;y391=y391+y251;y392=y392+y251;y393=y393+y251;y394=y394+y250;y395=y395+y248;y396=y396+y247;y422=y422+y281;y423=y423+y285;y424=y424+y276;y425=y425+y286;y426=y426+y287;y427=y427+y288;y428=y428+y284;y429=y429+y278;y430=y430+y287;y431=y431+y282;y432=y432+y279;y433=y433+y287;

y277;y434=y434+y283;y435=y435+y275;y436=y436+y288;y437=y437+y286;y438=y438+y284;y439=y439+y284;y440=y440+y278;y441=y441+y288;y442=y442+y288;y443=y443+y281;y444=y444+y275;y445=y445+y285;y446=y446+y285;y447=y447+y284;y448=y448+y288;y449=y449+y288;y450=y450+y287;y451=y451+y284;y452=y452+y284;y453=y453+y283;y454=y454+y274;y455=y455+y287;y456=y456+y286;y457=y457+y286;y458=y458+y285;y459=y459+y285;y460=y460+y283;y461=y461+y283;y462=y462+y283;y463=y463+y282;y464=y464+y280;y465=y465+y279;y491=y491+y313;y492=y492+y317;y493=y493+y308;y494=y494+y318;y495=y495+y319;y496=y496+y320;y497=y497+y316;y498=y498+y310;y499=y499+y319;y500=y500+y314;y501=y501+y311;y502=y502+y309;y503=y503+y315;y504=y504+y307;y505=y505+y320;y506=y506+y318;y507=y507+y316;y508=y508+y316;y509=y509+y310;y510=y510+y320;y511=y511+y320;y512=y512+y313;y513=y513+y307;y514=y514+y317;y515=y515+y317;y516=y516+y316;y517=y517+y320;y518=y518+y320;y519=y519+y319;y520=y520+y316;y521=y521+y316;y522=y522+y315;y523=y523+y306;y524=y524+y319;y525=y525+y318;y526=y526+y318;y527=y527+y317;y528=y528+y317;y529=y529+y315;y530=y530+y315;y531=y531+y315;y532=y532+y314;y533=y533+y312;y534=y534+y311;y560=y560+y345;y561=y561+y349;y562=y562+y340;y563=y563+y350;y564=y564+y351;y565=y565+y352;y566=y566+y348;y567=y567+y342;y568=y568+y351;y569=y569+y346;y570=y570+y343;y571=y571+y341;y572=y572+y347;y573=y573+y339;y574=y574+y352;y575=y575+y350;y576=y576+y348;y577=y577+y348;y578=y578+y342;y579=y579+y352;y580=y580+y352;y581=y581+y345;y582=y582+y339;y583=y583+y349;y584=y584+y349;y585=y585+y348;y586=y586+y352;y587=y587+y352;y588=y588+y351;y589=y589+y348;y590=y590+y348;y591=y591+y347;y592=y592+y338;y593=y593+y351;y594=y594+y350;y595=y595+y350;y596=y596+y349;y597=y597+y349;y598=y598+y347;y599=y599+y347;y600=y600+y347;y601=y601+y346;y602=y602+y344;y603=y603+y343;

#Full depth 17 and Toffoli depth 2

y353=y353+y247;y354=y354+y251;y355=y355+y242;y356=y356+y250;y357=y357+y243;y358=y358+y245;y359=y359+y248;y360=y360+y241;y361=y361+y254;y362=y362+y246;y363=y363+y244;y364=y364+y242;y365=y365+y249;y366=y366+y241;y367=y367+y241;y368=y368+y244;y369=y369+y247;y370=y370+y243;y371=y371+y245;y372=y372+y253;y373=y373+y244;y374=y374+y248;y375=y375+y242;y376=y376+y252;y377=y377+y250;y378=y378+y251;y379=y379+y255;y380=y380+y243;y381=y381+y248;y382=y382+y249;y383=y383+y245;y384=y384+y247;y385=y385+y241;y386=y386+y253;y387=y387+y246;y388=y388+y241;y389=y389+y248;y390=y390+y247;y391=y391+y250;y392=y392+y246;y393=y393+y244;y394=y394+y245;y395=y395+y246;y396=y396+y241;y422=y422+y279;y423=y423+y283;y424=y424+y274;y425=y425+y282;y426=y426+y275;y427=y427+y277;y428=y428+y280;y429=y429+y273;y430=y430+y286;y431=y431+y278;y432=y432+y276;y433=y433+y274;y434=y434+y281;y435=y435+y273;y436=y436+y273;y437=y437+y276;y438=y438+y279;y439=y439+y275;y440=y440+y277;y441=y441+y285;y442=y442+y276;y443=y443+y280;y444=y444+y274;y445=y445+y284;y446=y446+y282;y447=y447+y283;y448=y448+y287;y449=y449+y275;y450=y450+y280;y451=y451+y281;y452=y452+y277;y453=y453+y279;y454=y454+y273;y455=y455+y285;y456=y456+y278;y457=y457+y273;y458=y458+y280;y459=y459+y279;y460=y460+y282;y461=y461+y278;y462=y462+y276;y463=y463+y277;y464=y464+y278;y465=y465+y273;y491=y491+y311;y492=y492+y315;y493=y493+y306;y494=y494+y314;y495=y495+y307;y496=y496+y309;y497=y497+y312;y498=y498+y305;y499=y499+y318;y500=y500+y310;y501=y501+y308;y502=y502+y306;y503=y503+y313;y504=y504+y305;y505=y505+y305;y506=y506+y308;y507=y507+y311;y508=y508+y307;y509=y509+y309;y510=y510+y317;y511=y511+y308;y512=y512+y312;y513=y513+y306;y514=y514+y316;y515=y515+y314;y516=y516+y315;y517=y517+y319;y518=y518+y307;y519=y519+y312;y520=y520+y313;y521=y521+y309;y522=y522+y311;y523=y523+y305;y524=y524+y317;y525=y525+y310;y526=y526+y305;y527=y527+y312;y528=y528+y311;y529=y529+y314;y530=y530+y310;y531=y531+y308;y532=y532+y309;y533=y533+y310;y534=y534+y305;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y340;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y592+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y598=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+y337;

#Full depth 18 and Toffoli depth 3

y129=y129+y240*y405;y130=y130+y239*y406;y131=y131+y238*y398;y132=y132+y237*y629;y133=y133+y236*y407;y134=y134+y235*y630;y135=y135+y234*y413;y136=y136+y233*y631;y137=y

86=y1086+y259*y832;y1087=y1087+y258*y833;y1088=y1088+y257*y834;y1089=y1089+y304*y543;y1090=y1090+y303*y544;y1091=y1091+y302*y536;y1092=y1092+y301*y835;y1093=y1093+y300*y545;y1094=y1094+y299*y836;y1095=y1095+y298*y551;y1096=y1096+y297*y837;y1097=y1097+y296*y557;y1098=y1098+y295*y838;y1099=y1099+y294*y540;y1100=y1100+y293*y839;y1101=y1101+y292*y840;y1102=y1102+y291*y841;y1103=y1103+y290*y535;y1104=y1104+y289*y842;y1105=y1105+y304*y843;y1106=y1106+y303*y844;y1107=y1107+y302*y558;y1108=y1108+y301*y547;y1109=y1109+y300*y845;y1110=y1110+y299*y846;y1111=y1111+y298*y548;y1112=y1112+y297*y847;y1113=y1113+y296*y848;y1114=y1114+y295*y849;y1115=y1115+y294*y850;y1116=y1116+y293*y851;y1117=y1117+y292*y852;y1118=y1118+y291*y853;y1119=y1119+y290*y854;y1120=y1120+y289*y855;y1121=y1121+y304*y856;y1122=y1122+y303*y857;y1123=y1123+y302*y858;y1124=y1124+y301*y542;y1125=y1125+y300*y859;y1126=y1126+y299*y860;y1127=y1127+y298*y861;y1128=y1128+y297*y862;y1129=y1129+y296*y552;y1130=y1130+y295*y863;y1131=y1131+y294*y864;y1132=y1132+y293*y550;y1133=y1133+y292*y865;y1134=y1134+y291*y866;y1135=y1135+y290*y867;y1136=y1136+y289*y868;y1137=y1137+y304*y869;y1138=y1138+y303*y870;y1139=y1139+y302*y871;y1140=y1140+y301*y872;y1141=y1141+y300*y873;y1142=y1142+y299*y874;y1143=y1143+y298*y875;y1144=y1144+y297*y876;y1145=y1145+y296*y877;y1146=y1146+y295*y878;y1147=y1147+y294*y879;y1148=y1148+y293*y880;y1149=y1149+y292*y881;y1150=y1150+y291*y882;y1151=y1151+y290*y883;y1152=y1152+y289*y884;y1153=y1153+y304*y885;y1154=y1154+y303*y886;y1155=y1155+y302*y887;y1156=y1156+y301*y888;y1157=y1157+y300*y889;y1158=y1158+y299*y890;y1159=y1159+y298*y891;y1160=y1160+y297*y892;y1161=y1161+y296*y541;y1162=y1162+y295*y893;y1163=y1163+y294*y894;y1164=y1164+y293*y895;y1165=y1165+y292*y896;y1166=y1166+y291*y897;y1167=y1167+y290*y898;y1168=y1168+y289*y899;y1169=y1169+y304*y554;y1170=y1170+y303*y553;y1171=y1171+y302*y900;y1172=y1172+y301*y901;y1173=y1173+y300*y902;y1174=y1174+y299*y903;y1175=y1175+y298*y904;y1176=y1176+y297*y905;y1177=y1177+y296*y537;y1178=y1178+y295*y906;y1179=y1179+y294*y907;y1180=y1180+y293*y908;y1181=y1181+y292*y909;y1182=y1182+y291*y910;y1183=y1183+y290*y911;y1184=y1184+y289*y912;y1185=y1185+y304*y913;y1186=y1186+y303*y555;y1187=y1187+y302*y556;y1188=y1188+y301*y914;y1189=y1189+y300*y915;y1190=y1190+y299*y546;y1191=y1191+y298*y916;y1192=y1192+y297*y917;y1193=y1193+y296*y918;y1194=y1194+y295*y538;y1195=y1195+y294*y919;y1196=y1196+y293*y559;y1197=y1197+y292*y920;y1198=y1198+y291*y921;y1199=y1199+y290*y922;y1200=y1200+y289*y923;y1201=y1201+y304*y924;y1202=y1202+y303*y539;y1203=y1203+y302*y925;y1204=y1204+y301*y926;y1205=y1205+y300*y927;y1206=y1206+y299*y928;y1207=y1207+y298*y929;y1208=y1208+y297*y930;y1209=y1209+y296*y549;y1210=y1210+y295*y931;y1211=y1211+y294*y932;y1212=y1212+y293*y933;y1213=y1213+y292*y934;y1214=y1214+y291*y935;y1215=y1215+y290*y936;y1216=y1216+y289*y937;y1217=y1217+y336*y612;y1218=y1218+y335*y613;y1219=y1219+y334*y605;y1220=y1220+y333*y938;y1221=y1221+y332*y614;y1222=y1222+y331*y939;y1223=y1223+y330*y620;y1224=y1224+y329*y940;y1225=y1225+y328*y626;y1226=y1226+y327*y941;y1227=y1227+y326*y609;y1228=y1228+y325*y942;y1229=y1229+y324*y943;y1230=y1230+y323*y944;y1231=y1231+y322*y604;y1232=y1232+y321*y945;y1233=y1233+y336*y946;y1234=y1234+y335*y947;y1235=y1235+y334*y627;y1236=y1236+y333*y616;y1237=y1237+y332*y948;y1238=y1238+y331*y949;y1239=y1239+y330*y617;y1240=y1240+y329*y950;y1241=y1241+y328*y951;y1242=y1242+y327*y952;y1243=y1243+y326*y953;y1244=y1244+y325*y954;y1245=y1245+y324*y955;y1246=y1246+y323*y956;y1247=y1247+y322*y957;y1248=y1248+y321*y958;y1249=y1249+y336*y959;y1250=y1250+y335*y960;y1251=y1251+y334*y961;y1252=y1252+y333*y611;y1253=y1253+y332*y962;y1254=y1254+y331*y963;y1255=y1255+y330*y964;y1256=y1256+y329*y965;y1257=y1257+y328*y621;y1258=y1258+y327*y966;y1259=y1259+y326*y967;y1260=y1260+y325*y619;y1261=y1261+y324*y968;y1262=y1262+y323*y969;y1263=y1263+y322*y970;y1264=y1264+y321*y971;y1265=y1265+y336*y972;y1266=y1266+y335*y973;y1267=y1267+y334*y974;y1268=y1268+y333*y975;y1269=y1269+y332*y976;y1270=y1270+y331*y977;y1271=y1271+y330*y978;y1272=y1272+y329*y979;y1273=y1273+y328*y980;y1274=y1274+y327*y981;y1275=y1275+y326*y982;y1276=y1276+y325*y983;y1277=y1277+y324*y984;y1278=y1278+y323*y985;y1279=y1279+y322*y986;y1280=y1280+y321*y987;y1281=y1281+y336*y988;y1282=y1282+y335*y989;y1283=y1283+y334*y990;y1284=y1284+y333*y991;y1285=y1285+y332*y992;y1286=y1286+y331*y993;y1287=y1287+y330*y994;y1288=y1288+y329*y995;y1289=y1289+y328*y996;y1290=y1290+y327*y997;y1291=y1291+y326*y998;y1292=y1292+y325*y999;y1293=y1293+y324*y999;y1294=y1294+y323*y1000;y1295=y1295+y322*y1001;y1296=y1296+y321*y1002;y1297=y1297+y336*y623;y1298=y1298+y335*y622;y1299=y1299+y334*y1003;y1300=y1300+y333*y1004;y1301=y1301+y332*y1005;y1302=y1302+y331*y1006;y1303=y1303+y330*y1007;y1304=y1304+y329*y1008;y1305=y1305+y328*y606;y1306=y1306+y327*y1009;y1307=y1307+y326*y1010;y1308=y1308+y325*y1011;y1309=y1309+y324*y1012;y1310=y1310+y323*y1013;y1311=y1311+y322*y1

014; $y_{1312}=y_{1312}+y_{321}*y_{1015}$; $y_{1313}=y_{1313}+y_{336}*y_{1016}$; $y_{1314}=y_{1314}+y_{335}*y_{624}$; $y_{1315}=y_{1315}+y_{334}*y_{625}$; $y_{1316}=y_{1316}+y_{333}*y_{1017}$; $y_{1317}=y_{1317}+y_{332}*y_{1018}$; $y_{1318}=y_{1318}+y_{331}*y_{615}$; $y_{1319}=y_{1319}+y_{330}*y_{1019}$; $y_{1320}=y_{1320}+y_{329}*y_{1020}$; $y_{1321}=y_{1321}+y_{328}*y_{1021}$; $y_{1322}=y_{1322}+y_{327}*y_{607}$; $y_{1323}=y_{1323}+y_{326}*y_{1022}$; $y_{1324}=y_{1324}+y_{325}*y_{628}$; $y_{1325}=y_{1325}+y_{324}*y_{1023}$; $y_{1326}=y_{1326}+y_{323}*y_{1024}$; $y_{1327}=y_{1327}+y_{322}*y_{1025}$; $y_{1328}=y_{1328}+y_{321}*y_{1026}$; $y_{1329}=y_{1329}+y_{336}*y_{1027}$; $y_{1330}=y_{1330}+y_{335}*y_{608}$; $y_{1331}=y_{1331}+y_{334}*y_{1028}$; $y_{1332}=y_{1332}+y_{333}*y_{1029}$; $y_{1333}=y_{1333}+y_{332}*y_{1030}$; $y_{1334}=y_{1334}+y_{331}*y_{1031}$; $y_{1335}=y_{1335}+y_{330}*y_{1032}$; $y_{1336}=y_{1336}+y_{329}*y_{1033}$; $y_{1337}=y_{1337}+y_{328}*y_{618}$; $y_{1338}=y_{1338}+y_{327}*y_{1034}$; $y_{1339}=y_{1339}+y_{326}*y_{1035}$; $y_{1340}=y_{1340}+y_{325}*y_{1036}$; $y_{1341}=y_{1341}+y_{324}*y_{1037}$; $y_{1342}=y_{1342}+y_{323}*y_{1038}$; $y_{1343}=y_{1343}+y_{322}*y_{1039}$; $y_{1344}=y_{1344}+y_{321}*y_{1040}$;

#Full depth 19 and Toffoli depth 3

$y_{129}=y_{129}+y_{130}$; $y_{131}=y_{131}+y_{132}$; $y_{133}=y_{133}+y_{134}$; $y_{135}=y_{135}+y_{136}$; $y_{137}=y_{137}+y_{138}$; $y_{139}=y_{139}+y_{140}$; $y_{141}=y_{141}+y_{142}$; $y_{143}=y_{143}+y_{144}$; $y_{145}=y_{145}+y_{146}$; $y_{147}=y_{147}+y_{148}$; $y_{149}=y_{149}+y_{150}$; $y_{151}=y_{151}+y_{152}$; $y_{153}=y_{153}+y_{154}$; $y_{155}=y_{155}+y_{156}$; $y_{157}=y_{157}+y_{158}$; $y_{159}=y_{159}+y_{160}$; $y_{353}=y_{353}+y_{354}$; $y_{355}=y_{355}+y_{356}$; $y_{357}=y_{357}+y_{358}$; $y_{359}=y_{359}+y_{360}$; $y_{361}=y_{361}+y_{362}$; $y_{363}=y_{363}+y_{364}$; $y_{365}=y_{365}+y_{366}$; $y_{367}=y_{367}+y_{368}$; $y_{369}=y_{369}+y_{370}$; $y_{371}=y_{371}+y_{372}$; $y_{373}=y_{373}+y_{374}$; $y_{375}=y_{375}+y_{376}$; $y_{377}=y_{377}+y_{378}$; $y_{379}=y_{379}+y_{380}$; $y_{381}=y_{381}+y_{382}$; $y_{383}=y_{383}+y_{384}$; $y_{385}=y_{385}+y_{386}$; $y_{387}=y_{387}+y_{388}$; $y_{389}=y_{389}+y_{390}$; $y_{391}=y_{391}+y_{392}$; $y_{393}=y_{393}+y_{394}$; $y_{395}=y_{395}+y_{396}$; $y_{422}=y_{422}+y_{423}$; $y_{424}=y_{424}+y_{425}$; $y_{426}=y_{426}+y_{427}$; $y_{428}=y_{428}+y_{429}$; $y_{430}=y_{430}+y_{431}$; $y_{432}=y_{432}+y_{433}$; $y_{434}=y_{434}+y_{435}$; $y_{436}=y_{436}+y_{437}$; $y_{438}=y_{438}+y_{439}$; $y_{440}=y_{440}+y_{441}$; $y_{442}=y_{442}+y_{443}$; $y_{444}=y_{444}+y_{445}$; $y_{446}=y_{446}+y_{447}$; $y_{448}=y_{448}+y_{449}$; $y_{450}=y_{450}+y_{451}$; $y_{452}=y_{452}+y_{453}$; $y_{454}=y_{454}+y_{455}$; $y_{456}=y_{456}+y_{457}$; $y_{458}=y_{458}+y_{459}$; $y_{460}=y_{460}+y_{461}$; $y_{462}=y_{462}+y_{463}$; $y_{464}=y_{464}+y_{465}$; $y_{491}=y_{491}+y_{492}$; $y_{493}=y_{493}+y_{494}$; $y_{495}=y_{495}+y_{496}$; $y_{497}=y_{497}+y_{498}$; $y_{499}=y_{499}+y_{500}$; $y_{501}=y_{501}+y_{502}$; $y_{503}=y_{503}+y_{504}$; $y_{505}=y_{505}+y_{506}$; $y_{507}=y_{507}+y_{508}$; $y_{509}=y_{509}+y_{510}$; $y_{511}=y_{511}+y_{512}$; $y_{513}=y_{513}+y_{514}$; $y_{515}=y_{515}+y_{516}$; $y_{517}=y_{517}+y_{518}$; $y_{519}=y_{519}+y_{520}$; $y_{521}=y_{521}+y_{522}$; $y_{523}=y_{523}+y_{524}$; $y_{525}=y_{525}+y_{526}$; $y_{527}=y_{527}+y_{528}$; $y_{529}=y_{529}+y_{530}$; $y_{531}=y_{531}+y_{532}$; $y_{533}=y_{533}+y_{534}$; $y_{560}=y_{560}+y_{561}$; $y_{562}=y_{562}+y_{563}$; $y_{564}=y_{564}+y_{565}$; $y_{566}=y_{566}+y_{567}$; $y_{568}=y_{568}+y_{569}$; $y_{570}=y_{570}+y_{571}$; $y_{572}=y_{572}+y_{573}$; $y_{574}=y_{574}+y_{575}$; $y_{576}=y_{576}+y_{577}$; $y_{578}=y_{578}+y_{579}$; $y_{580}=y_{580}+y_{581}$; $y_{582}=y_{582}+y_{583}$; $y_{584}=y_{584}+y_{585}$; $y_{586}=y_{586}+y_{587}$; $y_{588}=y_{588}+y_{589}$; $y_{590}=y_{590}+y_{591}$; $y_{592}=y_{592}+y_{593}$; $y_{594}=y_{594}+y_{595}$; $y_{596}=y_{596}+y_{597}$; $y_{598}=y_{598}+y_{599}$; $y_{600}=y_{600}+y_{601}$; $y_{602}=y_{602}+y_{603}$; $y_{1041}=y_{1041}+y_{1042}$; $y_{1043}=y_{1043}+y_{1044}$; $y_{1045}=y_{1045}+y_{1046}$; $y_{1047}=y_{1047}+y_{1048}$; $y_{1049}=y_{1049}+y_{1050}$; $y_{1051}=y_{1051}+y_{1052}$; $y_{1053}=y_{1053}+y_{1054}$; $y_{1055}=y_{1055}+y_{1056}$; $y_{1057}=y_{1057}+y_{1058}$; $y_{1059}=y_{1059}+y_{1060}$; $y_{1061}=y_{1061}+y_{1062}$; $y_{1063}=y_{1063}+y_{1064}$; $y_{1065}=y_{1065}+y_{1066}$; $y_{1067}=y_{1067}+y_{1068}$; $y_{1069}=y_{1069}+y_{1070}$; $y_{1071}=y_{1071}+y_{1072}$; $y_{1073}=y_{1073}+y_{1074}$; $y_{1075}=y_{1075}+y_{1076}$; $y_{1077}=y_{1077}+y_{1078}$; $y_{1079}=y_{1079}+y_{1080}$; $y_{1081}=y_{1081}+y_{1082}$; $y_{1083}=y_{1083}+y_{1084}$; $y_{1085}=y_{1085}+y_{1086}$; $y_{1087}=y_{1087}+y_{1088}$; $y_{1089}=y_{1089}+y_{1090}$; $y_{1091}=y_{1091}+y_{1092}$; $y_{1093}=y_{1093}+y_{1094}$; $y_{1095}=y_{1095}+y_{1096}$; $y_{1097}=y_{1097}+y_{1098}$; $y_{1099}=y_{1099}+y_{1100}$; $y_{1101}=y_{1101}+y_{1102}$; $y_{1103}=y_{1103}+y_{1104}$; $y_{1105}=y_{1105}+y_{1106}$; $y_{1107}=y_{1107}+y_{1108}$; $y_{1109}=y_{1109}+y_{1110}$; $y_{1111}=y_{1111}+y_{1112}$; $y_{1113}=y_{1113}+y_{1114}$; $y_{1115}=y_{1115}+y_{1116}$; $y_{1117}=y_{1117}+y_{1118}$; $y_{1119}=y_{1119}+y_{1120}$; $y_{1121}=y_{1121}+y_{1122}$; $y_{1123}=y_{1123}+y_{1124}$; $y_{1125}=y_{1125}+y_{1126}$; $y_{1127}=y_{1127}+y_{1128}$; $y_{1129}=y_{1129}+y_{1130}$; $y_{1131}=y_{1131}+y_{1132}$; $y_{1133}=y_{1133}+y_{1134}$; $y_{1135}=y_{1135}+y_{1136}$; $y_{1137}=y_{1137}+y_{1138}$; $y_{1139}=y_{1139}+y_{1140}$; $y_{1141}=y_{1141}+y_{1142}$; $y_{1143}=y_{1143}+y_{1144}$; $y_{1145}=y_{1145}+y_{1146}$; $y_{1147}=y_{1147}+y_{1148}$; $y_{1149}=y_{1149}+y_{1150}$; $y_{1151}=y_{1151}+y_{1152}$; $y_{1153}=y_{1153}+y_{1154}$; $y_{1155}=y_{1155}+y_{1156}$; $y_{1157}=y_{1157}+y_{1158}$; $y_{1159}=y_{1159}+y_{1160}$; $y_{1161}=y_{1161}+y_{1162}$; $y_{1163}=y_{1163}+y_{1164}$; $y_{1165}=y_{1165}+y_{1166}$; $y_{1167}=y_{1167}+y_{1168}$; $y_{1169}=y_{1169}+y_{1170}$; $y_{1171}=y_{1171}+y_{1172}$; $y_{1173}=y_{1173}+y_{1174}$; $y_{1175}=y_{1175}+y_{1176}$; $y_{1177}=y_{1177}+y_{1178}$; $y_{1179}=y_{1179}+y_{1180}$; $y_{1181}=y_{1181}+y_{1182}$; $y_{1183}=y_{1183}+y_{1184}$; $y_{1185}=y_{1185}+y_{1186}$; $y_{1187}=y_{1187}+y_{1188}$; $y_{1189}=y_{1189}+y_{1190}$; $y_{1191}=y_{1191}+y_{1192}$; $y_{1193}=y_{1193}+y_{1194}$; $y_{1195}=y_{1195}+y_{1196}$; $y_{1197}=y_{1197}+y_{1198}$; $y_{1199}=y_{1199}+y_{1200}$; $y_{1201}=y_{1201}+y_{1202}$; $y_{1203}=y_{1203}+y_{1204}$; $y_{1205}=y_{1205}+y_{1206}$; $y_{1207}=y_{1207}+y_{1208}$; $y_{1209}=y_{1209}+y_{1210}$; $y_{1211}=y_{1211}+y_{1212}$; $y_{1213}=y_{1213}+y_{1214}$; $y_{1215}=y_{1215}+y_{1216}$; $y_{1217}=y_{1217}+y_{1218}$; $y_{1219}=y_{1219}+y_{1220}$; $y_{1221}=y_{1221}+y_{1222}$; $y_{1223}=y_{1223}+y_{1224}$; $y_{1225}=y_{1225}+y_{1226}$; $y_{1227}=y_{1227}+y_{1228}$; $y_{1229}=y_{1229}+y_{1230}$; $y_{1231}=y_{1231}+y_{1232}$; $y_{1233}=y_{1233}+y_{1234}$; $y_{1235}=y_{1235}+y_{1236}$; $y_{1237}=y_{1237}+y_{1238}$; $y_{1239}=y_{1239}+y_{1240}$; $y_{1241}=y_{1241}+y_{1242}$; $y_{1243}=y_{1243}+y_{1244}$; $y_{1245}=y_{1245}+y_{1246}$; $y_{1247}=y_{1247}+y_{1248}$; $y_{1249}=y_{1249}+y_{1250}$; $y_{1251}=y_{1251}+y_{1252}$; $y_{1253}=y_{1253}+y_{1254}$; $y_{1255}=y_{1255}+y_{1256}$; $y_{1257}=y_{1257}+y_{1258}$; $y_{1259}=y_{1259}+y_{1260}$; $y_{1261}=y_{1261}+y_{1262}$; $y_{1263}=y_{1263}+y_{1264}$; $y_{1265}=y_{1265}+y_{1266}$; $y_{1267}=y_{1267}+y_{1268}$; $y_{1269}=y_{1269}+y_{1270}$; $y_{1271}=y_{1271}+y_{1272}$; $y_{1273}=y_{1273}+y_{1274}$; $y_{1275}=y_{1275}+y_{1276}$; $y_{1277}=y_{1277}+y_{1278}$; $y_{1279}=y_{1279}+y_{1280}$; $y_{1281}=y_{1281}+y_{1282}$; $y_{1283}=y_{1283}+y_{1284}$; $y_{1285}=y_{1285}+y_{1286}$; $y_{1287}=y_{1287}+y_{1288}$; $y_{1289}=y_{1289}+y_{1290}$; $y_{1291}=y_{1291}+y_{1292}$; $y_{1293}=y_{1293}+y_{1294}$; $y_{1295}=y_{1295}+y_{1296}$; $y_{1297}=y_{1297}+y_{1298}$; $y_{1299}=y_{1299}+y_{1300}$; $y_{1301}=y_{1301}+y_{1302}$; $y_{1303}=y_{1303}+y_{1304}$; $y_{1305}=y_{1305}+y_{1306}$; $y_{1307}=y_{1307}+y_{1308}$; $y_{1309}=y_{1309}+y_{1310}$; $y_{1311}=y_{1311}+y_{1312}$; y_{1313}

=y1313+y1314;y1315=y1315+y1316;y1317=y1317+y1318;y1319=y1319+y1320;y1321=y1321+y1322;y1323=y1323+y1324;y1325=y1325+y1326;y1327=y1327+y1328;y1329=y1329+y1330;y1331=y1331+y1332;y1333=y1333+y1334;y1335=y1335+y1336;y1337=y1337+y1338;y1339=y1339+y1340;y1341=y1341+y1342;y1343=y1343+y1344;

#Full depth 20 and Toffoli depth 3

y129=y129+y131;y133=y133+y135;y137=y137+y139;y141=y141+y143;y145=y145+y147;y149=y149+y151;y153=y153+y155;y157=y157+y159;y353=y353+y355;y357=y357+y359;y361=y361+y363;y365=y365+y367;y369=y369+y371;y373=y373+y375;y377=y377+y379;y381=y381+y383;y385=y385+y387;y389=y389+y391;y393=y393+y395;y422=y422+y424;y426=y426+y428;y430=y430+y432;y434=y434+y436;y438=y438+y440;y442=y442+y444;y446=y446+y448;y450=y450+y452;y454=y454+y456;y458=y458+y460;y462=y462+y464;y491=y491+y493;y495=y495+y497;y499=y499+y501;y503=y503+y505;y507=y507+y509;y511=y511+y513;y515=y515+y517;y519=y519+y521;y523=y523+y525;y527=y527+y529;y531=y531+y533;y560=y560+y562;y564=y564+y566;y568=y568+y570;y572=y572+y574;y576=y576+y578;y580=y580+y582;y584=y584+y586;y588=y588+y590;y592=y592+y594;y596=y596+y598;y600=y600+y602;y1041=y1041+y1043;y1045=y1045+y1047;y1049=y1049+y1051;y1053=y1053+y1055;y1057=y1057+y1059;y1061=y1061+y1063;y1065=y1065+y1067;y1069=y1069+y1071;y1073=y1073+y1075;y1077=y1077+y1079;y1081=y1081+y1083;y1085=y1085+y1087;y1089=y1089+y1091;y1093=y1093+y1095;y1097=y1097+y1099;y1101=y1101+y1103;y1105=y1105+y1107;y1109=y1109+y1111;y1113=y1113+y1115;y1117=y1117+y1119;y1121=y1121+y1123;y1125=y1125+y1127;y1129=y1129+y1131;y1133=y1133+y1135;y1137=y1137+y1139;y1141=y1141+y1143;y1145=y1145+y1147;y1149=y1149+y1151;y1153=y1153+y1155;y1157=y1157+y1159;y1161=y1161+y1163;y1165=y1165+y1167;y1169=y1169+y1171;y1173=y1173+y1175;y1177=y1177+y1179;y1181=y1181+y1183;y1185=y1185+y1187;y1189=y1189+y1191;y1193=y1193+y1195;y1197=y1197+y1199;y1201=y1201+y1203;y1205=y1205+y1207;y1209=y1209+y1211;y1213=y1213+y1215;y1217=y1217+y1219;y1221=y1221+y1223;y1225=y1225+y1227;y1229=y1229+y1231;y1233=y1233+y1235;y1237=y1237+y1239;y1241=y1241+y1243;y1245=y1245+y1247;y1249=y1249+y1251;y1253=y1253+y1255;y1257=y1257+y1259;y1261=y1261+y1263;y1265=y1265+y1267;y1269=y1269+y1271;y1273=y1273+y1275;y1277=y1277+y1279;y1281=y1281+y1283;y1285=y1285+y1287;y1289=y1289+y1291;y1293=y1293+y1295;y1297=y1297+y1299;y1301=y1301+y1303;y1305=y1305+y1307;y1309=y1309+y1311;y1313=y1313+y1315;y1317=y1317+y1319;y1321=y1321+y1323;y1325=y1325+y1327;y1329=y1329+y1331;y1333=y1333+y1335;y1337=y1337+y1339;y1341=y1341+y1343;

#Full depth 21 and Toffoli depth 3

y129=y129+y133;y137=y137+y141;y145=y145+y149;y153=y153+y157;y353=y353+y357;y361=y361+y365;y369=y369+y373;y377=y377+y381;y385=y385+y389;y393=y393+y422;y426=y426+y430;y434=y434+y438;y442=y442+y446;y450=y450+y454;y458=y458+y462;y491=y491+y495;y499=y499+y503;y507=y507+y511;y515=y515+y519;y523=y523+y527;y531=y531+y560;y564=y564+y568;y572=y572+y576;y580=y580+y584;y588=y588+y592;y596=y596+y600;y1041=y1041+y1045;y1049=y1049+y1053;y1057=y1057+y1061;y1065=y1065+y1069;y1073=y1073+y1077;y1081=y1081+y1085;y1089=y1089+y1093;y1097=y1097+y1101;y1105=y1105+y1109;y1113=y1113+y1117;y1121=y1121+y1125;y1129=y1129+y1133;y1137=y1137+y1141;y1145=y1145+y1149;y1153=y1153+y1157;y1161=y1161+y1165;y1169=y1169+y1173;y1177=y1177+y1181;y1185=y1185+y1189;y1193=y1193+y1197;y1201=y1201+y1205;y1209=y1209+y1213;y1217=y1217+y1221;y1225=y1225+y1229;y1233=y1233+y1237;y1241=y1241+y1245;y1249=y1249+y1253;y1257=y1257+y1261;y1265=y1265+y1269;y1273=y1273+y1277;y1281=y1281+y1285;y1289=y1289+y1293;y1297=y1297+y1301;y1305=y1305+y1309;y1313=y1313+y1317;y1321=y1321+y1325;y1329=y1329+y1333;y1337=y1337+y1341;

#Full depth 22 and Toffoli depth 3

y129=y129+y137;y145=y145+y153;y353=y353+y361;y369=y369+y377;y385=y385+y393;y426=y426+y434;y442=y442+y450;y458=y458+y491;y499=y499+y507;y515=y515+y523;y531=y531+y564;y572=y572+y580;y588=y588+y596;y1041=y1041+y1049;y1057=y1057+y1065;y1073=y1073+y1081;y1089=y1089+y1097;y1105=y1105+y1113;y1121=y1121+y1129;y1137=y1137+y1145;y1153=y1153+y1161;y1169=y1169+y1177;y1185=y1185+y1193;y1201=y1201+y1209;y1217=y1217+y1225;y1233=y1233+y1241;y1249=y1249+y1257;y1265=y1265+y1273;y1281=y1281+y1289;y1297=y1297+y1305;y1313=y1313+y1321;y1329=y1329+y1337;

#Full depth 23 and Toffoli depth 3

y97=y97+y129;y98=y98+y145;y99=y99+y353;y100=y100+y129;y101=y101+y145;y102=y102+y353;y103=y103+y369;y104=y104+y385;y105=y105+y129;y106=y106+y145;y107=y107+y353;y108=y108+y109;

$8+y369; y109=y109+y385; y110=y110+y426; y111=y111+y129; y112=y112+y145; y113=y113+y353; y114=y114+y369; y115=y115+y129; y116=y116+y145; y117=y117+y353; y118=y118+y369; y119=y119+y129; y120=y120+y145; y121=y121+y353; y122=y122+y369; y123=y123+y385; y124=y124+y426; y125=y125+y442; y126=y126+y458; y127=y127+y129; y128=y128+y145;$

#Full depth 24 and Toffoli depth 3

$y97=y97+y353; y98=y98+y369; y99=y99+y385; y100=y100+y369; y101=y101+y385; y102=y102+y426; y103=y103+y442; y104=y104+y458; y105=y105+y426; y106=y106+y442; y107=y107+y458; y108=y108+y499; y109=y109+y515; y110=y110+y531; y111=y111+y442; y112=y112+y458; y113=y113+y499; y114=y114+y515; y115=y115+y385; y116=y116+y426; y117=y117+y442; y118=y118+y458; y119=y119+y385; y120=y120+y426; y121=y121+y442; y122=y122+y458; y123=y123+y499; y124=y124+y515; y125=y125+y531; y126=y126+y572; y127=y127+y499; y128=y128+y515;$

#Full depth 25 and Toffoli depth 3

$y97=y97+y531; y98=y98+y572; y99=y99+y588; y100=y100+y426; y101=y101+y442; y102=y102+y458; y103=y103+y499; y104=y104+y515; y105=y105+y499; y106=y106+y515; y107=y107+y531; y108=y108+y572; y109=y109+y588; y110=y110+y1041; y111=y111+y572; y112=y112+y588; y113=y113+y1041; y114=y114+y1057; y115=y115+y531; y116=y116+y572; y117=y117+y588; y118=y118+y1041; y119=y119+y499; y120=y120+y515; y121=y121+y531; y122=y122+y572; y123=y123+y588; y124=y124+y1041; y125=y125+y1057; y126=y126+y1073; y127=y127+y588; y128=y128+y1041;$

#Full depth 26 and Toffoli depth 3

$y97=y97+y1057; y98=y98+y1073; y99=y99+y1089; y100=y100+y1041; y101=y101+y1057; y102=y102+y1073; y103=y103+y1089; y104=y104+y1105; y105=y105+y531; y106=y106+y572; y107=y107+y588; y108=y108+y1041; y109=y109+y1057; y110=y110+y1073; y111=y111+y1057; y112=y112+y1073; y113=y113+y1089; y114=y114+y1105; y115=y115+y1073; y116=y116+y1089; y117=y117+y1105; y118=y118+y1121; y119=y119+y1057; y120=y120+y1073; y121=y121+y1089; y122=y122+y1105; y123=y123+y1121; y124=y124+y1137; y125=y125+y1153; y126=y126+y1169; y127=y127+y1089; y128=y128+y1105;$

#Full depth 27 and Toffoli depth 3

$y97=y97+y1121; y98=y98+y1137; y99=y99+y1153; y100=y100+y1105; y101=y101+y1121; y102=y102+y1137; y103=y103+y1153; y104=y104+y1169; y105=y105+y1121; y106=y106+y1137; y107=y107+y1153; y108=y108+y1169; y109=y109+y1185; y110=y110+y1201; y111=y111+y1089; y112=y112+y1105; y113=y113+y1121; y114=y114+y1137; y115=y115+y1121; y116=y116+y1137; y117=y117+y1153; y118=y118+y1169; y119=y119+y1137; y120=y120+y1153; y121=y121+y1169; y122=y122+y1185; y123=y123+y1201; y124=y124+y1217; y125=y125+y1233; y126=y126+y1249; y127=y127+y1185; y128=y128+y1201;$

#Full depth 28 and Toffoli depth 3

$y97=y97+y1217; y98=y98+y1233; y99=y99+y1249; y100=y100+y1169; y101=y101+y1185; y102=y102+y1201; y103=y103+y1217; y104=y104+y1233; y105=y105+y1185; y106=y106+y1201; y107=y107+y1217; y108=y108+y1233; y109=y109+y1249; y110=y110+y1265; y111=y111+y1217; y112=y112+y1233; y113=y113+y1249; y114=y114+y1265; y115=y115+y1153; y116=y116+y1169; y117=y117+y1185; y118=y118+y1201; y119=y119+y1185; y120=y120+y1201; y121=y121+y1217; y122=y122+y1233; y123=y123+y1249; y124=y124+y1265; y125=y125+y1281; y126=y126+y1297; y127=y127+y1265; y128=y128+y1281;$

#Full depth 29 and Toffoli depth 3

$y97=y97+y1297; y98=y98+y1313; y99=y99+y1329; y100=y100+y1265; y101=y101+y1281; y102=y102+y1297; y103=y103+y1313; y104=y104+y1329; y105=y105+y1249; y106=y106+y1265; y107=y107+y1281; y108=y108+y1297; y109=y109+y1313; y110=y110+y1329; y111=y111+y1281; y112=y112+y1297; y113=y113+y1313; y114=y114+y1329; y115=y115+y1281; y116=y116+y1297; y117=y117+y1313; y118=y118+y1329; y119=y119+y1217; y120=y120+y1233; y121=y121+y1249; y122=y122+y1265; y123=y123+y1281; y124=y124+y1297; y125=y125+y1313; y126=y126+y1329; y127=y127+y1313; y128=y128+y1329;$

#Full depth 30 and Toffoli depth 3

$y129=y129+y137; y145=y145+y153; y353=y353+y361; y369=y369+y377; y385=y385+y393; y426=y426+y434; y442=y442+y450; y458=y458+y491; y499=y499+y507; y515=y515+y523; y531=y531+y564; y5$

72=y572+y580;y588=y588+y596;y1041=y1041+y1049;y1057=y1057+y1065;y1073=y1073+y1081;y1089=y1089+y1097;y1105=y1105+y1113;y1121=y1121+y1129;y1137=y1137+y1145;y1153=y1153+y1161;y1169=y1169+y1177;y1185=y1185+y1193;y1201=y1201+y1209;y1217=y1217+y1225;y1233=y1233+y1241;y1249=y1249+y1257;y1265=y1265+y1273;y1281=y1281+y1289;y1297=y1297+y1305;y1313=y1313+y1321;y1329=y1329+y1337;

#Full depth 31 and Toffoli depth 3

y129=y129+y133;y137=y137+y141;y145=y145+y149;y153=y153+y157;y353=y353+y357;y361=y361+y365;y369=y369+y373;y377=y377+y381;y385=y385+y389;y393=y393+y422;y426=y426+y430;y434=y434+y438;y442=y442+y446;y450=y450+y454;y458=y458+y462;y491=y491+y495;y499=y499+y503;y507=y507+y511;y515=y515+y519;y523=y523+y527;y531=y531+y560;y564=y564+y568;y572=y572+y576;y580=y580+y584;y588=y588+y592;y596=y596+y600;y1041=y1041+y1045;y1049=y1049+y1053;y1057=y1057+y1061;y1065=y1065+y1069;y1073=y1073+y1077;y1081=y1081+y1085;y1089=y1089+y1093;y1097=y1097+y1101;y1105=y1105+y1109;y1113=y1113+y1117;y1121=y1121+y1125;y1129=y1129+y1133;y1137=y1137+y1141;y1145=y1145+y1149;y1153=y1153+y1157;y1161=y1161+y1165;y1169=y1169+y1173;y1177=y1177+y1181;y1185=y1185+y1189;y1193=y1193+y1197;y1201=y1201+y1205;y1209=y1209+y1213;y1217=y1217+y1221;y1225=y1225+y1229;y1233=y1233+y1237;y1241=y1241+y1245;y1249=y1249+y1253;y1257=y1257+y1261;y1265=y1265+y1269;y1273=y1273+y1277;y1281=y1281+y1285;y1289=y1289+y1293;y1297=y1297+y1301;y1305=y1305+y1309;y1313=y1313+y1317;y1321=y1321+y1325;y1329=y1329+y1333;y1337=y1337+y1341;

#Full depth 32 and Toffoli depth 3

y129=y129+y131;y133=y133+y135;y137=y137+y139;y141=y141+y143;y145=y145+y147;y149=y149+y151;y153=y153+y155;y157=y157+y159;y353=y353+y355;y357=y357+y359;y361=y361+y363;y365=y365+y367;y369=y369+y371;y373=y373+y375;y377=y377+y379;y381=y381+y383;y385=y385+y387;y389=y389+y391;y393=y393+y395;y422=y422+y424;y426=y426+y428;y430=y430+y432;y434=y434+y436;y438=y438+y440;y442=y442+y444;y446=y446+y448;y450=y450+y452;y454=y454+y456;y458=y458+y460;y462=y462+y464;y491=y491+y493;y495=y495+y497;y499=y499+y501;y503=y503+y505;y507=y507+y509;y511=y511+y513;y515=y515+y517;y519=y519+y521;y523=y523+y525;y527=y527+y529;y531=y531+y533;y560=y560+y562;y564=y564+y566;y568=y568+y570;y572=y572+y574;y576=y576+y578;y580=y580+y582;y584=y584+y586;y588=y588+y590;y592=y592+y594;y596=y596+y598;y600=y600+y602;y1041=y1041+y1043;y1045=y1045+y1047;y1049=y1049+y1051;y1053=y1053+y1055;y1057=y1057+y1059;y1061=y1061+y1063;y1065=y1065+y1067;y1069=y1069+y1071;y1073=y1073+y1075;y1077=y1077+y1079;y1081=y1081+y1083;y1085=y1085+y1087;y1089=y1089+y1091;y1093=y1093+y1095;y1097=y1097+y1099;y1101=y1101+y1103;y1105=y1105+y1107;y1109=y1109+y1111;y1113=y1113+y1115;y1117=y1117+y1119;y1121=y1121+y1123;y1125=y1125+y1127;y1129=y1129+y1131;y1133=y1133+y1135;y1137=y1137+y1139;y1141=y1141+y1143;y1145=y1145+y1147;y1149=y1149+y1151;y1153=y1153+y1155;y1157=y1157+y1159;y1161=y1161+y1163;y1165=y1165+y1167;y1169=y1169+y1171;y1173=y1173+y1175;y1177=y1177+y1179;y1181=y1181+y1183;y1185=y1185+y1187;y1189=y1189+y1191;y1193=y1193+y1195;y1197=y1197+y1199;y1201=y1201+y1203;y1205=y1205+y1207;y1209=y1209+y1211;y1213=y1213+y1215;y1217=y1217+y1219;y1221=y1221+y1223;y1225=y1225+y1227;y1229=y1229+y1231;y1233=y1233+y1235;y1237=y1237+y1239;y1241=y1241+y1243;y1245=y1245+y1247;y1249=y1249+y1251;y1253=y1253+y1255;y1257=y1257+y1259;y1261=y1261+y1263;y1265=y1265+y1267;y1269=y1269+y1271;y1273=y1273+y1275;y1277=y1277+y1279;y1281=y1281+y1283;y1285=y1285+y1287;y1289=y1289+y1291;y1293=y1293+y1295;y1297=y1297+y1299;y1301=y1301+y1303;y1305=y1305+y1307;y1309=y1309+y1311;y1313=y1313+y1315;y1317=y1317+y1319;y1321=y1321+y1323;y1325=y1325+y1327;y1329=y1329+y1331;y1333=y1333+y1335;y1337=y1337+y1339;y1341=y1341+y1343;

#Full depth 33 and Toffoli depth 3

y129=y129+y130;y131=y131+y132;y133=y133+y134;y135=y135+y136;y137=y137+y138;y139=y139+y140;y141=y141+y142;y143=y143+y144;y145=y145+y146;y147=y147+y148;y149=y149+y150;y151=y151+y152;y153=y153+y154;y155=y155+y156;y157=y157+y158;y159=y159+y160;y353=y353+y354;y355=y355+y356;y357=y357+y358;y359=y359+y360;y361=y361+y362;y363=y363+y364;y365=y365+y366;y367=y367+y368;y369=y369+y370;y371=y371+y372;y373=y373+y374;y375=y375+y376;y377=y377+y378;y379=y379+y380;y381=y381+y382;y383=y383+y384;y385=y385+y386;y387=y387+y388;y389=y389+y390;y391=y391+y392;y393=y393+y394;y395=y395+y396;y422=y422+y423;y424=y424+y425;y426=y426+y427;y428=y428+y429;y430=y430+y431;y432=y432+y433;y434=y434+y435;y436=y436+y437;y438=y438+y439;y440=y440+y441;y442=y442+y443;y444=y444+y445;y446=y446+y447;y448=y448+y449;y450=y450+y451;y452=y452+y453;y454=y454+y455;y456=y456+y457;

$y_{457}; y_{458}=y_{458}+y_{459}; y_{460}=y_{460}+y_{461}; y_{462}=y_{462}+y_{463}; y_{464}=y_{464}+y_{465}; y_{491}=y_{491}+y_{492}; y_{493}=y_{493}+y_{494}; y_{495}=y_{495}+y_{496}; y_{497}=y_{497}+y_{498}; y_{499}=y_{499}+y_{500}; y_{501}=y_{501}+y_{502}; y_{503}=y_{503}+y_{504}; y_{505}=y_{505}+y_{506}; y_{507}=y_{507}+y_{508}; y_{509}=y_{509}+y_{510}; y_{511}=y_{511}+y_{512}; y_{513}=y_{513}+y_{514}; y_{515}=y_{515}+y_{516}; y_{517}=y_{517}+y_{518}; y_{519}=y_{519}+y_{520}; y_{521}=y_{521}+y_{522}; y_{523}=y_{523}+y_{524}; y_{525}=y_{525}+y_{526}; y_{527}=y_{527}+y_{528}; y_{529}=y_{529}+y_{530}; y_{531}=y_{531}+y_{532}; y_{533}=y_{533}+y_{534}; y_{560}=y_{560}+y_{561}; y_{562}=y_{562}+y_{563}; y_{564}=y_{564}+y_{565}; y_{566}=y_{566}+y_{567}; y_{568}=y_{568}+y_{569}; y_{570}=y_{570}+y_{571}; y_{572}=y_{572}+y_{573}; y_{574}=y_{574}+y_{575}; y_{576}=y_{576}+y_{577}; y_{578}=y_{578}+y_{579}; y_{580}=y_{580}+y_{581}; y_{582}=y_{582}+y_{583}; y_{584}=y_{584}+y_{585}; y_{586}=y_{586}+y_{587}; y_{588}=y_{588}+y_{589}; y_{590}=y_{590}+y_{591}; y_{592}=y_{592}+y_{593}; y_{594}=y_{594}+y_{595}; y_{596}=y_{596}+y_{597}; y_{598}=y_{598}+y_{599}; y_{600}=y_{600}+y_{601}; y_{602}=y_{602}+y_{603}; y_{1041}=y_{1041}+y_{1042}; y_{1043}=y_{1043}+y_{1044}; y_{1045}=y_{1045}+y_{1046}; y_{1047}=y_{1047}+y_{1048}; y_{1049}=y_{1049}+y_{1050}; y_{1051}=y_{1051}+y_{1052}; y_{1053}=y_{1053}+y_{1054}; y_{1055}=y_{1055}+y_{1056}; y_{1057}=y_{1057}+y_{1058}; y_{1059}=y_{1059}+y_{1060}; y_{1061}=y_{1061}+y_{1062}; y_{1063}=y_{1063}+y_{1064}; y_{1065}=y_{1065}+y_{1066}; y_{1067}=y_{1067}+y_{1068}; y_{1069}=y_{1069}+y_{1070}; y_{1071}=y_{1071}+y_{1072}; y_{1073}=y_{1073}+y_{1074}; y_{1075}=y_{1075}+y_{1076}; y_{1077}=y_{1077}+y_{1078}; y_{1079}=y_{1079}+y_{1080}; y_{1081}=y_{1081}+y_{1082}; y_{1083}=y_{1083}+y_{1084}; y_{1085}=y_{1085}+y_{1086}; y_{1087}=y_{1087}+y_{1088}; y_{1089}=y_{1089}+y_{1090}; y_{1091}=y_{1091}+y_{1092}; y_{1093}=y_{1093}+y_{1094}; y_{1095}=y_{1095}+y_{1096}; y_{1097}=y_{1097}+y_{1098}; y_{1099}=y_{1099}+y_{1100}; y_{1101}=y_{1101}+y_{1102}; y_{1103}=y_{1103}+y_{1104}; y_{1105}=y_{1105}+y_{1106}; y_{1107}=y_{1107}+y_{1108}; y_{1109}=y_{1109}+y_{1110}; y_{1111}=y_{1111}+y_{1112}; y_{1113}=y_{1113}+y_{1114}; y_{1115}=y_{1115}+y_{1116}; y_{1117}=y_{1117}+y_{1118}; y_{1119}=y_{1119}+y_{1120}; y_{1121}=y_{1121}+y_{1122}; y_{1123}=y_{1123}+y_{1124}; y_{1125}=y_{1125}+y_{1126}; y_{1127}=y_{1127}+y_{1128}; y_{1129}=y_{1129}+y_{1130}; y_{1131}=y_{1131}+y_{1132}; y_{1133}=y_{1133}+y_{1134}; y_{1135}=y_{1135}+y_{1136}; y_{1137}=y_{1137}+y_{1138}; y_{1139}=y_{1139}+y_{1140}; y_{1141}=y_{1141}+y_{1142}; y_{1143}=y_{1143}+y_{1144}; y_{1145}=y_{1145}+y_{1146}; y_{1147}=y_{1147}+y_{1148}; y_{1149}=y_{1149}+y_{1150}; y_{1151}=y_{1151}+y_{1152}; y_{1153}=y_{1153}+y_{1154}; y_{1155}=y_{1155}+y_{1156}; y_{1157}=y_{1157}+y_{1158}; y_{1159}=y_{1159}+y_{1160}; y_{1161}=y_{1161}+y_{1162}; y_{1163}=y_{1163}+y_{1164}; y_{1165}=y_{1165}+y_{1166}; y_{1167}=y_{1167}+y_{1168}; y_{1169}=y_{1169}+y_{1170}; y_{1171}=y_{1171}+y_{1172}; y_{1173}=y_{1173}+y_{1174}; y_{1175}=y_{1175}+y_{1176}; y_{1177}=y_{1177}+y_{1178}; y_{1179}=y_{1179}+y_{1180}; y_{1181}=y_{1181}+y_{1182}; y_{1183}=y_{1183}+y_{1184}; y_{1185}=y_{1185}+y_{1186}; y_{1187}=y_{1187}+y_{1188}; y_{1189}=y_{1189}+y_{1190}; y_{1191}=y_{1191}+y_{1192}; y_{1193}=y_{1193}+y_{1194}; y_{1195}=y_{1195}+y_{1196}; y_{1197}=y_{1197}+y_{1198}; y_{1199}=y_{1199}+y_{1200}; y_{1201}=y_{1201}+y_{1202}; y_{1203}=y_{1203}+y_{1204}; y_{1205}=y_{1205}+y_{1206}; y_{1207}=y_{1207}+y_{1208}; y_{1209}=y_{1209}+y_{1210}; y_{1211}=y_{1211}+y_{1212}; y_{1213}=y_{1213}+y_{1214}; y_{1215}=y_{1215}+y_{1216}; y_{1217}=y_{1217}+y_{1218}; y_{1219}=y_{1219}+y_{1220}; y_{1221}=y_{1221}+y_{1222}; y_{1223}=y_{1223}+y_{1224}; y_{1225}=y_{1225}+y_{1226}; y_{1227}=y_{1227}+y_{1228}; y_{1229}=y_{1229}+y_{1230}; y_{1231}=y_{1231}+y_{1232}; y_{1233}=y_{1233}+y_{1234}; y_{1235}=y_{1235}+y_{1236}; y_{1237}=y_{1237}+y_{1238}; y_{1239}=y_{1239}+y_{1240}; y_{1241}=y_{1241}+y_{1242}; y_{1243}=y_{1243}+y_{1244}; y_{1245}=y_{1245}+y_{1246}; y_{1247}=y_{1247}+y_{1248}; y_{1249}=y_{1249}+y_{1250}; y_{1251}=y_{1251}+y_{1252}; y_{1253}=y_{1253}+y_{1254}; y_{1255}=y_{1255}+y_{1256}; y_{1257}=y_{1257}+y_{1258}; y_{1259}=y_{1259}+y_{1260}; y_{1261}=y_{1261}+y_{1262}; y_{1263}=y_{1263}+y_{1264}; y_{1265}=y_{1265}+y_{1266}; y_{1267}=y_{1267}+y_{1268}; y_{1269}=y_{1269}+y_{1270}; y_{1271}=y_{1271}+y_{1272}; y_{1273}=y_{1273}+y_{1274}; y_{1275}=y_{1275}+y_{1276}; y_{1277}=y_{1277}+y_{1278}; y_{1279}=y_{1279}+y_{1280}; y_{1281}=y_{1281}+y_{1282}; y_{1283}=y_{1283}+y_{1284}; y_{1285}=y_{1285}+y_{1286}; y_{1287}=y_{1287}+y_{1288}; y_{1289}=y_{1289}+y_{1290}; y_{1291}=y_{1291}+y_{1292}; y_{1293}=y_{1293}+y_{1294}; y_{1295}=y_{1295}+y_{1296}; y_{1297}=y_{1297}+y_{1298}; y_{1299}=y_{1299}+y_{1300}; y_{1301}=y_{1301}+y_{1302}; y_{1303}=y_{1303}+y_{1304}; y_{1305}=y_{1305}+y_{1306}; y_{1307}=y_{1307}+y_{1308}; y_{1309}=y_{1309}+y_{1310}; y_{1311}=y_{1311}+y_{1312}; y_{1313}=y_{1313}+y_{1314}; y_{1315}=y_{1315}+y_{1316}; y_{1317}=y_{1317}+y_{1318}; y_{1319}=y_{1319}+y_{1320}; y_{1321}=y_{1321}+y_{1322}; y_{1323}=y_{1323}+y_{1324}; y_{1325}=y_{1325}+y_{1326}; y_{1327}=y_{1327}+y_{1328}; y_{1329}=y_{1329}+y_{1330}; y_{1331}=y_{1331}+y_{1332}; y_{1333}=y_{1333}+y_{1334}; y_{1335}=y_{1335}+y_{1336}; y_{1337}=y_{1337}+y_{1338}; y_{1339}=y_{1339}+y_{1340}; y_{1341}=y_{1341}+y_{1342}; y_{1343}=y_{1343}+y_{1344};$

#Full depth 34 and Toffoli depth 4

$y_{129}=y_{129}+y_{240}*y_{405}; y_{130}=y_{130}+y_{239}*y_{406}; y_{131}=y_{131}+y_{238}*y_{398}; y_{132}=y_{132}+y_{237}*y_{629}; y_{133}=y_{133}+y_{236}*y_{407}; y_{134}=y_{134}+y_{235}*y_{630}; y_{135}=y_{135}+y_{234}*y_{413}; y_{136}=y_{136}+y_{233}*y_{631}; y_{137}=y_{137}+y_{232}*y_{419}; y_{138}=y_{138}+y_{231}*y_{632}; y_{139}=y_{139}+y_{230}*y_{402}; y_{140}=y_{140}+y_{229}*y_{633}; y_{141}=y_{141}+y_{228}*y_{634}; y_{142}=y_{142}+y_{227}*y_{635}; y_{143}=y_{143}+y_{226}*y_{397}; y_{144}=y_{144}+y_{225}*y_{636}; y_{145}=y_{145}+y_{240}*y_{637}; y_{146}=y_{146}+y_{239}*y_{638}; y_{147}=y_{147}+y_{238}*y_{420}; y_{148}=y_{148}+y_{237}*y_{409}; y_{149}=y_{149}+y_{236}*y_{639}; y_{150}=y_{150}+y_{235}*y_{640}; y_{151}=y_{151}+y_{234}*y_{410}; y_{152}=y_{152}+y_{233}*y_{641}; y_{153}=y_{153}+y_{232}*y_{642}; y_{154}=y_{154}+y_{231}*y_{643}; y_{155}=y_{155}+y_{230}*y_{644}; y_{156}=y_{156}+y_{229}*y_{645}; y_{157}=y_{157}+y_{228}*y_{646}; y_{158}=y_{158}+y_{227}*y_{647}; y_{159}=y_{159}+y_{226}*y_{648}; y_{160}=y_{160}+y_{225}*y_{649}; y_{353}=y_{353}+y_{240}*y_{650}; y_{354}=y_{354}+y_{239}*y_{651}; y_{355}=y_{355}+y_{238}*y_{652}; y_{356}=y_{356}+y_{237}*y_{404}; y_{357}=y_{357}+y_{236}*y_{653}; y_{358}=y_{358}+y_{235}*y_{654}; y_{359}=y_{359}+y_{234}*y_{655}; y_{360}=y_{360}+y_{233}*y_{656}; y_{361}=y_{361}+y_{232}*y_{414}; y_{362}=y_{362}+y_{231}*y_{657}; y_{363}=y_{363}+y_{230}*y_{658}; y_{364}=y_{364}+y_{229}*y_{412}; y_{365}=y_{365}+y_{228}*y_{659}; y_{366}=y_{366}+y_{227}*y_{660}; y_{367}=y_{367}+y_{226}*y_{661}; y_{368}=y_{368}+y_{225}*y_{662}; y_{369}=y_{369}+y_{240}*y_{663}; y_{370}=y_{370}+y_{239}*y_{664}; y_{371}=y_{371}+y_{238}*y_{665}; y_{372}=y_{372}+y_{237}*y_{666}; y_{373}=y_{373}+y_{236}*y_{667}; y_{374}=y_{374}+y_{235}*y_{668}; y_{375}=y_{375}+y_{234}*y_{669}; y_{376}=y_{376}+y_{233}*y_{670}; y_{377}=y_{377}+y_{232}*y_{671}; y_{378}=y_{378}+y_{231}*y_{672}; y_{379}=y_{379}+y_{230}*y_{673}; y_{380}=y_{380}+y_{229}*y_{674}; y_{381}=y_{381}+y_{228}*y_{675}; y_{382}=y_{382}+y_{227}*y_{676}; y_{383}=y_{383}+y_{226}*y_{677}; y_{384}=y_{384}+y_{225}*y_{678}; y_{385}=y_{385}+y_{240}*y_{679}; y_{386}=y_{386}+y_{239}*y_{680}; y_{387}=y_{387}+$

$y238*y681; y388=y388+y237*y682; y389=y389+y236*y683; y390=y390+y235*y684; y391=y391+y234*y685; y392=y392+y233*y686; y393=y393+y232*y687; y394=y394+y231*y688; y395=y395+y230*y689; y396=y396+y229*y690; y422=y422+y228*y691; y423=y423+y227*y692; y424=y424+y226*y693; y425=y425+y225*y694; y426=y426+y224*y695; y427=y427+y223*y696; y428=y428+y222*y697; y429=y429+y221*y698; y430=y430+y220*y699; y431=y431+y219*y700; y432=y432+y218*y701; y433=y433+y217*y702; y434=y434+y216*y703; y435=y435+y215*y704; y436=y436+y214*y705; y437=y437+y213*y706; y438=y438+y212*y707; y439=y439+y211*y708; y440=y440+y210*y709; y441=y441+y209*y710; y442=y442+y208*y711; y443=y443+y207*y712; y444=y444+y206*y713; y445=y445+y205*y714; y446=y446+y204*y715; y447=y447+y203*y716; y448=y448+y202*y717; y449=y449+y201*y718; y450=y450+y200*y719; y451=y451+y199*y720; y452=y452+y198*y721; y453=y453+y197*y722; y454=y454+y196*y723; y455=y455+y195*y724; y456=y456+y194*y725; y457=y457+y193*y726; y458=y458+y192*y727; y459=y459+y191*y728; y460=y460+y190*y729; y461=y461+y189*y730; y462=y462+y188*y731; y463=y463+y187*y732; y464=y464+y186*y733; y465=y465+y185*y734; y466=y466+y184*y735; y467=y467+y183*y736; y468=y468+y182*y737; y469=y469+y181*y738; y470=y470+y180*y739; y471=y471+y179*y740; y472=y472+y178*y741; y473=y473+y177*y742; y474=y474+y176*y743; y475=y475+y175*y744; y476=y476+y174*y745; y477=y477+y173*y746; y478=y478+y172*y747; y479=y479+y171*y748; y480=y480+y170*y749; y481=y481+y169*y750; y482=y482+y168*y751; y483=y483+y167*y752; y484=y484+y166*y753; y485=y485+y165*y754; y486=y486+y164*y755; y487=y487+y163*y756; y488=y488+y162*y757; y489=y489+y161*y758; y490=y490+y160*y759; y491=y491+y159*y760; y492=y492+y158*y761; y493=y493+y157*y762; y494=y494+y156*y763; y495=y495+y155*y764; y496=y496+y154*y765; y497=y497+y153*y766; y498=y498+y152*y767; y499=y499+y151*y768; y500=y500+y150*y769; y501=y501+y149*y770; y502=y502+y148*y771; y503=y503+y147*y772; y504=y504+y146*y773; y505=y505+y145*y774; y506=y506+y144*y775; y507=y507+y143*y776; y508=y508+y142*y777; y509=y509+y141*y778; y510=y510+y140*y779; y511=y511+y139*y780; y512=y512+y138*y781; y513=y513+y137*y782; y514=y514+y136*y783; y515=y515+y135*y784; y516=y516+y134*y785; y517=y517+y133*y786; y518=y518+y132*y787; y519=y519+y131*y788; y520=y520+y130*y789; y521=y521+y129*y790; y522=y522+y128*y791; y523=y523+y127*y792; y524=y524+y126*y793; y525=y525+y125*y794; y526=y526+y124*y795; y527=y527+y123*y796; y528=y528+y122*y797; y529=y529+y121*y798; y530=y530+y120*y799; y531=y531+y119*y800; y532=y532+y118*y801; y533=y533+y117*y802; y534=y534+y116*y803; y535=y535+y115*y804; y536=y536+y114*y805; y537=y537+y113*y806; y538=y538+y112*y807; y539=y539+y111*y808; y540=y540+y110*y809; y541=y541+y109*y810; y542=y542+y108*y811; y543=y543+y107*y812; y544=y544+y106*y813; y545=y545+y105*y814; y546=y546+y104*y815; y547=y547+y103*y816; y548=y548+y102*y817; y549=y549+y101*y818; y550=y550+y100*y819; y551=y551+y99*y820; y552=y552+y98*y821; y553=y553+y97*y822; y554=y554+y96*y823; y555=y555+y95*y824; y556=y556+y94*y825; y557=y557+y93*y826; y558=y558+y92*y827; y559=y559+y91*y828; y560=y560+y90*y829; y561=y561+y89*y830; y562=y562+y88*y831; y563=y563+y87*y832; y564=y564+y86*y833; y565=y565+y85*y834; y566=y566+y84*y835; y567=y567+y83*y836; y568=y568+y82*y837; y569=y569+y81*y838; y570=y570+y80*y839; y571=y571+y79*y840; y572=y572+y78*y841; y573=y573+y77*y842; y574=y574+y76*y843; y575=y575+y75*y844; y576=y576+y74*y845; y577=y577+y73*y846; y578=y578+y72*y847; y579=y579+y71*y848; y580=y580+y70*y849; y581=y581+y69*y850; y582=y582+y68*y851; y583=y583+y67*y852; y584=y584+y66*y853; y585=y585+y65*y854; y586=y586+y64*y855; y587=y587+y63*y856; y588=y588+y62*y857; y589=y589+y61*y858; y590=y590+y60*y859; y591=y591+y59*y860; y592=y592+y58*y861; y593=y593+y57*y862; y594=y594+y56*y863; y595=y595+y55*y864; y596=y596+y54*y865; y597=y597+y53*y866; y598=y598+y52*y867; y599=y599+y51*y868; y600=y600+y50*y869; y601=y601+y49*y870; y602=y602+y48*y871; y603=y603+y47*y872; y604=y604+y46*y873; y605=y605+y45*y874; y606=y606+y44*y875; y607=y607+y43*y876; y608=y608+y42*y877; y609=y609+y41*y878; y610=y610+y40*y879; y611=y611+y39*y880; y612=y612+y38*y881; y613=y613+y37*y882; y614=y614+y36*y883; y615=y615+y35*y884; y616=y616+y34*y885; y617=y617+y33*y886; y618=y618+y32*y887; y619=y619+y31*y888; y620=y620+y30*y889; y621=y621+y29*y890; y622=y622+y28*y891; y623=y623+y27*y892; y624=y624+y26*y893; y625=y625+y25*y894; y626=y626+y24*y895; y627=y627+y23*y896; y628=y628+y22*y897; y629=y629+y21*y898; y630=y630+y20*y899; y631=y631+y19*y900; y632=y632+y18*y901; y633=y633+y17*y902; y634=y634+y16*y903; y635=y635+y15*y904; y636=y636+y14*y905; y637=y637+y13*y906; y638=y638+y12*y907; y639=y639+y11*y908; y640=y640+y10*y909; y641=y641+y9*y910; y642=y642+y8*y911; y643=y643+y7*y912; y644=y644+y6*y913; y645=y645+y5*y914; y646=y646+y4*y915; y647=y647+y3*y916; y648=y648+y2*y917; y649=y649+y1*y918; y650=y650+y919; y651=y651+y920; y652=y652+y921; y653=y653+y922; y654=y654+y923; y655=y655+y924; y656=y656+y925; y657=y657+y926; y658=y658+y927; y659=y659+y928; y660=y660+y929; y661=y661+y930; y662=y662+y931; y663=y663+y932; y664=y664+y933; y665=y665+y934; y666=y666+y935; y667=y667+y936; y668=y668+y937; y669=y669+y938; y670=y670+y939; y671=y671+y940; y672=y672+y941; y673=y673+y942; y674=y674+y943; y675=y675+y944; y676=y676+y945; y677=y677+y946; y678=y678+y947; y679=y679+y948; y680=y680+y949; y681=y681+y950; y682=y682+y951; y683=y683+y952; y684=y684+y953; y685=y685+y954; y686=y686+y955; y687=y687+y956; y688=y688+y957; y689=y689+y958; y690=y690+y959; y691=y691+y960; y692=y692+y961; y693=y693+y962; y694=y694+y963; y695=y695+y964; y696=y696+y965; y697=y697+y966; y698=y698+y967; y699=y699+y968; y700=y700+y969; y701=y701+y970; y702=y702+y971; y703=y703+y972; y704=y704+y973; y705=y705+y974; y706=y706+y975; y707=y707+y976; y708=y708+y977; y709=y709+y978; y710=y710+y979; y711=y711+y980; y712=y712+y981; y713=y713+y982; y714=y714+y983; y715=y715+y984; y716=y716+y985; y717=y717+y986; y718=y718+y987; y719=y719+y988; y720=y720+y989; y721=y721+y990; y722=y722+y991; y723=y723+y992; y724=y724+y993; y725=y725+y994; y726=y726+y995; y727=y727+y996; y728=y728+y997; y729=y729+y998; y730=y730+y999; y731=y731+y1000; y732=y732+y1001; y733=y733+y1002; y734=y734+y1003; y735=y735+y1004; y736=y736+y1005; y737=y737+y1006; y738=y738+y1007; y739=y739+y1008; y740=y740+y1009; y741=y741+y1010; y742=y742+y1011; y743=y743+y1012; y744=y744+y1013; y745=y745+y1014; y746=y746+y1015; y747=y747+y1016; y748=y748+y1017; y749=y749+y1018; y750=y750+y1019; y751=y751+y1020; y752=y752+y1021; y753=y753+y1022; y754=y754+y1023; y755=y755+y1024; y756=y756+y1025; y757=y757+y1026; y758=y758+y1027; y759=y759+y1028; y760=y760+y1029; y761=y761+y1030; y762=y762+y1031; y763=y763+y1032; y764=y764+y1033; y765=y765+y1034; y766=y766+y1035; y767=y767+y1036; y768=y768+y1037; y769=y769+y1038; y770=y770+y1039; y771=y771+y1040; y772=y772+y1041; y773=y773+y1042; y774=y774+y1043; y775=y775+y1044; y776=y776+y1045; y777=y777+y1046; y778=y778+y1047; y779=y779+y1048; y780=y780+y1049; y781=y781+y1050; y782=y782+y1051; y783=y783+y1052; y784=y784+y1053; y785=y785+y1054; y786=y786+y1055; y787=y787+y1056; y788=y788+y1057; y789=y789+y1058; y790=y790+y1059; y791=y791+y1060; y792=y792+y1061; y793=y793+y1062; y794=y794+y1063; y795=y795+y1064; y796=y796+y1065; y797=y797+y1066; y798=y798+y1067; y799=y799+y1068; y800=y800+y1069; y801=y801+y1070; y802=y802+y1071; y803=y803+y1072; y804=y804+y1073; y805=y805+y1074; y806=y806+y1075; y807=y807+y1076; y808=y808+y1077; y809=y809+y1078; y810=y810+y1079; y811=y811+y1080; y812=y812+y1081; y813=y813+y1082; y814=y814+y1083; y815=y815+y1084; y816=y816+y1085; y817=y817+y1086; y818=y818+y1087; y819=y819+y1088; y820=y820+y1089; y821=y821+y1090; y822=y822+y1091; y823=y823+y1092; y824=y824+y1093; y825=y825+y1094; y826=y826+y1095; y827=y827+y1096; y828=y828+y1097; y829=y829+y1098; y830=y830+y1099; y831=y831+y1100; y832=y832+y1101; y833=y833+y1102; y834=y834+y1103; y835=y835+y1104; y836=y836+y1105; y837=y837+y1106; y838=y838+y1107; y839=y839+y1108; y840=y840+y1109; y841=y841+y1110; y842=y842+y1111; y843=y843+y1112; y844=y844+y1113; y845=y845+y1114; y846=y846+y1115; y847=y847+y1116; y848=y848+y1117; y849=y849+y1118; y850=y850+y1119; y851=y851+y1120; y852=y852+y1121; y853=y853+y1122; y854=y854+y1123; y855=y855+y1124; y856=y856+y1125; y857=y857+y1126; y858=y858+y1127; y859=y859+y1128; y860=y860+y1129; y861=y861+y1130; y862=y862+y1131; y863=y863+y1132; y864=y864+y1133; y865=y865+y1134; y866=y866+y1135; y867=y867+y1136; y868=y868+y1137; y869=y869+y1138; y870=y870+y1139; y871=y871+y1140; y872=y872+y1141; y873=y873+y1142; y874=y874+y1143; y875=y875+y1144; y876=y876+y1145; y877=y877+y1146; y878=y878+y1147; y879=y879+y1148; y880=y880+y1149; y881=y881+y1150; y882=y882+y1151; y883=y883+y1152; y884=y884+y1153; y885=y885+y1154; y886=y886+y1155; y887=y887+y1156; y888=y888+y1157; y889=y889+y1158; y890=y890+y1159; y891=y891+y1160; y892=y892+y1161; y893=y893+y1162; y894=y894+y1163; y895=y895+y1164; y896=y896+y1165; y897=y897+y1166; y898=y898+y1167; y899=y899+y1168; y900=y900+y1169; y901=y901+y1170; y902=y902+y1171; y903=y903+y1172; y904=y904+y1173; y905=y905+y1174; y906=y906+y1175; y907=y907+y1176; y908=y908+y1177; y909=y909+y1178; y910=y910+y1179; y911=y911+y1180; y912=y912+y1181; y913=y913+y1182; y914=y914+y1183; y915=y915+y1184; y916=y916+y1185; y917=y917+y1186; y918=y918+y1187; y919=y919+y1188; y920=y920+y1189; y921=y921+y1190; y922=y922+y1191; y923=y923+y1192; y924=y924+y1193; y925=y925+y1194; y926=y926+y1195; y927=y927+y1196; y928=y928+y1197; y929=y929+y1198; y930=y930+y1199; y931=y931+y1200; y932=y932+y1201; y933=y933+y1202; y934=y934+y1203; y935=y935+y1204; y936=y936+y1205; y937=y937+y1206; y938=y938+y1207; y939=y939+y1208; y940=y940+y1209; y941=y941+y1210; y942=y942+y1211; y943=y943+y1212; y944=y944+y1213; y945=y945+y1214; y946=y946+y1215; y947=y947+y1216; y948=y948+y1217; y949=y949+y1218; y950=y950+y1219; y951=y951+y1220; y952=y952+y1221; y953=y953+y1222; y954=y954+y1223; y955=y955+y1224; y956=y956+y1225; y957=y957+y1226; y958=y958+y1227; y959=y959+y1228; y960=y960+y1229; y961=y961+y1230; y962=y962+y1231; y963=y963+y1232; y964=y964+y1233; y965=y965+y1234; y966=y966+y1235; y967=y967+y1236; y968=y968+y1237; y969=y969+y1238; y970=y970+y1239; y971=y971+y1240; y972=y972+y1241; y973=y973+y1242; y974=y974+y1243; y975=y975+y1244; y976=y976+y1245; y977=y977+y1246; y978=y978+y1247; y979=y979+y1248; y980=y980+y1249; y981=y981+y1250; y982=y982+y1251; y983=y983+y1252; y984=y984+y1253; y985=y985+y1254; y986=y986+y1255; y987=y987+y1256; y988=y988+y1257; y989=y989+y1258; y990=y990+y1259; y991=y991+y1260; y992=y992+y1261; y993=y993+y1262; y994=y994+y1263; y995=y995+y1264; y996=y996+y1265; y997=y997+y1266; y998=y998+y1267; y999=y999+y1268; y1000=y1000+y1269; y1001=y1001+y1270; y1002=y1002+y1271; y1003=y1003+y1272; y1004=y1004+y1273; y1005=y1005+y1274; y1006=y1006+y1275; y1007=y1007+y1276; y1008=y1008+y1277; y1009=y1009+y1278; y1010=y1010+y1279; y1011=y1011+y1280; y1012=y1012+y1281; y1013=y1013+y1282; y1014=y1014+y1283; y1015=y1015+y1284; y1016=y1016+y1285; y1017=y1017+y1286; y1018=y1018+y1287; y1019=y1019+y1288; y1020=y1020+y1289; y1021=y1021+y1290; y1022=y1022+y1291; y1023=y1023+y1292; y1024=y1024+y1293; y1025=y1025+y1294; y1026=y1026+y1295; y1027=y1027+y1296; y1028=y1028+y1297; y1029=y1029+y1298; y1030=y1030+y1299; y1031=y1031+y1300; y1032=y1032+y1301; y1033=y1033+y1302; y1034=y1034+y1303; y1035=y1035+y1304; y1036=y1036+y1305; y1037=y1037+y1306; y1038=y1038+y1307; y1039=y1039+y1308; y1040=y1040+y1309; y1041=y1041+y1310; y1042=y1042+y1311; y1043=y1043+y1312; y1044=y1044+y1313; y1045=y1045+y1314; y1046=y1046+y1315; y1047=y1047+y1316; y1048=y1048+y1317; y1049=y1049+y1318; y1050=y1050+y1319; y1051=y1051+y1320; y1052=y1052+y1321; y1053=y1053+y1322; y1054=y1054+y1323; y1055=y1055+y1324; y1056=y1056+y1325; y1057=y1057+y1326; y1058=y1058+y1327; y1059=y1059+y1328; y1060=y1060+y1329; y1061=y1061+y1330; y1062=y1062+y1331; y1063=y1063+y1332; y1064=y1064+y1333; y1065=y1065+y1334; y1066=y1066+y1335; y1067=y1067+y1336; y1068=y1068+y1337; y1069=y1069+y1338; y1070=y1070+y1339; y1071=y1071+y1340; y1072=y1072+y1341; y1073=y1073+y1342; y1074=y1074+y1343; y1075=y1075+y1344; y1076=y1076+y1345; y1077=y1077+y1346; y1078=y1078+y1347; y1079=y1079+y1348; y1080=y1080+y1349; y1081=y1081+y1350; y1082=y1082+y1351; y1083=y1083+y1352; y1084=y1084+y1353; y1085=y1085+y1354; y1086=y1086+y1355; y1087=y1087+y1356; y1088=y1088+y1357; y1089=y1089+y1358; y1090=y1090+y1359; y1091=y1091+y1360; y1092=y1092+y1361; y1093=y1093+y1362; y1094=y1094+y1363; y1095=y1095+y1364; y1096=y1096+y1365; y1097=y1097+y1366; y1098=y1098+y1367; y1099=y1099+y1368; y1100=y1100+y1369; y1101=y1101+y1370; y1102=y1102+y1371; y1103=y1103+y1372; y1104=y1104+y1373; y1105=y1105+y1374; y1106=y1106+y1375; y1107=y1107+y1376; y1108=y1108+y1377; y1109=y1109+y1378; y1110=y1110+y1379; y1111=y1111+y1380; y1112=y1112+y1381; y1113=y1113+y1382; y1114=y1114+y1383; y1115=y1115+y1384; y1116=y1116+y1385; y1117=y1117+y1386; y1118=y1118+y1387; y1119=y1119+y1388; y1120=y1120+y1389; y1121=y1121+y1390; y1122=y1122+y1391; y1123=y1123+y1392; y1124=y1124+y1393; y1125=y1125+y1394; y1126=y1126+y1395; y1127=y1127+y1396; y1128=y1128+y1397; y1129=y1129+y1398; y1130=y1130+y1399; y1131=y1131+y1400; y1132=y1132+y1401; y1133=y1133+y1402; y1134=y1134+y1403; y1135=y1135+y1404; y1136=y1136+y1405; y1137=y1137+y1406; y1138=y1138+y1407; y1139=y1139+y1408; y1140=y1140+y1409; y1141=y1141+y1410; y1142=y1142+y1411; y1143=y1143+y1412; y1144=y1144+y1413; y1145=y1145+y1414; y1146=y1146+y1415; y1147=y1147+y1416; y1148=y1148+y1417; y1149=y1149+y1418; y1150=y1150+y1419; y1151=y1151+y1420; y1152=y1152+y1421; y1153=y1153+y1422; y1154=y1154+y1423; y1155=y1155+y1424; y1156=y1156+y1425; y1157=y1157+y1426; y1158=y1158+y1427; y1159=y1159+y1428; y1160=y1160+y1429; y1161=y1161+y1430; y1162=y1162+y1431; y1163=y1163+y1432; y1164=y1164+y1433; y1165=y1165+y1434; y1166=y1166+y1435; y1167=y1167+y1436; y1168=y1168+y1437; y1169=y1169+y1438; y1170=y1170+y1439; y1171=y1171+y1440; y1172=y1172+y1441; y1173=y1173+y1442; y1174=y1174+y1443; y1175=y1175+y1444; y1176=y1176+y1445; y1177=y1177+y1446; y1178=y1178+y1447; y1179=y1179+y1448; y1180=y1180+y1449; y1181=y1181+y1450; y1182=y1182+y1451; y1183=y1183+y1452; y1184=y1184+y1453; y1185=y1185+y1454; y1186=y1186+y1455; y1187=y1187+y1456; y1188=y1188+y1457; y1189=y1189+y1458; y1190=y1190+y1459; y1191=y1191+y1460; y1192=y1192+y1461; y1193=y1193+y1462; y1194=y1194+y1463; y1195=y1195+y1464; y1196=y1196+y1465; y1197=y1197+y1466; y1198=y1198+y1467; y1199=y1199+y1468; y1200=y1200+y1469; y1201=y1201+y1470; y1202=y1202+y1471; y1203=y1203+y1472; y1204=y1204+y1473; y1205=y1205+y1474; y1206=y1206+y1475; y1207=y1207+y1476; y1208=y1208+y1477; y1209=y1209+y1478; y1210=y1210+y1479; y1211=y1211+y1480; y1212=y1212+y1481; y1213=y1213+y1482; y1214=y1214+y1483; y1215=y1215+y1484; y1216=y1216+y1485; y1217=y1217+y1486; y1218=y1218+y1487; y1219=y1219+y1488; y1220=y1220+y1489; y1221=y1221+y1490$

;y1139=y1139+y302*y871;y1140=y1140+y301*y872;y1141=y1141+y300*y873;y1142=y1142+y299*y874;y1143=y1143+y298*y875;y1144=y1144+y297*y876;y1145=y1145+y296*y877;y1146=y1146+y295*y878;y1147=y1147+y294*y879;y1148=y1148+y293*y880;y1149=y1149+y292*y881;y1150=y1150+y291*y882;y1151=y1151+y290*y883;y1152=y1152+y289*y884;y1153=y1153+y304*y885;y1154=y1154+y303*y886;y1155=y1155+y302*y887;y1156=y1156+y301*y888;y1157=y1157+y300*y889;y1158=y1158+y299*y890;y1159=y1159+y298*y891;y1160=y1160+y297*y892;y1161=y1161+y296*y893;y1162=y1162+y295*y894;y1163=y1163+y294*y895;y1164=y1164+y293*y896;y1165=y1165+y292*y897;y1166=y1166+y291*y898;y1167=y1167+y290*y899;y1168=y1168+y289*y900;y1169=y1169+y304*y901;y1170=y1170+y303*y902;y1171=y1171+y302*y903;y1172=y1172+y301*y904;y1173=y1173+y300*y905;y1174=y1174+y299*y906;y1175=y1175+y298*y907;y1176=y1176+y297*y908;y1177=y1177+y296*y909;y1178=y1178+y295*y910;y1179=y1179+y294*y911;y1180=y1180+y293*y912;y1181=y1181+y292*y913;y1182=y1182+y291*y914;y1183=y1183+y290*y915;y1184=y1184+y289*y916;y1185=y1185+y304*y917;y1186=y1186+y303*y918;y1187=y1187+y302*y919;y1188=y1188+y301*y920;y1189=y1189+y300*y921;y1190=y1190+y299*y922;y1191=y1191+y298*y923;y1192=y1192+y297*y924;y1193=y1193+y296*y925;y1194=y1194+y295*y926;y1195=y1195+y294*y927;y1196=y1196+y293*y928;y1197=y1197+y292*y929;y1198=y1198+y291*y930;y1199=y1199+y304*y931;y1200=y1200+y303*y932;y1201=y1201+y302*y933;y1202=y1202+y301*y934;y1203=y1203+y300*y935;y1204=y1204+y299*y936;y1205=y1205+y298*y937;y1206=y1206+y297*y938;y1207=y1207+y296*y939;y1208=y1208+y295*y940;y1209=y1209+y294*y941;y1210=y1210+y293*y942;y1211=y1211+y292*y943;y1212=y1212+y291*y944;y1213=y1213+y290*y945;y1214=y1214+y289*y946;y1215=y1215+y304*y947;y1216=y1216+y303*y948;y1217=y1217+y302*y949;y1218=y1218+y301*y950;y1219=y1219+y300*y951;y1220=y1220+y299*y952;y1221=y1221+y298*y953;y1222=y1222+y297*y954;y1223=y1223+y296*y955;y1224=y1224+y295*y956;y1225=y1225+y294*y957;y1226=y1226+y293*y958;y1227=y1227+y292*y959;y1228=y1228+y291*y960;y1229=y1229+y304*y961;y1230=y1230+y303*y962;y1231=y1231+y302*y963;y1232=y1232+y301*y964;y1233=y1233+y300*y965;y1234=y1234+y299*y966;y1235=y1235+y298*y967;y1236=y1236+y297*y968;y1237=y1237+y296*y969;y1238=y1238+y295*y970;y1239=y1239+y294*y971;y1240=y1240+y293*y972;y1241=y1241+y292*y973;y1242=y1242+y291*y974;y1243=y1243+y290*y975;y1244=y1244+y289*y976;y1245=y1245+y304*y977;y1246=y1246+y303*y978;y1247=y1247+y302*y979;y1248=y1248+y301*y980;y1249=y1249+y300*y981;y1250=y1250+y299*y982;y1251=y1251+y298*y983;y1252=y1252+y297*y984;y1253=y1253+y296*y985;y1254=y1254+y295*y986;y1255=y1255+y294*y987;y1256=y1256+y293*y988;y1257=y1257+y292*y989;y1258=y1258+y291*y990;y1259=y1259+y304*y991;y1260=y1260+y303*y992;y1261=y1261+y302*y993;y1262=y1262+y301*y994;y1263=y1263+y300*y995;y1264=y1264+y299*y996;y1265=y1265+y298*y997;y1266=y1266+y297*y998;y1267=y1267+y296*y999;y1268=y1268+y295*y1000;y1269=y1269+y294*y1001;y1270=y1270+y293*y1002;y1271=y1271+y292*y1003;y1272=y1272+y291*y1004;y1273=y1273+y290*y1005;y1274=y1274+y289*y1006;y1275=y1275+y304*y1007;y1276=y1276+y303*y1008;y1277=y1277+y302*y1009;y1278=y1278+y301*y1010;y1279=y1279+y300*y1011;y1280=y1280+y299*y1012;y1281=y1281+y298*y1013;y1282=y1282+y297*y1014;y1283=y1283+y296*y1015;y1284=y1284+y295*y1016;y1285=y1285+y294*y1017;y1286=y1286+y293*y1018;y1287=y1287+y292*y1019;y1288=y1288+y291*y1020;y1289=y1289+y304*y1021;y1290=y1290+y303*y1022;y1291=y1291+y302*y1023;y1292=y1292+y301*y1024;y1293=y1293+y300*y1025;y1294=y1294+y299*y1026;y1295=y1295+y298*y1027;y1296=y1296+y297*y1028;y1297=y1297+y296*y1029;y1298=y1298+y295*y1030;y1299=y1299+y294*y1031;y1300=y1300+y293*y1032;y1301=y1301+y292*y1033;y1302=y1302+y291*y1034;y1303=y1303+y290*y1035;y1304=y1304+y289*y1036;y1305=y1305+y304*y1037;y1306=y1306+y303*y1038;y1307=y1307+y302*y1039;y1308=y1308+y301*y1040;

#Full depth 35 and Toffoli depth 4

y353=y353+y247;y354=y354+y251;y355=y355+y242;y356=y356+y250;y357=y357+y243;y358=y358+y244;y359=y359+y248;y360=y360+y241;y361=y361+y254;y362=y362+y246;y363=y363+y244;y3

64=y364+y242;y365=y365+y249;y366=y366+y241;y367=y367+y241;y368=y368+y244;y369=y369+y247;y370=y370+y243;y371=y371+y245;y372=y372+y253;y373=y373+y244;y374=y374+y248;y375=y375+y242;y376=y376+y252;y377=y377+y250;y378=y378+y251;y379=y379+y255;y380=y380+y243;y381=y381+y248;y382=y382+y249;y383=y383+y245;y384=y384+y247;y385=y385+y241;y386=y386+y253;y387=y387+y246;y388=y388+y241;y389=y389+y248;y390=y390+y247;y391=y391+y250;y392=y392+y246;y393=y393+y244;y394=y394+y245;y395=y395+y246;y396=y396+y241;y422=y422+y279;y423=y423+y283;y424=y424+y274;y425=y425+y282;y426=y426+y275;y427=y427+y277;y428=y428+y280;y429=y429+y273;y430=y430+y286;y431=y431+y278;y432=y432+y276;y433=y433+y274;y434=y434+y281;y435=y435+y273;y436=y436+y273;y437=y437+y276;y438=y438+y279;y439=y439+y275;y440=y440+y277;y441=y441+y285;y442=y442+y276;y443=y443+y280;y444=y444+y274;y445=y445+y284;y446=y446+y282;y447=y447+y283;y448=y448+y287;y449=y449+y275;y450=y450+y280;y451=y451+y281;y452=y452+y277;y453=y453+y279;y454=y454+y273;y455=y455+y285;y456=y456+y278;y457=y457+y273;y458=y458+y280;y459=y459+y279;y460=y460+y282;y461=y461+y278;y462=y462+y276;y463=y463+y277;y464=y464+y278;y465=y465+y273;y491=y491+y311;y492=y492+y315;y493=y493+y306;y494=y494+y314;y495=y495+y307;y496=y496+y309;y497=y497+y312;y498=y498+y305;y499=y499+y318;y500=y500+y310;y501=y501+y308;y502=y502+y306;y503=y503+y313;y504=y504+y305;y505=y505+y305;y506=y506+y308;y507=y507+y311;y508=y508+y307;y509=y509+y309;y510=y510+y317;y511=y511+y308;y512=y512+y312;y513=y513+y306;y514=y514+y316;y515=y515+y314;y516=y516+y315;y517=y517+y319;y518=y518+y307;y519=y519+y312;y520=y520+y313;y521=y521+y309;y522=y522+y311;y523=y523+y305;y524=y524+y317;y525=y525+y310;y526=y526+y305;y527=y527+y312;y528=y528+y311;y529=y529+y314;y530=y530+y310;y531=y531+y308;y532=y532+y309;y533=y533+y310;y534=y534+y305;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y340;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y592+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y598=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+y337;

#Full depth 36 and Toffoli depth 4

y353=y353+y249;y354=y354+y253;y355=y355+y244;y356=y356+y254;y357=y357+y255;y358=y358+y256;y359=y359+y252;y360=y360+y246;y361=y361+y255;y362=y362+y250;y363=y363+y247;y364=y364+y245;y365=y365+y251;y366=y366+y243;y367=y367+y256;y368=y368+y254;y369=y369+y252;y370=y370+y252;y371=y371+y246;y372=y372+y256;y373=y373+y256;y374=y374+y249;y375=y375+y243;y376=y376+y253;y377=y377+y253;y378=y378+y252;y379=y379+y256;y380=y380+y256;y381=y381+y255;y382=y382+y252;y383=y383+y252;y384=y384+y251;y385=y385+y242;y386=y386+y255;y387=y387+y254;y388=y388+y254;y389=y389+y253;y390=y390+y253;y391=y391+y251;y392=y392+y251;y393=y393+y251;y394=y394+y250;y395=y395+y248;y396=y396+y247;y422=y422+y281;y423=y423+y285;y424=y424+y276;y425=y425+y286;y426=y426+y287;y427=y427+y288;y428=y428+y284;y429=y429+y278;y430=y430+y287;y431=y431+y282;y432=y432+y279;y433=y433+y277;y434=y434+y283;y435=y435+y275;y436=y436+y288;y437=y437+y286;y438=y438+y284;y439=y439+y284;y440=y440+y278;y441=y441+y288;y442=y442+y288;y443=y443+y281;y444=y444+y275;y445=y445+y285;y446=y446+y285;y447=y447+y284;y448=y448+y288;y449=y449+y288;y450=y450+y287;y451=y451+y284;y452=y452+y284;y453=y453+y283;y454=y454+y274;y455=y455+y287;y456=y456+y286;y457=y457+y286;y458=y458+y285;y459=y459+y285;y460=y460+y283;y461=y461+y283;y462=y462+y283;y463=y463+y282;y464=y464+y280;y465=y465+y279;y491=y491+y313;y492=y492+y317;y493=y493+y308;y494=y494+y318;y495=y495+y319;y496=y496+y320;y497=y497+y316;y498=y498+y310;y499=y499+y319;y500=y500+y314;y501=y501+y311;y502=y502+y309;y503=y503+y315;y504=y504+y307;y505=y505+y320;y506=y506+y318;y507=y507+y316;y508=y508+y316;y509=y509+y310;y510=y510+y320;y511=y511+y320;y512=y512+y313;y513=y513+y307;y514=y514+y317;y515=y515+y317;y516=y516+y316;y517=y517+y320;y518=y518+y320;y519=y519+y319;y520=y520+y316;y521=y521+y316;y522=y522+y315;y523=y523+y306;y524=y524+y319;y525=y525+y318;y526=y526+y318;y527=y527+y317;y528=y528+y317;y529=y529+y315;y530=y530+y315;y531=y531+y315;y532=y532+y314;y533=y533+y312;y534=y534+y311;y560=y560+y345;y561=y561+y349;y562=y562+y340;y563=y563+y350;y564=y564+y351;y565=y565+y352;y566=y566+y348;y567=y567+y342;y568=y568+y351;y569=y569+y346;y570=y570+y343;y571=y571+y341;y572=y572+y347;y573=y573+y339;y574=y574+y352;y575=y575+y350;y576=y576+y348;y577=y577+y348;y578=y578+y342;y579=y579+y352;y580=y580+y352;y581=y581+y345;y582=y582+y339;y583=y583+y349;

$y584=y584+y349; y585=y585+y348; y586=y586+y352; y587=y587+y352; y588=y588+y351; y589=y589+y348; y590=y590+y348; y591=y591+y347; y592=y592+y338; y593=y593+y351; y594=y594+y350; y595=y595+y350; y596=y596+y349; y597=y597+y349; y598=y598+y347; y599=y599+y347; y600=y600+y347; y601=y601+y346; y602=y602+y344; y603=y603+y343;$

#Full depth 37 and Toffoli depth 4

$y675=y675+y362; y693=y693+y375; y778=y778+y431; y796=y796+y444; y881=y881+y500; y899=y899+y513; y984=y984+y569; y1002=y1002+y582;$

#Full depth 38 and Toffoli depth 4

$y629=y629+y377; y645=y645+y369; y649=y649+y363; y675=y675+y359; y690=y690+y379; y693=y693+y360; y713=y713+y364; y721=y721+y384; y727=y727+y381; y732=y732+y446; y748=y748+y438; y752=y752+y432; y778=y778+y428; y793=y793+y448; y796=y796+y429; y816=y816+y433; y824=y824+y453; y830=y830+y450; y835=y835+y515; y851=y851+y507; y855=y855+y501; y881=y881+y497; y896=y896+y517; y899=y899+y498; y919=y919+y502; y927=y927+y522; y933=y933+y519; y938=y938+y584; y954=y954+y576; y958=y958+y570; y984=y984+y566; y999=y999+y586; y1002=y1002+y567; y1022=y1022+y571; y1030=y1030+y591; y1036=y1036+y588;$

#Full depth 39 and Toffoli depth 4

$y629=y629+y360; y419=y419+y361; y632=y632+y396; y635=y635+y382; y640=y640+y391; y641=y641+y375; y642=y642+y370; y644=y644+y385; y645=y645+y360; y648=y648+y374; y649=y649+y360; y650=y650+y356; y654=y654+y367; y655=y655+y364; y657=y657+y381; y658=y658+y361; y660=y660+y366; y662=y662+y364; y663=y663+y372; y672=y672+y378; y674=y674+y385; y675=y675+y357; y676=y676+y360; y678=y678+y392; y682=y682+y369; y684=y684+y381; y689=y689+y378; y690=y690+y378; y693=y693+y359; y695=y695+y380; y696=y696+y372; y699=y699+y390; y702=y702+y373; y705=y705+y366; y417=y417+y387; y709=y709+y379; y711=y711+y383; y712=y712+y395; y713=y713+y360; y721=y721+y359; y722=y722+y364; y727=y727+y367; y728=y728+y393; y730=y730+y381; y732=y732+y429; y488=y488+y430; y735=y735+y465; y738=y738+y451; y743=y743+y460; y744=y744+y444; y745=y745+y439; y747=y747+y454; y748=y748+y429; y751=y751+y443; y752=y752+y429; y753=y753+y425; y757=y757+y436; y758=y758+y433; y760=y760+y450; y761=y761+y430; y763=y763+y435; y765=y765+y433; y766=y766+y441; y775=y775+y447; y777=y777+y454; y778=y778+y426; y779=y779+y429; y781=y781+y461; y785=y785+y438; y787=y787+y450; y792=y792+y447; y793=y793+y447; y796=y796+y428; y798=y798+y449; y799=y799+y441; y802=y802+y459; y805=y805+y442; y808=y808+y435; y486=y486+y456; y812=y812+y448; y814=y814+y452; y815=y815+y464; y816=y816+y429; y824=y824+y428; y825=y825+y433; y830=y830+y436; y831=y831+y462; y833=y833+y450; y835=y835+y498; y557=y557+y499; y838=y838+y534; y841=y841+y520; y846=y846+y529; y847=y847+y513; y848=y848+y508; y850=y850+y523; y851=y851+y498; y854=y854+y512; y855=y855+y498; y856=y856+y494; y860=y860+y505; y861=y861+y502; y863=y863+y519; y864=y864+y499; y866=y866+y504; y868=y868+y502; y869=y869+y510; y878=y878+y516; y880=y880+y523; y881=y881+y495; y882=y882+y498; y884=y884+y530; y888=y888+y507; y890=y890+y519; y895=y895+y516; y896=y896+y516; y899=y899+y497; y901=y901+y518; y902=y902+y510; y905=y905+y528; y908=y908+y511; y911=y911+y504; y555=y555+y525; y915=y915+y517; y917=y917+y521; y918=y918+y533; y919=y919+y498; y927=y927+y497; y928=y928+y502; y933=y933+y505; y934=y934+y531; y936=y936+y519; y938=y938+y567; y626=y626+y568; y941=y941+y603; y944=y944+y589; y949=y949+y598; y950=y950+y582; y951=y951+y577; y953=y953+y592; y954=y954+y567; y957=y957+y581; y958=y958+y567; y959=y959+y563; y963=y963+y574; y964=y964+y571; y966=y966+y588; y967=y967+y568; y969=y969+y573; y971=y971+y571; y972=y972+y579; y981=y981+y585; y983=y983+y592; y984=y984+y564; y985=y985+y567; y987=y987+y599; y991=y991+y576; y993=y993+y588; y998=y998+y585; y999=y999+y585; y1002=y1002+y566; y1004=y1004+y587; y1005=y1005+y579; y1008=y1008+y597; y1011=y1011+y580; y1014=y1014+y573; y624=y624+y594; y1018=y1018+y586; y1020=y1020+y590; y1021=y1021+y602; y1022=y1022+y567; y1030=y1030+y566; y1031=y1031+y571; y1036=y1036+y574; y1037=y1037+y600; y1039=y1039+y588;$

#Full depth 40 and Toffoli depth 4

$y405=y405+y358; y629=y629+y359; y630=y630+y389; y631=y631+y378; y419=y419+y360; y632=y632+y372; y634=y634+y369; y635=y635+y360; y637=y637+y370; y420=y420+y368; y639=y639+y369; y640=y640+y363; y641=y641+y374; y642=y642+y363; y644=y644+y365; y645=y645+y358; y646=y646+y364; y648=y648+y362; y649=y649+y357; y650=y650+y355; y651=y651+y359; y652=y652+y374; y653=y653+y382; y654=y654+y363; y655=y655+y359; y414=y414+y373; y657=y657+y360; y658=y658+y360; y660=y660+y359; y661=y661+y362; y662=y662+y361; y663=y663+y362; y665=y665+y374; y666=y666+y371; y667=y667+y383; y668=y668+y377; y669=y669+y362; y670=y670+y367; y671=y671+y386;$

$y672=y672+y371; y674=y674+y369; y675=y675+y355; y676=y676+y356; y678=y678+y361; y679=y679+y370; y680=y680+y368; y681=y681+y380; y682=y682+y364; y683=y683+y364; y684=y684+y362; y685=y685+y387; y688=y688+y396; y689=y689+y362; y690=y690+y360; y691=y691+y374; y692=y692+y376; y693=y693+y358; y416=y416+y372; y694=y694+y380; y695=y695+y360; y696=y696+y357; y698=y698+y372; y699=y699+y366; y700=y700+y374; y701=y701+y379; y702=y702+y370; y703=y703+y373; y704=y704+y375; y705=y705+y365; y706=y706+y371; y707=y707+y389; y417=y417+y359; y709=y709+y365; y710=y710+y388; y711=y711+y360; y712=y712+y361; y713=y713+y357; y421=y421+y364; y714=y714+y376; y716=y716+y375; y719=y719+y393; y720=y720+y376; y721=y721+y358; y722=y722+y361; y723=y723+y394; y411=y411+y357; y725=y725+y388; y726=y726+y363; y727=y727+y355; y728=y728+y361; y729=y729+y365; y730=y730+y363; y731=y731+y395; y474=y474+y427; y732=y732+y428; y733=y733+y458; y734=y734+y447; y488=y488+y429; y735=y735+y441; y737=y737+y438; y738=y738+y429; y740=y740+y439; y489=y489+y437; y742=y742+y438; y743=y743+y432; y744=y744+y443; y745=y745+y432; y747=y747+y434; y748=y748+y427; y749=y749+y433; y751=y751+y431; y752=y752+y426; y753=y753+y424; y754=y754+y428; y755=y755+y443; y756=y756+y451; y757=y757+y432; y758=y758+y428; y483=y483+y442; y760=y760+y429; y761=y761+y429; y763=y763+y428; y764=y764+y431; y765=y765+y430; y766=y766+y431; y768=y768+y443; y769=y769+y440; y770=y770+y452; y771=y771+y446; y772=y772+y431; y773=y773+y436; y774=y774+y455; y775=y775+y440; y777=y777+y438; y778=y778+y424; y779=y779+y425; y781=y781+y430; y782=y782+y439; y783=y783+y437; y784=y784+y449; y785=y785+y433; y786=y786+y433; y787=y787+y431; y788=y788+y456; y791=y791+y465; y792=y792+y431; y793=y793+y429; y794=y794+y443; y795=y795+y445; y796=y796+y427; y485=y485+y441; y797=y797+y449; y798=y798+y429; y799=y799+y426; y801=y801+y441; y802=y802+y435; y803=y803+y443; y804=y804+y448; y805=y805+y439; y806=y806+y442; y807=y807+y444; y808=y808+y434; y809=y809+y440; y810=y810+y458; y486=y486+y428; y812=y812+y434; y813=y813+y457; y814=y814+y429; y815=y815+y430; y816=y816+y426; y490=y490+y433; y817=y817+y445; y819=y819+y444; y822=y822+y462; y823=y823+y445; y824=y824+y427; y825=y825+y430; y826=y826+y463; y480=y480+y426; y828=y828+y457; y829=y829+y432; y830=y830+y424; y831=y831+y430; y832=y832+y434; y833=y833+y432; y834=y834+y464; y543=y543+y496; y835=y835+y497; y836=y836+y527; y837=y837+y516; y557=y557+y498; y838=y838+y510; y840=y840+y507; y841=y841+y498; y843=y843+y508; y558=y558+y506; y845=y845+y507; y846=y846+y501; y847=y847+y512; y848=y848+y501; y850=y850+y503; y851=y851+y496; y852=y852+y502; y854=y854+y500; y855=y855+y495; y856=y856+y493; y857=y857+y497; y858=y858+y512; y859=y859+y520; y860=y860+y501; y861=y861+y497; y552=y552+y511; y863=y863+y498; y864=y864+y498; y866=y866+y497; y867=y867+y500; y868=y868+y499; y869=y869+y500; y871=y871+y512; y872=y872+y509; y873=y873+y521; y874=y874+y515; y875=y875+y500; y876=y876+y505; y877=y877+y524; y878=y878+y509; y880=y880+y507; y881=y881+y493; y882=y882+y494; y884=y884+y499; y885=y885+y508; y886=y886+y506; y887=y887+y518; y888=y888+y502; y889=y889+y502; y890=y890+y500; y891=y891+y525; y894=y894+y534; y895=y895+y500; y896=y896+y498; y897=y897+y512; y898=y898+y514; y899=y899+y496; y554=y554+y510; y900=y900+y518; y901=y901+y498; y902=y902+y495; y904=y904+y510; y905=y905+y504; y906=y906+y512; y907=y907+y517; y908=y908+y508; y909=y909+y511; y910=y910+y513; y911=y911+y503; y912=y912+y509; y913=y913+y527; y555=y555+y497; y915=y915+y503; y916=y916+y526; y917=y917+y498; y918=y918+y499; y919=y919+y495; y559=y559+y502; y920=y920+y514; y922=y922+y513; y925=y925+y531; y926=y926+y514; y927=y927+y496; y928=y928+y499; y929=y929+y532; y549=y549+y495; y931=y931+y526; y932=y932+y501; y933=y933+y493; y934=y934+y499; y935=y935+y503; y936=y936+y501; y937=y937+y533; y612=y612+y565; y938=y938+y566; y939=y939+y596; y940=y940+y585; y626=y626+y567; y941=y941+y579; y943=y943+y576; y944=y944+y567; y946=y946+y577; y627=y627+y575; y948=y948+y576; y949=y949+y570; y950=y950+y581; y951=y951+y570; y953=y953+y572; y954=y954+y565; y955=y955+y571; y957=y957+y569; y958=y958+y564; y959=y959+y562; y960=y960+y566; y961=y961+y581; y962=y962+y589; y963=y963+y570; y964=y964+y566; y621=y621+y580; y966=y966+y567; y967=y967+y567; y969=y969+y566; y970=y970+y569; y971=y971+y568; y972=y972+y569; y974=y974+y581; y975=y975+y578; y976=y976+y590; y977=y977+y584; y978=y978+y569; y979=y979+y574; y980=y980+y593; y981=y981+y578; y983=y983+y576; y984=y984+y562; y985=y985+y563; y987=y987+y568; y988=y988+y577; y989=y989+y575; y990=y990+y587; y991=y991+y571; y992=y992+y571; y993=y993+y569; y994=y994+y594; y997=y997+y603; y998=y998+y569; y999=y999+y567; y1000=y1000+y581; y1001=y1001+y583; y1002=y1002+y565; y623=y623+y579; y1003=y1003+y587; y1004=y1004+y567; y1005=y1005+y564; y1007=y1007+y579; y1008=y1008+y573; y1009=y1009+y581; y1010=y1010+y586; y1011=y1011+y577; y1012=y1012+y580; y1013=y1013+y582; y1014=y1014+y572; y1015=y1015+y578; y1016=y1016+y596; y624=y624+y566; y1018=y1018+y572; y1019=y1019+y595; y1020=y1020+y567; y1021=y1021+y568; y1022=y1022+y564; y628=y628+y571; y1023=y1023+y583; y1025=y1025+y582; y1028=y1028+y600; y1029=y1029+y583; y1030=y1030+y565; y1031=y1031+y568; y1032=y1032+y601; y618=y618+y564; y1034=y1034+y595; y1035=y1035+y570; y1036=y1036+y562; y1037=y1037+y568; y10$

38=y1038+y572;y1039=y1039+y570;y1040=y1040+y602;

#Full depth 41 and Toffoli depth 4

y405=y405+y357;y398=y398+y370;y629=y629+y358;y407=y407+y368;y630=y630+y356;y631=y631+y366;y419=y419+y358;y632=y632+y359;y633=y633+y391;y634=y634+y367;y635=y635+y357;y397=y397+y384;y636=y636+y363;y637=y637+y358;y638=y638+y386;y420=y420+y357;y639=y639+y355;y640=y640+y359;y641=y641+y358;y642=y642+y358;y643=y643+y368;y644=y644+y357;y645=y645+y357;y646=y646+y360;y647=y647+y384;y648=y648+y357;y649=y649+y356;y650=y650+y354;y651=y651+y358;y652=y652+y355;y404=y404+y377;y653=y653+y368;y654=y654+y359;y655=y655+y357;y414=y414+y360;y657=y657+y358;y658=y658+y354;y412=y412+y371;y660=y660+y356;y661=y661+y357;y662=y662+y359;y663=y663+y357;y664=y664+y360;y665=y665+y372;y666=y666+y361;y667=y667+y362;y668=y668+y366;y669=y669+y358;y670=y670+y354;y671=y671+y362;y672=y672+y356;y674=y674+y361;y675=y675+y354;y676=y676+y355;y677=y677+y392;y678=y678+y359;y679=y679+y356;y680=y680+y367;y681=y681+y365;y682=y682+y356;y683=y683+y360;y684=y684+y358;y685=y685+y358;y687=y687+y359;y688=y688+y377;y689=y689+y357;y690=y690+y356;y691=y691+y373;y692=y692+y358;y693=y693+y356;y416=y416+y360;y415=y415+y364;y694=y694+y355;y695=y695+y355;y696=y696+y356;y697=y697+y375;y698=y698+y361;y699=y699+y359;y700=y700+y368;y701=y701+y355;y702=y702+y356;y703=y703+y365;y704=y704+y359;y705=y705+y359;y706=y706+y363;y707=y707+y358;y417=y417+y358;y418=y418+y365;y708=y708+y396;y709=y709+y359;y408=y408+y362;y710=y710+y376;y711=y711+y356;y712=y712+y358;y713=y713+y356;y421=y421+y359;y714=y714+y361;y716=y716+y356;y717=y717+y371;y718=y718+y366;y719=y719+y371;y720=y720+y357;y721=y721+y357;y722=y722+y360;y723=y723+y366;y724=y724+y383;y411=y411+y355;y725=y725+y359;y726=y726+y360;y727=y727+y354;y728=y728+y359;y729=y729+y359;y730=y730+y358;y731=y731+y356;y474=y474+y426;y467=y467+y439;y732=y732+y427;y476=y476+y437;y733=y733+y425;y734=y734+y435;y488=y488+y427;y735=y735+y428;y736=y736+y460;y737=y737+y436;y738=y738+y426;y466=y466+y453;y739=y739+y432;y740=y740+y427;y741=y741+y455;y489=y489+y426;y742=y742+y424;y743=y743+y428;y744=y744+y427;y745=y745+y427;y746=y746+y437;y747=y747+y426;y748=y748+y426;y749=y749+y429;y750=y750+y453;y751=y751+y426;y752=y752+y425;y753=y753+y423;y754=y754+y427;y755=y755+y424;y473=y473+y446;y756=y756+y437;y757=y757+y428;y758=y758+y426;y483=y483+y429;y760=y760+y427;y761=y761+y423;y481=y481+y440;y763=y763+y425;y764=y764+y426;y765=y765+y428;y766=y766+y426;y767=y767+y429;y768=y768+y441;y769=y769+y430;y770=y770+y431;y771=y771+y435;y772=y772+y427;y773=y773+y423;y774=y774+y431;y775=y775+y425;y777=y777+y430;y778=y778+y423;y779=y779+y424;y780=y780+y461;y781=y781+y428;y782=y782+y425;y783=y783+y436;y784=y784+y434;y785=y785+y425;y786=y786+y429;y787=y787+y427;y788=y788+y427;y790=y790+y428;y791=y791+y446;y792=y792+y426;y793=y793+y425;y794=y794+y442;y795=y795+y427;y796=y796+y425;y485=y485+y429;y484=y484+y433;y797=y797+y424;y798=y798+y424;y799=y799+y425;y800=y800+y444;y801=y801+y430;y802=y802+y428;y803=y803+y437;y804=y804+y424;y805=y805+y425;y806=y806+y434;y807=y807+y428;y808=y808+y428;y809=y809+y432;y810=y810+y427;y486=y486+y427;y487=y487+y434;y811=y811+y465;y812=y812+y428;y477=y477+y431;y813=y813+y445;y814=y814+y425;y815=y815+y427;y816=y816+y425;y490=y490+y428;y817=y817+y430;y819=y819+y425;y820=y820+y440;y821=y821+y435;y822=y822+y440;y823=y823+y426;y824=y824+y426;y825=y825+y429;y826=y826+y435;y827=y827+y452;y480=y480+y424;y828=y828+y428;y829=y829+y429;y830=y830+y423;y831=y831+y428;y832=y832+y428;y833=y833+y427;y834=y834+y425;y543=y543+y495;y536=y536+y508;y835=y835+y496;y545=y545+y506;y836=y836+y494;y837=y837+y504;y557=y557+y496;y838=y838+y497;y839=y839+y529;y840=y840+y505;y841=y841+y495;y535=y535+y522;y842=y842+y501;y843=y843+y496;y844=y844+y524;y558=y558+y495;y845=y845+y493;y846=y846+y497;y847=y847+y496;y848=y848+y496;y849=y849+y506;y850=y850+y495;y851=y851+y495;y852=y852+y498;y853=y853+y522;y854=y854+y495;y855=y855+y494;y856=y856+y492;y857=y857+y496;y858=y858+y493;y542=y542+y515;y859=y859+y506;y860=y860+y497;y861=y861+y495;y552=y552+y498;y863=y863+y496;y864=y864+y492;y550=y550+y509;y866=y866+y494;y867=y867+y495;y868=y868+y497;y869=y869+y495;y870=y870+y498;y871=y871+y510;y872=y872+y499;y873=y873+y500;y874=y874+y504;y875=y875+y496;y876=y876+y492;y877=y877+y500;y878=y878+y494;y880=y880+y499;y881=y881+y492;y882=y882+y493;y883=y883+y530;y884=y884+y497;y885=y885+y494;y886=y886+y505;y887=y887+y503;y888=y888+y494;y889=y889+y498;y890=y890+y496;y891=y891+y496;y893=y893+y497;y894=y894+y515;y895=y895+y495;y896=y896+y494;y897=y897+y511;y898=y898+y496;y899=y899+y494;y554=y554+y498;y553=y553+y502;y900=y900+y493;y901=y901+y493;y902=y902+y494;y903=y903+y513;y904=y904+y499;y905=y905+y497;y906=y906+y506;y907=y907+y493;y908=y908+y494;y909=y909+y503;y910=y910+y497;y911=y911+y497;y912=y912+y501;y913=y913+y496;y555=y555+y496;y556=y556+y503;y914=y914+y534;y915=y915+y501

5+y497;y546=y546+y500;y916=y916+y514;y917=y917+y494;y918=y918+y496;y919=y919+y494;y559=y559+y497;y920=y920+y499;y922=y922+y494;y923=y923+y509;y924=y924+y504;y925=y925+y509;y926=y926+y495;y927=y927+y495;y928=y928+y498;y929=y929+y504;y930=y930+y521;y549=y549+y493;y931=y931+y497;y932=y932+y498;y933=y933+y492;y934=y934+y497;y935=y935+y497;y936=y936+y496;y937=y937+y494;y612=y612+y564;y605=y605+y577;y938=y938+y565;y614=y614+y575;y939=y939+y563;y940=y940+y573;y626=y626+y565;y941=y941+y566;y942=y942+y598;y943=y943+y574;y944=y944+y564;y604=y604+y591;y945=y945+y570;y946=y946+y565;y947=y947+y593;y627=y627+y564;y948=y948+y562;y949=y949+y566;y950=y950+y565;y951=y951+y565;y952=y952+y575;y953=y953+y564;y954=y954+y564;y955=y955+y567;y956=y956+y591;y957=y957+y564;y958=y958+y563;y959=y959+y561;y960=y960+y565;y961=y961+y562;y611=y611+y584;y962=y962+y575;y963=y963+y566;y964=y964+y564;y621=y621+y567;y966=y966+y565;y967=y967+y561;y619=y619+y578;y969=y969+y563;y970=y970+y564;y971=y971+y566;y972=y972+y564;y973=y973+y567;y974=y974+y579;y975=y975+y568;y976=y976+y569;y977=y977+y573;y978=y978+y565;y979=y979+y561;y980=y980+y569;y981=y981+y563;y983=y983+y568;y984=y984+y561;y985=y985+y562;y986=y986+y599;y987=y987+y566;y988=y988+y563;y989=y989+y574;y990=y990+y572;y991=y991+y563;y992=y992+y567;y993=y993+y565;y994=y994+y565;y996=y996+y566;y997=y997+y584;y998=y998+y564;y999=y999+y563;y1000=y1000+y580;y1001=y1001+y565;y1002=y1002+y563;y623=y623+y567;y622=y622+y571;y1003=y1003+y562;y1004=y1004+y562;y1005=y1005+y563;y1006=y1006+y582;y1007=y1007+y568;y1008=y1008+y566;y1009=y1009+y575;y1010=y1010+y562;y1011=y1011+y563;y1012=y1012+y572;y1013=y1013+y566;y1014=y1014+y566;y1015=y1015+y570;y1016=y1016+y565;y624=y624+y565;y625=y625+y572;y1017=y1017+y603;y1018=y1018+y566;y615=y615+y569;y1019=y1019+y583;y1020=y1020+y563;y1021=y1021+y565;y1022=y1022+y563;y628=y628+y566;y1023=y1023+y568;y1025=y1025+y563;y1026=y1026+y578;y1027=y1027+y573;y1028=y1028+y578;y1029=y1029+y564;y1030=y1030+y564;y1031=y1031+y567;y1032=y1032+y573;y1033=y1033+y590;y618=y618+y562;y1034=y1034+y566;y1035=y1035+y567;y1036=y1036+y561;y1037=y1037+y566;y1038=y1038+y566;y1039=y1039+y565;y1040=y1040+y563;

#Full depth 42 and Toffoli depth 4

y405=y405+y354;y406=y406+y366;y398=y398+y355;y629=y629+y357;y407=y407+y366;y630=y630+y353;y631=y631+y358;y419=y419+y354;y632=y632+y357;y402=y402+y359;y633=y633+y379;y634=y634+y356;y635=y635+y355;y397=y397+y360;y636=y636+y357;y637=y637+y354;y638=y638+y363;y420=y420+y354;y409=y409+y363;y639=y639+y354;y640=y640+y358;y410=y410+y362;y641=y641+y354;y642=y642+y354;y643=y643+y367;y644=y644+y356;y645=y645+y355;y646=y646+y353;y647=y647+y362;y648=y648+y355;y649=y649+y354;y650=y650+y353;y651=y651+y354;y652=y652+y354;y404=y404+y358;y653=y653+y360;y654=y654+y354;y655=y655+y353;y656=y656+y394;y414=y414+y357;y657=y657+y355;y658=y658+y353;y412=y412+y368;y659=y659+y382;y660=y660+y355;y661=y661+y353;y662=y662+y353;y663=y663+y353;y664=y664+y354;y665=y665+y369;y666=y666+y355;y667=y667+y354;y668=y668+y359;y669=y669+y354;y670=y670+y353;y671=y671+y355;y672=y672+y355;y673=y673+y357;y674=y674+y358;y675=y675+y353;y676=y676+y353;y677=y677+y356;y678=y678+y358;y679=y679+y355;y680=y680+y359;y681=y681+y362;y682=y682+y354;y683=y683+y353;y684=y684+y353;y685=y685+y354;y686=y686+y385;y403=y403+y358;y687=y687+y357;y688=y688+y358;y689=y689+y355;y690=y690+y355;y691=y691+y354;y692=y692+y357;y693=y693+y354;y416=y416+y359;y415=y415+y357;y694=y694+y354;y695=y695+y353;y696=y696+y355;y697=y697+y360;y698=y698+y360;y699=y699+y358;y399=y399+y364;y700=y700+y357;y701=y701+y354;y702=y702+y354;y703=y703+y356;y704=y704+y353;y705=y705+y356;y706=y706+y356;y707=y707+y353;y417=y417+y357;y418=y418+y360;y708=y708+y364;y709=y709+y356;y408=y408+y357;y710=y710+y358;y711=y711+y354;y712=y712+y354;y400=y400+y369;y713=y713+y354;y421=y421+y356;y714=y714+y355;y715=y715+y358;y716=y716+y354;y717=y717+y370;y718=y718+y365;y401=y401+y373;y719=y719+y359;y720=y720+y355;y721=y721+y356;y722=y722+y353;y723=y723+y359;y724=y724+y354;y411=y411+y353;y725=y725+y355;y726=y726+y358;y727=y727+y353;y728=y728+y358;y729=y729+y357;y730=y730+y356;y731=y731+y355;y474=y474+y423;y475=y475+y435;y467=y467+y424;y732=y732+y426;y476=y476+y435;y733=y733+y422;y734=y734+y427;y488=y488+y423;y735=y735+y426;y471=y471+y428;y736=y736+y448;y737=y737+y425;y738=y738+y424;y466=y466+y429;y739=y739+y426;y740=y740+y423;y741=y741+y432;y489=y489+y423;y478=y478+y432;y742=y742+y423;y743=y743+y427;y479=y479+y431;y744=y744+y423;y745=y745+y423;y746=y746+y436;y747=y747+y425;y748=y748+y424;y749=y749+y422;y750=y750+y431;y751=y751+y424;y752=y752+y423;y753=y753+y422;y754=y754+y423;y755=y755+y423;y473=y473+y427;y756=y756+y429;y757=y757+y423;y758=y758+y422;y759=y759+y463;y483=y483+y426;y760=y760+y424;y761=y761+y422;y481=y481+y437;y762=y762+y451;y763=y763+y424;y764=y764+y422;y765=y765+y422;y766=y766+y422;y767=y767+y423;y768=y768+y438;y769=y769+y424;y770

=y770+y423;y771=y771+y428;y772=y772+y423;y773=y773+y422;y774=y774+y424;y775=y775+y424;y776=y776+y426;y777=y777+y427;y778=y778+y422;y779=y779+y422;y780=y780+y425;y781=y781+y427;y782=y782+y424;y783=y783+y428;y784=y784+y431;y785=y785+y423;y786=y786+y422;y787=y787+y422;y788=y788+y423;y789=y789+y454;y472=y472+y427;y790=y790+y426;y791=y791+y427;y792=y792+y424;y793=y793+y424;y794=y794+y423;y795=y795+y426;y796=y796+y423;y485=y485+y428;y484=y484+y426;y797=y797+y423;y798=y798+y422;y799=y799+y424;y800=y800+y429;y801=y801+y429;y802=y802+y427;y468=y468+y433;y803=y803+y426;y804=y804+y423;y805=y805+y423;y806=y806+y425;y807=y807+y422;y808=y808+y425;y809=y809+y425;y810=y810+y422;y486=y486+y426;y487=y487+y429;y811=y811+y433;y812=y812+y425;y477=y477+y426;y813=y813+y427;y814=y814+y423;y815=y815+y423;y469=y469+y438;y816=y816+y423;y490=y490+y425;y817=y817+y424;y818=y818+y427;y819=y819+y423;y820=y820+y439;y821=y821+y434;y470=y470+y442;y822=y822+y428;y823=y823+y424;y824=y824+y425;y825=y825+y422;y826=y826+y428;y827=y827+y423;y480=y480+y422;y828=y828+y424;y829=y829+y427;y830=y830+y422;y831=y831+y427;y832=y832+y426;y833=y833+y425;y834=y834+y424;y543=y543+y492;y544=y544+y504;y536=y536+y493;y835=y835+y495;y545=y545+y504;y836=y836+y491;y837=y837+y496;y557=y557+y492;y838=y838+y495;y540=y540+y497;y839=y839+y517;y840=y840+y494;y841=y841+y493;y535=y535+y498;y842=y842+y495;y843=y843+y492;y844=y844+y501;y558=y558+y492;y547=y547+y501;y845=y845+y492;y846=y846+y496;y548=y548+y500;y847=y847+y492;y848=y848+y492;y849=y849+y505;y850=y850+y494;y851=y851+y493;y852=y852+y491;y853=y853+y500;y854=y854+y493;y855=y855+y492;y856=y856+y491;y857=y857+y492;y858=y858+y492;y542=y542+y496;y859=y859+y498;y860=y860+y492;y861=y861+y491;y862=y862+y532;y552=y552+y495;y863=y863+y493;y864=y864+y491;y550=y550+y506;y865=y865+y520;y866=y866+y493;y867=y867+y491;y868=y868+y491;y869=y869+y491;y870=y870+y492;y871=y871+y507;y872=y872+y493;y873=y873+y492;y874=y874+y497;y875=y875+y492;y876=y876+y491;y877=y877+y493;y878=y878+y493;y879=y879+y495;y880=y880+y496;y881=y881+y491;y882=y882+y491;y883=y883+y494;y884=y884+y496;y885=y885+y493;y886=y886+y497;y887=y887+y500;y888=y888+y492;y889=y889+y491;y890=y890+y491;y891=y891+y492;y892=y892+y523;y541=y541+y496;y893=y893+y495;y894=y894+y496;y895=y895+y493;y896=y896+y493;y897=y897+y492;y898=y898+y495;y899=y899+y492;y554=y554+y497;y553=y553+y495;y900=y900+y492;y901=y901+y491;y902=y902+y493;y903=y903+y498;y904=y904+y498;y905=y905+y496;y537=y537+y502;y906=y906+y495;y907=y907+y492;y908=y908+y492;y909=y909+y494;y910=y910+y491;y911=y911+y494;y912=y912+y494;y913=y913+y491;y555=y555+y495;y556=y556+y498;y914=y914+y502;y915=y915+y494;y546=y546+y495;y916=y916+y496;y917=y917+y492;y918=y918+y492;y538=y538+y507;y919=y919+y492;y559=y559+y494;y920=y920+y493;y921=y921+y496;y922=y922+y492;y923=y923+y508;y924=y924+y503;y539=y539+y511;y925=y925+y497;y926=y926+y493;y927=y927+y494;y928=y928+y491;y929=y929+y497;y930=y930+y492;y549=y549+y491;y931=y931+y493;y932=y932+y496;y933=y933+y491;y934=y934+y496;y935=y935+y495;y936=y936+y494;y937=y937+y493;y612=y612+y561;y613=y613+y573;y605=y605+y562;y938=y938+y564;y614=y614+y573;y939=y939+y560;y940=y940+y565;y626=y626+y561;y941=y941+y564;y609=y609+y566;y942=y942+y586;y943=y943+y563;y944=y944+y562;y604=y604+y567;y945=y945+y564;y946=y946+y561;y947=y947+y570;y627=y627+y561;y616=y616+y570;y948=y948+y561;y949=y949+y565;y617=y617+y569;y950=y950+y561;y951=y951+y561;y952=y952+y574;y953=y953+y563;y954=y954+y562;y955=y955+y560;y956=y956+y569;y957=y957+y562;y958=y958+y561;y959=y959+y560;y960=y960+y561;y961=y961+y561;y611=y611+y565;y962=y962+y567;y963=y963+y561;y964=y964+y560;y965=y965+y601;y621=y621+y564;y966=y966+y562;y967=y967+y560;y619=y619+y575;y968=y968+y589;y969=y969+y562;y970=y970+y560;y971=y971+y560;y972=y972+y560;y973=y973+y561;y974=y974+y576;y975=y975+y562;y976=y976+y561;y977=y977+y566;y978=y978+y561;y979=y979+y560;y980=y980+y562;y981=y981+y562;y982=y982+y564;y983=y983+y565;y984=y984+y560;y985=y985+y560;y986=y986+y563;y987=y987+y565;y988=y988+y562;y989=y989+y566;y990=y990+y569;y991=y991+y561;y992=y992+y560;y993=y993+y560;y994=y994+y561;y995=y995+y592;y610=y610+y565;y996=y996+y564;y997=y997+y565;y998=y998+y562;y999=y999+y562;y1000=y1000+y561;y1001=y1001+y564;y1002=y1002+y561;y623=y623+y566;y622=y622+y564;y1003=y1003+y561;y1004=y1004+y560;y1005=y1005+y562;y1006=y1006+y567;y1007=y1007+y567;y1008=y1008+y565;y606=y606+y571;y1009=y1009+y564;y1010=y1010+y561;y1011=y1011+y561;y1012=y1012+y563;y1013=y1013+y560;y1014=y1014+y563;y1015=y1015+y563;y1016=y1016+y560;y624=y624+y564;y625=y625+y567;y1017=y1017+y571;y1018=y1018+y563;y615=y615+y564;y1019=y1019+y565;y1020=y1020+y561;y1021=y1021+y561;y607=y607+y576;y1022=y1022+y561;y628=y628+y563;y1023=y1023+y562;y1024=y1024+y565;y1025=y1025+y561;y1026=y1026+y577;y1027=y1027+y572;y608=y608+y580;y1028=y1028+y566;y1029=y1029+y562;y1030=y1030+y563;y1031=y1031+y560;y1032=y1032+y566;y1033=y1033+y561;y618=y618+y560;y1034=y1034+y562;y1035=y1035+y565;y1036=y1036+y560;y1037=y1037+y565;y1038=y1038+y564;y1039=y1039+y563;y1040

$0=y1040+y562;$

#Full depth 43 and Toffoli depth 4

$y405=y405+y353; y406=y406+y365; y398=y398+y354; y629=y629+y353; y407=y407+y359; y630=y630+y242;$
 $y413=y413+y241; y631=y631+y355; y419=y419+y353; y632=y632+y355; y402=y402+y356; y633=y633+y369;$
 $y634=y634+y354; y635=y635+y354; y397=y397+y355; y636=y636+y354; y637=y637+y254;$
 $y638=y638+y249; y420=y420+y353; y409=y409+y358; y639=y639+y256; y640=y640+y357; y410=y410+y354;$
 $y641=y641+y254; y642=y642+y246; y643=y643+y249; y644=y644+y248; y645=y645+y354;$
 $y646=y646+y251; y647=y647+y357; y648=y648+y253; y649=y649+y248; y650=y650+y243; y651=y651+y255;$
 $y652=y652+y241; y404=y404+y355; y653=y653+y354; y654=y654+y255; y655=y655+y246;$
 $y656=y656+y363; y414=y414+y353; y657=y657+y354; y658=y658+y245; y412=y412+y353; y659=y659+y356;$
 $y660=y660+y353; y661=y661+y248; y662=y662+y253; y663=y663+y248; y664=y664+y254; y665=y665+y356;$
 $y666=y666+y353; y667=y667+y353; y668=y668+y355; y669=y669+y242; y670=y670+y252;$
 $y671=y671+y353; y672=y672+y353; y673=y673+y256; y674=y674+y251; y675=y675+y241; y676=y676+y245;$
 $y677=y677+y353; y678=y678+y355; y679=y679+y253; y680=y680+y251; y681=y681+y355;$
 $y682=y682+y243; y683=y683+y253; y684=y684+y242; y685=y685+y242; y686=y686+y359; y403=y403+y356;$
 $y687=y687+y356; y688=y688+y355; y689=y689+y353; y690=y690+y353; y691=y691+y250;$
 $y692=y692+y353; y693=y693+y247; y416=y416+y353; y415=y415+y354; y694=y694+y247; y695=y695+y254;$
 $y696=y696+y246; y697=y697+y251; y698=y698+y248; y699=y699+y356; y399=y399+y353; y700=y700+y354;$
 $y701=y701+y241; y702=y702+y353; y703=y703+y246; y704=y704+y251; y705=y705+y355;$
 $y706=y706+y354; y707=y707+y243; y417=y417+y354; y418=y418+y357; y708=y708+y357; y709=y709+y355;$
 $y408=y408+y353; y710=y710+y353; y711=y711+y353; y712=y712+y247; y400=y400+y360;$
 $y713=y713+y252; y421=y421+y353; y714=y714+y249; y715=y715+y250; y716=y716+y249; y717=y717+y363;$
 $y718=y718+y356; y401=y401+y360; y719=y719+y243; y720=y720+y353; y721=y721+y242;$
 $y722=y722+y252; y723=y723+y244; y724=y724+y250; y411=y411+y241; y725=y725+y353; y726=y726+y243;$
 $y727=y727+y254; y728=y728+y353; y729=y729+y356; y730=y730+y246; y731=y731+y243; y474=y474+y422;$
 $y475=y475+y434; y467=y467+y423; y732=y732+y422; y476=y476+y428; y733=y733+y274;$
 $y482=y482+y273; y734=y734+y424; y488=y488+y422; y735=y735+y424; y471=y471+y425; y736=y736+y438;$
 $y737=y737+y423; y738=y738+y423; y466=y466+y424; y739=y739+y423; y740=y740+y286;$
 $y741=y741+y281; y489=y489+y422; y478=y478+y427; y742=y742+y288; y743=y743+y426; y479=y479+y423;$
 $y744=y744+y286; y745=y745+y278; y746=y746+y281; y747=y747+y280; y748=y748+y423;$
 $y749=y749+y283; y750=y750+y426; y751=y751+y285; y752=y752+y280; y753=y753+y275; y754=y754+y287;$
 $y755=y755+y273; y473=y473+y424; y756=y756+y423; y757=y757+y287; y758=y758+y278; y759=y759+y432;$
 $y483=y483+y422; y760=y760+y423; y761=y761+y277; y481=y481+y422; y762=y762+y425;$
 $y763=y763+y422; y764=y764+y280; y765=y765+y285; y766=y766+y280; y767=y767+y286; y768=y768+y425;$
 $y769=y769+y422; y770=y770+y422; y771=y771+y424; y772=y772+y274; y773=y773+y284;$
 $y774=y774+y422; y775=y775+y422; y776=y776+y288; y777=y777+y283; y778=y778+y273; y779=y779+y277;$
 $y780=y780+y422; y781=y781+y424; y782=y782+y285; y783=y783+y283; y784=y784+y424;$
 $y785=y785+y275; y786=y786+y285; y787=y787+y274; y788=y788+y274; y789=y789+y428; y472=y472+y425;$
 $y790=y790+y425; y791=y791+y424; y792=y792+y422; y793=y793+y422; y794=y794+y282; y795=y795+y422;$
 $y796=y796+y279; y485=y485+y422; y484=y484+y423; y797=y797+y279; y798=y798+y286;$
 $y799=y799+y278; y800=y800+y283; y801=y801+y280; y802=y802+y425; y468=y468+y422; y803=y803+y423;$
 $y804=y804+y273; y805=y805+y422; y806=y806+y278; y807=y807+y283; y808=y808+y424;$
 $y809=y809+y423; y810=y810+y275; y486=y486+y423; y487=y487+y426; y811=y811+y426; y812=y812+y424;$
 $y477=y477+y422; y813=y813+y422; y814=y814+y422; y815=y815+y279; y469=y469+y429;$
 $y816=y816+y284; y490=y490+y422; y817=y817+y281; y818=y818+y282; y819=y819+y281; y820=y820+y432;$
 $y821=y821+y425; y470=y470+y429; y822=y822+y275; y823=y823+y422; y824=y824+y274; y825=y825+y284;$
 $y826=y826+y276; y827=y827+y282; y480=y480+y273; y828=y828+y422; y829=y829+y275;$
 $y830=y830+y286; y831=y831+y422; y832=y832+y425; y833=y833+y278; y834=y834+y275; y543=y543+y491;$
 $y544=y544+y503; y536=y536+y492; y835=y835+y491; y545=y545+y497; y836=y836+y306;$
 $y551=y551+y305; y837=y837+y493; y557=y557+y491; y838=y838+y493; y540=y540+y494; y839=y839+y507;$
 $y840=y840+y492; y841=y841+y492; y535=y535+y493; y842=y842+y492; y843=y843+y318;$
 $y844=y844+y313; y558=y558+y491; y547=y547+y496; y845=y845+y320; y846=y846+y495; y548=y548+y492;$
 $y847=y847+y318; y848=y848+y310; y849=y849+y313; y850=y850+y312; y851=y851+y492; y852=y852+y315;$
 $y853=y853+y495; y854=y854+y317; y855=y855+y312; y856=y856+y307; y857=y857+y319;$
 $y858=y858+y305; y542=y542+y493; y859=y859+y492; y860=y860+y319; y861=y861+y310; y862=y862+y501;$
 $y552=y552+y491; y863=y863+y492; y864=y864+y309; y550=y550+y491; y865=y865+y494;$
 $y866=y866+y491; y867=y867+y312; y868=y868+y317; y869=y869+y312; y870=y870+y318; y871=y871+y494;$
 $y872=y872+y491; y873=y873+y491; y874=y874+y493; y875=y875+y306; y876=y876+y316;$
 $y877=y877+y491; y878=y878+y491; y879=y879+y320; y880=y880+y315; y881=y881+y305; y882=y882+y88$

$2+y309;y883=y883+y491;y884=y884+y493;y885=y885+y317;y886=y886+y315;y887=y887+y493;y888=y888+y307;y889=y889+y317;y890=y890+y306;y891=y891+y306;y892=y892+y497;y541=y541+y494;y893=y893+y494;y894=y894+y493;y895=y895+y491;y896=y896+y491;y897=y897+y314;y898=y898+y491;y899=y899+y311;y554=y554+y491;y553=y553+y492;y900=y900+y311;y901=y901+y318;y902=y902+y310;y903=y903+y315;y904=y904+y312;y905=y905+y494;y537=y537+y491;y906=y906+y492;y907=y907+y305;y908=y908+y491;y909=y909+y310;y910=y910+y315;y911=y911+y493;y912=y912+y492;y913=y913+y307;y555=y555+y492;y556=y556+y495;y914=y914+y495;y915=y915+y493;y546=y546+y491;y916=y916+y491;y917=y917+y491;y918=y918+y311;y538=y538+y498;y919=y919+y316;y559=y559+y491;y920=y920+y313;y921=y921+y314;y922=y922+y313;y923=y923+y501;y924=y924+y494;y539=y539+y498;y925=y925+y307;y926=y926+y491;y927=y927+y306;y928=y928+y316;y929=y929+y308;y930=y930+y314;y549=y549+y305;y931=y931+y491;y932=y932+y307;y933=y933+y318;y934=y934+y491;y935=y935+y494;y936=y936+y310;y937=y937+y307;y612=y612+y560;y613=y613+y572;y605=y605+y561;y938=y938+y560;y614=y614+y566;y939=y939+y338;y620=y620+y337;y940=y940+y562;y626=y626+y560;y941=y941+y562;y609=y609+y563;y942=y942+y576;y943=y943+y561;y944=y944+y561;y604=y604+y562;y945=y945+y561;y946=y946+y350;y947=y947+y345;y627=y627+y560;y616=y616+y565;y948=y948+y352;y949=y949+y564;y617=y617+y561;y950=y950+y350;y951=y951+y342;y952=y952+y345;y953=y953+y344;y954=y954+y561;y955=y955+y347;y956=y956+y564;y957=y957+y349;y958=y958+y344;y959=y959+y339;y960=y960+y351;y961=y961+y337;y611=y611+y562;y962=y962+y561;y963=y963+y351;y964=y964+y342;y965=y965+y570;y621=y621+y560;y966=y966+y561;y967=y967+y341;y619=y619+y560;y968=y968+y563;y969=y969+y560;y970=y970+y344;y971=y971+y349;y972=y972+y344;y973=y973+y350;y974=y974+y563;y975=y975+y560;y976=y976+y560;y977=y977+y562;y978=y978+y338;y979=y979+y348;y980=y980+y560;y981=y981+y560;y982=y982+y352;y983=y983+y347;y984=y984+y337;y985=y985+y341;y986=y986+y560;y987=y987+y562;y988=y988+y349;y989=y989+y347;y990=y990+y562;y991=y991+y339;y992=y992+y349;y993=y993+y338;y994=y994+y338;y995=y995+y566;y610=y610+y563;y996=y996+y563;y997=y997+y562;y998=y998+y560;y999=y999+y560;y1000=y1000+y346;y1001=y1001+y560;y1002=y1002+y343;y623=y623+y560;y622=y622+y561;y1003=y1003+y343;y1004=y1004+y350;y1005=y1005+y342;y1006=y1006+y347;y1007=y1007+y344;y1008=y1008+y563;y606=y606+y560;y1009=y1009+y561;y1010=y1010+y337;y1011=y1011+y560;y1012=y1012+y342;y1013=y1013+y347;y1014=y1014+y562;y1015=y1015+y561;y1016=y1016+y339;y624=y624+y561;y625=y625+y564;y1017=y1017+y564;y1018=y1018+y562;y615=y615+y560;y1019=y1019+y560;y1020=y1020+y560;y1021=y1021+y343;y607=y607+y567;y1022=y1022+y348;y628=y628+y560;y1023=y1023+y345;y1024=y1024+y346;y1025=y1025+y345;y1026=y1026+y570;y1027=y1027+y563;y608=y608+y567;y1028=y1028+y339;y1029=y1029+y560;y1030=y1030+y338;y1031=y1031+y348;y1032=y1032+y340;y1033=y1033+y346;y618=y618+y337;y1034=y1034+y560;y1035=y1035+y339;y1036=y1036+y350;y1037=y1037+y560;y1038=y1038+y563;y1039=y1039+y342;y1040=y1040+y339;$

#Full depth 44 and Toffoli depth 4

$y353=y353+y247;y354=y354+y251;y355=y355+y242;y356=y356+y250;y357=y357+y243;y358=y358+y245;y359=y359+y248;y360=y360+y241;y361=y361+y254;y362=y362+y246;y363=y363+y244;y364=y364+y242;y365=y365+y249;y366=y366+y241;y367=y367+y241;y368=y368+y244;y369=y369+y247;y370=y370+y243;y371=y371+y245;y372=y372+y253;y373=y373+y244;y374=y374+y248;y375=y375+y242;y376=y376+y252;y377=y377+y250;y378=y378+y251;y379=y379+y255;y380=y380+y243;y381=y381+y248;y382=y382+y249;y383=y383+y245;y384=y384+y247;y385=y385+y241;y386=y386+y253;y387=y387+y246;y388=y388+y241;y389=y389+y248;y390=y390+y247;y391=y391+y250;y392=y392+y246;y393=y393+y244;y394=y394+y245;y395=y395+y246;y396=y396+y241;y397=y397+y254;y398=y398+y250;y399=y399+y251;y400=y400+y250;y401=y401+y249;y402=y402+y245;y403=y403+y244;y404=y404+y241;y405=y405+y252;y406=y406+y248;y407=y407+y249;y408=y408+y245;y409=y409+y250;y410=y410+y242;y411=y411+y248;y412=y412+y243;y413=y413+y249;y414=y414+y248;y415=y415+y247;y416=y416+y242;y417=y417+y242;y418=y418+y245;y419=y419+y244;y420=y420+y241;y421=y421+y246;y422=y422+y279;y423=y423+y283;y424=y424+y274;y425=y425+y282;y426=y426+y275;y427=y427+y277;y428=y428+y280;y429=y429+y273;y430=y430+y286;y431=y431+y278;y432=y432+y276;y433=y433+y274;y434=y434+y281;y435=y435+y273;y436=y436+y273;y437=y437+y276;y438=y438+y279;y439=y439+y275;y440=y440+y277;y441=y441+y285;y442=y442+y276;y443=y443+y280;y444=y444+y274;y445=y445+y284;y446=y446+y282;y447=y447+y283;y448=y448+y287;y449=y449+y275;y450=y450+y280;y451=y451+y281;y452=y452+y277;y453=y453+y279;y454=y454+y273;y455=y455+y285;y456=y456+y278;y457=y457+y273;y458=y458+y280;y459=y459+y279;y460=y460+y282;y461=y461+y278;y462=y462+y276;y463=y463+y277;y464=y464+y278;y465=y465+y273;y466=y466+y286;y467=y467+y282;y468=y468+y283;y469=y469+y282;y470=y470+y281;y471=y471+y277;y472=y472+y276;y473=y473+y273;y474=y474+y284;y475=y475+y277;$

5+y280;y476=y476+y281;y477=y477+y277;y478=y478+y282;y479=y479+y274;y480=y480+y280;y481=y481+y275;y482=y482+y281;y483=y483+y280;y484=y484+y279;y485=y485+y274;y486=y486+y274;y487=y487+y277;y488=y488+y276;y489=y489+y273;y490=y490+y278;y491=y491+y311;y492=y492+y315;y493=y493+y306;y494=y494+y314;y495=y495+y307;y496=y496+y309;y497=y497+y312;y498=y498+y305;y499=y499+y318;y500=y500+y310;y501=y501+y308;y502=y502+y306;y503=y503+y313;y504=y504+y305;y505=y505+y305;y506=y506+y308;y507=y507+y311;y508=y508+y307;y509=y509+y309;y510=y510+y317;y511=y511+y308;y512=y512+y312;y513=y513+y306;y514=y514+y316;y515=y515+y314;y516=y516+y315;y517=y517+y319;y518=y518+y307;y519=y519+y312;y520=y520+y313;y521=y521+y309;y522=y522+y311;y523=y523+y305;y524=y524+y317;y525=y525+y310;y526=y526+y305;y527=y527+y312;y528=y528+y311;y529=y529+y314;y530=y530+y310;y531=y531+y308;y532=y532+y309;y533=y533+y310;y534=y534+y305;y535=y535+y318;y536=y536+y314;y537=y537+y315;y538=y538+y314;y539=y539+y313;y540=y540+y309;y541=y541+y308;y542=y542+y305;y543=y543+y316;y544=y544+y312;y545=y545+y313;y546=y546+y309;y547=y547+y314;y548=y548+y306;y549=y549+y312;y550=y550+y307;y551=y551+y313;y552=y552+y312;y553=y553+y311;y554=y554+y306;y555=y555+y306;y556=y556+y309;y557=y557+y308;y558=y558+y305;y559=y559+y310;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y340;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y592+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y598=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+y337;y604=y604+y350;y605=y605+y346;y606=y606+y347;y607=y607+y346;y608=y608+y345;y609=y609+y341;y610=y610+y340;y611=y611+y337;y612=y612+y348;y613=y613+y344;y614=y614+y345;y615=y615+y341;y616=y616+y346;y617=y617+y338;y618=y618+y344;y619=y619+y339;y620=y620+y345;y621=y621+y344;y622=y622+y343;y623=y623+y338;y624=y624+y338;y625=y625+y341;y626=y626+y340;y627=y627+y337;y628=y628+y342;

#Full depth 45 and Toffoli depth 4

y353=y353+y249;y354=y354+y253;y355=y355+y244;y356=y356+y254;y357=y357+y255;y358=y358+y256;y359=y359+y252;y360=y360+y246;y361=y361+y255;y362=y362+y250;y363=y363+y247;y364=y364+y245;y365=y365+y251;y366=y366+y243;y367=y367+y256;y368=y368+y254;y369=y369+y252;y370=y370+y252;y371=y371+y246;y372=y372+y256;y373=y373+y256;y374=y374+y249;y375=y375+y243;y376=y376+y253;y377=y377+y253;y378=y378+y252;y379=y379+y256;y380=y380+y256;y381=y381+y255;y382=y382+y252;y383=y383+y252;y384=y384+y251;y385=y385+y242;y386=y386+y255;y387=y387+y254;y388=y388+y254;y389=y389+y253;y390=y390+y253;y391=y391+y251;y392=y392+y251;y393=y393+y251;y394=y394+y250;y395=y395+y248;y396=y396+y247;y397=y397+y256;y398=y398+y256;y399=y399+y255;y400=y400+y255;y401=y401+y255;y402=y402+y255;y403=y403+y255;y404=y404+y255;y405=y405+y254;y406=y406+y254;y407=y407+y253;y408=y408+y253;y409=y409+y252;y410=y410+y252;y411=y411+y251;y412=y412+y251;y413=y413+y250;y414=y414+y250;y415=y415+y250;y416=y416+y250;y417=y417+y249;y418=y418+y248;y419=y419+y248;y420=y420+y245;y421=y421+y255;y422=y422+y281;y423=y423+y285;y424=y424+y276;y425=y425+y286;y426=y426+y287;y427=y427+y288;y428=y428+y284;y429=y429+y278;y430=y430+y287;y431=y431+y282;y432=y432+y279;y433=y433+y277;y434=y434+y283;y435=y435+y275;y436=y436+y288;y437=y437+y286;y438=y438+y284;y439=y439+y284;y440=y440+y278;y441=y441+y288;y442=y442+y288;y443=y443+y281;y444=y444+y275;y445=y445+y285;y446=y446+y285;y447=y447+y284;y448=y448+y288;y449=y449+y288;y450=y450+y287;y451=y451+y284;y452=y452+y284;y453=y453+y283;y454=y454+y274;y455=y455+y287;y456=y456+y286;y457=y457+y286;y458=y458+y285;y459=y459+y285;y460=y460+y283;y461=y461+y283;y462=y462+y283;y463=y463+y282;y464=y464+y280;y465=y465+y279;y466=y466+y288;y467=y467+y288;y468=y468+y287;y469=y469+y287;y470=y470+y287;y471=y471+y287;y472=y472+y287;y473=y473+y287;y474=y474+y286;y475=y475+y286;y476=y476+y285;y477=y477+y285;y478=y478+y284;y479=y479+y284;y480=y480+y283;y481=y481+y283;y482=y482+y282;y483=y483+y282;y484=y484+y282;y485=y485+y282;y486=y486+y281;y487=y487+y280;y488=y488+y280;y489=y489+y277;y490=y490+y287;y491=y491+y313;y492=y492+y317;y493=y493+y308;y494=y494+y318;y495=y495+y319;y496=y496+y320;y497=y497+y316;y498=y498+y310;y499=y499+y319;y500=y500+y314;y501=y501+y311;y502=y502+y309;y503=y503+y315;y504=y504+y307;y505=y505+y320;y506=y506+y318;y507=y507+y316;y508=y508+y316;y509=y509+y310;y510=y510+y320;y511=y511+y320;y512=y512+y313;y513=y513+y307;y514=y514+y317;y515=y515+y317;y516=y516+y316;y517=y517+y320;y518=y518+y320;y519=y519+y319;y5

20=y520+y316;y521=y521+y316;y522=y522+y315;y523=y523+y306;y524=y524+y319;y525=y525+y318;y526=y526+y318;y527=y527+y317;y528=y528+y317;y529=y529+y315;y530=y530+y315;y531=y531+y315;y532=y532+y314;y533=y533+y312;y534=y534+y311;y535=y535+y320;y536=y536+y320;y537=y537+y319;y538=y538+y319;y539=y539+y319;y540=y540+y319;y541=y541+y319;y542=y542+y319;y543=y543+y318;y544=y544+y318;y545=y545+y317;y546=y546+y317;y547=y547+y316;y548=y548+y316;y549=y549+y315;y550=y550+y315;y551=y551+y314;y552=y552+y314;y553=y553+y314;y554=y554+y314;y555=y555+y313;y556=y556+y312;y557=y557+y312;y558=y558+y309;y559=y559+y319;y560=y560+y345;y561=y561+y349;y562=y562+y340;y563=y563+y350;y564=y564+y351;y565=y565+y352;y566=y566+y348;y567=y567+y342;y568=y568+y351;y569=y569+y346;y570=y570+y343;y571=y571+y341;y572=y572+y347;y573=y573+y339;y574=y574+y352;y575=y575+y350;y576=y576+y348;y577=y577+y348;y578=y578+y342;y579=y579+y352;y580=y580+y352;y581=y581+y345;y582=y582+y339;y583=y583+y349;y584=y584+y349;y585=y585+y348;y586=y586+y352;y587=y587+y352;y588=y588+y351;y589=y589+y348;y590=y590+y348;y591=y591+y347;y592=y592+y338;y593=y593+y351;y594=y594+y350;y595=y595+y350;y596=y596+y349;y597=y597+y349;y598=y598+y347;y599=y599+y347;y600=y600+y347;y601=y601+y346;y602=y602+y344;y603=y603+y343;y604=y604+y352;y605=y605+y352;y606=y606+y351;y607=y607+y351;y608=y608+y351;y609=y609+y351;y610=y610+y351;y611=y611+y351;y612=y612+y350;y613=y613+y350;y614=y614+y349;y615=y615+y349;y616=y616+y348;y617=y617+y348;y618=y618+y347;y619=y619+y347;y620=y620+y346;y621=y621+y346;y622=y622+y346;y623=y623+y346;y624=y624+y345;y625=y625+y344;y626=y626+y344;y627=y627+y341;y628=y628+y351;y129=y129+y1;y130=y130+y2;y131=y131+y3;y132=y132+y4;y133=y133+y5;y134=y134+y6;y135=y135+y7;y136=y136+y8;y137=y137+y9;y138=y138+y10;y139=y139+y11;y140=y140+y12;y141=y141+y13;y142=y142+y14;y143=y143+y15;y144=y144+y16;y145=y145+y17;y146=y146+y18;y147=y147+y19;y148=y148+y20;y149=y149+y21;y150=y150+y22;y151=y151+y23;y152=y152+y24;y153=y153+y25;y154=y154+y26;y155=y155+y27;y156=y156+y28;y157=y157+y29;y158=y158+y30;y159=y159+y31;y160=y160+y32;

#Full depth 46 and Toffoli depth 5

y225=y225+y161*y165;y226=y226+y161*y166;y227=y227+y161*y167;y228=y228+y161*y168;y229=y229+y162*y165;y230=y230+y162*y166;y231=y231+y162*y167;y232=y232+y162*y168;y233=y233+y163*y165;y234=y234+y163*y166;y235=y235+y163*y167;y236=y236+y163*y168;y237=y237+y164*y165;y238=y238+y164*y166;y239=y239+y164*y167;y240=y240+y164*y168;y241=y241+y169*y173;y242=y242+y169*y174;y243=y243+y169*y175;y244=y244+y169*y176;y245=y245+y170*y173;y246=y246+y170*y174;y247=y247+y170*y175;y248=y248+y170*y176;y249=y249+y171*y173;y250=y250+y171*y174;y251=y251+y171*y175;y252=y252+y171*y176;y253=y253+y172*y173;y254=y254+y172*y174;y255=y255+y172*y175;y256=y256+y172*y176;y257=y257+y177*y181;y258=y258+y177*y182;y259=y259+y177*y183;y260=y260+y177*y184;y261=y261+y178*y181;y262=y262+y178*y182;y263=y263+y178*y183;y264=y264+y178*y184;y265=y265+y179*y181;y266=y266+y179*y182;y267=y267+y179*y183;y268=y268+y179*y184;y269=y269+y180*y181;y270=y270+y180*y182;y271=y271+y180*y183;y272=y272+y180*y184;y273=y273+y185*y189;y274=y274+y185*y190;y275=y275+y185*y191;y276=y276+y185*y192;y277=y277+y186*y189;y278=y278+y186*y190;y279=y279+y186*y191;y280=y280+y186*y192;y281=y281+y187*y189;y282=y282+y187*y190;y283=y283+y187*y191;y284=y284+y187*y192;y285=y285+y188*y189;y286=y286+y188*y190;y287=y287+y188*y191;y288=y288+y188*y192;y289=y289+y193*y197;y290=y290+y193*y198;y291=y291+y193*y199;y292=y292+y193*y200;y293=y293+y194*y197;y294=y294+y194*y198;y295=y295+y194*y199;y296=y296+y194*y200;y297=y297+y195*y197;y298=y298+y195*y198;y299=y299+y195*y199;y300=y300+y195*y200;y301=y301+y196*y197;y302=y302+y196*y198;y303=y303+y196*y199;y304=y304+y196*y200;y305=y305+y201*y205;y306=y306+y201*y206;y307=y307+y201*y207;y308=y308+y201*y208;y309=y309+y202*y205;y310=y310+y202*y206;y311=y311+y202*y207;y312=y312+y202*y208;y313=y313+y203*y205;y314=y314+y203*y206;y315=y315+y203*y207;y316=y316+y203*y208;y317=y317+y204*y205;y318=y318+y204*y206;y319=y319+y204*y207;y320=y320+y204*y208;y321=y321+y209*y213;y322=y322+y209*y214;y323=y323+y209*y215;y324=y324+y209*y216;y325=y325+y210*y213;y326=y326+y210*y214;y327=y327+y210*y215;y328=y328+y210*y216;y329=y329+y211*y213;y330=y330+y211*y214;y331=y331+y211*y215;y332=y332+y211*y216;y333=y333+y212*y213;y334=y334+y212*y214;y335=y335+y212*y215;y336=y336+y212*y216;y337=y337+y217*y221;y338=y338+y217*y222;y339=y339+y217*y223;y340=y340+y217*y224;y341=y341+y218*y221;y342=y342+y218*y222;y343=y343+y218*y223;y344=y344+y218*y224;y345=y345+y219*y221;y346=y346+y219*y222;y347=y347+y219*y223;y348=y348+y219*y224;y349=y349+y220*y221;y350=y350+y220*y222;y351=y351+y220*y223;y352=y352+y220*y224;y129=y129+1;y130=y130+1;y131=y131+1;y132=y132+1;y133=y133+1;y134=y134+1;y135=y135+1;y136=y136+1;y137=y137+1;y138=y138+1;y139=y139+1;y140=y140+1;y141=y141+1;y142=y142+1;y143=y143+1;y144=y

144+1;y145=y145+1;y146=y146+1;y147=y147+1;y148=y148+1;y149=y149+1;y150=y150+1;y151=y151+1;y152=y152+1;y153=y153+1;y154=y154+1;y155=y155+1;y156=y156+1;y157=y157+1;y158=y158+1;y159=y159+1;y160=y160+1;

#Full depth 47 and Toffoli depth 6

y161=y161+y1*y2;y162=y162+y1*y130;y163=y163+y129*y2;y164=y164+y129*y130;y165=y165+y3*y4;y166=y166+y3*y132;y167=y167+y131*y4;y168=y168+y131*y132;y169=y169+y5*y6;y170=y170+y5*y134;y171=y171+y133*y6;y172=y172+y133*y134;y173=y173+y7*y8;y174=y174+y7*y136;y175=y175+y135*y8;y176=y176+y135*y136;y177=y177+y9*y10;y178=y178+y9*y138;y179=y179+y137*y10;y180=y180+y137*y138;y181=y181+y11*y12;y182=y182+y11*y140;y183=y183+y139*y12;y184=y184+y139*y140;y185=y185+y13*y14;y186=y186+y13*y142;y187=y187+y141*y14;y188=y188+y141*y142;y189=y189+y15*y16;y190=y190+y15*y144;y191=y191+y143*y16;y192=y192+y143*y144;y193=y193+y17*y18;y194=y194+y17*y146;y195=y195+y145*y18;y196=y196+y145*y146;y197=y197+y19*y20;y198=y198+y19*y148;y199=y199+y147*y20;y200=y200+y147*y148;y201=y201+y21*y22;y202=y202+y21*y150;y203=y203+y149*y22;y204=y204+y149*y150;y205=y205+y23*y24;y206=y206+y23*y152;y207=y207+y151*y24;y208=y208+y151*y152;y209=y209+y25*y26;y210=y210+y25*y154;y211=y211+y153*y26;y212=y212+y153*y154;y213=y213+y27*y28;y214=y214+y27*y156;y215=y215+y155*y28;y216=y216+y155*y156;y217=y217+y29*y30;y218=y218+y29*y158;y219=y219+y157*y30;y220=y220+y157*y158;y221=y221+y31*y32;y222=y222+y31*y160;y223=y223+y159*y32;y224=y224+y159*y160;

#Full depth 48 and Toffoli depth 6

y129=y129+1;y130=y130+1;y131=y131+1;y132=y132+1;y133=y133+1;y134=y134+1;y135=y135+1;y136=y136+1;y137=y137+1;y138=y138+1;y139=y139+1;y140=y140+1;y141=y141+1;y142=y142+1;y143=y143+1;y144=y144+1;y145=y145+1;y146=y146+1;y147=y147+1;y148=y148+1;y149=y149+1;y150=y150+1;y151=y151+1;y152=y152+1;y153=y153+1;y154=y154+1;y155=y155+1;y156=y156+1;y157=y157+1;y158=y158+1;y159=y159+1;y160=y160+1;

#Full depth 49 and Toffoli depth 6

y129=y129+y1;y130=y130+y2;y131=y131+y3;y132=y132+y4;y133=y133+y5;y134=y134+y6;y135=y135+y7;y136=y136+y8;y137=y137+y9;y138=y138+y10;y139=y139+y11;y140=y140+y12;y141=y141+y13;y142=y142+y14;y143=y143+y15;y144=y144+y16;y145=y145+y17;y146=y146+y18;y147=y147+y19;y148=y148+y20;y149=y149+y21;y150=y150+y22;y151=y151+y23;y152=y152+y24;y153=y153+y25;y154=y154+y26;y155=y155+y27;y156=y156+y28;y157=y157+y29;y158=y158+y30;y159=y159+y31;y160=y160+y32;

#Full depth 50 and Toffoli depth 6

y1=y1+y65;y2=y2+y66;y3=y3+y67;y4=y4+y68;y5=y5+y69;y6=y6+y70;y7=y7+y71;y8=y8+y72;y9=y9+y73;y10=y10+y74;y11=y11+y75;y12=y12+y76;y13=y13+y77;y14=y14+y78;y15=y15+y79;y16=y16+y80;y17=y17+y81;y18=y18+y82;y19=y19+y83;y20=y20+y84;y21=y21+y85;y22=y22+y86;y23=y23+y87;y24=y24+y88;y25=y25+y89;y26=y26+y90;y27=y27+y91;y28=y28+y92;y29=y29+y93;y30=y30+y94;y31=y31+y95;y32=y32+y96;

#Full depth 51 and Toffoli depth 6

y1=y1+y33;y2=y2+y34;y3=y3+y35;y4=y4+y36;y5=y5+y37;y6=y6+y38;y7=y7+y39;y8=y8+y40;y9=y9+y41;y10=y10+y42;y11=y11+y43;y12=y12+y44;y13=y13+y45;y14=y14+y46;y15=y15+y47;y16=y16+y48;y17=y17+y49;y18=y18+y50;y19=y19+y51;y20=y20+y52;y21=y21+y53;y22=y22+y54;y23=y23+y55;y24=y24+y56;y25=y25+y57;y26=y26+y58;y27=y27+y59;y28=y28+y60;y29=y29+y61;y30=y30+y62;y31=y31+y63;y32=y32+y64;

源码 3.公因式提取

语言：C++

运行环境：Visual Studio 2019

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
#define MAXSIZE 1000

//定义表达式结构体
```

```

typedef struct _bds
{
    string ys[8]; //存储 8 个字符串
    int deal; //是否已经处理标记
    _bds()
    {
        deal = 0; //0 表示未处理，1 表示已处理
    }
} StBds;

typedef struct _subbds
{
    string sub[4];
} StSub;

StSub g_zh[] = {
    {"(A+1)","(B+1)","(C+1)","(D+1)"}, //1
    {"(A+1)","(B+1)","(C+1)","D"}, //2
    {"(A+1)","(B+1)","C","(D+1)"}, //3
    {"(A+1)","(B+1)","C","D"}, //4
    {"(A+1)","B","(C+1)","(D+1)"}, //5
    {"(A+1)","B","(C+1)","D"}, //6
    {"(A+1)","B","C","(D+1)"}, //7
    {"(A+1)","B","C","D"}, //8
    {"A","(B+1)","(C+1)","(D+1)"}, //9
    {"A","(B+1)","(C+1)","D"}, //10
    {"A","(B+1)","C","(D+1)"}, //11
    {"A","(B+1)","C","D"}, //12
    {"A","B","(C+1)","(D+1)"}, //13
    {"A","B","(C+1)","D"}, //14
    {"A","B","C","(D+1)"}, //15
    {"A","B","C","D"} //16
};

/*将 str 字符串插入表达式对应的数组中*/
void InsertStr(StBds& bds, char str[])
{
    if (str[0] == '(')
    {
        int t = str[1] - 'A';
        bds.ys[t] = str;
    }
    else
    {
        int t = str[0] - 'A';
        bds.ys[t] = str;
    }
}

//读取文件，+前的部分作为一个表达式，每个表达式包含 8 个子式，表达式按照首字母插入数组
void ReadFile(const char* fileName, StBds allbds[], int& n)
{
    ifstream is(fileName);
    char buf[10] = { 0 };
    int i = 0;
    int flag = 0;
    int ret = 0;
    n = 0;

```

```

if (is.is_open())
{
    while (!is.eof())
    {
        buf[i] = is.get();
        if (buf[i] == EOF)
            break;

        if (buf[i] == '*') //表达式的一个子式
        {
            buf[i] = '\0';
            InsertStr(allbds[n], buf);
            i = 0;
        }
        else if (flag == 0 && buf[i] == '+') //+前的部分结束
        {
            buf[i] = '\0';
            InsertStr(allbds[n], buf);
            i = 0;
            n++;
        }
        else if ((buf[i] >= 'A' && buf[i] <= 'H') || buf[i] == 'I' || buf[i] == 'C' || buf[i] == 'J' ||
buf[i] == '+')
        {
            if (buf[i] == '(')
                flag = 1;
            else if (buf[i] == ')')
                flag = 0;
            i++;
        }
    }
    //如果表达式最后不是以'+'结束，需要用这部分处理
    if (i > 0 && (buf[0] != EOF) && (buf[0] != '+'))
    {
        buf[i] = '\0';
        InsertStr(allbds[n], buf);
        n++;
    }
    is.close();
}
else
    cout << "文件打开失败！" << endl;
}

//判断前 4 项是否一致
int IsCompete(StSub st, StBds bds)
{
    for (int i = 0; i < 4; i++)
    {
        if (st.sub[i].compare(bds.ys[i]) != 0)
            return 0;
    }
    return 1;
}

//处理数据
void DealData(const char* filename, StBds allbds[], int n)
{

```

```

ofstream os(filename); //结果保存到文件
int flag = 0;

for (int i = 0; i < 16; i++)
{
    flag = 0;
    for (int j = 0; j < n; j++)
    {
        if (allbds[j].deal == 0 && IsCompete(g_zh[i], allbds[j]))
        {
            allbds[j].deal = 1;
            if (flag != 0)
            {
                os << "+" << endl;
                cout << "+" << endl;
            }
            else
                flag = 1;

            //写文件
            for (int k = 0; k < 8; k++)
            {
                if (k < 7)
                {
                    os << allbds[j].ys[k] << "*";
                    cout << allbds[j].ys[k] << "*";
                }
                else
                {
                    os << allbds[j].ys[k];
                    cout << allbds[j].ys[k];
                }
            }
        }
    }
    if (i < 15)
    {
        os << "+" << endl << endl; //每个公因式之间加 2 个换行符
        cout << "+" << endl << endl; //每个公因式之间加 2 个换行符
    }
}
os.close();
}

int main()
{
    char src[200] = { 0 }; // "data.txt"; //源数据文件
    char dst[200] = { 0 }; // "result.txt"; //结果输出文件
    StBds allbds[MAXSIZE]; //保存所有表达式
    int n = 0; //表达式的数量

    cout << "请输入表达式所在的文件（TXT 格式）： ";
    cin >> src;
    cout << "请输入结果保存文件（TXT 格式）： ";
    cin >> dst;

    ReadFile(src, allbds, n);
    DealData(dst, allbds, n);
    cout << endl << "处理完成，结果保存在" << dst << "文件中！" << endl;
}

```

<pre> return 0; } </pre>	
源码 4.四变量元素组合情况统计	
语言：C++	运行环境：Visual Studio 2019
<pre> #include <iostream> #include <fstream> #include <string> using namespace std; #define MAXSIZE 1000 //定义表达式结构体 typedef struct _bds { string ys[8]; //存储 8 个字符串 } StBds; //定义情况结构体 typedef struct _datainfo { string gs[4]; //提取公因式后的表达式 } StDataInfo; //16 种可能的情况 StDataInfo g_qk[] = { {"E","F","G","H"}, //1 {"E","F","G","(H+1)"}, //2 {"E","F","(G+1)","H"}, //3 {"E","(F+1)","G","H"}, //4 {"(E+1)","F","G","H"}, //5 {"E","F","(G+1)","(H+1)"}, //6 {"E","(F+1)","G","(H+1)"}, //7 {"(E+1)","F","G","(H+1)"}, //8 {"E","(F+1)","(G+1)","H"}, //9 {"(E+1)","F","(G+1)","H"}, //10 {"(E+1)","(F+1)","G","H"}, //11 {"(E+1)","(F+1)","(G+1)","H"}, //12 {"(E+1)","(F+1)","G","(H+1)"}, //13 {"(E+1)","F","(G+1)","(H+1)"}, //14 {"E","(F+1)","(G+1)","(H+1)"}, //15 {"(E+1)","(F+1)","(G+1)","(H+1)"}, //16 }; /*将 str 字符串插入表达式对应的数组中*/ void InsertStr(StBds& bds, char str[]) { if (str[0] == '(') { int t = str[1] - 'A'; bds.ys[t] = str; } else { int t = str[0] - 'A'; bds.ys[t] = str; } } </pre>	

```
//读取文件（文件为提取公因式后的文件）
void ReadFile(const char* fileName, StBds allbds[], int& n)
{
    ifstream is(fileName);
    char buf[10] = { 0 };
    int i = 0;
    int flag = 0;
    int ret = 0;
    n = 0;
    if (is.is_open())
    {
        while (!is.eof())
        {
            buf[i] = is.get();
            if (buf[i] == EOF)
                break;

            if (buf[i] == '*') //表达式的一个子式
            {
                buf[i] = '\0';
                InsertStr(allbds[n], buf);
                i = 0;
            }
            else if (flag == 0 && buf[i] == '+') //+前的部分结束
            {
                buf[i] = '\0';
                InsertStr(allbds[n], buf);
                i = 0;
                n++;
            }
            else if ((buf[i] >= 'A' && buf[i] <= 'H') || buf[i] == 'I' || buf[i] == 'C' || buf[i] == 'J' ||
buf[i] == '+')
            {
                if (buf[i] == 'C')
                    flag = 1;
                else if (buf[i] == 'J')
                    flag = 0;
                i++;
            }
        }
        //如果表达式最后不是以+结束，需要用这部分处理
        if (i > 0 && (buf[0] != EOF) && (buf[0] != '+'))
        {
            buf[i] = '\0';
            InsertStr(allbds[n], buf);
            n++;
        }
        is.close();
    }
    else
        cout << "文件打开失败！" << endl;
}

//显示后 4 位
void show(StDataInfo qk, ofstream& os)
{
    for (int i = 0; i < 4; i++)
    {
        if (i < 3)
```



```

        {
            cout << qk.gs[i] << "x";
            os << qk.gs[i] << "x";
        }
        else
        {
            cout << qk.gs[i];
            os << qk.gs[i];
        }
    }
}

//判断两个表达式的公因式部分是否一致
int IsCompete(StBds s1, StBds s2)
{
    for (int i = 0; i < 4; i++)
    {
        if (s1.ys[i].compare(s2.ys[i]) != 0)
            return 0;
    }
    return 1;
}

//判断表达式的后 4 个子式是否是需要的
int IsNeeded(StBds s, StDataInfo info)
{
    for (int i = 0; i < 4; i++)
    {
        if (s.ys[i + 4].compare(info.gs[i]) != 0)
            return 0;
    }
    return 1;
}

//处理数据
void DealData(StBds allbds[], int n, const char* dst)
{
    ofstream os(dst); //结果保存到文件
    int cnt = 0;
    for (int i = 0; i < 16; i++)
    {
        for (int j = i + 1; j < 16; j++)
        {
            cnt = 0; //次数置 0
            int start = 0;
            int end = 0;
            while (start < n)
            {
                //找出公因式相同的表达式
                for (end = start + 1; end < n; end++)
                {
                    if (IsCompete(allbds[start], allbds[end]))
                        continue;
                    else
                        break;
                }
                //找到 start 到 end 区间, g_qk[i]与 g_qk[j]搭配的组合出现的次数
                int tmp = 0;
            }
        }
    }
}

```

<pre> for (int k = start; k < end; k++) { if (IsNeeded(allbds[k], g_qk[i]) IsNeeded(allbds[k], g_qk[j])) tmp++; } if (tmp == 2) //两种情况都出现了 { cnt++; //次数+1 } start = end; //继续判断下一组 } //输出--只输出出现过的组合，如果未出现过的组合也要输出的话，就把 if(cnt>0)这一句注释掉 if (cnt > 0) { show(g_qk[i], os); cout << "与"; os << "与"; show(g_qk[j], os); cout << "组合，出现的次数：" << cnt << endl; os << "组合，出现的次数：" << cnt << endl; } } } os.close(); cout << "结果已保存到" << dst << "文件！" << endl; } int main() { char src[200] = { 0 }; // "result.txt"; //提取公因式后的表达式文件 char dst[200] = { 0 }; // "tongji.txt"; //统计结果保存文件 StBds allbds[MAXSIZE]; int n = 0; cout << "请输入提取公因式后的表达式所在文件："; cin >> src; cout << "请输入统计结果保存文件："; cin >> dst; ReadFile(src, allbds, n); DealData(allbds, n, dst); return 0; } </pre>	
源码 5.两两组合后的合并元素替换两两组合前的双元素	
语言：C++	运行环境：Visual Studio 2019
<pre> #include <iostream> #include <fstream> using namespace std; #define MAXSIZE 1000 typedef struct _datainfo { int x; //值 1 int y; //值 2 </pre>	

```

        int v; //替换值
    } StDataInfo;

//插入数组
void AddData(StDataInfo data[], int& n, int x, int y, int v)
{
    int t;
    if (x > y) //保证存入数组中的 x-y 对, x 是小的
    {
        t = x;
        x = y;
        y = t;
    }
    data[n].x = x;
    data[n].y = y;
    data[n].v = v;
    n++;
}

//读取降序文件表, 存储的辅助位序号
void ReadFile(const char* filename, StDataInfo data[], int& n)
{
    ifstream is(filename);
    int x, y, v;
    n = 0;
    if (is.is_open())
    {
        while (!is.eof())
        {
            if (is >> x >> y >> v)
                AddData(data, n, x, y, v);
        }
        is.close();
    }
    else
        cout << "文件读取失败!" << endl;
}

/**判断 x,y 是否在数组 input 中, 如在, 从数组中删除, 并返回 1
 * input 数组
 * n 数组元素个数
 * x x 值, 保证 y>x
 * y y 值
 */
int IsInArray(int input[], int& n, int x, int y)
{
    int i = 0, j = 0;
    for (i = 0; i < n; i++)
    {
        if (input[i] == x) //x 在 input 中
        {
            for (j = i + 1; j < n; j++)
            {
                if (input[j] == y) //y 在 input 中
                {
                    int k = 0, m = 0;

```

```

        //将 x, y 从 input 中删除
        for (; k < n; k++)
        {
            if (k != i && k != j)
                input[m++] = input[k];
        }
        n -= 2;
        return 1;
    }
}

return 0;
}

/**处理数据
 * data 替换组合对
 * n data 数组元素个数
 * input 需要处理的数组
 * nmb input 数组元素个数
 */
void DealData(StDataInfo data[], int n, int input[], int nmb)
{
    int out[MAXSIZE];
    int cnt = 0; //保存个数
    cout << "    >> 数字移除顺序: " << endl;
    while (nmb >= 2)
    {
        for (int i = 0; i < n; i++)
        {
            if (IsInArray(input, nmb, data[i].x, data[i].y))
            {
                cout << "        ->移除: " << data[i].x << " " << data[i].y << endl;
                out[cnt++] = data[i].v;
                //cout << data[i].v << " ";
            }
        }
    }
    cout << "    >> 输出结果: ";
    if (nmb == 1)
        cout << input[0] << " ";
    for (int i = 0; i < cnt; i++)
    {
        if (i < cnt - 1)
            cout << out[i] << " ";
        else
            cout << out[i];
    }
    cout << endl;
}

int main()
{
    StDataInfo data[MAXSIZE]; //保存替换数值对
    int n = 0; //记录条数
    const char* filename = "data.txt";
    ReadFile(filename, data, n);
}

```

<pre> int input[20]; //保存需要处理的数字 int k = 0; //需要处理的数据组数 int nmb = 0; //每组数据的个数 cout << "请输入需要处理的数据组数："; cin >> k; for (int i = 0; i < k; i++) { cout << "请输入第" << i + 1 << "组数据，数据以空格间隔，回车结束："; nmb = 0; while (1) { cin >> input[nmb]; nmb++; char ch = cin.get(); if (ch == '\n') break; } //处理数据 DealData(data, n, input, nmb); } return 0; } </pre>	
源码 6.SageMath 直接验证输出 ANF 正确性	
语言：python	运行环境：SageMath9.3
<pre> #向量布尔函数二元域空间及自变量定义 P.<x1,x2,x3,x4,x5,x6,x7,x8,x9,x10,x11,x12,x13,x14,x15,x16,x17,x18,x19,x20,x21,x22,x23,x24,x25,x 26,x27,x28,x29,x30,x31,x32,x33,x34,x35,x36,x37,x38,x39,x40,x41,x42,x43,x44,x45,x46,x47,x48,x49, x50,x51,x52,x53,x54,x55,x56,x57,x58,x59,x60,x61,x62,x63,x64,x65,x66,x67,x68,x69,x70,x71,x72,x7 3,x74,x75,x76,x77,x78,x79,x80,x81,x82,x83,x84,x85,x86,x87,x88,x89,x90,x91,x92,x93,x94,x95,x96,x 97,x98,x99,x100,x101,x102,x103,x104,x105,x106,x107,x108,x109,x110,x111,x112,x113,x114,x115,x 116,x117,x118,x119,x120,x121,x122,x123,x124,x125,x126,x127,x128>= BooleanPolynomialRing(128) #向量布尔函数目标原位输出内容 f1=x33 f2=x34 f3=x35 f4=x36 f5=x37 f6=x38 f7=x39 f8=x40 f9=x41 f10=x42 f11=x43 f12=x44 f13=x45 f14=x46 f15=x47 f16=x48 f17=x49 f18=x50 f19=x51 f20=x52 f21=x53 f22=x54 </pre>	

$f_{23}=x_{55}$
 $f_{24}=x_{56}$
 $f_{25}=x_{57}$
 $f_{26}=x_{58}$
 $f_{27}=x_{59}$
 $f_{28}=x_{60}$
 $f_{29}=x_{61}$
 $f_{30}=x_{62}$
 $f_{31}=x_{63}$
 $f_{32}=x_{64}$
 $f_{33}=x_{65}$
 $f_{34}=x_{66}$
 $f_{35}=x_{67}$
 $f_{36}=x_{68}$
 $f_{37}=x_{69}$
 $f_{38}=x_{70}$
 $f_{39}=x_{71}$
 $f_{40}=x_{72}$
 $f_{41}=x_{73}$
 $f_{42}=x_{74}$
 $f_{43}=x_{75}$
 $f_{44}=x_{76}$
 $f_{45}=x_{77}$
 $f_{46}=x_{78}$
 $f_{47}=x_{79}$
 $f_{48}=x_{80}$
 $f_{49}=x_{81}$
 $f_{50}=x_{82}$
 $f_{51}=x_{83}$
 $f_{52}=x_{84}$
 $f_{53}=x_{85}$
 $f_{54}=x_{86}$
 $f_{55}=x_{87}$
 $f_{56}=x_{88}$
 $f_{57}=x_{89}$
 $f_{58}=x_{90}$
 $f_{59}=x_{91}$
 $f_{60}=x_{92}$
 $f_{61}=x_{93}$
 $f_{62}=x_{94}$
 $f_{63}=x_{95}$
 $f_{64}=x_{96}$
 $f_{65}=x_{97}$
 $f_{66}=x_{98}$
 $f_{67}=x_{99}$
 $f_{68}=x_{100}$
 $f_{69}=x_{101}$
 $f_{70}=x_{102}$
 $f_{71}=x_{103}$
 $f_{72}=x_{104}$
 $f_{73}=x_{105}$
 $f_{74}=x_{106}$
 $f_{75}=x_{107}$
 $f_{76}=x_{108}$
 $f_{77}=x_{109}$
 $f_{78}=x_{110}$
 $f_{79}=x_{111}$
 $f_{80}=x_{112}$
 $f_{81}=x_{113}$
 $f_{82}=x_{114}$

```

f83=x115
f84=x116
f85=x117
f86=x118
f87=x119
f88=x120
f89=x121
f90=x122
f91=x123
f92=x124
f93=x125
f94=x126
f95=x127
f96=x128
f97=x1+.....详见赛题二附录 SM4_ANF 中 SM4out1
F98=x2+.....详见赛题二附录 SM4_ANF 中 SM4out2
f99=x3+.....详见赛题二附录 SM4_ANF 中 SM4out3
f100=x4+.....详见赛题二附录 SM4_ANF 中 SM4out4
f101=x5+.....详见赛题二附录 SM4_ANF 中 SM4out5
f102=x6+.....详见赛题二附录 SM4_ANF 中 SM4out6
f103=x7+.....详见赛题二附录 SM4_ANF 中 SM4out7
f104=x8+.....详见赛题二附录 SM4_ANF 中 SM4out8
f105=x9+.....详见赛题二附录 SM4_ANF 中 SM4out9
f106=x10+.....详见赛题二附录 SM4_ANF 中 SM4out10
f107=x11+.....详见赛题二附录 SM4_ANF 中 SM4out11
f108=x12+.....详见赛题二附录 SM4_ANF 中 SM4out12
f109=x13+.....详见赛题二附录 SM4_ANF 中 SM4out13
f110=x14+.....详见赛题二附录 SM4_ANF 中 SM4out14
f111=x15+.....详见赛题二附录 SM4_ANF 中 SM4out15
f112=x16+.....详见赛题二附录 SM4_ANF 中 SM4out16
f113=x17+.....详见赛题二附录 SM4_ANF 中 SM4out17
f114=x18+.....详见赛题二附录 SM4_ANF 中 SM4out18
f115=x19+.....详见赛题二附录 SM4_ANF 中 SM4out19
f116=x20+.....详见赛题二附录 SM4_ANF 中 SM4out20
f117=x21+.....详见赛题二附录 SM4_ANF 中 SM4out21
f118=x22+.....详见赛题二附录 SM4_ANF 中 SM4out22
f119=x23+.....详见赛题二附录 SM4_ANF 中 SM4out23
f120=x24+.....详见赛题二附录 SM4_ANF 中 SM4out24
f121=x25+.....详见赛题二附录 SM4_ANF 中 SM4out25
f122=x26+.....详见赛题二附录 SM4_ANF 中 SM4out26
f123=x27+.....详见赛题二附录 SM4_ANF 中 SM4out27
f124=x28+.....详见赛题二附录 SM4_ANF 中 SM4out28
f125=x29+.....详见赛题二附录 SM4_ANF 中 SM4out29
f126=x30+.....详见赛题二附录 SM4_ANF 中 SM4out30
f127=x31+.....详见赛题二附录 SM4_ANF 中 SM4out31
f128=x32+.....详见赛题二附录 SM4_ANF 中 SM4out32

# Initialization
y1=x33;y2=x34;y3=x35;y4=x36;y5=x37;y6=x38;y7=x39;y8=x40;y9=x41;y10=x42;y11=x43;y12=x44
;y13=x45;y14=x46;y15=x47;y16=x48;y17=x49;y18=x50;y19=x51;y20=x52;y21=x53;y22=x54;y23=x
55;y24=x56;y25=x57;y26=x58;y27=x59;y28=x60;y29=x61;y30=x62;y31=x63;y32=x64;y33=x65;y34
=x66;y35=x67;y36=x68;y37=x69;y38=x70;y39=x71;y40=x72;y41=x73;y42=x74;y43=x75;y44=x76;y
45=x77;y46=x78;y47=x79;y48=x80;y49=x81;y50=x82;y51=x83;y52=x84;y53=x85;y54=x86;y55=x87
;y56=x88;y57=x89;y58=x90;y59=x91;y60=x92;y61=x93;y62=x94;y63=x95;y64=x96;y65=x97;y66=x
98;y67=x99;y68=x100;y69=x101;y70=x102;y71=x103;y72=x104;y73=x105;y74=x106;y75=x107;y76
=x108;y77=x109;y78=x110;y79=x111;y80=x112;y81=x113;y82=x114;y83=x115;y84=x116;y85=x11

```

7;y86=x118;y87=x119;y88=x120;y89=x121;y90=x122;y91=x123;y92=x124;y93=x125;y94=x126;y95=x127;y96=x128;y97=x1;y98=x2;y99=x3;y100=x4;y101=x5;y102=x6;y103=x7;y104=x8;y105=x9;y106=x10;y107=x11;y108=x12;y109=x13;y110=x14;y111=x15;y112=x16;y113=x17;y114=x18;y115=x19;y116=x20;y117=x21;y118=x22;y119=x23;y120=x24;y121=x25;y122=x26;y123=x27;y124=x28;y125=x29;y126=x30;y127=x31;y128=x32;y129=0;y130=0;y131=0;y132=0;y133=0;y134=0;y135=0;y136=0;y137=0;y138=0;y139=0;y140=0;y141=0;y142=0;y143=0;y144=0;y145=0;y146=0;y147=0;y148=0;y149=0;y150=0;y151=0;y152=0;y153=0;y154=0;y155=0;y156=0;y157=0;y158=0;y159=0;y160=0;y161=0;y162=0;y163=0;y164=0;y165=0;y166=0;y167=0;y168=0;y169=0;y170=0;y171=0;y172=0;y173=0;y174=0;y175=0;y176=0;y177=0;y178=0;y179=0;y180=0;y181=0;y182=0;y183=0;y184=0;y185=0;y186=0;y187=0;y188=0;y189=0;y190=0;y191=0;y192=0;y193=0;y194=0;y195=0;y196=0;y197=0;y198=0;y199=0;y200=0;y201=0;y202=0;y203=0;y204=0;y205=0;y206=0;y207=0;y208=0;y209=0;y210=0;y211=0;y212=0;y213=0;y214=0;y215=0;y216=0;y217=0;y218=0;y219=0;y220=0;y221=0;y222=0;y223=0;y224=0;y225=0;y226=0;y227=0;y228=0;y229=0;y230=0;y231=0;y232=0;y233=0;y234=0;y235=0;y236=0;y237=0;y238=0;y239=0;y240=0;y241=0;y242=0;y243=0;y244=0;y245=0;y246=0;y247=0;y248=0;y249=0;y250=0;y251=0;y252=0;y253=0;y254=0;y255=0;y256=0;y257=0;y258=0;y259=0;y260=0;y261=0;y262=0;y263=0;y264=0;y265=0;y266=0;y267=0;y268=0;y269=0;y270=0;y271=0;y272=0;y273=0;y274=0;y275=0;y276=0;y277=0;y278=0;y279=0;y280=0;y281=0;y282=0;y283=0;y284=0;y285=0;y286=0;y287=0;y288=0;y289=0;y290=0;y291=0;y292=0;y293=0;y294=0;y295=0;y296=0;y297=0;y298=0;y299=0;y300=0;y301=0;y302=0;y303=0;y304=0;y305=0;y306=0;y307=0;y308=0;y309=0;y310=0;y311=0;y312=0;y313=0;y314=0;y315=0;y316=0;y317=0;y318=0;y319=0;y320=0;y321=0;y322=0;y323=0;y324=0;y325=0;y326=0;y327=0;y328=0;y329=0;y330=0;y331=0;y332=0;y333=0;y334=0;y335=0;y336=0;y337=0;y338=0;y339=0;y340=0;y341=0;y342=0;y343=0;y344=0;y345=0;y346=0;y347=0;y348=0;y349=0;y350=0;y351=0;y352=0;y353=0;y354=0;y355=0;y356=0;y357=0;y358=0;y359=0;y360=0;y361=0;y362=0;y363=0;y364=0;y365=0;y366=0;y367=0;y368=0;y369=0;y370=0;y371=0;y372=0;y373=0;y374=0;y375=0;y376=0;y377=0;y378=0;y379=0;y380=0;y381=0;y382=0;y383=0;y384=0;y385=0;y386=0;y387=0;y388=0;y389=0;y390=0;y391=0;y392=0;y393=0;y394=0;y395=0;y396=0;y397=0;y398=0;y399=0;y400=0;y401=0;y402=0;y403=0;y404=0;y405=0;y406=0;y407=0;y408=0;y409=0;y410=0;y411=0;y412=0;y413=0;y414=0;y415=0;y416=0;y417=0;y418=0;y419=0;y420=0;y421=0;y422=0;y423=0;y424=0;y425=0;y426=0;y427=0;y428=0;y429=0;y430=0;y431=0;y432=0;y433=0;y434=0;y435=0;y436=0;y437=0;y438=0;y439=0;y440=0;y441=0;y442=0;y443=0;y444=0;y445=0;y446=0;y447=0;y448=0;y449=0;y450=0;y451=0;y452=0;y453=0;y454=0;y455=0;y456=0;y457=0;y458=0;y459=0;y460=0;y461=0;y462=0;y463=0;y464=0;y465=0;y466=0;y467=0;y468=0;y469=0;y470=0;y471=0;y472=0;y473=0;y474=0;y475=0;y476=0;y477=0;y478=0;y479=0;y480=0;y481=0;y482=0;y483=0;y484=0;y485=0;y486=0;y487=0;y488=0;y489=0;y490=0;y491=0;y492=0;y493=0;y494=0;y495=0;y496=0;y497=0;y498=0;y499=0;y500=0;y501=0;y502=0;y503=0;y504=0;y505=0;y506=0;y507=0;y508=0;y509=0;y510=0;y511=0;y512=0;y513=0;y514=0;y515=0;y516=0;y517=0;y518=0;y519=0;y520=0;y521=0;y522=0;y523=0;y524=0;y525=0;y526=0;y527=0;y528=0;y529=0;y530=0;y531=0;y532=0;y533=0;y534=0;y535=0;y536=0;y537=0;y538=0;y539=0;y540=0;y541=0;y542=0;y543=0;y544=0;y545=0;y546=0;y547=0;y548=0;y549=0;y550=0;y551=0;y552=0;y553=0;y554=0;y555=0;y556=0;y557=0;y558=0;y559=0;y560=0;y561=0;y562=0;y563=0;y564=0;y565=0;y566=0;y567=0;y568=0;y569=0;y570=0;y571=0;y572=0;y573=0;y574=0;y575=0;y576=0;y577=0;y578=0;y579=0;y580=0;y581=0;y582=0;y583=0;y584=0;y585=0;y586=0;y587=0;y588=0;y589=0;y590=0;y591=0;y592=0;y593=0;y594=0;y595=0;y596=0;y597=0;y598=0;y599=0;y600=0;y601=0;y602=0;y603=0;y604=0;y605=0;y606=0;y607=0;y608=0;y609=0;y610=0;y611=0;y612=0;y613=0;y614=0;y615=0;y616=0;y617=0;y618=0;y619=0;y620=0;y621=0;y622=0;y623=0;y624=0;y625=0;y626=0;y627=0;y628=0;y629=0;y630=0;y631=0;y632=0;y633=0;y634=0;y635=0;y636=0;y637=0;y638=0;y639=0;y640=0;y641=0;y642=0;y643=0;y644=0;y645=0;y646=0;y647=0;y648=0;y649=0;y650=0;y651=0;y652=0;y653=0;y654=0;y655=0;y656=0;y657=0;y658=0;y659=0;y660=0;y661=0;y662=0;y663=0;y664=0;y665=0;y666=0;y667=0;y668=0;y669=0;y670=0;y671=0;y672=0;y673=0;y674=0;y675=0;y676=0;y677=0;y678=0;y679=0;y680=0;y681=0;y682=0;y683=0;y684=0;y685=0;y686=0;y687=0;y688=0;y689=0;y690=0;y691=0;y692=0;y693=0;y694=0;y695=0;y696=0;y697=0;y698=0;y699=0;y700=0;y701=0;y702=0;y703=0;y704=0;y705=0;y706=0;y707=0;y708=0;y709=0;y710=0;y711=0;y712=0;y713=0;y714=0;y715=0;y716=0;y717=0;y718=0;y719=0;y720=0;y721=0;y722=0;y723=0;y724=0;y725=0;y726=0;y727=0;y728=0;y729=0;y730=0;y731=0;y732=0;y733=0;y734=0;y735=0;y736=0;y737=0;y738=0;y739=0;y740=0;y741=0;y742=0;y743=0;y744=0;y745=0;y746=0;y747=0;y748=0;y749=0;y750=0;y751=0;y752=0;y753=0;y754=0;y755=0;y756=0;y757=0;y758=0;y759=0;y760=0;y761=0;y762=0;y763=0;y764=0;y765=0;y766=0;y767=0;y768=0;y769=0;y770=0;y771=0;y772=0;y773=0;y774=0;y775=0;y776=0;y777=0;y778=0;y779=0;y780=0;y781=0;y782=0;y783=0;y784=0;y785=0;y786=0;y787=0;y788=0;y789=0;y790=0;y791=0;y792=0;y793=0;y794=0;y795=0;y796=0;y797=0;y798=0;y799=0;y800=0;y801=0;y802=0;y803=0;y804=0;y805=0;y806=0;y807=0;y808=0;y809=0;y810=0;y811=0;y812=0;y813=0;y814=

0;y815=0;y816=0;y817=0;y818=0;y819=0;y820=0;y821=0;y822=0;y823=0;y824=0;y825=0;y826=0;y827=0;y828=0;y829=0;y830=0;y831=0;y832=0;y833=0;y834=0;y835=0;y836=0;y837=0;y838=0;y839=0;y840=0;y841=0;y842=0;y843=0;y844=0;y845=0;y846=0;y847=0;y848=0;y849=0;y850=0;y851=0;y852=0;y853=0;y854=0;y855=0;y856=0;y857=0;y858=0;y859=0;y860=0;y861=0;y862=0;y863=0;y864=0;y865=0;y866=0;y867=0;y868=0;y869=0;y870=0;y871=0;y872=0;y873=0;y874=0;y875=0;y876=0;y877=0;y878=0;y879=0;y880=0;y881=0;y882=0;y883=0;y884=0;y885=0;y886=0;y887=0;y888=0;y889=0;y890=0;y891=0;y892=0;y893=0;y894=0;y895=0;y896=0;y897=0;y898=0;y899=0;y900=0;y901=0;y902=0;y903=0;y904=0;y905=0;y906=0;y907=0;y908=0;y909=0;y910=0;y911=0;y912=0;y913=0;y914=0;y915=0;y916=0;y917=0;y918=0;y919=0;y920=0;y921=0;y922=0;y923=0;y924=0;y925=0;y926=0;y927=0;y928=0;y929=0;y930=0;y931=0;y932=0;y933=0;y934=0;y935=0;y936=0;y937=0;y938=0;y939=0;y940=0;y941=0;y942=0;y943=0;y944=0;y945=0;y946=0;y947=0;y948=0;y949=0;y950=0;y951=0;y952=0;y953=0;y954=0;y955=0;y956=0;y957=0;y958=0;y959=0;y960=0;y961=0;y962=0;y963=0;y964=0;y965=0;y966=0;y967=0;y968=0;y969=0;y970=0;y971=0;y972=0;y973=0;y974=0;y975=0;y976=0;y977=0;y978=0;y979=0;y980=0;y981=0;y982=0;y983=0;y984=0;y985=0;y986=0;y987=0;y988=0;y989=0;y990=0;y991=0;y992=0;y993=0;y994=0;y995=0;y996=0;y997=0;y998=0;y999=0;y1000=0;y1001=0;y1002=0;y1003=0;y1004=0;y1005=0;y1006=0;y1007=0;y1008=0;y1009=0;y1010=0;y1011=0;y1012=0;y1013=0;y1014=0;y1015=0;y1016=0;y1017=0;y1018=0;y1019=0;y1020=0;y1021=0;y1022=0;y1023=0;y1024=0;y1025=0;y1026=0;y1027=0;y1028=0;y1029=0;y1030=0;y1031=0;y1032=0;y1033=0;y1034=0;y1035=0;y1036=0;y1037=0;y1038=0;y1039=0;y1040=0;y1041=0;y1042=0;y1043=0;y1044=0;y1045=0;y1046=0;y1047=0;y1048=0;y1049=0;y1050=0;y1051=0;y1052=0;y1053=0;y1054=0;y1055=0;y1056=0;y1057=0;y1058=0;y1059=0;y1060=0;y1061=0;y1062=0;y1063=0;y1064=0;y1065=0;y1066=0;y1067=0;y1068=0;y1069=0;y1070=0;y1071=0;y1072=0;y1073=0;y1074=0;y1075=0;y1076=0;y1077=0;y1078=0;y1079=0;y1080=0;y1081=0;y1082=0;y1083=0;y1084=0;y1085=0;y1086=0;y1087=0;y1088=0;y1089=0;y1090=0;y1091=0;y1092=0;y1093=0;y1094=0;y1095=0;y1096=0;y1097=0;y1098=0;y1099=0;y1100=0;y1101=0;y1102=0;y1103=0;y1104=0;y1105=0;y1106=0;y1107=0;y1108=0;y1109=0;y1110=0;y1111=0;y1112=0;y1113=0;y1114=0;y1115=0;y1116=0;y1117=0;y1118=0;y1119=0;y1120=0;y1121=0;y1122=0;y1123=0;y1124=0;y1125=0;y1126=0;y1127=0;y1128=0;y1129=0;y1130=0;y1131=0;y1132=0;y1133=0;y1134=0;y1135=0;y1136=0;y1137=0;y1138=0;y1139=0;y1140=0;y1141=0;y1142=0;y1143=0;y1144=0;y1145=0;y1146=0;y1147=0;y1148=0;y1149=0;y1150=0;y1151=0;y1152=0;y1153=0;y1154=0;y1155=0;y1156=0;y1157=0;y1158=0;y1159=0;y1160=0;y1161=0;y1162=0;y1163=0;y1164=0;y1165=0;y1166=0;y1167=0;y1168=0;y1169=0;y1170=0;y1171=0;y1172=0;y1173=0;y1174=0;y1175=0;y1176=0;y1177=0;y1178=0;y1179=0;y1180=0;y1181=0;y1182=0;y1183=0;y1184=0;y1185=0;y1186=0;y1187=0;y1188=0;y1189=0;y1190=0;y1191=0;y1192=0;y1193=0;y1194=0;y1195=0;y1196=0;y1197=0;y1198=0;y1199=0;y1200=0;y1201=0;y1202=0;y1203=0;y1204=0;y1205=0;y1206=0;y1207=0;y1208=0;y1209=0;y1210=0;y1211=0;y1212=0;y1213=0;y1214=0;y1215=0;y1216=0;y1217=0;y1218=0;y1219=0;y1220=0;y1221=0;y1222=0;y1223=0;y1224=0;y1225=0;y1226=0;y1227=0;y1228=0;y1229=0;y1230=0;y1231=0;y1232=0;y1233=0;y1234=0;y1235=0;y1236=0;y1237=0;y1238=0;y1239=0;y1240=0;y1241=0;y1242=0;y1243=0;y1244=0;y1245=0;y1246=0;y1247=0;y1248=0;y1249=0;y1250=0;y1251=0;y1252=0;y1253=0;y1254=0;y1255=0;y1256=0;y1257=0;y1258=0;y1259=0;y1260=0;y1261=0;y1262=0;y1263=0;y1264=0;y1265=0;y1266=0;y1267=0;y1268=0;y1269=0;y1270=0;y1271=0;y1272=0;y1273=0;y1274=0;y1275=0;y1276=0;y1277=0;y1278=0;y1279=0;y1280=0;y1281=0;y1282=0;y1283=0;y1284=0;y1285=0;y1286=0;y1287=0;y1288=0;y1289=0;y1290=0;y1291=0;y1292=0;y1293=0;y1294=0;y1295=0;y1296=0;y1297=0;y1298=0;y1299=0;y1300=0;y1301=0;y1302=0;y1303=0;y1304=0;y1305=0;y1306=0;y1307=0;y1308=0;y1309=0;y1310=0;y1311=0;y1312=0;y1313=0;y1314=0;y1315=0;y1316=0;y1317=0;y1318=0;y1319=0;y1320=0;y1321=0;y1322=0;y1323=0;y1324=0;y1325=0;y1326=0;y1327=0;y1328=0;y1329=0;y1330=0;y1331=0;y1332=0;y1333=0;y1334=0;y1335=0;y1336=0;y1337=0;y1338=0;y1339=0;y1340=0;y1341=0;y1342=0;y1343=0;y1344=0;

#Full depth 1 and Toffoli depth 0

#对位相加 1

y1=y1+y33;y2=y2+y34;y3=y3+y35;y4=y4+y36;y5=y5+y37;y6=y6+y38;y7=y7+y39;y8=y8+y40;y9=y9+y41;y10=y10+y42;y11=y11+y43;y12=y12+y44;y13=y13+y45;y14=y14+y46;y15=y15+y47;y16=y16+y48;y17=y17+y49;y18=y18+y50;y19=y19+y51;y20=y20+y52;y21=y21+y53;y22=y22+y54;y23=y23+y55;y24=y24+y56;y25=y25+y57;y26=y26+y58;y27=y27+y59;y28=y28+y60;y29=y29+y61;y30=y30+y62;y31=y31+y63;y32=y32+y64;

#Full depth 2 and Toffoli depth 0

#对位相加 2

y1=y1+y65;y2=y2+y66;y3=y3+y67;y4=y4+y68;y5=y5+y69;y6=y6+y70;y7=y7+y71;y8=y8+y72;y9=y9+y73;

$y_{73}=y_{10}+y_{74}; y_{11}=y_{11}+y_{75}; y_{12}=y_{12}+y_{76}; y_{13}=y_{13}+y_{77}; y_{14}=y_{14}+y_{78}; y_{15}=y_{15}+y_{79}; y_{16}=y_{16}+y_{80}; y_{17}=y_{17}+y_{81}; y_{18}=y_{18}+y_{82}; y_{19}=y_{19}+y_{83}; y_{20}=y_{20}+y_{84}; y_{21}=y_{21}+y_{85}; y_{22}=y_{22}+y_{86}; y_{23}=y_{23}+y_{87}; y_{24}=y_{24}+y_{88}; y_{25}=y_{25}+y_{89}; y_{26}=y_{26}+y_{90}; y_{27}=y_{27}+y_{91}; y_{28}=y_{28}+y_{92}; y_{29}=y_{29}+y_{93}; y_{30}=y_{30}+y_{94}; y_{31}=y_{31}+y_{95}; y_{32}=y_{32}+y_{96};$

#Full depth 3 and Toffoli depth 0

#添加辅助位

$y_{129}=y_{129}+y_1; y_{130}=y_{130}+y_2; y_{131}=y_{131}+y_3; y_{132}=y_{132}+y_4; y_{133}=y_{133}+y_5; y_{134}=y_{134}+y_6; y_{135}=y_{135}+y_7; y_{136}=y_{136}+y_8; y_{137}=y_{137}+y_9; y_{138}=y_{138}+y_{10}; y_{139}=y_{139}+y_{11}; y_{140}=y_{140}+y_{12}; y_{141}=y_{141}+y_{13}; y_{142}=y_{142}+y_{14}; y_{143}=y_{143}+y_{15}; y_{144}=y_{144}+y_{16}; y_{145}=y_{145}+y_{17}; y_{146}=y_{146}+y_{18}; y_{147}=y_{147}+y_{19}; y_{148}=y_{148}+y_{20}; y_{149}=y_{149}+y_{21}; y_{150}=y_{150}+y_{22}; y_{151}=y_{151}+y_{23}; y_{152}=y_{152}+y_{24}; y_{153}=y_{153}+y_{25}; y_{154}=y_{154}+y_{26}; y_{155}=y_{155}+y_{27}; y_{156}=y_{156}+y_{28}; y_{157}=y_{157}+y_{29}; y_{158}=y_{158}+y_{30}; y_{159}=y_{159}+y_{31}; y_{160}=y_{160}+y_{32};$

#Full depth 4 and Toffoli depth 0

#映射

$y_{129}=y_{129}+1; y_{130}=y_{130}+1; y_{131}=y_{131}+1; y_{132}=y_{132}+1; y_{133}=y_{133}+1; y_{134}=y_{134}+1; y_{135}=y_{135}+1; y_{136}=y_{136}+1; y_{137}=y_{137}+1; y_{138}=y_{138}+1; y_{139}=y_{139}+1; y_{140}=y_{140}+1; y_{141}=y_{141}+1; y_{142}=y_{142}+1; y_{143}=y_{143}+1; y_{144}=y_{144}+1; y_{145}=y_{145}+1; y_{146}=y_{146}+1; y_{147}=y_{147}+1; y_{148}=y_{148}+1; y_{149}=y_{149}+1; y_{150}=y_{150}+1; y_{151}=y_{151}+1; y_{152}=y_{152}+1; y_{153}=y_{153}+1; y_{154}=y_{154}+1; y_{155}=y_{155}+1; y_{156}=y_{156}+1; y_{157}=y_{157}+1; y_{158}=y_{158}+1; y_{159}=y_{159}+1; y_{160}=y_{160}+1;$

#Full depth 5 and Toffoli depth 1

#2 变量元素组合情况

$y_{161}=y_{161}+y_1*y_2; y_{162}=y_{162}+y_1*y_{130}; y_{163}=y_{163}+y_{129}*y_2; y_{164}=y_{164}+y_{129}*y_{130}; y_{165}=y_{165}+y_3*y_4; y_{166}=y_{166}+y_3*y_{132}; y_{167}=y_{167}+y_{131}*y_4; y_{168}=y_{168}+y_{131}*y_{132}; y_{169}=y_{169}+y_5*y_6; y_{170}=y_{170}+y_5*y_{134}; y_{171}=y_{171}+y_{133}*y_6; y_{172}=y_{172}+y_{133}*y_{134}; y_{173}=y_{173}+y_7*y_8; y_{174}=y_{174}+y_7*y_{136}; y_{175}=y_{175}+y_{135}*y_8; y_{176}=y_{176}+y_{135}*y_{136}; y_{177}=y_{177}+y_9*y_{10}; y_{178}=y_{178}+y_9*y_{138}; y_{179}=y_{179}+y_{137}*y_{10}; y_{180}=y_{180}+y_{137}*y_{138}; y_{181}=y_{181}+y_{11}*y_{12}; y_{182}=y_{182}+y_{11}*y_{140}; y_{183}=y_{183}+y_{139}*y_{12}; y_{184}=y_{184}+y_{139}*y_{140}; y_{185}=y_{185}+y_{13}*y_{14}; y_{186}=y_{186}+y_{13}*y_{142}; y_{187}=y_{187}+y_{141}*y_{14}; y_{188}=y_{188}+y_{141}*y_{142}; y_{189}=y_{189}+y_{15}*y_{16}; y_{190}=y_{190}+y_{15}*y_{144}; y_{191}=y_{191}+y_{143}*y_{16}; y_{192}=y_{192}+y_{143}*y_{144}; y_{193}=y_{193}+y_{17}*y_{18}; y_{194}=y_{194}+y_{17}*y_{146}; y_{195}=y_{195}+y_{145}*y_{18}; y_{196}=y_{196}+y_{145}*y_{146}; y_{197}=y_{197}+y_{19}*y_{20}; y_{198}=y_{198}+y_{19}*y_{148}; y_{199}=y_{199}+y_{147}*y_{20}; y_{200}=y_{200}+y_{147}*y_{148}; y_{201}=y_{201}+y_{21}*y_{22}; y_{202}=y_{202}+y_{21}*y_{150}; y_{203}=y_{203}+y_{149}*y_{22}; y_{204}=y_{204}+y_{149}*y_{150}; y_{205}=y_{205}+y_{23}*y_{24}; y_{206}=y_{206}+y_{23}*y_{152}; y_{207}=y_{207}+y_{151}*y_{24}; y_{208}=y_{208}+y_{151}*y_{152}; y_{209}=y_{209}+y_{25}*y_{26}; y_{210}=y_{210}+y_{25}*y_{154}; y_{211}=y_{211}+y_{153}*y_{26}; y_{212}=y_{212}+y_{153}*y_{154}; y_{213}=y_{213}+y_{27}*y_{28}; y_{214}=y_{214}+y_{27}*y_{156}; y_{215}=y_{215}+y_{155}*y_{28}; y_{216}=y_{216}+y_{155}*y_{156}; y_{217}=y_{217}+y_{29}*y_{30}; y_{218}=y_{218}+y_{29}*y_{158}; y_{219}=y_{219}+y_{157}*y_{30}; y_{220}=y_{220}+y_{157}*y_{158}; y_{221}=y_{221}+y_{31}*y_{32}; y_{222}=y_{222}+y_{31}*y_{160}; y_{223}=y_{223}+y_{159}*y_{32}; y_{224}=y_{224}+y_{159}*y_{160};$

#Full depth 6 and Toffoli depth 2

#4 变量元素组合情况

$y_{225}=y_{225}+y_{161}*y_{165}; y_{226}=y_{226}+y_{161}*y_{166}; y_{227}=y_{227}+y_{161}*y_{167}; y_{228}=y_{228}+y_{161}*y_{168}; y_{229}=y_{229}+y_{162}*y_{165}; y_{230}=y_{230}+y_{162}*y_{166}; y_{231}=y_{231}+y_{162}*y_{167}; y_{232}=y_{232}+y_{162}*y_{168}; y_{233}=y_{233}+y_{163}*y_{165}; y_{234}=y_{234}+y_{163}*y_{166}; y_{235}=y_{235}+y_{163}*y_{167}; y_{236}=y_{236}+y_{163}*y_{168}; y_{237}=y_{237}+y_{164}*y_{165}; y_{238}=y_{238}+y_{164}*y_{166}; y_{239}=y_{239}+y_{164}*y_{167}; y_{240}=y_{240}+y_{164}*y_{168}; y_{241}=y_{241}+y_{169}*y_{173}; y_{242}=y_{242}+y_{169}*y_{174}; y_{243}=y_{243}+y_{169}*y_{175}; y_{244}=y_{244}+y_{169}*y_{176}; y_{245}=y_{245}+y_{170}*y_{173}; y_{246}=y_{246}+y_{170}*y_{174}; y_{247}=y_{247}+y_{170}*y_{175}; y_{248}=y_{248}+y_{170}*y_{176}; y_{249}=y_{249}+y_{171}*y_{173}; y_{250}=y_{250}+y_{171}*y_{174}; y_{251}=y_{251}+y_{171}*y_{175}; y_{252}=y_{252}+y_{171}*y_{176}; y_{253}=y_{253}+y_{172}*y_{173}; y_{254}=y_{254}+y_{172}*y_{174}; y_{255}=y_{255}+y_{172}*y_{175}; y_{256}=y_{256}+y_{172}*y_{176}; y_{257}=y_{257}+y_{177}*y_{181}; y_{258}=y_{258}+y_{177}*y_{182}; y_{259}=y_{259}+y_{177}*y_{183}; y_{260}=y_{260}+y_{177}*y_{184}; y_{261}=y_{261}+y_{178}*y_{181}; y_{262}=y_{262}+y_{178}*y_{182}; y_{263}=y_{263}+y_{178}*y_{183}; y_{264}=y_{264}+y_{178}*y_{184}; y_{265}=y_{265}+y_{179}*y_{181}; y_{266}=y_{266}+y_{179}*y_{182}; y_{267}=y_{267}+y_{179}*y_{183}; y_{268}=y_{268}+y_{179}*y_{184}; y_{269}=y_{269}+y_{180}*y_{181}; y_{270}=y_{270}+y_{180}*y_{182}; y_{271}=y_{271}+y_{180}*y_{183}; y_{272}=y_{272}+y_{180}*y_{184}; y_{273}=y_{273}+y_{185}*y_{189}; y_{274}=y_{274}+y_{185}*y_{190}; y_{275}=y_{275}+y_{185}*y_{191}; y_{276}=y_{276}+y_{185}*y_{192}; y_{277}=y_{277}+y_{186}*y_{189}; y_{278}=y_{278}+y_{186}*y_{190}; y_{279}=y_{279}+y_{186}*y_{191}; y_{280}=y_{280}+y_{186}*y_{192}; y_{281}=y_{281}+y_{187}*y_{189}; y_{282}=y_{282}+y_{187}*y_{190}; y_{283}=y_{283}+y_{187}*y_{191}; y_{284}=y_{284}+y_{187}*y_{192}; y_{285}=y_{285}+y_{188}*y_{189}; y_{286}=y_{286}+y_{188}*y_{190}; y_{287}=y_{287}+y_{188}*y_{191}; y_{288}=y_{288}+y_{188}*y_{192}; y_{289}=y_{289}+y_{193}*y_{197}; y_{290}=y_{290}+y_{193}*y_{198}; y_{291}=y_{291}+y_{193}*y_{199}; y_{292}=y_{292}+y_{193}*y_{200}; y_{293}=y_{293}+y_{194}*y_{197}; y_{294}=y_{294}+y_{194}*y_{198}; y_{295}=y_{295}+y_{194}*y_{199};$

$4*y199;y296=y296+y194*y200;y297=y297+y195*y197;y298=y298+y195*y198;y299=y299+y195*y199;$
 $y300=y300+y195*y200;y301=y301+y196*y197;y302=y302+y196*y198;y303=y303+y196*y199;$
 $y304=y304+y196*y200;y305=y305+y201*y205;y306=y306+y201*y206;y307=y307+y201*y207;y308=y308+y201*y208;$
 $y309=y309+y202*y205;y310=y310+y202*y206;y311=y311+y202*y207;y312=y312+y202*y208;$
 $y313=y313+y203*y205;y314=y314+y203*y206;y315=y315+y203*y207;y316=y316+y203*y208;$
 $y317=y317+y204*y205;y318=y318+y204*y206;y319=y319+y204*y207;y320=y320+y204*y208;$
 $y321=y321+y209*y213;y322=y322+y209*y214;y323=y323+y209*y215;y324=y324+y209*y216;$
 $y325=y325+y210*y213;y326=y326+y210*y214;y327=y327+y210*y215;y328=y328+y210*y216;$
 $y329=y329+y211*y213;y330=y330+y211*y214;y331=y331+y211*y215;y332=y332+y211*y216;$
 $y333=y333+y212*y213;y334=y334+y212*y214;y335=y335+y212*y215;y336=y336+y212*y216;y337=y337+y217*y221;$
 $y338=y338+y217*y222;y339=y339+y217*y223;y340=y340+y217*y224;y341=y341+y218*y221;$
 $y342=y342+y218*y222;y343=y343+y218*y223;y344=y344+y218*y224;y345=y345+y219*y221;$
 $y346=y346+y219*y222;y347=y347+y219*y223;y348=y348+y219*y224;y349=y349+y220*y221;$
 $y350=y350+y220*y222;y351=y351+y220*y223;y352=y352+y220*y224;y129=y129+1;y130=y130+1;$
 $y131=y131+1;y132=y132+1;y133=y133+1;y134=y134+1;y135=y135+1;y136=y136+1;y137=y137+1;$
 $y138=y138+1;y139=y139+1;y140=y140+1;y141=y141+1;y142=y142+1;y143=y143+1;y144=y144+1;$
 $y145=y145+1;y146=y146+1;y147=y147+1;y148=y148+1;y149=y149+1;y150=y150+1;y151=y151+1;$
 $y152=y152+1;y153=y153+1;y154=y154+1;y155=y155+1;y156=y156+1;y157=y157+1;y158=y158+1;$
 $y159=y159+1;y160=y160+1;$

#Full depth 7 and Toffoli depth 2

#4 变量元素两两搭配组合情况 1+辅助位清空

$y353=y353+y249;y354=y354+y253;y355=y355+y244;y356=y356+y254;y357=y357+y255;y358=y358+y256;$
 $y359=y359+y252;y360=y360+y246;y361=y361+y255;y362=y362+y250;y363=y363+y247;y364=y364+y245;$
 $y365=y365+y251;y366=y366+y243;y367=y367+y256;y368=y368+y254;y369=y369+y252;y370=y370+y252;$
 $y371=y371+y246;y372=y372+y256;y373=y373+y256;y374=y374+y249;y375=y375+y243;y376=y376+y253;$
 $y377=y377+y253;y378=y378+y252;y379=y379+y256;y380=y380+y256;y381=y381+y255;y382=y382+y252;$
 $y383=y383+y252;y384=y384+y251;y385=y385+y242;y386=y386+y255;y387=y387+y254;y388=y388+y254;$
 $y389=y389+y253;y390=y390+y253;y391=y391+y251;y392=y392+y251;y393=y393+y251;$
 $y394=y394+y250;y395=y395+y248;y396=y396+y247;y397=y397+y256;y398=y398+y256;y399=y399+y255;$
 $y400=y400+y255;y401=y401+y255;y402=y402+y255;y403=y403+y255;y404=y404+y255;$
 $y405=y405+y254;y406=y406+y254;y407=y407+y253;y408=y408+y253;y409=y409+y252;$
 $y410=y410+y252;y411=y411+y251;y412=y412+y251;y413=y413+y250;y414=y414+y250;$
 $y415=y415+y250;y416=y416+y250;y417=y417+y249;y418=y418+y248;y419=y419+y248;$
 $y420=y420+y245;y421=y421+y255;y422=y422+y281;y423=y423+y285;y424=y424+y276;y425=y425+y286;$
 $y426=y426+y287;y427=y427+y288;y428=y428+y284;y429=y429+y278;y430=y430+y287;y431=y431+y282;$
 $y432=y432+y279;y433=y433+y277;y434=y434+y283;y435=y435+y275;y436=y436+y288;y437=y437+y286;$
 $y438=y438+y284;y439=y439+y284;y440=y440+y278;y441=y441+y288;y442=y442+y288;y443=y443+y281;$
 $y444=y444+y275;y445=y445+y285;y446=y446+y285;y447=y447+y284;y448=y448+y288;y449=y449+y288;$
 $y450=y450+y287;y451=y451+y284;y452=y452+y284;y453=y453+y283;y454=y454+y274;$
 $y455=y455+y287;y456=y456+y286;y457=y457+y286;y458=y458+y285;y459=y459+y285;$
 $y460=y460+y283;y461=y461+y283;y462=y462+y283;y463=y463+y282;y464=y464+y280;$
 $y465=y465+y279;y466=y466+y288;y467=y467+y288;y468=y468+y287;y469=y469+y287;y470=y470+y287;$
 $y471=y471+y287;y472=y472+y287;y473=y473+y287;y474=y474+y286;y475=y475+y286;$
 $y476=y476+y285;y477=y477+y285;y478=y478+y284;y479=y479+y284;y480=y480+y283;y481=y481+y283;$
 $y482=y482+y282;y483=y483+y282;y484=y484+y282;y485=y485+y282;y486=y486+y281;y487=y487+y280;$
 $y488=y488+y280;y489=y489+y277;y490=y490+y287;y491=y491+y313;y492=y492+y317;$
 $y493=y493+y308;y494=y494+y318;y495=y495+y319;y496=y496+y320;y497=y497+y316;y498=y498+y310;$
 $y499=y499+y319;y500=y500+y314;y501=y501+y311;y502=y502+y309;y503=y503+y315;y504=y504+y307;$
 $y505=y505+y320;y506=y506+y318;y507=y507+y316;y508=y508+y316;y509=y509+y310;y510=y510+y320;$
 $y511=y511+y320;y512=y512+y313;y513=y513+y307;y514=y514+y317;y515=y515+y317;$
 $y516=y516+y316;y517=y517+y320;y518=y518+y320;y519=y519+y319;y520=y520+y316;y521=y521+y316;$
 $y522=y522+y315;y523=y523+y306;y524=y524+y319;y525=y525+y318;y526=y526+y318;$
 $y527=y527+y317;y528=y528+y317;y529=y529+y315;y530=y530+y315;y531=y531+y315;$
 $y532=y532+y314;y533=y533+y312;y534=y534+y311;y535=y535+y320;y536=y536+y320;$
 $y537=y537+y319;y538=y538+y319;y539=y539+y319;y540=y540+y319;y541=y541+y319;y542=y542+y319;$
 $y543=y543+y318;y544=y544+y318;y545=y545+y317;y546=y546+y317;y547=y547+y316;y548=y548+y316;$
 $y549=y549+y315;y550=y550+y315;y551=y551+y314;y552=y552+y314;y553=y553+y314;$
 $y554=y554+y314;y555=y555+y313;y556=y556+y312;y557=y557+y312;y558=y558+y309;y559=y559+y319;$
 $y560=y560+y345;y561=y561+y349;y562=y562+y340;y563=y563+y350;y564=y564+y340;$

y351;y565=y565+y352;y566=y566+y348;y567=y567+y342;y568=y568+y351;y569=y569+y346;y570=y570+y343;y571=y571+y341;y572=y572+y347;y573=y573+y339;y574=y574+y352;y575=y575+y350;y576=y576+y348;y577=y577+y348;y578=y578+y342;y579=y579+y352;y580=y580+y352;y581=y581+y345;y582=y582+y339;y583=y583+y349;y584=y584+y349;y585=y585+y348;y586=y586+y352;y587=y587+y352;y588=y588+y351;y589=y589+y348;y590=y590+y348;y591=y591+y347;y592=y592+y338;y593=y593+y351;y594=y594+y350;y595=y595+y350;y596=y596+y349;y597=y597+y349;y598=y598+y347;y599=y599+y347;y600=y600+y347;y601=y601+y346;y602=y602+y344;y603=y603+y343;y604=y604+y352;y605=y605+y352;y606=y606+y351;y607=y607+y351;y608=y608+y351;y609=y609+y351;y610=y610+y351;y611=y611+y351;y612=y612+y350;y613=y613+y350;y614=y614+y349;y615=y615+y349;y616=y616+y348;y617=y617+y348;y618=y618+y347;y619=y619+y347;y620=y620+y346;y621=y621+y346;y622=y622+y346;y623=y623+y346;y624=y624+y345;y625=y625+y344;y626=y626+y344;y627=y627+y341;y628=y628+y351;y129=y129+y1;y130=y130+y2;y131=y131+y3;y132=y132+y4;y133=y133+y5;y134=y134+y6;y135=y135+y7;y136=y136+y8;y137=y137+y9;y138=y138+y10;y139=y139+y11;y140=y140+y12;y141=y141+y13;y142=y142+y14;y143=y143+y15;y144=y144+y16;y145=y145+y17;y146=y146+y18;y147=y147+y19;y148=y148+y20;y149=y149+y21;y150=y150+y22;y151=y151+y23;y152=y152+y24;y153=y153+y25;y154=y154+y26;y155=y155+y27;y156=y156+y28;y157=y157+y29;y158=y158+y30;y159=y159+y31;y160=y160+y32;

#Full depth 8 and Toffoli depth 2

#4 变量元素两两搭配组合情况 2+辅助位映射

y353=y353+y247;y354=y354+y251;y355=y355+y242;y356=y356+y250;y357=y357+y243;y358=y358+y245;y359=y359+y248;y360=y360+y241;y361=y361+y254;y362=y362+y246;y363=y363+y244;y364=y364+y242;y365=y365+y249;y366=y366+y241;y367=y367+y241;y368=y368+y244;y369=y369+y247;y370=y370+y243;y371=y371+y245;y372=y372+y253;y373=y373+y244;y374=y374+y248;y375=y375+y242;y376=y376+y252;y377=y377+y250;y378=y378+y251;y379=y379+y255;y380=y380+y243;y381=y381+y248;y382=y382+y249;y383=y383+y245;y384=y384+y247;y385=y385+y241;y386=y386+y253;y387=y387+y246;y388=y388+y241;y389=y389+y248;y390=y390+y247;y391=y391+y250;y392=y392+y246;y393=y393+y244;y394=y394+y245;y395=y395+y246;y396=y396+y241;y397=y397+y254;y398=y398+y250;y399=y399+y251;y400=y400+y250;y401=y401+y249;y402=y402+y245;y403=y403+y244;y404=y404+y241;y405=y405+y252;y406=y406+y248;y407=y407+y249;y408=y408+y245;y409=y409+y250;y410=y410+y242;y411=y411+y248;y412=y412+y243;y413=y413+y249;y414=y414+y248;y415=y415+y247;y416=y416+y242;y417=y417+y242;y418=y418+y245;y419=y419+y244;y420=y420+y241;y421=y421+y246;y422=y422+y279;y423=y423+y283;y424=y424+y274;y425=y425+y282;y426=y426+y275;y427=y427+y277;y428=y428+y280;y429=y429+y273;y430=y430+y286;y431=y431+y278;y432=y432+y276;y433=y433+y274;y434=y434+y281;y435=y435+y273;y436=y436+y273;y437=y437+y276;y438=y438+y279;y439=y439+y275;y440=y440+y277;y441=y441+y285;y442=y442+y276;y443=y443+y280;y444=y444+y274;y445=y445+y284;y446=y446+y282;y447=y447+y283;y448=y448+y287;y449=y449+y275;y450=y450+y280;y451=y451+y281;y452=y452+y277;y453=y453+y279;y454=y454+y273;y455=y455+y285;y456=y456+y278;y457=y457+y273;y458=y458+y280;y459=y459+y279;y460=y460+y282;y461=y461+y278;y462=y462+y276;y463=y463+y277;y464=y464+y278;y465=y465+y273;y466=y466+y286;y467=y467+y282;y468=y468+y283;y469=y469+y282;y470=y470+y281;y471=y471+y277;y472=y472+y276;y473=y473+y273;y474=y474+y284;y475=y475+y280;y476=y476+y281;y477=y477+y277;y478=y478+y282;y479=y479+y274;y480=y480+y280;y481=y481+y275;y482=y482+y281;y483=y483+y280;y484=y484+y279;y485=y485+y274;y486=y486+y274;y487=y487+y277;y488=y488+y276;y489=y489+y273;y490=y490+y278;y491=y491+y311;y492=y492+y315;y493=y493+y306;y494=y494+y314;y495=y495+y307;y496=y496+y309;y497=y497+y312;y498=y498+y305;y499=y499+y318;y500=y500+y310;y501=y501+y308;y502=y502+y306;y503=y503+y313;y504=y504+y305;y505=y505+y305;y506=y506+y308;y507=y507+y311;y508=y508+y307;y509=y509+y309;y510=y510+y317;y511=y511+y308;y512=y512+y312;y513=y513+y306;y514=y514+y316;y515=y515+y314;y516=y516+y315;y517=y517+y319;y518=y518+y307;y519=y519+y312;y520=y520+y313;y521=y521+y309;y522=y522+y311;y523=y523+y305;y524=y524+y317;y525=y525+y310;y526=y526+y305;y527=y527+y312;y528=y528+y311;y529=y529+y314;y530=y530+y310;y531=y531+y308;y532=y532+y309;y533=y533+y310;y534=y534+y305;y535=y535+y318;y536=y536+y314;y537=y537+y315;y538=y538+y314;y539=y539+y313;y540=y540+y309;y541=y541+y308;y542=y542+y305;y543=y543+y316;y544=y544+y312;y545=y545+y313;y546=y546+y309;y547=y547+y314;y548=y548+y306;y549=y549+y312;y550=y550+y307;y551=y551+y313;y552=y552+y312;y553=y553+y311;y554=y554+y306;y555=y555+y306;y556=y556+y309;y557=y557+y308;y558=y558+y305;y559=y559+y310;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y3

40;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y592+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y598=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+y337;y604=y604+y350;y605=y605+y346;y606=y606+y347;y607=y607+y346;y608=y608+y345;y609=y609+y341;y610=y610+y340;y611=y611+y337;y612=y612+y348;y613=y613+y344;y614=y614+y345;y615=y615+y341;y616=y616+y346;y617=y617+y338;y618=y618+y344;y619=y619+y339;y620=y620+y345;y621=y621+y344;y622=y622+y343;y623=y623+y338;y624=y624+y338;y625=y625+y341;y626=y626+y340;y627=y627+y337;y628=y628+y342;

#Full depth 9 and Toffoli depth 2

#剩余因式的加法多项式 1

y405=y405+y353;y406=y406+y365;y398=y398+y354;y629=y629+y353;y407=y407+y359;y630=y630+y242;y413=y413+y241;y631=y631+y355;y419=y419+y353;y632=y632+y355;y402=y402+y356;y633=y633+y369;y634=y634+y354;y635=y635+y354;y397=y397+y355;y636=y636+y354;y637=y637+y254;y638=y638+y249;y420=y420+y353;y409=y409+y358;y639=y639+y256;y640=y640+y357;y410=y410+y354;y641=y641+y254;y642=y642+y246;y643=y643+y249;y644=y644+y248;y645=y645+y354;y646=y646+y251;y647=y647+y357;y648=y648+y253;y649=y649+y248;y650=y650+y243;y651=y651+y255;y652=y652+y241;y404=y404+y355;y653=y653+y354;y654=y654+y255;y655=y655+y246;y656=y656+y363;y414=y414+y353;y657=y657+y354;y658=y658+y245;y412=y412+y353;y659=y659+y356;y660=y660+y353;y661=y661+y248;y662=y662+y253;y663=y663+y248;y664=y664+y254;y665=y665+y356;y666=y666+y353;y667=y667+y353;y668=y668+y355;y669=y669+y242;y670=y670+y252;y671=y671+y353;y672=y672+y353;y673=y673+y256;y674=y674+y251;y675=y675+y241;y676=y676+y245;y677=y677+y353;y678=y678+y355;y679=y679+y253;y680=y680+y251;y681=y681+y355;y682=y682+y243;y683=y683+y253;y684=y684+y242;y685=y685+y242;y686=y686+y359;y403=y403+y356;y687=y687+y356;y688=y688+y355;y689=y689+y353;y690=y690+y353;y691=y691+y250;y692=y692+y353;y693=y693+y247;y416=y416+y353;y415=y415+y354;y694=y694+y247;y695=y695+y254;y696=y696+y246;y697=y697+y251;y698=y698+y248;y699=y699+y356;y399=y399+y353;y700=y700+y354;y701=y701+y241;y702=y702+y353;y703=y703+y246;y704=y704+y251;y705=y705+y355;y706=y706+y354;y707=y707+y243;y417=y417+y354;y418=y418+y357;y708=y708+y357;y709=y709+y355;y408=y408+y353;y710=y710+y353;y711=y711+y353;y712=y712+y247;y400=y400+y360;y713=y713+y252;y421=y421+y353;y714=y714+y249;y715=y715+y250;y716=y716+y249;y717=y717+y363;y718=y718+y356;y401=y401+y360;y719=y719+y243;y720=y720+y353;y721=y721+y242;y722=y722+y252;y723=y723+y244;y724=y724+y250;y411=y411+y241;y725=y725+y353;y726=y726+y243;y727=y727+y254;y728=y728+y353;y729=y729+y356;y730=y730+y246;y731=y731+y243;y474=y474+y227;y475=y475+y434;y467=y467+y423;y732=y732+y422;y476=y476+y428;y733=y733+y274;y482=y482+y273;y734=y734+y424;y488=y488+y422;y735=y735+y424;y471=y471+y425;y736=y736+y438;y737=y737+y423;y738=y738+y423;y466=y466+y424;y739=y739+y423;y740=y740+y286;y741=y741+y281;y489=y489+y422;y478=y478+y427;y742=y742+y288;y743=y743+y426;y479=y479+y423;y744=y744+y286;y745=y745+y278;y746=y746+y281;y747=y747+y280;y748=y748+y423;y749=y749+y283;y750=y750+y426;y751=y751+y285;y752=y752+y280;y753=y753+y275;y754=y754+y287;y755=y755+y273;y473=y473+y424;y756=y756+y423;y757=y757+y287;y758=y758+y278;y759=y759+y432;y483=y483+y422;y760=y760+y423;y761=y761+y277;y481=y481+y422;y762=y762+y425;y763=y763+y422;y764=y764+y280;y765=y765+y285;y766=y766+y280;y767=y767+y286;y768=y768+y425;y769=y769+y422;y770=y770+y422;y771=y771+y424;y772=y772+y274;y773=y773+y284;y774=y774+y422;y775=y775+y422;y776=y776+y288;y777=y777+y283;y778=y778+y273;y779=y779+y277;y780=y780+y422;y781=y781+y424;y782=y782+y285;y783=y783+y283;y784=y784+y424;y785=y785+y275;y786=y786+y285;y787=y787+y274;y788=y788+y274;y789=y789+y428;y472=y472+y425;y790=y790+y425;y791=y791+y424;y792=y792+y422;y793=y793+y422;y794=y794+y282;y795=y795+y422;y796=y796+y279;y485=y485+y422;y484=y484+y423;y797=y797+y279;y798=y798+y286;y799=y799+y278;y800=y800+y283;y801=y801+y280;y802=y802+y425;y468=y468+y422;y803=y803+y423;y804=y804+y273;y805=y805+y422;y806=y806+y278;y807=y807+y283;y808=y808+y424;y809=y809+y423;y810=y810+y275;y486=y486+y423;y487=y487+y426;y811=y811+y426;y812=y812+y424;y477=y477+y422;y813=y813+y422;y814=y814+y422;y815=y815+y279;y469=y469+y429;y816=y816+y284;y490=y490+y422;y817=y817+y281;y818=y818+y282;y819=y819+y281;y820=y820+y432;y821=y821+y425;y470=y470+y429;y822=y822+y275;y823=y823+y422;y824=y824+y274;y825=y825+y284;y826=y826+y276;y827=y827+y282;y480=y480+y273;y828=y828+y422;y829=y829+y275;y830=y830+y286;y831=y831+y422;y832=y832+y425;y833=y833+y278;y834=y834+y275;y543=y543+y491;y544=y544+y503;y536=y536+y492;y835=y835+y491;y545=y545+y497;y836=y836+y3

06;y551=y551+y305;y837=y837+y493;y557=y557+y491;y838=y838+y493;y540=y540+y494;y839=y839+y507;y840=y840+y492;y841=y841+y492;y535=y535+y493;y842=y842+y492;y843=y843+y318;y844=y844+y313;y558=y558+y491;y547=y547+y496;y845=y845+y320;y846=y846+y495;y548=y548+y492;y847=y847+y318;y848=y848+y310;y849=y849+y313;y850=y850+y312;y851=y851+y492;y852=y852+y315;y853=y853+y495;y854=y854+y317;y855=y855+y312;y856=y856+y307;y857=y857+y319;y858=y858+y305;y542=y542+y493;y859=y859+y492;y860=y860+y319;y861=y861+y310;y862=y862+y501;y552=y552+y491;y863=y863+y492;y864=y864+y309;y550=y550+y491;y865=y865+y494;y866=y866+y491;y867=y867+y312;y868=y868+y317;y869=y869+y312;y870=y870+y318;y871=y871+y494;y872=y872+y491;y873=y873+y491;y874=y874+y493;y875=y875+y306;y876=y876+y316;y877=y877+y491;y878=y878+y491;y879=y879+y320;y880=y880+y315;y881=y881+y305;y882=y882+y309;y883=y883+y491;y884=y884+y493;y885=y885+y317;y886=y886+y315;y887=y887+y493;y888=y888+y307;y889=y889+y317;y890=y890+y306;y891=y891+y306;y892=y892+y497;y541=y541+y494;y893=y893+y494;y894=y894+y493;y895=y895+y491;y896=y896+y491;y897=y897+y314;y898=y898+y491;y899=y899+y311;y554=y554+y491;y553=y553+y492;y900=y900+y311;y901=y901+y318;y902=y902+y310;y903=y903+y315;y904=y904+y312;y905=y905+y494;y537=y537+y491;y906=y906+y492;y907=y907+y305;y908=y908+y491;y909=y909+y310;y910=y910+y315;y911=y911+y493;y912=y912+y492;y913=y913+y307;y555=y555+y492;y556=y556+y495;y914=y914+y495;y915=y915+y493;y546=y546+y491;y916=y916+y491;y917=y917+y491;y918=y918+y311;y538=y538+y498;y919=y919+y316;y559=y559+y491;y920=y920+y313;y921=y921+y314;y922=y922+y313;y923=y923+y501;y924=y924+y494;y539=y539+y498;y925=y925+y307;y926=y926+y491;y927=y927+y306;y928=y928+y316;y929=y929+y308;y930=y930+y314;y549=y549+y305;y931=y931+y491;y932=y932+y307;y933=y933+y318;y934=y934+y491;y935=y935+y494;y936=y936+y310;y937=y937+y307;y612=y612+y560;y613=y613+y572;y605=y605+y561;y938=y938+y560;y614=y614+y566;y939=y939+y338;y620=y620+y337;y940=y940+y562;y626=y626+y560;y941=y941+y562;y609=y609+y563;y942=y942+y576;y943=y943+y561;y944=y944+y561;y604=y604+y562;y945=y945+y561;y946=y946+y350;y947=y947+y345;y627=y627+y560;y616=y616+y565;y948=y948+y352;y949=y949+y564;y617=y617+y561;y950=y950+y350;y951=y951+y342;y952=y952+y345;y953=y953+y344;y954=y954+y561;y955=y955+y347;y956=y956+y564;y957=y957+y349;y958=y958+y344;y959=y959+y339;y960=y960+y351;y961=y961+y337;y611=y611+y562;y962=y962+y561;y963=y963+y351;y964=y964+y342;y965=y965+y570;y621=y621+y560;y966=y966+y561;y967=y967+y341;y619=y619+y560;y968=y968+y563;y969=y969+y560;y970=y970+y344;y971=y971+y349;y972=y972+y344;y973=y973+y350;y974=y974+y563;y975=y975+y560;y976=y976+y560;y977=y977+y562;y978=y978+y338;y979=y979+y348;y980=y980+y560;y981=y981+y560;y982=y982+y352;y983=y983+y347;y984=y984+y337;y985=y985+y341;y986=y986+y560;y987=y987+y562;y988=y988+y349;y989=y989+y347;y990=y990+y562;y991=y991+y339;y992=y992+y349;y993=y993+y338;y994=y994+y338;y995=y995+y566;y610=y610+y563;y996=y996+y563;y997=y997+y562;y998=y998+y560;y999=y999+y560;y1000=y1000+y346;y1001=y1001+y560;y1002=y1002+y343;y623=y623+y560;y622=y622+y561;y1003=y1003+y343;y1004=y1004+y350;y1005=y1005+y342;y1006=y1006+y347;y1007=y1007+y344;y1008=y1008+y563;y606=y606+y560;y1009=y1009+y561;y1010=y1010+y337;y1011=y1011+y560;y1012=y1012+y342;y1013=y1013+y347;y1014=y1014+y562;y1015=y1015+y561;y1016=y1016+y339;y624=y624+y561;y625=y625+y564;y1017=y1017+y564;y1018=y1018+y562;y615=y615+y560;y1019=y1019+y560;y1020=y1020+y560;y1021=y1021+y343;y607=y607+y567;y1022=y1022+y348;y628=y628+y560;y1023=y1023+y345;y1024=y1024+y346;y1025=y1025+y345;y1026=y1026+y570;y1027=y1027+y563;y608=y608+y567;y1028=y1028+y339;y1029=y1029+y560;y1030=y1030+y338;y1031=y1031+y348;y1032=y1032+y340;y1033=y1033+y346;y618=y618+y337;y1034=y1034+y560;y1035=y1035+y339;y1036=y1036+y350;y1037=y1037+y560;y1038=y1038+y563;y1039=y1039+y342;y1040=y1040+y339;

#Full depth 10 and Toffoli depth 2

#剩余因式的加法多项式 2

y405=y405+y354;y406=y406+y366;y398=y398+y355;y629=y629+y357;y407=y407+y366;y630=y630+y353;y631=y631+y358;y419=y419+y354;y632=y632+y357;y402=y402+y359;y633=y633+y379;y634=y634+y356;y635=y635+y355;y397=y397+y360;y636=y636+y357;y637=y637+y354;y638=y638+y363;y420=y420+y354;y409=y409+y363;y639=y639+y354;y640=y640+y358;y410=y410+y362;y641=y641+y354;y642=y642+y354;y643=y643+y367;y644=y644+y356;y645=y645+y355;y646=y646+y353;y647=y647+y362;y648=y648+y355;y649=y649+y354;y650=y650+y353;y651=y651+y354;y652=y652+y354;y404=y404+y358;y653=y653+y360;y654=y654+y354;y655=y655+y353;y656=y656+y394;y414=y414+y357;y657=y657+y355;y658=y658+y353;y412=y412+y368;y659=y659+y382;y660=y660+y355;y661=y661+y353;y662=y662+y353;y663=y663+y353;y664=y664+y354;y665=y665+y369;y666=y666+y355;y667=y667+y354;y668=y668+y359;y669=y669+y354;y670=y670+y353;y671=y671+y355;y672=y672+y355;y673=y673+y357;y674=y674+y358;y675=y675+y353;y676=y676+y353;y677

=y677+y356;y678=y678+y358;y679=y679+y355;y680=y680+y359;y681=y681+y362;y682=y682+y354;y683=y683+y353;y684=y684+y353;y685=y685+y354;y686=y686+y385;y403=y403+y358;y687=y687+y357;y688=y688+y358;y689=y689+y355;y690=y690+y355;y691=y691+y354;y692=y692+y357;y693=y693+y354;y416=y416+y359;y415=y415+y357;y694=y694+y354;y695=y695+y353;y696=y696+y355;y697=y697+y360;y698=y698+y360;y699=y699+y358;y399=y399+y364;y700=y700+y357;y701=y701+y354;y702=y702+y354;y703=y703+y356;y704=y704+y353;y705=y705+y356;y706=y706+y356;y707=y707+y353;y417=y417+y357;y418=y418+y360;y708=y708+y364;y709=y709+y356;y408=y408+y357;y710=y710+y358;y711=y711+y354;y712=y712+y354;y400=y400+y369;y713=y713+y354;y421=y421+y356;y714=y714+y355;y715=y715+y358;y716=y716+y354;y717=y717+y370;y718=y718+y365;y401=y401+y373;y719=y719+y359;y720=y720+y355;y721=y721+y356;y722=y722+y353;y723=y723+y359;y724=y724+y354;y411=y411+y353;y725=y725+y355;y726=y726+y358;y727=y727+y353;y728=y728+y358;y729=y729+y357;y730=y730+y356;y731=y731+y355;y474=y474+y423;y475=y475+y435;y467=y467+y424;y732=y732+y426;y476=y476+y435;y733=y733+y422;y734=y734+y427;y488=y488+y423;y735=y735+y426;y471=y471+y428;y736=y736+y448;y737=y737+y425;y738=y738+y424;y466=y466+y429;y739=y739+y426;y740=y740+y423;y741=y741+y432;y489=y489+y423;y478=y478+y432;y742=y742+y423;y743=y743+y427;y479=y479+y431;y744=y744+y423;y745=y745+y423;y746=y746+y436;y747=y747+y425;y748=y748+y424;y749=y749+y422;y750=y750+y431;y751=y751+y424;y752=y752+y423;y753=y753+y422;y754=y754+y423;y755=y755+y423;y473=y473+y427;y756=y756+y429;y757=y757+y423;y758=y758+y422;y759=y759+y463;y483=y483+y426;y760=y760+y424;y761=y761+y422;y481=y481+y437;y762=y762+y451;y763=y763+y424;y764=y764+y422;y765=y765+y422;y766=y766+y422;y767=y767+y423;y768=y768+y438;y769=y769+y424;y770=y770+y423;y771=y771+y428;y772=y772+y423;y773=y773+y422;y774=y774+y424;y775=y775+y424;y776=y776+y426;y777=y777+y427;y778=y778+y422;y779=y779+y422;y780=y780+y425;y781=y781+y427;y782=y782+y424;y783=y783+y428;y784=y784+y431;y785=y785+y423;y786=y786+y422;y787=y787+y422;y788=y788+y423;y789=y789+y454;y472=y472+y427;y790=y790+y426;y791=y791+y427;y792=y792+y424;y793=y793+y424;y794=y794+y423;y795=y795+y426;y796=y796+y423;y485=y485+y428;y484=y484+y426;y797=y797+y423;y798=y798+y422;y799=y799+y424;y800=y800+y429;y801=y801+y429;y802=y802+y427;y468=y468+y433;y803=y803+y426;y804=y804+y423;y805=y805+y423;y806=y806+y425;y807=y807+y422;y808=y808+y425;y809=y809+y425;y810=y810+y422;y486=y486+y426;y487=y487+y429;y811=y811+y433;y812=y812+y425;y477=y477+y426;y813=y813+y427;y814=y814+y423;y815=y815+y423;y469=y469+y438;y816=y816+y423;y490=y490+y425;y817=y817+y424;y818=y818+y427;y819=y819+y423;y820=y820+y439;y821=y821+y434;y470=y470+y442;y822=y822+y428;y823=y823+y424;y824=y824+y425;y825=y825+y422;y826=y826+y428;y827=y827+y423;y480=y480+y422;y828=y828+y424;y829=y829+y427;y830=y830+y422;y831=y831+y427;y832=y832+y426;y833=y833+y425;y834=y834+y424;y543=y543+y492;y544=y544+y504;y536=y536+y493;y835=y835+y495;y545=y545+y504;y836=y836+y491;y837=y837+y496;y557=y557+y492;y838=y838+y495;y540=y540+y497;y839=y839+y517;y840=y840+y494;y841=y841+y493;y535=y535+y498;y842=y842+y495;y843=y843+y492;y844=y844+y501;y558=y558+y492;y547=y547+y501;y845=y845+y492;y846=y846+y496;y548=y548+y500;y847=y847+y492;y848=y848+y492;y849=y849+y505;y850=y850+y494;y851=y851+y493;y852=y852+y491;y853=y853+y500;y854=y854+y493;y855=y855+y492;y856=y856+y491;y857=y857+y492;y858=y858+y492;y542=y542+y496;y859=y859+y498;y860=y860+y492;y861=y861+y491;y862=y862+y532;y552=y552+y495;y863=y863+y493;y864=y864+y491;y550=y550+y506;y865=y865+y520;y866=y866+y493;y867=y867+y491;y868=y868+y491;y869=y869+y491;y870=y870+y492;y871=y871+y507;y872=y872+y493;y873=y873+y492;y874=y874+y497;y875=y875+y492;y876=y876+y491;y877=y877+y493;y878=y878+y493;y879=y879+y495;y880=y880+y496;y881=y881+y491;y882=y882+y491;y883=y883+y494;y884=y884+y496;y885=y885+y493;y886=y886+y497;y887=y887+y500;y888=y888+y492;y889=y889+y491;y890=y890+y491;y891=y891+y492;y892=y892+y523;y541=y541+y496;y893=y893+y495;y894=y894+y496;y895=y895+y493;y896=y896+y493;y897=y897+y492;y898=y898+y495;y899=y899+y492;y554=y554+y497;y553=y553+y495;y900=y900+y492;y901=y901+y491;y902=y902+y493;y903=y903+y498;y904=y904+y498;y905=y905+y496;y537=y537+y502;y906=y906+y495;y907=y907+y492;y908=y908+y492;y909=y909+y494;y910=y910+y491;y911=y911+y494;y912=y912+y494;y913=y913+y491;y555=y555+y495;y556=y556+y498;y914=y914+y502;y915=y915+y494;y546=y546+y495;y916=y916+y496;y917=y917+y492;y918=y918+y492;y538=y538+y507;y919=y919+y492;y559=y559+y494;y920=y920+y493;y921=y921+y496;y922=y922+y492;y923=y923+y508;y924=y924+y503;y539=y539+y511;y925=y925+y497;y926=y926+y493;y927=y927+y494;y928=y928+y491;y929=y929+y497;y930=y930+y492;y549=y549+y491;y931=y931+y493;y932=y932+y496;y933=y933+y491;y934=y934+y496;y935=y935+y495;y936=y936+y494;y937=y937+y493;y612=y612+y561;y613=y613+y573;y605=y605+y562;y938=y938+y564;y614=y614+y573;y939=y939+y560;y940=y940+y565;y626=y626+y561;y941=y941+y564;y609=y609+y566;y942=y942+y586;y943=y943+y563;y944=y944+y562;y604=y604+y567;y945=y945+y94

5+y564;y946=y946+y561;y947=y947+y570;y627=y627+y561;y616=y616+y570;y948=y948+y561;y949=y949+y565;y617=y617+y569;y950=y950+y561;y951=y951+y561;y952=y952+y574;y953=y953+y563;y954=y954+y562;y955=y955+y560;y956=y956+y569;y957=y957+y562;y958=y958+y561;y959=y959+y560;y960=y960+y561;y961=y961+y561;y611=y611+y565;y962=y962+y567;y963=y963+y561;y964=y964+y560;y965=y965+y601;y621=y621+y564;y966=y966+y562;y967=y967+y560;y619=y619+y575;y968=y968+y589;y969=y969+y562;y970=y970+y560;y971=y971+y560;y972=y972+y560;y973=y973+y561;y974=y974+y576;y975=y975+y562;y976=y976+y561;y977=y977+y566;y978=y978+y561;y979=y979+y560;y980=y980+y562;y981=y981+y562;y982=y982+y564;y983=y983+y565;y984=y984+y560;y985=y985+y560;y986=y986+y563;y987=y987+y565;y988=y988+y562;y989=y989+y566;y990=y990+y569;y991=y991+y561;y992=y992+y560;y993=y993+y560;y994=y994+y561;y995=y995+y592;y610=y610+y565;y996=y996+y564;y997=y997+y565;y998=y998+y562;y999=y999+y562;y1000=y1000+y561;y1001=y1001+y564;y1002=y1002+y561;y623=y623+y566;y622=y622+y564;y1003=y1003+y561;y1004=y1004+y560;y1005=y1005+y562;y1006=y1006+y567;y1007=y1007+y567;y1008=y1008+y565;y606=y606+y571;y1009=y1009+y564;y1010=y1010+y561;y1011=y1011+y561;y1012=y1012+y563;y1013=y1013+y560;y1014=y1014+y563;y1015=y1015+y563;y1016=y1016+y560;y624=y624+y564;y625=y625+y567;y1017=y1017+y571;y1018=y1018+y563;y615=y615+y564;y1019=y1019+y565;y1020=y1020+y561;y1021=y1021+y561;y607=y607+y576;y1022=y1022+y561;y628=y628+y563;y1023=y1023+y562;y1024=y1024+y565;y1025=y1025+y561;y1026=y1026+y577;y1027=y1027+y572;y608=y608+y580;y1028=y1028+y566;y1029=y1029+y562;y1030=y1030+y563;y1031=y1031+y560;y1032=y1032+y566;y1033=y1033+y561;y618=y618+y560;y1034=y1034+y562;y1035=y1035+y565;y1036=y1036+y560;y1037=y1037+y565;y1038=y1038+y564;y1039=y1039+y563;y1040=y1040+y562;

#Full depth 11 and Toffoli depth 2

#剩余因式的加法多项式 3

y405=y405+y357;y398=y398+y370;y629=y629+y358;y407=y407+y368;y630=y630+y356;y631=y631+y366;y419=y419+y358;y632=y632+y359;y633=y633+y391;y634=y634+y367;y635=y635+y357;y397=y397+y384;y636=y636+y363;y637=y637+y358;y638=y638+y386;y420=y420+y357;y639=y639+y355;y640=y640+y359;y641=y641+y358;y642=y642+y358;y643=y643+y368;y644=y644+y357;y645=y645+y357;y646=y646+y360;y647=y647+y384;y648=y648+y357;y649=y649+y356;y650=y650+y354;y651=y651+y358;y652=y652+y355;y404=y404+y377;y653=y653+y368;y654=y654+y359;y655=y655+y357;y414=y414+y360;y657=y657+y358;y658=y658+y354;y412=y412+y371;y660=y660+y356;y661=y661+y357;y662=y662+y359;y663=y663+y357;y664=y664+y360;y665=y665+y372;y666=y666+y361;y667=y667+y362;y668=y668+y366;y669=y669+y358;y670=y670+y354;y671=y671+y362;y672=y672+y356;y674=y674+y361;y675=y675+y354;y676=y676+y355;y677=y677+y392;y678=y678+y359;y679=y679+y356;y680=y680+y367;y681=y681+y365;y682=y682+y356;y683=y683+y360;y684=y684+y358;y685=y685+y358;y687=y687+y359;y688=y688+y377;y689=y689+y357;y690=y690+y356;y691=y691+y373;y692=y692+y358;y693=y693+y356;y416=y416+y360;y415=y415+y364;y694=y694+y355;y695=y695+y355;y696=y696+y356;y697=y697+y375;y698=y698+y361;y699=y699+y359;y700=y700+y368;y701=y701+y355;y702=y702+y356;y703=y703+y365;y704=y704+y359;y705=y705+y359;y706=y706+y363;y707=y707+y358;y417=y417+y358;y418=y418+y365;y708=y708+y396;y709=y709+y359;y408=y408+y362;y710=y710+y376;y711=y711+y356;y712=y712+y358;y713=y713+y356;y421=y421+y359;y714=y714+y361;y716=y716+y356;y717=y717+y371;y718=y718+y366;y719=y719+y371;y720=y720+y357;y721=y721+y357;y722=y722+y360;y723=y723+y366;y724=y724+y383;y411=y411+y355;y725=y725+y359;y726=y726+y360;y727=y727+y354;y728=y728+y359;y729=y729+y359;y730=y730+y358;y731=y731+y356;y474=y474+y426;y467=y467+y439;y732=y732+y427;y476=y476+y437;y733=y733+y425;y734=y734+y435;y488=y488+y427;y735=y735+y428;y736=y736+y460;y737=y737+y436;y738=y738+y426;y466=y466+y453;y739=y739+y432;y740=y740+y427;y741=y741+y455;y489=y489+y426;y742=y742+y424;y743=y743+y428;y744=y744+y427;y745=y745+y427;y746=y746+y437;y747=y747+y426;y748=y748+y426;y749=y749+y429;y750=y750+y453;y751=y751+y426;y752=y752+y425;y753=y753+y423;y754=y754+y427;y755=y755+y424;y473=y473+y446;y756=y756+y437;y757=y757+y428;y758=y758+y426;y483=y483+y429;y760=y760+y427;y761=y761+y423;y481=y481+y440;y763=y763+y425;y764=y764+y426;y765=y765+y428;y766=y766+y426;y767=y767+y429;y768=y768+y441;y769=y769+y430;y770=y770+y431;y771=y771+y435;y772=y772+y427;y773=y773+y423;y774=y774+y431;y775=y775+y425;y777=y777+y430;y778=y778+y423;y779=y779+y424;y780=y780+y461;y781=y781+y428;y782=y782+y425;y783=y783+y436;y784=y784+y434;y785=y785+y425;y786=y786+y429;y787=y787+y427;y788=y788+y427;y790=y790+y428;y791=y791+y446;y792=y792+y426;y793=y793+y425;y794=y794+y442;y795=y795+y427;y796=y796+y425;y485=y485+y429;y484=y484+y433;y797=y797+y424;y798=y798+y424;y799=y799+y425;y800=y800+y444;y801=y801+y430;y802=y802+y428;y803=y803+y437;y804=y804+y424;y805=y805+y425;

$y806=y806+y434; y807=y807+y428; y808=y808+y428; y809=y809+y432; y810=y810+y427; y486=y486+y427; y487=y487+y434; y811=y811+y465; y812=y812+y428; y477=y477+y431; y813=y813+y445; y814=y814+y425; y815=y815+y427; y816=y816+y425; y490=y490+y428; y817=y817+y430; y819=y819+y425; y820=y820+y440; y821=y821+y435; y822=y822+y440; y823=y823+y426; y824=y824+y426; y825=y825+y429; y826=y826+y435; y827=y827+y452; y480=y480+y424; y828=y828+y428; y829=y829+y429; y830=y830+y423; y831=y831+y428; y832=y832+y428; y833=y833+y427; y834=y834+y425; y543=y543+y495; y536=y536+y508; y835=y835+y496; y545=y545+y506; y836=y836+y494; y837=y837+y504; y557=y557+y496; y838=y838+y497; y839=y839+y529; y840=y840+y505; y841=y841+y495; y535=y535+y522; y842=y842+y501; y843=y843+y496; y844=y844+y524; y558=y558+y495; y845=y845+y493; y846=y846+y497; y847=y847+y496; y848=y848+y496; y849=y849+y506; y850=y850+y495; y851=y851+y495; y852=y852+y498; y853=y853+y522; y854=y854+y495; y855=y855+y494; y856=y856+y492; y857=y857+y496; y858=y858+y493; y542=y542+y515; y859=y859+y506; y860=y860+y497; y861=y861+y495; y552=y552+y498; y863=y863+y496; y864=y864+y492; y550=y550+y509; y866=y866+y494; y867=y867+y495; y868=y868+y497; y869=y869+y495; y870=y870+y498; y871=y871+y510; y872=y872+y499; y873=y873+y500; y874=y874+y504; y875=y875+y496; y876=y876+y492; y877=y877+y500; y878=y878+y494; y880=y880+y499; y881=y881+y492; y882=y882+y493; y883=y883+y530; y884=y884+y497; y885=y885+y494; y886=y886+y505; y887=y887+y503; y888=y888+y494; y889=y889+y498; y890=y890+y496; y891=y891+y496; y893=y893+y497; y894=y894+y515; y895=y895+y495; y896=y896+y494; y897=y897+y511; y898=y898+y496; y899=y899+y494; y554=y554+y498; y553=y553+y502; y900=y900+y493; y901=y901+y493; y902=y902+y494; y903=y903+y513; y904=y904+y499; y905=y905+y497; y906=y906+y506; y907=y907+y493; y908=y908+y494; y909=y909+y503; y910=y910+y497; y911=y911+y497; y912=y912+y501; y913=y913+y496; y555=y555+y496; y556=y556+y503; y914=y914+y534; y915=y915+y497; y546=y546+y500; y916=y916+y514; y917=y917+y494; y918=y918+y496; y919=y919+y494; y559=y559+y497; y920=y920+y499; y922=y922+y494; y923=y923+y509; y924=y924+y504; y925=y925+y509; y926=y926+y495; y927=y927+y495; y928=y928+y498; y929=y929+y504; y930=y930+y521; y549=y549+y493; y931=y931+y497; y932=y932+y498; y933=y933+y492; y934=y934+y497; y935=y935+y497; y936=y936+y496; y937=y937+y494; y612=y612+y564; y605=y605+y577; y938=y938+y565; y614=y614+y575; y939=y939+y563; y940=y940+y573; y626=y626+y565; y941=y941+y566; y942=y942+y598; y943=y943+y574; y944=y944+y564; y604=y604+y591; y945=y945+y570; y946=y946+y565; y947=y947+y593; y627=y627+y564; y948=y948+y562; y949=y949+y566; y950=y950+y565; y951=y951+y565; y952=y952+y575; y953=y953+y564; y954=y954+y564; y955=y955+y567; y956=y956+y591; y957=y957+y564; y958=y958+y563; y959=y959+y561; y960=y960+y565; y961=y961+y562; y611=y611+y584; y962=y962+y575; y963=y963+y566; y964=y964+y564; y621=y621+y567; y966=y966+y565; y967=y967+y561; y619=y619+y578; y969=y969+y563; y970=y970+y564; y971=y971+y566; y972=y972+y564; y973=y973+y567; y974=y974+y579; y975=y975+y568; y976=y976+y569; y977=y977+y573; y978=y978+y565; y979=y979+y561; y980=y980+y569; y981=y981+y563; y983=y983+y568; y984=y984+y561; y985=y985+y562; y986=y986+y599; y987=y987+y566; y988=y988+y563; y989=y989+y574; y990=y990+y572; y991=y991+y563; y992=y992+y567; y993=y993+y565; y994=y994+y565; y996=y996+y566; y997=y997+y584; y998=y998+y564; y999=y999+y563; y1000=y1000+y580; y1001=y1001+y565; y1002=y1002+y563; y623=y623+y567; y622=y622+y571; y1003=y1003+y562; y1004=y1004+y562; y1005=y1005+y563; y1006=y1006+y582; y1007=y1007+y568; y1008=y1008+y566; y1009=y1009+y575; y1010=y1010+y562; y1011=y1011+y563; y1012=y1012+y572; y1013=y1013+y566; y1014=y1014+y566; y1015=y1015+y570; y1016=y1016+y565; y624=y624+y565; y625=y625+y572; y1017=y1017+y603; y1018=y1018+y566; y615=y615+y569; y1019=y1019+y583; y1020=y1020+y563; y1021=y1021+y565; y1022=y1022+y563; y628=y628+y566; y1023=y1023+y568; y1025=y1025+y563; y1026=y1026+y578; y1027=y1027+y573; y1028=y1028+y578; y1029=y1029+y564; y1030=y1030+y564; y1031=y1031+y567; y1032=y1032+y573; y1033=y1033+y590; y618=y618+y562; y1034=y1034+y566; y1035=y1035+y567; y1036=y1036+y561; y1037=y1037+y566; y1038=y1038+y566; y1039=y1039+y565; y1040=y1040+y563;$

#Full depth 12 and Toffoli depth 2

#剩余因式的加法多项式 4

$y405=y405+y358; y629=y629+y359; y630=y630+y389; y631=y631+y378; y419=y419+y360; y632=y632+y372; y634=y634+y369; y635=y635+y360; y637=y637+y370; y420=y420+y368; y639=y639+y369; y640=y640+y363; y641=y641+y374; y642=y642+y363; y644=y644+y365; y645=y645+y358; y646=y646+y364; y648=y648+y362; y649=y649+y357; y650=y650+y355; y651=y651+y359; y652=y652+y374; y653=y653+y382; y654=y654+y363; y655=y655+y359; y414=y414+y373; y657=y657+y360; y658=y658+y360; y660=y660+y359; y661=y661+y362; y662=y662+y361; y663=y663+y362; y665=y665+y374; y666=y666+y371; y667=y667+y383; y668=y668+y377; y669=y669+y362; y670=y670+y367; y671=y671+y386; y672=y672+y371; y674=y674+y369; y675=y675+y355; y676=y676+y356; y678=y678+y361; y679=y679+y370; y680=y680+y368; y681=y681+y380; y682=y682+y364; y683=y683+y364; y684=y684+y362; y6$

85=y685+y387;y688=y688+y396;y689=y689+y362;y690=y690+y360;y691=y691+y374;y692=y692+y376;y693=y693+y358;y416=y416+y372;y694=y694+y380;y695=y695+y360;y696=y696+y357;y698=y698+y372;y699=y699+y366;y700=y700+y374;y701=y701+y379;y702=y702+y370;y703=y703+y373;y704=y704+y375;y705=y705+y365;y706=y706+y371;y707=y707+y389;y417=y417+y359;y709=y709+y365;y710=y710+y388;y711=y711+y360;y712=y712+y361;y713=y713+y357;y421=y421+y364;y714=y714+y376;y716=y716+y375;y719=y719+y393;y720=y720+y376;y721=y721+y358;y722=y722+y361;y723=y723+y394;y411=y411+y357;y725=y725+y388;y726=y726+y363;y727=y727+y355;y728=y728+y361;y729=y729+y365;y730=y730+y363;y731=y731+y395;y474=y474+y427;y732=y732+y428;y733=y733+y458;y734=y734+y447;y488=y488+y429;y735=y735+y441;y737=y737+y438;y738=y738+y429;y740=y740+y439;y489=y489+y437;y742=y742+y438;y743=y743+y432;y744=y744+y443;y745=y745+y432;y747=y747+y434;y748=y748+y427;y749=y749+y433;y751=y751+y431;y752=y752+y426;y753=y753+y424;y754=y754+y428;y755=y755+y443;y756=y756+y451;y757=y757+y432;y758=y758+y428;y483=y483+y442;y760=y760+y429;y761=y761+y429;y763=y763+y428;y764=y764+y431;y765=y765+y430;y766=y766+y431;y768=y768+y443;y769=y769+y440;y770=y770+y452;y771=y771+y446;y772=y772+y431;y773=y773+y436;y774=y774+y455;y775=y775+y440;y777=y777+y438;y778=y778+y424;y779=y779+y425;y781=y781+y430;y782=y782+y439;y783=y783+y437;y784=y784+y449;y785=y785+y433;y786=y786+y433;y787=y787+y431;y788=y788+y456;y791=y791+y465;y792=y792+y431;y793=y793+y429;y794=y794+y443;y795=y795+y445;y796=y796+y427;y485=y485+y441;y797=y797+y449;y798=y798+y429;y799=y799+y426;y801=y801+y441;y802=y802+y435;y803=y803+y443;y804=y804+y448;y805=y805+y439;y806=y806+y442;y807=y807+y444;y808=y808+y434;y809=y809+y440;y810=y810+y458;y486=y486+y428;y812=y812+y434;y813=y813+y457;y814=y814+y429;y815=y815+y430;y816=y816+y426;y490=y490+y433;y817=y817+y445;y819=y819+y444;y822=y822+y462;y823=y823+y445;y824=y824+y427;y825=y825+y430;y826=y826+y463;y480=y480+y426;y828=y828+y457;y829=y829+y432;y830=y830+y424;y831=y831+y430;y832=y832+y434;y833=y833+y432;y834=y834+y464;y543=y543+y496;y835=y835+y497;y836=y836+y527;y837=y837+y516;y557=y557+y498;y838=y838+y510;y840=y840+y507;y841=y841+y498;y843=y843+y508;y558=y558+y506;y845=y845+y507;y846=y846+y501;y847=y847+y512;y848=y848+y501;y850=y850+y503;y851=y851+y496;y852=y852+y502;y854=y854+y500;y855=y855+y495;y856=y856+y493;y857=y857+y497;y858=y858+y512;y859=y859+y520;y860=y860+y501;y861=y861+y497;y552=y552+y511;y863=y863+y498;y864=y864+y498;y866=y866+y497;y867=y867+y500;y868=y868+y499;y869=y869+y500;y871=y871+y512;y872=y872+y509;y873=y873+y521;y874=y874+y515;y875=y875+y500;y876=y876+y505;y877=y877+y524;y878=y878+y509;y880=y880+y507;y881=y881+y493;y882=y882+y494;y884=y884+y499;y885=y885+y508;y886=y886+y506;y887=y887+y518;y888=y888+y502;y889=y889+y502;y890=y890+y500;y891=y891+y525;y894=y894+y534;y895=y895+y500;y896=y896+y498;y897=y897+y512;y898=y898+y514;y899=y899+y496;y554=y554+y510;y900=y900+y518;y901=y901+y498;y902=y902+y495;y904=y904+y510;y905=y905+y504;y906=y906+y512;y907=y907+y517;y908=y908+y508;y909=y909+y511;y910=y910+y513;y911=y911+y503;y912=y912+y509;y913=y913+y527;y555=y555+y497;y915=y915+y503;y916=y916+y526;y917=y917+y498;y918=y918+y499;y919=y919+y495;y559=y559+y502;y920=y920+y514;y922=y922+y513;y925=y925+y531;y926=y926+y514;y927=y927+y496;y928=y928+y499;y929=y929+y532;y549=y549+y495;y931=y931+y526;y932=y932+y501;y933=y933+y493;y934=y934+y499;y935=y935+y503;y936=y936+y501;y937=y937+y533;y612=y612+y565;y938=y938+y566;y939=y939+y596;y940=y940+y585;y626=y626+y567;y941=y941+y579;y943=y943+y576;y944=y944+y567;y946=y946+y577;y627=y627+y575;y948=y948+y576;y949=y949+y570;y950=y950+y581;y951=y951+y570;y953=y953+y572;y954=y954+y565;y955=y955+y571;y957=y957+y569;y958=y958+y564;y959=y959+y562;y960=y960+y566;y961=y961+y581;y962=y962+y589;y963=y963+y570;y964=y964+y566;y621=y621+y580;y966=y966+y567;y967=y967+y567;y969=y969+y566;y970=y970+y569;y971=y971+y568;y972=y972+y569;y974=y974+y581;y975=y975+y578;y976=y976+y590;y977=y977+y584;y978=y978+y569;y979=y979+y574;y980=y980+y593;y981=y981+y578;y983=y983+y576;y984=y984+y562;y985=y985+y563;y987=y987+y568;y988=y988+y577;y989=y989+y575;y990=y990+y587;y991=y991+y571;y992=y992+y571;y993=y993+y569;y994=y994+y594;y997=y997+y603;y998=y998+y569;y999=y999+y567;y1000=y1000+y581;y1001=y1001+y583;y1002=y1002+y565;y623=y623+y579;y1003=y1003+y587;y1004=y1004+y567;y1005=y1005+y564;y1007=y1007+y579;y1008=y1008+y573;y1009=y1009+y581;y1010=y1010+y586;y1011=y1011+y577;y1012=y1012+y580;y1013=y1013+y582;y1014=y1014+y572;y1015=y1015+y578;y1016=y1016+y596;y624=y624+y566;y1018=y1018+y572;y1019=y1019+y595;y1020=y1020+y567;y1021=y1021+y568;y1022=y1022+y564;y628=y628+y571;y1023=y1023+y583;y1025=y1025+y582;y1028=y1028+y600;y1029=y1029+y583;y1030=y1030+y565;y1031=y1031+y568;y1032=y1032+y601;y618=y618+y564;y1034=y1034+y595;y1035=y1035+y570;y1036=y1036+y562;y1037=y1037+y568;y1038=y1038+y572;y1039=y1039+y570;y1040=y1040+y602;

#Full depth 13 and Toffoli depth 2

#剩余因式的加法多项式 5

$y_{629}=y_{629}+y_{360}; y_{419}=y_{419}+y_{361}; y_{632}=y_{632}+y_{396}; y_{635}=y_{635}+y_{382}; y_{640}=y_{640}+y_{391}; y_{641}=y_{641}+y_{375}; y_{642}=y_{642}+y_{370}; y_{644}=y_{644}+y_{385}; y_{645}=y_{645}+y_{360}; y_{648}=y_{648}+y_{374}; y_{649}=y_{649}+y_{360}; y_{650}=y_{650}+y_{356}; y_{654}=y_{654}+y_{367}; y_{655}=y_{655}+y_{364}; y_{657}=y_{657}+y_{381}; y_{658}=y_{658}+y_{361}; y_{660}=y_{660}+y_{366}; y_{662}=y_{662}+y_{364}; y_{663}=y_{663}+y_{372}; y_{672}=y_{672}+y_{378}; y_{674}=y_{674}+y_{385}; y_{675}=y_{675}+y_{357}; y_{676}=y_{676}+y_{360}; y_{678}=y_{678}+y_{392}; y_{682}=y_{682}+y_{369}; y_{684}=y_{684}+y_{381}; y_{689}=y_{689}+y_{378}; y_{690}=y_{690}+y_{378}; y_{693}=y_{693}+y_{359}; y_{695}=y_{695}+y_{380}; y_{696}=y_{696}+y_{372}; y_{699}=y_{699}+y_{390}; y_{702}=y_{702}+y_{373}; y_{705}=y_{705}+y_{366}; y_{417}=y_{417}+y_{387}; y_{709}=y_{709}+y_{379}; y_{711}=y_{711}+y_{383}; y_{712}=y_{712}+y_{395}; y_{713}=y_{713}+y_{360}; y_{721}=y_{721}+y_{359}; y_{722}=y_{722}+y_{364}; y_{727}=y_{727}+y_{367}; y_{728}=y_{728}+y_{393}; y_{730}=y_{730}+y_{381}; y_{732}=y_{732}+y_{429}; y_{488}=y_{488}+y_{430}; y_{735}=y_{735}+y_{465}; y_{738}=y_{738}+y_{451}; y_{743}=y_{743}+y_{460}; y_{744}=y_{744}+y_{444}; y_{745}=y_{745}+y_{439}; y_{747}=y_{747}+y_{454}; y_{748}=y_{748}+y_{429}; y_{751}=y_{751}+y_{443}; y_{752}=y_{752}+y_{429}; y_{753}=y_{753}+y_{425}; y_{757}=y_{757}+y_{436}; y_{758}=y_{758}+y_{433}; y_{760}=y_{760}+y_{450}; y_{761}=y_{761}+y_{430}; y_{763}=y_{763}+y_{435}; y_{765}=y_{765}+y_{433}; y_{766}=y_{766}+y_{441}; y_{775}=y_{775}+y_{447}; y_{777}=y_{777}+y_{454}; y_{778}=y_{778}+y_{426}; y_{779}=y_{779}+y_{429}; y_{781}=y_{781}+y_{461}; y_{785}=y_{785}+y_{438}; y_{787}=y_{787}+y_{450}; y_{792}=y_{792}+y_{447}; y_{793}=y_{793}+y_{447}; y_{796}=y_{796}+y_{428}; y_{798}=y_{798}+y_{449}; y_{799}=y_{799}+y_{441}; y_{802}=y_{802}+y_{459}; y_{805}=y_{805}+y_{442}; y_{808}=y_{808}+y_{435}; y_{486}=y_{486}+y_{456}; y_{812}=y_{812}+y_{448}; y_{814}=y_{814}+y_{452}; y_{815}=y_{815}+y_{464}; y_{816}=y_{816}+y_{429}; y_{824}=y_{824}+y_{428}; y_{825}=y_{825}+y_{433}; y_{830}=y_{830}+y_{436}; y_{831}=y_{831}+y_{462}; y_{833}=y_{833}+y_{450}; y_{835}=y_{835}+y_{498}; y_{557}=y_{557}+y_{499}; y_{838}=y_{838}+y_{534}; y_{841}=y_{841}+y_{520}; y_{846}=y_{846}+y_{529}; y_{847}=y_{847}+y_{513}; y_{848}=y_{848}+y_{508}; y_{850}=y_{850}+y_{523}; y_{851}=y_{851}+y_{498}; y_{854}=y_{854}+y_{512}; y_{855}=y_{855}+y_{498}; y_{856}=y_{856}+y_{494}; y_{860}=y_{860}+y_{505}; y_{861}=y_{861}+y_{502}; y_{863}=y_{863}+y_{519}; y_{864}=y_{864}+y_{499}; y_{866}=y_{866}+y_{504}; y_{868}=y_{868}+y_{502}; y_{869}=y_{869}+y_{510}; y_{878}=y_{878}+y_{516}; y_{880}=y_{880}+y_{523}; y_{881}=y_{881}+y_{495}; y_{882}=y_{882}+y_{498}; y_{884}=y_{884}+y_{530}; y_{888}=y_{888}+y_{507}; y_{890}=y_{890}+y_{519}; y_{895}=y_{895}+y_{516}; y_{896}=y_{896}+y_{516}; y_{899}=y_{899}+y_{497}; y_{901}=y_{901}+y_{518}; y_{902}=y_{902}+y_{510}; y_{905}=y_{905}+y_{528}; y_{908}=y_{908}+y_{511}; y_{911}=y_{911}+y_{504}; y_{555}=y_{555}+y_{525}; y_{915}=y_{915}+y_{517}; y_{917}=y_{917}+y_{521}; y_{918}=y_{918}+y_{533}; y_{919}=y_{919}+y_{498}; y_{927}=y_{927}+y_{497}; y_{928}=y_{928}+y_{502}; y_{933}=y_{933}+y_{505}; y_{934}=y_{934}+y_{531}; y_{936}=y_{936}+y_{519}; y_{938}=y_{938}+y_{567}; y_{626}=y_{626}+y_{568}; y_{941}=y_{941}+y_{603}; y_{944}=y_{944}+y_{589}; y_{949}=y_{949}+y_{598}; y_{950}=y_{950}+y_{582}; y_{951}=y_{951}+y_{577}; y_{953}=y_{953}+y_{592}; y_{954}=y_{954}+y_{567}; y_{957}=y_{957}+y_{581}; y_{958}=y_{958}+y_{567}; y_{959}=y_{959}+y_{563}; y_{963}=y_{963}+y_{574}; y_{964}=y_{964}+y_{571}; y_{966}=y_{966}+y_{588}; y_{967}=y_{967}+y_{568}; y_{969}=y_{969}+y_{573}; y_{971}=y_{971}+y_{571}; y_{972}=y_{972}+y_{579}; y_{981}=y_{981}+y_{585}; y_{983}=y_{983}+y_{592}; y_{984}=y_{984}+y_{564}; y_{985}=y_{985}+y_{567}; y_{987}=y_{987}+y_{599}; y_{991}=y_{991}+y_{576}; y_{993}=y_{993}+y_{588}; y_{998}=y_{998}+y_{585}; y_{999}=y_{999}+y_{585}; y_{1002}=y_{1002}+y_{566}; y_{1004}=y_{1004}+y_{587}; y_{1005}=y_{1005}+y_{579}; y_{1008}=y_{1008}+y_{597}; y_{1011}=y_{1011}+y_{580}; y_{1014}=y_{1014}+y_{573}; y_{624}=y_{624}+y_{594}; y_{1018}=y_{1018}+y_{586}; y_{1020}=y_{1020}+y_{590}; y_{1021}=y_{1021}+y_{602}; y_{1022}=y_{1022}+y_{567}; y_{1030}=y_{1030}+y_{566}; y_{1031}=y_{1031}+y_{571}; y_{1036}=y_{1036}+y_{574}; y_{1037}=y_{1037}+y_{600}; y_{1039}=y_{1039}+y_{588};$

#Full depth 14 and Toffoli depth 2

#剩余因式的加法多项式 6

$y_{629}=y_{629}+y_{377}; y_{645}=y_{645}+y_{369}; y_{649}=y_{649}+y_{363}; y_{675}=y_{675}+y_{359}; y_{690}=y_{690}+y_{379}; y_{693}=y_{693}+y_{360}; y_{713}=y_{713}+y_{364}; y_{721}=y_{721}+y_{384}; y_{727}=y_{727}+y_{381}; y_{732}=y_{732}+y_{446}; y_{748}=y_{748}+y_{438}; y_{752}=y_{752}+y_{432}; y_{778}=y_{778}+y_{428}; y_{793}=y_{793}+y_{448}; y_{796}=y_{796}+y_{429}; y_{816}=y_{816}+y_{433}; y_{824}=y_{824}+y_{453}; y_{830}=y_{830}+y_{450}; y_{835}=y_{835}+y_{515}; y_{851}=y_{851}+y_{507}; y_{855}=y_{855}+y_{501}; y_{881}=y_{881}+y_{497}; y_{896}=y_{896}+y_{517}; y_{899}=y_{899}+y_{498}; y_{919}=y_{919}+y_{502}; y_{927}=y_{927}+y_{522}; y_{933}=y_{933}+y_{519}; y_{938}=y_{938}+y_{584}; y_{954}=y_{954}+y_{576}; y_{958}=y_{958}+y_{570}; y_{984}=y_{984}+y_{566}; y_{999}=y_{999}+y_{586}; y_{1002}=y_{1002}+y_{567}; y_{1022}=y_{1022}+y_{571}; y_{1030}=y_{1030}+y_{591}; y_{1036}=y_{1036}+y_{588};$

#Full depth 15 and Toffoli depth 2

#剩余因式的加法多项式 7

$y_{675}=y_{675}+y_{362}; y_{693}=y_{693}+y_{375}; y_{778}=y_{778}+y_{431}; y_{796}=y_{796}+y_{444}; y_{881}=y_{881}+y_{500}; y_{899}=y_{899}+y_{513}; y_{984}=y_{984}+y_{569}; y_{1002}=y_{1002}+y_{582};$

#Full depth 16 and Toffoli depth 2

#4 变量元素两两搭配组合情况部分位置清空 1

$y_{353}=y_{353}+y_{249}; y_{354}=y_{354}+y_{253}; y_{355}=y_{355}+y_{244}; y_{356}=y_{356}+y_{254}; y_{357}=y_{357}+y_{255}; y_{358}=y_{358}+y_{256}; y_{359}=y_{359}+y_{252}; y_{360}=y_{360}+y_{246}; y_{361}=y_{361}+y_{255}; y_{362}=y_{362}+y_{250}; y_{363}=y_{363}+y_{247}; y_{364}=y_{364}+y_{245}; y_{365}=y_{365}+y_{251}; y_{366}=y_{366}+y_{243}; y_{367}=y_{367}+y_{256}; y_{368}=y_{368}+y_{254}; y_{369}=y_{369}+y_{252}; y_{370}=y_{370}+y_{252}; y_{371}=y_{371}+y_{246}; y_{372}=y_{372}+y_{256}; y_{373}=y_{373}+y_{256}; y_{374}=y_{374}+y_{249}; y_{375}=y_{375}+y_{243}; y_{376}=y_{376}+y_{253}; y_{377}=y_{377}+y_{253}; y_{378}=y_{378}+y_{252}; y_{379}=y_{379}+y_{256}; y_{380}=y_{380}+y_{256}; y_{381}=y_{381}+y_{255}; y_{382}=y_{382}+y_{252}; y_{383}=y_{383}+y_{252}; y_{384}=y_{384}+y_{251}; y_{385}=y_{385}+y_{242}; y_{386}=y_{386}+y_{255}; y_{387}=y_{387}+y_{254}; y_{388}=y_{388}+y_{254}; y_{389}=y_{389}+y_{253}; y_{390}=y_{390}+y_{253}; y_{391}=y_{391}+y_{251}; y_{392}=y_{392}+y_{251}; y_{393}=y_{393}+y_{251}; y_{394}=y_{394}+y_{250}; y_{395}=y_{395}+y_{248}; y_{396}=y_{396}+y_{247}; y_{422}=y_{422}+y_{422};$

2+y281;y423=y423+y285;y424=y424+y276;y425=y425+y286;y426=y426+y287;y427=y427+y288;y428=y428+y284;y429=y429+y278;y430=y430+y287;y431=y431+y282;y432=y432+y279;y433=y433+y277;y434=y434+y283;y435=y435+y275;y436=y436+y288;y437=y437+y286;y438=y438+y284;y439=y439+y284;y440=y440+y278;y441=y441+y288;y442=y442+y288;y443=y443+y281;y444=y444+y275;y445=y445+y285;y446=y446+y285;y447=y447+y284;y448=y448+y288;y449=y449+y288;y450=y450+y287;y451=y451+y284;y452=y452+y284;y453=y453+y283;y454=y454+y274;y455=y455+y287;y456=y456+y286;y457=y457+y286;y458=y458+y285;y459=y459+y285;y460=y460+y283;y461=y461+y283;y462=y462+y283;y463=y463+y282;y464=y464+y280;y465=y465+y279;y491=y491+y313;y492=y492+y317;y493=y493+y308;y494=y494+y318;y495=y495+y319;y496=y496+y320;y497=y497+y316;y498=y498+y310;y499=y499+y319;y500=y500+y314;y501=y501+y311;y502=y502+y309;y503=y503+y315;y504=y504+y307;y505=y505+y320;y506=y506+y318;y507=y507+y316;y508=y508+y316;y509=y509+y310;y510=y510+y320;y511=y511+y320;y512=y512+y313;y513=y513+y307;y514=y514+y317;y515=y515+y317;y516=y516+y316;y517=y517+y320;y518=y518+y320;y519=y519+y319;y520=y520+y316;y521=y521+y316;y522=y522+y315;y523=y523+y306;y524=y524+y319;y525=y525+y318;y526=y526+y318;y527=y527+y317;y528=y528+y317;y529=y529+y315;y530=y530+y315;y531=y531+y315;y532=y532+y314;y533=y533+y312;y534=y534+y311;y560=y560+y345;y561=y561+y349;y562=y562+y340;y563=y563+y350;y564=y564+y351;y565=y565+y352;y566=y566+y348;y567=y567+y342;y568=y568+y351;y569=y569+y346;y570=y570+y343;y571=y571+y341;y572=y572+y347;y573=y573+y339;y574=y574+y352;y575=y575+y350;y576=y576+y348;y577=y577+y348;y578=y578+y342;y579=y579+y352;y580=y580+y352;y581=y581+y345;y582=y582+y339;y583=y583+y349;y584=y584+y349;y585=y585+y348;y586=y586+y352;y587=y587+y352;y588=y588+y351;y589=y589+y348;y590=y590+y348;y591=y591+y347;y592=y592+y338;y593=y593+y351;y594=y594+y350;y595=y595+y350;y596=y596+y349;y597=y597+y349;y598=y598+y347;y599=y599+y347;y600=y600+y347;y601=y601+y346;y602=y602+y344;y603=y603+y343;

#Full depth 17 and Toffoli depth 2

#4 变量元素两两搭配组合情况部分位置清空 2

y353=y353+y247;y354=y354+y251;y355=y355+y242;y356=y356+y250;y357=y357+y243;y358=y358+y245;y359=y359+y248;y360=y360+y241;y361=y361+y254;y362=y362+y246;y363=y363+y244;y364=y364+y242;y365=y365+y249;y366=y366+y241;y367=y367+y241;y368=y368+y244;y369=y369+y247;y370=y370+y243;y371=y371+y245;y372=y372+y253;y373=y373+y244;y374=y374+y248;y375=y375+y242;y376=y376+y252;y377=y377+y250;y378=y378+y251;y379=y379+y255;y380=y380+y243;y381=y381+y248;y382=y382+y249;y383=y383+y245;y384=y384+y247;y385=y385+y241;y386=y386+y253;y387=y387+y246;y388=y388+y241;y389=y389+y248;y390=y390+y247;y391=y391+y250;y392=y392+y246;y393=y393+y244;y394=y394+y245;y395=y395+y246;y396=y396+y241;y422=y422+y279;y423=y423+y283;y424=y424+y274;y425=y425+y282;y426=y426+y275;y427=y427+y277;y428=y428+y280;y429=y429+y273;y430=y430+y286;y431=y431+y278;y432=y432+y276;y433=y433+y274;y434=y434+y281;y435=y435+y273;y436=y436+y273;y437=y437+y276;y438=y438+y279;y439=y439+y275;y440=y440+y277;y441=y441+y285;y442=y442+y276;y443=y443+y280;y444=y444+y274;y445=y445+y284;y446=y446+y282;y447=y447+y283;y448=y448+y287;y449=y449+y275;y450=y450+y280;y451=y451+y281;y452=y452+y277;y453=y453+y279;y454=y454+y273;y455=y455+y285;y456=y456+y278;y457=y457+y273;y458=y458+y280;y459=y459+y279;y460=y460+y282;y461=y461+y278;y462=y462+y276;y463=y463+y277;y464=y464+y278;y465=y465+y273;y491=y491+y311;y492=y492+y315;y493=y493+y306;y494=y494+y314;y495=y495+y307;y496=y496+y309;y497=y497+y312;y498=y498+y305;y499=y499+y318;y500=y500+y310;y501=y501+y308;y502=y502+y306;y503=y503+y313;y504=y504+y305;y505=y505+y305;y506=y506+y308;y507=y507+y311;y508=y508+y307;y509=y509+y309;y510=y510+y317;y511=y511+y308;y512=y512+y312;y513=y513+y306;y514=y514+y316;y515=y515+y314;y516=y516+y315;y517=y517+y319;y518=y518+y307;y519=y519+y312;y520=y520+y313;y521=y521+y309;y522=y522+y311;y523=y523+y305;y524=y524+y317;y525=y525+y310;y526=y526+y305;y527=y527+y312;y528=y528+y311;y529=y529+y314;y530=y530+y310;y531=y531+y308;y532=y532+y309;y533=y533+y310;y534=y534+y305;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y340;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y592+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y598=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+y337;

$1071=y1071+y258*y819; y1072=y1072+y257*y820; y1073=y1073+y272*y821; y1074=y1074+y271*y$
 $470; y1075=y1075+y270*y822; y1076=y1076+y269*y823; y1077=y1077+y268*y824; y1078=y1078+y$
 $267*y825; y1079=y1079+y266*y826; y1080=y1080+y265*y827; y1081=y1081+y264*y480; y1082=y1$
 $082+y263*y828; y1083=y1083+y262*y829; y1084=y1084+y261*y830; y1085=y1085+y260*y831; y10$
 $86=y1086+y259*y832; y1087=y1087+y258*y833; y1088=y1088+y257*y834; y1089=y1089+y304*y54$
 $3; y1090=y1090+y303*y544; y1091=y1091+y302*y536; y1092=y1092+y301*y835; y1093=y1093+y30$
 $0*y545; y1094=y1094+y299*y836; y1095=y1095+y298*y551; y1096=y1096+y297*y837; y1097=y109$
 $7+y296*y557; y1098=y1098+y295*y838; y1099=y1099+y294*y540; y1100=y1100+y293*y839; y1101$
 $=y1101+y292*y840; y1102=y1102+y291*y841; y1103=y1103+y290*y535; y1104=y1104+y289*y842;$
 $y1105=y1105+y304*y843; y1106=y1106+y303*y844; y1107=y1107+y302*y558; y1108=y1108+y301*$
 $y547; y1109=y1109+y300*y845; y1110=y1110+y299*y846; y1111=y1111+y298*y548; y1112=y1112+$
 $y297*y847; y1113=y1113+y296*y848; y1114=y1114+y295*y849; y1115=y1115+y294*y850; y1116=y$
 $1116+y293*y851; y1117=y1117+y292*y852; y1118=y1118+y291*y853; y1119=y1119+y290*y854; y1$
 $120=y1120+y289*y855; y1121=y1121+y304*y856; y1122=y1122+y303*y857; y1123=y1123+y302*y8$
 $58; y1124=y1124+y301*y542; y1125=y1125+y300*y859; y1126=y1126+y299*y860; y1127=y1127+y2$
 $98*y861; y1128=y1128+y297*y862; y1129=y1129+y296*y552; y1130=y1130+y295*y863; y1131=y11$
 $31+y294*y864; y1132=y1132+y293*y550; y1133=y1133+y292*y865; y1134=y1134+y291*y866; y113$
 $5=y1135+y290*y867; y1136=y1136+y289*y868; y1137=y1137+y304*y869; y1138=y1138+y303*y870$
 $; y1139=y1139+y302*y871; y1140=y1140+y301*y872; y1141=y1141+y300*y873; y1142=y1142+y299$
 $*y874; y1143=y1143+y298*y875; y1144=y1144+y297*y876; y1145=y1145+y296*y877; y1146=y1146$
 $+y295*y878; y1147=y1147+y294*y879; y1148=y1148+y293*y880; y1149=y1149+y292*y881; y1150=$
 $y1150+y291*y882; y1151=y1151+y290*y883; y1152=y1152+y289*y884; y1153=y1153+y304*y885; y$
 $1154=y1154+y303*y886; y1155=y1155+y302*y887; y1156=y1156+y301*y888; y1157=y1157+y300*y$
 $889; y1158=y1158+y299*y890; y1159=y1159+y298*y891; y1160=y1160+y297*y892; y1161=y1161+y$
 $296*y541; y1162=y1162+y295*y893; y1163=y1163+y294*y894; y1164=y1164+y293*y895; y1165=y1$
 $165+y292*y896; y1166=y1166+y291*y897; y1167=y1167+y290*y898; y1168=y1168+y289*y899; y11$
 $69=y1169+y304*y554; y1170=y1170+y303*y553; y1171=y1171+y302*y900; y1172=y1172+y301*y90$
 $1; y1173=y1173+y300*y902; y1174=y1174+y299*y903; y1175=y1175+y298*y904; y1176=y1176+y29$
 $7*y905; y1177=y1177+y296*y537; y1178=y1178+y295*y906; y1179=y1179+y294*y907; y1180=y118$
 $0+y293*y908; y1181=y1181+y292*y909; y1182=y1182+y291*y910; y1183=y1183+y290*y911; y1184$
 $=y1184+y289*y912; y1185=y1185+y304*y913; y1186=y1186+y303*y555; y1187=y1187+y302*y556;$
 $y1188=y1188+y301*y914; y1189=y1189+y300*y915; y1190=y1190+y299*y546; y1191=y1191+y298*$
 $y916; y1192=y1192+y297*y917; y1193=y1193+y296*y918; y1194=y1194+y295*y538; y1195=y1195+$
 $y294*y919; y1196=y1196+y293*y559; y1197=y1197+y292*y920; y1198=y1198+y291*y921; y1199=y$
 $1199+y290*y922; y1200=y1200+y289*y923; y1201=y1201+y304*y924; y1202=y1202+y303*y539; y1$
 $203=y1203+y302*y925; y1204=y1204+y301*y926; y1205=y1205+y300*y927; y1206=y1206+y299*y9$
 $28; y1207=y1207+y298*y929; y1208=y1208+y297*y930; y1209=y1209+y296*y549; y1210=y1210+y2$
 $95*y931; y1211=y1211+y294*y932; y1212=y1212+y293*y933; y1213=y1213+y292*y934; y1214=y12$
 $14+y291*y935; y1215=y1215+y290*y936; y1216=y1216+y289*y937; y1217=y1217+y336*y612; y121$
 $8=y1218+y335*y613; y1219=y1219+y334*y605; y1220=y1220+y333*y938; y1221=y1221+y332*y614$
 $; y1222=y1222+y331*y939; y1223=y1223+y330*y620; y1224=y1224+y329*y940; y1225=y1225+y328$
 $*y626; y1226=y1226+y327*y941; y1227=y1227+y326*y609; y1228=y1228+y325*y942; y1229=y1229$
 $+y324*y943; y1230=y1230+y323*y944; y1231=y1231+y322*y604; y1232=y1232+y321*y945; y1233=$
 $y1233+y336*y946; y1234=y1234+y335*y947; y1235=y1235+y334*y627; y1236=y1236+y333*y616; y$
 $1237=y1237+y332*y948; y1238=y1238+y331*y949; y1239=y1239+y330*y617; y1240=y1240+y329*y$
 $950; y1241=y1241+y328*y951; y1242=y1242+y327*y952; y1243=y1243+y326*y953; y1244=y1244+y$
 $325*y954; y1245=y1245+y324*y955; y1246=y1246+y323*y956; y1247=y1247+y322*y957; y1248=y1$
 $248+y321*y958; y1249=y1249+y336*y959; y1250=y1250+y335*y960; y1251=y1251+y334*y961; y12$
 $52=y1252+y333*y611; y1253=y1253+y332*y962; y1254=y1254+y331*y963; y1255=y1255+y330*y96$
 $4; y1256=y1256+y329*y965; y1257=y1257+y328*y621; y1258=y1258+y327*y966; y1259=y1259+y32$
 $6*y967; y1260=y1260+y325*y619; y1261=y1261+y324*y968; y1262=y1262+y323*y969; y1263=y126$
 $3+y322*y970; y1264=y1264+y321*y971; y1265=y1265+y336*y972; y1266=y1266+y335*y973; y1267$
 $=y1267+y334*y974; y1268=y1268+y333*y975; y1269=y1269+y332*y976; y1270=y1270+y331*y977;$
 $y1271=y1271+y330*y978; y1272=y1272+y329*y979; y1273=y1273+y328*y980; y1274=y1274+y327*$
 $y981; y1275=y1275+y326*y982; y1276=y1276+y325*y983; y1277=y1277+y324*y984; y1278=y1278+$
 $y323*y985; y1279=y1279+y322*y986; y1280=y1280+y321*y987; y1281=y1281+y336*y988; y1282=y$
 $1282+y335*y989; y1283=y1283+y334*y990; y1284=y1284+y333*y991; y1285=y1285+y332*y992; y1$
 $286=y1286+y331*y993; y1287=y1287+y330*y994; y1288=y1288+y329*y995; y1289=y1289+y328*y6$
 $10; y1290=y1290+y327*y996; y1291=y1291+y326*y997; y1292=y1292+y325*y998; y1293=y1293+y3$
 $24*y999; y1294=y1294+y323*y1000; y1295=y1295+y322*y1001; y1296=y1296+y321*y1002; y1297=$

$y_{1297}+y_{336}*y_{623}; y_{1298}=y_{1298}+y_{335}*y_{622}; y_{1299}=y_{1299}+y_{334}*y_{1003}; y_{1300}=y_{1300}+y_{333}*y_{1004};$
 $y_{1301}=y_{1301}+y_{332}*y_{1005}; y_{1302}=y_{1302}+y_{331}*y_{1006}; y_{1303}=y_{1303}+y_{330}*y_{1007}; y_{1304}=y_{1304}+y_{329}*y_{1008};$
 $y_{1305}=y_{1305}+y_{328}*y_{606}; y_{1306}=y_{1306}+y_{327}*y_{1009}; y_{1307}=y_{1307}+y_{326}*y_{1010}; y_{1308}=y_{1308}+y_{325}*y_{1011};$
 $y_{1309}=y_{1309}+y_{324}*y_{1012}; y_{1310}=y_{1310}+y_{323}*y_{1013}; y_{1311}=y_{1311}+y_{322}*y_{1014}; y_{1312}=y_{1312}+y_{321}*y_{1015};$
 $y_{1313}=y_{1313}+y_{336}*y_{1016}; y_{1314}=y_{1314}+y_{335}*y_{624}; y_{1315}=y_{1315}+y_{334}*y_{625}; y_{1316}=y_{1316}+y_{333}*y_{1017};$
 $y_{1317}=y_{1317}+y_{332}*y_{1018}; y_{1318}=y_{1318}+y_{331}*y_{615}; y_{1319}=y_{1319}+y_{330}*y_{1019}; y_{1320}=y_{1320}+y_{329}*y_{1020};$
 $y_{1321}=y_{1321}+y_{328}*y_{1021}; y_{1322}=y_{1322}+y_{327}*y_{607}; y_{1323}=y_{1323}+y_{326}*y_{1022};$
 $y_{1324}=y_{1324}+y_{325}*y_{628}; y_{1325}=y_{1325}+y_{324}*y_{1023}; y_{1326}=y_{1326}+y_{323}*y_{1024}; y_{1327}=y_{1327}+y_{322}*y_{1025};$
 $y_{1328}=y_{1328}+y_{321}*y_{1026}; y_{1329}=y_{1329}+y_{336}*y_{1027}; y_{1330}=y_{1330}+y_{335}*y_{608}; y_{1331}=y_{1331}+y_{334}*y_{1028};$
 $y_{1332}=y_{1332}+y_{333}*y_{1029}; y_{1333}=y_{1333}+y_{332}*y_{1030}; y_{1334}=y_{1334}+y_{331}*y_{1031}; y_{1335}=y_{1335}+y_{330}*y_{1032};$
 $y_{1336}=y_{1336}+y_{329}*y_{1033}; y_{1337}=y_{1337}+y_{328}*y_{618}; y_{1338}=y_{1338}+y_{327}*y_{1034}; y_{1339}=y_{1339}+y_{326}*y_{1035};$
 $y_{1340}=y_{1340}+y_{325}*y_{1036}; y_{1341}=y_{1341}+y_{324}*y_{1037}; y_{1342}=y_{1342}+y_{323}*y_{1038}; y_{1343}=y_{1343}+y_{322}*y_{1039};$
 $y_{1344}=y_{1344}+y_{321}*y_{1040};$

#Full depth 19 and Toffoli depth 3

#S 函数构造 1

$y_{129}=y_{129}+y_{130}; y_{131}=y_{131}+y_{132}; y_{133}=y_{133}+y_{134}; y_{135}=y_{135}+y_{136}; y_{137}=y_{137}+y_{138}; y_{139}=y_{139}+y_{140};$
 $y_{141}=y_{141}+y_{142}; y_{143}=y_{143}+y_{144}; y_{145}=y_{145}+y_{146}; y_{147}=y_{147}+y_{148}; y_{149}=y_{149}+y_{150}; y_{151}=y_{151}+y_{152};$
 $y_{153}=y_{153}+y_{154}; y_{155}=y_{155}+y_{156}; y_{157}=y_{157}+y_{158}; y_{159}=y_{159}+y_{160}; y_{353}=y_{353}+y_{354}; y_{355}=y_{355}+y_{356};$
 $y_{357}=y_{357}+y_{358}; y_{359}=y_{359}+y_{360}; y_{361}=y_{361}+y_{362}; y_{363}=y_{363}+y_{364}; y_{365}=y_{365}+y_{366}; y_{367}=y_{367}+y_{368};$
 $y_{369}=y_{369}+y_{370}; y_{371}=y_{371}+y_{372}; y_{373}=y_{373}+y_{374}; y_{375}=y_{375}+y_{376}; y_{377}=y_{377}+y_{378}; y_{379}=y_{379}+y_{380};$
 $y_{381}=y_{381}+y_{382}; y_{383}=y_{383}+y_{384}; y_{385}=y_{385}+y_{386}; y_{387}=y_{387}+y_{388}; y_{389}=y_{389}+y_{390}; y_{391}=y_{391}+y_{392};$
 $y_{393}=y_{393}+y_{394}; y_{395}=y_{395}+y_{396}; y_{422}=y_{422}+y_{423}; y_{424}=y_{424}+y_{425}; y_{426}=y_{426}+y_{427}; y_{428}=y_{428}+y_{429};$
 $y_{430}=y_{430}+y_{431}; y_{432}=y_{432}+y_{433}; y_{434}=y_{434}+y_{435}; y_{436}=y_{436}+y_{437}; y_{438}=y_{438}+y_{439}; y_{440}=y_{440}+y_{441};$
 $y_{442}=y_{442}+y_{443}; y_{444}=y_{444}+y_{445}; y_{446}=y_{446}+y_{447}; y_{448}=y_{448}+y_{449}; y_{450}=y_{450}+y_{451}; y_{452}=y_{452}+y_{453};$
 $y_{454}=y_{454}+y_{455}; y_{456}=y_{456}+y_{457}; y_{458}=y_{458}+y_{459}; y_{460}=y_{460}+y_{461}; y_{462}=y_{462}+y_{463}; y_{464}=y_{464}+y_{465};$
 $y_{491}=y_{491}+y_{492}; y_{493}=y_{493}+y_{494}; y_{495}=y_{495}+y_{496}; y_{497}=y_{497}+y_{498}; y_{499}=y_{499}+y_{500}; y_{501}=y_{501}+y_{502};$
 $y_{503}=y_{503}+y_{504}; y_{505}=y_{505}+y_{506}; y_{507}=y_{507}+y_{508}; y_{509}=y_{509}+y_{510}; y_{511}=y_{511}+y_{512}; y_{513}=y_{513}+y_{514};$
 $y_{515}=y_{515}+y_{516}; y_{517}=y_{517}+y_{518}; y_{519}=y_{519}+y_{520}; y_{521}=y_{521}+y_{522}; y_{523}=y_{523}+y_{524}; y_{525}=y_{525}+y_{526};$
 $y_{527}=y_{527}+y_{528}; y_{529}=y_{529}+y_{530}; y_{531}=y_{531}+y_{532}; y_{533}=y_{533}+y_{534}; y_{560}=y_{560}+y_{561}; y_{562}=y_{562}+y_{563};$
 $y_{564}=y_{564}+y_{565}; y_{566}=y_{566}+y_{567}; y_{568}=y_{568}+y_{569}; y_{570}=y_{570}+y_{571}; y_{572}=y_{572}+y_{573}; y_{574}=y_{574}+y_{575};$
 $y_{576}=y_{576}+y_{577}; y_{578}=y_{578}+y_{579}; y_{580}=y_{580}+y_{581}; y_{582}=y_{582}+y_{583}; y_{584}=y_{584}+y_{585}; y_{586}=y_{586}+y_{587};$
 $y_{588}=y_{588}+y_{589}; y_{590}=y_{590}+y_{591}; y_{592}=y_{592}+y_{593}; y_{594}=y_{594}+y_{595}; y_{596}=y_{596}+y_{597}; y_{598}=y_{598}+y_{599};$
 $y_{600}=y_{600}+y_{601}; y_{602}=y_{602}+y_{603}; y_{1041}=y_{1041}+y_{1042}; y_{1043}=y_{1043}+y_{1044}; y_{1045}=y_{1045}+y_{1046};$
 $y_{1047}=y_{1047}+y_{1048}; y_{1049}=y_{1049}+y_{1050}; y_{1051}=y_{1051}+y_{1052}; y_{1053}=y_{1053}+y_{1054}; y_{1055}=y_{1055}+y_{1056};$
 $y_{1057}=y_{1057}+y_{1058}; y_{1059}=y_{1059}+y_{1060}; y_{1061}=y_{1061}+y_{1062}; y_{1063}=y_{1063}+y_{1064}; y_{1065}=y_{1065}+y_{1066};$
 $y_{1067}=y_{1067}+y_{1068}; y_{1069}=y_{1069}+y_{1070}; y_{1071}=y_{1071}+y_{1072}; y_{1073}=y_{1073}+y_{1074}; y_{1075}=y_{1075}+y_{1076};$
 $y_{1077}=y_{1077}+y_{1078}; y_{1079}=y_{1079}+y_{1080}; y_{1081}=y_{1081}+y_{1082}; y_{1083}=y_{1083}+y_{1084}; y_{1085}=y_{1085}+y_{1086};$
 $y_{1087}=y_{1087}+y_{1088}; y_{1089}=y_{1089}+y_{1090}; y_{1091}=y_{1091}+y_{1092}; y_{1093}=y_{1093}+y_{1094}; y_{1095}=y_{1095}+y_{1096};$
 $y_{1097}=y_{1097}+y_{1098}; y_{1099}=y_{1099}+y_{1100}; y_{1101}=y_{1101}+y_{1102}; y_{1103}=y_{1103}+y_{1104}; y_{1105}=y_{1105}+y_{1106};$
 $y_{1107}=y_{1107}+y_{1108}; y_{1109}=y_{1109}+y_{1110}; y_{1111}=y_{1111}+y_{1112}; y_{1113}=y_{1113}+y_{1114}; y_{1115}=y_{1115}+y_{1116};$
 $y_{1117}=y_{1117}+y_{1118}; y_{1119}=y_{1119}+y_{1120}; y_{1121}=y_{1121}+y_{1122}; y_{1123}=y_{1123}+y_{1124}; y_{1125}=y_{1125}+y_{1126};$
 $y_{1127}=y_{1127}+y_{1128}; y_{1129}=y_{1129}+y_{1130}; y_{1131}=y_{1131}+y_{1132}; y_{1133}=y_{1133}+y_{1134}; y_{1135}=y_{1135}+y_{1136};$
 $y_{1137}=y_{1137}+y_{1138}; y_{1139}=y_{1139}+y_{1140}; y_{1141}=y_{1141}+y_{1142}; y_{1143}=y_{1143}+y_{1144}; y_{1145}=y_{1145}+y_{1146};$
 $y_{1147}=y_{1147}+y_{1148}; y_{1149}=y_{1149}+y_{1150}; y_{1151}=y_{1151}+y_{1152}; y_{1153}=y_{1153}+y_{1154}; y_{1155}=y_{1155}+y_{1156};$
 $y_{1157}=y_{1157}+y_{1158}; y_{1159}=y_{1159}+y_{1160}; y_{1161}=y_{1161}+y_{1162}; y_{1163}=y_{1163}+y_{1164}; y_{1165}=y_{1165}+y_{1166};$
 $y_{1167}=y_{1167}+y_{1168}; y_{1169}=y_{1169}+y_{1170}; y_{1171}=y_{1171}+y_{1172}; y_{1173}=y_{1173}+y_{1174}; y_{1175}=y_{1175}+y_{1176};$
 $y_{1177}=y_{1177}+y_{1178}; y_{1179}=y_{1179}+y_{1180}; y_{1181}=y_{1181}+y_{1182}; y_{1183}=y_{1183}+y_{1184}; y_{1185}=y_{1185}+y_{1186};$
 $y_{1187}=y_{1187}+y_{1188}; y_{1189}=y_{1189}+y_{1190}; y_{1191}=y_{1191}+y_{1192}; y_{1193}=y_{1193}+y_{1194}; y_{1195}=y_{1195}+y_{1196};$
 $y_{1197}=y_{1197}+y_{1198}; y_{1199}=y_{1199}+y_{1200}; y_{1201}=y_{1201}+y_{1202}; y_{1203}=y_{1203}+y_{1204}; y_{1205}=y_{1205}+y_{1206};$
 $y_{1207}=y_{1207}+y_{1208}; y_{1209}=y_{1209}+y_{1210}; y_{1211}=y_{1211}+y_{1212}; y_{1213}=y_{1213}+y_{1214}; y_{1215}=y_{1215}+y_{1216};$
 $y_{1217}=y_{1217}+y_{1218}; y_{1219}=y_{1219}+y_{1220}; y_{1221}=y_{1221}+y_{1222}; y_{1223}=y_{1223}+y_{1224}; y_{1225}=y_{1225}+y_{1226};$
 $y_{1227}=y_{1227}+y_{1228}; y_{1229}=y_{1229}+y_{1230}; y_{1231}=y_{1231}+y_{1232}; y_{1233}=y_{1233}+y_{1234}; y_{1235}=y_{1235}+y_{1236};$
 $y_{1237}=y_{1237}+y_{1238}; y_{1239}=y_{1239}+y_{1240}; y_{1241}=y_{1241}+y_{1242}; y_{1243}=y_{1243}+y_{1244}; y_{1245}=y_{1245}+y_{1246};$
 $y_{1247}=y_{1247}+y_{1248}; y_{1249}=y_{1249}+y_{1250}; y_{1251}=y_{1251}+y_{1252}; y_{1253}=y_{1253}+y_{1254}; y_{1255}=y_{1255}+y_{1256};$
 $y_{1257}=y_{1257}+y_{1258}; y_{1259}=y_{1259}+y_{1260}; y_{1261}=y_{1261}+y_{1262}; y_{1263}=y_{1263}+y_{1264}; y_{1265}=y_{1265}+y_{1266};$

1267=y1267+y1268;y1269=y1269+y1270;y1271=y1271+y1272;y1273=y1273+y1274;y1275=y1275+y1276;y1277=y1277+y1278;y1279=y1279+y1280;y1281=y1281+y1282;y1283=y1283+y1284;y1285=y1285+y1286;y1287=y1287+y1288;y1289=y1289+y1290;y1291=y1291+y1292;y1293=y1293+y1294;y1295=y1295+y1296;y1297=y1297+y1298;y1299=y1299+y1300;y1301=y1301+y1302;y1303=y1303+y1304;y1305=y1305+y1306;y1307=y1307+y1308;y1309=y1309+y1310;y1311=y1311+y1312;y1313=y1313+y1314;y1315=y1315+y1316;y1317=y1317+y1318;y1319=y1319+y1320;y1321=y1321+y1322;y1323=y1323+y1324;y1325=y1325+y1326;y1327=y1327+y1328;y1329=y1329+y1330;y1331=y1331+y1332;y1333=y1333+y1334;y1335=y1335+y1336;y1337=y1337+y1338;y1339=y1339+y1340;y1341=y1341+y1342;y1343=y1343+y1344;

#Full depth 20 and Toffoli depth 3

#S 函数构造 2

y129=y129+y131;y133=y133+y135;y137=y137+y139;y141=y141+y143;y145=y145+y147;y149=y149+y151;y153=y153+y155;y157=y157+y159;y353=y353+y355;y357=y357+y359;y361=y361+y363;y365=y365+y367;y369=y369+y371;y373=y373+y375;y377=y377+y379;y381=y381+y383;y385=y385+y387;y389=y389+y391;y393=y393+y395;y422=y422+y424;y426=y426+y428;y430=y430+y432;y434=y434+y436;y438=y438+y440;y442=y442+y444;y446=y446+y448;y450=y450+y452;y454=y454+y456;y458=y458+y460;y462=y462+y464;y491=y491+y493;y495=y495+y497;y499=y499+y501;y503=y503+y505;y507=y507+y509;y511=y511+y513;y515=y515+y517;y519=y519+y521;y523=y523+y525;y527=y527+y529;y531=y531+y533;y560=y560+y562;y564=y564+y566;y568=y568+y570;y572=y572+y574;y576=y576+y578;y580=y580+y582;y584=y584+y586;y588=y588+y590;y592=y592+y594;y596=y596+y598;y600=y600+y602;y1041=y1041+y1043;y1045=y1045+y1047;y1049=y1049+y1051;y1053=y1053+y1055;y1057=y1057+y1059;y1061=y1061+y1063;y1065=y1065+y1067;y1069=y1069+y1071;y1073=y1073+y1075;y1077=y1077+y1079;y1081=y1081+y1083;y1085=y1085+y1087;y1089=y1089+y1091;y1093=y1093+y1095;y1097=y1097+y1099;y1101=y1101+y1103;y1105=y1105+y1107;y1109=y1109+y1111;y1113=y1113+y1115;y1117=y1117+y1119;y1121=y1121+y1123;y1125=y1125+y1127;y1129=y1129+y1131;y1133=y1133+y1135;y1137=y1137+y1139;y1141=y1141+y1143;y1145=y1145+y1147;y1149=y1149+y1151;y1153=y1153+y1155;y1157=y1157+y1159;y1161=y1161+y1163;y1165=y1165+y1167;y1169=y1169+y1171;y1173=y1173+y1175;y1177=y1177+y1179;y1181=y1181+y1183;y1185=y1185+y1187;y1189=y1189+y1191;y1193=y1193+y1195;y1197=y1197+y1199;y1201=y1201+y1203;y1205=y1205+y1207;y1209=y1209+y1211;y1213=y1213+y1215;y1217=y1217+y1219;y1221=y1221+y1223;y1225=y1225+y1227;y1229=y1229+y1231;y1233=y1233+y1235;y1237=y1237+y1239;y1241=y1241+y1243;y1245=y1245+y1247;y1249=y1249+y1251;y1253=y1253+y1255;y1257=y1257+y1259;y1261=y1261+y1263;y1265=y1265+y1267;y1269=y1269+y1271;y1273=y1273+y1275;y1277=y1277+y1279;y1281=y1281+y1283;y1285=y1285+y1287;y1289=y1289+y1291;y1293=y1293+y1295;y1297=y1297+y1299;y1301=y1301+y1303;y1305=y1305+y1307;y1309=y1309+y1311;y1313=y1313+y1315;y1317=y1317+y1319;y1321=y1321+y1323;y1325=y1325+y1327;y1329=y1329+y1331;y1333=y1333+y1335;y1337=y1337+y1339;y1341=y1341+y1343;

#Full depth 21 and Toffoli depth 3

#S 函数构造 3

y129=y129+y133;y137=y137+y141;y145=y145+y149;y153=y153+y157;y353=y353+y357;y361=y361+y365;y369=y369+y373;y377=y377+y381;y385=y385+y389;y393=y393+y422;y426=y426+y430;y434=y434+y438;y442=y442+y446;y450=y450+y454;y458=y458+y462;y491=y491+y495;y499=y499+y503;y507=y507+y511;y515=y515+y519;y523=y523+y527;y531=y531+y560;y564=y564+y568;y572=y572+y576;y580=y580+y584;y588=y588+y592;y596=y596+y600;y1041=y1041+y1045;y1049=y1049+y1053;y1057=y1057+y1061;y1065=y1065+y1069;y1073=y1073+y1077;y1081=y1081+y1085;y1089=y1089+y1093;y1097=y1097+y1101;y1105=y1105+y1109;y1113=y1113+y1117;y1121=y1121+y1125;y1129=y1129+y1133;y1137=y1137+y1141;y1145=y1145+y1149;y1153=y1153+y1157;y1161=y1161+y1165;y1169=y1169+y1173;y1177=y1177+y1181;y1185=y1185+y1189;y1193=y1193+y1197;y1201=y1201+y1205;y1209=y1209+y1213;y1217=y1217+y1221;y1225=y1225+y1229;y1233=y1233+y1237;y1241=y1241+y1245;y1249=y1249+y1253;y1257=y1257+y1261;y1265=y1265+y1269;y1273=y1273+y1277;y1281=y1281+y1285;y1289=y1289+y1293;y1297=y1297+y1301;y1305=y1305+y1309;y1313=y1313+y1317;y1321=y1321+y1325;y1329=y1329+y1333;y1337=y1337+y1341;

#Full depth 22 and Toffoli depth 3

#S 函数构造 4

y129=y129+y137;y145=y145+y153;y353=y353+y361;y369=y369+y377;y385=y385+y393;y426=y426+y434;y442=y442+y450;y458=y458+y491;y499=y499+y507;y515=y515+y523;y531=y531+y564;y572=y572+y580;y588=y588+y596;y1041=y1041+y1049;y1057=y1057+y1065;y1073=y1073+y1081;y1

089=y1089+y1097;y1105=y1105+y1113;y1121=y1121+y1129;y1137=y1137+y1145;y1153=y1153+y1161;y1169=y1169+y1177;y1185=y1185+y1193;y1201=y1201+y1209;y1217=y1217+y1225;y1233=y1233+y1241;y1249=y1249+y1257;y1265=y1265+y1273;y1281=y1281+y1289;y1297=y1297+y1305;y1313=y1313+y1321;y1329=y1329+y1337;

#Full depth 23 and Toffoli depth 3

#L 函数构造 1

y97=y97+y129;y98=y98+y145;y99=y99+y353;y100=y100+y129;y101=y101+y145;y102=y102+y353;y103=y103+y369;y104=y104+y385;y105=y105+y129;y106=y106+y145;y107=y107+y353;y108=y108+y369;y109=y109+y385;y110=y110+y426;y111=y111+y129;y112=y112+y145;y113=y113+y353;y114=y114+y369;y115=y115+y129;y116=y116+y145;y117=y117+y353;y118=y118+y369;y119=y119+y129;y120=y120+y145;y121=y121+y353;y122=y122+y369;y123=y123+y385;y124=y124+y426;y125=y125+y442;y126=y126+y458;y127=y127+y129;y128=y128+y145;

#Full depth 24 and Toffoli depth 3

#L 函数构造 2

y97=y97+y353;y98=y98+y369;y99=y99+y385;y100=y100+y369;y101=y101+y385;y102=y102+y426;y103=y103+y442;y104=y104+y458;y105=y105+y426;y106=y106+y442;y107=y107+y458;y108=y108+y499;y109=y109+y515;y110=y110+y531;y111=y111+y442;y112=y112+y458;y113=y113+y499;y114=y114+y515;y115=y115+y385;y116=y116+y426;y117=y117+y442;y118=y118+y458;y119=y119+y385;y120=y120+y426;y121=y121+y442;y122=y122+y458;y123=y123+y499;y124=y124+y515;y125=y125+y531;y126=y126+y572;y127=y127+y499;y128=y128+y515;

#Full depth 25 and Toffoli depth 3

#L 函数构造 3

y97=y97+y531;y98=y98+y572;y99=y99+y588;y100=y100+y426;y101=y101+y442;y102=y102+y458;y103=y103+y499;y104=y104+y515;y105=y105+y499;y106=y106+y515;y107=y107+y531;y108=y108+y572;y109=y109+y588;y110=y110+y1041;y111=y111+y572;y112=y112+y588;y113=y113+y1041;y114=y114+y1057;y115=y115+y531;y116=y116+y572;y117=y117+y588;y118=y118+y1041;y119=y119+y499;y120=y120+y515;y121=y121+y531;y122=y122+y572;y123=y123+y588;y124=y124+y1041;y125=y125+y1057;y126=y126+y1073;y127=y127+y588;y128=y128+y1041;

#Full depth 26 and Toffoli depth 3

#L 函数构造 4

y97=y97+y1057;y98=y98+y1073;y99=y99+y1089;y100=y100+y1041;y101=y101+y1057;y102=y102+y1073;y103=y103+y1089;y104=y104+y1105;y105=y105+y531;y106=y106+y572;y107=y107+y588;y108=y108+y1041;y109=y109+y1057;y110=y110+y1073;y111=y111+y1057;y112=y112+y1073;y113=y113+y1089;y114=y114+y1105;y115=y115+y1073;y116=y116+y1089;y117=y117+y1105;y118=y118+y1121;y119=y119+y1057;y120=y120+y1073;y121=y121+y1089;y122=y122+y1105;y123=y123+y1121;y124=y124+y1137;y125=y125+y1153;y126=y126+y1169;y127=y127+y1089;y128=y128+y1105;

#Full depth 27 and Toffoli depth 3

#L 函数构造 5

y97=y97+y1121;y98=y98+y1137;y99=y99+y1153;y100=y100+y1105;y101=y101+y1121;y102=y102+y1137;y103=y103+y1153;y104=y104+y1169;y105=y105+y1121;y106=y106+y1137;y107=y107+y1153;y108=y108+y1169;y109=y109+y1185;y110=y110+y1201;y111=y111+y1089;y112=y112+y1105;y113=y113+y1121;y114=y114+y1137;y115=y115+y1121;y116=y116+y1137;y117=y117+y1153;y118=y118+y1169;y119=y119+y1137;y120=y120+y1153;y121=y121+y1169;y122=y122+y1185;y123=y123+y1201;y124=y124+y1217;y125=y125+y1233;y126=y126+y1249;y127=y127+y1185;y128=y128+y1201;

#Full depth 28 and Toffoli depth 3

#L 函数构造 6

y97=y97+y1217;y98=y98+y1233;y99=y99+y1249;y100=y100+y1169;y101=y101+y1185;y102=y102+y1201;y103=y103+y1217;y104=y104+y1233;y105=y105+y1185;y106=y106+y1201;y107=y107+y1217;y108=y108+y1233;y109=y109+y1249;y110=y110+y1265;y111=y111+y1217;y112=y112+y1233;y113=y113+y1249;y114=y114+y1265;y115=y115+y1153;y116=y116+y1169;y117=y117+y1185;y118=y118+y1201;y119=y119+y1185;y120=y120+y1201;y121=y121+y1217;y122=y122+y1233;y123=y1

23+y1249;y124=y124+y1265;y125=y125+y1281;y126=y126+y1297;y127=y127+y1265;y128=y128+y1281;

#Full depth 29 and Toffoli depth 3

#L 函数构造 7

y97=y97+y1297;y98=y98+y1313;y99=y99+y1329;y100=y100+y1265;y101=y101+y1281;y102=y102+y1297;y103=y103+y1313;y104=y104+y1329;y105=y105+y1249;y106=y106+y1265;y107=y107+y1281;y108=y108+y1297;y109=y109+y1313;y110=y110+y1329;y111=y111+y1281;y112=y112+y1297;y113=y113+y1313;y114=y114+y1329;y115=y115+y1281;y116=y116+y1297;y117=y117+y1313;y118=y118+y1329;y119=y119+y1217;y120=y120+y1233;y121=y121+y1249;y122=y122+y1265;y123=y123+y1281;y124=y124+y1297;y125=y125+y1313;y126=y126+y1329;y127=y127+y1313;y128=y128+y1329;

#电路可逆（第 1 层电路-第 22 层电路可逆运行）

#Full depth 30 and Toffoli depth 3

y129=y129+y137;y145=y145+y153;y353=y353+y361;y369=y369+y377;y385=y385+y393;y426=y426+y434;y442=y442+y450;y458=y458+y491;y499=y499+y507;y515=y515+y523;y531=y531+y564;y572=y572+y580;y588=y588+y596;y1041=y1041+y1049;y1057=y1057+y1065;y1073=y1073+y1081;y1089=y1089+y1097;y1105=y1105+y1113;y1121=y1121+y1129;y1137=y1137+y1145;y1153=y1153+y1161;y1169=y1169+y1177;y1185=y1185+y1193;y1201=y1201+y1209;y1217=y1217+y1225;y1233=y1233+y1241;y1249=y1249+y1257;y1265=y1265+y1273;y1281=y1281+y1289;y1297=y1297+y1305;y1313=y1313+y1321;y1329=y1329+y1337;

#Full depth 31 and Toffoli depth 3

y129=y129+y133;y137=y137+y141;y145=y145+y149;y153=y153+y157;y353=y353+y357;y361=y361+y365;y369=y369+y373;y377=y377+y381;y385=y385+y389;y393=y393+y422;y426=y426+y430;y434=y434+y438;y442=y442+y446;y450=y450+y454;y458=y458+y462;y491=y491+y495;y499=y499+y503;y507=y507+y511;y515=y515+y519;y523=y523+y527;y531=y531+y560;y564=y564+y568;y572=y572+y576;y580=y580+y584;y588=y588+y592;y596=y596+y600;y1041=y1041+y1045;y1049=y1049+y1053;y1057=y1057+y1061;y1065=y1065+y1069;y1073=y1073+y1077;y1081=y1081+y1085;y1089=y1089+y1093;y1097=y1097+y1101;y1105=y1105+y1109;y1113=y1113+y1117;y1121=y1121+y1125;y1129=y1129+y1133;y1137=y1137+y1141;y1145=y1145+y1149;y1153=y1153+y1157;y1161=y1161+y1165;y1169=y1169+y1173;y1177=y1177+y1181;y1185=y1185+y1189;y1193=y1193+y1197;y1201=y1201+y1205;y1209=y1209+y1213;y1217=y1217+y1221;y1225=y1225+y1229;y1233=y1233+y1237;y1241=y1241+y1245;y1249=y1249+y1253;y1257=y1257+y1261;y1265=y1265+y1269;y1273=y1273+y1277;y1281=y1281+y1285;y1289=y1289+y1293;y1297=y1297+y1301;y1305=y1305+y1309;y1313=y1313+y1317;y1321=y1321+y1325;y1329=y1329+y1333;y1337=y1337+y1341;

#Full depth 32 and Toffoli depth 3

y129=y129+y131;y133=y133+y135;y137=y137+y139;y141=y141+y143;y145=y145+y147;y149=y149+y151;y153=y153+y155;y157=y157+y159;y353=y353+y355;y357=y357+y359;y361=y361+y363;y365=y365+y367;y369=y369+y371;y373=y373+y375;y377=y377+y379;y381=y381+y383;y385=y385+y387;y389=y389+y391;y393=y393+y395;y422=y422+y424;y426=y426+y428;y430=y430+y432;y434=y434+y436;y438=y438+y440;y442=y442+y444;y446=y446+y448;y450=y450+y452;y454=y454+y456;y458=y458+y460;y462=y462+y464;y491=y491+y493;y495=y495+y497;y499=y499+y501;y503=y503+y505;y507=y507+y509;y511=y511+y513;y515=y515+y517;y519=y519+y521;y523=y523+y525;y527=y527+y529;y531=y531+y533;y560=y560+y562;y564=y564+y566;y568=y568+y570;y572=y572+y574;y576=y576+y578;y580=y580+y582;y584=y584+y586;y588=y588+y590;y592=y592+y594;y596=y596+y598;y600=y600+y602;y1041=y1041+y1043;y1045=y1045+y1047;y1049=y1049+y1051;y1053=y1053+y1055;y1057=y1057+y1059;y1061=y1061+y1063;y1065=y1065+y1067;y1069=y1069+y1071;y1073=y1073+y1075;y1077=y1077+y1079;y1081=y1081+y1083;y1085=y1085+y1087;y1089=y1089+y1091;y1093=y1093+y1095;y1097=y1097+y1099;y1101=y1101+y1103;y1105=y1105+y1107;y1109=y1109+y1111;y1113=y1113+y1115;y1117=y1117+y1119;y1121=y1121+y1123;y1125=y1125+y1127;y1129=y1129+y1131;y1133=y1133+y1135;y1137=y1137+y1139;y1141=y1141+y1143;y1145=y1145+y1147;y1149=y1149+y1151;y1153=y1153+y1155;y1157=y1157+y1159;y1161=y1161+y1163;y1165=y1165+y1167;y1169=y1169+y1171;y1173=y1173+y1175;y1177=y1177+y1179;y1181=y1181+y1183;y1185=y1185+y1187;y1189=y1189+y1191;y1193=y1193+y1195;y1197=y1197+y1199;y1201=y1201+y1203;y1205=y1205+y1207;y1209=y1209+y1211;y1213=y1213+y1215;y1217=y1217+y1219;y1221=y1221+y1223;y1225=y1225+y1227;y1229=y1229+y1231;y1233=y1233+y1235;y1237=y1237+y1239;y1241=y1241+y1243;y1245=y1245+y1247;y1249=y1249+y1251;y1253=y1253+y1255;y1257=y1257+y1259;

7=y1257+y1259;y1261=y1261+y1263;y1265=y1265+y1267;y1269=y1269+y1271;y1273=y1273+y1275;y1277=y1277+y1279;y1281=y1281+y1283;y1285=y1285+y1287;y1289=y1289+y1291;y1293=y1293+y1295;y1297=y1297+y1299;y1301=y1301+y1303;y1305=y1305+y1307;y1309=y1309+y1311;y1313=y1313+y1315;y1317=y1317+y1319;y1321=y1321+y1323;y1325=y1325+y1327;y1329=y1329+y1331;y1333=y1333+y1335;y1337=y1337+y1339;y1341=y1341+y1343;

#Full depth 33 and Toffoli depth 3

y129=y129+y130;y131=y131+y132;y133=y133+y134;y135=y135+y136;y137=y137+y138;y139=y139+y140;y141=y141+y142;y143=y143+y144;y145=y145+y146;y147=y147+y148;y149=y149+y150;y151=y151+y152;y153=y153+y154;y155=y155+y156;y157=y157+y158;y159=y159+y160;y353=y353+y354;y355=y355+y356;y357=y357+y358;y359=y359+y360;y361=y361+y362;y363=y363+y364;y365=y365+y366;y367=y367+y368;y369=y369+y370;y371=y371+y372;y373=y373+y374;y375=y375+y376;y377=y377+y378;y379=y379+y380;y381=y381+y382;y383=y383+y384;y385=y385+y386;y387=y387+y388;y389=y389+y390;y391=y391+y392;y393=y393+y394;y395=y395+y396;y422=y422+y423;y424=y424+y425;y426=y426+y427;y428=y428+y429;y430=y430+y431;y432=y432+y433;y434=y434+y435;y436=y436+y437;y438=y438+y439;y440=y440+y441;y442=y442+y443;y444=y444+y445;y446=y446+y447;y448=y448+y449;y450=y450+y451;y452=y452+y453;y454=y454+y455;y456=y456+y457;y458=y458+y459;y460=y460+y461;y462=y462+y463;y464=y464+y465;y491=y491+y492;y493=y493+y494;y495=y495+y496;y497=y497+y498;y499=y499+y500;y501=y501+y502;y503=y503+y504;y505=y505+y506;y507=y507+y508;y509=y509+y510;y511=y511+y512;y513=y513+y514;y515=y515+y516;y517=y517+y518;y519=y519+y520;y521=y521+y522;y523=y523+y524;y525=y525+y526;y527=y527+y528;y529=y529+y530;y531=y531+y532;y533=y533+y534;y560=y560+y561;y562=y562+y563;y564=y564+y565;y566=y566+y567;y568=y568+y569;y570=y570+y571;y572=y572+y573;y574=y574+y575;y576=y576+y577;y578=y578+y579;y580=y580+y581;y582=y582+y583;y584=y584+y585;y586=y586+y587;y588=y588+y589;y590=y590+y591;y592=y592+y593;y594=y594+y595;y596=y596+y597;y598=y598+y599;y600=y600+y601;y602=y602+y603;y1041=y1041+y1042;y1043=y1043+y1044;y1045=y1045+y1046;y1047=y1047+y1048;y1049=y1049+y1050;y1051=y1051+y1052;y1053=y1053+y1054;y1055=y1055+y1056;y1057=y1057+y1058;y1059=y1059+y1060;y1061=y1061+y1062;y1063=y1063+y1064;y1065=y1065+y1066;y1067=y1067+y1068;y1069=y1069+y1070;y1071=y1071+y1072;y1073=y1073+y1074;y1075=y1075+y1076;y1077=y1077+y1078;y1079=y1079+y1080;y1081=y1081+y1082;y1083=y1083+y1084;y1085=y1085+y1086;y1087=y1087+y1088;y1089=y1089+y1090;y1091=y1091+y1092;y1093=y1093+y1094;y1095=y1095+y1096;y1097=y1097+y1098;y1099=y1099+y1100;y1101=y1101+y1102;y1103=y1103+y1104;y1105=y1105+y1106;y1107=y1107+y1108;y1109=y1109+y1110;y1111=y1111+y1112;y1113=y1113+y1114;y1115=y1115+y1116;y1117=y1117+y1118;y1119=y1119+y1120;y1121=y1121+y1122;y1123=y1123+y1124;y1125=y1125+y1126;y1127=y1127+y1128;y1129=y1129+y1130;y1131=y1131+y1132;y1133=y1133+y1134;y1135=y1135+y1136;y1137=y1137+y1138;y1139=y1139+y1140;y1141=y1141+y1142;y1143=y1143+y1144;y1145=y1145+y1146;y1147=y1147+y1148;y1149=y1149+y1150;y1151=y1151+y1152;y1153=y1153+y1154;y1155=y1155+y1156;y1157=y1157+y1158;y1159=y1159+y1160;y1161=y1161+y1162;y1163=y1163+y1164;y1165=y1165+y1166;y1167=y1167+y1168;y1169=y1169+y1170;y1171=y1171+y1172;y1173=y1173+y1174;y1175=y1175+y1176;y1177=y1177+y1178;y1179=y1179+y1180;y1181=y1181+y1182;y1183=y1183+y1184;y1185=y1185+y1186;y1187=y1187+y1188;y1189=y1189+y1190;y1191=y1191+y1192;y1193=y1193+y1194;y1195=y1195+y1196;y1197=y1197+y1198;y1199=y1199+y1200;y1201=y1201+y1202;y1203=y1203+y1204;y1205=y1205+y1206;y1207=y1207+y1208;y1209=y1209+y1210;y1211=y1211+y1212;y1213=y1213+y1214;y1215=y1215+y1216;y1217=y1217+y1218;y1219=y1219+y1220;y1221=y1221+y1222;y1223=y1223+y1224;y1225=y1225+y1226;y1227=y1227+y1228;y1229=y1229+y1230;y1231=y1231+y1232;y1233=y1233+y1234;y1235=y1235+y1236;y1237=y1237+y1238;y1239=y1239+y1240;y1241=y1241+y1242;y1243=y1243+y1244;y1245=y1245+y1246;y1247=y1247+y1248;y1249=y1249+y1250;y1251=y1251+y1252;y1253=y1253+y1254;y1255=y1255+y1256;y1257=y1257+y1258;y1259=y1259+y1260;y1261=y1261+y1262;y1263=y1263+y1264;y1265=y1265+y1266;y1267=y1267+y1268;y1269=y1269+y1270;y1271=y1271+y1272;y1273=y1273+y1274;y1275=y1275+y1276;y1277=y1277+y1278;y1279=y1279+y1280;y1281=y1281+y1282;y1283=y1283+y1284;y1285=y1285+y1286;y1287=y1287+y1288;y1289=y1289+y1290;y1291=y1291+y1292;y1293=y1293+y1294;y1295=y1295+y1296;y1297=y1297+y1298;y1299=y1299+y1300;y1301=y1301+y1302;y1303=y1303+y1304;y1305=y1305+y1306;y1307=y1307+y1308;y1309=y1309+y1310;y1311=y1311+y1312;y1313=y1313+y1314;y1315=y1315+y1316;y1317=y1317+y1318;y1319=y1319+y1320;y1321=y1321+y1322;y1323=y1323+y1324;y1325=y1325+y1326;y1327=y1327+y1328;y1329=y1329+y1330;y1331=y1331+y1332;y1333=y1333+y1334;y1335=y1335+y1336;y1337=y1337+y1338;y1339=y1339+y1340;y1341=y1341+y1342;y1343=y1343+y1344;

#Full depth 34 and Toffoli depth 4

$y_{129}=y_{129}+y_{240}*y_{405}; y_{130}=y_{130}+y_{239}*y_{406}; y_{131}=y_{131}+y_{238}*y_{398}; y_{132}=y_{132}+y_{237}*y_{629}; y_{133}=y_{133}+y_{236}*y_{407}; y_{134}=y_{134}+y_{235}*y_{630}; y_{135}=y_{135}+y_{234}*y_{413}; y_{136}=y_{136}+y_{233}*y_{631}; y_{137}=y_{137}+y_{232}*y_{419}; y_{138}=y_{138}+y_{231}*y_{632}; y_{139}=y_{139}+y_{230}*y_{402}; y_{140}=y_{140}+y_{229}*y_{633}; y_{141}=y_{141}+y_{228}*y_{634}; y_{142}=y_{142}+y_{227}*y_{635}; y_{143}=y_{143}+y_{226}*y_{397}; y_{144}=y_{144}+y_{225}*y_{636}; y_{145}=y_{145}+y_{240}*y_{637}; y_{146}=y_{146}+y_{239}*y_{638}; y_{147}=y_{147}+y_{238}*y_{420}; y_{148}=y_{148}+y_{237}*y_{409}; y_{149}=y_{149}+y_{236}*y_{639}; y_{150}=y_{150}+y_{235}*y_{640}; y_{151}=y_{151}+y_{234}*y_{410}; y_{152}=y_{152}+y_{233}*y_{641}; y_{153}=y_{153}+y_{232}*y_{642}; y_{154}=y_{154}+y_{231}*y_{643}; y_{155}=y_{155}+y_{230}*y_{644}; y_{156}=y_{156}+y_{229}*y_{645}; y_{157}=y_{157}+y_{228}*y_{646}; y_{158}=y_{158}+y_{227}*y_{647}; y_{159}=y_{159}+y_{226}*y_{648}; y_{160}=y_{160}+y_{225}*y_{649}; y_{353}=y_{353}+y_{240}*y_{650}; y_{354}=y_{354}+y_{239}*y_{651}; y_{355}=y_{355}+y_{238}*y_{652}; y_{356}=y_{356}+y_{237}*y_{404}; y_{357}=y_{357}+y_{236}*y_{653}; y_{358}=y_{358}+y_{235}*y_{654}; y_{359}=y_{359}+y_{234}*y_{655}; y_{360}=y_{360}+y_{233}*y_{656}; y_{361}=y_{361}+y_{232}*y_{414}; y_{362}=y_{362}+y_{231}*y_{657}; y_{363}=y_{363}+y_{230}*y_{658}; y_{364}=y_{364}+y_{229}*y_{412}; y_{365}=y_{365}+y_{228}*y_{659}; y_{366}=y_{366}+y_{227}*y_{660}; y_{367}=y_{367}+y_{226}*y_{661}; y_{368}=y_{368}+y_{225}*y_{662}; y_{369}=y_{369}+y_{240}*y_{663}; y_{370}=y_{370}+y_{239}*y_{664}; y_{371}=y_{371}+y_{238}*y_{665}; y_{372}=y_{372}+y_{237}*y_{666}; y_{373}=y_{373}+y_{236}*y_{667}; y_{374}=y_{374}+y_{235}*y_{668}; y_{375}=y_{375}+y_{234}*y_{669}; y_{376}=y_{376}+y_{233}*y_{670}; y_{377}=y_{377}+y_{232}*y_{671}; y_{378}=y_{378}+y_{231}*y_{672}; y_{379}=y_{379}+y_{230}*y_{673}; y_{380}=y_{380}+y_{229}*y_{674}; y_{381}=y_{381}+y_{228}*y_{675}; y_{382}=y_{382}+y_{227}*y_{676}; y_{383}=y_{383}+y_{226}*y_{677}; y_{384}=y_{384}+y_{225}*y_{678}; y_{385}=y_{385}+y_{240}*y_{679}; y_{386}=y_{386}+y_{239}*y_{680}; y_{387}=y_{387}+y_{238}*y_{681}; y_{388}=y_{388}+y_{237}*y_{682}; y_{389}=y_{389}+y_{236}*y_{683}; y_{390}=y_{390}+y_{235}*y_{684}; y_{391}=y_{391}+y_{234}*y_{685}; y_{392}=y_{392}+y_{233}*y_{686}; y_{393}=y_{393}+y_{232}*y_{403}; y_{394}=y_{394}+y_{231}*y_{687}; y_{395}=y_{395}+y_{230}*y_{688}; y_{396}=y_{396}+y_{229}*y_{689}; y_{422}=y_{422}+y_{228}*y_{690}; y_{423}=y_{423}+y_{227}*y_{691}; y_{424}=y_{424}+y_{226}*y_{692}; y_{425}=y_{425}+y_{225}*y_{693}; y_{426}=y_{426}+y_{240}*y_{416}; y_{427}=y_{427}+y_{239}*y_{415}; y_{428}=y_{428}+y_{238}*y_{694}; y_{429}=y_{429}+y_{237}*y_{695}; y_{430}=y_{430}+y_{236}*y_{696}; y_{431}=y_{431}+y_{235}*y_{697}; y_{432}=y_{432}+y_{234}*y_{698}; y_{433}=y_{433}+y_{233}*y_{699}; y_{434}=y_{434}+y_{232}*y_{399}; y_{435}=y_{435}+y_{231}*y_{700}; y_{436}=y_{436}+y_{230}*y_{701}; y_{437}=y_{437}+y_{229}*y_{702}; y_{438}=y_{438}+y_{228}*y_{703}; y_{439}=y_{439}+y_{227}*y_{704}; y_{440}=y_{440}+y_{226}*y_{705}; y_{441}=y_{441}+y_{225}*y_{706}; y_{442}=y_{442}+y_{240}*y_{707}; y_{443}=y_{443}+y_{239}*y_{417}; y_{444}=y_{444}+y_{238}*y_{418}; y_{445}=y_{445}+y_{237}*y_{708}; y_{446}=y_{446}+y_{236}*y_{709}; y_{447}=y_{447}+y_{235}*y_{408}; y_{448}=y_{448}+y_{234}*y_{710}; y_{449}=y_{449}+y_{233}*y_{711}; y_{450}=y_{450}+y_{232}*y_{712}; y_{451}=y_{451}+y_{231}*y_{400}; y_{452}=y_{452}+y_{230}*y_{713}; y_{453}=y_{453}+y_{229}*y_{421}; y_{454}=y_{454}+y_{228}*y_{714}; y_{455}=y_{455}+y_{227}*y_{715}; y_{456}=y_{456}+y_{226}*y_{716}; y_{457}=y_{457}+y_{225}*y_{717}; y_{458}=y_{458}+y_{240}*y_{718}; y_{459}=y_{459}+y_{239}*y_{401}; y_{460}=y_{460}+y_{238}*y_{719}; y_{461}=y_{461}+y_{237}*y_{720}; y_{462}=y_{462}+y_{236}*y_{721}; y_{463}=y_{463}+y_{235}*y_{722}; y_{464}=y_{464}+y_{234}*y_{723}; y_{465}=y_{465}+y_{233}*y_{724}; y_{491}=y_{491}+y_{232}*y_{411}; y_{492}=y_{492}+y_{231}*y_{725}; y_{493}=y_{493}+y_{230}*y_{726}; y_{494}=y_{494}+y_{229}*y_{727}; y_{495}=y_{495}+y_{228}*y_{728}; y_{496}=y_{496}+y_{227}*y_{729}; y_{497}=y_{497}+y_{226}*y_{730}; y_{498}=y_{498}+y_{225}*y_{731}; y_{499}=y_{499}+y_{272}*y_{474}; y_{500}=y_{500}+y_{271}*y_{475}; y_{501}=y_{501}+y_{270}*y_{467}; y_{502}=y_{502}+y_{269}*y_{732}; y_{503}=y_{503}+y_{268}*y_{476}; y_{504}=y_{504}+y_{267}*y_{733}; y_{505}=y_{505}+y_{266}*y_{482}; y_{506}=y_{506}+y_{265}*y_{734}; y_{507}=y_{507}+y_{264}*y_{488}; y_{508}=y_{508}+y_{263}*y_{735}; y_{509}=y_{509}+y_{262}*y_{471}; y_{510}=y_{510}+y_{261}*y_{736}; y_{511}=y_{511}+y_{260}*y_{737}; y_{512}=y_{512}+y_{259}*y_{738}; y_{513}=y_{513}+y_{258}*y_{466}; y_{514}=y_{514}+y_{257}*y_{739}; y_{515}=y_{515}+y_{272}*y_{740}; y_{516}=y_{516}+y_{271}*y_{741}; y_{517}=y_{517}+y_{270}*y_{489}; y_{518}=y_{518}+y_{269}*y_{478}; y_{519}=y_{519}+y_{268}*y_{742}; y_{520}=y_{520}+y_{267}*y_{743}; y_{521}=y_{521}+y_{266}*y_{479}; y_{522}=y_{522}+y_{265}*y_{744}; y_{523}=y_{523}+y_{264}*y_{745}; y_{524}=y_{524}+y_{263}*y_{746}; y_{525}=y_{525}+y_{262}*y_{747}; y_{526}=y_{526}+y_{261}*y_{748}; y_{527}=y_{527}+y_{260}*y_{749}; y_{528}=y_{528}+y_{259}*y_{750}; y_{529}=y_{529}+y_{258}*y_{751}; y_{530}=y_{530}+y_{257}*y_{752}; y_{531}=y_{531}+y_{272}*y_{753}; y_{532}=y_{532}+y_{271}*y_{754}; y_{533}=y_{533}+y_{270}*y_{755}; y_{534}=y_{534}+y_{269}*y_{473}; y_{560}=y_{560}+y_{268}*y_{756}; y_{561}=y_{561}+y_{267}*y_{757}; y_{562}=y_{562}+y_{266}*y_{758}; y_{563}=y_{563}+y_{265}*y_{759}; y_{564}=y_{564}+y_{264}*y_{483}; y_{565}=y_{565}+y_{263}*y_{760}; y_{566}=y_{566}+y_{262}*y_{761}; y_{567}=y_{567}+y_{261}*y_{481}; y_{568}=y_{568}+y_{260}*y_{762}; y_{569}=y_{569}+y_{259}*y_{763}; y_{570}=y_{570}+y_{258}*y_{764}; y_{571}=y_{571}+y_{257}*y_{765}; y_{572}=y_{572}+y_{272}*y_{766}; y_{573}=y_{573}+y_{271}*y_{767}; y_{574}=y_{574}+y_{270}*y_{768}; y_{575}=y_{575}+y_{269}*y_{769}; y_{576}=y_{576}+y_{268}*y_{770}; y_{577}=y_{577}+y_{267}*y_{771}; y_{578}=y_{578}+y_{266}*y_{772}; y_{579}=y_{579}+y_{265}*y_{773}; y_{580}=y_{580}+y_{264}*y_{774}; y_{581}=y_{581}+y_{263}*y_{775}; y_{582}=y_{582}+y_{262}*y_{776}; y_{583}=y_{583}+y_{261}*y_{777}; y_{584}=y_{584}+y_{260}*y_{778}; y_{585}=y_{585}+y_{259}*y_{779}; y_{586}=y_{586}+y_{258}*y_{780}; y_{587}=y_{587}+y_{257}*y_{781}; y_{588}=y_{588}+y_{272}*y_{782}; y_{589}=y_{589}+y_{271}*y_{783}; y_{590}=y_{590}+y_{270}*y_{784}; y_{591}=y_{591}+y_{269}*y_{785}; y_{592}=y_{592}+y_{268}*y_{786}; y_{593}=y_{593}+y_{267}*y_{787}; y_{594}=y_{594}+y_{266}*y_{788}; y_{595}=y_{595}+y_{265}*y_{789}; y_{596}=y_{596}+y_{264}*y_{472}; y_{597}=y_{597}+y_{263}*y_{790}; y_{598}=y_{598}+y_{262}*y_{791}; y_{599}=y_{599}+y_{261}*y_{792}; y_{600}=y_{600}+y_{260}*y_{793}; y_{601}=y_{601}+y_{259}*y_{794}; y_{602}=y_{602}+y_{258}*y_{795}; y_{603}=y_{603}+y_{257}*y_{796}; y_{1041}=y_{1041}+y_{272}*y_{485}; y_{1042}=y_{1042}+y_{271}*y_{484}; y_{1043}=y_{1043}+y_{270}*y_{797}; y_{1044}=y_{1044}+y_{269}*y_{798}; y_{1045}=y_{1045}+y_{268}*y_{799}; y_{1046}=y_{1046}+y_{267}*y_{800}; y_{1047}=y_{1047}+y_{266}*y_{801}; y_{1048}=y_{1048}+y_{265}*y_{802}; y_{1049}=y_{1049}+y_{264}*y_{468}; y_{1050}=y_{1050}+y_{263}*y_{803}; y_{1051}=y_{1051}+y_{262}*y_{804}; y_{1052}=y_{1052}+y_{261}*y_{805}; y_{1053}=y_{1053}+y_{260}*y_{806}; y_{1054}=y_{1054}+y_{259}*y_{807}; y_{1055}=y_{1055}+y_{258}*y_{808}; y_{1056}=y_{1056}+y_{257}*y_{809}; y_{1057}=y_{1057}+y_{272}*y_{810}; y_{1058}=y_{1058}+y_{271}*y_{486}; y_{1059}=y_{1059}+y_{270}*y_{487}; y_{1060}=y_{1060}+y_{269}*y_{811}; y_{1061}=y_{1061}+y_{268}*y_{812}; y_{1062}=y_{1062}+y_{267}*y_{477}; y_{1063}=y_{1063}+y_{266}*y_{813}; y_{1064}=y_{1064}+y_{265}*y_{814}; y_{1065}=y_{1065}+y_{264}*y_{815}; y_{1066}=y_{1066}+y_{263}*y_{469}; y_{1067}=y_{1067}+y_{262}*y_{816}; y_{1068}=y_{1068}+y_{261}*y_{490}; y_{1069}=y_{1069}+y_{260}*y_{817}; y_{1070}=y_{1070}+y_{259}*y_{818}; y_{1071}=y_{1071}+y_{258}*y_{819}; y_{1072}=y_{1072}+y_{257}*y_{820}; y_{1073}=y_{1073}+y_{272}*y_{821}; y_{1074}=y_{1074}+y_{271}*y_{822}$

;y1301=y1301+y332*y1005;y1302=y1302+y331*y1006;y1303=y1303+y330*y1007;y1304=y1304+y329*y1008;y1305=y1305+y328*y606;y1306=y1306+y327*y1009;y1307=y1307+y326*y1010;y1308=y1308+y325*y1011;y1309=y1309+y324*y1012;y1310=y1310+y323*y1013;y1311=y1311+y322*y1014;y1312=y1312+y321*y1015;y1313=y1313+y336*y1016;y1314=y1314+y335*y624;y1315=y1315+y334*y625;y1316=y1316+y333*y1017;y1317=y1317+y332*y1018;y1318=y1318+y331*y615;y1319=y1319+y330*y1019;y1320=y1320+y329*y1020;y1321=y1321+y328*y1021;y1322=y1322+y327*y607;y1323=y1323+y326*y1022;y1324=y1324+y325*y628;y1325=y1325+y324*y1023;y1326=y1326+y323*y1024;y1327=y1327+y322*y1025;y1328=y1328+y321*y1026;y1329=y1329+y336*y1027;y1330=y1330+y335*y608;y1331=y1331+y334*y1028;y1332=y1332+y333*y1029;y1333=y1333+y332*y1030;y1334=y1334+y331*y1031;y1335=y1335+y330*y1032;y1336=y1336+y329*y1033;y1337=y1337+y328*y618;y1338=y1338+y327*y1034;y1339=y1339+y326*y1035;y1340=y1340+y325*y1036;y1341=y1341+y324*y1037;y1342=y1342+y323*y1038;y1343=y1343+y322*y1039;y1344=y1344+y321*y1040;

#Full depth 35 and Toffoli depth 4

y353=y353+y247;y354=y354+y251;y355=y355+y242;y356=y356+y250;y357=y357+y243;y358=y358+y245;y359=y359+y248;y360=y360+y241;y361=y361+y254;y362=y362+y246;y363=y363+y244;y364=y364+y242;y365=y365+y249;y366=y366+y241;y367=y367+y241;y368=y368+y244;y369=y369+y247;y370=y370+y243;y371=y371+y245;y372=y372+y253;y373=y373+y244;y374=y374+y248;y375=y375+y242;y376=y376+y252;y377=y377+y250;y378=y378+y251;y379=y379+y255;y380=y380+y243;y381=y381+y248;y382=y382+y249;y383=y383+y245;y384=y384+y247;y385=y385+y241;y386=y386+y253;y387=y387+y246;y388=y388+y241;y389=y389+y248;y390=y390+y247;y391=y391+y250;y392=y392+y246;y393=y393+y244;y394=y394+y245;y395=y395+y246;y396=y396+y241;y422=y422+y279;y423=y423+y283;y424=y424+y274;y425=y425+y282;y426=y426+y275;y427=y427+y277;y428=y428+y280;y429=y429+y273;y430=y430+y286;y431=y431+y278;y432=y432+y276;y433=y433+y274;y434=y434+y281;y435=y435+y273;y436=y436+y273;y437=y437+y276;y438=y438+y279;y439=y439+y275;y440=y440+y277;y441=y441+y285;y442=y442+y276;y443=y443+y280;y444=y444+y274;y445=y445+y284;y446=y446+y282;y447=y447+y283;y448=y448+y287;y449=y449+y275;y450=y450+y280;y451=y451+y281;y452=y452+y277;y453=y453+y279;y454=y454+y273;y455=y455+y285;y456=y456+y278;y457=y457+y273;y458=y458+y280;y459=y459+y279;y460=y460+y282;y461=y461+y278;y462=y462+y276;y463=y463+y277;y464=y464+y278;y465=y465+y273;y491=y491+y311;y492=y492+y315;y493=y493+y306;y494=y494+y314;y495=y495+y307;y496=y496+y309;y497=y497+y312;y498=y498+y305;y499=y499+y318;y500=y500+y310;y501=y501+y308;y502=y502+y306;y503=y503+y313;y504=y504+y305;y505=y505+y305;y506=y506+y308;y507=y507+y311;y508=y508+y307;y509=y509+y309;y510=y510+y317;y511=y511+y308;y512=y512+y312;y513=y513+y306;y514=y514+y316;y515=y515+y314;y516=y516+y315;y517=y517+y319;y518=y518+y307;y519=y519+y312;y520=y520+y313;y521=y521+y309;y522=y522+y311;y523=y523+y305;y524=y524+y317;y525=y525+y310;y526=y526+y305;y527=y527+y312;y528=y528+y311;y529=y529+y314;y530=y530+y310;y531=y531+y308;y532=y532+y309;y533=y533+y310;y534=y534+y305;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y340;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y592+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y598=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+y337;

#Full depth 36 and Toffoli depth 4

y353=y353+y249;y354=y354+y253;y355=y355+y244;y356=y356+y254;y357=y357+y255;y358=y358+y256;y359=y359+y252;y360=y360+y246;y361=y361+y255;y362=y362+y250;y363=y363+y247;y364=y364+y245;y365=y365+y251;y366=y366+y243;y367=y367+y256;y368=y368+y254;y369=y369+y252;y370=y370+y252;y371=y371+y246;y372=y372+y256;y373=y373+y256;y374=y374+y249;y375=y375+y243;y376=y376+y253;y377=y377+y253;y378=y378+y252;y379=y379+y256;y380=y380+y256;y381=y381+y255;y382=y382+y252;y383=y383+y252;y384=y384+y251;y385=y385+y242;y386=y386+y255;y387=y387+y254;y388=y388+y254;y389=y389+y253;y390=y390+y253;y391=y391+y251;y392=y392+y251;y393=y393+y251;y394=y394+y250;y395=y395+y248;y396=y396+y247;y422=y422+y281;y423=y423+y285;y424=y424+y276;y425=y425+y286;y426=y426+y287;y427=y427+y288;y428=y428+y284;y429=y429+y278;y430=y430+y287;y431=y431+y282;y432=y432+y279;y433=y433+y277;y434=y434+y283;y435=y435+y275;y436=y436+y288;y437=y437+y286;y438=y438+y284;y439=y439+y286;

=y439+y284;y440=y440+y278;y441=y441+y288;y442=y442+y288;y443=y443+y281;y444=y444+y275;y445=y445+y285;y446=y446+y285;y447=y447+y284;y448=y448+y288;y449=y449+y288;y450=y450+y287;y451=y451+y284;y452=y452+y284;y453=y453+y283;y454=y454+y274;y455=y455+y287;y456=y456+y286;y457=y457+y286;y458=y458+y285;y459=y459+y285;y460=y460+y283;y461=y461+y283;y462=y462+y283;y463=y463+y282;y464=y464+y280;y465=y465+y279;y491=y491+y313;y492=y492+y317;y493=y493+y308;y494=y494+y318;y495=y495+y319;y496=y496+y320;y497=y497+y316;y498=y498+y310;y499=y499+y319;y500=y500+y314;y501=y501+y311;y502=y502+y309;y503=y503+y315;y504=y504+y307;y505=y505+y320;y506=y506+y318;y507=y507+y316;y508=y508+y316;y509=y509+y310;y510=y510+y320;y511=y511+y320;y512=y512+y313;y513=y513+y307;y514=y514+y317;y515=y515+y317;y516=y516+y316;y517=y517+y320;y518=y518+y320;y519=y519+y319;y520=y520+y316;y521=y521+y316;y522=y522+y315;y523=y523+y306;y524=y524+y319;y525=y525+y318;y526=y526+y318;y527=y527+y317;y528=y528+y317;y529=y529+y315;y530=y530+y315;y531=y531+y315;y532=y532+y314;y533=y533+y312;y534=y534+y311;y560=y560+y345;y561=y561+y349;y562=y562+y340;y563=y563+y350;y564=y564+y351;y565=y565+y352;y566=y566+y348;y567=y567+y342;y568=y568+y351;y569=y569+y346;y570=y570+y343;y571=y571+y341;y572=y572+y347;y573=y573+y339;y574=y574+y352;y575=y575+y350;y576=y576+y348;y577=y577+y348;y578=y578+y342;y579=y579+y352;y580=y580+y352;y581=y581+y345;y582=y582+y339;y583=y583+y349;y584=y584+y349;y585=y585+y348;y586=y586+y352;y587=y587+y352;y588=y588+y351;y589=y589+y348;y590=y590+y348;y591=y591+y347;y592=y592+y338;y593=y593+y351;y594=y594+y350;y595=y595+y350;y596=y596+y349;y597=y597+y349;y598=y598+y347;y599=y599+y347;y600=y600+y347;y601=y601+y346;y602=y602+y344;y603=y603+y343;

#Full depth 37 and Toffoli depth 4

y675=y675+y362;y693=y693+y375;y778=y778+y431;y796=y796+y444;y881=y881+y500;y899=y899+y513;y984=y984+y569;y1002=y1002+y582;

#Full depth 38 and Toffoli depth 4

y629=y629+y377;y645=y645+y369;y649=y649+y363;y675=y675+y359;y690=y690+y379;y693=y693+y360;y713=y713+y364;y721=y721+y384;y727=y727+y381;y732=y732+y446;y748=y748+y438;y752=y752+y432;y778=y778+y428;y793=y793+y448;y796=y796+y429;y816=y816+y433;y824=y824+y453;y830=y830+y450;y835=y835+y515;y851=y851+y507;y855=y855+y501;y881=y881+y497;y896=y896+y517;y899=y899+y498;y919=y919+y502;y927=y927+y522;y933=y933+y519;y938=y938+y584;y954=y954+y576;y958=y958+y570;y984=y984+y566;y999=y999+y586;y1002=y1002+y567;y1022=y1022+y571;y1030=y1030+y591;y1036=y1036+y588;

#Full depth 39 and Toffoli depth 4

y629=y629+y360;y419=y419+y361;y632=y632+y396;y635=y635+y382;y640=y640+y391;y641=y641+y375;y642=y642+y370;y644=y644+y385;y645=y645+y360;y648=y648+y374;y649=y649+y360;y650=y650+y356;y654=y654+y367;y655=y655+y364;y657=y657+y381;y658=y658+y361;y660=y660+y366;y662=y662+y364;y663=y663+y372;y672=y672+y378;y674=y674+y385;y675=y675+y357;y676=y676+y360;y678=y678+y392;y682=y682+y369;y684=y684+y381;y689=y689+y378;y690=y690+y378;y693=y693+y359;y695=y695+y380;y696=y696+y372;y699=y699+y390;y702=y702+y373;y705=y705+y366;y417=y417+y387;y709=y709+y379;y711=y711+y383;y712=y712+y395;y713=y713+y360;y721=y721+y359;y722=y722+y364;y727=y727+y367;y728=y728+y393;y730=y730+y381;y732=y732+y429;y488=y488+y430;y735=y735+y465;y738=y738+y451;y743=y743+y460;y744=y744+y444;y745=y745+y439;y747=y747+y454;y748=y748+y429;y751=y751+y443;y752=y752+y429;y753=y753+y425;y757=y757+y436;y758=y758+y433;y760=y760+y450;y761=y761+y430;y763=y763+y435;y765=y765+y433;y766=y766+y441;y775=y775+y447;y777=y777+y454;y778=y778+y426;y779=y779+y429;y781=y781+y461;y785=y785+y438;y787=y787+y450;y792=y792+y447;y793=y793+y447;y796=y796+y428;y798=y798+y449;y799=y799+y441;y802=y802+y459;y805=y805+y442;y808=y808+y435;y486=y486+y456;y812=y812+y448;y814=y814+y452;y815=y815+y464;y816=y816+y429;y824=y824+y428;y825=y825+y433;y830=y830+y436;y831=y831+y462;y833=y833+y450;y835=y835+y498;y557=y557+y499;y838=y838+y534;y841=y841+y520;y846=y846+y529;y847=y847+y513;y848=y848+y508;y850=y850+y523;y851=y851+y498;y854=y854+y512;y855=y855+y498;y856=y856+y494;y860=y860+y505;y861=y861+y502;y863=y863+y519;y864=y864+y499;y866=y866+y504;y868=y868+y502;y869=y869+y510;y878=y878+y516;y880=y880+y523;y881=y881+y495;y882=y882+y498;y884=y884+y530;y888=y888+y507;y890=y890+y519;y895=y895+y516;y896=y896+y516;y899=y899+y497;y901=y901+y518;y902=y902+y510;y905=y905+y528;y908=y908+y511;y911=y911+y504;y555=y555+y525;y915=y915+y517;y917=y917+y521;y918=y918+y533;y919=y919+y498;y927=y927+y497;y928=y928+y502;y933=y933+y505;y934=y934+y531;y936=y936+y519;y938=y938+y567;y626=y626+

y568;y941=y941+y603;y944=y944+y589;y949=y949+y598;y950=y950+y582;y951=y951+y577;y953=y953+y592;y954=y954+y567;y957=y957+y581;y958=y958+y567;y959=y959+y563;y963=y963+y574;y964=y964+y571;y966=y966+y588;y967=y967+y568;y969=y969+y573;y971=y971+y571;y972=y972+y579;y981=y981+y585;y983=y983+y592;y984=y984+y564;y985=y985+y567;y987=y987+y599;y991=y991+y576;y993=y993+y588;y998=y998+y585;y999=y999+y585;y1002=y1002+y566;y1004=y1004+y587;y1005=y1005+y579;y1008=y1008+y597;y1011=y1011+y580;y1014=y1014+y573;y624=y624+y594;y1018=y1018+y586;y1020=y1020+y590;y1021=y1021+y602;y1022=y1022+y567;y1030=y1030+y566;y1031=y1031+y571;y1036=y1036+y574;y1037=y1037+y600;y1039=y1039+y588;

#Full depth 40 and Toffoli depth 4

y405=y405+y358;y629=y629+y359;y630=y630+y389;y631=y631+y378;y419=y419+y360;y632=y632+y372;y634=y634+y369;y635=y635+y360;y637=y637+y370;y420=y420+y368;y639=y639+y369;y640=y640+y363;y641=y641+y374;y642=y642+y363;y644=y644+y365;y645=y645+y358;y646=y646+y364;y648=y648+y362;y649=y649+y357;y650=y650+y355;y651=y651+y359;y652=y652+y374;y653=y653+y382;y654=y654+y363;y655=y655+y359;y414=y414+y373;y657=y657+y360;y658=y658+y360;y660=y660+y359;y661=y661+y362;y662=y662+y361;y663=y663+y362;y665=y665+y374;y666=y666+y371;y667=y667+y383;y668=y668+y377;y669=y669+y362;y670=y670+y367;y671=y671+y386;y672=y672+y371;y674=y674+y369;y675=y675+y355;y676=y676+y356;y678=y678+y361;y679=y679+y370;y680=y680+y368;y681=y681+y380;y682=y682+y364;y683=y683+y364;y684=y684+y362;y685=y685+y387;y688=y688+y396;y689=y689+y362;y690=y690+y360;y691=y691+y374;y692=y692+y376;y693=y693+y358;y416=y416+y372;y694=y694+y380;y695=y695+y360;y696=y696+y357;y698=y698+y372;y699=y699+y366;y700=y700+y374;y701=y701+y379;y702=y702+y370;y703=y703+y373;y704=y704+y375;y705=y705+y365;y706=y706+y371;y707=y707+y389;y417=y417+y359;y709=y709+y365;y710=y710+y388;y711=y711+y360;y712=y712+y361;y713=y713+y357;y421=y421+y364;y714=y714+y376;y716=y716+y375;y719=y719+y393;y720=y720+y376;y721=y721+y358;y722=y722+y361;y723=y723+y394;y411=y411+y357;y725=y725+y388;y726=y726+y363;y727=y727+y355;y728=y728+y361;y729=y729+y365;y730=y730+y363;y731=y731+y395;y474=y474+y427;y732=y732+y428;y733=y733+y458;y734=y734+y447;y488=y488+y429;y735=y735+y441;y737=y737+y438;y738=y738+y429;y740=y740+y439;y489=y489+y437;y742=y742+y438;y743=y743+y432;y744=y744+y443;y745=y745+y432;y747=y747+y434;y748=y748+y427;y749=y749+y433;y751=y751+y431;y752=y752+y426;y753=y753+y424;y754=y754+y428;y755=y755+y443;y756=y756+y451;y757=y757+y432;y758=y758+y428;y483=y483+y442;y760=y760+y429;y761=y761+y429;y763=y763+y428;y764=y764+y431;y765=y765+y430;y766=y766+y431;y768=y768+y443;y769=y769+y440;y770=y770+y452;y771=y771+y446;y772=y772+y431;y773=y773+y436;y774=y774+y455;y775=y775+y440;y777=y777+y438;y778=y778+y424;y779=y779+y425;y781=y781+y430;y782=y782+y439;y783=y783+y437;y784=y784+y449;y785=y785+y433;y786=y786+y433;y787=y787+y431;y788=y788+y456;y791=y791+y465;y792=y792+y431;y793=y793+y429;y794=y794+y443;y795=y795+y445;y796=y796+y427;y485=y485+y441;y797=y797+y449;y798=y798+y429;y799=y799+y426;y801=y801+y441;y802=y802+y435;y803=y803+y443;y804=y804+y448;y805=y805+y439;y806=y806+y442;y807=y807+y444;y808=y808+y434;y809=y809+y440;y810=y810+y458;y486=y486+y428;y812=y812+y434;y813=y813+y457;y814=y814+y429;y815=y815+y430;y816=y816+y426;y490=y490+y433;y817=y817+y445;y819=y819+y444;y822=y822+y462;y823=y823+y445;y824=y824+y427;y825=y825+y430;y826=y826+y463;y480=y480+y426;y828=y828+y457;y829=y829+y432;y830=y830+y424;y831=y831+y430;y832=y832+y434;y833=y833+y432;y834=y834+y464;y543=y543+y496;y835=y835+y497;y836=y836+y527;y837=y837+y516;y557=y557+y498;y838=y838+y510;y840=y840+y507;y841=y841+y498;y843=y843+y508;y558=y558+y506;y845=y845+y507;y846=y846+y501;y847=y847+y512;y848=y848+y501;y850=y850+y503;y851=y851+y496;y852=y852+y502;y854=y854+y500;y855=y855+y495;y856=y856+y493;y857=y857+y497;y858=y858+y512;y859=y859+y520;y860=y860+y501;y861=y861+y497;y552=y552+y511;y863=y863+y498;y864=y864+y498;y866=y866+y497;y867=y867+y500;y868=y868+y499;y869=y869+y500;y871=y871+y512;y872=y872+y509;y873=y873+y521;y874=y874+y515;y875=y875+y500;y876=y876+y505;y877=y877+y524;y878=y878+y509;y880=y880+y507;y881=y881+y493;y882=y882+y494;y884=y884+y499;y885=y885+y508;y886=y886+y506;y887=y887+y518;y888=y888+y502;y889=y889+y502;y890=y890+y500;y891=y891+y525;y894=y894+y534;y895=y895+y500;y896=y896+y498;y897=y897+y512;y898=y898+y514;y899=y899+y496;y554=y554+y510;y900=y900+y518;y901=y901+y498;y902=y902+y495;y904=y904+y510;y905=y905+y504;y906=y906+y512;y907=y907+y517;y908=y908+y508;y909=y909+y511;y910=y910+y513;y911=y911+y503;y912=y912+y509;y913=y913+y527;y555=y555+y497;y915=y915+y503;y916=y916+y526;y917=y917+y498;y918=y918+y499;y919=y919+y495;y559=y559+y502;y920=y920+y514;y922=y922+y513;y925=y925+y531;y926=y926+y514;y927=y927+y496;y928=y928+y499;y929=y929+y532;y549=y549+y495;y931=y931+y526;y932=y932+y501;y933=y933+y493;y934=y934+y499;y935=y935+y503;y936=y936+y501;y937=y937

7+y533;y612=y612+y565;y938=y938+y566;y939=y939+y596;y940=y940+y585;y626=y626+y567;y941=y941+y579;y943=y943+y576;y944=y944+y567;y946=y946+y577;y627=y627+y575;y948=y948+y576;y949=y949+y570;y950=y950+y581;y951=y951+y570;y953=y953+y572;y954=y954+y565;y955=y955+y571;y957=y957+y569;y958=y958+y564;y959=y959+y562;y960=y960+y566;y961=y961+y581;y962=y962+y589;y963=y963+y570;y964=y964+y566;y621=y621+y580;y966=y966+y567;y967=y967+y567;y969=y969+y566;y970=y970+y569;y971=y971+y568;y972=y972+y569;y974=y974+y581;y975=y975+y578;y976=y976+y590;y977=y977+y584;y978=y978+y569;y979=y979+y574;y980=y980+y593;y981=y981+y578;y983=y983+y576;y984=y984+y562;y985=y985+y563;y987=y987+y568;y988=y988+y577;y989=y989+y575;y990=y990+y587;y991=y991+y571;y992=y992+y571;y993=y993+y569;y994=y994+y594;y997=y997+y603;y998=y998+y569;y999=y999+y567;y1000=y1000+y581;y1001=y1001+y583;y1002=y1002+y565;y623=y623+y579;y1003=y1003+y587;y1004=y1004+y567;y1005=y1005+y564;y1007=y1007+y579;y1008=y1008+y573;y1009=y1009+y581;y1010=y1010+y586;y1011=y1011+y577;y1012=y1012+y580;y1013=y1013+y582;y1014=y1014+y572;y1015=y1015+y578;y1016=y1016+y596;y624=y624+y566;y1018=y1018+y572;y1019=y1019+y595;y1020=y1020+y567;y1021=y1021+y568;y1022=y1022+y564;y628=y628+y571;y1023=y1023+y583;y1025=y1025+y582;y1028=y1028+y600;y1029=y1029+y583;y1030=y1030+y565;y1031=y1031+y568;y1032=y1032+y601;y618=y618+y564;y1034=y1034+y595;y1035=y1035+y570;y1036=y1036+y562;y1037=y1037+y568;y1038=y1038+y572;y1039=y1039+y570;y1040=y1040+y602;

#Full depth 41 and Toffoli depth 4

y405=y405+y357;y398=y398+y370;y629=y629+y358;y407=y407+y368;y630=y630+y356;y631=y631+y366;y419=y419+y358;y632=y632+y359;y633=y633+y391;y634=y634+y367;y635=y635+y357;y397=y397+y384;y636=y636+y363;y637=y637+y358;y638=y638+y386;y420=y420+y357;y639=y639+y355;y640=y640+y359;y641=y641+y358;y642=y642+y358;y643=y643+y368;y644=y644+y357;y645=y645+y357;y646=y646+y360;y647=y647+y384;y648=y648+y357;y649=y649+y356;y650=y650+y354;y651=y651+y358;y652=y652+y355;y404=y404+y377;y653=y653+y368;y654=y654+y359;y655=y655+y357;y414=y414+y360;y657=y657+y358;y658=y658+y354;y412=y412+y371;y660=y660+y356;y661=y661+y357;y662=y662+y359;y663=y663+y357;y664=y664+y360;y665=y665+y372;y666=y666+y361;y667=y667+y362;y668=y668+y366;y669=y669+y358;y670=y670+y354;y671=y671+y362;y672=y672+y356;y674=y674+y361;y675=y675+y354;y676=y676+y355;y677=y677+y392;y678=y678+y359;y679=y679+y356;y680=y680+y367;y681=y681+y365;y682=y682+y356;y683=y683+y360;y684=y684+y358;y685=y685+y358;y687=y687+y359;y688=y688+y377;y689=y689+y357;y690=y690+y356;y691=y691+y373;y692=y692+y358;y693=y693+y356;y416=y416+y360;y415=y415+y364;y694=y694+y355;y695=y695+y355;y696=y696+y356;y697=y697+y375;y698=y698+y361;y699=y699+y359;y700=y700+y368;y701=y701+y355;y702=y702+y356;y703=y703+y365;y704=y704+y359;y705=y705+y359;y706=y706+y363;y707=y707+y358;y417=y417+y358;y418=y418+y365;y708=y708+y396;y709=y709+y359;y408=y408+y362;y710=y710+y376;y711=y711+y356;y712=y712+y358;y713=y713+y356;y421=y421+y359;y714=y714+y361;y716=y716+y356;y717=y717+y371;y718=y718+y366;y719=y719+y371;y720=y720+y357;y721=y721+y357;y722=y722+y360;y723=y723+y366;y724=y724+y383;y411=y411+y355;y725=y725+y359;y726=y726+y360;y727=y727+y354;y728=y728+y359;y729=y729+y359;y730=y730+y358;y731=y731+y356;y474=y474+y426;y467=y467+y439;y732=y732+y427;y476=y476+y437;y733=y733+y425;y734=y734+y435;y488=y488+y427;y735=y735+y428;y736=y736+y460;y737=y737+y436;y738=y738+y426;y466=y466+y453;y739=y739+y432;y740=y740+y427;y741=y741+y455;y489=y489+y426;y742=y742+y424;y743=y743+y428;y744=y744+y427;y745=y745+y427;y746=y746+y437;y747=y747+y426;y748=y748+y426;y749=y749+y429;y750=y750+y453;y751=y751+y426;y752=y752+y425;y753=y753+y423;y754=y754+y427;y755=y755+y424;y756=y756+y437;y757=y757+y428;y758=y758+y426;y483=y483+y429;y760=y760+y427;y761=y761+y423;y481=y481+y440;y763=y763+y425;y764=y764+y426;y765=y765+y428;y766=y766+y426;y767=y767+y429;y768=y768+y441;y769=y769+y430;y770=y770+y431;y771=y771+y435;y772=y772+y427;y773=y773+y423;y774=y774+y431;y775=y775+y425;y777=y777+y430;y778=y778+y423;y779=y779+y424;y780=y780+y461;y781=y781+y428;y782=y782+y425;y783=y783+y436;y784=y784+y434;y785=y785+y425;y786=y786+y429;y787=y787+y427;y788=y788+y427;y790=y790+y428;y791=y791+y446;y792=y792+y426;y793=y793+y425;y794=y794+y442;y795=y795+y427;y796=y796+y425;y485=y485+y429;y484=y484+y433;y797=y797+y424;y798=y798+y424;y799=y799+y425;y800=y800+y444;y801=y801+y430;y802=y802+y428;y803=y803+y437;y804=y804+y424;y805=y805+y425;y806=y806+y434;y807=y807+y428;y808=y808+y428;y809=y809+y432;y810=y810+y427;y486=y486+y427;y487=y487+y434;y811=y811+y465;y812=y812+y428;y477=y477+y431;y813=y813+y445;y814=y814+y425;y815=y815+y427;y816=y816+y425;y490=y490+y428;y817=y817+y430;y819=y819+y425;y820=y820+y440;y821=y821+y435;y822=y822+y440;y823=y823+y426;y824=y824+y426;y825=y825+y429;y826=y826+y435;y827=y827+y452;y480=y480+y424;y828=y828+y428;y829=y829+y4

29;y830=y830+y423;y831=y831+y428;y832=y832+y428;y833=y833+y427;y834=y834+y425;y543=y543+y495;y536=y536+y508;y835=y835+y496;y545=y545+y506;y836=y836+y494;y837=y837+y504;y557=y557+y496;y838=y838+y497;y839=y839+y529;y840=y840+y505;y841=y841+y495;y535=y535+y522;y842=y842+y501;y843=y843+y496;y844=y844+y524;y558=y558+y495;y845=y845+y493;y846=y846+y497;y847=y847+y496;y848=y848+y496;y849=y849+y506;y850=y850+y495;y851=y851+y495;y852=y852+y498;y853=y853+y522;y854=y854+y495;y855=y855+y494;y856=y856+y492;y857=y857+y496;y858=y858+y493;y542=y542+y515;y859=y859+y506;y860=y860+y497;y861=y861+y495;y552=y552+y498;y863=y863+y496;y864=y864+y492;y550=y550+y509;y866=y866+y494;y867=y867+y495;y868=y868+y497;y869=y869+y495;y870=y870+y498;y871=y871+y510;y872=y872+y499;y873=y873+y500;y874=y874+y504;y875=y875+y496;y876=y876+y492;y877=y877+y500;y878=y878+y494;y880=y880+y499;y881=y881+y492;y882=y882+y493;y883=y883+y530;y884=y884+y497;y885=y885+y494;y886=y886+y505;y887=y887+y503;y888=y888+y494;y889=y889+y498;y890=y890+y496;y891=y891+y496;y893=y893+y497;y894=y894+y515;y895=y895+y495;y896=y896+y494;y897=y897+y511;y898=y898+y496;y899=y899+y494;y554=y554+y498;y553=y553+y502;y900=y900+y493;y901=y901+y493;y902=y902+y494;y903=y903+y513;y904=y904+y499;y905=y905+y497;y906=y906+y506;y907=y907+y493;y908=y908+y494;y909=y909+y503;y910=y910+y497;y911=y911+y497;y912=y912+y501;y913=y913+y496;y555=y555+y496;y556=y556+y503;y914=y914+y534;y915=y915+y497;y546=y546+y500;y916=y916+y514;y917=y917+y494;y918=y918+y496;y919=y919+y494;y559=y559+y497;y920=y920+y499;y922=y922+y494;y923=y923+y509;y924=y924+y504;y925=y925+y509;y926=y926+y495;y927=y927+y495;y928=y928+y498;y929=y929+y504;y930=y930+y521;y549=y549+y493;y931=y931+y497;y932=y932+y498;y933=y933+y492;y934=y934+y497;y935=y935+y497;y936=y936+y496;y937=y937+y494;y612=y612+y564;y605=y605+y577;y938=y938+y565;y614=y614+y575;y939=y939+y563;y940=y940+y573;y626=y626+y565;y941=y941+y566;y942=y942+y598;y943=y943+y574;y944=y944+y564;y604=y604+y591;y945=y945+y570;y946=y946+y565;y947=y947+y593;y627=y627+y564;y948=y948+y562;y949=y949+y566;y950=y950+y565;y951=y951+y565;y952=y952+y575;y953=y953+y564;y954=y954+y564;y955=y955+y567;y956=y956+y591;y957=y957+y564;y958=y958+y563;y959=y959+y561;y960=y960+y565;y961=y961+y562;y611=y611+y584;y962=y962+y575;y963=y963+y566;y964=y964+y564;y621=y621+y567;y966=y966+y565;y967=y967+y561;y619=y619+y578;y969=y969+y563;y970=y970+y564;y971=y971+y566;y972=y972+y564;y973=y973+y567;y974=y974+y579;y975=y975+y568;y976=y976+y569;y977=y977+y573;y978=y978+y565;y979=y979+y561;y980=y980+y569;y981=y981+y563;y983=y983+y568;y984=y984+y561;y985=y985+y562;y986=y986+y599;y987=y987+y566;y988=y988+y563;y989=y989+y574;y990=y990+y572;y991=y991+y563;y992=y992+y567;y993=y993+y565;y994=y994+y565;y996=y996+y566;y997=y997+y584;y998=y998+y564;y999=y999+y563;y1000=y1000+y580;y1001=y1001+y565;y1002=y1002+y563;y623=y623+y567;y622=y622+y571;y1003=y1003+y562;y1004=y1004+y562;y1005=y1005+y563;y1006=y1006+y582;y1007=y1007+y568;y1008=y1008+y566;y1009=y1009+y575;y1010=y1010+y562;y1011=y1011+y563;y1012=y1012+y572;y1013=y1013+y566;y1014=y1014+y566;y1015=y1015+y570;y1016=y1016+y565;y624=y624+y565;y625=y625+y572;y1017=y1017+y603;y1018=y1018+y566;y615=y615+y569;y1019=y1019+y583;y1020=y1020+y563;y1021=y1021+y565;y1022=y1022+y563;y628=y628+y566;y1023=y1023+y568;y1025=y1025+y563;y1026=y1026+y578;y1027=y1027+y573;y1028=y1028+y578;y1029=y1029+y564;y1030=y1030+y564;y1031=y1031+y567;y1032=y1032+y573;y1033=y1033+y590;y618=y618+y562;y1034=y1034+y566;y1035=y1035+y567;y1036=y1036+y561;y1037=y1037+y566;y1038=y1038+y566;y1039=y1039+y565;y1040=y1040+y563;

#Full depth 42 and Toffoli depth 4

y405=y405+y354;y406=y406+y366;y398=y398+y355;y629=y629+y357;y407=y407+y366;y630=y630+y353;y631=y631+y358;y419=y419+y354;y632=y632+y357;y402=y402+y359;y633=y633+y379;y634=y634+y356;y635=y635+y355;y397=y397+y360;y636=y636+y357;y637=y637+y354;y638=y638+y363;y420=y420+y354;y409=y409+y363;y639=y639+y354;y640=y640+y358;y410=y410+y362;y641=y641+y354;y642=y642+y354;y643=y643+y367;y644=y644+y356;y645=y645+y355;y646=y646+y353;y647=y647+y362;y648=y648+y355;y649=y649+y354;y650=y650+y353;y651=y651+y354;y652=y652+y354;y404=y404+y358;y653=y653+y360;y654=y654+y354;y655=y655+y353;y656=y656+y394;y414=y414+y357;y657=y657+y355;y658=y658+y353;y412=y412+y368;y659=y659+y382;y660=y660+y355;y661=y661+y353;y662=y662+y353;y663=y663+y353;y664=y664+y354;y665=y665+y369;y666=y666+y355;y667=y667+y354;y668=y668+y359;y669=y669+y354;y670=y670+y353;y671=y671+y355;y672=y672+y355;y673=y673+y357;y674=y674+y358;y675=y675+y353;y676=y676+y353;y677=y677+y356;y678=y678+y358;y679=y679+y355;y680=y680+y359;y681=y681+y362;y682=y682+y354;y683=y683+y353;y684=y684+y353;y685=y685+y354;y686=y686+y385;y403=y403+y358;y687=y687+y357;y688=y688+y358;y689=y689+y355;y690=y690+y355;y691=y691+y354;y692=y692+y357;y693=y693+y354;y416=y416+y359;y415=y415+y357;y694=y694+y354;y695=y695+y353;y696=y696+y353;

6+y355;y697=y697+y360;y698=y698+y360;y699=y699+y358;y399=y399+y364;y700=y700+y357;y701=y701+y354;y702=y702+y354;y703=y703+y356;y704=y704+y353;y705=y705+y356;y706=y706+y356;y707=y707+y353;y417=y417+y357;y418=y418+y360;y708=y708+y364;y709=y709+y356;y408=y408+y357;y710=y710+y358;y711=y711+y354;y712=y712+y354;y400=y400+y369;y713=y713+y354;y421=y421+y356;y714=y714+y355;y715=y715+y358;y716=y716+y354;y717=y717+y370;y718=y718+y365;y401=y401+y373;y719=y719+y359;y720=y720+y355;y721=y721+y356;y722=y722+y353;y723=y723+y359;y724=y724+y354;y411=y411+y353;y725=y725+y355;y726=y726+y358;y727=y727+y353;y728=y728+y358;y729=y729+y357;y730=y730+y356;y731=y731+y355;y474=y474+y423;y475=y475+y435;y467=y467+y424;y732=y732+y426;y476=y476+y435;y733=y733+y422;y734=y734+y427;y488=y488+y423;y735=y735+y426;y471=y471+y428;y736=y736+y448;y737=y737+y425;y738=y738+y424;y466=y466+y429;y739=y739+y426;y740=y740+y423;y741=y741+y432;y489=y489+y423;y478=y478+y432;y742=y742+y423;y743=y743+y427;y479=y479+y431;y744=y744+y423;y745=y745+y423;y746=y746+y436;y747=y747+y425;y748=y748+y424;y749=y749+y422;y750=y750+y431;y751=y751+y424;y752=y752+y423;y753=y753+y422;y754=y754+y423;y755=y755+y423;y473=y473+y427;y756=y756+y429;y757=y757+y423;y758=y758+y422;y759=y759+y463;y483=y483+y426;y760=y760+y424;y761=y761+y422;y481=y481+y437;y762=y762+y451;y763=y763+y424;y764=y764+y422;y765=y765+y422;y766=y766+y422;y767=y767+y423;y768=y768+y438;y769=y769+y424;y770=y770+y423;y771=y771+y428;y772=y772+y423;y773=y773+y422;y774=y774+y424;y775=y775+y424;y776=y776+y426;y777=y777+y427;y778=y778+y422;y779=y779+y422;y780=y780+y425;y781=y781+y427;y782=y782+y424;y783=y783+y428;y784=y784+y431;y785=y785+y423;y786=y786+y422;y787=y787+y422;y788=y788+y423;y789=y789+y454;y472=y472+y427;y790=y790+y426;y791=y791+y427;y792=y792+y424;y793=y793+y424;y794=y794+y423;y795=y795+y426;y796=y796+y423;y485=y485+y428;y484=y484+y426;y797=y797+y423;y798=y798+y422;y799=y799+y424;y800=y800+y429;y801=y801+y429;y802=y802+y427;y468=y468+y433;y803=y803+y426;y804=y804+y423;y805=y805+y423;y806=y806+y425;y807=y807+y422;y808=y808+y425;y809=y809+y425;y810=y810+y422;y486=y486+y426;y487=y487+y429;y811=y811+y433;y812=y812+y425;y477=y477+y426;y813=y813+y427;y814=y814+y423;y815=y815+y423;y469=y469+y438;y816=y816+y423;y490=y490+y425;y817=y817+y424;y818=y818+y427;y819=y819+y423;y820=y820+y439;y821=y821+y434;y470=y470+y442;y822=y822+y428;y823=y823+y424;y824=y824+y425;y825=y825+y422;y826=y826+y428;y827=y827+y423;y480=y480+y422;y828=y828+y424;y829=y829+y427;y830=y830+y422;y831=y831+y427;y832=y832+y426;y833=y833+y425;y834=y834+y424;y543=y543+y492;y544=y544+y504;y536=y536+y493;y835=y835+y495;y545=y545+y504;y836=y836+y491;y837=y837+y496;y557=y557+y492;y838=y838+y495;y540=y540+y497;y839=y839+y517;y840=y840+y494;y841=y841+y493;y535=y535+y498;y842=y842+y495;y843=y843+y492;y844=y844+y501;y558=y558+y492;y547=y547+y501;y845=y845+y492;y846=y846+y496;y548=y548+y500;y847=y847+y492;y848=y848+y492;y849=y849+y505;y850=y850+y494;y851=y851+y493;y852=y852+y491;y853=y853+y500;y854=y854+y493;y855=y855+y492;y856=y856+y491;y857=y857+y492;y858=y858+y492;y542=y542+y496;y859=y859+y498;y860=y860+y492;y861=y861+y491;y862=y862+y532;y552=y552+y495;y863=y863+y493;y864=y864+y491;y550=y550+y506;y865=y865+y520;y866=y866+y493;y867=y867+y491;y868=y868+y491;y869=y869+y491;y870=y870+y492;y871=y871+y507;y872=y872+y493;y873=y873+y492;y874=y874+y497;y875=y875+y492;y876=y876+y491;y877=y877+y493;y878=y878+y493;y879=y879+y495;y880=y880+y496;y881=y881+y491;y882=y882+y491;y883=y883+y494;y884=y884+y496;y885=y885+y493;y886=y886+y497;y887=y887+y500;y888=y888+y492;y889=y889+y491;y890=y890+y491;y891=y891+y492;y892=y892+y523;y541=y541+y496;y893=y893+y495;y894=y894+y496;y895=y895+y493;y896=y896+y493;y897=y897+y492;y898=y898+y495;y899=y899+y492;y554=y554+y497;y553=y553+y495;y900=y900+y492;y901=y901+y491;y902=y902+y493;y903=y903+y498;y904=y904+y498;y905=y905+y496;y537=y537+y502;y906=y906+y495;y907=y907+y492;y908=y908+y492;y909=y909+y494;y910=y910+y491;y911=y911+y494;y912=y912+y494;y913=y913+y491;y555=y555+y495;y556=y556+y498;y914=y914+y502;y915=y915+y494;y546=y546+y495;y916=y916+y496;y917=y917+y492;y918=y918+y492;y538=y538+y507;y919=y919+y492;y559=y559+y494;y920=y920+y493;y921=y921+y496;y922=y922+y492;y923=y923+y508;y924=y924+y503;y539=y539+y511;y925=y925+y497;y926=y926+y493;y927=y927+y494;y928=y928+y491;y929=y929+y497;y930=y930+y492;y549=y549+y491;y931=y931+y493;y932=y932+y496;y933=y933+y491;y934=y934+y496;y935=y935+y495;y936=y936+y494;y937=y937+y493;y612=y612+y561;y613=y613+y573;y605=y605+y562;y938=y938+y564;y614=y614+y573;y939=y939+y560;y940=y940+y565;y626=y626+y561;y941=y941+y564;y609=y609+y566;y942=y942+y586;y943=y943+y563;y944=y944+y562;y604=y604+y567;y945=y945+y564;y946=y946+y561;y947=y947+y570;y627=y627+y561;y616=y616+y570;y948=y948+y561;y949=y949+y565;y617=y617+y569;y950=y950+y561;y951=y951+y561;y952=y952+y574;y953=y953+y563;y954=y954+y562;y955=y955+y560;y956=y956+y569;y957=y957+y562;y958=y958+y561;y959=y959+y560;y960=y960+y561;y961=y961+y561;y611=y611+y565;y962=y962+y567;y963=y963+y561

61;y964=y964+y560;y965=y965+y601;y621=y621+y564;y966=y966+y562;y967=y967+y560;y619=y619+y575;y968=y968+y589;y969=y969+y562;y970=y970+y560;y971=y971+y560;y972=y972+y560;y973=y973+y561;y974=y974+y576;y975=y975+y562;y976=y976+y561;y977=y977+y566;y978=y978+y561;y979=y979+y560;y980=y980+y562;y981=y981+y562;y982=y982+y564;y983=y983+y565;y984=y984+y560;y985=y985+y560;y986=y986+y563;y987=y987+y565;y988=y988+y562;y989=y989+y566;y990=y990+y569;y991=y991+y561;y992=y992+y560;y993=y993+y560;y994=y994+y561;y995=y995+y592;y610=y610+y565;y996=y996+y564;y997=y997+y565;y998=y998+y562;y999=y999+y562;y1000=y1000+y561;y1001=y1001+y564;y1002=y1002+y561;y623=y623+y566;y622=y622+y564;y1003=y1003+y561;y1004=y1004+y560;y1005=y1005+y562;y1006=y1006+y567;y1007=y1007+y567;y1008=y1008+y565;y606=y606+y571;y1009=y1009+y564;y1010=y1010+y561;y1011=y1011+y561;y1012=y1012+y563;y1013=y1013+y560;y1014=y1014+y563;y1015=y1015+y563;y1016=y1016+y560;y624=y624+y564;y625=y625+y567;y1017=y1017+y571;y1018=y1018+y563;y615=y615+y564;y1019=y1019+y565;y1020=y1020+y561;y1021=y1021+y561;y607=y607+y576;y1022=y1022+y561;y628=y628+y563;y1023=y1023+y562;y1024=y1024+y565;y1025=y1025+y561;y1026=y1026+y577;y1027=y1027+y572;y608=y608+y580;y1028=y1028+y566;y1029=y1029+y562;y1030=y1030+y563;y1031=y1031+y560;y1032=y1032+y566;y1033=y1033+y561;y618=y618+y560;y1034=y1034+y562;y1035=y1035+y565;y1036=y1036+y560;y1037=y1037+y565;y1038=y1038+y564;y1039=y1039+y563;y1040=y1040+y562;

#Full depth 43 and Toffoli depth 4

y405=y405+y353;y406=y406+y365;y398=y398+y354;y629=y629+y353;y407=y407+y359;y630=y630+y242;y413=y413+y241;y631=y631+y355;y419=y419+y353;y632=y632+y355;y402=y402+y356;y633=y633+y369;y634=y634+y354;y635=y635+y354;y397=y397+y355;y636=y636+y354;y637=y637+y254;y638=y638+y249;y420=y420+y353;y409=y409+y358;y639=y639+y256;y640=y640+y357;y410=y410+y354;y641=y641+y254;y642=y642+y246;y643=y643+y249;y644=y644+y248;y645=y645+y354;y646=y646+y251;y647=y647+y357;y648=y648+y253;y649=y649+y248;y650=y650+y243;y651=y651+y255;y652=y652+y241;y404=y404+y355;y653=y653+y354;y654=y654+y255;y655=y655+y246;y656=y656+y363;y414=y414+y353;y657=y657+y354;y658=y658+y245;y412=y412+y353;y659=y659+y356;y660=y660+y353;y661=y661+y248;y662=y662+y253;y663=y663+y248;y664=y664+y254;y665=y665+y356;y666=y666+y353;y667=y667+y353;y668=y668+y355;y669=y669+y242;y670=y670+y252;y671=y671+y353;y672=y672+y353;y673=y673+y256;y674=y674+y251;y675=y675+y241;y676=y676+y245;y677=y677+y353;y678=y678+y355;y679=y679+y253;y680=y680+y251;y681=y681+y355;y682=y682+y243;y683=y683+y253;y684=y684+y242;y685=y685+y242;y686=y686+y359;y403=y403+y356;y687=y687+y356;y688=y688+y355;y689=y689+y353;y690=y690+y353;y691=y691+y250;y692=y692+y353;y693=y693+y247;y416=y416+y353;y415=y415+y354;y694=y694+y247;y695=y695+y254;y696=y696+y246;y697=y697+y251;y698=y698+y248;y699=y699+y356;y399=y399+y353;y700=y700+y354;y701=y701+y241;y702=y702+y353;y703=y703+y246;y704=y704+y251;y705=y705+y355;y706=y706+y354;y707=y707+y243;y417=y417+y354;y418=y418+y357;y708=y708+y357;y709=y709+y355;y408=y408+y353;y710=y710+y353;y711=y711+y353;y712=y712+y247;y400=y400+y360;y713=y713+y252;y421=y421+y353;y714=y714+y249;y715=y715+y250;y716=y716+y249;y717=y717+y363;y718=y718+y356;y401=y401+y360;y719=y719+y243;y720=y720+y353;y721=y721+y242;y722=y722+y252;y723=y723+y244;y724=y724+y250;y411=y411+y241;y725=y725+y353;y726=y726+y243;y727=y727+y254;y728=y728+y353;y729=y729+y356;y730=y730+y246;y731=y731+y243;y474=y474+y422;y475=y475+y434;y467=y467+y423;y732=y732+y422;y476=y476+y428;y733=y733+y274;y482=y482+y273;y734=y734+y424;y488=y488+y422;y735=y735+y424;y471=y471+y425;y736=y736+y438;y737=y737+y423;y738=y738+y423;y466=y466+y424;y739=y739+y423;y740=y740+y286;y741=y741+y281;y489=y489+y422;y478=y478+y427;y742=y742+y288;y743=y743+y426;y479=y479+y423;y744=y744+y286;y745=y745+y278;y746=y746+y281;y747=y747+y280;y748=y748+y423;y749=y749+y283;y750=y750+y426;y751=y751+y285;y752=y752+y280;y753=y753+y275;y754=y754+y287;y755=y755+y273;y473=y473+y424;y756=y756+y423;y757=y757+y287;y758=y758+y278;y759=y759+y432;y483=y483+y422;y760=y760+y423;y761=y761+y277;y481=y481+y422;y762=y762+y425;y763=y763+y422;y764=y764+y280;y765=y765+y285;y766=y766+y280;y767=y767+y286;y768=y768+y425;y769=y769+y422;y770=y770+y422;y771=y771+y424;y772=y772+y274;y773=y773+y284;y774=y774+y422;y775=y775+y422;y776=y776+y288;y777=y777+y283;y778=y778+y273;y779=y779+y277;y780=y780+y422;y781=y781+y424;y782=y782+y285;y783=y783+y283;y784=y784+y424;y785=y785+y275;y786=y786+y285;y787=y787+y274;y788=y788+y274;y789=y789+y428;y472=y472+y425;y790=y790+y425;y791=y791+y424;y792=y792+y422;y793=y793+y422;y794=y794+y282;y795=y795+y422;y796=y796+y279;y485=y485+y422;y484=y484+y423;y797=y797+y279;y798=y798+y286;y799=y799+y278;y800=y800+y283;y801=y801+y280;y802=y802+y425;y468=y468+y422;y803=y803+y423;y804=y804+y273;y805=y805+y422;y806=y806+y278;y807=y807+y283;y808=y808+y4

24;y809=y809+y423;y810=y810+y275;y486=y486+y423;y487=y487+y426;y811=y811+y426;y812=y812+y424;y477=y477+y422;y813=y813+y422;y814=y814+y422;y815=y815+y279;y469=y469+y429;y816=y816+y284;y490=y490+y422;y817=y817+y281;y818=y818+y282;y819=y819+y281;y820=y820+y432;y821=y821+y425;y470=y470+y429;y822=y822+y275;y823=y823+y422;y824=y824+y274;y825=y825+y284;y826=y826+y276;y827=y827+y282;y480=y480+y273;y828=y828+y422;y829=y829+y275;y830=y830+y286;y831=y831+y422;y832=y832+y425;y833=y833+y278;y834=y834+y275;y543=y543+y491;y544=y544+y503;y536=y536+y492;y835=y835+y491;y545=y545+y497;y836=y836+y306;y551=y551+y305;y837=y837+y493;y557=y557+y491;y838=y838+y493;y540=y540+y494;y839=y839+y507;y840=y840+y492;y841=y841+y492;y535=y535+y493;y842=y842+y492;y843=y843+y318;y844=y844+y313;y558=y558+y491;y547=y547+y496;y845=y845+y320;y846=y846+y495;y548=y548+y492;y847=y847+y318;y848=y848+y310;y849=y849+y313;y850=y850+y312;y851=y851+y492;y852=y852+y315;y853=y853+y495;y854=y854+y317;y855=y855+y312;y856=y856+y307;y857=y857+y319;y858=y858+y305;y542=y542+y493;y859=y859+y492;y860=y860+y319;y861=y861+y310;y862=y862+y501;y552=y552+y491;y863=y863+y492;y864=y864+y309;y550=y550+y491;y865=y865+y494;y866=y866+y491;y867=y867+y312;y868=y868+y317;y869=y869+y312;y870=y870+y318;y871=y871+y494;y872=y872+y491;y873=y873+y491;y874=y874+y493;y875=y875+y306;y876=y876+y316;y877=y877+y491;y878=y878+y491;y879=y879+y320;y880=y880+y315;y881=y881+y305;y882=y882+y309;y883=y883+y491;y884=y884+y493;y885=y885+y317;y886=y886+y315;y887=y887+y493;y888=y888+y307;y889=y889+y317;y890=y890+y306;y891=y891+y306;y892=y892+y497;y541=y541+y494;y893=y893+y494;y894=y894+y493;y895=y895+y491;y896=y896+y491;y897=y897+y314;y898=y898+y491;y899=y899+y311;y554=y554+y491;y553=y553+y492;y900=y900+y311;y901=y901+y318;y902=y902+y310;y903=y903+y315;y904=y904+y312;y905=y905+y494;y537=y537+y491;y906=y906+y492;y907=y907+y305;y908=y908+y491;y909=y909+y310;y910=y910+y315;y911=y911+y493;y912=y912+y492;y913=y913+y307;y555=y555+y492;y556=y556+y495;y914=y914+y495;y915=y915+y493;y546=y546+y491;y916=y916+y491;y917=y917+y491;y918=y918+y311;y538=y538+y498;y919=y919+y316;y559=y559+y491;y920=y920+y313;y921=y921+y314;y922=y922+y313;y923=y923+y501;y924=y924+y494;y539=y539+y498;y925=y925+y307;y926=y926+y491;y927=y927+y306;y928=y928+y316;y929=y929+y308;y930=y930+y314;y549=y549+y305;y931=y931+y491;y932=y932+y307;y933=y933+y318;y934=y934+y491;y935=y935+y494;y936=y936+y310;y937=y937+y307;y612=y612+y560;y613=y613+y572;y605=y605+y561;y938=y938+y560;y614=y614+y566;y939=y939+y338;y620=y620+y337;y940=y940+y562;y626=y626+y560;y941=y941+y562;y609=y609+y563;y942=y942+y576;y943=y943+y561;y944=y944+y561;y604=y604+y562;y945=y945+y561;y946=y946+y350;y947=y947+y345;y627=y627+y560;y616=y616+y565;y948=y948+y352;y949=y949+y564;y617=y617+y561;y950=y950+y350;y951=y951+y342;y952=y952+y345;y953=y953+y344;y954=y954+y561;y955=y955+y347;y956=y956+y564;y957=y957+y349;y958=y958+y344;y959=y959+y339;y960=y960+y351;y961=y961+y337;y611=y611+y562;y962=y962+y561;y963=y963+y351;y964=y964+y342;y965=y965+y570;y621=y621+y560;y966=y966+y561;y967=y967+y341;y619=y619+y560;y968=y968+y563;y969=y969+y560;y970=y970+y344;y971=y971+y349;y972=y972+y344;y973=y973+y350;y974=y974+y563;y975=y975+y560;y976=y976+y560;y977=y977+y562;y978=y978+y338;y979=y979+y348;y980=y980+y560;y981=y981+y560;y982=y982+y352;y983=y983+y347;y984=y984+y337;y985=y985+y341;y986=y986+y560;y987=y987+y562;y988=y988+y349;y989=y989+y347;y990=y990+y562;y991=y991+y339;y992=y992+y349;y993=y993+y338;y994=y994+y338;y995=y995+y566;y610=y610+y563;y996=y996+y563;y997=y997+y562;y998=y998+y560;y999=y999+y560;y1000=y1000+y346;y1001=y1001+y560;y1002=y1002+y343;y623=y623+y560;y622=y622+y561;y1003=y1003+y343;y1004=y1004+y350;y1005=y1005+y342;y1006=y1006+y347;y1007=y1007+y344;y1008=y1008+y563;y606=y606+y560;y1009=y1009+y561;y1010=y1010+y337;y1011=y1011+y560;y1012=y1012+y342;y1013=y1013+y347;y1014=y1014+y562;y1015=y1015+y561;y1016=y1016+y339;y624=y624+y561;y625=y625+y564;y1017=y1017+y564;y1018=y1018+y562;y615=y615+y560;y1019=y1019+y560;y1020=y1020+y560;y1021=y1021+y343;y607=y607+y567;y1022=y1022+y348;y628=y628+y560;y1023=y1023+y345;y1024=y1024+y346;y1025=y1025+y345;y1026=y1026+y570;y1027=y1027+y563;y608=y608+y567;y1028=y1028+y339;y1029=y1029+y560;y1030=y1030+y338;y1031=y1031+y348;y1032=y1032+y340;y1033=y1033+y346;y618=y618+y337;y1034=y1034+y560;y1035=y1035+y339;y1036=y1036+y350;y1037=y1037+y560;y1038=y1038+y563;y1039=y1039+y342;y1040=y1040+y339;

#Full depth 44 and Toffoli depth 4

y353=y353+y247;y354=y354+y251;y355=y355+y242;y356=y356+y250;y357=y357+y243;y358=y358+y245;y359=y359+y248;y360=y360+y241;y361=y361+y254;y362=y362+y246;y363=y363+y244;y364=y364+y242;y365=y365+y249;y366=y366+y241;y367=y367+y241;y368=y368+y244;y369=y369+y247;y370=y370+y243;y371=y371+y245;y372=y372+y253;y373=y373+y244;y374=y374+y248;y375=y375+y242;y376=y376+y252;y377=y377+y250;y378=y378+y251;y379=y379+y255;y380=y380+y2

43;y381=y381+y248;y382=y382+y249;y383=y383+y245;y384=y384+y247;y385=y385+y241;y386=y386+y253;y387=y387+y246;y388=y388+y241;y389=y389+y248;y390=y390+y247;y391=y391+y250;y392=y392+y246;y393=y393+y244;y394=y394+y245;y395=y395+y246;y396=y396+y241;y397=y397+y254;y398=y398+y250;y399=y399+y251;y400=y400+y250;y401=y401+y249;y402=y402+y245;y403=y403+y244;y404=y404+y241;y405=y405+y252;y406=y406+y248;y407=y407+y249;y408=y408+y245;y409=y409+y250;y410=y410+y242;y411=y411+y248;y412=y412+y243;y413=y413+y249;y414=y414+y248;y415=y415+y247;y416=y416+y242;y417=y417+y242;y418=y418+y245;y419=y419+y244;y420=y420+y241;y421=y421+y246;y422=y422+y279;y423=y423+y283;y424=y424+y274;y425=y425+y282;y426=y426+y275;y427=y427+y277;y428=y428+y280;y429=y429+y273;y430=y430+y286;y431=y431+y278;y432=y432+y276;y433=y433+y274;y434=y434+y281;y435=y435+y273;y436=y436+y273;y437=y437+y276;y438=y438+y279;y439=y439+y275;y440=y440+y277;y441=y441+y285;y442=y442+y276;y443=y443+y280;y444=y444+y274;y445=y445+y284;y446=y446+y282;y447=y447+y283;y448=y448+y287;y449=y449+y275;y450=y450+y280;y451=y451+y281;y452=y452+y277;y453=y453+y279;y454=y454+y273;y455=y455+y285;y456=y456+y278;y457=y457+y273;y458=y458+y280;y459=y459+y279;y460=y460+y282;y461=y461+y278;y462=y462+y276;y463=y463+y277;y464=y464+y278;y465=y465+y273;y466=y466+y286;y467=y467+y282;y468=y468+y283;y469=y469+y282;y470=y470+y281;y471=y471+y277;y472=y472+y276;y473=y473+y273;y474=y474+y284;y475=y475+y280;y476=y476+y281;y477=y477+y277;y478=y478+y282;y479=y479+y274;y480=y480+y280;y481=y481+y275;y482=y482+y281;y483=y483+y280;y484=y484+y279;y485=y485+y274;y486=y486+y274;y487=y487+y277;y488=y488+y276;y489=y489+y273;y490=y490+y278;y491=y491+y311;y492=y492+y315;y493=y493+y306;y494=y494+y314;y495=y495+y307;y496=y496+y309;y497=y497+y312;y498=y498+y305;y499=y499+y318;y500=y500+y310;y501=y501+y308;y502=y502+y306;y503=y503+y313;y504=y504+y305;y505=y505+y305;y506=y506+y308;y507=y507+y311;y508=y508+y307;y509=y509+y309;y510=y510+y317;y511=y511+y308;y512=y512+y312;y513=y513+y306;y514=y514+y316;y515=y515+y314;y516=y516+y315;y517=y517+y319;y518=y518+y307;y519=y519+y312;y520=y520+y313;y521=y521+y309;y522=y522+y311;y523=y523+y305;y524=y524+y317;y525=y525+y310;y526=y526+y305;y527=y527+y312;y528=y528+y311;y529=y529+y314;y530=y530+y310;y531=y531+y308;y532=y532+y309;y533=y533+y310;y534=y534+y305;y535=y535+y318;y536=y536+y314;y537=y537+y315;y538=y538+y314;y539=y539+y313;y540=y540+y309;y541=y541+y308;y542=y542+y305;y543=y543+y316;y544=y544+y312;y545=y545+y313;y546=y546+y309;y547=y547+y314;y548=y548+y306;y549=y549+y312;y550=y550+y307;y551=y551+y313;y552=y552+y312;y553=y553+y311;y554=y554+y306;y555=y555+y306;y556=y556+y309;y557=y557+y308;y558=y558+y305;y559=y559+y310;y560=y560+y343;y561=y561+y347;y562=y562+y338;y563=y563+y346;y564=y564+y339;y565=y565+y341;y566=y566+y344;y567=y567+y337;y568=y568+y350;y569=y569+y342;y570=y570+y340;y571=y571+y338;y572=y572+y345;y573=y573+y337;y574=y574+y337;y575=y575+y340;y576=y576+y343;y577=y577+y339;y578=y578+y341;y579=y579+y349;y580=y580+y340;y581=y581+y344;y582=y582+y338;y583=y583+y348;y584=y584+y346;y585=y585+y347;y586=y586+y351;y587=y587+y339;y588=y588+y344;y589=y589+y345;y590=y590+y341;y591=y591+y343;y592=y592+y337;y593=y593+y349;y594=y594+y342;y595=y595+y337;y596=y596+y344;y597=y597+y343;y598=y598+y346;y599=y599+y342;y600=y600+y340;y601=y601+y341;y602=y602+y342;y603=y603+y337;y604=y604+y350;y605=y605+y346;y606=y606+y347;y607=y607+y346;y608=y608+y345;y609=y609+y341;y610=y610+y340;y611=y611+y337;y612=y612+y348;y613=y613+y344;y614=y614+y345;y615=y615+y341;y616=y616+y346;y617=y617+y338;y618=y618+y344;y619=y619+y339;y620=y620+y345;y621=y621+y344;y622=y622+y343;y623=y623+y338;y624=y624+y338;y625=y625+y341;y626=y626+y340;y627=y627+y337;y628=y628+y342;

#Full depth 45 and Toffoli depth 4

y353=y353+y249;y354=y354+y253;y355=y355+y244;y356=y356+y254;y357=y357+y255;y358=y358+y256;y359=y359+y252;y360=y360+y246;y361=y361+y255;y362=y362+y250;y363=y363+y247;y364=y364+y245;y365=y365+y251;y366=y366+y243;y367=y367+y256;y368=y368+y254;y369=y369+y252;y370=y370+y252;y371=y371+y246;y372=y372+y256;y373=y373+y256;y374=y374+y249;y375=y375+y243;y376=y376+y253;y377=y377+y253;y378=y378+y252;y379=y379+y256;y380=y380+y256;y381=y381+y255;y382=y382+y252;y383=y383+y252;y384=y384+y251;y385=y385+y242;y386=y386+y255;y387=y387+y254;y388=y388+y254;y389=y389+y253;y390=y390+y253;y391=y391+y251;y392=y392+y251;y393=y393+y251;y394=y394+y250;y395=y395+y248;y396=y396+y247;y397=y397+y256;y398=y398+y256;y399=y399+y255;y400=y400+y255;y401=y401+y255;y402=y402+y255;y403=y403+y255;y404=y404+y255;y405=y405+y254;y406=y406+y254;y407=y407+y253;y408=y408+y253;y409=y409+y252;y410=y410+y252;y411=y411+y251;y412=y412+y251;y413=y413+y250;y414=y414+y250;y415=y415+y250;y416=y416+y250;y417=y417+y249;y418=y418+y248;y419=y419+y248;y420=y420+y245;y421=y421+y255;y422=y422+y281;y423=y423+y285;y424=y424+y276;y425=y

425+y286;y426=y426+y287;y427=y427+y288;y428=y428+y284;y429=y429+y278;y430=y430+y287;
y431=y431+y282;y432=y432+y279;y433=y433+y277;y434=y434+y283;y435=y435+y275;y436=y43
6+y288;y437=y437+y286;y438=y438+y284;y439=y439+y284;y440=y440+y278;y441=y441+y288;y4
42=y442+y288;y443=y443+y281;y444=y444+y275;y445=y445+y285;y446=y446+y285;y447=y447+y
284;y448=y448+y288;y449=y449+y288;y450=y450+y287;y451=y451+y284;y452=y452+y284;y453
=y453+y283;y454=y454+y274;y455=y455+y287;y456=y456+y286;y457=y457+y286;y458=y458+y2
85;y459=y459+y285;y460=y460+y283;y461=y461+y283;y462=y462+y283;y463=y463+y282;y464=y
464+y280;y465=y465+y279;y466=y466+y288;y467=y467+y288;y468=y468+y287;y469=y469+y287;
y470=y470+y287;y471=y471+y287;y472=y472+y287;y473=y473+y287;y474=y474+y286;y475=y47
5+y286;y476=y476+y285;y477=y477+y285;y478=y478+y284;y479=y479+y284;y480=y480+y283;y4
81=y481+y283;y482=y482+y282;y483=y483+y282;y484=y484+y282;y485=y485+y282;y486=y486+y
281;y487=y487+y280;y488=y488+y280;y489=y489+y277;y490=y490+y287;y491=y491+y313;y492
=y492+y317;y493=y493+y308;y494=y494+y318;y495=y495+y319;y496=y496+y320;y497=y497+y3
16;y498=y498+y310;y499=y499+y319;y500=y500+y314;y501=y501+y311;y502=y502+y309;y503=y
503+y315;y504=y504+y307;y505=y505+y320;y506=y506+y318;y507=y507+y316;y508=y508+y316;
y509=y509+y310;y510=y510+y320;y511=y511+y320;y512=y512+y313;y513=y513+y307;y514=y51
4+y317;y515=y515+y317;y516=y516+y316;y517=y517+y320;y518=y518+y320;y519=y519+y319;y5
20=y520+y316;y521=y521+y316;y522=y522+y315;y523=y523+y306;y524=y524+y319;y525=y525+y
318;y526=y526+y318;y527=y527+y317;y528=y528+y317;y529=y529+y315;y530=y530+y315;y531
=y531+y315;y532=y532+y314;y533=y533+y312;y534=y534+y311;y535=y535+y320;y536=y536+y3
20;y537=y537+y319;y538=y538+y319;y539=y539+y319;y540=y540+y319;y541=y541+y319;y542=y
542+y319;y543=y543+y318;y544=y544+y318;y545=y545+y317;y546=y546+y317;y547=y547+y316;
y548=y548+y316;y549=y549+y315;y550=y550+y315;y551=y551+y314;y552=y552+y314;y553=y55
3+y314;y554=y554+y314;y555=y555+y313;y556=y556+y312;y557=y557+y312;y558=y558+y309;y5
59=y559+y319;y560=y560+y345;y561=y561+y349;y562=y562+y340;y563=y563+y350;y564=y564+y
351;y565=y565+y352;y566=y566+y348;y567=y567+y342;y568=y568+y351;y569=y569+y346;y570
=y570+y343;y571=y571+y341;y572=y572+y347;y573=y573+y339;y574=y574+y352;y575=y575+y3
50;y576=y576+y348;y577=y577+y348;y578=y578+y342;y579=y579+y352;y580=y580+y352;y581=y
581+y345;y582=y582+y339;y583=y583+y349;y584=y584+y349;y585=y585+y348;y586=y586+y352;
y587=y587+y352;y588=y588+y351;y589=y589+y348;y590=y590+y348;y591=y591+y347;y592=y59
2+y338;y593=y593+y351;y594=y594+y350;y595=y595+y350;y596=y596+y349;y597=y597+y349;y5
98=y598+y347;y599=y599+y347;y600=y600+y347;y601=y601+y346;y602=y602+y344;y603=y603+y
343;y604=y604+y352;y605=y605+y352;y606=y606+y351;y607=y607+y351;y608=y608+y351;y609
=y609+y351;y610=y610+y351;y611=y611+y351;y612=y612+y350;y613=y613+y350;y614=y614+y3
49;y615=y615+y349;y616=y616+y348;y617=y617+y348;y618=y618+y347;y619=y619+y347;y620=y
620+y346;y621=y621+y346;y622=y622+y346;y623=y623+y346;y624=y624+y345;y625=y625+y344;
y626=y626+y344;y627=y627+y341;y628=y628+y351;y129=y129+y1;y130=y130+y2;y131=y131+y3;
y132=y132+y4;y133=y133+y5;y134=y134+y6;y135=y135+y7;y136=y136+y8;y137=y137+y9;y138=y
138+y10;y139=y139+y11;y140=y140+y12;y141=y141+y13;y142=y142+y14;y143=y143+y15;y144=y
144+y16;y145=y145+y17;y146=y146+y18;y147=y147+y19;y148=y148+y20;y149=y149+y21;y150=y
150+y22;y151=y151+y23;y152=y152+y24;y153=y153+y25;y154=y154+y26;y155=y155+y27;y156=y
156+y28;y157=y157+y29;y158=y158+y30;y159=y159+y31;y160=y160+y32;

#Full depth 46 and Toffoli depth 5

y225=y225+y161*y165;y226=y226+y161*y166;y227=y227+y161*y167;y228=y228+y161*y168;y22
9=y229+y162*y165;y230=y230+y162*y166;y231=y231+y162*y167;y232=y232+y162*y168;y233=y
233+y163*y165;y234=y234+y163*y166;y235=y235+y163*y167;y236=y236+y163*y168;y237=y237
+y164*y165;y238=y238+y164*y166;y239=y239+y164*y167;y240=y240+y164*y168;y241=y241+y1
69*y173;y242=y242+y169*y174;y243=y243+y169*y175;y244=y244+y169*y176;y245=y245+y170*
y173;y246=y246+y170*y174;y247=y247+y170*y175;y248=y248+y170*y176;y249=y249+y171*y17
3;y250=y250+y171*y174;y251=y251+y171*y175;y252=y252+y171*y176;y253=y253+y172*y173;y2
54=y254+y172*y174;y255=y255+y172*y175;y256=y256+y172*y176;y257=y257+y177*y181;y258=
y258+y177*y182;y259=y259+y177*y183;y260=y260+y177*y184;y261=y261+y178*y181;y262=y26
2+y178*y182;y263=y263+y178*y183;y264=y264+y178*y184;y265=y265+y179*y181;y266=y266+y
179*y182;y267=y267+y179*y183;y268=y268+y179*y184;y269=y269+y180*y181;y270=y270+y180
*y182;y271=y271+y180*y183;y272=y272+y180*y184;y273=y273+y185*y189;y274=y274+y185*y1
90;y275=y275+y185*y191;y276=y276+y185*y192;y277=y277+y186*y189;y278=y278+y186*y190;y
279=y279+y186*y191;y280=y280+y186*y192;y281=y281+y187*y189;y282=y282+y187*y190;y283
=y283+y187*y191;y284=y284+y187*y192;y285=y285+y188*y189;y286=y286+y188*y190;y287=y2
87+y188*y191;y288=y288+y188*y192;y289=y289+y193*y197;y290=y290+y193*y198;y291=y291+

$y_{193}=y_{199}; y_{292}=y_{292}+y_{193}*y_{200}; y_{293}=y_{293}+y_{194}*y_{197}; y_{294}=y_{294}+y_{194}*y_{198}; y_{295}=y_{295}+y_{194}*y_{199}; y_{296}=y_{296}+y_{194}*y_{200}; y_{297}=y_{297}+y_{195}*y_{197}; y_{298}=y_{298}+y_{195}*y_{198}; y_{299}=y_{299}+y_{195}*y_{199}; y_{300}=y_{300}+y_{195}*y_{200}; y_{301}=y_{301}+y_{196}*y_{197}; y_{302}=y_{302}+y_{196}*y_{198}; y_{303}=y_{303}+y_{196}*y_{199}; y_{304}=y_{304}+y_{196}*y_{200}; y_{305}=y_{305}+y_{201}*y_{205}; y_{306}=y_{306}+y_{201}*y_{206}; y_{307}=y_{307}+y_{201}*y_{207}; y_{308}=y_{308}+y_{201}*y_{208}; y_{309}=y_{309}+y_{202}*y_{205}; y_{310}=y_{310}+y_{202}*y_{206}; y_{311}=y_{311}+y_{202}*y_{207}; y_{312}=y_{312}+y_{202}*y_{208}; y_{313}=y_{313}+y_{203}*y_{205}; y_{314}=y_{314}+y_{203}*y_{206}; y_{315}=y_{315}+y_{203}*y_{207}; y_{316}=y_{316}+y_{203}*y_{208}; y_{317}=y_{317}+y_{204}*y_{205}; y_{318}=y_{318}+y_{204}*y_{206}; y_{319}=y_{319}+y_{204}*y_{207}; y_{320}=y_{320}+y_{204}*y_{208}; y_{321}=y_{321}+y_{209}*y_{213}; y_{322}=y_{322}+y_{209}*y_{214}; y_{323}=y_{323}+y_{209}*y_{215}; y_{324}=y_{324}+y_{209}*y_{216}; y_{325}=y_{325}+y_{210}*y_{213}; y_{326}=y_{326}+y_{210}*y_{214}; y_{327}=y_{327}+y_{210}*y_{215}; y_{328}=y_{328}+y_{210}*y_{216}; y_{329}=y_{329}+y_{211}*y_{213}; y_{330}=y_{330}+y_{211}*y_{214}; y_{331}=y_{331}+y_{211}*y_{215}; y_{332}=y_{332}+y_{211}*y_{216}; y_{333}=y_{333}+y_{212}*y_{213}; y_{334}=y_{334}+y_{212}*y_{214}; y_{335}=y_{335}+y_{212}*y_{215}; y_{336}=y_{336}+y_{212}*y_{216}; y_{337}=y_{337}+y_{217}*y_{221}; y_{338}=y_{338}+y_{217}*y_{222}; y_{339}=y_{339}+y_{217}*y_{223}; y_{340}=y_{340}+y_{217}*y_{224}; y_{341}=y_{341}+y_{218}*y_{221}; y_{342}=y_{342}+y_{218}*y_{222}; y_{343}=y_{343}+y_{218}*y_{223}; y_{344}=y_{344}+y_{218}*y_{224}; y_{345}=y_{345}+y_{219}*y_{221}; y_{346}=y_{346}+y_{219}*y_{222}; y_{347}=y_{347}+y_{219}*y_{223}; y_{348}=y_{348}+y_{219}*y_{224}; y_{349}=y_{349}+y_{220}*y_{221}; y_{350}=y_{350}+y_{220}*y_{222}; y_{351}=y_{351}+y_{220}*y_{223}; y_{352}=y_{352}+y_{220}*y_{224}; y_{129}=y_{129}+1; y_{130}=y_{130}+1; y_{131}=y_{131}+1; y_{132}=y_{132}+1; y_{133}=y_{133}+1; y_{134}=y_{134}+1; y_{135}=y_{135}+1; y_{136}=y_{136}+1; y_{137}=y_{137}+1; y_{138}=y_{138}+1; y_{139}=y_{139}+1; y_{140}=y_{140}+1; y_{141}=y_{141}+1; y_{142}=y_{142}+1; y_{143}=y_{143}+1; y_{144}=y_{144}+1; y_{145}=y_{145}+1; y_{146}=y_{146}+1; y_{147}=y_{147}+1; y_{148}=y_{148}+1; y_{149}=y_{149}+1; y_{150}=y_{150}+1; y_{151}=y_{151}+1; y_{152}=y_{152}+1; y_{153}=y_{153}+1; y_{154}=y_{154}+1; y_{155}=y_{155}+1; y_{156}=y_{156}+1; y_{157}=y_{157}+1; y_{158}=y_{158}+1; y_{159}=y_{159}+1; y_{160}=y_{160}+1;$

#Full depth 47 and Toffoli depth 6

$y_{161}=y_{161}+y_1*y_2; y_{162}=y_{162}+y_1*y_{130}; y_{163}=y_{163}+y_{129}*y_2; y_{164}=y_{164}+y_{129}*y_{130}; y_{165}=y_{165}+y_3*y_4; y_{166}=y_{166}+y_3*y_{132}; y_{167}=y_{167}+y_{131}*y_4; y_{168}=y_{168}+y_{131}*y_{132}; y_{169}=y_{169}+y_5*y_6; y_{170}=y_{170}+y_5*y_{134}; y_{171}=y_{171}+y_{133}*y_6; y_{172}=y_{172}+y_{133}*y_{134}; y_{173}=y_{173}+y_7*y_8; y_{174}=y_{174}+y_7*y_{136}; y_{175}=y_{175}+y_{135}*y_8; y_{176}=y_{176}+y_{135}*y_{136}; y_{177}=y_{177}+y_9*y_{10}; y_{178}=y_{178}+y_9*y_{138}; y_{179}=y_{179}+y_{137}*y_{10}; y_{180}=y_{180}+y_{137}*y_{138}; y_{181}=y_{181}+y_{11}*y_{12}; y_{182}=y_{182}+y_{11}*y_{140}; y_{183}=y_{183}+y_{139}*y_{12}; y_{184}=y_{184}+y_{139}*y_{140}; y_{185}=y_{185}+y_{13}*y_{14}; y_{186}=y_{186}+y_{13}*y_{142}; y_{187}=y_{187}+y_{141}*y_{14}; y_{188}=y_{188}+y_{141}*y_{142}; y_{189}=y_{189}+y_{15}*y_{16}; y_{190}=y_{190}+y_{15}*y_{144}; y_{191}=y_{191}+y_{143}*y_{16}; y_{192}=y_{192}+y_{143}*y_{144}; y_{193}=y_{193}+y_{17}*y_{18}; y_{194}=y_{194}+y_{17}*y_{146}; y_{195}=y_{195}+y_{145}*y_{18}; y_{196}=y_{196}+y_{145}*y_{146}; y_{197}=y_{197}+y_{19}*y_{20}; y_{198}=y_{198}+y_{19}*y_{148}; y_{199}=y_{199}+y_{147}*y_{20}; y_{200}=y_{200}+y_{147}*y_{148}; y_{201}=y_{201}+y_{21}*y_{22}; y_{202}=y_{202}+y_{21}*y_{150}; y_{203}=y_{203}+y_{149}*y_{22}; y_{204}=y_{204}+y_{149}*y_{150}; y_{205}=y_{205}+y_{23}*y_{24}; y_{206}=y_{206}+y_{23}*y_{152}; y_{207}=y_{207}+y_{151}*y_{24}; y_{208}=y_{208}+y_{151}*y_{152}; y_{209}=y_{209}+y_{25}*y_{26}; y_{210}=y_{210}+y_{25}*y_{154}; y_{211}=y_{211}+y_{153}*y_{26}; y_{212}=y_{212}+y_{153}*y_{154}; y_{213}=y_{213}+y_{27}*y_{28}; y_{214}=y_{214}+y_{27}*y_{156}; y_{215}=y_{215}+y_{155}*y_{28}; y_{216}=y_{216}+y_{155}*y_{156}; y_{217}=y_{217}+y_{29}*y_{30}; y_{218}=y_{218}+y_{29}*y_{158}; y_{219}=y_{219}+y_{157}*y_{30}; y_{220}=y_{220}+y_{157}*y_{158}; y_{221}=y_{221}+y_{31}*y_{32}; y_{222}=y_{222}+y_{31}*y_{160}; y_{223}=y_{223}+y_{159}*y_{32}; y_{224}=y_{224}+y_{159}*y_{160};$

#Full depth 48 and Toffoli depth 6

$y_{129}=y_{129}+1; y_{130}=y_{130}+1; y_{131}=y_{131}+1; y_{132}=y_{132}+1; y_{133}=y_{133}+1; y_{134}=y_{134}+1; y_{135}=y_{135}+1; y_{136}=y_{136}+1; y_{137}=y_{137}+1; y_{138}=y_{138}+1; y_{139}=y_{139}+1; y_{140}=y_{140}+1; y_{141}=y_{141}+1; y_{142}=y_{142}+1; y_{143}=y_{143}+1; y_{144}=y_{144}+1; y_{145}=y_{145}+1; y_{146}=y_{146}+1; y_{147}=y_{147}+1; y_{148}=y_{148}+1; y_{149}=y_{149}+1; y_{150}=y_{150}+1; y_{151}=y_{151}+1; y_{152}=y_{152}+1; y_{153}=y_{153}+1; y_{154}=y_{154}+1; y_{155}=y_{155}+1; y_{156}=y_{156}+1; y_{157}=y_{157}+1; y_{158}=y_{158}+1; y_{159}=y_{159}+1; y_{160}=y_{160}+1;$

#Full depth 49 and Toffoli depth 6

$y_{129}=y_{129}+y_1; y_{130}=y_{130}+y_2; y_{131}=y_{131}+y_3; y_{132}=y_{132}+y_4; y_{133}=y_{133}+y_5; y_{134}=y_{134}+y_6; y_{135}=y_{135}+y_7; y_{136}=y_{136}+y_8; y_{137}=y_{137}+y_9; y_{138}=y_{138}+y_{10}; y_{139}=y_{139}+y_{11}; y_{140}=y_{140}+y_{12}; y_{141}=y_{141}+y_{13}; y_{142}=y_{142}+y_{14}; y_{143}=y_{143}+y_{15}; y_{144}=y_{144}+y_{16}; y_{145}=y_{145}+y_{17}; y_{146}=y_{146}+y_{18}; y_{147}=y_{147}+y_{19}; y_{148}=y_{148}+y_{20}; y_{149}=y_{149}+y_{21}; y_{150}=y_{150}+y_{22}; y_{151}=y_{151}+y_{23}; y_{152}=y_{152}+y_{24}; y_{153}=y_{153}+y_{25}; y_{154}=y_{154}+y_{26}; y_{155}=y_{155}+y_{27}; y_{156}=y_{156}+y_{28}; y_{157}=y_{157}+y_{29}; y_{158}=y_{158}+y_{30}; y_{159}=y_{159}+y_{31}; y_{160}=y_{160}+y_{32};$

#Full depth 50 and Toffoli depth 6

$y_1=y_1+y_{65}; y_2=y_2+y_{66}; y_3=y_3+y_{67}; y_4=y_4+y_{68}; y_5=y_5+y_{69}; y_6=y_6+y_{70}; y_7=y_7+y_{71}; y_8=y_8+y_{72}; y_9=y_9+y_{73}; y_{10}=y_{10}+y_{74}; y_{11}=y_{11}+y_{75}; y_{12}=y_{12}+y_{76}; y_{13}=y_{13}+y_{77}; y_{14}=y_{14}+y_{78}; y_{15}=y_{15}+y_{79}; y_{16}=y_{16}+y_{80}; y_{17}=y_{17}+y_{81}; y_{18}=y_{18}+y_{82}; y_{19}=y_{19}+y_{83}; y_{20}=y_{20}+y_{84}; y_{21}=y_{21}+y_{85}; y_{22}=y_{22}+y_{86}; y_{23}=y_{23}+y_{87}; y_{24}=y_{24}+y_{88}; y_{25}=y_{25}+y_{89}; y_{26}=y_{26}+y_{90}; y_{27}=y_{27}+y_{91}; y_{28}=y_{28}+y_{92}; y_{29}=y_{29}+y_{93}; y_{30}=y_{30}+y_{94}; y_{31}=y_{31}+y_{95}; y_{32}=y_{32}+y_{96};$

#向量布尔函数输出与目标结果验证

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